

Appendix F

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C2
		Application Number	
Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module		
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number		NETL.046C2	
		Application Number			
Title of Invention		Apparatus and Method for Self-Test in a Multi-Rank Memory Module			
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Title of the Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module				
Attorney Docket Number	NETL.046C2	Small Entity Status Claimed ☒			
Application Type	Nonprovisional				
Subject Matter	Utility				
Suggested Class (if any)		Sub Class (if any)			
Suggested Technology Center (if any)					
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)	2		

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C2
		Application Number	
Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module		
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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	13183253	2011-07-14
Prior Application Status		Remove	
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13183253	Continuation of	12422925	2009-04-13
Prior Application Status		Remove	
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12422925	non provisional of	61044801	2008-04-14
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C2
		Application Number	
Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module		

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C2
		Application Number	
Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module		

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Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module		

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Apparatus and Method for Self-Test in a Multi-Rank Memory Module

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. Patent Application No. 13/183,253, filed July 14, 2011, which is a continuation of U.S. Patent Application No. 12/422,925, filed April 13, 2009, which claims the benefit of priority from U.S. Provisional Application No. 61/044,801, filed April 14, 2008, U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and U.S. Provisional Application No. 61/044,839, filed April 14, 2008. Each of the foregoing applications is incorporated in their entirety by reference herein. This application is related to US. Patent Application No. 12/422,912, filed on April 13, 2009 and entitled "Self-Adjusting Damper", and to US. Application No. 12/422,853, filed on April 13, 2009 and entitled "Circuit Providing Load Isolation and Noise Reduction", both of which are incorporated in their entirety by reference herein.

BACKGROUND

Field

[0002] The present invention relates to self-testing electronic modules and, more particularly, to self-testing electronic memory modules.

Description of the Related Art

[0003] The failure of memory components in an electronic system may result in the loss of valid data. Therefore, it is important to ensure proper memory operation in an electronic system. Memory integrated circuits ("memory chips") often go through a series of tests at various stages of system manufacture. Once memory chips are deployed in a system, they also generally go through a system level memory test each time the system is booted. In addition, memory chips may undergo a parity checking process during normal system operation.

[0004] There are typically at least three test phases which memories undergo during system manufacture. Each phase generally tests for memory defects and for the correct operation of the input/output interface. The first test phase is typically conducted by the memory chip manufacturer and generally involves checking for bit failures, correct memory access speed, etc. The second test phase is typically done by memory module manufacturers and generally involves testing the signal quality, the noise susceptibility, and the operational

speed of the memory module as a single unit. The second test phase may also include checking for bit failures in individual memory chips. The third phase is usually carried out by the system manufacturer. During the third phase, the interaction of the memory subsystem with other components in the system is tested. During the third phase, the individual memory module operation is also tested again and the memory array is checked for defects. Because of the significant amount of testing that memories undergo during the manufacturing process, there is generally substantial test cost and test time associated with ensuring the proper memory operation. This test cost and test time translate into an increase in system cost and a decrease in system performance.

[0005] There are a number of memory test methodologies that employ either external test hardware, embedded self-test logic (“MBIST”), or both. However, the usefulness of these test methodologies is limited due to the high cost and other limitations associated with them. For instance, external test hardware such as automatic test equipment (“ATE”) is very expensive. Moreover, the development time and cost associated with implementing MBIST is relatively high. These costs and limitations are especially significant when testing dynamic random access memory (“DRAM”). For example, technological developments, such as increases in DRAM speed, may require manufacturers to upgrade ATE machines relatively frequently. In addition, MBIST in DRAM chips generally cannot be fully utilized for system level testing of memory boards.

[0006] Because of the increasing cost, complexity, and time involved with fully testing DRAM chips, DRAM manufacturers often provide “effectively tested” (“ETT”) DRAM chips to memory module manufacturers at a lower price rather than providing fully tested DRAM chips. Memory module manufacturers often prefer the ETT DRAM chips mainly due to their greater availability. Memory module manufacturers who receive ETT DRAM chips then have to assume a part of the responsibility of validating the DRAM chips, adding to the complexity of the memory module test process.

SUMMARY

[0007] In certain embodiments, a self-testing memory module is provided comprising a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module may further comprise a plurality of

memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprises data, address, and control ports. In certain embodiments, the memory module includes a control module configured to generate address and control signals for testing the memory devices and a data module comprising a plurality of data handlers. Each data handler in certain embodiments is operable independently from each of the other data handlers of the plurality of data handlers. Each data handler may be operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports.

[0008] A self-testing memory module is provided in certain embodiments comprising a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module may further comprise a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprises data, address, and control ports in some embodiments. The memory module of certain embodiments includes a control module configured to generate address and control signals for testing the memory devices. In certain embodiments, the memory module comprises a data module comprising at least one data handler operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate cyclic data for writing to the corresponding plurality of data ports.

[0009] In certain embodiments, a method of self-testing a memory module includes providing a self-testing memory module. The memory module comprises a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module further comprises a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprising data, address, and control ports. In certain embodiments, the memory module comprises a control module configured to generate address and control signals for testing the memory devices. The memory module may further comprise a data module comprising a plurality of data handlers. Each data handler may be operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports. The method of certain embodiments further includes generating, by each of the data handlers, data for writing to the corresponding plurality of data ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein.

[0011] Figure 2 is a block diagram of an example self-testing memory module including eight memory devices and eight data handlers in accordance with certain embodiments described herein.

[0012] Figure 3 is a block diagram of an example data handler module and an example control module in accordance with certain embodiments described herein.

[0013] Figure 4 is a flowchart of an example method of self-testing a memory module in accordance with certain embodiments described herein.

[0014] Figure 5 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

[0015] Figure 6 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0016] Certain embodiments described herein present a self-testing electronic system, such as for example, a self-testing electronic memory module. Some embodiments described herein present a self-testing memory module that is populated with ETT DRAM chips. Certain embodiments described herein present a self-testing registered dual in-line memory module (“RDIMM”). Some embodiments described herein present a self-testing RDIMM that does not require any additional pins other than the pins on the standard JEDEC RDIMM connector. For example, the memory module may utilize the address and control signals (e.g., address and control signals generated by the memory module to test the memory module) along with a test signal to enable and execute a self testing function.

[0017] Certain embodiments described herein present a self-testing memory module that includes a control module and a data module which can generate memory addresses and data according to the JEDEC standard memory protocol.

[0018] Some embodiments described herein presents a self-testing memory module that can be configured through an I²C interface and that allows test results to be read out through the I²C interface. Certain embodiments described herein present a self-testing memory

module that allows a test function to be configured, controlled, and/or executed without substantial system memory controller involvement. Some embodiments present a self-testing memory module that can be tested without any external test equipment. For example, certain embodiments present a self-testing memory module that can be tested without any system driven test procedure. Various embodiments described herein present a self-testing memory module that can be self-tested at a target system speed. Certain embodiments described herein present a self-testing memory module that can generate data (DQ) and data-strobe (DQS) signals with wave characteristics that resemble the wave characteristics of DQ and DQS signals from a system memory controller.

[0019] MBIST is commonly used to test memories embedded in application specific integrated circuits (“ASICs”) or system on chip integrated circuits (“SoCs”) such as, for example, advanced memory buffers (“AMBs”). The MBIST implementation usually includes three distinct functional blocks: the address/control generator, the data generator/checker (sometimes referred to as a signature analyzer), and the test interface controller/register (“JTAG”). The test instructions and test patterns are generally loaded to the address/control generator and the data generator/checker through the JTAG interface. The test results are generally read out through the JTAG interface.

[0020] In memory module applications, there are obstacles associated with using MBIST. One of these obstacles is the large number of signals used to interface the three MBIST functional blocks. The inter-block timing constraints can present another obstacle. In addition, the ability to control the test is limited. For example, synchronizing the address/or control signals with the data signal can be difficult. The relatively large amount of information to be gathered and stored while testing the memory presents yet another obstacle.

[0021] Unlike on an ASIC, in which the three MBIST functional blocks are in a single chip, the MBIST functional blocks on a memory module generally would be segregated into multiple chips on the memory module due to physical and electrical limitations and requirements. This makes implementing MBIST on a memory module difficult because, while there is virtually no limit on the number of available interface signals among the three MBIST functional blocks in an ASIC, the memory module can support only a limited amount of interface signals between memory chips. In addition, because the MBIST functional blocks are

spread out to multiple chips on a memory module, the inter-block signal delay is generally much longer on a memory module than on an ASIC. It is also generally not feasible to implement cross-checking logic that operates the three MBIST functional blocks in lock-operational block. This is due to the limitation on the number of interface signals and to the relatively long inter-block signal delay time on a memory module versus on an ASIC.

[0022] In some cases such as where a memory module includes an AMB, the self-test logic (MBIST) implemented in the AMB includes command and address generation logic in addition to a data generator and checker. Each of these functional blocks may be implemented on a single physical AMB device (e.g., a single integrated circuit package). However, because all of the self-test command, address and data signals are combined in one physical area of the memory module, a memory module (e.g., DIMM) level routing problem can occur, making it difficult to route the self-test signals on the memory module and resulting in performance degradation and/or implementation difficulty. In addition, the data width of self-test logic of the memory module will be limited to the data width of the AMB (e.g., to the data width of the self-test logic implemented on the AMB, to the number of available ports on the AMB, etc.). As such, implementing memory module test logic on an AMB is not flexible (e.g., to changes in the data width of the memory module) and generally supports only memory modules having certain predetermined, fixed data widths.

[0023] Finally, in most cases, since an ASIC cannot be repaired, the ASIC MBIST is generally only capable of detecting and reporting the pass/fail status of memory tests. For memory module test results, on the other hand, it is generally advantageous to include both addresses of the memory locations where failures occur and the data patterns that were read back from the failed memory locations. This type of reporting can help to facilitate the repair of the memory module by, for example, allowing for the identification and replacement of failed components.

[0024] Figure 1 is a block diagram of an example self-testing memory module 10 in accordance with certain embodiments described herein. The memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board (PCB) 12, each memory device 20 of the plurality of memory

devices 18 comprising data, address, and control ports. The memory module 10 comprises a control module 22 configured to generate address and control signals 24 for testing the memory devices 18. The memory module 10 also includes a data module 28 comprising a plurality of data handlers 30. Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports of one or more of the plurality of memory devices 18. For example, each of the data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports. Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. The memory module 10 may further include an I²C interface 15 in certain embodiments.

[0025] As described more fully below, in certain embodiments the data module 28 generates test data patterns to write to the plurality of memory devices 18 of the memory module 12 and checks the data patterns read or received back from the plurality of memory devices 18 for agreement with corresponding data patterns that are expected to be read back from the plurality of memory devices 18. For example, in one embodiment, the data module 28 generates cyclic patterns to write to the plurality of memory devices 18. In some embodiments, the data module 18 also isolates the data path from the system board of the computer system 16 to the plurality of memory devices 18 while the memory module 10 is not accessed by the computer system 16. For example, the data module 28 may isolate the data path from the system board to the plurality of memory devices 18 when the memory module 10 is in a self-testing mode. The control module 22 may include, for example, a dual input register (e.g., the memory device controller 32 described more fully below) for registering address and control signals coming from either self-testing logic (e.g., from the test controller 36 described more fully below) or from the memory controller 14 on the system board. In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20. For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation. The data module 28 may then receive a read command to read back the data from

those locations and check the read data for agreement with the expected data. If there is a mismatch between the read data and the expected data, the data module 28 may, for example, store the failure information (e.g., the failed data word) and inform the control module 22 about the failure. The control module may save the address of the memory location where the failure occurred.

[0026] In certain embodiments, the memory module 10 is configured to be operated in a test mode in which the control module 22 selectively inputs the address and control signals to the address and control ports of the plurality of memory devices 18. Moreover, in the test mode, each of the data handlers 30 write the data generated by the data handler 30 to the corresponding plurality of data ports by selectively inputting data signals to the data ports of the plurality of memory devices 18. The data module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20. For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according the operating specification of the memory devices 20. In some embodiments, the control module 22 and the data module 28 produce memory addresses, control and/or data signals according to the JEDEC standard memory protocol. In some embodiments, for example, the control module 22 and the data module 28 generate the memory interface signals with proper edge relationships based on the JEDEC standard. In certain embodiments, the test speed, for example, may be defined by the speed of the clock (e.g., the system clock). The address sequences and/or the data patterns of certain embodiments may be programmable either through the I²C interface or they may be defaulted to pre-defined values.

[0027] In certain embodiments, data module 28 is configured to generate data signals with programmable slew rates and/or with variable peak values. In one embodiment, for example, the data module 28 is also able to generate data (DQ) and data-strobe (DQS) signals with programmable slew rates and programmable peak values so that the characteristics of the signals generated by the data module 28 generally correspond to the characteristics DQ and DQS signals generated by the system memory controller 14.

[0028] In some embodiments, the data module 28 and/or the control module 22 are configured to test the plurality of memory devices 18 under non-normal conditions. For example, the data module 28 and/or control module 22 may be configured to provide signals having frequencies which are higher or lower than the normal operating frequencies of the memory devices 20.

[0029] In certain embodiments, the memory module 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 12 has an industry-standard form factor. For example, the PCB 12 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 12 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 12 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0030] In certain embodiments, the plurality of memory devices 18 of the memory module 10 may be arranged as ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 20 of the plurality of memory devices 18 that can be accessed by the memory controller 14 to access the full bit-width of the memory bus of the memory module 10. For example, a memory module 10 in which each rank of the memory module is 64 bits wide is described as having an “x 64” organization. Similarly, a memory module 10 having 72-bit-wide ranks is described as having an “x 72” organization. The number of memory devices 20 and corresponding memory capacity of a memory module 10 can be increased by increasing the number of memory

devices 20 per rank or by increasing the number of ranks. For example, a memory module with four ranks with each rank having N 512-MB memory devices 20 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 20 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices 20. During operation, the ranks of a memory module 10 may be selected or activated by control signals that are received from a component of the system (e.g., a system memory controller 14 or a local memory controller of the memory module 10). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 10 comprises only one rank of memory devices 20.

[0031] As discussed, the PCB 12 may include at least one connector (not shown) configured to operatively couple the memory module 10 to the memory controller 14 of the computer system 16. The computer system 16 may include a host computer system. For example, the memory module is electrically coupled, logically coupled, or both, with the memory controller 14. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 14 may comprise a disk controller of the computer system 16, for example. The memory controller 14 may be mounted on a system board of the host computer 16. The connector can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system 16. The connector of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 10 and the host system 16. For example, the connector can comprise a standard DDR2, DDR3, and other future generation edge connectors. Additionally, in certain embodiments, more than one memory module 10 is coupled to the host system 16.

[0032] The plurality of memory devices 18 on the PCB 12 may include one or more volatile memory components. For example, the plurality of memory devices 18 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 20. Types of DRAM devices 20 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM

(SDRAM). The memory devices 18 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 20 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 20 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP). The plurality of memory devices 18 may further include one or more non-volatile memory devices 20, such as, for example, flash memories. The plurality of memory devices 18 of certain embodiments may include both volatile and non-volatile memory devices 20. For example, the plurality of memory devices 18 may include one or more of DRAM, SRAM, and/or flash memory devices in some embodiments.

[0033] Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28. For example, each data handler 30 is configured to write to and/or read from the corresponding plurality of data ports of one or more of the memory devices 20 without being in communication any of the other data handlers 30 or other data ports of the memory devices 20. As such, each data handler 30 can be used to generally independently test a portion of the memory space of the memory module 10. For example, each data handler 30 may be used to independently test one memory device 20 of the memory module 10. In such a configuration, the corresponding plurality of data ports of each data handler 30 may comprise each data port of the corresponding memory device 20. In other embodiments, each data handler 30 may be used to test a segment of one memory device 20, more than one memory device 20, segments more than one memory device 20, or any combination or sub-combination thereof. Because each of the data handlers 30 is operable independently of each of the other data handlers 30, the data handlers 30 are generally modular. As such, modifications in the configuration of the memory module 10 (e.g., changes in the bit-width of the memory bus, changes in the number of memory devices 20, etc.) may be less complicated to accommodate than in other types of self-testing memory modules 10. For example, where a new memory device 20 or set of memory devices 20 is added to the memory module 10, the change may be generally accommodated by adding a corresponding data handler 30. The change may be accommodated without having to

implement a major reorganization of the memory module 10 or the self-testing logic of the memory module 10, for example.

[0034] Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. Figure 2 is a block diagram of an example self-testing memory module 10 including eight memory devices 20 (e.g., memory devices 40a-40h) and a data module 28 comprising eight data handlers 30 (e.g., data handlers 30a-30h) in accordance with certain embodiments described herein. Each of the memory devices 20 includes an eight bit output data word and eight corresponding data ports. In addition, the system memory bus 50 between the memory controller 14 and the example memory module 10 is 64 bits wide and each of the data handlers 30 receives an eight bit segment of the system memory bus 50. Each of the data handlers 30 is operatively coupled to a corresponding plurality of data ports 21 of a corresponding one of the memory devices 20. As such, the data handlers 30 may be operatively coupled (e.g., electrically and/or logically coupled or connected) to the eight data ports 21 of one of the corresponding memory devices 20. For example, the data handler 30a may be operatively coupled to the eight data ports 21 of the memory device 40a of Figure 2.

[0035] The configuration shown in Figure 2 is for the purposes of illustration and is not intended to be limiting. For example, while the example memory module 10 of Figure 2 includes an equal number of memory devices 20 and data handlers 30, other configurations are possible. In some configurations there are more memory devices 20 than data handlers 30 or vice versa. Moreover, the one or more data handlers 30 may be operatively coupled to a subset of the data ports 21 of one the memory devices 20 instead of all of the data ports 21 of one of the memory devices 20. In other embodiments, one or more data handlers 30 may be operatively coupled to a subset or all of the data ports 21 of more than one of the memory devices 20. For example, in one embodiment, each of the data handlers 30 are operatively coupled to all of the data ports of two memory devices 20.

[0036] In certain embodiments, the plurality of data handlers 28 comprises at least two physically separate components mounted on the PCB 12. For example, the plurality of data handlers 28 may include at least two physically separate integrated circuit packages. The physically separate integrated circuit packages are mounted on different portions of the

PCB 12 in some embodiments. For example, each of the eight data handlers 30a-30h shown in Figure 2 may include physically separate integrated circuit packages mounted on different portions of the PCB 12. While eight data handlers 30 are shown in Figure 2, other numbers of data handlers 30 are possible including fewer or more than eight.

[0037] In certain embodiments, each of the plurality of data handlers 30 is positioned on the PCB 12 proximate to the corresponding plurality of data ports. For example, each data handler 30 of certain embodiments is positioned closer to the corresponding plurality of data ports 21 than the data handler 30 is to the other data ports 21 of the plurality of memory devices 18. For example, the data handler 30a is positioned closer to the corresponding plurality of data ports 21 of the memory device 40a than to the other data ports 21 of the other memory devices 40b-40h.

[0038] Figure 3 is a block diagram an example data module 28 and an example control module 22 in accordance with certain embodiments described herein. The control module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18. In some embodiments, the control module 22 includes a control mixer element 32. The control mixer element 32 may include a memory device controller 34 (e.g., a DRAM controller) and a test controller 36. In certain embodiments, the control mixer element 32 generally controls the address and the control signals for the self-testing function.

[0039] In certain embodiments, the memory device controller 34 generally pre-processes address and control information before it sends the information to a register 40. In one embodiment, the memory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36. The control module 22 of certain embodiments is configured to selectively input to the address and control ports of the plurality memory devices 18 either the address and control signals 38 from the system memory controller 14 or the address and control signals 42 from the control module 22 (e.g., from the test controller 36). For example, the memory device controller 34 may send either the signals 38 from the system memory controller 14 or, alternatively, the signals 42 from the test controller 36, to the register 40 depending on whether the memory module 10 is in normal (non-test) mode or in a

test mode, respectively. In one embodiment, the memory device controller 34 generates the address and control signals for memory device (e.g., DRAM device) operations. The test controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28. The control module 22 may be implemented in the control register of the memory module 10 in certain embodiments. In various embodiments, the control module 22 includes discrete logic, one or more application-specific integrated circuit (ASICs), one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic device (CPLDs).

[0040] The data module 28 and the subcomponents thereof (e.g., the data handlers 30) may be in communication with one or more of the memory devices 20, the control module 22, and the memory controller 14. In certain embodiments, the data module 28 comprises a plurality of data handlers 30. In other embodiments the data module 28 includes at least one data handler 30. Each of the data handlers 30 of certain embodiments comprises a switch 44. For example, the switch 44 may include a data multiplexer/demultiplexer (“data mux/demux”). The switch 44 may provide a bi-directional data multiplexer function. In certain embodiments, the switch 44 is configured to selectively input to the corresponding plurality of data ports either data signals 48 from the system memory controller 14 or data signals 50 from the data handler logic element 46. The switch 44 of certain embodiments may further be configured to receive data signals 52 (e.g., during a read operation) from the plurality of memory devices 18 and to propagate the data signals 52 to the data handler logic element 46 and/or the memory controller 14. In some embodiments, for example, the switch 44 selectively inputs the data signals 48 to be written to the plurality of memory devices 18 from the system memory controller 14 when the memory module 10 is a normal (non-test mode) mode and, alternatively, inputs the data signals 50 from the data handler logic element 46 during a test mode. While the switch 44 is shown as being included in the data handler 30 in the example of Figure 3, other configurations are possible. For example, in other embodiments the switch 44 may be logically and/or physically separated from the data handler module 28 and/or the data handlers 30.

[0041] Each of the data handlers 30 of certain embodiments further includes a data handler logic element 46. The data handler logic element 46 of certain embodiments comprises a data generation element 54 and a verification element 56. The data generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports, for example. The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information) the data handler logic element 46 receives from the control module 22, for example. The data may be cyclic data in some embodiments or non-cyclic data in other embodiments. For example, the cyclic data may comprise at least one predetermined pattern of data which repeats or is cycled two or more times. In various embodiments, the data comprises one or more incrementing patterns or decrementing patterns, for example. In other embodiments, the data comprises a pattern which alternates each bit on successive memory writes. For example, a memory write comprising one or more hexadecimal “A” characters (each corresponding to a four-bit binary word of “1010”) may be followed by a memory write comprising one or more hexadecimal “5” characters (each corresponding to a four-bit binary word of “0101”). The data may be generated in a variety of ways. In one embodiment, the data is generated based on a current write address value. For example, in one example configuration, on a first write cycle, hexadecimal “A’s” are generated and written to even address locations and hexadecimal “5’s” are generated and written to odd address locations, and on a second write cycle, “5’s” are written to even addresses and “A’s” are written to odd addresses, and this pattern repeats in subsequent cycles. The data may be generated based previously written data (e.g., inverting each of the bits of a previously written data word) in some embodiments. In general, any manner of generating a cyclic or otherwise deterministic data pattern may be compatible with embodiments described herein. In other embodiments, random or pseudorandom data may be generated and written to the corresponding plurality of data ports. For example, a linear feedback shift register (LFSR) may be used in some embodiments. In addition, the data patterns may be programmable. For example, the data patterns may be programmable based on information received by the data generation element 54 from the memory controller 14 (e.g., through the control module 22), or from the control module 22.

[0042] The plurality of data handlers 30 are further configured to read data from the corresponding plurality of data ports. For example, the verification element 56 may be configured to receive data from the corresponding plurality of data ports (e.g., through the switch 44 during a test mode). The verification element 56 may further be configured to check for failures in the operation of the plurality memory devices 18 by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler 30 and written to the corresponding plurality of data ports.

[0043] In certain embodiments, the verification element 56 is configured to perform the verification without storing a copy of the data written to the corresponding plurality of data ports or accessing a stored copy beyond the data read from the plurality of memory devices 18. For example, the verification element 56 does not store or access a copy of the data that is written to the corresponding plurality of data ports except for the data stored and read back from the plurality of memory devices 18. As such, the memory module 10 of certain embodiments advantageously does not require separate memory for storing duplicate copies of test data that is written to the plurality of memory devices 18 for later comparison. For example, the verification element 56 may calculate comparison data and may compares the comparison data to the data read from the corresponding plurality of data ports. In certain embodiments, the comparison data comprises data which expresses the data or values expected to be received from the plurality of memory devices 18 if the write, store, and read processes of the data using the data module 28 and the plurality of memory devices 18 are performed correctly or as expected. The calculation may be performed simultaneously or substantially simultaneously with receiving the data read from the corresponding plurality of data ports in certain embodiments. In other embodiments, the calculation is performed either before or after receiving the data. Other configurations are possible. For example, in one embodiment, the verification element 56 does store a separate copy of the data written to the corresponding plurality of data ports upon writing the data and compares the separate copy to the read data received from the plurality of memory devices 18.

[0044] The verification element 56 of certain embodiments calculates the comparison data based on the cyclic data. For example, in one embodiment, the verification element 56 calculates the comparison data in substantially the same manner that the data

generation element 54 generates the data as described above (e.g., based on a current write address, using an LFSR, etc.). As such, the comparison data of certain embodiments is substantially a repeat of the data written. In one example embodiment, the data handler 30 is configured (e.g., is programmed by the control module 22) to write an alternating series of “A’s” and “5’s” to the data ports of the corresponding plurality of data ports as described herein. For example, the data handler 30 may be configured to write one or more “A’s” to the first address location of an N-word memory device 20 including the corresponding plurality of data ports. The data handler 30 may then write one or more “5’s” to the second address location, one or more “A’s” to the third address location and so on until the data handler 30 has written to all N memory locations. The verification element 56 of the example embodiment then calculates the comparison data based on the cyclic data written to the corresponding plurality of data ports. For example, the verification element 56 calculates a comparison word including one or more “A’s”, “5’s,” and “A’s,” respectively, to be compared to the data read from the first, second, and third address locations based on the cyclic data (e.g., based on the known cycle of the data). In some embodiments, the comparison data is calculated based on a current read address.

[0045] In certain embodiments, data associated with failures in the operation of the plurality of memory devices 18 are stored in the data module 28. For example, data read from the corresponding plurality of data ports which do not correspond to (e.g., match) the comparison data calculated by the verification element 56 may be stored in the data module. Moreover, in some embodiments, memory addresses associated with the failures in the operation of the plurality of memory devices 18 are stored in the control module 22. For example, the data handler 30 may communicate data failures (e.g., when data read from the corresponding plurality of data ports does not correspond to calculated comparison data) to the control module 22 which may then store the addresses corresponding to the data failure. In certain embodiments, the memory module 12 is configured to report failures (e.g., the failed data, the address corresponding to the memory location of the failed data, and/or expected data) via the I²C interface 15 to the memory controller 14. In addition, in certain embodiments the test controller 36 and/or the data handlers 30 may be updated through the

I²C interface 15 with new data patterns and/or with alternative memory access sequences to conduct AC tests (e.g., tests of the power, current, I/O speed, etc.).

[0046] In various embodiments, the components of the data module 28 (e.g., the switch 44, the data handlers 30, the data handler logic element 46, the data generation element 54, and/or verification element 56) may include discrete logic, one or more application-specific integrated circuits (ASICs) one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic devices (CPLDs). Additionally, one or more of the various functional blocks (e.g., the switch 44) of the data module 28 of Figure 3 may not be included. In some embodiments, additional functional blocks may be included. Moreover, some of the functional blocks are described as separate functional blocks for illustration purposes and may comprise one physical component. For example, in one embodiment, each of the data handlers 30 and the corresponding switch 44, data generation element 54, and verification element 56 comprise one physical component (e.g., are included in one integrated circuit package). In another embodiment, the data module 28 comprises one physical component.

[0047] Referring again to Figure 1, a self-testing memory module 10 of certain embodiments comprises a printed circuit board (PCB) 12 and is configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board 12 where each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. The memory module 10 further comprises a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. In certain embodiments, the memory module 10 further comprises a data module 28 comprising at least one data handler 30 and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices 20. The data handler 30 is configured to generate cyclic data for writing to the corresponding plurality of data ports. In certain embodiments, there may be one data handler 30, for example. In other embodiments, there may be more than one data handler 30. The elements of the memory module 10 may be compatible with any of the embodiments described herein.

[0048] Figure 4 is a flowchart of an example method 70 of self-testing a memory module 10 in accordance with certain embodiments described herein. While the method 70 is described herein by reference to the memory module 10, other memory modules, electronic systems or subsystems, and/or circuits are also compatible with the embodiments described herein. The method 70 of certain embodiments comprises providing a self-testing memory module 10 at operational block 72. The memory module 10 may comprise a printed circuit board (PCB) 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 may further comprise a plurality of memory devices 18 on the printed circuit board 12. Each memory device 20 of the plurality of memory devices 18 may comprise data, address, and control ports. The memory module 10 may comprise a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. The memory module 10 may further comprise a data module 28 comprising a plurality of data handlers 30. In certain embodiments, the data module 28 comprises at least one data handler 30. Each data handler 30 of certain embodiments is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports. At operational block 74, the method 70 further comprises generating, by each of the data handlers 30, data for writing to the corresponding plurality of data ports.

[0049] Figure 5 is a flow diagram 100 illustrating self-testing operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. At operational block 110 the memory module 10 enters an idle state after power up or, in some embodiments, when the memory board of the computer system 16 is connected to a mother board or to a test board. In one embodiment, at operational block 110, the contents of the register 40 are undefined and the controllers (e.g., the test controller 36 and the memory device controller 34) are in unknown state. After power up, the memory module 10 is reset (e.g., a reset command is executed) at operational block 111. At operational block 111, the logic of the data handler 30 (e.g., data generation element 54 and/or verification element 56) and the logic of the control mixer 32 are set to default values and/or states. For example, the memory module 10 may default to a non-test mode (e.g., normal operational mode) and the switches 44 of the data handlers 30 and the memory device controller 34 are not configured in

a test mode. At operational block 112, the test mode is configured. For example, the computer system 16 (e.g., through the memory controller 14) may configure the test mode. The configuration may be through the I²C interface 15, for example, and may include configuring the test controller 36 for test mode. For example, configuring the test mode may include switching (e.g., by configuring the test controller 36) the mode of the memory module 10 from a normal operation mode to the test mode.

[0050] At operational block 113, the test mode is initiated. Initiating the test mode may comprise configuring a particular test case (e.g., particular test data or data patterns, particular write and/or read address sequences, etc.). The initiation may be achieved by activating the test controller 36 to initiate the test mode. For example, at operational block 113, the data handlers 30 and the control mixer 32 are readied (e.g., via a tester) beginning the self-testing of the memory module 10. For example, the tester may comprise an ATM tester, a server, a specialized tester. The tester may ready the data handlers 30 and the control mixer through the I²C interface 15 or through the control signals 38, for example. For example, the data handlers 30 are configured to input data generated by the data handlers 30 to the corresponding plurality of data ports of the plurality of memory devices 18 and the control mixer 32 is configured to input address and control signals from the test controller 36 to the plurality of memory devices 18. At operational block 113 the test controller 36 updates each of the data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.).

[0051] In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal (“Par-in”) is un-asserted (e.g., set to a “low” value), it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. The memory module 10 executes the test mode (e.g., writes and reads test data patterns) at operational block 114. The memory module 10 continues executing the test mode at operational block 114 until the self-test is complete or until the memory module 10 detects a certain number of memory failures such that a failure count exceeds a preset number.

[0052] If the failure count exceeds the preset number, the memory module 10 enters operational block 115 and “errors out” of the test mode. If the self-test is completed at

operational block 114 and the failure count does not exceed the preset number, the memory module 10 enters operational block 115. At operational block 115, the memory module 10 sends out a test completion indication signal (e.g., through the test controller 36 to the memory controller 14). At operational block 116, the fail status of the test may be read out through, for example, the I²C interface 15. For example, one or more address values associated with failed memory locations, and/or the data read from those locations, and/or expected data may be read. In other cases, where the test fails, the memory module 10 may report the failure to the memory controller 14 through any available signal (e.g., bidirectional data or data strobe signal) between the memory module 10 and the memory controller 14.

[0053] In some embodiments, the memory module 10 is generally interruptible. For example, the memory module 10 may exit the self-test and return to operational block 110 if the signal going into the parity-in pin is, for example, de-asserted or removed at operational block 155. In other embodiments, the interrupt signal or condition may be different and may not be a parity-in signal. For example, in one embodiment, the memory module 10 may exit the self-test and return to operational block 110 if a timer having a pre-determined count expires. When the self-testing is interrupted, the memory module 10 will return to operational block 110 and the bit failure information in the control mixer 32 and the data handler 30 becomes invalid.

[0054] Figure 6 is a flow diagram 200 illustrating the operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. For example, one or more of the operational blocks of the flow diagram 200 of Figure 6 may correspond to one or more of the operational blocks of Figure 5. At operational block 210, the memory module 10 may be generally inactive. For example, the testing logic (e.g., the control module 22 and/or the data module 28) may be generally inactive and the memory module 10 may default to a functional (e.g., non-test) mode. In certain embodiments, the memory module 10 executes one or more tasks which configure the self-test mode of the self-testing memory module using an I²C interface 15. For example, the memory module 200 may configure the test controller 36 through the I²C interface 15 (e.g., as described with respect to Figure 5). In other embodiments, the test mode is configured using the control signals 38. For

example, the test mode may be updated at operational block 211 in certain embodiments. In certain embodiments, updating the test mode comprises programming the test controller 36 with test configuration information such as test data patterns, test data sequences, test failure conditions, etc. After updating the test mode at operational block 211, the test controller 36 may then be activated through the I²C interface 15 (e.g., as described with respect to Figure 5) in certain embodiments. Alternatively, as shown, the test controller 36 may be activated through the I²C interface 15 without first updating the test mode.

[0055] At operational block 212, the memory module test mode is ready for execution. In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal is low, it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. At operational block 113, a set of status registers (e.g., one or more registers of the control module 22 or the data module 28) are cleared.

[0056] The memory module 200 generates address and data signals for testing the plurality of memory devices 18 at operational block 214. For example, the data generation elements 54 of the data handlers 30 may generate the data for writing to the plurality of memory devices 18 as described herein. In addition, the control module 22 may generate address and control signals for testing the plurality of memory devices 18 as described herein. At operational block 215, the memory module 10 performs a burst write function. For example, in certain embodiments, the memory module 10 writes multiple memory locations in the memory module 10 that are to be tested at operational block 215 in a relatively short amount of time (or in a burst). In certain embodiments, the memory module 10 can perform multiple write bursts at operational block 215. Next, at operational block 216, the memory module 10 performs a read & compare function. For example, in certain embodiments, the memory module 10 reads back certain memory locations at operational block 216 that have been written at operational block 215 and compares the values with certain expected data. For example, a verification element 56 of each of the data handlers 30 may calculate the expected data and/or compare the values as described herein. In some embodiments, the memory module 200 reads from multiple memory locations in the memory module 10 in a relatively

short amount of time (or in a burst). In certain embodiments, the memory module 10 can execute multiple read bursts at operational block 216 as it compares the read data with expected data. In some embodiments, the memory module 200 (e.g., the data handlers 30 of the memory module 200) stores any failed read data and the associated expected data at operational block 217. In some embodiments, the memory module sends one or more failure indications to the memory controller 14 over one or more available signals between memory module 10 and the memory controller 14. For example, the I²C interface 15 or one or more bi-directional data pins (e.g., one or more of the data pins 48) may be used.

[0057] The memory module 200 stores the addresses of the memory locations associated with the failed read data using the test controller 32 at operational block 217. In other embodiments the failed read data, expected data, and the addresses of the memory locations associated with the failed read data may be stored using the test controller 32. For example, in one embodiment, the test controller 32 stores the failed read data, expected data, and the addresses of the memory locations associated with the failed read data at operational block 217. In some embodiments, the test controller 32 is generally inaccessible through the I²C interface 15 during self-test operation. For example, the test controller is inaccessible through the I²C interface 15 until a test failure occurs at operational block 217 or until the test completes at operational block 219. In some embodiments, when the test is complete, the test results can be read out of the memory module 10 (e.g., through the I²C interface 15). In some cases, the test can be interrupted using the I²C interface 15 or using the parity-in signal (e.g., by de-asserting the parity-in signal).

[0058] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects

or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A method performed by a memory module having a plurality of data ports that are coupled to a system memory controller, the memory module comprising a plurality of memory devices organized in ranks, comprising:
 - initiating a test mode for the memory module in response to control signals received from the system memory controller;
 - generating data at each of a plurality of data handlers on the memory module, each data handler being coupled to a respective set of data ports of the plurality of data ports and to a respective set of memory devices of the plurality of memory devices, each set of memory devices including at least one memory device from each rank;
 - obtaining test results by writing the data generated in each respective data handler into memory locations in the set of memory devices coupled to the respective data handler and by reading from the memory locations; and
 - providing the test results to the system memory controller.
2. The method of claim 1 further comprising:
 - generating control and address signals at a module controller of the memory module, the control and address signals indicating the memory locations; and
 - providing the control and address signals to the plurality of memory devices.
3. The method of claim 1 further comprising:
 - updating the test mode by programming a test controller of the memory module with test configuration information, the test configuration information including test data patterns, test data sequences, test failure conditions.
4. The method of claim 1 further comprising:
 - exiting the test mode in response to a signal from the system memory controller.
5. The method of claim 4 wherein the signal from the memory controller is received via a parity-in pin of the memory module.
6. The method of claim 1 further comprising:

sending out a test completion indication signal to the memory controller.

7. The method of claim 1 wherein the test results are provided to the memory controller via bidirectional data signal channels between the memory module and system memory controller.
8. The method of claim 1 wherein the test results are provided to the memory controller via bidirectional data strobe signal channels between the memory module and system memory controller.
9. The method of claim 1 wherein the test results are provided to the memory controller in response to a memory failure count exceeding a preset number.
10. The method of claim 1 wherein initiating the test mode comprises configuring a test case according to specified test data patterns and read/write address sequences.
11. The method of claim 1 further comprising programming a data pattern for the test mode based on information received from the system memory controller.
12. The method of claim 1 wherein initiating the test mode comprises configuring at least one switch in each of the plurality of data handlers.
13. A memory module for operating with a system memory controller, comprising:
 - a plurality of data ports;
 - a plurality of memory devices organized in ranks; and
 - a plurality of data handlers, each respective data handler being coupled to a respective set of data ports of the plurality of data ports and to a respective set of memory devices of the plurality of memory devices, each set of memory devices including at least one memory device from each rank; andwherein each respective data handler is configured to provide write data received from the system memory controller via the respective data ports to the respective set of memory devices in a normal mode, and to provide test data generated in the respective data handler to the respective set of memory devices in a test mode.

14. The memory module of claim 13, further comprising
a module controller coupled to the plurality of memory devices and to the plurality of data handlers, the module controller is configured to provide control and address information received from the system memory controller to the plurality of memory devices in the normal mode, and to provide control and address information generated in the module controller to the plurality of memory devices in the test mode.
15. The memory module of claim 14 wherein the plurality of data handlers are capable of being operated independently of each other.

Apparatus and Method for Self-Test in a Multi-Rank Memory Module

ABSTRACT OF THE DISCLOSURE

A memory module for operating with a system memory controller comprises a plurality of data ports, a plurality of memory devices organized in ranks, and a plurality of data handlers. Each respective data handler is coupled to a respective set of data ports of the plurality of data ports and to a respective set of memory devices of the plurality of memory devices. Each set of memory devices include at least one memory device from each rank. In a normal mode, each respective data handler is configured to provide write data received from the system memory controller via the respective data ports to the respective set of memory devices. In a test mode, each respective data handler is configured to provide test data generated in the respective data handler to the respective set of memory devices.

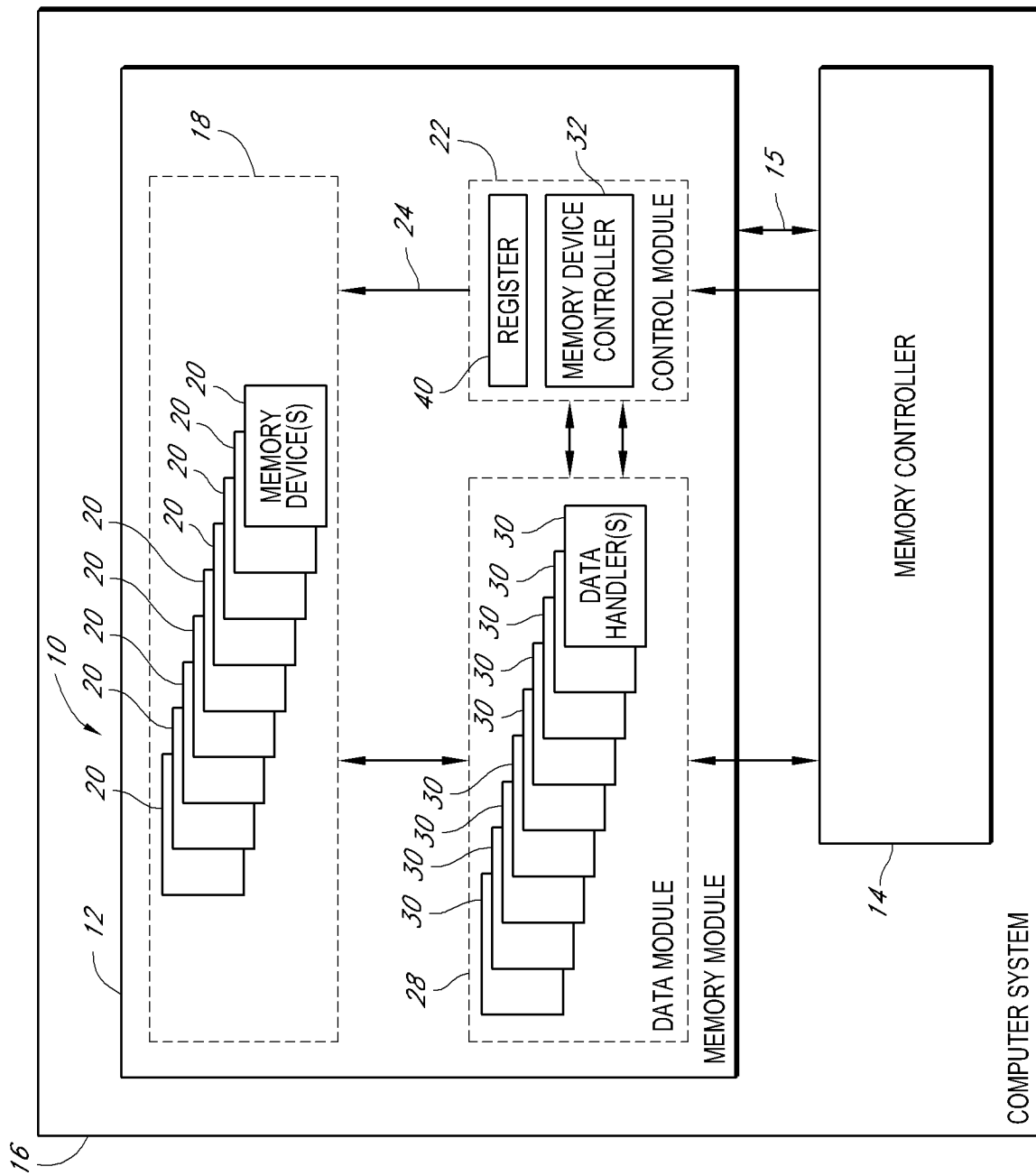
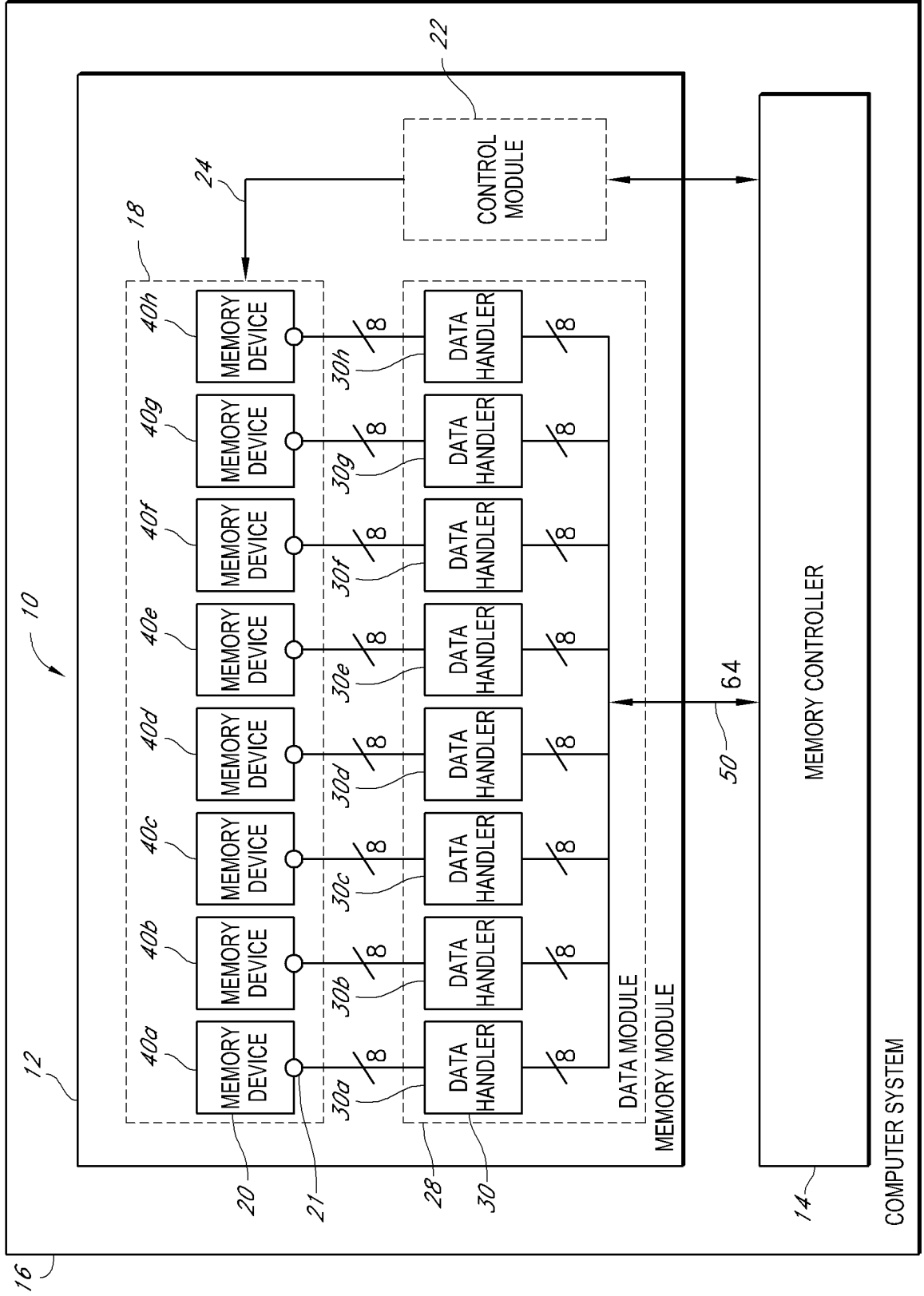


FIG. 1



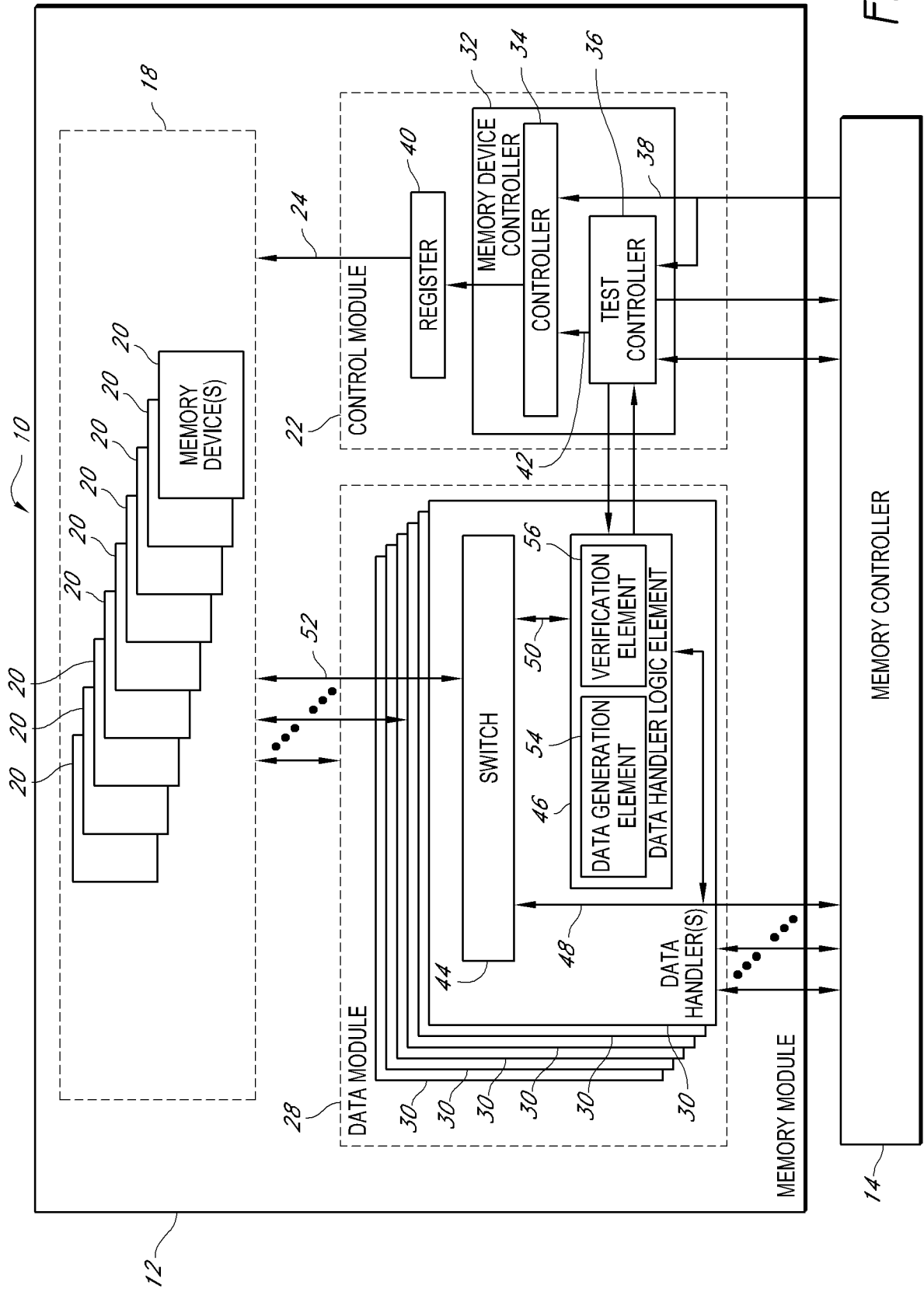


FIG. 3

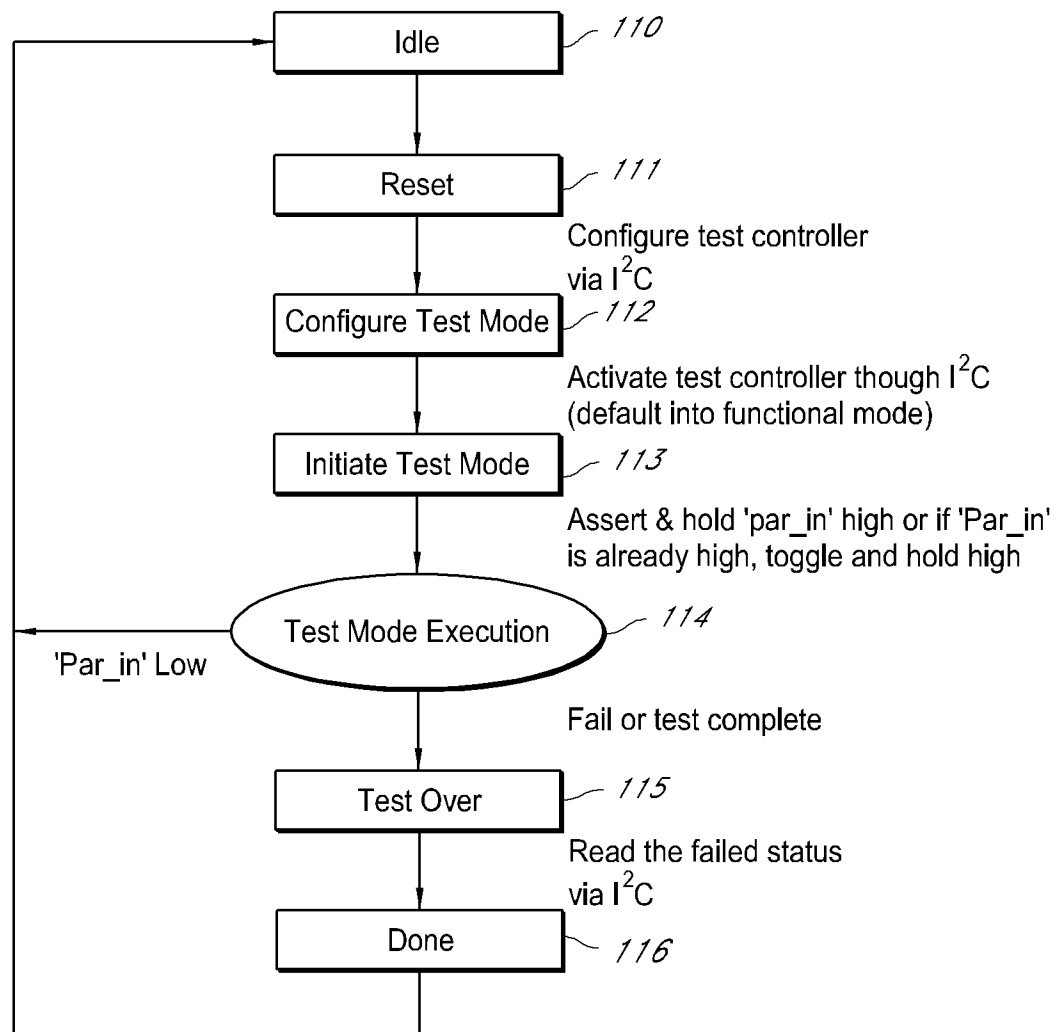
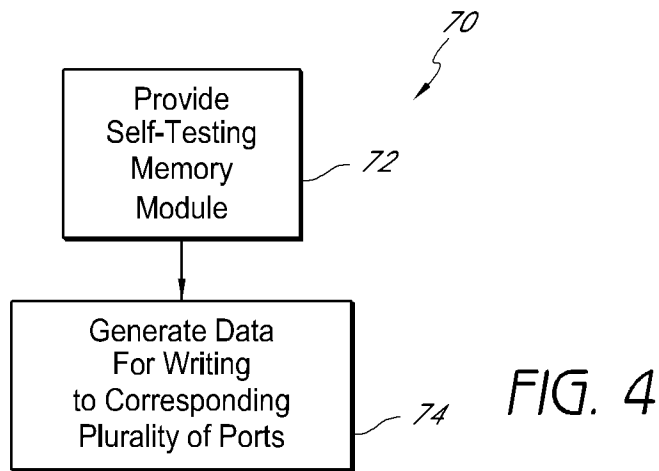
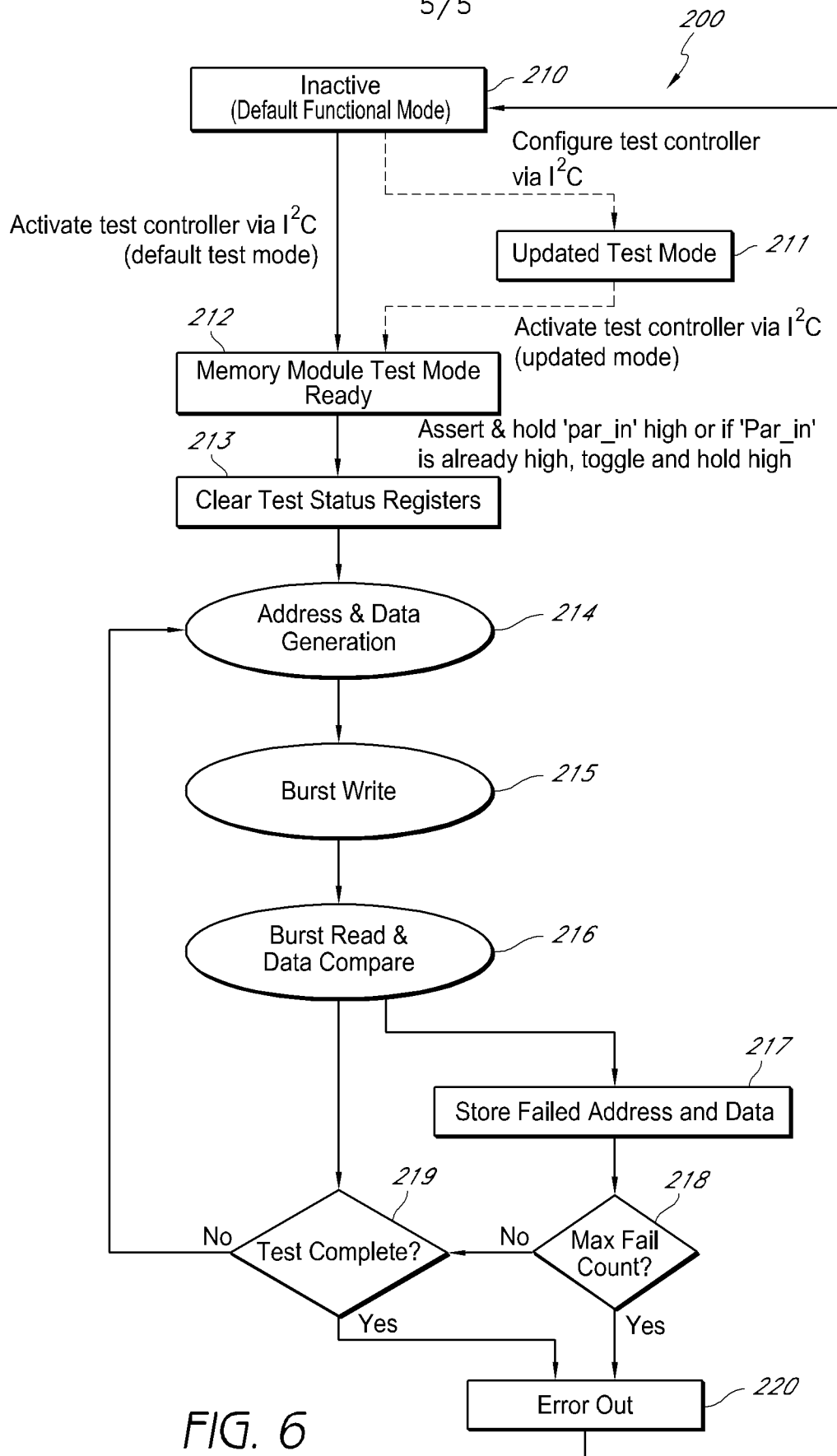


FIG. 5



Electronic Patent Application Fee Transmittal				
Application Number:				
Filing Date:				
Title of Invention:		Apparatus and Method for Self-Test in a Multi-Rank Memory Module		
First Named Inventor/Applicant Name:		Hyun Lee		
Filer:		Jamie Jie Zheng		
Attorney Docket Number:		NETL.046C2		
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	98	98
Utility Search Fee	2111	1	310	310
Utility Examination Fee	2311	1	125	125
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				533

Electronic Acknowledgement Receipt

EFS ID:	14745719
Application Number:	13745790
International Application Number:	
Confirmation Number:	4041
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C2
Receipt Date:	19-JAN-2013
Filing Date:	
Time Stamp:	19:12:51
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	2013-01-19_ApplicationDataSheet.pdf	1437595 8d48b15a2eaf62fa8db142b7d46afbe24f8c0e32	no	7

Warnings:

Information:

2	Specification	2013-01-19_Specification- NETL046C2.pdf	15495072 57bbf1b13fdce561b26760533a63bb05499a6c5d	no	24
Warnings:					
Information:					
3		2013-01-19_Claims_Abstract_N ETL046C2.pdf	70995 dbcfc4a45fd99df7ca97746d59412ba9e8e1da71	yes	4
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Claims		1	3	
	Abstract		4	4	
Warnings:					
Information:					
4	Drawings-only black and white line drawings	2013-01-19_Drawings- NETL046C2.pdf	102122 de74d99741f676bc55026863da2a0f47fed26e9b	no	5
Warnings:					
Information:					
5	Fee Worksheet (SB06)	fee-info.pdf	32696 a54e768f605e1f9163f58ce5832512311a20dc01	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			17138480		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/745,790

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	15 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	98
N/A	310
N/A	125
x 31 =	0.00
x 125 =	0.00
	0.00
	0.00
TOTAL	533

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/745,790	01/19/2013	Hyun Lee	NETL.046C2

CONFIRMATION NO. 4041

FORMALITIES LETTER



OC000000059243879

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 02/15/2013

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
Applicant must submit \$98 to complete the basic filing fee for a small entity.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- A surcharge (for late submission of the basic filing fee, search fee, examination fee or inventor's oath or declaration) as set forth in 37 CFR 1.16(f) of \$ **65** for a small entity in compliance with 37 CFR 1.27, must be submitted.

SUMMARY OF FEES DUE:

Total fee(s) required within **TWO MONTHS** from the date of this Notice is \$ **598** for a small entity

- \$ **98** Statutory basic filing fee.
- \$ **65** Surcharge.
- The application search fee has not been paid. Applicant must submit \$ **310** to complete the search fee.
- The application examination fee has not been paid. Applicant must submit \$ **125** to complete the examination fee for a small entity in compliance with 37 CFR 1.27.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

All

Applicant may submit the inventor's oath or declaration at any time before the Notice of Allowance and Fee(s) Due, PTOL-85, is mailed.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

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Alexandria VA 22313-1450

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/745,790	01/19/2013	2113	0.00	NETL.046C2	15	2

CONFIRMATION NO. 4041

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

FILING RECEIPT



OC000000059243878

Date Mailed: 02/15/2013

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CS;
Soonju Choi, Irvine, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/183,253 07/14/2011 PAT 8359501 *
which is a CON of 12/422,925 04/13/2009 PAT 8001434
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

(*)Data provided by applicant is not consistent with PTO records.

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 02/11/2013

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/745,790**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Apparatus and Method for Self-Test in a Multi-Rank Memory Module

Preliminary Class

714

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

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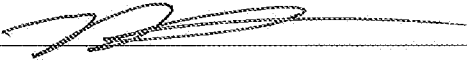
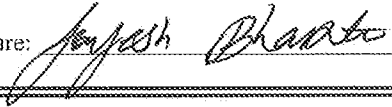
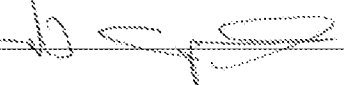
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DECLARATION FOR UTILITY OR DESIGN APPLICATION	
Title of Invention	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
As a below named inventor, I hereby declare that: This declaration is directed to: ☐ The attached application, or ☒ United States application or PCT international application number <u>13/745,790</u> filed on <u>January 19, 2013</u>. The above-identified application was made or authorized to be made by me. I believe that I am the original inventor or an original joint inventor of a claimed invention in the application. I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both. Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."	
First Inventor (Legal Name): <u>Hyun Lee</u>	
Signature: <u></u>	Date (Optional): <u>4/17/2013</u>
Second Inventor (Legal Name): <u>Jayesh R. Bhakta</u>	
Signature: <u></u>	Date (Optional): <u>4/17/2013</u>
Third Inventor (Legal Name): <u>Soonju Choi</u>	
Signature: <u></u>	Date (Optional): <u>4/17/2013</u>

Electronic Acknowledgement Receipt

EFS ID:	15547519
Application Number:	13745790
International Application Number:	
Confirmation Number:	4041
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C2
Receipt Date:	17-APR-2013
Filing Date:	19-JAN-2013
Time Stamp:	19:48:25
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Oath or Declaration filed	2013-04-17_Declaration_Signed_NETL046C2.pdf	381396 c44cf7f9988fae05e4b03b32f230809360c177b8	no	1

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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

APR 17 2013

Docket No.: NETL046C2

Customer No. 79141

RESPONSE TO NOTICE TO FILE MISSING PARTS

Applicant : Netlist, Inc.
App. No. : 13/745,790
Filed : 01/19/2013
For : APPARATUS AND METHOD FOR
SELF-TEST IN A MULTI-RANK
MEMORY MODULE

Examiner : Unknown
Art Unit : 2113
Conf No. : 4041

CERTIFICATE OF FACSIMILE

TRANSMISSION

I hereby certify that this correspondence and any
other attachment noted on the Automated
Acknowledgement Receipt, is being transmitted
from within the Pacific Time zone to the
Commissioner for Patents via
Facsimile on:

April 17, 2013

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

04/18/2013 APEREZAM 00000007 505963 13745790

01 FC:4011		70.00 OP
02 FC:2111		300.00 OP
03 FC:2311	32.00 DA	320.00 OP
04 FC:2051	70.00 DA	

Dear Sir:

Applicant herewith submit the following as required in the Notice to File Missing Parts of
Nonprovisional Application mailed on February 15, 2013:

04/18/2013 APEREZAM 00000007 505963 13745790
05 FC:2251 100.00 DA

- (X) \$98 Statutory basic filing fee
- (X) \$65 Surcharge
- (X) \$310 Search fee
- (X) \$125 Examination fee
- (X) \$100 Petition fee for one-month extension of time
- (X) Inventor's oath and declaration

The Commissioner is hereby authorized to charge any additional fees to Deposit Account
No. 50-5963.

Date: April 17, 2013


Jamie J. Zheng
Customer No. 79141
(650) 273-6008

51167
(Reg. No.)

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/745,790

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	15 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/745,790	01/19/2013	2113	800	NETL.046C2	15	2

CONFIRMATION NO. 4041

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

UPDATED FILING RECEIPT



Date Mailed: 05/06/2013

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CS;
Soonju Choi, Irvine, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Assignment For Published Patent Application

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/183,253 07/14/2011 PAT 8359501 *
which is a CON of 12/422,925 04/13/2009 PAT 8001434
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

(*)Data provided by applicant is not consistent with PTO records.

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 02/11/2013

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/745,790**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Apparatus and Method for Self-Test in a Multi-Rank Memory Module

Preliminary Class

714

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications:

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/745,790

01/19/2013

Hyun Lee

NETL.046C2

4041

79141 7590 06/13/2013
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

CHUNG, PHUNG M

ART UNIT

PAPER NUMBER

2117

NOTIFICATION DATE

DELIVERY MODE

06/13/2013

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com

Office Action Summary	Application No. 13/745,790	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 19 January 2013.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 1-15 is/are pending in the application.
 5a) Of the above claim(s) _____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) _____ is/are allowed.
- 7) ☒ Claim(s) 1,2 and 13-15 is/are rejected.
- 8) ☒ Claim(s) 3-12 is/are objected to.
- 9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 19 January 2013 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
 Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
 Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Interim copies:

- a) ☐ All b) ☐ Some c) ☐ None of the: Interim copies of the priority documents have been received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____ |
| 2) ☐ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date _____ | 4) ☐ Other: _____ |

Double Patenting

1. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory obviousness-type double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the conflicting application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement.

Effective January 1, 1994, a registered attorney or agent of record may sign a terminal disclaimer. A terminal disclaimer signed by the assignee must fully comply with 37 CFR 3.73(b).

Although the conflicting claims are not identical, they are not patentably distinct from each other. For example:

a) Claims 1, 2 and 13-15 are rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 7, 19, 20, 21, 29, 30 and 35 of U.S. Patent No. 8,001,434 in view of Han et al (patent No. 7,965,530).

Claims 1, 7, 8 and 19; claims 20, 21; and claims 29, 30 and 35 of Patent no. 8,001,434 recites all the limitations of claims 1, 2 and 13-15 of the present application except memory devices including at least one memory device including at least one memory device from each rank. However, this limitation is obvious and well known in the art, evidenced by Han et al, abstract, Fig. 19, col. 23, lines 18-47, discloses a system includes a memory module (1800) and a memory controller (1870). The memory module (1800) has a multi-rank architecture. The memory module (1800) includes first memory devices (1810) having memory device (M11, M12, M13), the first memory devices (1810) constitute a first rank (RANK0); and a second memory devices (1820) having memory devices (M21, M22, M23), the second memory devices (1820) constitute a second rank (RANK1);...; Data ports (D1, D2 and D3 of the memory controller (1870) is connected to the data ports (1801, 1802 and 1803) through data buses (1851-1853) respectively , and a command/address port of the memory controller (1870) is connected to the command/address port (1804) of the memory module (1800) through a command/address bus (1854). The data ports (1801-1803) of the memory module (1800) transmit/receive data with the memory controller (1870) through the data buses (1851-1853). That is, the memory controller (1800) receives write data (WR)

Art Unit: 2117

from the memory controller (1870), and transmits read data (RD) to the memory controller (1870) through the data ports (1851-1853). Therefore, it would have been obvious to a person of ordinary skill in the art, at the time the invention was made to add each memory of a plurality of memories with a predetermined RANK in claims 1 and 7 of the patent No. 8,001,434, as disclosed by Han et al, so that memory controller can transmits data to a memory device via at least one other memory device instead of all memory devices in a memory module, and thus, a capacitive loading effect due to the data bus with respect to the memory controller may be greatly reduced. In addition, the memory controller provides a command/address signal to at least one specific memory device instead of all memory devices in a memory module, and the specific transfers the command/address signal to other memory devices in the memory module. As a result, a capacitive loading effect due to connections to the command/address bus may be reduced.

b) Claims 1, 2 and 13-15 are rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 2, 5, 6 and 16-17 of U.S. Patent No. 8,359,501 in view of Han et al (patent No. 7,965,530).

Claims 1, 2, 5, 6 and 16-17 of Patent no. 8,359,501 recites all the limitations of claims 1, 2, 7, 8 and 13-15 of the present application except memory devices including at least one memory device including at least one memory device from each rank. However, this limitation is obvious and well known in the art, evidenced by Han et al, abstract, Fig. 19, col. 23, lines 18-47, discloses a system includes a memory module (1800) and a memory controller (1870). The memory module (1800) has a multi-rank

Art Unit: 2117

architecture. The memory module (1800) includes first memory devices (1810) having memory device (M11, M12, M13), the first memory devices (1810) constitute a first rank (RANK0); and a second memory devices (1820) having memory devices (M21, M22, M23), the second memory devices (1820) constitute a second rank (RANK1);...; Data ports (D1, D2 and D3 of the memory controller (1870) is connected to the data ports (1801, 1802 and 1803) through data buses (1851-1853) respectively , and a command/address port of the memory controller (1870) is connected to the command/address port (1804) of the memory module (1800) through a command/address bus (1854). The data ports (1801-1803) of the memory module (1800) transmit/receive data with the memory controller (1870) through the data buses (1851-1853). That is, the memory controller (1800) receives write data (WR) from the memory controller (1870), and transmits read data (RD) to the memory controller (1870) through the data ports (1851-1853). Therefore, it would have been obvious to a person of ordinary skill in the art, at the time the invention was made to add each memory of a plurality of memories with a predetermined RANK in claims 1 and 7 of the patent No. 8,001,434, as disclosed by Han et al, so that memory controller can transmits data to a memory device via at least one other memory device instead of all memory devices in a memory module, and thus, a capacitive loading effect due to the data bus with respect to the memory controller may be greatly reduced. In addition, the memory controller provides a command/address signal to at least one specific memory device instead of all memory devices in a memory module, and the specific transfers the command/address signal to other memory devices in the memory module. As a result,

Art Unit: 2117

a capacitive loading effect due to connections to the command/address bus may be reduced.

c) Claims 1, 2 and 13-15 are rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 7, 19, 20, 21, 29, 30 and 35 of U.S. Patent No. 8,001,434 in view of Welker (patent No. 7,957,218).

Claims 1, 7 and 19; claims 20, 21; and claims 29, 30 and 35 of Patent no. 8,001,434 recites all the limitations of claims 1, 2 and 13-15 of the present application except memory devices including at least one memory device including at least one memory device from each rank. However, this limitation is obvious and well known in the art, evidenced by Welker (col. 4, lines 46-67 to col. 5, lines 1-34, where a first memory having a first rank, a second memory having a second rank and a third memory having a third rank; and col. 6, lines 10-14, "each memory of the plurality of memories belongs to a predetermined rank). (Also see abstract, Fig. 1, col. 2, lines 17-49). Therefore, it would have been obvious to a person of ordinary skill in the art, at the time the invention was made to add each memory of a plurality of memories with a predetermined rank in claims 1 and 7 of the patent No. 8,001,434, as disclosed by Welker, so that memory controller can ensure that data transfers between tween platform and memory are performed efficiently which includes consideration of speed and reliability.

d) Claims 1, 2 and 13-15 are rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 2, 5, 6 and 16-17 of U.S. Patent No. 8,359,501 in view of Welker (patent No. 7,957,218).

Claims 1, 2, 5, 6 and 16-17 of Patent no. 8,359,501 recites all the limitations of claims 1, 2 and 13-15 of the present application except memory devices including at least one memory device including at least one memory device from each rank. However, this limitation is obvious and well known in the art, evidenced by Welker (col. 4, lines 46-67 to col. 5, lines 1-34, where a first memory having a first rank, a second memory having a second rank and a third memory having a third rank; and col. 6, lines 10-14, "each memory of the plurality of memories belongs to a predetermined rank). (Also see abstract, Fig. 1, col. 2, lines 17-49). Therefore, it would have been obvious to a person of ordinary skill in the art, at the time the invention was made to add each memory of a plurality of memories with a predetermined rank in claims 1 and 7 of the patent No. 8,001,434, as disclosed by Welker, so that memory controller can ensure that data transfers between tween platform and memory are performed efficiently which includes consideration of speed and reliability.

Claim Objections

2. Claims 3-12 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday.

Art Unit: 2117

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ajay M. Bhatia can be reached on 571-272-3906. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

Notice of References Cited	Application/Control No. 13/745,790	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-7,957,218	06-2011	Welker, James A.	365/233.13
*	B	US-7,965,530	06-2011	Han et al.	365/51
*	C	US-5,732,241	03-1998	Chan, Alfred K.	711/131
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	122857	data handler or data management or data controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:19
L2	6356	(organiz\$3 or set\$4 or arrang\$3) same (memory devices) same (ranks or sizes or levels)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:19
L3	84	(obtain\$3 or compar\$3 or determin\$3) and (((written data or write data) and data handler) and (read data and memory location\$1) and (test result\$1 or result\$1 or comparison result\$1))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:19
L4	62	(provid\$3 or send\$3 or forward\$3 or transmit\$4 or propagat\$3) same (test result\$1 or result\$1 or comparison result\$1) same (system memory controller)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:19
L5	5	L4 and (L1 or L2 or L3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:19
L6	84	L3 and (L1 or L2)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:21
L7	408	L1 and (L2 or L6)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:21
L8	3	(provid\$3) adj (control and address	US-PGPUB;	ADJ	ON	2013/06/06

		signal\$1) same (memory device\$1)	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			11:21
L9	55901	memory rank\$1 or memory size\$1 or bit width or full bit-with or bi-wide rank\$1 or rank\$1 with memory device\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:24
L10	5	l5 and (l6 or l7 or l8 or l9)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:25
L11	175518	system memory controller or module controller or memory controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:25
L12	5	l10 and l11	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:26
L13	10717	(organiz\$3 or set\$4 or arrang\$3) same (memory devices) same (ranks or sizes or levels or bit width or bit\$1 or bit-wide or bit-wide rank\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:27
L14	4	l12 and l13	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 11:28
L17	7	(compar\$3 or test\$3 or determin\$3) and (((writ\$3 with data generated with data handler) same (read\$3 with memory location\$1) or (data written in and data read out)) and (test result\$1 or result\$1 or comparison result\$1 or error\$1 or failure or fault\$1))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 15:56
L20	5	l17 and (l1 or l2 or l3 or l4 or l9 or l11 or l13)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	ADJ	ON	2013/06/06 15:59

			IBM_TDB			
L21	8714	(714/719,736,819;365/201).CCLS.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 16:02
L22	1	L21 AND L20	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 16:02
L23	1	L21 AND L17	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 16:02
L24	2	L21 AND L6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/06/06 16:02

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C:\ Users\ pchung\ Documents\ EAST\ Workspaces\ 13745790.wsp

<i>Index of Claims</i> 	Application/Control No. 13745790	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	06/07/2013								
	1	v								
	2	v								
	3	o								
	4	o								
	5	o								
	6	o								
	7	o								
	8	o								
	9	o								
	10	o								
	11	o								
	12	O								
	13	V								
	14	V								
	15	V								

Search Notes 	Application/Control No. 13745790	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
714	719,736,819	6/6/13	pmc
365	201	6/6/13	pmc

SEARCH NOTES		
Search Notes	Date	Examiner

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

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BIB DATA SHEET

CONFIRMATION NO. 4041

SERIAL NUMBER 13/745,790	FILING or 371(c) DATE 01/19/2013 RULE	CLASS 714	GROUP ART UNIT 2117	ATTORNEY DOCKET NO. NETL.046C2		
APPLICANTS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CS; Soonju Choi, Irvine, CA; Netlist, Inc., Irvine, CA ** CONTINUING DATA ***** This application is a CON of 13/183,253 07/14/2011 PAT 8359501 * which is a CON of 12/422,925 04/13/2009 PAT 8001434 which claims benefit of 61/044,801 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,839 04/14/2008 (*)Data provided by applicant is not consistent with PTO records. ** FOREIGN APPLICATIONS ***** ** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** ** SMALL ENTITY ** 02/11/2013						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/PHUNG M CHUNG/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 5	TOTAL CLAIMS 15	INDEPENDENT CLAIMS 2
ADDRESS The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES						
TITLE Apparatus and Method for Self-Test in a Multi-Rank Memory Module						
FILING FEE RECEIVED 800	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit			

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist
App. No. : 13/745,790
Filed : January 19, 2013
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Examiner : CHUNG, PHUNG M
Art Unit : 2117
Conf. No. : 4041
Docket No. : NETL.046C2

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September 11, 2013

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Office Action mailed June 13, 2013
for the above-identified patent application.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit
Account No. 50-5963.

AMENDMENTS to the CLAIMS begin on Page 2 of this paper.

REMARKS begin on Page 6 of this paper.

IN THE CLAIMS:

Rewrite the pending claims as follows:

1. (Currently Amended) A method performed by a memory module ~~having a plurality of data ports that are coupled to a system memory controller~~, the memory module comprising a plurality of memory devices organized in ~~ranks~~ at least one rank, comprising:

initiating a test mode for the memory module in response to control signals received from the system memory controller;

generating data at each of a plurality of data handlers on the memory module, ~~each the data generated at each respective data handler being coupled to a respective set of data ports of the plurality of data ports and~~ written to a respective set of memory devices at least one memory device of the plurality of memory devices, each set of ~~memory devices at least one memory device~~ including at least one memory device from each of the at least one rank;

obtaining test results by checking data read from the plurality of memory devices ~~writing the data generated in each respective data handler into memory locations in the set of memory devices coupled to the respective data handler and by reading from the memory locations~~; and

providing the test results to the system memory controller.

2. (Currently Amended) The method of claim 1 further comprising:

generating control and address signals at a module controller of the memory module, the control and address signals indicating the memory locations in the respective set of at least one memory device into which data generated at each respective data handler is written; and

providing the control and address signals to the plurality of memory devices.

3. (Original) The method of claim 1 further comprising:

updating the test mode by programming a test controller of the memory module with test configuration information, the test configuration information including test data patterns, test data sequences, test failure conditions.

4. (Original) The method of claim 1 further comprising:

exiting the test mode in response to a signal from the system memory controller.

5. (Original) The method of claim 4 wherein the signal from the memory controller is received via a parity-in pin of the memory module.

6. (Original) The method of claim 1 further comprising:
sending out a test completion indication signal to the memory controller.
7. (Original) The method of claim 1 wherein the test results are provided to the memory controller via bidirectional data signal channels between the memory module and system memory controller.
8. (Original) The method of claim 1 wherein the test results are provided to the memory controller via bidirectional data strobe signal channels between the memory module and system memory controller.
9. (Original) The method of claim 1 wherein the test results are provided to the memory controller in response to a memory failure count exceeding a preset number.
10. (Original) The method of claim 1 wherein initiating the test mode comprises configuring a test case according to specified test data patterns and read/write address sequences.
11. (Original) The method of claim 1 further comprising programming a data pattern for the test mode based on information received from the system memory controller.
12. (Original) The method of claim 1 wherein initiating the test mode comprises configuring at least one switch in each of the plurality of data handlers.
13. (Currently Amended) A memory module for operating with a system memory controller, comprising:
 - a plurality of data ports;
 - a plurality of memory devices organized in ~~ranks~~at least one rank; and
 - a plurality of data handlers, each respective data handler being operatively coupled to a respective set of data ports of the plurality of data ports and to a respective set of ~~memory devices~~at least one memory device of the plurality of memory devices, each set of ~~memory devices~~at least one memory device including at least one memory device from each rank; andwherein each respective data handler is configured to provide write data received from the system memory controller via the respective data ports to the respective set of memory devices in a normal mode, and to provide test data generated in the respective data handler to the respective set of memory devices in a test mode.

14. (Original) The memory module of claim 13, further comprising
a module controller coupled to the plurality of memory devices and to the plurality of data handlers, the module controller is configured to provide control and address information received from the system memory controller to the plurality of memory devices in the normal mode, and to provide control and address information generated in the module controller to the plurality of memory devices in the test mode.
15. (Original) The memory module of claim 14 wherein the plurality of data handlers are capable of being operated independently of each other.
16. (New) A memory module for operating with a system memory controller, comprising:
a module controller to process input control signals from the system memory controller and to generate output control signals;
a plurality of memory devices configured to perform memory operations in response to signals from the module controller; and
a plurality of data handlers, each respective data handler being configured to generate test data and to provide the test data to a respective set of at least one memory device of the plurality of memory devices in response to signals from the module controller; and
wherein the memory module is configured to obtain test results by reading from the respective set of at least one memory device in response to signals from the module controller and by comparing data read from the respective set of at least one memory device with the test data provided to the respective set of at least one memory device.
17. (New) The memory module of claim 16, wherein the memory module is configurable to operate in either a normal mode or a test mode, wherein the module controller is configured to provide control and address information received from the system memory controller to the plurality of memory devices in the normal mode, and to provide control and address information generated in the module controller to the plurality of memory devices in the test mode.
18. (New) The memory module of claim 17, wherein the module controller is further configured to program a data pattern for the test mode based on information received from the system memory controller

19. (New) The memory module of claim 17, wherein the respective data handler is further configured to provide write data received from the system memory controller to the respective set of at least one memory device in the normal mode.
20. (New) The memory module of claim 16 wherein the plurality of data handlers are capable of being operated independently of each other.

REMARKS

This amendment responds to the office action mailed June 13, 2013.

I. MCKESSON STATEMENT

The pending application is a continuation of U.S. Patent Application No. 13/183,253, filed July 14, 2011, now U.S. Pat. No. 8,359,501, which is a continuation of U.S. Patent Application No. 12/422,925, filed April 13, 2009, now U.S. Pat. No. 8,001,434. The present application is also related to US. Patent Application No. 12/422,912, filed on April 13, 2009, and entitled "Self-Adjusting Damper", which is now abandoned, and to U.S. Application No. 13/412,243, filed on March 5, 2012 and entitled "Circuit Providing Load Isolation and Noise Reduction," which is a continuation of U.S. Application No. 12/422,853, filed on April 13, 2009 and entitled "Circuit Providing Load Isolation and Noise Reduction," now U.S. Pat. No. 8,154,901. The Examiner is encouraged to review the art made of record, Office Action(s) and the Notice of Allowance in each of the above-mentioned related applications, all of which are available on PAIR.

II. AMENDMENTS TO THE CLAIMS

Claims 1, 2 and 13 are amended, and new claims 16-20 are added. Support for the amendments made in claims 1, 2, and 13 and for the new claims can be found throughout the specification and drawings. No new matter is added.

III. DOUBLE PATENTING REJECTION

Claims 1, 2 and 13-15 are rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 7, 19, 20, 21, 29, 30 and 35 of U.S. Patent No. 8,001,434 in view of Han et al (patent No. 7,965,530).

Claims 1, 2 and 13-15 are further rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 2, 5, 6 and 16-17 of U.S. Patent No. 8,359,501 in view of Han et al (patent No. 7,965,530).

Claims 1, 2 and 13-15 are further rejected on the ground of nonstatutory obviousness-type double patenting as being unpatentable over claims 1, 7, 19, 20, 21, 29, 30 and 35 of U.S. Patent No. 8,001,434 in view of Welker (patent No. 7,957,218).

Based on the terminal disclaimer filed with this response, the double patenting rejections should be withdrawn.

In light of the above amendments and remarks, the Applicants respectfully request that the Examiner reconsider this application with a view towards allowance. The Examiner is invited to call the undersigned attorney at (650) 273-6008, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: September 11, 2013

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

Electronic Acknowledgement Receipt

EFS ID:	16830636
Application Number:	13745790
International Application Number:	
Confirmation Number:	4041
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C2
Receipt Date:	12-SEP-2013
Filing Date:	19-JAN-2013
Time Stamp:	01:12:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		2013-09-11_Amendment_NETL046C2.pdf	97260 ef972e0962b1a46f08ec8301e3a794840c7a1d7f	yes	7

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Claims	2	5
	Applicant Arguments/Remarks Made in an Amendment	6	7
Warnings:			
Information:			
Total Files Size (in bytes):		97260	
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Application Number	13/745,790
Filing Date	January 19, 2013
First Named Inventor	Hyun Lee
Title	MEMORY BOARD WITH SELF-TESTING CAPABILITY
Art Unit	2117
Examiner Name	CHUNG, PHUNG M
Attorney Docket Number	NETL.046C2

SIGNATURE of Applicant or Patent Practitioner

Signature	/Jamie J. Zheng/	Date	September 11, 2013
Name	Jamie J. Zheng	Telephone	650-273-6008
Registration Number	51167		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications.

☐ *Total of _____ forms are submitted.

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I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☐ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document)

SIGNATURE of Applicant for Patent

Signature	Date
Name	Telephone
Title and Company	

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- ☐ *Total of _____ forms are submitted.

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The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

GENERAL POWER OF ATTORNEY BY APPLICANT FOR UNITED STATES PATENT

I hereby revoke all previous powers of attorney given in the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

I hereby appoint the practitioner associated with the following Customer Number
:

79141

as my/our attorney or agent, and to transact all business in the United States Patent and Trademark Office in connection with the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

Please recognize or change the **correspondence address** for the application referenced in the attached transmittal letter to the address associated with the above-mentioned **Customer Number**.

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is currently being filed in this document)

SIGNATURE of Applicant for Patent

Signature

Date: January 9, 2013

Name

Gail Sasaki

Title

Vice President and Chief Financial Officer

Company

Netlist, Inc.

Note: Signature - this form must be signed by the Applicant or a person authorized to act on behalf of Applicant.

Submit multiple forms for more than one signature

Total of ___ forms are submitted.

**TERMINAL DISCLAIMER TO OBVIATE A DOUBLE PATENTING
REJECTION OVER A "PRIOR" PATENT**

Docket Number (Optional)

In re Application of: Netlist, Inc.

Application No.: 13/745,790

Filed: January 19, 2013

For: MEMORY BOARD WITH SELF-TESTING CAPABILITY

The applicant, Netlist, Inc., owner of 100 percent interest in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of **prior patent** No. 8,001,434; 8,359,501 as the term of said **prior patent** is presently shortened by any terminal disclaimer. The applicant hereby agrees that any patent so granted on the instant application shall be enforceable only for any during such period that it and the **prior patent** are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the applicant does not disclaim the terminal part of the term of any patent granted on the instant application that would extend to the expiration date of the full statutory term of the **prior patent**, "as the term of said **prior patent** is presently shortened by any terminal disclaimer," in the event that said **prior patent** later:

- expires for failure to pay a maintenance fee;
- is held unenforceable;
- is found invalid by a court of competent jurisdiction;
- is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321;
- has all claims canceled by a reexamination certificate;
- is reissued; or
- is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer.

Check either box 1 or 2 below, if appropriate.

1. ☐ The undersigned is the applicant. If the applicant is an assignee, the undersigned is authorized to act on behalf of the assignee.

I hereby acknowledge that any willful false statements made are punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

2. ☒ The undersigned is an attorney or agent of record. Reg. No. 51,167

/Jamie J. Zheng/

Signature

September 11, 2013

Date

Jamie J. Zheng

Typed or printed name

Attorney of Record

Title

650-273-6008

Telephone Number

- ☒ Terminal disclaimer fee under 37 CFR 1.20(d) included.

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

This collection of information is required by 37 CFR 1.321. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13745790			
Filing Date:	19-Jan-2013			
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C2			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Statutory or Terminal Disclaimer	1814	1	160	160
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	16830654
Application Number:	13745790
International Application Number:	
Confirmation Number:	4041
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C2
Receipt Date:	12-SEP-2013
Filing Date:	19-JAN-2013
Time Stamp:	01:47:14
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 160
RAM confirmation Number	10115
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Power of Attorney	2013-09-11_POA_Transmittal_ NETL046C2.pdf	195870 8f056bd4a09774e07cdcfefb3b8b85ec7f17 9d30	no	3
Warnings:					
Information:					
2	Power of Attorney	AIA_General_Power_of_Attorn ey.pdf	133545 1f355baa450b5635a767788f0161daaf2767 3c4f	no	1
Warnings:					
The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing					
Information:					
3	Terminal Disclaimer Filed	2013-09-11_Terminal_Disclaim er_NETL046C2.pdf	147583 453a0abf88b49db04e5c65149ae7d8df1e4 161de	no	2
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	30035 cf42a7b2138df085a6638cbdef345b1efe31 0b7c	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			507033		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 13/745,790	Filing Date 01/19/2013	☐ To be Mailed
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 ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	09/12/2013	CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$40 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	X \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						0	

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
		CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE							

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

 LIE
/DORRETTA BROOKS/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Application Number 	Application/Control No. 13/745,790	Applicant(s)/Patent under Reexamination LEE ET AL.	

Document Code - DISQ	Internal Document – DO NOT MAIL
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TERMINAL DISCLAIMER	☒ APPROVED	☐ DISAPPROVED
Date Filed : 9/12/13	This patent is subject to a Terminal Disclaimer	

Approved/Disapproved by:

Felicia D. Roberts 8,001,434 and 8,359,501



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/745,790	01/19/2013	Hyun Lee	NETL.046C2

79141
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

CONFIRMATION NO. 4041
POA ACCEPTANCE LETTER



Date Mailed: 09/18/2013

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 09/12/2013.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/sleutchit/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 11/15/2013
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

CHUNG, PHUNG M

ART UNIT

PAPER NUMBER

2117

DATE MAILED: 11/15/2013

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

13/745,790

01/19/2013

Hyun Lee

NETL.046C2

4041

TITLE OF INVENTION: Apparatus and Method for Self-Test in a Multi-Rank Memory Module

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$890	\$0	\$0	\$890	02/18/2014

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 11/15/2013
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/745,790	01/19/2013	Hyun Lee	NETL.046C2	4041

TITLE OF INVENTION: Apparatus and Method for Self-Test in a Multi-Rank Memory Module

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$890	\$0	\$0	\$890	02/18/2014

EXAMINER	ART UNIT	CLASS-SUBCLASS
CHUNG, PHUNG M	2117	714-719000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 1 _____ 2 _____ 3 _____
---	---

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

NOTE: Absent a valid certification of Micro Entity Status (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

☐ Applicant asserting small entity status. See 37 CFR 1.27

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

☐ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/745,790	01/19/2013	Hyun Lee	NETL.046C2	4041

79141 7590 11/15/2013
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

CHUNG, PHUNG M

ART UNIT	PAPER NUMBER
----------	--------------

2117

DATE MAILED: 11/15/2013

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 0 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 0 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

**Notices of Allowance and Fee(s) Due mailed between October 1, 2013 and
December 31, 2013**

(Addendum to PTOL-85)

If the “Notice of Allowance and Fee(s) Due” has a mailing date on or after October 1, 2013 and before January 1, 2014, the following information is applicable to this application.

If the issue fee is being timely paid on or after January 1, 2014, the amount due is the issue fee and publication fee in effect January 1, 2014. On January 1, 2014, the issue fees set forth in 37 CFR 1.18 decrease significantly and the publication fee set forth in 37 CFR 1.18(d)(1) decreases to \$0.

If an issue fee or publication fee has been previously paid in this application, applicant is not entitled to a refund of the difference between the amount paid and the amount in effect on January 1, 2014.

Notice of Allowability	Application No. 13/745,790	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 9/12/13.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|---|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____ | 6. ☐ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/Phung M. Chung/
Primary Examiner
Art Unit: 2117

The present application is being examined under the pre-AIA first to invent provisions.

EXAMINER'S AMENDMENT

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

The application has been amended as follows:

Specification: Paragraph [0001], line 2, after "July 14, 2011," inserted - - now U.S. Patent No. 8,359,501, - -; and

Line 3, after "April 13, 2009," inserted - -now U.S. Patent No. 8,001,434, - -.

//Phung M. Chung//

Primary Examiner, Art Unit 2117

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	3	((("8359501") or ("8001434") or ("8154901")).PN.	US-PGPUB; USPAT; USOCR; JPO	OR	OFF	2013/10/21 09:01
L2	128203	data handler or data management or data controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:01
L3	184020	system memory controller or module controller or memory controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:01
L4	79440	memory module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:02
L5	268383	memory devices	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:02
L6	138880	(test data or test vector\$1 or test pattern\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:03
L7	4243	l2 and l6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:03
L8	521	l7 and l5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:03
L9	70	l8 and l4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:03
L10	42	l9 and l3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:03
L13	268	(writ\$3 or provid\$1) same (test	US-PGPUB;	ADJ	ON	2013/10/21

		data or test vector\$1 or test pattern\$1) same (memory device\$1) same (test mode)	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			10:06
L14	389	(writ\$3 or provid\$3) same (data) same (memory device\$1) same (normal mode)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:06
L15	4	l10 and (l13 or l14)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:07
L16	8550	compar\$3 and (((data) adj (written or stored)) and (data read out)) and (memory device\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:08
L17	7084	(714/719,718,799).ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:33
L18	9	l17 and l10	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:34
L19	2	"7844868"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2013/10/21 10:53

10/21/2013 11:53:22 AM
C:\Users\pchung\Documents\EAST\Workspaces\13745790.wsp

Issue Classification 	Application/Control No. 13745790	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☒ T.D. ☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		17												
	2		18												
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	14														
	15														
	16														

NONE		Total Claims Allowed: 20	
(Assistant Examiner)	(Date)	O.G. Print Claim(s)	O.G. Print Figure
/PHUNG M CHUNG/ Primary Examiner.Art Unit 2117	11/9/13	1	1
(Primary Examiner)	(Date)		

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L2	0	(system memory controller AND memory module AND memory devices AND test mode AND data handlers AND test result AND checking data AND test data AND write data AND read data).clm.	US-PGPUB; USPAT; UPAD	ADJ	ON	2013/11/09 22:22

11/ 9/ 2013 10:38:43 PM
C:\ Users\ pchung\ Documents\ EAST\ Workspaces\ 13745790.wsp

Search Notes 	Application/Control No. 13745790	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
714	719,736,819	10/21/13	pmc
365	201	10/21/13	pmc

SEARCH NOTES		
Search Notes	Date	Examiner
EAST searched.	10/21/13	pmc

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
714	719	11/9/13	pmc

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The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306



**Courtesy Reminder for
Application Serial No: 13/745,790**

Attorney Docket No: NETL.046C2

Customer Number: 79141

Date of Electronic Notification: 11/15/2013

This is a courtesy reminder that new correspondence is available for this application. If you have not done so already, please review the correspondence. The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

An email notification regarding the correspondence was sent to the following email address(es) associated with your customer number:

jz@jzpatent.com

To view your correspondence online or update your email addresses, please visit us anytime at <https://sportal.uspto.gov/secure/myportal/privatepair>. If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov or call 1-866-217-9197.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax **(571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Customer No. 79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/745,790	01/19/2013	Hyun Lee	NETL.046C2	4041
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TITLE OF INVENTION:

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional	SMALL	\$890	\$0	\$0	\$890	02/18/2014
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EXAMINER	ART UNIT	CLASS-SUBCLASS
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1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,

1 JAMIE J. ZHENG, Ph.D, Esq.

(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☒ Payment by credit card. Form PTO-2038 is attached.
- ☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 50-5963 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☒ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /Jamie J. Zheng/

Date February 11, 2014

Typed or printed name Jamie J. Zheng

Registration No. 51,167

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13745790			
Filing Date:	19-Jan-2013			
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C2			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl Issue Fee	2501	1	480	480
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				480

Electronic Acknowledgement Receipt

EFS ID:	18181014
Application Number:	13745790
International Application Number:	
Confirmation Number:	4041
Title of Invention:	Apparatus and Method for Self-Test in a Multi-Rank Memory Module
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Issue Fee Payment (PTO-85B)	2014-02-11_Issue_Fee_Transmission_NETL046C2.pdf	74203 d420e9dc870e18ae57848b37e57e1742ea7d13f5	no	2
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30029 e7c05f838400fe657d3f8beb596e21b7206ec173	no	2
Warnings:					
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/745,790	04/01/2014	8689064	NETL.046C2	4041

79141 7590 03/12/2014
The Law Office of Jamie Zheng, Ph.D Esq.
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ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

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AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO. 8:16-cv-01605	DATE FILED 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,756,364	June 17, 2014	Netlist, Inc.
2 8,516,185	August 20, 2013	Netlist, Inc.
3 8,001,434	August 16, 2011	Netlist, Inc.
4 8,359,501	January 22, 2013	Netlist, Inc.
5 8,689,064	April 1, 2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK	
1			
2			
3			
4			
5			

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX
MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00560
Patent 8,689,064 B1

Before STEPHEN C. SIU, MATTHEW R. CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

SIU, *Administrative Patent Judge*.

DECISION
Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

SK Hynix Inc. (“Petitioner”) requests *inter partes* review of claim 16, of U.S. Patent No. 8,689,064 B1 (“the ’064 patent,” Ex. 1001) pursuant to 35 U.S.C. §§ 311 *et seq.* Paper 1 (“Pet.”). Netlist, Inc. (“Patent Owner”) filed a preliminary response. Paper 6 (“Prelim. Resp.”). Institution of an *inter partes* review is authorized by statute when “the information presented in the petition . . . and any response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a); *see* 37 C.F.R. § 42.108. Upon consideration of the Petition and Preliminary Response, we conclude the information presented shows there is a reasonable likelihood that Petitioner would prevail in establishing the unpatentability of claim 16 of the ’064 patent.

A. Related Matters

We are informed that the ’064 patent is presently the subject of the following: *Netlist, Inc. v. Smart Modular Technologies, Inc. et al.*, Case No. 4:13-cv-05889-YGR (N.D. Cal.); *Netlist, Inc. v. Smart Modular Technologies, Inc. et al.*, Case No. 2:13-cv-02613-TLN (E.D. Cal.); *SanDisk Corp. et al. v. Netlist, Inc.*, Case No. IPR2014-00970 (PTAB); *SanDisk Corp. et al. v. Netlist, Inc.*, Case No. IPR2014-00971 (PTAB); *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. IPR2014-01372 (PTAB); *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. IPR2014-01373 (PTAB); *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. IPR2014-01374 (PTAB); *Smart Modular Technologies, Inc. v. Netlist, Inc.*, Case No. IPR2014-01375 (PTAB); *Netlist, Inc. v. SanDisk LLC et al.*, Case Nos. 16-2274, -2338, -2339 (Fed. Cir.); *Smart Modular*

Technologies, Inc. v. Netlist, Inc., Case No. 16-2666 (Fed. Cir.); *Netlist, Inc. v. SK Hynix Inc. et al.*, Case No. 8:16-cv-01605-JLS (C.D. Cal.); and *In re Certain Memory Modules & Components Thereof*, Inv. No. 337-TA-1023 (ITC). *See* Pet. 2.

B. The '064 Patent

The '064 patent describes self-testing memory modules. Ex. 1001, 1:27–29.

C. Illustrative Claim

Independent claim 16, reproduced below, is illustrative of the claimed subject matter:

16. A memory module for operating with a system memory controller, comprising:
- a module controller to process input control signals from the system memory controller and to generate output control signals;
 - a plurality of memory devices configured to perform memory operations in response to signals from the module controller; and
 - a plurality of data handlers, each respective data handler being configured to generate test data and to provide the test data to a respective set of at least one memory device of the plurality of memory devices in response to signals from the module controller; and
- wherein the memory module is configured to obtain test results by reading from the respective set of at least one memory device in response to signals from the module controller and by comparing data read from the respective set of at least one memory device with the test data provided to the respective set of at least one memory device.

Id. at 18:8–28.

D. Asserted Grounds of Unpatentability

Petitioner asserts that claim 16 is unpatentable under 35 U.S.C. § 102 or § 103 as anticipated by or obvious over Averbuj.¹

II. DISCUSSION

A. Claim Construction

In an *inter partes* review, we construe claim terms in an unexpired patent according to their broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b). Consistent with the broadest reasonable construction, claim terms are presumed to have their ordinary and customary meaning as understood by a person of ordinary skill in the art in the context of the entire patent disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007).

“Memory module”

Petitioner does not provide a proposed construction for this term. Patent Owner contends that “the term ‘memory module’ is known to the industry and standards bodies as a removable printed circuit board that contains one or more memory chips—the term ‘memory module’ is not used to refer to a portion of an integrated circuit (chip).” Prelim. Resp. 19–20.

Claim 1 recites a “memory module” operating with a system memory controller and comprising a controller, devices, and data handlers. Claim 1 does not recite additional defining characteristics of a memory module.

¹ US Patent Publication 2005/0257109 A1, published Nov. 17, 2005 (“Averbuj,” Ex. 1005).

Hence, based on the context in which the term “memory module” is used in the claim, one of skill in the art would have broadly but reasonably understood the term “memory module” to comprise various claimed components. *Phillips v. AWH Corp.*, 415 F.2d 1301, 1314 (Fed. Cir. 2005), citing *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996) (stating that “[t]o begin with, the context in which a term is used in the asserted claim can be highly instructive,” “the claims themselves provide substantial guidance as to the meaning of particular claim terms”).

One of skill in the art in attempting to construe the meaning of the claim term “memory module,” after having examined the context of the claim itself, would have turned to the Specification for guidance. The Specification “is always highly relevant to the claim construction analysis.” *Vitronics Corp.*, 90 F.3d at 1582. Patent Owner does not point out a specific definition of the term “memory module” disclosed in the Specification. We observe that the Specification does not appear to provide a definition of the term “memory module.” While providing an *example* of one embodiment of a “memory module” that “includes a printed circuit board” (Ex. 1001 5:6–7), the Specification does not also disclose that a “memory module” must include a “printed circuit board” or that any of the disclosed functions of the “memory module” rely in any way on the presence of a “printed circuit board.” Indeed, the Specification explicitly discloses that “the inventive subject matter [disclosed in the Specification] extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof” and further suggests that the disclosure is intended to include any system or apparatus capable of performing “the acts or operations making up the

method/process,” as disclosed. Ex. 1001, 16:16–19, 22–23. The Specification does not disclose also that the “acts or operations making up the method/process,” as disclosed in the Specification cannot be performed without the “printed circuit board.”

Hence, one of ordinary skill in the art would have broadly but reasonably construed the term “memory module,” in light of the Specification as potentially including a “printed circuit board,” but would also have “extended” the interpretation of “memory module” “beyond the specifically disclosed” example in the Specification to exclude features that are not disclosed as required in performing the disclosed “acts or operations making up the method/process.” Because the Specification does not appear to disclose that a “printed circuit board” is a required component in performing the “acts or operations making up the method/process” as disclosed, we conclude that one of ordinary skill in the art would not have unreasonably construed the term “memory module” overly *narrowly* as requiring a “printed circuit board,” but, instead, would have broadly but reasonably construed the term “memory module” as potentially including a “printed circuit board,” but not requiring a “printed circuit board” (or any other form of “carrier”).

Extrinsic evidence has been considered “less significant than the intrinsic record” but “can shed useful light on the relevant art.” *Phillips*, 415 F.3d at 1317–1318 (citing *Renishaw PLC v. Marposs Societa per Azioni*, 158 F.3d 1243, 1250 (Fed. Cir. 1998) and *Rexnord Corp. v. Laitram Corp.*, 274 F.3d 1336, 1344 (Fed. Cir. 2001)). Even though extrinsic evidence is “less significant,” Patent Owner argues that “Figure 7.3 . . . of

Petitioner's Exhibit 1008² . . . confirms that a POSITA would have known that a 'memory module' is one or more memory chips . . . on a printed circuit board carrier." Prelim. Resp. 21. The cited reference states that "computer memory" "appears in most computers in the form shown in Figure 7.1 – the ubiquitous *memory module*, a small computer board (a *printed circuit board*, or PCB) that has a handful of chips attached to it." Ex. 1008, 315. Figure 7.1 of Exhibit 1008 illustrates the appearance "in most computers" of a "memory module." *Id.* Patent Owner does not indicate, nor can we discern, that the cited reference provides an explicit definition of the term "memory module" or that a "memory module" must include a "computer board." Indeed, the cited reference explicitly states that such a "memory module" may "appear" as such "in most computers" but does not indicate how a "memory module" "appears" in other computers, for example.

Patent Owner argues that "Averbuj 'uses the term 'memory module' to refer to an embedded memory portion of chip [i.e., not requiring a printed circuit board or other "carrier"]" but "Averbuj does not use the term 'memory module' in accordance [with] its conventional meaning [of requiring] memory chips on a carrier such as a printed circuit board." Prelim. Resp. 5. In other words, Patent Owner argues that even though Averbuj discloses a "memory module" that is supposedly "embedded in" a chip and does not include a "carrier such as a printed circuit board," one of ordinary skill in the art would not consider such a device to be a "memory module."

² Bruce Jacob, et al., "Memory Systems – Cache, DRAM, Disk," Ex. 1008.

In the absence of any persuasive evidence that Averbuj is mistaken that a “memory module” can lack a “carrier such as a printed circuit board” and that, in fact, a “memory module” cannot be “embedded [in] memory,” as Patent Owner alleges, we are not persuaded by Patent Owner’s argument at least because Patent Owner does not provide sufficient evidence to support the contention that Averbuj (as a representative of one of ordinary skill in the art) is, in fact, mistaken that a “memory module” can lack a “carrier such as a printed circuit board.” In other words, Patent Owner does not explain how one of ordinary skill in the art would have imported a non-recited claim requirement of a “memory module” *not* being embedded in an integrated circuit and to require a carrier such as a printed circuit board when Averbuj (as a representative of one of ordinary skill in the art) explicitly discloses a “memory module” to be embedded in an integrated circuit (and lacking a “carrier”).

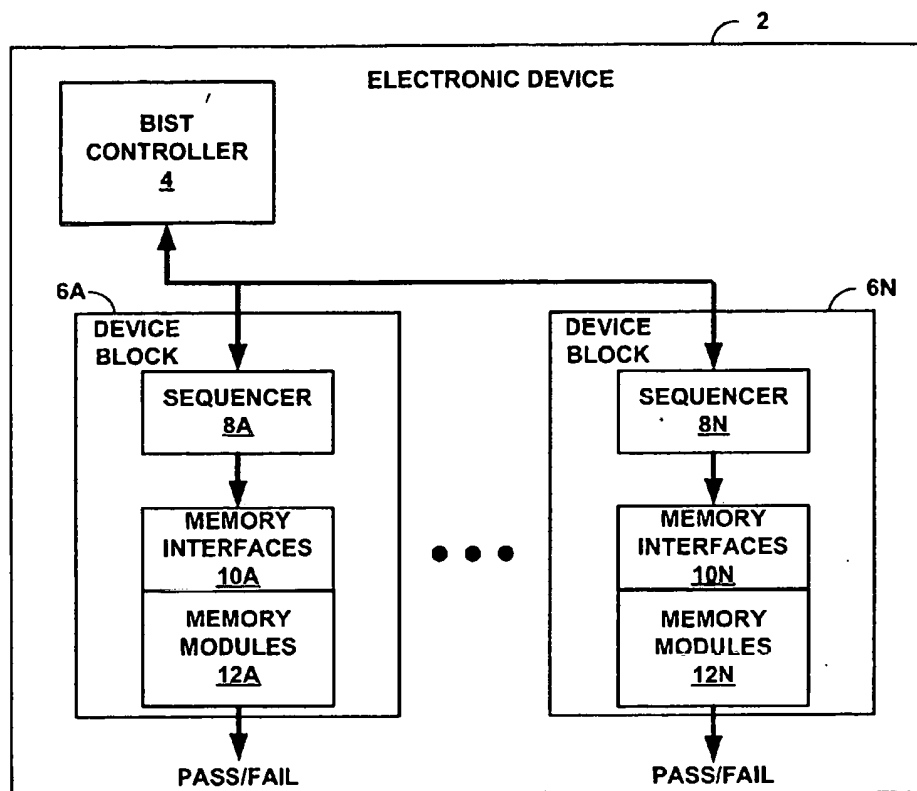
Hence, at this preliminary stage of the proceeding, after having examined the context of the claim, the Specification, and extrinsic evidence and in view of the apparent understanding of those of ordinary skill in the art at the time of the invention, based at least on Averbuj, that a “memory module” may be embedded in an integrated circuit, we decline to import the alleged additional requirements that memory modules are never “embedded in an integrated circuit” and must always include a “carrier such as a printed circuit board” into the claims.

At this juncture of the proceeding, with the exception noted above, we determine that it is not necessary to provide an express interpretation of any term of the claims.

B. Averbuj Reference

Averbuj describes a system for testing memory modules. Ex. 1005

¶ 7. Figure 1 of Averbuj is reproduced below.



As illustrated in Figure 1 of Averbuj, the system includes a built-in self-test (BIST) controller that controls a plurality of device blocks, each of which includes a sequencer, memory interfaces, and memory modules. Ex. 1005 ¶ 28.

Output from the BIST controller (e.g., test algorithms or high-level commands) is received and executed by the sequencer of the device block, which issues sequences of memory operations to a respective memory interface to perform commands. Ex. 1005 ¶ 30.

C. Asserted Unpatentability based on Averbuj

Petitioner contends claim 16 is unpatentable under 35 U.S.C. § 102 or § 103 over Averbuj. Pet. 5. Relying on the testimony of Dr. Pinaki Mazumder, Petitioner explains how Averbuj describes or suggests all of the claim limitations. *Id.* at 13–25 (citing Ex. 1003).

Claim 16 recites a memory module that operates with a system memory controller and comprises a module controller to process input control signals from the system memory controller and to generate output control signals. Petitioner argues that Averbuj discloses these features. Pet. 13–15. As Petitioner explains, Averbuj discloses a plurality of “device blocks” containing “memory modules” that operate with a “BIST controller” (i.e., system memory controller). *Id.* at 13–14 (citing Ex. 1005 ¶¶ 8, 29, 32–37, 48, Figs. 1, 2). As Petitioner also points out, each of the “device blocks” of Averbuj comprises a sequencer (i.e., a “module controller”) that “receives . . . signals . . . from the BIST controller” (i.e., from the “system memory controller”)³ and produces (or “generates”) “control signals for the memory modules . . . and the BIST circuitry” (i.e., “system memory controller”). *Id.* at 16. In other words, Averbuj discloses a “module controller” (or “sequencer”) of a “memory module” (or “memory device”) that processes input control signals and generates output control signals.

³ Averbuj discloses, for example, that the “BIST controller . . . provides and communicates test algorithms to sequencers . . . as a set of commands” (¶ 29) and that “in response to the commands, each of [the] sequencers . . . issues one or more sequences of memory operations to their respective memory interfaces . . . to perform the commands.” Ex. 1005 ¶¶ 29–30.

As Petitioner further points out, Averbuj discloses an “electronic device” that “may be a computer [or] server” that includes “memory modules” and may receive signals from an external device. Pet. 24. Petitioner also argues that it was known in the art that such “electronic devices” (e.g., computers or servers) receive and process signals as a generally accepted and known function of such electronic devices. Petitioner also argues that memory controllers were widely known by those of ordinary skill in the art to be used in such electronic devices (e.g., computers or servers) and that such use was known in the art. *Id.* (citing Ex. 1005 ¶ 48, Figs. 5, 6; Ex. 1003 ¶¶ 101–103).

Hence, Petitioner argues that it would have been obvious to one of ordinary skill in the art to have provided a “module controller” to process input control signals from a “system memory controller” in a “computer” or “server” at least because “memory controllers were commonly used in computers and servers in order to manage access to the memory systems” such that the combination of known memory controllers performing their known function of managing access to memory systems in a known memory system (e.g., the known memory system of Averbuj) would have resulted in no more than the predictable and expected result of managing access in a known fashion in a known memory system of a known computer or server. Pet. 23–25. “The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *KSR Int’l Co. v. Teleflex, Inc.*, 550 U.S. 398, 416 (2007).

Patent Owner argues that Averbuj fails to disclose or suggest “a module controller to process [these] input control signals” or that the “alleged ‘module controller’ . . . processes the alleged ‘input control

signals’.” Prelim. Resp. 31, 39. We are not persuaded by Patent Owner’s argument at least because Patent Owner does not provide sufficient evidence at this preliminary stage of the proceeding that would sufficiently support the contention that one of ordinary skill in the art, being of ordinary creativity and not an automaton, would not have understood that computers or servers receive and process signals and that a “controller” would be used to process the received signals.

Claim 16 recites that the memory module comprises a plurality of memory devices configured to perform memory operations in response to signals from the module controller. Petitioner argues that Averbuj discloses this feature. Pet. 17–18. As Petitioner explains, Averbuj discloses that the “device block” (or “memory module”) includes “any type of memory . . . each of which is a different memory device [i.e., “plurality of memory devices”].” *Id.* at 18 (citing Ex. 1005, Figs. 4–7, ¶¶ 32, 48–49, 55–57). In other words, Averbuj discloses a “memory module” (or “device block” of Averbuj) that includes a plurality of “memory devices” (or “memory module” of Averbuj).

Claim 16 recites that the memory module comprises a plurality of data handlers configured to generate test data and to provide the test data to a respective set of at least one memory device of the plurality of memory devices in response to signals from the module controller. Petitioner argues that Averbuj discloses this feature. Pet. 19–21. As Petitioner explains, Averbuj discloses a “device block” (or “memory module”) that contains a plurality of “memory interfaces” (or “data handlers”) that “writes data to and read[s] data from [a] corresponding memory module” and “is designed to produce . . . BIST_DATA_T signal [or “test data”] . . . by transforming the

[received] signal . . . *in response to* . . . signals from the sequencer.” *Id.* at 19–20. As Petitioner also points out, Averbuj discloses that the memory interface (or “data handler”) is “designed to . . . select the generated . . . ‘*test data*’ . . . and provide it to the memory module” “in response to” a “signal from the sequencer [or “module controller”].” *Id.* at 20–21. Hence, Averbuj discloses that the memory module (or “device block”) comprises a plurality of data handlers (or “memory interfaces”) configured to generate test data (e.g., a “BIST_DATA_T signal”) and to provide the test data to a respective set of at least one memory device (or a “memory module”) of the plurality of memory devices in response to signals from the module controller (or “sequencer”).

Claim 16 recites that the memory module is configured to obtain test results by reading from the respective set of at least one memory device in response to signals from the module controller and by comparing data read from the respective set of at least one memory device with the test data provided to the respective set of at least one memory device. Petitioner argues that Averbuj discloses this feature. Pet. 21–23. As Petitioner explains, Averbuj discloses that the memory interface reads data from a corresponding memory module (or “at least one memory device”) in response to signals from the sequencer (or “module controller”) and compares data read from the memory device (or a “memory module”) with the test data (e.g., data “previously written to the same address”) provided to the memory module (or “memory device”). *Id.* at 22–23.

Patent Owner argues that “Averbuj[] uses the term ‘memory [module]’ to refer to one or more memory arrays that are . . . *embedded* in . . . an integrated circuit” but that one of ordinary skill in the art would have

understood a “memory module” “**not . . . to refer to a portion of an integrated circuit (chip).**” Prelim. Resp. 20, 23, 26. Hence, Patent Owner argues that Averbuj fails to disclose or suggest a “memory module” that is not embedded in an integrated circuit, as one of ordinary skill in the art would have understood to be inherent, or at least a required feature of, a “memory module.” We are not persuaded by Patent Owner’s arguments for at least the previously stated reasons. *See above.*

Patent Owner also argues that Averbuj fails to disclose a system memory controller, as recited in claim 16. Prelim. Resp. 27–31. In particular, Patent Owner argues that the “BIST controller” of Averbuj differs from the “system memory controller,” as recited in claim 16, because “[t]he Petitioner . . . fails to show that Averbuj’s BIST controller manages the flow of data to and from Averbuj’s alleged memory modules (the alleged ‘*plurality of memory devices*’).” Prelim. Resp. 28.

Claim 16 recites a “system memory controller” and a module controller that processes signals from the system memory controller. Claim 16 does not recite additional features or characteristics of the “system memory controller.” Hence, based on the context of the claim, a “system memory controller,” as recited in claim 16, merely operates with a memory module and emits a signal. Claim 16 further recites that the signal emitted by the “system memory controller” is received at a module controller.

As previously discussed, Petitioner states that Averbuj discloses a “BIST Controller” (that Petitioner equates to the claimed “system memory controller”) and a “Sequencer” (that Petitioner equates to the claimed “module controller”), as a component of a “device block” (that Petitioner equates to the claimed memory module). Petitioner also states that the

“Sequencer” “receive[s] high-level commands from BIST controller” and, in response, “issues one or more sequences of memory operations to their respective memory interfaces . . . to perform the commands.” Ex. 1005 ¶ 30. We do not identify any meaningful difference, and Patent Owner does not point out any meaningful difference, between the “BIST Controller” of Averbuj operating with a “device block” and emitting a signal that is received by a “Sequencer” (of the device block) and the claimed “system memory controller” operating with a “memory module” and emitting a signal that is received by a “module controller.” In both cases, a component operates with another component and emits a signal that is received by a component.

Patent Owner argues that Averbuj fails to disclose or suggest that the “BIST Controller” “manages the flow of data to and from” memory devices. Prelim. Resp. 28. As previously discussed, claim 16 recites a “system memory controller that “operates with” a memory module and provides a signal to a “module controller.” Claim 16 does not also recite that the “system memory controller” must also “manage the flow of data to and from” memory devices. For at least this reason, we are not persuaded by Patent Owner’s argument.

Even assuming that Patent Owner is correct that one of ordinary skill in the art would have somehow understood a *broadest* reasonable construction of the term “system memory controller” to *narrowly* require this alleged additional (non-recited) feature, we disagree with Patent Owner that Averbuj fails to disclose or suggest this alleged feature.

For example, as Petitioner points out, Averbuj discloses a “BIST controller” (i.e., the claimed “system memory controller”) that sends “high-

level commands” to a “sequencer” and, based on the “high-level commands” from the “BIST controller,” the “application of . . . operations” are controlled such that the sequencer “receives data from the . . . memory modules [which Petitioner equates with the claimed “memory devices].” Ex. 1005 ¶¶ 40, 44, 45. Hence, Averbuj discloses that the “BIST controller” (or “system memory controller”) manages the flow of data to and from both the “sequencer” (which constitutes a “memory device” as a “device” associated with “memory”) and “memory modules” (which Petitioner equates with the *claimed* “memory devices” – c.g., data received from “memory devices”).

Patent Owner also argues that Averbuj fails to disclose or suggest a plurality of memory devices configured to perform memory operations in response to signals from the module controller, as recited in claim 16. Prelim. Resp. 32–33. In particular, Patent Owner argues that “reasoning in the [Final Written Decision of a separate matter] . . . is demonstrably incorrect and the subject of appeal to the Federal Circuit” and “misapprehended paragraph 56 of Averbuj.” Prelim. Resp. 34. We are not persuaded by Patent Owner’s argument at least because Patent Owner does not demonstrate how the alleged incorrect “reasoning” provided in a different matter is relevant to the present case.

For example, Patent Owner argues that with respect to the separate matter (i.e., IPR2014-00970), Averbuj fails to disclose or suggest “a control module” and “does not contain the text ‘without modification or transformation, directly to a memory module, irrespective of its physical configuration.’” Prelim. Resp. 33–34. However, Patent Owner does not sufficiently explain the relevance of these statements to the present

proceeding, even if assumed to be true, given that claim 16 of the '064 patent (i.e., the present proceeding) does not recite either one of “a control module” or “the text ‘without modification or transformation, directly to a memory module, irrespective of its physical configuration.’”

Patent Owner also argues that “[t]he Petition . . . fails because it . . . relies on a single feature of Averbuj to disclose two distinct limitations. . . .” Prelim. Resp. 36. In particular, Patent Owner argues that Averbuj discloses a data interface (equated to the claimed “data handler”) that generates test data in response to signals from the module controller and a plurality of memory modules (equated to the claimed “memory devices” by Petitioner) that perform memory operations in response to the same signals from the sequencer (or “memory controller”). Hence, Patent Owner argues that Averbuj fails to disclose or suggest a data interface and memory modules that operate in response to different signals from a memory controller. We note that claim 16 recites memory devices configured to perform memory operations and a data handler that generates test data in response to signals from the module controller. Claim 16 does not recite additional qualities or characteristics of any “signals from the module controller.” Specifically, Patent Owner does not assert or demonstrate persuasively that claim 16 requires that the “signals” are different from (or the same as) each other. Hence, we determine Patent Owner’s arguments not to be persuasive in this regard.

III. CONCLUSION

For the foregoing reasons, we determine that the information presented establishes a reasonable likelihood that Petitioner would prevail in

showing that claim 16 of the '064 patent is unpatentable. At this preliminary stage, we have not made a final determination with respect to the patentability of the challenged claims or any underlying factual and legal issues.

IV. ORDER

Accordingly, it is:

ORDERED that pursuant to 35 U.S.C. § 314(a), an *inter partes* review is hereby instituted as to claim 16 of the '064 patent as being unpatentable under 35 U.S.C. § 102 and § 103 over Averbuj:

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial, which commences on the entry date of this decision; and

FURTHER ORDERED that the trial is limited to the ground identified immediately above, and no other ground is authorized.

IPR2017-00560
Patent 8,689,064 B1

PETITIONER:

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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY.DOCKET NO./TITLE	REQUEST ID
13/745,790	01/19/2013	Hyun Lee	N3044.10010US03	51851

Acknowledgement of Loss of Entitlement to Entity Status Discount

The entity status change request below filed through Private PAIR on 11/22/2017 has been accepted.

CERTIFICATIONS:

Change of Entity Status:

✕ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be notification of loss of entitlement to small or micro entity status, as applicable.

This portion must be completed by the signatory or signatories making the entity status change in accordance with 37 CFR 1.4(d)(4).

Signature:	/JAMIE J. ZHENG/
Name:	Jamie Zheng
Registration Number:	51167

NOTE: This disposition is nonprecedential.

**United States Court of Appeals
for the Federal Circuit**

NETLIST, INC.,
Appellant

v.

**SK HYNIX INC., SK HYNIX AMERICA INC., SK
HYNIX MEMORY SOLUTIONS INC.,**
Appellees

2018-2123

Appeal from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in No. IPR2017-
00560.

JUDGMENT

THOMAS J. WIMBISCUS, McAndrews, Held & Malloy,
Ltd., Chicago, IL, argued for appellant. Also represented
by WAYNE BRADLEY, RONALD SPUHLER.

JOSEPH A. MICALLEF, Sidley Austin LLP, Washington,
DC, argued for appellees. Also represented by CARTER
GLASGOW PHILLIPS, RYAN C. MORRIS, BRIAN R. NESTER,
WONJOO SUH.

THIS CAUSE having been heard and considered, it is

ORDERED and ADJUDGED:

PER CURIAM (PROST, *Chief Judge*, NEWMAN and
MOORE, *Circuit Judges*).

AFFIRMED. See Fed. Cir. R. 36.

ENTERED BY ORDER OF THE COURT

December 12, 2019
Date

/s/ Peter R. Marksteiner
Peter R. Marksteiner
Clerk of Court

**United States Court of Appeals
for the Federal Circuit**

NETLIST, INC.,
Appellant

v.

**SK HYNIX INC., SK HYNIX AMERICA INC., SK
HYNIX MEMORY SOLUTIONS INC.,**
Appellees

2018-2123

Appeal from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in No. IPR2017-
00560.

MANDATE

In accordance with the judgment of this Court, entered
December 12, 2019, and pursuant to Rule 41 of the Federal
Rules of Appellate Procedure, the formal mandate is
hereby issued.

FOR THE COURT

January 21, 2020

/s/ Peter R. Marksteiner
Peter R. Marksteiner
Clerk of Court

(12) **INTER PARTES REVIEW CERTIFICATE** (1962nd)

United States Patent
Lee et al.

(10) **Number:** **US 8,689,064 K1**
(45) **Certificate Issued:** **Mar. 12, 2021**

(54) **APPARATUS AND METHOD FOR
SELF-TEST IN A MULTI-RANK MEMORY
MODULE**

(71) Applicants: **Hyun Lee; Jayesh R. Bhakta;
Soonju Choi**

(72) Inventors: **Hyun Lee; Jayesh R. Bhakta;
Soonju Choi**

Trial Number:

IPR2017-00560 filed Jan. 3, 2017

Inter Partes Review Certificate for:

Patent No.: **8,689,064**
Issued: **Apr. 1, 2014**
Appl. No.: **13/745,790**
Filed: **Jan. 19, 2013**

The results of IPR2017-00560 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,689,064 K1
Trial No. IPR2017-00560
Certificate Issued Mar. 12, 2021

1

2

AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claim 16 is cancelled.

5

* * * * *

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292).

DOCKET NO. 8:16-cv-01605 JLS	DATE FILED 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,756,364	June 17, 2014	Netlist, Inc.
2 8,516,185	August 20, 2013	Netlist, Inc.
3 8,001,434	August 16, 2011	Netlist, Inc.
4 8,359,501	January 22, 2013	Netlist, Inc.
5 8,689,064	April 1, 2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY	☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
2		
3		
4		
5		

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT "SEE ATTACHED COPY OF ORDER DISMISSING CASE"

CLERK Kiry Gray	(BY) DEPUTY CLERK /s/ J. Lam	DATE 04/05/2021
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Copy 3—Upon termination of action,

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292).

<u>DOCKET NO.</u> 8:16-cv-01605	<u>DATE FILED</u> 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,489,837	July 16, 2013	Netlist, Inc.
2		
3		
4		
5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

<u>DATE INCLUDED</u>	<u>INCLUDED BY</u> ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT "SEE ATTACHED COPY OF ORDER DISMISSING CASE"

CLERK Kiry Gray	(BY) DEPUTY CLERK /s/ J. Lam	DATE 04/05/2021
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Copy 3—Upon termination of action,



UNITED STATES
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Alexandria, Virginia 22313-1450
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FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
13/745,790	01/19/2013	411121

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, Virginia 22313-1450
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CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
13/745,790	01/19/2013	411121

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine , CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



United States Patent and Trademark Office

Office of the Chief Financial Officer

Document Code:WFEE

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Sale Accounting Date:04/07/2026

Sale Item Reference Number

13745790

Effective Date

03/31/2026

Document Number	Fee Code	Fee Code Description	Amount Paid	Payment Method
I202647052277054	1553	MAINTENANCE FEE DUE AT 11.5 YEARS	\$8,280.00	Wire
I202647052277054	1556	MAINTENANCE FEE SURCHARGE-11.5 YR LATE PYMT WITHIN SIX MONTH	\$540.00	Wire

MEMORY BOARD WITH SELF-TESTING CAPABILITY**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application is a continuation of U.S. Patent Application No. 12/422,925, filed April 13, 2009, which claims the benefit of priority from U.S. Provisional Application No. 61/044,801, filed April 14, 2008, U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and U.S. Provisional Application No. 61/044,839, filed April 14, 2008. Each of the foregoing applications are incorporated in their entirety by reference herein. This application is related to U.S. Patent Application No. 12/422,912, filed on even date herewith and entitled “Self-Adjusting Damper”, and to U.S. Application No. 12/422,853, filed on even date herewith and entitled “Circuit Providing Load Isolation and Noise Reduction”, both of which are incorporated in their entirety by reference herein.

BACKGROUNDField

[0002] The present invention relates to self-testing electronic modules and, more particularly, to self-testing electronic memory modules.

Description of the Related Art

[0003] The failure of memory components in an electronic system may result in the loss of valid data. Therefore, it is important to ensure proper memory operation in an electronic system. Memory integrated circuits (“memory chips”) often go through a series of tests at various stages of system manufacture. Once memory chips are deployed in a system, they also generally go through a system level memory test each time the system is booted. In addition, memory chips may undergo a parity checking process during normal system operation.

[0004] There are typically at least three test phases which memories undergo during system manufacture. Each phase generally tests for memory defects and for the correct operation of the input/output interface. The first test phase is typically conducted by the memory chip manufacturer and generally involves checking for bit failures, correct memory access speed, etc. The second test phase is typically done by memory module manufacturers and generally involves testing the signal quality, the noise susceptibility, and the operational

speed of the memory module as a single unit. The second test phase may also include checking for bit failures in individual memory chips. The third phase is usually carried out by the system manufacturer. During the third phase, the interaction of the memory subsystem with other components in the system is tested. During the third phase, the individual memory module operation is also tested again and the memory array is checked for defects. Because of the significant amount of testing that memories undergo during the manufacturing process, there is generally substantial test cost and test time associated with ensuring the proper memory operation. This test cost and test time translate into an increase in system cost and a decrease in system performance.

[0005] There are a number of memory test methodologies that employ either external test hardware, embedded self-test logic (“MBIST”), or both. However, the usefulness of these test methodologies is limited due to the high cost and other limitations associated with them. For instance, external test hardware such as automatic test equipment (“ATE”) is very expensive. Moreover, the development time and cost associated with implementing MBIST is relatively high. These costs and limitations are especially significant when testing dynamic random access memory (“DRAM”). For example, technological developments, such as increases in DRAM speed, may require manufacturers to upgrade ATE machines relatively frequently. In addition, MBIST in DRAM chips generally cannot be fully utilized for system level testing of memory boards.

[0006] Because of the increasing cost, complexity, and time involved with fully testing DRAM chips, DRAM manufacturers often provide “effectively tested” (“ETT”) DRAM chips to memory module manufacturers at a lower price rather than providing fully tested DRAM chips. Memory module manufacturers often prefer the ETT DRAM chips mainly due to their greater availability. Memory module manufacturers who receive ETT DRAM chips then have to assume a part of the responsibility of validating the DRAM chips, adding to the complexity of the memory module test process.

SUMMARY

[0007] In certain embodiments, a self-testing memory module is provided comprising a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module may further comprise a plurality of

memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprises data, address, and control ports. In certain embodiments, the memory module includes a control module configured to generate address and control signals for testing the memory devices and a data module comprising a plurality of data handlers. Each data handler in certain embodiments is operable independently from each of the other data handlers of the plurality of data handlers. Each data handler may be operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports.

[0008] A self-testing memory module is provided in certain embodiments comprising a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module may further comprise a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprises data, address, and control ports in some embodiments. The memory module of certain embodiments includes a control module configured to generate address and control signals for testing the memory devices. In certain embodiments, the memory module comprises a data module comprising at least one data handler operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate cyclic data for writing to the corresponding plurality of data ports.

[0009] In certain embodiments, a method of self-testing a memory module includes providing a self-testing memory module. The memory module comprises a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module further comprises a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprising data, address, and control ports. In certain embodiments, the memory module comprises a control module configured to generate address and control signals for testing the memory devices. The memory module may further comprise a data module comprising a plurality of data handlers. Each data handler may be operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports. The method of certain embodiments further includes generating, by each of the data handlers, data for writing to the corresponding plurality of data ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein.

[0011] Figure 2 is a block diagram of an example self-testing memory module including eight memory devices and eight data handlers in accordance with certain embodiments described herein.

[0012] Figure 3 is a block diagram of an example data handler module and an example control module in accordance with certain embodiments described herein.

[0013] Figure 4 is a flowchart of an example method of self-testing a memory module in accordance with certain embodiments described herein.

[0014] Figure 5 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

[0015] Figure 6 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0016] Certain embodiments described herein present a self-testing electronic system, such as for example, a self-testing electronic memory module. Some embodiments described herein present a self-testing memory module that is populated with ETT DRAM chips. Certain embodiments described herein present a self-testing registered dual in-line memory module (“RDIMM”). Some embodiments described herein present a self-testing RDIMM that does not require any additional pins other than the pins on the standard JEDEC RDIMM connector. For example, the memory module may utilize the address and control signals (e.g., address and control signals generated by the memory module to test the memory module) along with a test signal to enable and execute a self testing function.

[0017] Certain embodiments described herein present a self-testing memory module that includes a control module and a data module which can generate memory addresses and data according to the JEDEC standard memory protocol.

[0018] Some embodiments described herein presents a self-testing memory module that can be configured through an I²C interface and that allows test results to be read out through the I²C interface. Certain embodiments described herein present a self-testing memory

module that allows a test function to be configured, controlled, and/or executed without substantial system memory controller involvement. Some embodiments present a self-testing memory module that can be tested without any external test equipment. For example, certain embodiments present a self-testing memory module that can be tested without any system driven test procedure. Various embodiments described herein present a self-testing memory module that can be self-tested at a target system speed. Certain embodiments described herein present a self-testing memory module that can generate data (DQ) and data-strobe (DQS) signals with wave characteristics that resemble the wave characteristics of DQ and DQS signals from a system memory controller.

[0019] MBIST is commonly used to test memories embedded in application specific integrated circuits (“ASICs”) or system on chip integrated circuits (“SoCs”) such as, for example, advanced memory buffers (“AMBs”). The MBIST implementation usually includes three distinct functional blocks: the address/control generator, the data generator/checker (sometimes referred to as a signature analyzer), and the test interface controller/register (“JTAG”). The test instructions and test patterns are generally loaded to the address/control generator and the data generator/checker through the JTAG interface. The test results are generally read out through the JTAG interface.

[0020] In memory module applications, there are obstacles associated with using MBIST. One of these obstacles is the large number of signals used to interface the three MBIST functional blocks. The inter-block timing constraints can present another obstacle. In addition, the ability to control the test is limited. For example, synchronizing the address/or control signals with the data signal can be difficult. The relatively large amount of information to be gathered and stored while testing the memory presents yet another obstacle.

[0021] Unlike on an ASIC, in which the three MBIST functional blocks are in a single chip, the MBIST functional blocks on a memory module generally would be segregated into multiple chips on the memory module due to physical and electrical limitations and requirements. This makes implementing MBIST on a memory module difficult because, while there is virtually no limit on the number of available interface signals among the three MBIST functional blocks in an ASIC, the memory module can support only a limited amount of interface signals between memory chips. In addition, because the MBIST functional blocks are

spread out to multiple chips on a memory module, the inter-block signal delay is generally much longer on a memory module than on an ASIC. It is also generally not feasible to implement cross-checking logic that operates the three MBIST functional blocks in lock-operational block. This is due to the limitation on the number of interface signals and to the relatively long inter-block signal delay time on a memory module versus on an ASIC.

[0022] In some cases such as where a memory module includes an AMB, the self-test logic (MBIST) implemented in the AMB includes command and address generation logic in addition to a data generator and checker. Each of these functional blocks may be implemented on a single physical AMB device (e.g., a single integrated circuit package). However, because all of the self-test command, address and data signals are combined in one physical area of the memory module, a memory module (e.g., DIMM) level routing problem can occur, making it difficult to route the self-test signals on the memory module and resulting in performance degradation and/or implementation difficulty. In addition, the data width of self-test logic of the memory module will be limited to the data width of the AMB (e.g., to the data width of the self-test logic implemented on the AMB, to the number of available ports on the AMB, etc.). As such, implementing memory module test logic on an AMB is not flexible (e.g., to changes in the data width of the memory module) and generally supports only memory modules having certain predetermined, fixed data widths.

[0023] Finally, in most cases, since an ASIC cannot be repaired, the ASIC MBIST is generally only capable of detecting and reporting the pass/fail status of memory tests. For memory module test results, on the other hand, it is generally advantageous to include both addresses of the memory locations where failures occur and the data patterns that were read back from the failed memory locations. This type of reporting can help to facilitate the repair of the memory module by, for example, allowing for the identification and replacement of failed components.

[0024] Figure 1 is a block diagram of an example self-testing memory module 10 in accordance with certain embodiments described herein. The memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board (PCB) 12, each memory device 20 of the plurality of memory

devices 18 comprising data, address, and control ports. The memory module 10 comprises a control module 22 configured to generate address and control signals 24 for testing the memory devices 18. The memory module 10 also includes a data module 28 comprising a plurality of data handlers 30. Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports of one or more of the plurality of memory devices 18. For example, each of the data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports. Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. The memory module 10 may further include an I²C interface 15 in certain embodiments.

[0025] As described more fully below, in certain embodiments the data module 28 generates test data patterns to write to the plurality of memory devices 18 of the memory module 12 and checks the data patterns read or received back from the plurality of memory devices 18 for agreement with corresponding data patterns that are expected to be read back from the plurality of memory devices 18. For example, in one embodiment, the data module 28 generates cyclic patterns to write to the plurality of memory devices 18. In some embodiments, the data module 18 also isolates the data path from the system board of the computer system 16 to the plurality of memory devices 18 while the memory module 10 is not accessed by the computer system 16. For example, the data module 28 may isolate the data path from the system board to the plurality of memory devices 18 when the memory module 10 is in a self-testing mode. The control module 22 may include, for example, a dual input register (e.g., the memory device controller 32 described more fully below) for registering address and control signals coming from either self-testing logic (e.g., from the test controller 36 described more fully below) or from the memory controller 14 on the system board. In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20. For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation. The data module 28 may then receive a read command to read back the data from

those locations and check the read data for agreement with the expected data. If there is a mismatch between the read data and the expected data, the data module 28 may, for example, store the failure information (e.g., the failed data word) and inform the control module 22 about the failure. The control module may save the address of the memory location where the failure occurred.

[0026] In certain embodiments, the memory module 10 is configured to be operated in a test mode in which the control module 22 selectively inputs the address and control signals to the address and control ports of the plurality of memory devices 18. Moreover, in the test mode, each of the data handlers 30 write the data generated by the data handler 30 to the corresponding plurality of data ports by selectively inputting data signals to the data ports of the plurality of memory devices 18. The data module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20. For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according the operating specification of the memory devices 20. In some embodiments, the control module 22 and the data module 28 produce memory addresses, control and/or data signals according to the JEDEC standard memory protocol. In some embodiments, for example, the control module 22 and the data module 28 generate the memory interface signals with proper edge relationships based on the JEDEC standard. In certain embodiments, the test speed, for example, may be defined by the speed of the clock (e.g., the system clock). The address sequences and/or the data patterns of certain embodiments may be programmable either through the I²C interface or they may be defaulted to pre-defined values.

[0027] In certain embodiments, data module 28 is configured to generate data signals with programmable slew rates and/or with variable peak values. In one embodiment, for example, the data module 28 is also able to generate data (DQ) and data-strobe (DQS) signals with programmable slew rates and programmable peak values so that the characteristics of the signals generated by the data module 28 generally correspond to the characteristics DQ and DQS signals generated by the system memory controller 14.

[0028] In some embodiments, the data module 28 and/or the control module 22 are configured to test the plurality of memory devices 18 under non-normal conditions. For example, the data module 28 and/or control module 22 may be configured to provide signals having frequencies which are higher or lower than the normal operating frequencies of the memory devices 20.

[0029] In certain embodiments, the memory module 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 12 has an industry-standard form factor. For example, the PCB 12 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 12 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 12 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0030] In certain embodiments, the plurality of memory devices 18 of the memory module 10 may be arranged as ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 20 of the plurality of memory devices 18 that can be accessed by the memory controller 14 to access the full bit-width of the memory bus of the memory module 10. For example, a memory module 10 in which each rank of the memory module is 64 bits wide is described as having an “x 64” organization. Similarly, a memory module 10 having 72-bit-wide ranks is described as having an “x 72” organization. The number of memory devices 20 and corresponding memory capacity of a memory module 10 can be increased by increasing the number of memory

devices 20 per rank or by increasing the number of ranks. For example, a memory module with four ranks with each rank having N 512-MB memory devices 20 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 20 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices 20. During operation, the ranks of a memory module 10 may be selected or activated by control signals that are received from a component of the system (e.g., a system memory controller 14 or a local memory controller of the memory module 10). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 10 comprises only one rank of memory devices 20.

[0031] As discussed, the PCB 12 may include at least one connector (not shown) configured to operatively couple the memory module 10 to the memory controller 14 of the computer system 16. The computer system 16 may include a host computer system. For example, the memory module is electrically coupled, logically coupled, or both, with the memory controller 14. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 14 may comprise a disk controller of the computer system 16, for example. The memory controller 14 may be mounted on a system board of the host computer 16. The connector can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system 16. The connector of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 10 and the host system 16. For example, the connector can comprise a standard DDR2, DDR3, and other future generation edge connectors. Additionally, in certain embodiments, more than one memory module 10 is coupled to the host system 16.

[0032] The plurality of memory devices 18 on the PCB 12 may include one or more volatile memory components. For example, the plurality of memory devices 18 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 20. Types of DRAM devices 20 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM

(SDRAM). The memory devices 18 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 20 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 20 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP). The plurality of memory devices 18 may further include one or more non-volatile memory devices 20, such as, for example, flash memories. The plurality of memory devices 18 of certain embodiments may include both volatile and non-volatile memory devices 20. For example, the plurality of memory devices 18 may include one or more of DRAM, SRAM, and/or flash memory devices in some embodiments.

[0033] Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28. For example, each data handler 30 is configured to write to and/or read from the corresponding plurality of data ports of one or more of the memory devices 20 without being in communication any of the other data handlers 30 or other data ports of the memory devices 20. As such, each data handler 30 can be used to generally independently test a portion of the memory space of the memory module 10. For example, each data handler 30 may be used to independently test one memory device 20 of the memory module 10. In such a configuration, the corresponding plurality of data ports of each data handler 30 may comprise each data port of the corresponding memory device 20. In other embodiments, each data handler 30 may be used to test a segment of one memory device 20, more than one memory device 20, segments more than one memory device 20, or any combination or sub-combination thereof. Because each of the data handlers 30 is operable independently of each of the other data handlers 30, the data handlers 30 are generally modular. As such, modifications in the configuration of the memory module 10 (e.g., changes in the bit-width of the memory bus, changes in the number of memory devices 20, etc.) may be less complicated to accommodate than in other types of self-testing memory modules 10. For example, where a new memory device 20 or set of memory devices 20 is added to the memory module 10, the change may be generally accommodated by adding a corresponding data handler 30. The change may be accommodated without having to

implement a major reorganization of the memory module 10 or the self-testing logic of the memory module 10, for example.

[0034] Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. Figure 2 is a block diagram of an example self-testing memory module 10 including eight memory devices 20 (e.g., memory devices 40a-40h) and a data module 28 comprising eight data handlers 30 (e.g., data handlers 30a-30h) in accordance with certain embodiments described herein. Each of the memory devices 20 includes an eight bit output data word and eight corresponding data ports. In addition, the system memory bus 50 between the memory controller 14 and the example memory module 10 is 64 bits wide and each of the data handlers 30 receives an eight bit segment of the system memory bus 50. Each of the data handlers 30 is operatively coupled to a corresponding plurality of data ports 21 of a corresponding one of the memory devices 20. As such, the data handlers 30 may be operatively coupled (e.g., electrically and/or logically coupled or connected) to the eight data ports 21 of one of the corresponding memory devices 20. For example, the data handler 30a may be operatively coupled to the eight data ports 21 of the memory device 40a of Figure 2.

[0035] The configuration shown in Figure 2 is for the purposes of illustration and is not intended to be limiting. For example, while the example memory module 10 of Figure 2 includes an equal number of memory devices 20 and data handlers 30, other configurations are possible. In some configurations there are more memory devices 20 than data handlers 30 or vice versa. Moreover, the one or more data handlers 30 may be operatively coupled to a subset of the data ports 21 of one of the memory devices 20 instead of all of the data ports 21 of one of the memory devices 20. In other embodiments, one or more data handlers 30 may be operatively coupled to a subset or all of the data ports 21 of more than one of the memory devices 20. For example, in one embodiment, each of the data handlers 30 are operatively coupled to all of the data ports of two memory devices 20.

[0036] In certain embodiments, the plurality of data handlers 28 comprises at least two physically separate components mounted on the PCB 12. For example, the plurality of data handlers 28 may include at least two physically separate integrated circuit packages. The physically separate integrated circuit packages are mounted on different portions of the

PCB 12 in some embodiments. For example, each of the eight data handlers 30a-30h shown in Figure 2 may include physically separate integrated circuit packages mounted on different portions of the PCB 12. While eight data handlers 30 are shown in Figure 2, other numbers of data handlers 30 are possible including fewer or more than eight.

[0037] In certain embodiments, each of the plurality of data handlers 30 is positioned on the PCB 12 proximate to the corresponding plurality of data ports. For example, each data handler 30 of certain embodiments is positioned closer to the corresponding plurality of data ports 21 than the data handler 30 is to the other data ports 21 of the plurality of memory devices 18. For example, the data handler 30a is positioned closer to the corresponding plurality of data ports 21 of the memory device 40a than to the other data ports 21 of the other memory devices 40b-40h.

[0038] Figure 3 is a block diagram an example data module 28 and an example control module 22 in accordance with certain embodiments described herein. The control module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18. In some embodiments, the control module 22 includes a control mixer element 32. The control mixer element 32 may include a memory device controller 34 (e.g., a DRAM controller) and a test controller 36. In certain embodiments, the control mixer element 32 generally controls the address and the control signals for the self-testing function.

[0039] In certain embodiments, the memory device controller 34 generally pre-processes address and control information before it sends the information to a register 40. In one embodiment, the memory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36. The control module 22 of certain embodiments is configured to selectively input to the address and control ports of the plurality memory devices 18 either the address and control signals 38 from the system memory controller 14 or the address and control signals 42 from the control module 22 (e.g., from the test controller 36). For example, the memory device controller 34 may send either the signals 38 from the system memory controller 14 or, alternatively, the signals 42 from the test controller 36, to the register 40 depending on whether the memory module 10 is in normal (non-test) mode or in a

test mode, respectively. In one embodiment, the memory device controller 34 generates the address and control signals for memory device (e.g., DRAM device) operations. The test controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28. The control module 22 may be implemented in the control register of the memory module 10 in certain embodiments. In various embodiments, the control module 22 includes discrete logic, one or more application-specific integrated circuit (ASICs), one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic device (CPLDs).

[0040] The data module 28 and the subcomponents thereof (e.g., the data handlers 30) may be in communication with one or more of the memory devices 20, the control module 22, and the memory controller 14. In certain embodiments, the data module 28 comprises a plurality of data handlers 30. In other embodiments the data module 28 includes at least one data handler 30. Each of the data handlers 30 of certain embodiments comprises a switch 44. For example, the switch 44 may include a data multiplexer/demultiplexer (“data mux/demux”). The switch 44 may provide a bi-directional data multiplexer function. In certain embodiments, the switch 44 is configured to selectively input to the corresponding plurality of data ports either data signals 48 from the system memory controller 14 or data signals 50 from the data handler logic element 46. The switch 44 of certain embodiments may further be configured to receive data signals 52 (e.g., during a read operation) from the plurality of memory devices 18 and to propagate the data signals 52 to the data handler logic element 46 and/or the memory controller 14. In some embodiments, for example, the switch 44 selectively inputs the data signals 48 to be written to the plurality of memory devices 18 from the system memory controller 14 when the memory module 10 is a normal (non-test mode) mode and, alternatively, inputs the data signals 50 from the data handler logic element 46 during a test mode. While the switch 44 is shown as being included in the data handler 30 in the example of Figure 3, other configurations are possible. For example, in other embodiments the switch 44 may be logically and/or physically separated from the data handler module 28 and/or the data handlers 30.

[0041] Each of the data handlers 30 of certain embodiments further includes a data handler logic element 46. The data handler logic element 46 of certain embodiments comprises a data generation element 54 and a verification element 56. The data generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports, for example. The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information) the data handler logic element 46 receives from the control module 22, for example. The data may be cyclic data in some embodiments or non-cyclic data in other embodiments. For example, the cyclic data may comprise at least one predetermined pattern of data which repeats or is cycled two or more times. In various embodiments, the data comprises one or more incrementing patterns or decrementing patterns, for example. In other embodiments, the data comprises a pattern which alternates each bit on successive memory writes. For example, a memory write comprising one or more hexadecimal “A” characters (each corresponding to a four-bit binary word of “1010”) may be followed by a memory write comprising one or more hexadecimal “5” characters (each corresponding to a four-bit binary word of “0101”). The data may be generated in a variety of ways. In one embodiment, the data is generated based on a current write address value. For example, in one example configuration, on a first write cycle, hexadecimal “A’s” are generated and written to even address locations and hexadecimal “5’s” are generated and written to odd address locations, and on a second write cycle, “5’s” are written to even addresses and “A’s” are written to odd addresses, and this pattern repeats in subsequent cycles. The data may be generated based previously written data (e.g., inverting each of the bits of a previously written data word) in some embodiments. In general, any manner of generating a cyclic or otherwise deterministic data pattern may be compatible with embodiments described herein. In other embodiments, random or pseudorandom data may be generated and written to the corresponding plurality of data ports. For example, a linear feedback shift register (LFSR) may be used in some embodiments. In addition, the data patterns may be programmable. For example, the data patterns may be programmable based on information received by the data generation element 54 from the memory controller 14 (e.g., through the control module 22), or from the control module 22.

[0042] The plurality of data handlers 30 are further configured to read data from the corresponding plurality of data ports. For example, the verification element 56 may be configured to receive data from the corresponding plurality of data ports (e.g., through the switch 44 during a test mode). The verification element 56 may further be configured to check for failures in the operation of the plurality memory devices 18 by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler 30 and written to the corresponding plurality of data ports.

[0043] In certain embodiments, the verification element 56 is configured to perform the verification without storing a copy of the data written to the corresponding plurality of data ports or accessing a stored copy beyond the data read from the plurality of memory devices 18. For example, the verification element 56 does not store or access a copy of the data that is written to the corresponding plurality of data ports except for the data stored and read back from the plurality of memory devices 18. As such, the memory module 10 of certain embodiments advantageously does not require separate memory for storing duplicate copies of test data that is written to the plurality of memory devices 18 for later comparison. For example, the verification element 56 may calculate comparison data and may compares the comparison data to the data read from the corresponding plurality of data ports. In certain embodiments, the comparison data comprises data which expresses the data or values expected to be received from the plurality of memory devices 18 if the write, store, and read processes of the data using the data module 28 and the plurality of memory devices 18 are performed correctly or as expected. The calculation may be performed simultaneously or substantially simultaneously with receiving the data read from the corresponding plurality of data ports in certain embodiments. In other embodiments, the calculation is performed either before or after receiving the data. Other configurations are possible. For example, in one embodiment, the verification element 56 does store a separate copy of the data written to the corresponding plurality of data ports upon writing the data and compares the separate copy to the read data received from the plurality of memory devices 18.

[0044] The verification element 56 of certain embodiments calculates the comparison data based on the cyclic data. For example, in one embodiment, the verification element 56 calculates the comparison data in substantially the same manner that the data

generation element 54 generates the data as described above (e.g., based on a current write address, using an LFSR, etc.). As such, the comparison data of certain embodiments is substantially a repeat of the data written. In one example embodiment, the data handler 30 is configured (e.g., is programmed by the control module 22) to write an alternating series of “A’s” and “5’s” to the data ports of the corresponding plurality of data ports as described herein. For example, the data handler 30 may be configured to write one or more “A’s” to the first address location of an N-word memory device 20 including the corresponding plurality of data ports. The data handler 30 may then write one or more “5’s” to the second address location, one or more “A’s” to the third address location and so on until the data handler 30 has written to all N memory locations. The verification element 56 of the example embodiment then calculates the comparison data based on the cyclic data written to the corresponding plurality of data ports. For example, the verification element 56 calculates a comparison word including one or more “A’s”, “5’s,” and “A’s,” respectively, to be compared to the data read from the first, second, and third address locations based on the cyclic data (e.g., based on the known cycle of the data). In some embodiments, the comparison data is calculated based on a current read address.

[0045] In certain embodiments, data associated with failures in the operation of the plurality of memory devices 18 are stored in the data module 28. For example, data read from the corresponding plurality of data ports which do not correspond to (e.g., match) the comparison data calculated by the verification element 56 may be stored in the data module. Moreover, in some embodiments, memory addresses associated with the failures in the operation of the plurality of memory devices 18 are stored in the control module 22. For example, the data handler 30 may communicate data failures (e.g., when data read from the corresponding plurality of data ports does not correspond to calculated comparison data) to the control module 22 which may then store the addresses corresponding to the data failure. In certain embodiments, the memory module 12 is configured to report failures (e.g., the failed data, the address corresponding to the memory location of the failed data, and/or expected data) via the I²C interface 15 to the memory controller 14. In addition, in certain embodiments the test controller 36 and/or the data handlers 30 may be updated through the

I²C interface 15 with new data patterns and/or with alternative memory access sequences to conduct AC tests (e.g., tests of the power, current, I/O speed, etc.).

[0046] In various embodiments, the components of the data module 28 (e.g., the switch 44, the data handlers 30, the data handler logic element 46, the data generation element 54, and/or verification element 56) may include discrete logic, one or more application-specific integrated circuits (ASICs) one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic devices (CPLDs). Additionally, one or more of the various functional blocks (e.g., the switch 44) of the data module 28 of Figure 3 may not be included. In some embodiments, additional functional blocks may be included. Moreover, some of the functional blocks are described as separate functional blocks for illustration purposes and may comprise one physical component. For example, in one embodiment, each of the data handlers 30 and the corresponding switch 44, data generation element 54, and verification element 56 comprise one physical component (e.g., are included in one integrated circuit package). In another embodiment, the data module 28 comprises one physical component.

[0047] Referring again to Figure 1, a self-testing memory module 10 of certain embodiments comprises a printed circuit board (PCB) 12 and is configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board 12 where each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. The memory module 10 further comprises a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. In certain embodiments, the memory module 10 further comprises a data module 28 comprising at least one data handler 30 and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices 20. The data handler 30 is configured to generate cyclic data for writing to the corresponding plurality of data ports. In certain embodiments, there may be one data handler 30, for example. In other embodiments, there may be more than one data handler 30. The elements of the memory module 10 may be compatible with any of the embodiments described herein.

[0048] Figure 4 is a flowchart of an example method 70 of self-testing a memory module 10 in accordance with certain embodiments described herein. While the method 70 is described herein by reference to the memory module 10, other memory modules, electronic systems or subsystems, and/or circuits are also compatible with the embodiments described herein. The method 70 of certain embodiments comprises providing a self-testing memory module 10 at operational block 72. The memory module 10 may comprise a printed circuit board (PCB) 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 may further comprise a plurality of memory devices 18 on the printed circuit board 12. Each memory device 20 of the plurality of memory devices 18 may comprise data, address, and control ports. The memory module 10 may comprise a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. The memory module 10 may further comprise a data module 28 comprising a plurality of data handlers 30. In certain embodiments, the data module 28 comprises at least one data handler 30. Each data handler 30 of certain embodiments is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports. At operational block 74, the method 70 further comprises generating, by each of the data handlers 30, data for writing to the corresponding plurality of data ports.

[0049] Figure 5 is a flow diagram 100 illustrating self-testing operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. At operational block 110 the memory module 10 enters an idle state after power up or, in some embodiments, when the memory board of the computer system 16 is connected to a mother board or to a test board. In one embodiment, at operational block 110, the contents of the register 40 are undefined and the controllers (e.g., the test controller 36 and the memory device controller 34) are in unknown state. After power up, the memory module 10 is reset (e.g., a reset command is executed) at operational block 111. At operational block 111, the logic of the data handler 30 (e.g., data generation element 54 and/or verification element 56) and the logic of the control mixer 32 are set to default values and/or states. For example, the memory module 10 may default to a non-test mode (e.g., normal operational mode) and the switches 44 of the data handlers 30 and the memory device controller 34 are not configured in

a test mode. At operational block 112, the test mode is configured. For example, the computer system 16 (e.g., through the memory controller 14) may configure the test mode. The configuration may be through the I²C interface 15, for example, and may include configuring the test controller 36 for test mode. For example, configuring the test mode may include switching (e.g., by configuring the test controller 36) the mode of the memory module 10 from a normal operation mode to the test mode.

[0050] At operational block 113, the test mode is initiated. Initiating the test mode may comprise configuring a particular test case (e.g., particular test data or data patterns, particular write and/or read address sequences, etc.). The initiation may be achieved by activating the test controller 36 to initiate the test mode. For example, at operational block 113, the data handlers 30 and the control mixer 32 are readied (e.g., via a tester) beginning the self-testing of the memory module 10. For example, the tester may comprise an ATM tester, a server, a specialized tester. The tester may ready the data handlers 30 and the control mixer 32 through the I²C interface 15 or through the control signals 38, for example. For example, the data handlers 30 are configured to input data generated by the data handlers 30 to the corresponding plurality of data ports of the plurality of memory devices 18 and the control mixer 32 is configured to input address and control signals from the test controller 36 to the plurality of memory devices 18. At operational block 113 the test controller 36 updates each of the data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.).

[0051] In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal (“Par-in”) is un-asserted (e.g., set to a “low” value), it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. The memory module 10 executes the test mode (e.g., writes and reads test data patterns) at operational block 114. The memory module 10 continues executing the test mode at operational block 114 until the self-test is complete or until the memory module 10 detects a certain number of memory failures such that a failure count exceeds a preset number.

[0052] If the failure count exceeds the preset number, the memory module 10 enters operational block 115 and “errors out” of the test mode. If the self-test is completed at

operational block 114 and the failure count does not exceed the preset number, the memory module 10 enters operational block 115. At operational block 115, the memory module 10 sends out a test completion indication signal (e.g., through the test controller 36 to the memory controller 14). At operational block 116, the fail status of the test may be read out through, for example, the I²C interface 15. For example, one or more address values associated with failed memory locations, and/or the data read from those locations, and/or expected data may be read. In other cases, where the test fails, the memory module 10 may report the failure to the memory controller 14 through any available signal (e.g., bidirectional data or data strobe signal) between the memory module 10 and the memory controller 14.

[0053] In some embodiments, the memory module 10 is generally interruptible. For example, the memory module 10 may exit the self-test and return to operational block 110 if the signal going into the parity-in pin is, for example, de-asserted or removed at operational block 155. In other embodiments, the interrupt signal or condition may be different and may not be a parity-in signal. For example, in one embodiment, the memory module 10 may exit the self-test and return to operational block 110 if a timer having a pre-determined count expires. When the self-testing is interrupted, the memory module 10 will return to operational block 110 and the bit failure information in the control mixer 32 and the data handler 30 becomes invalid.

[0054] Figure 6 is a flow diagram 200 illustrating the operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. For example, one or more of the operational blocks of the flow diagram 200 of Figure 6 may correspond to one or more of the operational blocks of Figure 5. At operational block 210, the memory module 10 may be generally inactive. For example, the testing logic (e.g., the control module 22 and/or the data module 28) may be generally inactive and the memory module 10 may default to a functional (e.g., non-test) mode. In certain embodiments, the memory module 10 executes one or more tasks which configure the self-test mode of the self-testing memory module using an I²C interface 15. For example, the memory module 200 may configure the test controller 36 through the I²C interface 15 (e.g., as described with respect to Figure 5). In other embodiments, the test mode is configured using the control signals 38. For

example, the test mode may be updated at operational block 211 in certain embodiments. In certain embodiments, updating the test mode comprises programming the test controller 36 with test configuration information such as test data patterns, test data sequences, test failure conditions, etc. After updating the test mode at operational block 211, the test controller 36 may then be activated through the I²C interface 15 (e.g., as described with respect to Figure 5) in certain embodiments. Alternatively, as shown, the test controller 36 may be activated through the I²C interface 15 without first updating the test mode.

[0055] At operational block 212, the memory module test mode is ready for execution. In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal is low, it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. At operational block 113, a set of status registers (e.g., one or more registers of the control module 22 or the data module 28) are cleared.

[0056] The memory module 200 generates address and data signals for testing the plurality of memory devices 18 at operational block 214. For example, the data generation elements 54 of the data handlers 30 may generate the data for writing to the plurality of memory devices 18 as described herein. In addition, the control module 22 may generate address and control signals for testing the plurality of memory devices 18 as described herein. At operational block 215, the memory module 10 performs a burst write function. For example, in certain embodiments, the memory module 10 writes multiple memory locations in the memory module 10 that are to be tested at operational block 215 in a relatively short amount of time (or in a burst). In certain embodiments, the memory module 10 can perform multiple write bursts at operational block 215. Next, at operational block 216, the memory module 10 performs a read & compare function. For example, in certain embodiments, the memory module 10 reads back certain memory locations at operational block 216 that have been written at operational block 215 and compares the values with certain expected data. For example, a verification element 56 of each of the data handlers 30 may calculate the expected data and/or compare the values as described herein. In some embodiments, the memory module 200 reads from multiple memory locations in the memory module 10 in a relatively

short amount of time (or in a burst). In certain embodiments, the memory module 10 can execute multiple read bursts at operational block 216 as it compares the read data with expected data. In some embodiments, the memory module 200 (e.g., the data handlers 30 of the memory module 200) stores any failed read data and the associated expected data at operational block 217. In some embodiments, the memory module sends one or more failure indications to the memory controller 14 over one or more available signals between memory module 10 and the memory controller 14. For example, the I²C interface 15 or one or more bi-directional data pins (e.g., one or more of the data pins 48) may be used.

[0057] The memory module 200 stores the addresses of the memory locations associated with the failed read data using the test controller 32 at operational block 217. In other embodiments the failed read data, expected data, and the addresses of the memory locations associated with the failed read data may be stored using the test controller 32. For example, in one embodiment, the test controller 32 stores the failed read data, expected data, and the addresses of the memory locations associated with the failed read data at operational block 217. In some embodiments, the test controller 32 is generally inaccessible through the I²C interface 15 during self-test operation. For example, the test controller is inaccessible through the I²C interface 15 until a test failure occurs at operational block 217 or until the test completes at operational block 219. In some embodiments, when the test is complete, the test results can be read out of the memory module 10 (e.g., through the I²C interface 15). In some cases, the test can be interrupted using the I²C interface 15 or using the parity-in signal (e.g., by de-asserting the parity-in signal).

[0058] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects

or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports.

MEMORY BOARD WITH SELF-TESTING CAPABILITY

ABSTRACT

A self-testing memory module includes a printed circuit board configured to be operatively coupled to a memory controller of a computer system and includes a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports. The memory module also includes a control module configured to generate address and control signals for testing the memory devices. The memory module includes a data module comprising a plurality of data handlers. Each data handler is operable independently from each of the other data handlers of the plurality of data handlers. Each data handler is operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and is configured to generate data for writing to the corresponding plurality of data ports.

11557833

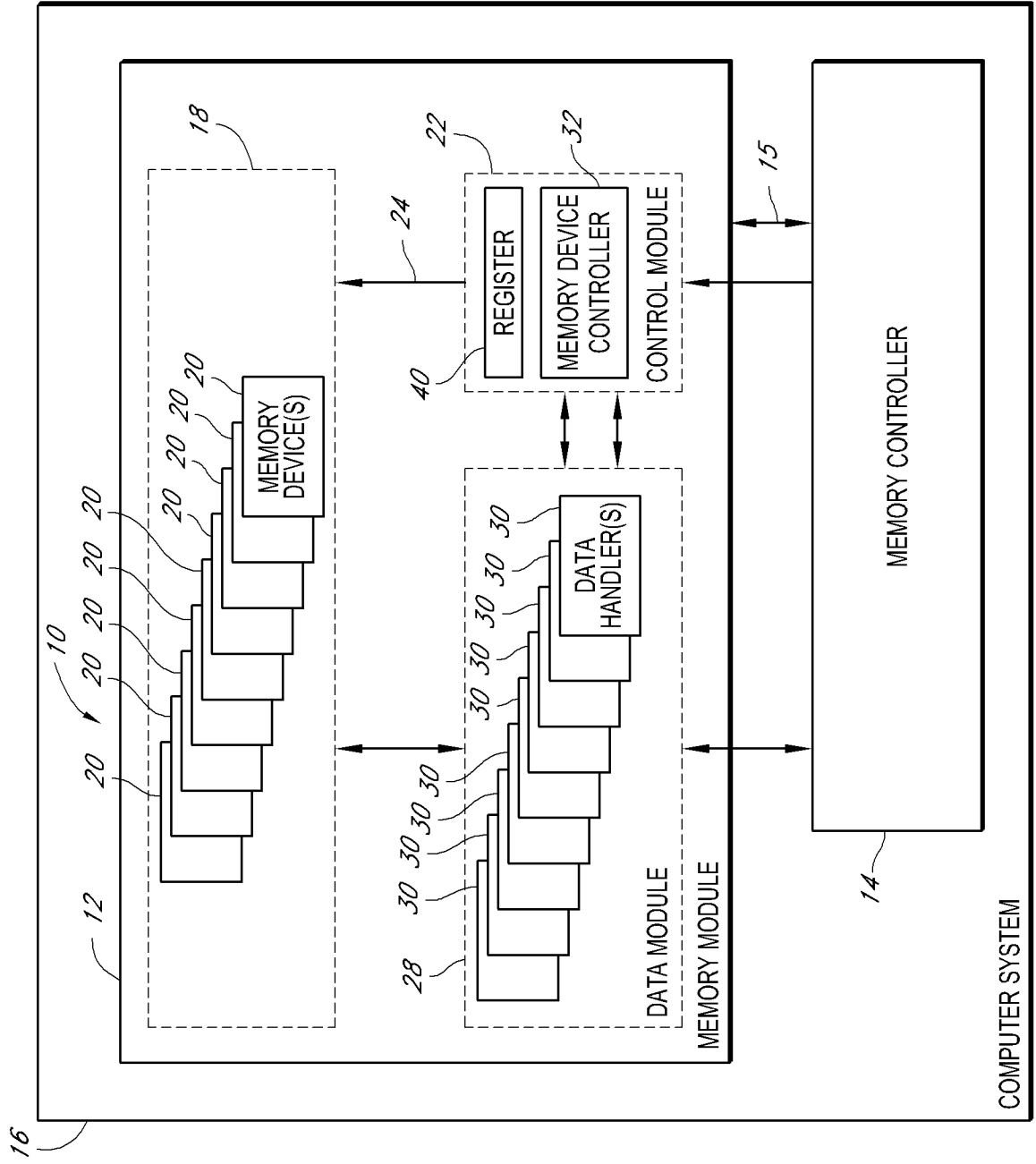
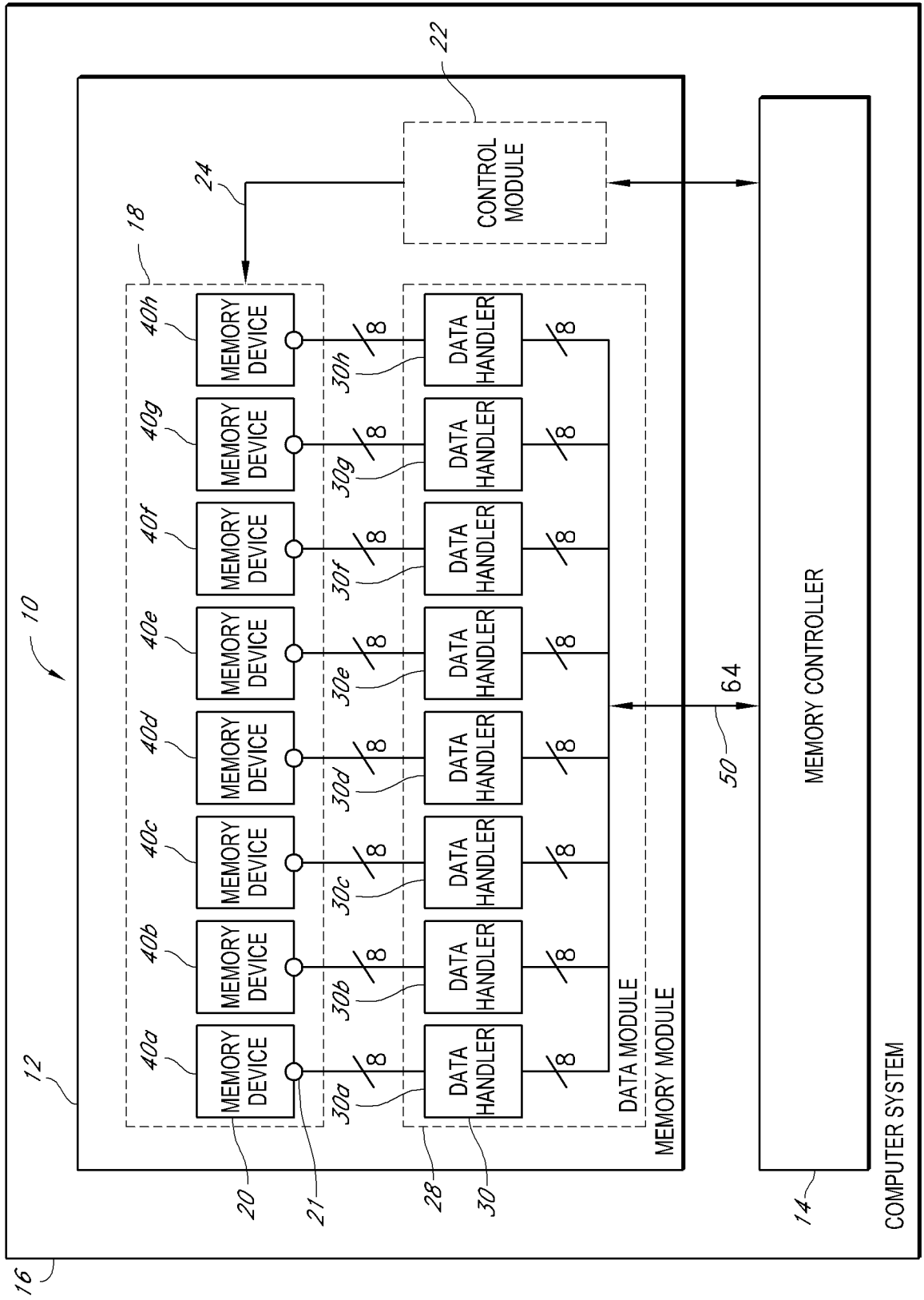


FIG. 1



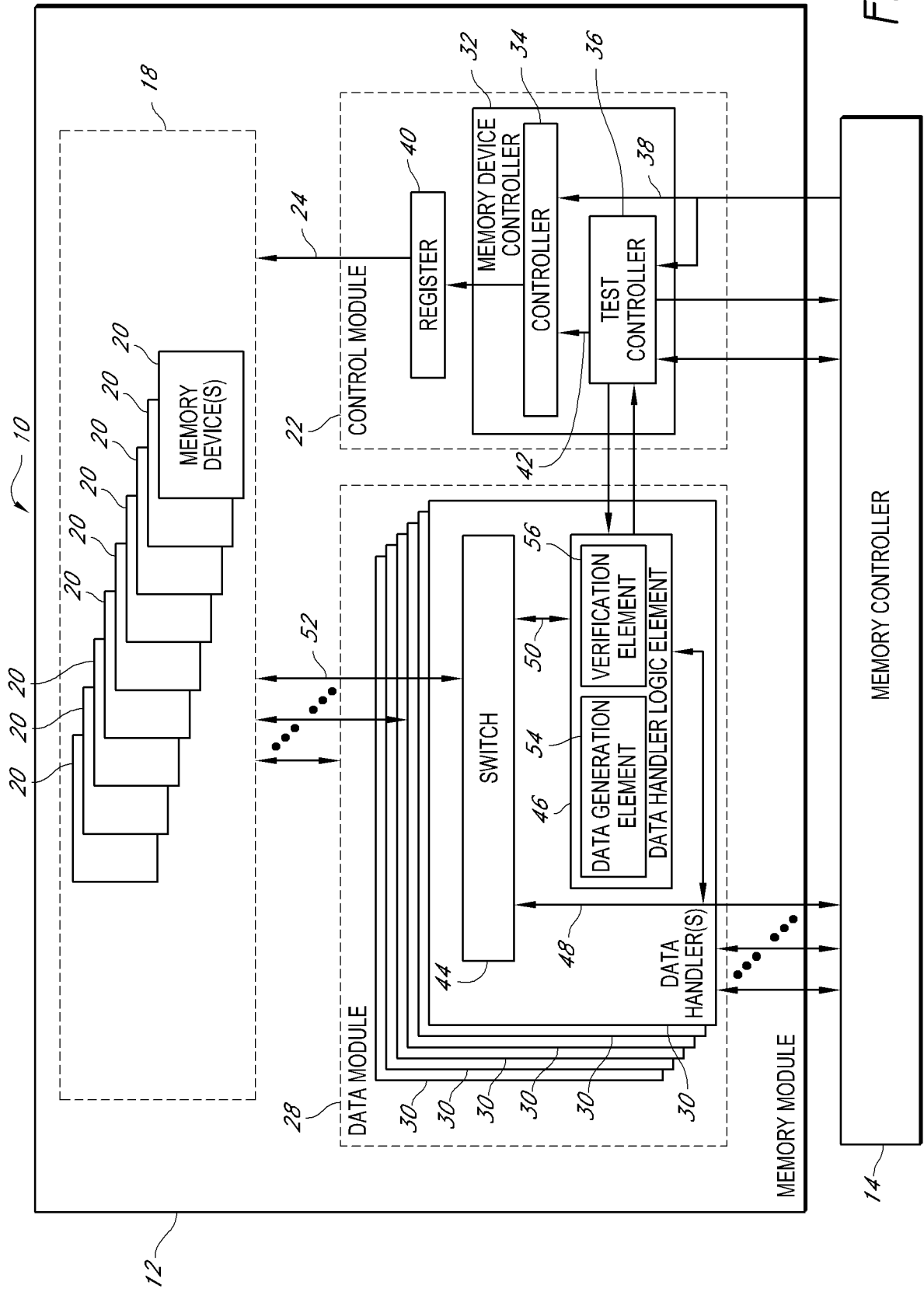


FIG. 3

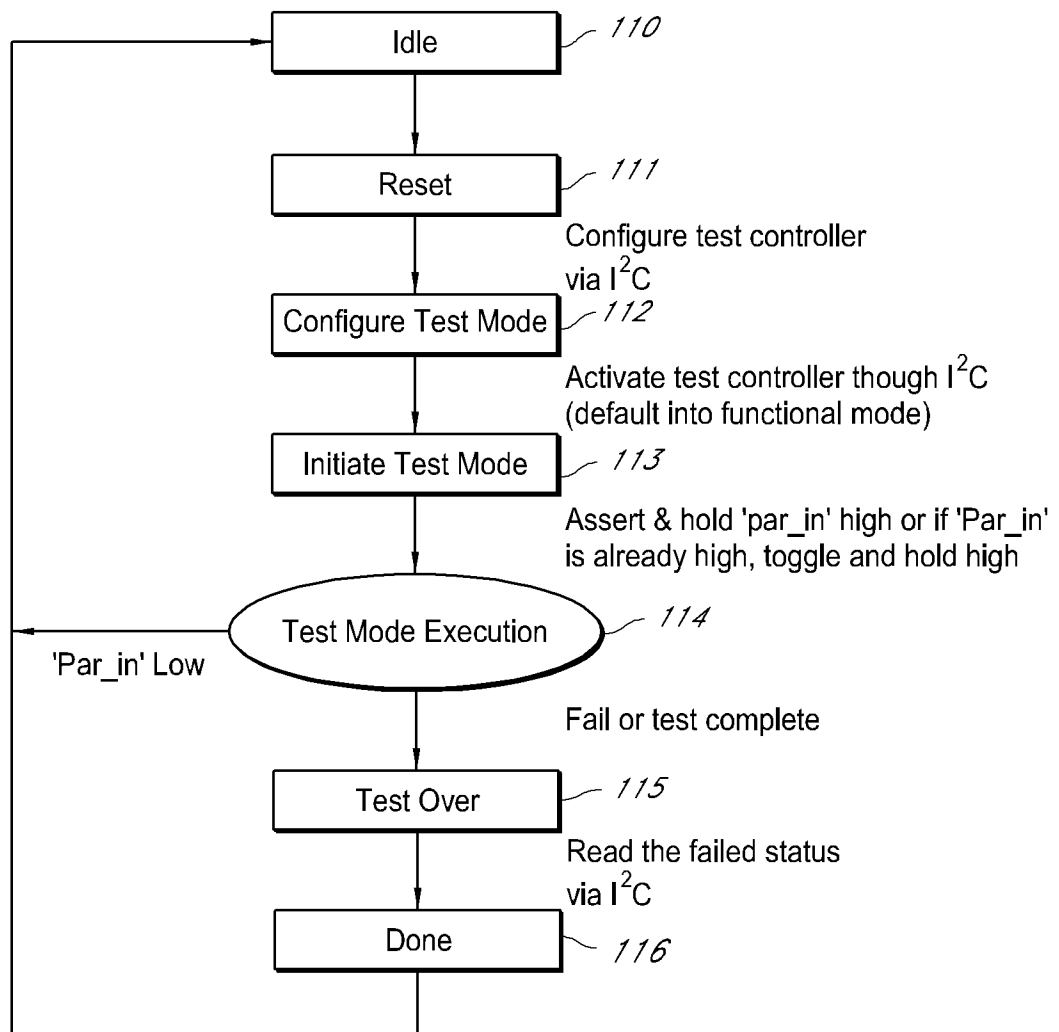
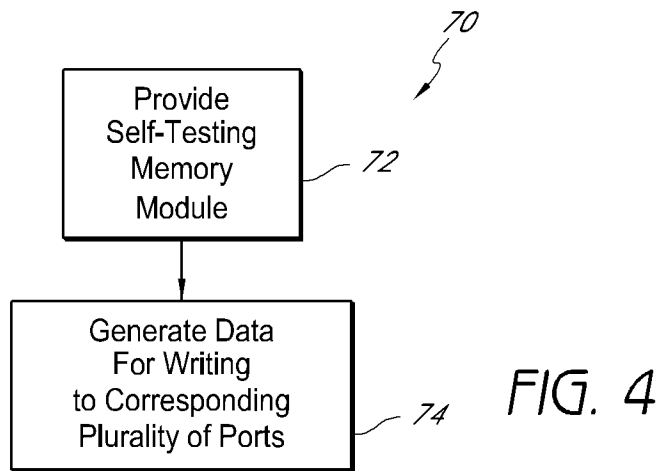
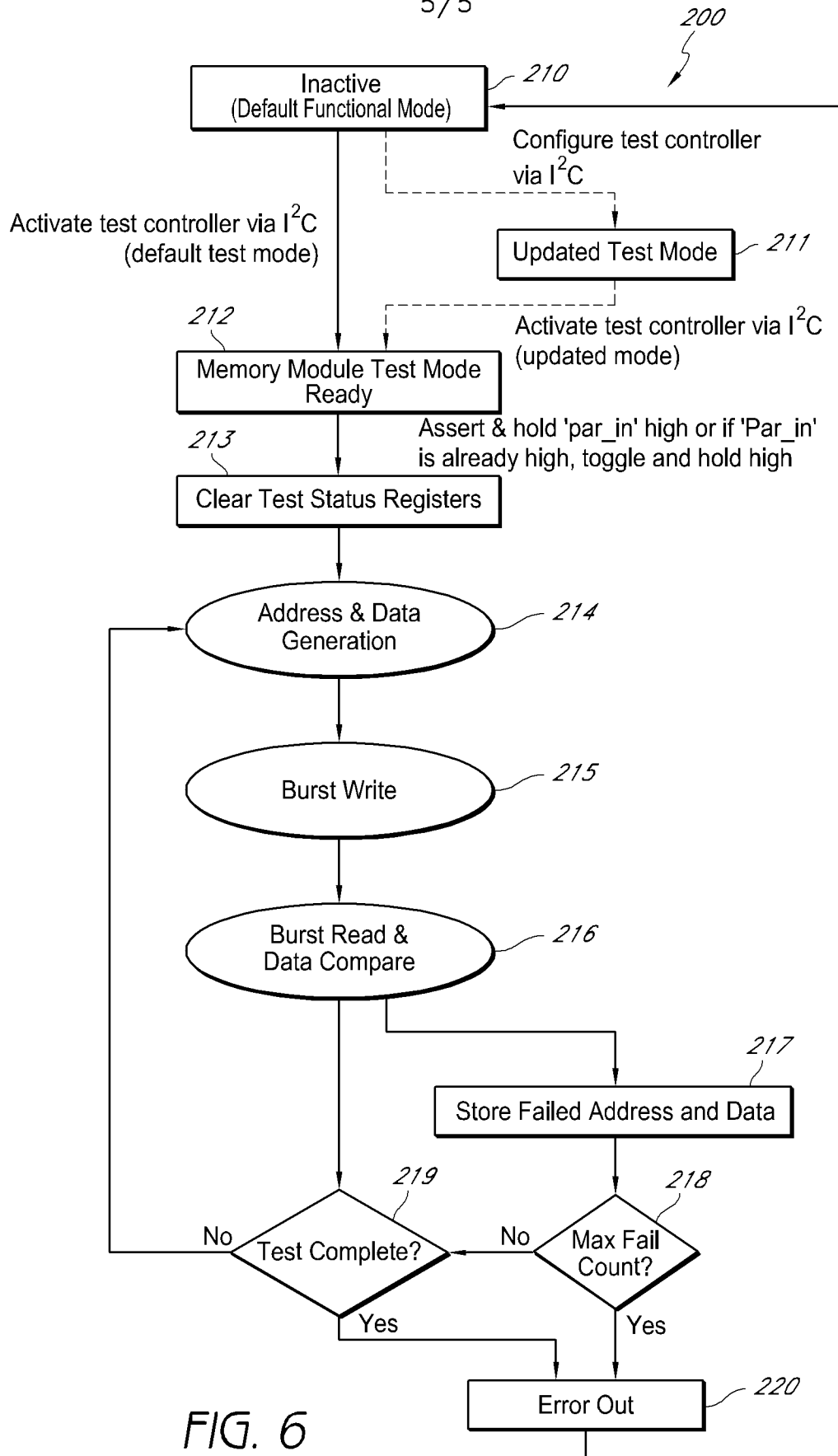


FIG. 5



Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C1
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1					Remove
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	To be Provided Later		To be Provided Later		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City		State/Province		Country of Residence i	
Citizenship under 37 CFR 1.41(b) i					
Mailing Address of Applicant:					
Address 1					
Address 2					
City		State/Province			
Postal Code		Country i			
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
Attorney Docket Number	NETL.046C1	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C1
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

Publication Information:
☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☒ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	12/422925	2009-04-13
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/422925	non provisional of	61/044801	2008-04-14
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/422925	non provisional of	61/044825	2008-04-14
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/422925	non provisional of	61/044839	2008-04-14
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button. Add			

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046C1
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

			Remove
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.			
Assignee ¹			Remove
If the Assignee is an Organization check here. ☒			
Organization Name	NETLIST, Inc.		
Mailing Address Information:			
Address 1	51 Discovery, #150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number		Fax Number	
Email Address			
Additional Assignee Data may be generated within this form by selecting the Add button.			Add

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature	/Sean Ambrosius/			Date (YYYY-MM-DD)	2011-07-14
First Name	Sean	Last Name	Ambrosius	Registration Number	65290

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	10520708
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	To be Provided Later
Customer Number:	20995
Filer:	Sean Patrick Ambrosius/Valerie Jones
Filer Authorized By:	Sean Patrick Ambrosius
Attorney Docket Number:	NETL.046C1
Receipt Date:	14-JUL-2011
Filing Date:	
Time Stamp:	18:03:30
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		Spec-NETL.pdf	1344491 01004eb8ab069f6b0a04ce8d48e4d19a249a3e3c	yes	26

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Specification	1	24
	Claims	25	25
	Abstract	26	26

Warnings:

Information:

2	Drawings-only black and white line drawings	Drawings-NETL.pdf	101263	no	5
			1efb3dcd27e8d1162d4d509505b6bc55c775ffe2		

Warnings:

Information:

3	Application Data Sheet	NETL-ADS.pdf	838064	no	4
			f92961cd0007da928f6d1826d227d16ab32cd6d1		

Warnings:

Information:

Total Files Size (in bytes):			2283818
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/183,253	07/14/2011		NETL.046C1

CONFIRMATION NO. 7935

FORMALITIES LETTER



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 07/28/2011

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
Applicant must submit \$330 to complete the basic filing fee for a non-small entity. If appropriate, applicant may make a written assertion of entitlement to small entity status and pay the small entity filing fee (37 CFR 1.27).
- The oath or declaration is missing.
A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.
Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- A surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of \$130 for a non-small entity, must be submitted.

SUMMARY OF FEES DUE:

Total fee(s) required within **TWO MONTHS** from the date of this Notice is \$1220 for a non-small entity

- \$330 Statutory basic filing fee.
- \$130 Surcharge.
- The application search fee has not been paid. Applicant must submit \$540 to complete the search fee.
- The application examination fee has not been paid. Applicant must submit \$220 to complete the examination fee for a non-small entity.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
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<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

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If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/etadesse/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/183,253	07/14/2011	2189	0.00	NETL.046C1	1	1

CONFIRMATION NO. 7935

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000048972886

Date Mailed: 07/28/2011

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 12/422,925 04/13/2009
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 07/26/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/183,253**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

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NOT GRANTED

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/183,253

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	1 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$270 (\$135 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	330
N/A	540
N/A	220
x 52 =	0.00
x 220 =	0.00
	0.00
	0.00
TOTAL	1090

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant	:	Lee et al.	Group Art Unit: 2189
Appl. No.	:	13/183,253	
Filed	:	July 14, 2011	
For	:	MEMORY BOARD WITH SELF-TESTING CAPABILITY	
Examiner	:	Not yet known	
Conf. No.	:	7935	

PRELIMINARY AMENDMENT**Mail Stop Amendment**

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

Prior to examination on the merits, please amend the above-captioned application as described herein.

Amendments to the Claims: Begin on page 2 of this Preliminary Amendment.

Remarks: Begin on page 6 of this Preliminary Amendment.

AMENDMENTS TO THE CLAIMS

Please cancel Claim 1 without prejudice, as indicated below.

Please add Claims 2-21, as indicated below.

1. (Canceled)
2. (New) A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.
3. (New) The memory system of Claim 2, wherein each of the one or more memory chips comprises address and control ports configured to receive the address and control signals generated by the control circuit and data ports configured to receive the data generated by the one or more data handlers.
4. (New) The memory system of Claim 3, wherein the one or more data handlers are positioned closer to the data ports of the one or more memory chips than to data ports of other memory chips of the plurality of memory chips.
5. (New) The memory system of Claim 2, wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.
6. (New) The memory system of Claim 2, wherein the memory system is configured to test the one or more memory chips by writing the data generated by the one or more data handlers to one or more memory locations of the one or more memory chips.

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Filed : **July 14, 2011**

7. (New) The memory system of Claim 6, wherein the one or more data handlers are further configured to receive data from the one or more memory chips and to identify failures in the operation of the one or more memory chips by verifying that data received from the one or more memory locations of the one or more memory chips corresponds to the data generated and written by the one or more data handlers to the one or more memory locations of the one or more memory chips.

8. (New) The memory system of Claim 7, wherein the one or more data handlers are further configured to verify that the data read from the one or more memory locations of the one or more memory chips corresponds to the data generated and written by the one or more data handlers to the one or more memory locations of the one or more memory chips without storing a copy of the data generated and written by the one or more data handlers.

9. (New) The memory system of Claim 7, wherein the one or more data handlers are configured to calculate comparison data and to compare the comparison data to the data read from the one or more memory locations of the one or more memory chips.

10. (New) The memory system of Claim 2, wherein the data generated by the one or more data handlers for writing to the one or more memory chips comprises cyclic data.

11. (New) The memory system of Claim 10, wherein the one or more data handlers are further configured to calculate comparison data based on the cyclic data and to compare the comparison data to data read from the one or more memory locations of the one or more memory chips.

12. (New) The memory system of Claim 2, wherein the one or more data handlers are further configured to calculate comparison data based on a current read address and to compare the comparison data to data read from the one or more memory locations of the one or more memory chips.

13. (New) The memory system of Claim 6, wherein the control circuit is further configured to store memory addresses associated with failures in the operation of the one or more memory locations of the one or more memory chips.

14. (New) The memory system of Claim 6, wherein the memory system is further configured to store data associated with failures in the operation of the one or more memory locations of the one or more memory chips.

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Filed : **July 14, 2011**

15. (New) The memory system of Claim 2, wherein the one or more circuits are further configured to write data to the one or more memory chips by selectively inputting to the one or more memory chips either data signals from the memory controller of the computer system or data signals from the one or more circuits.

16. (New) The memory system of Claim 2, wherein the control circuit is further configured to selectively input to the address and control ports of the one or more memory chips either the address and control signals from the memory controller of the computer system or the address and control signals from the control circuit.

17. (New) A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:

operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;

operating a control circuit to generate address and control signals; and

testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

18. (New) The method of Claim 17, further comprising:

receiving, by the plurality of data handlers, data from the one or more memory locations of the one or more memory chips; and

verifying that the data received from the one or more memory locations of the one or more memory chips corresponds to the data transmitted to the one or more memory locations of the one or more memory chips.

19. (New) The method of Claim 18, wherein said verifying is performed without storing a copy of the data transmitted to the one or more memory locations of the one or more memory chips.

20. (New) The method of Claim 18, wherein said verifying comprises:

calculating comparison data; and

comparing the comparison data to the data received from the one or more memory locations of the one or more memory chips.

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Filed : **July 14, 2011**

21. (New) The method of Claim 18, wherein the data transmitted to the one or more memory locations of the one or more memory chips comprises cyclic data.

Appl. No. : 13/183,253
Filed : July 14, 2011

REMARKS

Original Claim 1 has been canceled without prejudice, and new Claims 2-21 have been added. Applicants believe that none of the foregoing amendments have added new matter to the application.

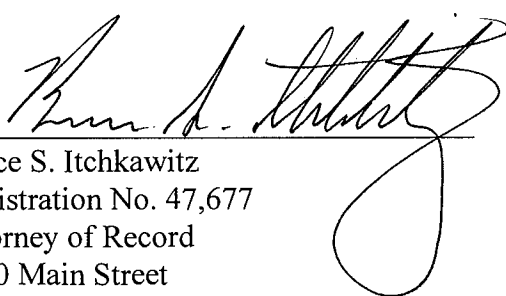
All of the claims currently pending in the present application are believed to be allowable over the prior art, and such action is earnestly solicited. If, however, any matters remain which could be resolved by Examiner's Amendment, the Examiner is cordially invited to contact the undersigned by telephone so that any such matters can be promptly resolved.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: 11/28/11

By:


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
2040 Main Street
Fourteenth Floor
Irvine, CA 92614
(949) 760-0404

12333895
112811

Doc Code: OATH

Document Description: Oath or declaration filed

PTO/SB/01 (10-08)

Approved for use through 06/30/2010. OMB 0651-0032
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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**DECLARATION FOR UTILITY OR
DESIGN
PATENT APPLICATION
(37 CFR 1.63)**

☐ Declaration
Submitted
With Initial
Filing

OR

☒ Declaration
Submitted after Initial
Filing (surcharge
(37 CFR 1.16 (f))
required)

Attorney Docket
Number NETL.046A

First Named Inventor Hyun Lee

COMPLETE IF KNOWN

Application Number 12/422,925

Filing Date April 13, 2009

Art Unit Unknown

Examiner Name Unknown

I hereby declare that: (1) Each inventor's residence, mailing address, and citizenship are as stated below next to their name; and (2) I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is claimed and for which a patent is sought on the invention entitled:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

(Title of the Invention)

the application of which

☐ is attached hereto

OR

☒ was filed on (MM/DD/YYYY) April 13, 2009

as United States Application Number or PCT International

Application Number 12/422,925 and was amended on (MM/DD/YYYY) (if applicable).

I hereby state that I have reviewed and understand the contents of the above identified application, including the claims, as amended by any amendment specifically referred to above.

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56, including for continuation-in-part applications, material information which became available between the filing date of the prior application and the national or PCT international filing date of the continuation-in-part application.

Authorization To Permit Access To Application by Participating Offices

☐ If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), and any other intellectual property offices in which a foreign application claiming priority to the above-identified application is filed access to the above-identified patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, or other intellectual property office in which a foreign application claiming priority to the above-identified application is filed to have access to the application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the application-as-filed with respect to: 1) the above-identified application, 2) any foreign application to which the above-identified application claims priority under 35 USC 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the above-identified US application, and 3) any U.S. application from which benefit is sought in the above-identified application.

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[Page 1 of 3]

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DECLARATION — Utility or Design Patent Application

Claim of Foreign Priority Benefits

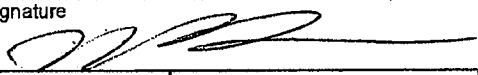
I hereby claim foreign priority benefits under 35 U.S.C. 119(a)-(d) or (f), or 365(b) of any foreign application(s) for patent, inventor's or plant breeder's rights certificate(s), or 365(a) of any PCT international application which designated at least one country other than the United States of America, listed below and have also identified below, by checking the box, any foreign application for patent, inventor's or plant breeder's rights certificate(s), or any PCT international application having a filing date before that of the application on which priority is claimed.

Prior Foreign Application Number(s)	Country	Foreign Filing Date (MM/DD/YYYY)	Priority Not Claimed	Certified Copy Attached?	
				YES	NO
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐

☐ Additional foreign application numbers are listed on a supplemental priority data sheet PTO/SB/02B attached hereto.

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DECLARATION — Utility or Design Patent Application

Direct all correspondence to:	☒ The address associated with Customer Number.	20995	OR	☐ Correspondence address below
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City State			ZIP	
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NAME OF SOLE OR FIRST INVENTOR:		☐ A petition has been filed for this unsigned inventor		
Given Name (first and middle (if any))		Family Name or Surname		
Hyun		Lee		
Inventor's Signature 			Date 10/1/09	
Residence: City	State	Country	Citizenship	
Ladera Ranch	CA	US	US	
Mailing Address				
21 Thalia				
City State		Zip	Country	
Ladera Ranch	CA	92694	US	
☒ Additional inventors or a legal representative are being named on the 1 supplemental sheet(s) PTO/SB/02A or 02LR attached hereto.				

Doc Code: OATH

Document Description: Oath or declaration filed

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**DECLARATION FOR UTILITY OR
DESIGN
PATENT APPLICATION
(37 CFR 1.63)**

☐ Declaration
Submitted
With Initial
Filing

OR

☒ Declaration
Submitted after Initial
Filing (surcharge
(37 CFR 1.16 (f))
required)

Attorney Docket Number	NETL.046A
First Named Inventor	Hyun Lee
COMPLETE IF KNOWN	
Application Number	12/422,925
Filing Date	April 13, 2009
Art Unit	Unknown
Examiner Name	Unknown

I hereby declare that: (1) Each inventor's residence, mailing address, and citizenship are as stated below next to their name; and (2) I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is claimed and for which a patent is sought on the invention entitled:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

(Title of the Invention)

the application of which

☐ is attached hereto

OR

☒ was filed on (MM/DD/YYYY) April 13, 2009

as United States Application Number or PCT International

Application Number 12/422,925 and was amended on (MM/DD/YYYY) (if applicable).

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[Page 1 of 3]

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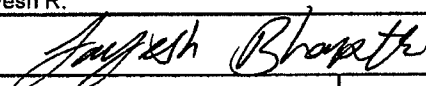
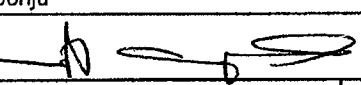
Claim of Foreign Priority Benefits

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Prior Foreign Application Number(s)	Country	Foreign Filing Date (MM/DD/YYYY)	Priority Not Claimed	Certified Copy Attached?	
				YES	NO
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐

☐ Additional foreign application numbers are listed on a supplemental priority data sheet PTO/SB/02B attached hereto.

DECLARATION**ADDITIONAL INVENTOR(S)**
Supplemental SheetPage 1 of 1

Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Jayesh R.		Bhakta	
Inventor's Signature 		Date 10-1-09	
Residence: City Cerritos	State CA	Country US	Citizenship US
Mailing Address 12220 Rose Street			
City Cerritos	State CA	Zip 90703	Country US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Soonju		Choi	
Inventor's Signature 		Date 10/1/2009	
Residence: City Irvine	State CA	Country US	Citizenship KR
Mailing Address 195 Woodbury			
City Irvine	State CA	Zip 92620	Country US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Inventor's Signature		Date	
Residence: City	State	Country	Citizenship
Mailing Address			
City	State	Zip	Country

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1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
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RESPONSE TO FORMALITIES NOTICE

Applicant	: Lee et al.
App. No.	: 13/183,253
Filed	: July 14, 2011
For	: MEMORY BOARD WITH SELF- TESTING CAPABILITY
Art Unit	: 2189
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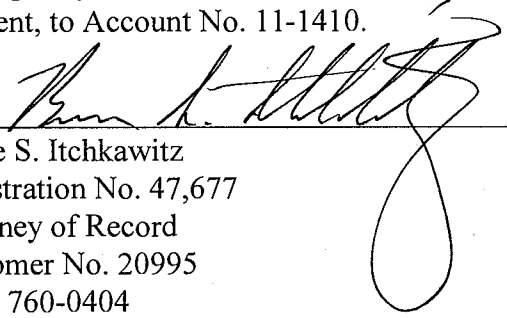
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without a Declaration and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following.

- (X) A Declaration in 7 pages.
- (X) Preliminary Amendment in 6 pages.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.



Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Electronic Patent Application Fee Transmittal

Application Number:	13183253			
Filing Date:	14-Jul-2011			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/Nicole Jevsevar			
Attorney Docket Number:	NETL.046C1			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	380	380
Utility Search Fee	1111	1	620	620
Utility Examination Fee	1311	1	250	250
Pages:				
Claims:				
Miscellaneous-Filing:				
Late filing fee for oath or declaration	1051	1	130	130
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 2 months with \$0 paid	1252	1	560	560
Miscellaneous:				
Total in USD (\$)				1940

Electronic Acknowledgement Receipt

EFS ID:	11491974
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Erica Van Sciver
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046C1
Receipt Date:	28-NOV-2011
Filing Date:	14-JUL-2011
Time Stamp:	18:37:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1940
RAM confirmation Number	5623
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		amendment.pdf	209772	yes	6
			b5672dbce374744be3f37fa3311e506d1cd27fd2		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Claims		2	5	
	Applicant Arguments/Remarks Made in an Amendment		6	6	
Warnings:					
Information:					
2	Oath or Declaration filed	declaration.pdf	383789	no	7
			9fc38399ab08e16c4ee2c5bf6b8c1395bfa2adba		
Warnings:					
Information:					
3	Applicant Response to Pre-Exam Formalities Notice	response.pdf	32651	no	1
			63fda2f172a8ad0fe11b8de2827d9d4f900291ac		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	38965	no	2
			1370f1ae8a440cee7dbac4159225dfe5ce78e14b		
Warnings:					
Information:					
Total Files Size (in bytes):			665177		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Electronic Acknowledgement Receipt

EFS ID:	11491974
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Erica Van Sciver
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046C1
Receipt Date:	28-NOV-2011
Filing Date:	14-JUL-2011
Time Stamp:	18:37:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1940
RAM confirmation Number	5623
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		amendment.pdf	209772	yes	6
			b5672dbce374744be3f37fa3311e506d1cd27fd2		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Claims		2	5	
	Applicant Arguments/Remarks Made in an Amendment		6	6	
Warnings:					
Information:					
2	Oath or Declaration filed	declaration.pdf	383789	no	7
			9fc38399ab08e16c4ee2c5bf6b8c1395bfa2adba		
Warnings:					
Information:					
3	Applicant Response to Pre-Exam Formalities Notice	response.pdf	32651	no	1
			63fda2f172a8ad0fe11b8de2827d9d4f900291ac		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	38965	no	2
			1370f1ae8a440cee7dbac4159225dfe5ce78e14b		
Warnings:					
Information:					
Total Files Size (in bytes):			665177		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

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National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 13/183,253		Filing Date 07/14/2011		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	11/28/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$	=	OR	X \$60=	0
	Independent (37 CFR 1.16(h))	* 2	Minus	***3	= 0	X \$	=	OR	X \$250=	0
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	0
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=	OR	X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=	OR	X \$	=
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/CAROLYN THOMAS/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/183,253

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	380
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	620
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	250
TOTAL CLAIMS (37 CFR 1.16(j))	20	minus 20 = *			OR	x 60 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2	minus 3 = *				x 250 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
			TOTAL			TOTAL	1250

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/183,253	07/14/2011	2189	1380	NETL.046C1	20	2

CONFIRMATION NO. 7935

UPDATED FILING RECEIPT



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 12/13/2011

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 12/422,925 04/13/2009 PAT 8001434
which claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 07/26/2011

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/183,253**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

Preliminary Class

711

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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NOT GRANTED

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
	Filing Date	July 14, 2011
	First Named Inventor	Lee et al.
	Art Unit	2117
(Multiple sheets used when necessary)	Examiner	Chung, Phung M.
SHEET 1 OF 4	Attorney Docket No.	NETL.046C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
	11	4,910,597	03-20-1990	Harada et al.	
	12	4,942,556	07-17-1990	Sasaki et al.	
	13	4,987,321	01-22-1991	Toohey	
	14	5,033,048	07-16-1991	Pierce et al.	
	15	5,051,997	09-24-1991	Sakashita et al.	
	16	5,138,619	08-11-1992	Fasang et al.	
	17	5,173,906	12-22-1992	Dreibelbis et al.	
	18	5,222,066	06-22-1993	Grula et al.	
	19	5,241,503	08-31-1993	Cheng	
	20	5,304,856	04-19-1994	Rainal	
	21	5,337,254	08-09-1994	Knee et al.	
	22	5,359,235	10-25-1994	Coyle et al.	
	23	5,394,037	02-28-1995	Josephson et al.	
	24	5,430,335	07-04-1995	Tanoi	
	25	5,525,917	06-11-1996	Wong et al.	
	26	5,841,296	11-24-1998	Churcher et al.	
	27	5,914,543	06-22-1999	Scherpenberg, et al.	
	28	6,000,048	12-07-1999	Krishna et al.	
	29	6,044,481	03-28-2000	Kornachuk et al.	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
	Filing Date	July 14, 2011
	First Named Inventor	Lee et al.
	Art Unit	2117
(Multiple sheets used when necessary)	Examiner	Chung, Phung M.
SHEET 2 OF 4	Attorney Docket No.	NETL.046C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	30	6,070,217	05-30-2000	Connolly et al.	
	31	6,169,696	01-02-2001	Bissey	
	32	6,194,959	02-27-2001	Kamoshida et al.	
	33	6,216,240	04-10-2001	Won et al.	
	34	6,467,056	10-15-2002	Satou et al.	
	35	6,560,740	05-06-2003	Zuraski, Jr. et al.	
	36	6,614,702	09-02-2003	Lee	
	37	6,721,150	04-13-2004	Guerrero, Jr.	
	38	6,812,869	11-02-2004	Rahman et al.	
	39	6,829,728	12-07-2004	Cheng et al.	
	40	6,918,072	07-12-2005	Cowels et al.	
	41	6,928,024	08-09-2005	Pfeiffer et al.	
	42	6,928,593	08-09-2005	Halbert et al.	
	43	6,930,509	08-16-2005	Banik	
	44	6,934,900	08-23-2005	Cheng et al.	
	45	7,036,064	04-25-2006	Kebichi et al.	
	46	7,053,470	05-30-2006	Sellers et al.	
	47	7,062,696	06-13-2006	Barry et al.	
	48	7,184,915	02-27-2007	Hansquine et al.	
	49	7,190,210	03-13-2007	Azrai et al.	
	50	7,203,873	04-10-2007	Adams	
	51	7,253,652	08-07-2007	Azimi et al.	
	52	7,284,166	10-16-2007	Zappa et al.	
	53	2001/0048342	12-06-2001	Yoshida et al.	
	54	2002/0000847	01-03-2002	Taguchi	
	55	2002/0131535	09-19-2002	Huber	
	56	2002/0140523	10-03-2002	Park et al.	
	57	2003/0098742	05-29-2003	Nakagawa et al.	
	58	2003/0126346	07-03-2003	Kuo	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
	Filing Date	July 14, 2011
	First Named Inventor	Lee et al.
	Art Unit	2117
(Multiple sheets used when necessary)	Examiner	Chung, Phung M.
SHEET 3 OF 4	Attorney Docket No.	NETL.046C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	59	2003/0197797	10-23-2003	Segura	
	60	2003/0218491	11-27-2003	Nagasue	
	61	2004/0155702	08-12-2004	Danielsson	
	62	2004/0199843	10-07-2004	Hansquine et al.	
	63	2005/0093620	05-05-2005	Ho et al.	
	64	2005/0127989	06-16-2005	Miyagi	
	65	2005/0289423	12-29-2005	Yabuta	
	66	2006/0082383	04-20-2006	Choi	
	67	2006/0107156	05-18-2006	Lee et al.	
	68	2006/0140015	06-26-2006	Kasamsetty	
	69	2006/0144015	07-06-2006	Cash et al.	
	70	2006/0147217	07-06-2006	Hahin et al.	
	71	2006/0192653	08-31-2006	Atkinson et al.	
	72	2006/0242458	10-26-2006	Feldman et al.	
	73	2006/0262586	11-23-2006	Solomon et al.	
	74	2006/0271748	11-30-2006	Jain et al.	
	75	2007/0030814	02-08-2007	Shin et al.	
	76	2007/0058471	03-15-2007	Rajan et al.	
	77	2007/0079199	04-05-2007	Chorn et al.	
	78	2007/0109707	05-17-2007	Honda	
	79	2007/0152743	07-05-2007	Keeth et al.	
	80	2007/0204075	08-30-2007	Rajan et al.	
	81	2007/0223296	09-27-2007	Miller et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

NON PATENT LITERATURE DOCUMENTS

Examiner Signature	Date Considered
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
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	Art Unit	2117
(Multiple sheets used when necessary)	Examiner	Chung, Phung M.
SHEET 4 OF 4	Attorney Docket No.	NETL.046C1

Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
	82	DER-CHANG ET AL. "A <i>parallel built-in self-diagnostic method for embedded memory arrays</i> ", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, April 2002, Volume 21, Issue 4, Pages 449-465	
	83	BAYARD "On the LTI properties of adaptive feed forward systems with tapdelay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
	84	MUTOH et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, November 2-6, 2004, Volume 1, pages 963-968	
	85	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

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Electronic Acknowledgement Receipt

EFS ID:	12552816
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Heide Young
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046C1
Receipt Date:	16-APR-2012
Filing Date:	14-JUL-2011
Time Stamp:	18:09:58
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		IDS.pdf	275217 bc4423d82f1432daaa03dc6403d72aab4ad 9febd	yes	5

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Transmittal Letter	1	1
	Information Disclosure Statement (IDS) Form (SB08)	2	5
Warnings:			
Information:			
Total Files Size (in bytes):		275217	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

INFORMATION DISCLOSURE STATEMENT

Applicant : Lee et al.
App. No : 13/183,253
Filed : July 14, 2011
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Examiner : Chung, Phung M.
Art Unit : 2117
Conf No. : 7935

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

References and Listing

Submitted herewith in the above-identified application is an Information Disclosure Statement listing references for consideration. References numbered 1-85 are of record in U.S. Patent Application No. 12/422,925, filed April 13, 2009, which is relied upon for an earlier filing date under 35 USC 120. Accordingly, copies of references numbered 1-85 are not submitted pursuant to 37 CFR 1.98(d).

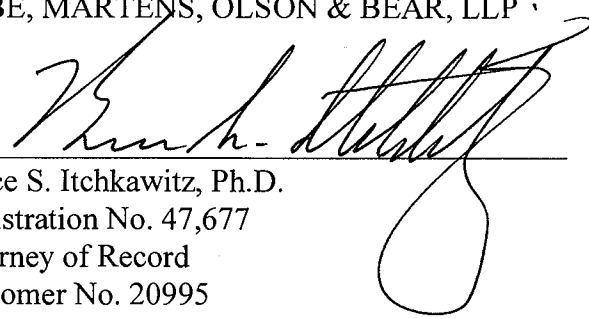
Timing of Disclosure

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 CFR 1.17(p) to Deposit Account No. 11-1410.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: 4/16/12

By: 

Bruce S. Itchkawitz, Ph.D.
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/183,253	07/14/2011	Hyun Lee	NETL.046C1	7935

20995 7590 05/29/2012
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

EXAMINER

CHUNG, PHUNG M

ART UNIT	PAPER NUMBER
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2117

NOTIFICATION DATE	DELIVERY MODE
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05/29/2012

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOAPilot@kmob.com
jayna.cartee@knobbe.com
efiling@knobbe.com

Office Action Summary	Application No. 13/183,253	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 28 November 2011.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 5) ☒ Claim(s) 2-21 is/are pending in the application.
- 5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 2-21 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

Application Papers

- 10) ☒ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 14 July 2011 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 12) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 13) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).
- * See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413) |
| 2) ☐ Notice of Draftperson's Patent Drawing Review (PTO-948) | Paper No(s)/Mail Date. ____. |
| 3) ☒ Information Disclosure Statement(s) (PTO/SB/08) | 5) ☐ Notice of Informal Patent Application |
| Paper No(s)/Mail Date <u>4/16/12</u> . | 6) ☐ Other: ____. |

Double Patenting

1. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory obviousness-type double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the conflicting application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement.

Effective January 1, 1994, a registered attorney or agent of record may sign a terminal disclaimer. A terminal disclaimer signed by the assignee must fully comply with 37 CFR 3.73(b).

Claim(s) 1-35 of patent # 8,001,434 contain(s) every element of claim(s) 2-21 of the instant application and as such anticipate(s) claim(s) 2-21 of the instant application.

“A later patent claim is not patentably distinct from an earlier patent claim if the later claim is obvious over, or anticipated by, the earlier claim. In re Longi, 759 F.2d at 896, 225 USPQ at 651 (affirming a holding of obviousness-type double patenting because the claims at issue were obvious over claims in four prior art patents); In re Berg, 140 F.3d at 1437, 46 USPQ2d at 1233 (Fed. Cir. 1998) (affirming a holding of obviousness-type double patenting where a patent application claim to a genus is anticipated by a patent claim to a species within that genus). “ ELI LILLY AND COMPANY v BARR LABORATORIES, INC., United States Court of Appeals for the Federal Circuit, ON PETITION FOR REHEARING EN BANC (DECIDED: May 30, 2001).

Specification

2. The disclosure is objected to because of the following informalities:

Paragraph (0001), lines 2, 6 and 7, the U.S. Patent Application No. 12/422,925, 12/422,912 and 12/422,853 are not up to today date, for example, now US Patent No. or now abandoned or still pending...Appropriate correction is required.

Art Unit: 2117

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ajay M. Bhatia can be reached on 571-272-3906. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

Notice of References Cited	Application/Control No. 13/183,253	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-4,958,346	09-1990	Fujisaki, Kenichi	714/720
*	B	US-6,681,358	01-2004	Karimi et al.	714/733
*	C	US-7,774,667	08-2010	Saito et al.	714/733
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 13183253	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

SEARCHED			
Class	Subclass	Date	Examiner
714	718,719,736,733,734	4/22/12	pmc
365	201	4/22/12	pmc

SEARCH NOTES		
Search Notes	Date	Examiner

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
	Filing Date	July 14, 2011
	First Named Inventor	Lee et al.
	Art Unit	2117
(Multiple sheets used when necessary)	Examiner	Chung, Phung M.
SHEET 1 OF 4	Attorney Docket No.	NETL.046C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
	11	4,910,597	03-20-1990	Harada et al.	
	12	4,942,556	07-17-1990	Sasaki et al.	
	13	4,987,321	01-22-1991	Toohey	
	14	5,033,048	07-16-1991	Pierce et al.	
	15	5,051,997	09-24-1991	Sakashita et al.	
	16	5,138,619	08-11-1992	Fasang et al.	
	17	5,173,906	12-22-1992	Dreibelbis et al.	
	18	5,222,066	06-22-1993	Grula et al.	
	19	5,241,503	08-31-1993	Cheng	
	20	5,304,856	04-19-1994	Rainal	
	21	5,337,254	08-09-1994	Knee et al.	
	22	5,359,235	10-25-1994	Coyle et al.	
	23	5,394,037	02-28-1995	Josephson et al.	
	24	5,430,335	07-04-1995	Tanoi	
	25	5,525,917	06-11-1996	Wong et al.	
	26	5,841,296	11-24-1998	Churcher et al.	
	27	5,914,543	06-22-1999	Scherpenberg, et al.	
	28	6,000,048	12-07-1999	Krishna et al.	
	29	6,044,481	03-28-2000	Kornachuk et al.	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
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SHEET 2 OF 4	Attorney Docket No.	NETL.046C1

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	30	6,070,217	05-30-2000	Connolly et al.	
	31	6,169,696	01-02-2001	Bissey	
	32	6,194,959	02-27-2001	Kamoshida et al.	
	33	6,216,240	04-10-2001	Won et al.	
	34	6,467,056	10-15-2002	Satou et al.	
	35	6,560,740	05-06-2003	Zuraski, Jr. et al.	
	36	6,614,702	09-02-2003	Lee	
	37	6,721,150	04-13-2004	Guerrero, Jr.	
	38	6,812,869	11-02-2004	Rahman et al.	
	39	6,829,728	12-07-2004	Cheng et al.	
	40	6,918,072	07-12-2005	Cowels et al.	
	41	6,928,024	08-09-2005	Pfeiffer et al.	
	42	6,928,593	08-09-2005	Halbert et al.	
	43	6,930,509	08-16-2005	Banik	
	44	6,934,900	08-23-2005	Cheng et al.	
	45	7,036,064	04-25-2006	Kebichi et al.	
	46	7,053,470	05-30-2006	Sellers et al.	
	47	7,062,696	06-13-2006	Barry et al.	
	48	7,184,915	02-27-2007	Hansquine et al.	
	49	7,190,210	03-13-2007	Azrai et al.	
	50	7,203,873	04-10-2007	Adams	
	51	7,253,652	08-07-2007	Azimi et al.	
	52	7,284,166	10-16-2007	Zappa et al.	
	53	2001/0048342	12-06-2001	Yoshida et al.	
	54	2002/0000847	01-03-2002	Taguchi	
	55	2002/0131535	09-19-2002	Huber	
	56	2002/0140523	10-03-2002	Park et al.	
	57	2003/0098742	05-29-2003	Nakagawa et al.	
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	63	2005/0093620	05-05-2005	Ho et al.	
	64	2005/0127989	06-16-2005	Miyagi	
	65	2005/0289423	12-29-2005	Yabuta	
	66	2006/0082383	04-20-2006	Choi	
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	68	2006/0140015	06-26-2006	Kasamsetty	
	69	2006/0144015	07-06-2006	Cash et al.	
	70	2006/0147217	07-06-2006	Hahin et al.	
	71	2006/0192653	08-31-2006	Atkinson et al.	
	72	2006/0242458	10-26-2006	Feldman et al.	
	73	2006/0262586	11-23-2006	Solomon et al.	
	74	2006/0271748	11-30-2006	Jain et al.	
	75	2007/0030814	02-08-2007	Shin et al.	
	76	2007/0058471	03-15-2007	Rajan et al.	
	77	2007/0079199	04-05-2007	Chorn et al.	
	78	2007/0109707	05-17-2007	Honda	
	79	2007/0152743	07-05-2007	Keeth et al.	
	80	2007/0204075	08-30-2007	Rajan et al.	
	81	2007/0223296	09-27-2007	Miller et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

NON PATENT LITERATURE DOCUMENTS
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Examiner Signature /Phung Chung/	Date Considered 04/21/2012
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*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

T¹ - Place a check mark in this area when an English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/183,253
	Filing Date.	July 14, 2011
	First Named Inventor	Lee et al.
	Art Unit	2117
(Multiple sheets used when necessary)	Examiner	Chung, Phung M.
SHEET 4 OF 4	Attorney Docket No.	NETL.046C1

Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
	82	DER-CHANG ET AL. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, April 2002, Volume 21, Issue 4, Pages 449-465	
	83	BAYARD "On the LTI properties of adaptive feed forward systems with tapdelay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
	84	MUTOH et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, November 2-6, 2004, Volume 1, pages 963-968	
	85	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

13105533
041112

Examiner Signature	/Phung Chung/	Date Considered	04/21/2012
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			

T¹ - Place a check mark in this area when an English language translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /PC/

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L2	1518244	memory chips or chips	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:42
L3	143571	memory controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:42
L4	1	(data handler\$1) same ((data generation element or data generator) and (verification element or verifier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:44
L5	2046	(data handler\$1) or ((data generation element or data generator) and (verification element or verifier))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:44
L6	1	data handler logic element or data handler logic controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:45
L7	16577	data module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:45
L8	74	I5 and I7	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:45
L9	45	control circuit same (address generator	US-PGPUB;	ADJ	ON	2012/04/22

		same control signal generator)	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			10:47
L11	19324	data generator	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:48
L12	22	I11 and (I8 or I9)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:48
L13	1	I12 and I2 and I3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:48
L14	16	I12 and (I2 or I3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:48
L18	8989	(test\$3) and ((memory chips) and ((data or data handlers) and (address and control signals) or (control circuit)))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:51
L19	1	I14 and I18	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:51
L20	444	I18 and (I7 or I11)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 10:54
L21	2	control module and (memory device controller and test controller)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	ADJ	ON	2012/04/22 10:54

			IBM_TDB			
L24	444	I20 and (I2 or I3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:04
L25	142518	control module or memory device controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:04
L26	84	I24 and I25	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:05
L27	4820	test controller or memory test controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:05
L28	2	I26 and I27	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:05
L29	16	I14 and (I7 or I11)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:14
L30	1	I29 and I25 and (I2 or I3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:15
L31	16	I29 and (I2 or I3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:15
L32	1	I27 and I31	US-PGPUB; USPAT; USOCR; FPRS;	ADJ	ON	2012/04/22 11:15

			EPO; JPO; DERWENT; IBM_TDB			
L33	191	self-test memory module or memory self-test	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:17
L35	335	(identify\$3 or verify\$3 or determin\$3 or test\$3 or compar\$3 or analyz\$3) same ((comparison data or calculated data) same (read data))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:18
L36	1	I14 and I35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:18
L37	1	I31 and I35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 11:19
L40	12550	(714/718,719,736,733,734;365/201).ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 12:29
L41	853	(identify\$3 or verify\$3 or determin\$3 or test\$3 or compar\$3 or analyz\$3) same ((comparison data or calculated data or computed data) same (read data or received data or transmitted data))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 12:31
L42	11	I41 and I18	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 12:31
L43	8	I42 and I35	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/04/22 12:32
L44	5	I40 and I43	US-PGPUB; USPAT;	ADJ	ON	2012/04/22 12:32

				USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
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BIB DATA SHEET

CONFIRMATION NO. 7935

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
13/183,253	07/14/2011	714	2117	NETL.046C1
APPLICANTS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Soonju Choi, Irvine, CA;				
** CONTINUING DATA ***** This application is a CON of 12/422,925 04/13/2009 PAT 8,001,434 which claims benefit of 61/044,801 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,839 04/14/2008				
** FOREIGN APPLICATIONS *****				
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 07/26/2011				
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/PHUNG M CHUNG/</u> Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 5	TOTAL CLAIMS 20
INDEPENDENT CLAIMS 2				
ADDRESS KNOBBE MARTENS OLSON & BEAR LLP 2040 MAIN STREET FOURTEENTH FLOOR IRVINE, CA 92614 UNITED STATES				
TITLE MEMORY BOARD WITH SELF-TESTING CAPABILITY				
FILING FEE RECEIVED 1380	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit	

<i>Index of Claims</i> 	Application/Control No. 13183253	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47					
CLAIM		DATE									
Final	Original	04/22/2012									
-	1										
v	2										
v	3										
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v	5										
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v	21										



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POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	13/183253
	Filing Date	14-Jul-2011
	First Named Inventor	Lee, Hyun
	Title	MEMORY BOARD WITH SELF-TESTING CAPABILITY
	Art Unit	2117
	Examiner Name	CHUNG, PHUNG M
	Attorney Docket Number	NETL.046C1

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

79141

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

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I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.
Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record			
Signature		Date	August 24, 2012
Name	Hyun Lee	Telephone	949-435-0025
Title and Company	VP and CTO		

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

☐ *Total of _____ forms are submitted.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Lee, et al.	Confirmation No.:	7935
Serial No.:	13/183253	Art Unit:	2117
Filed:	14-Jul-2011	Examiner:	CHUNG, PHUNG M
For:	<i>MEMORY BOARD WITH SELF-TESTING CAPABILITY</i>	Attorney Docket No.:	NETL.046C1

Date: August 29, 2011

AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Office Action dated May 29, 2012 for the above identified patent application.

AMENDMENTS to the SPECIFICATION begin on Page 2 of this paper.

A LIST OF ALL PENDING CLAIMS begin on Page 3 of this paper

REMARKS begin on Page 6 of this paper.

AMENDMENTS TO THE SPECIFICATION

Please replace the first paragraph on page 1 of the Specification, lines 10-25 with the following replacement paragraph.

[0001] The present application is a continuation of US. Patent Application No. 12/422,925, filed April 13, 2009, now U.S. Patent No. 8,001,434, which claims the benefit of priority from US. Provisional Application No. 61/044,801, filed April 14, 2008, US. Provisional Application No. 61/044,825, filed April 14, 2008, and US. Provisional Application No. 61/044,839, filed April 14, 2008. Each of the foregoing applications ~~are~~ is incorporated in their entirety by reference herein. This application is related to US. Patent Application No. 12/422,912, filed on April 13, 2009 ~~even-date herewith~~ and entitled "Self-Adjusting Damper", which is still pending at the United States Patent and Trademark Office, and to US. Application No. 12/422,853, filed on April 13, 2009, ~~even-date herewith~~ and entitled "Circuit Providing Load Isolation and Noise Reduction", now U.S. Patent No. 8,154,901, both of which are incorporated in their entirety by reference herein.

IN THE CLAIMS:

The following claims are pending in the present application. No amendment to the claims is currently made.

2. (Previously Presented) A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:

a plurality of memory chips;

a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;

a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

3. (Previously Presented) The memory system of Claim 2, wherein each of the one or more memory chips comprises address and control ports configured to receive the address and control signals generated by the control circuit and data ports configured to receive the data generated by the one or more data handlers.

4. (Previously Presented) The memory system of Claim 3, wherein the one or more data handlers are positioned closer to the data ports of the one or more memory chips than to data ports of other memory chips of the plurality of memory chips.

5. (Previously Presented) The memory system of Claim 2, wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.

6. (Previously Presented) The memory system of Claim 2, wherein the memory system is configured to test the one or more memory chips by writing the data generated by the one or more data handlers to one or more memory locations of the one or more memory chips.

7. (Previously Presented) The memory system of Claim 6, wherein the one or more data handlers are further configured to receive data from the one or more memory chips and to identify failures in the operation of the one or more memory chips by verifying that data received from the one or more memory locations of the one or more memory chips corresponds

to the data generated and written by the one or more data handlers to the one or more memory locations of the one or more memory chips.

8. (Previously Presented) The memory system of Claim 7, wherein the one or more data handlers are further configured to verify that the data read from the one or more memory locations of the one or more memory chips corresponds to the data generated and written by the one or more data handlers to the one or more memory locations of the one or more memory chips without storing a copy of the data generated and written by the one or more data handlers.

9. (Previously Presented) The memory system of Claim 7, wherein the one or more data handlers are configured to calculate comparison data and to compare the comparison data to the data read from the one or more memory locations of the one or more memory chips.

10. (Previously Presented) The memory system of Claim 2, wherein the data generated by the one or more data handlers for writing to the one or more memory chips comprises cyclic data.

11. (Previously Presented) The memory system of Claim 10, wherein the one or more data handlers are further configured to calculate comparison data based on the cyclic data and to compare the comparison data to data read from the one or more memory locations of the one or more memory chips.

12. (Previously Presented) The memory system of Claim 2, wherein the one or more data handlers are further configured to calculate comparison data based on a current read address and to compare the comparison data to data read from the one or more memory locations of the one or more memory chips.

13. (Previously Presented) The memory system of Claim 6, wherein the control circuit is further configured to store memory addresses associated with failures in the operation of the one or more memory locations of the one or more memory chips.

14. (Previously Presented) The memory system of Claim 6, wherein the memory system is further configured to store data associated with failures in the operation of the one or more memory locations of the one or more memory chips.

15. (Previously Presented) The memory system of Claim 2, wherein the one or more circuits are further configured to write data to the one or more memory chips by selectively inputting to the one or more memory chips either data signals from the memory controller of the computer system or data signals from the one or more circuits.

16. (Previously Presented) The memory system of Claim 2, wherein the control circuit is further configured to selectively input to the address and control ports of the one or

more memory chips either the address and control signals from the memory controller of the computer system or the address and control signals from the control circuit.

17. (Previously Presented) A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:

operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;

operating a control circuit to generate address and control signals; and

testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

18. (Previously Presented) The method of Claim 17, further comprising:

receiving, by the plurality of data handlers, data from the one or more memory locations of the one or more memory chips; and

verifying that the data received from the one or more memory locations of the one or more memory chips corresponds to the data transmitted to the one or more memory locations of the one or more memory chips.

19. (Previously Presented) The method of Claim 18, wherein said verifying is performed without storing a copy of the data transmitted to the one or more memory locations of the one or more memory chips.

20. (Previously Presented) The method of Claim 18, wherein said verifying comprises:

calculating comparison data; and

comparing the comparison data to the data received from the one or more memory locations of the one or more memory chips.

21. (Previously Presented) The method of Claim 18, wherein the data transmitted to the one or more memory locations of the one or more memory chips comprises cyclic data.

REMARKS

This amendment responds to the office action mailed May 29, 2012. In the office action the Examiner:

- rejected claims 2-21 on the ground of non-statutory obviousness-type double patenting over claims 1-35 of US Patent No. 8,001,434;
- objected to the disclosure because of informalities.

After entry of this amendment, the pending claims are: claims 2-21.

I. MCKESSON STATEMENT

The pending application is related to U.S. Patent No. 8,001,434 and U.S. Patent No. 8,154,901. The Examiner is encouraged to review the art made of record, Office Action(s) and the Notice of Allowance in the above-mentioned related patents, all of which are available on PAIR.

II. DOUBLE PATENTING REJECTION OF CLAIMS 2-21

Claims 2-21 are rejected on the ground of non-statutory obviousness-type double patenting over claims 1-35 of US Patent No. 8,001,434. Applicants do not agree with the Examiner's statement that "Claim(s) 1-35 patent # 8,001,434 contain(s) every element of claim(s) 2-21 of the instant application and as such anticipate(s) claim(s) 2-21 of (*sic*) the instant application." Nonetheless, to expedite prosecution, Applicants herewith submit a terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d). Based on the terminal disclaimer filed with this response, the double patenting rejection should be withdrawn.

III. OBJECTION TO THE DISCLOSURE

The Examiner objected to the disclosure because of informalities in Paragraph [0001]. Applicants have amended Paragraph [0001] accordingly. Thus, the objection should be withdrawn.

In light of the above amendments and remarks, Applicants respectfully request that the Examiner reconsider this application with a view towards allowance. The Examiner is invited to call the undersigned attorney at (650) 852-9432, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: August 29, 2012

/Jamie J. Zheng/

51167

Jamie J. Zheng
3524 Waverley Street
Palo Alto, CA 94306
(650) 842-9432

(Reg. No.)

**TERMINAL DISCLAIMER TO OBIATE A DOUBLE PATENTING
REJECTION OVER A "PRIOR" PATENT**Docket Number (Optional)
NETL.046C1

In re Application of: Lee, et al.

Application No.: 13/183,253

Filed: July 14, 2011

For: Memory Board with Self-Testing Capability

The owner*, Netlist, Inc., of 100 percent interest in the instant application hereby disclaims, except as provided below, the terminal part of the statutory term of any patent granted on the instant application which would extend beyond the expiration date of the full statutory term of **prior patent** No. 8,001,434 as the term of said **prior patent** is presently shortened by any terminal disclaimer. The owner hereby agrees that any patent so granted on the instant application shall be enforceable only for and during such period that it and the **prior patent** are commonly owned. This agreement runs with any patent granted on the instant application and is binding upon the grantee, its successors or assigns.

In making the above disclaimer, the owner does not disclaim the terminal part of the term of any patent granted on the instant application that would extend to the expiration date of the full statutory term of the **prior patent**, "as the term of said **prior patent** is presently shortened by any terminal disclaimer," in the event that said **prior patent** later:

- expires for failure to pay a maintenance fee;
- is held unenforceable;
- is found invalid by a court of competent jurisdiction;
- is statutorily disclaimed in whole or terminally disclaimed under 37 CFR 1.321;
- has all claims canceled by a reexamination certificate;
- is reissued; or
- is in any manner terminated prior to the expiration of its full statutory term as presently shortened by any terminal disclaimer.

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I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under Section 1001 of Title 18 of the United States Code and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.

2. ☐ The undersigned is an attorney or agent of record. Reg. No. _____

Signature
August 29, 2012
Date
Gail Sasaki
Typed or printed name

949-435-0025
Telephone Number

- ☐ Terminal disclaimer fee under 37 CFR 1.20(d) included.

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Form PTO/SB/96 may be used for making this certification. See MPEP § 324.

This collection of information is required by 37 CFR 1.321. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 13/183,253Filed/Issue Date: 14-Jul-2011

Titled:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

Netlist, Inc. _____, a corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 025230, Frame 0797, or for which a copy therefore is attached.

OR

- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

- ☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (*i.e.*, a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

Signature

Gail Sasaki

Printed or Typed Name

August 29, 2012

Date

VP and CFO

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 1

ASSIGNMENT

WHEREAS, We, (1) Hyun Lee, a United States citizen, residing at 21 Thalia, Ladera Ranch CA 92694, (2) Jayesh R. Bhakta, a United States citizen, residing at 12220 Rose Street, Cerritos CA 90703, and (3) Soonju Choi, a Korean citizen, residing at 195 Woodbury, Irvine CA 92620, have invented certain new and useful improvements in a MEMORY BOARD WITH SELF-TESTING CAPABILITY, for which we have filed an application for Letters Patent in the United States, Application No. 12/422,925, filed on April 13, 2009.

AND WHEREAS, Netlist, Inc., a Delaware Corporation, with its principal place of business at 51 Discovery # 150, Irvine CA 92618, (hereinafter "ASSIGNEE") desires to acquire the entire right, title, and interest in and to said improvements and said Application:

NOW, THEREFORE, in consideration of good and valuable consideration, the receipt of which is hereby acknowledged, we, the said inventors, do hereby acknowledge that we have sold, assigned, transferred and set over, and by these presents do hereby sell, assign, transfer and set over, unto said ASSIGNEE, its successors, legal representatives and assigns, the entire right, title, and interest throughout the world in, and its territories and possessions in, to and under said improvements, and any Patent Applications in the United States of America and all divisions, renewals and continuations thereof, and all Patents of the United States of America which may be granted thereon and all reissues and extensions thereof, and all rights of priority under International Conventions; and we hereby authorize and request the Commissioner of Patents of the United States of America to issue all Patents for said improvements to ASSIGNEE, its successors, legal representatives and assigns, in accordance with the terms of this instrument.

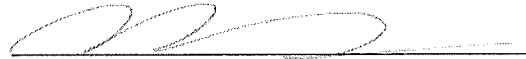
AND WE DO HEREBY sell, assign, transfer, and convey to ASSIGNEE, its successors, legal representatives, and assigns all claims for damages and all remedies arising out of any violation of the rights assigned hereby that may have accrued prior to the date of assignment to ASSIGNEE, or may accrue hereafter, including, but not limited to, the right to sue for, collect, and retain damages for past infringements of said Letters Patent before or after issuance.

AND WE HEREBY covenant and agree that we will communicate to said ASSIGNEE, its successors, legal representatives and assigns, any facts known to us respecting said improvements, and testify in any legal proceeding, sign all lawful papers, execute all divisional, continuing and reissue applications, make all rightful oaths and generally do everything possible to aid said ASSIGNEE, its successors, legal representatives and assigns, to obtain and enforce proper patent protection for said improvements in all countries.

Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 2

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 19 day of Oct,
2010


Hyun Lee

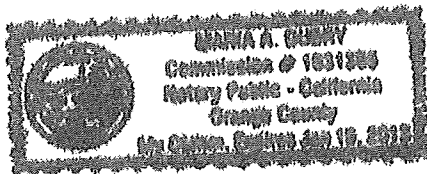
STATE OF CA }
COUNTY OF Orange } ss.

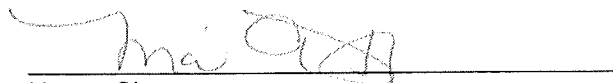
On Oct 19, 2010, before me, Maria A Dubry, notary public, personally appeared Hyun Lee who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]




Notary Signature

Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 3

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 19 day of Oct,
2010

Jayesh R. Bhakta
Jayesh R. Bhakta

STATE OF CA }
COUNTY OF Orange } ss.

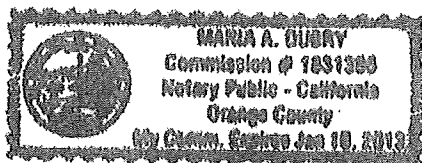
On Oct 19, 2010, before me, Maria A DuBry, notary public, personally appeared Jayesh R. Bhakta who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]

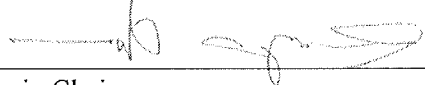
Maria A DuBry
Notary Signature



Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 4

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 19 day of Oct, 2010.



Soonju Choi

STATE OF CA }
COUNTY OF Orange } ss.

On Oct 19, 2010, before me, Maria A DuBry, notary public, personally appeared Soonju Choi who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

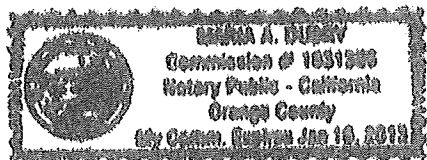
WITNESS my hand and official seal.

[SEAL]



Notary Signature

6992023/jlm



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POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	13/183253
	Filing Date	14-Jul-2011
	First Named Inventor	Lee, Hyun
	Title	MEMORY BOARD WITH SELF-TESTING CAPABILITY
	Art Unit	2117
	Examiner Name	CHUNG, PHUNG M
	Attorney Docket Number	NETL.046C1

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

79141

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

I am the:

☐ Applicant/Inventor.

OR

☒ Assignee of record of the entire interest. See 37 CFR 3.71.
Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____.

SIGNATURE of Applicant or Assignee of Record

Signature	Date	August 29, 2012
Name	Telephone	949-435-0025
Title and Company		

VP and CFO

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: **Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Acknowledgement Receipt

EFS ID:	13620647
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C1
Receipt Date:	29-AUG-2012
Filing Date:	14-JUL-2011
Time Stamp:	18:23:50
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		RspstoOAmailed5292012.pdf	117108 fea66d1c7b66a83767d8e495ac6909a34bc73ec6	yes	7

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Amendment/Req. Reconsideration-After Non-Final Reject	1	1
	Specification	2	2
	Claims	3	5
	Applicant Arguments/Remarks Made in an Amendment	6	7

Warnings:

Information:

2	Terminal Disclaimer Filed	TerminalDisclaimer.pdf	661536	no	6
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Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

3	Power of Attorney	PowerofAttorney.pdf	179496	no	1
			27fc36b9e68b987770bbdbedbd7c1c8032a9621		

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Information:

Total Files Size (in bytes):		958140
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 13/183,253		Filing Date 07/14/2011		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL				
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	08/29/2012	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	=	0		X \$60=	0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	=	0		X \$250=	0	
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE	0	
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=		X \$	=		
	Independent (37 CFR 1.16(h))	*	Minus	***	=		X \$	=		
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
					TOTAL ADD'L FEE			TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/BURNELL ROSS/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Document code: WFEE

United States Patent and Trademark Office
Sales Receipt for Accounting Date: 09/10/2012

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UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/183,253	07/14/2011	Hyun Lee	NETL.046C1

CONFIRMATION NO. 7935

MISCELLANEOUS NOTICE



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 09/06/2012

A communication which cannot be delivered in electronic form has been mailed to the applicant.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
 United States Patent and Trademark Office
 Address: COMMISSIONER FOR PATENTS
 P.O. Box 1450
 Alexandria, Virginia 22313-1450
 www.uspto.gov

APPLICATION NUMBER	FILING DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/183,253	07/14/2011	Hyun Lee	NETL.046C1

CONFIRMATION NO. 7935

20995
 KNOBBE MARTENS OLSON & BEAR LLP
 2040 MAIN STREET
 FOURTEENTH FLOOR
 IRVINE, CA 92614



OC000000056369925

Cc: THE LAW OFFICE OF JAMIE ZHENG, PH.D ESQ.
 P.O. BOX 60573
 PALO ALTO, CA 94306

Date Mailed: 09/06/12

DENIAL OF REQUEST FOR POWER OF ATTORNEY

The request for Power of Attorney filed 08/29/12 is acknowledged. However, the request cannot be granted at this time for the reason stated below.

- ☐ The Power of Attorney you provided did not comply with the new Power of Attorney rules that became effective on June 25, 2004. See 37 CFR 1.32.
- ☐ The revocation is not signed by the applicant, the assignee of the entire interest, or one particular principal attorney having the authority to revoke.
- ☒ The Power of Attorney is from an assignee and the Certificate required by 37 CFR 3.73(b) has not been received.
- ☐ The person signing for the assignee has omitted their empowerment to sign on behalf of the assignee.
- ☐ The inventor(s) is without authority to appoint attorneys since the assignee has intervened as provided by 37 CFR 3.71.
- ☐ The signature(s) of _____, a co-inventor in this application, has been omitted. The Power of Attorney will be entered upon receipt of confirmation signed by said co-inventor(s).
- ☐ The person(s) appointed in the Power of Attorney is not registered to practice before the U.S. Patent and Trademark Office.

Questions relating to this Notice should be directed to the Application Assistance Unit.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	13/183253
	Filing Date	14-Jul-2011
	First Named Inventor	Lee, Hyun
	Title	MEMORY BOARD WITH SELF-TESTING CAPABILITY
	Art Unit	2117
	Examiner Name	CHUNG, PHUNG M
	Attorney Docket Number	NETL.046C1

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

79141

OR

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Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

I am the:

☐ Applicant/Inventor.

OR

☒ Assignee of record of the entire interest. See 37 CFR 3.71.
Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____.

SIGNATURE of Applicant or Assignee of Record

Signature	Date	August 29, 2012
Name	Telephone	949-435-0025
Title and Company		

VP and CFO

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

☐ *Total of _____ forms are submitted.

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If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 13/183,253Filed/Issue Date: 14-Jul-2011

Titled:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

Netlist, Inc. _____, a corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 025230, Frame 0797, or for which a copy therefore is attached.

OR

- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

- ☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (*i.e.*, a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

Signature

Gail Sasaki

Printed or Typed Name

August 29, 2012

Date

VP and CFO

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

 *** TX REPORT ***

TRANSMISSION OK

TX/RX NO 1158
 CONNECTION TEL 915712730140
 SUBADDRESS
 CONNECTION ID
 ST. TIME 11/01 11:31
 USAGE T 01'36
 PGS. SENT 5
 RESULT OK

Client Code: NETL.046A

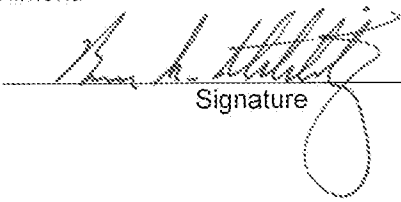
RECORDATION FORM COVER SHEET PATENTS ONLY

To the Director, U.S. Patent and Trademark Office: Please record the attached original documents or copy thereof.

1. Name of conveying party(ies): (List using letters or numbers for multiple parties) 1) Hyun Lee 2) Jayesh R. Bhakta 3) Soonju Choi Additional name(s) of conveying party(ies) attached? () Yes (X) No	2. Name and address of receiving party(ies): Name: Netlist, Inc. Street Address: 51 Discovery #150 City: Irvine State: CA ZIP: 92618 Additional name(s) of receiving party(ies) attached? () Yes (X) No
3. Nature of conveyance: (X) Assignment () Security Agreement () Merger () Change of Name () Other: Execution Date: (List as in section 1 if multiple signatures) 1) October 19, 2010 2) October 19, 2010 3) October 19, 2010	4. US or PCT Application number(s) or US Patent number(s): (X) Patent Application No.: 12/422,925 Filing Date: April 13, 2009 Additional numbers attached? () Yes (X) No
5. Party to whom correspondence concerning document should be mailed: Customer No. 20,995 Address: Knobbe, Martens, Olson & Bear, LLP 2040 Main Street, 14th Floor Irvine, CA 92614 Return Fax: (949) 760-9502 Attorney's Docket No.: NETL.046A	6. Total number of applications and patents involved: 1
7. Total fee (37 CFR 1.21(h)): \$40 (X) Authorized to be charged to deposit account	8. Deposit account number: 11-1410 Please charge this account for any additional fees which may be required, or credit any overpayment to this account.
9. Statement and signature. To the best of my knowledge and belief, the foregoing information is true and correct, and any attached copy is a true copy of the original document.	

RECORDATION FORM COVER SHEET **PATENTS ONLY**

To the Director, U.S. Patent and Trademark Office: Please record the attached original documents or copy thereof.

1. Name of conveying party(ies): (List using letters or numbers for multiple parties) 1) Hyun Lee 2) Jayesh R. Bhakta 3) Soonju Choi Additional name(s) of conveying party(ies) attached? () Yes (X) No	2. Name and address of receiving party(ies): Name: Netlist, Inc. Street Address: 51 Discovery #150 City: Irvine State: CA ZIP: 92618 Additional name(s) of receiving party(ies) attached? () Yes (X) No
3. Nature of conveyance: (X) Assignment () Security Agreement () Merger () Change of Name () Other: Execution Date: (List as in section 1 if multiple signatures) 1) October 19, 2010 2) October 19, 2010 3) October 19, 2010	4. US or PCT Application number(s) or US Patent number(s): (X) Patent Application No.: 12/422,925 Filing Date: April 13, 2009 Additional numbers attached? () Yes (X) No
5. Party to whom correspondence concerning document should be mailed: Customer No. 20,995 Address: Knobbe, Martens, Olson & Bear, LLP 2040 Main Street, 14th Floor Irvine, CA 92614 Return Fax: (949) 760-9502 Attorney's Docket No.: NETL.046A	6. Total number of applications and patents involved: 1
7. Total fee (37 CFR 1.21(h)): \$40 (X) Authorized to be charged to deposit account	8. Deposit account number: 11-1410 Please charge this account for any additional fees which may be required, or credit any overpayment to this account.
9. Statement and signature. To the best of my knowledge and belief, the foregoing information is true and correct, and any attached copy is a true copy of the original document. <u>Bruce S. Itchkawitz</u> Name of Person Signing <u>47,677</u> Registration No.  Signature <u>11/1/10</u> Date Total number of pages including cover sheet, attachments and document: 5	

Documents transmitted via Facsimile to be recorded with required cover sheet information to:

Mail Stop Assignment Recordation Services
Director, U.S. Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450
Facsimile Number: (571) 273-0140

Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 1

ASSIGNMENT

WHEREAS, We, (1) Hyun Lee, a United States citizen, residing at 21 Thalia, Ladera Ranch CA 92694, (2) Jayesh R. Bhakta, a United States citizen, residing at 12220 Rose Street, Cerritos CA 90703, and (3) Soonju Choi, a Korean citizen, residing at 195 Woodbury, Irvine CA 92620, have invented certain new and useful improvements in a MEMORY BOARD WITH SELF-TESTING CAPABILITY, for which we have filed an application for Letters Patent in the United States, Application No. 12/422,925, filed on April 13, 2009.

AND WHEREAS, Netlist, Inc., a Delaware Corporation, with its principal place of business at 51 Discovery # 150, Irvine CA 92618, (hereinafter "ASSIGNEE") desires to acquire the entire right, title, and interest in and to said improvements and said Application:

NOW, THEREFORE, in consideration of good and valuable consideration, the receipt of which is hereby acknowledged, we, the said inventors, do hereby acknowledge that we have sold, assigned, transferred and set over, and by these presents do hereby sell, assign, transfer and set over, unto said ASSIGNEE, its successors, legal representatives and assigns, the entire right, title, and interest throughout the world in, and its territories and possessions in, to and under said improvements, and any Patent Applications in the United States of America and all divisions, renewals and continuations thereof, and all Patents of the United States of America which may be granted thereon and all reissues and extensions thereof, and all rights of priority under International Conventions; and we hereby authorize and request the Commissioner of Patents of the United States of America to issue all Patents for said improvements to ASSIGNEE, its successors, legal representatives and assigns, in accordance with the terms of this instrument.

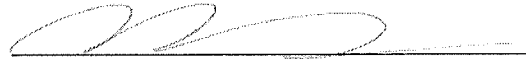
AND WE DO HEREBY sell, assign, transfer, and convey to ASSIGNEE, its successors, legal representatives, and assigns all claims for damages and all remedies arising out of any violation of the rights assigned hereby that may have accrued prior to the date of assignment to ASSIGNEE, or may accrue hereafter, including, but not limited to, the right to sue for, collect, and retain damages for past infringements of said Letters Patent before or after issuance.

AND WE HEREBY covenant and agree that we will communicate to said ASSIGNEE, its successors, legal representatives and assigns, any facts known to us respecting said improvements, and testify in any legal proceeding, sign all lawful papers, execute all divisional, continuing and reissue applications, make all rightful oaths and generally do everything possible to aid said ASSIGNEE, its successors, legal representatives and assigns, to obtain and enforce proper patent protection for said improvements in all countries.

Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 2

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 19 day of Oct,
2010


Hyun Lee

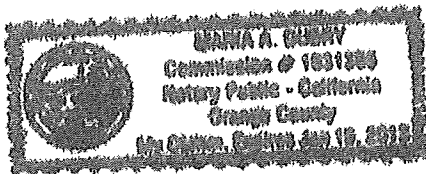
STATE OF CA }
COUNTY OF Orange } ss.

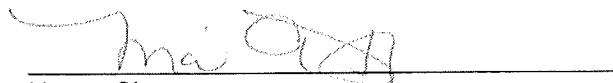
On Oct 19, 2010, before me, Maria A Dubry, notary public, personally appeared Hyun Lee who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]




Notary Signature

Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 3

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 19 day of Oct,
2010

Jayesh R. Bhakta
Jayesh R. Bhakta

STATE OF CA }
COUNTY OF Orange } ss.

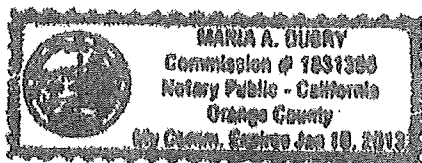
On Oct 19, 2010, before me, Maria A DuBry, notary public, personally appeared Jayesh R. Bhakta who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

WITNESS my hand and official seal.

[SEAL]

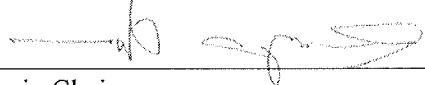
Maria A DuBry
Notary Signature



Application No.: 12/422,925
Filing Date: April 13, 2009

PATENT
Client Code: NETL.046A
Page 4

IN TESTIMONY WHEREOF, I hereunto set my hand and seal this 19 day of Oct, 2010.



Soonju Choi

STATE OF CA }
COUNTY OF Orange } ss.

On Oct 19, 2010, before me, Maria A DuBry, notary public, personally appeared Soonju Choi who proved to me on the basis of satisfactory evidence to be the person(s) whose name(s) is/are subscribed to the within instrument, and acknowledged to me that he executed the same in his authorized capacity(ies), and that by his signature(s) on the instrument the person(s), or the entity upon behalf of which the person(s) acted, executed the instrument.

I certify under PENALTY OF PERJURY under the laws of the State of California that the foregoing paragraph is true and correct.

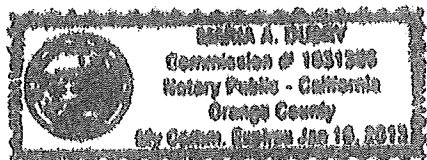
WITNESS my hand and official seal.

[SEAL]



Notary Signature

6992023/jlm



Electronic Acknowledgement Receipt

EFS ID:	13691088
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C1
Receipt Date:	07-SEP-2012
Filing Date:	14-JUL-2011
Time Stamp:	18:48:36
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	PowerofAttorney.pdf	1303601 ebeeaa31551bdf5a3ec76139883e98b41c1bf9e36	no	8

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):

1303601

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Application Number 	Application/Control No. 13/183,253	Applicant(s)/Patent under Reexamination LEE ET AL.	
Document Code - DISQ		Internal Document – DO NOT MAIL	

TERMINAL DISCLAIMER	☒ APPROVED	☐ DISAPPROVED
Date Filed : August 29, 2012	This patent is subject to a Terminal Disclaimer	

Approved/Disapproved by:

Henry D. Jefferson



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/183,253	07/14/2011	Hyun Lee	NETL.046C1

79141
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

CONFIRMATION NO. 7935
POA ACCEPTANCE LETTER



Date Mailed: 09/13/2012

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 09/07/2012.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/hchristian/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 09/20/2012
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

CHUNG, PHUNG M

ART UNIT

PAPER NUMBER

2117

DATE MAILED: 09/20/2012

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/183,253

07/14/2011

Hyun Lee

NETL.046C1

7935

TITLE OF INVENTION: MEMORY BOARD WITH SELF-TESTING CAPABILITY

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$0	\$0	\$1740	12/20/2012

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 09/20/2012
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
-----------------	-------------	----------------------	---------------------	------------------

13/183,253 07/14/2011 Hyun Lee NETL.046C1 7935

TITLE OF INVENTION: MEMORY BOARD WITH SELF-TESTING CAPABILITY

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional NO \$1740 \$0 \$0 \$1740 12/20/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
----------	----------	----------------

CHUNG, PHUNG M 2117 714-719000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/183,253	07/14/2011	Hyun Lee	NETL.046C1	7935
79141	7590	09/20/2012	EXAMINER	
The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306			CHUNG, PHUNG M	
			ART UNIT	PAPER NUMBER
			2117	
DATE MAILED: 09/20/2012				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 0 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 0 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	13/183,253	LEE ET AL.	
	Examiner	Art Unit	
	/Phung M. Chung/	2117	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment dated on 8/29/12.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 2-21, which are now as 1-20.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
 6. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
7. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. ☒ Notice of References Cited (PTO-892) 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) 3. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____ 4. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | <ol style="list-style-type: none"> 5. ☐ Notice of Informal Patent Application 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. 7. ☐ Examiner's Amendment/Comment 8. ☒ Examiner's Statement of Reasons for Allowance 9. ☐ Other ____. |
|---|--|

Allowable Subject Matter

The following is an examiner's statement of reasons for allowance: Claims 1-20 are allowable over the art of record. This is because the art of record does not teach or disclose the memory system and method as recited in claims 1 and 16, and including, for example:

Claim 1, including: Control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

Claim 16, including: testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

//Phung M. Chung//

Primary Examiner, Art Unit 2117

Notice of References Cited	Application/Control No. 13/183,253	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner /Phung M. Chung/	Art Unit 2117	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-6,910,162	06-2005	Co et al.	714/718
	B	US-			
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L23	0	(memory controller AND memory system AND computer system AND memory chips AND data handlers AND control circuit AND address AND signal AND data AND memory test).clm.	USPAT; UPAD	ADJ	ON	2012/09/13 11:48

9/ 13/ 2012 11:49:18 AM
C:\ Users\ pchung\ Documents\ EAST\ Workspaces\ 13183253.wsp

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	111938	memory system	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 10:55
L2	18	generat\$3 adj data same data handler\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 10:57
L3	5688	(control circuit or controller) same generat\$3 adj address and control signal\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 10:58
L5	1	I3 and I2	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 10:59
L6	1000	I1 and (I2 or I3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 10:59
L7	7	test\$3 same (memory location\$1 same memory chip\$1) same (address and control signal\$1 and data)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:00
L9	1	I7 and (I2 or I3 or I1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:00
L10	8	verify\$3 and ((data received same	US-PGPUB;	ADJ	ON	2012/09/13

		memory chip\$1) same (data transmitted))	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			11:15
L13	2307	calculat\$3 same comparison data	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:16
L14	3	compar\$3 same comparison data same data received same memory chip\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:16
L15	522775	comparator	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:26
L16	224669	test circuit or tester	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:26
L17	13791	l15 and l16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:26
L18	109	l17 and l6	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:26
L19	42296	data generator or pattern generator	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:27
L20	85	l18 and l19	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT;	ADJ	ON	2012/09/13 11:27

			IBM_TDB			
L21	13547	(714/718,719,736,738,819;365/201).cds.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:27
L22	6	I20 and I21	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2012/09/13 11:27

9/ 13/ 2012 11:29:24 AM
C:\ Users\ pchung\ Documents\ EAST\ Workspaces\ 13183253.wsp

Search Notes 	Application/Control No. 13183253	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner PHUNG M CHUNG	Art Unit 2117

SEARCHED			
Class	Subclass	Date	Examiner
714	718,719,736,738,819	9/13/12	pmc
365	201	9/13/12	pmc

SEARCH NOTES		
Search Notes	Date	Examiner
EAST searched.	9/13/12	pmc

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
714	718,719	9/13/12	pmc

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Electronic Patent Application Fee Transmittal

Application Number:	13183253			
Filing Date:	14-Jul-2011			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C1			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Statutory or terminal disclaimer	1814	1	160	160
Total in USD (\$)				160

Electronic Acknowledgement Receipt

EFS ID:	13869962
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C1
Receipt Date:	28-SEP-2012
Filing Date:	14-JUL-2011
Time Stamp:	16:24:06
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 160
RAM confirmation Number	7516
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Fee Worksheet (SB06)	fee-info.pdf	30185 608a4d7efb9025c3bc7d2c6bee91863df23 1ead5	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			30185		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Customer No. 79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/183,253	07/14/2011	Hyun Lee	NETL.046C1	7935
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TITLE OF INVENTION:

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional	NO	\$1740	\$0	\$0	\$1740	12/20/2012
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EXAMINER	ART UNIT	CLASS-SUBCLASS
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1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
- (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
- 3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☐ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☒ Payment by credit card. Form PTO-2038 is attached.
- ☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 50-5963 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☒ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /Jamie J. Zheng/

Date December 17, 2012

Typed or printed name Jamie J. Zheng

Registration No. 51,167

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13183253			
Filing Date:	14-Jul-2011			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.046C1			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	2501	1	885	885
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				885

Electronic Acknowledgement Receipt

EFS ID:	14486100
Application Number:	13183253
International Application Number:	
Confirmation Number:	7935
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046C1
Receipt Date:	17-DEC-2012
Filing Date:	14-JUL-2011
Time Stamp:	14:36:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$885
RAM confirmation Number	1234
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Issue Fee Payment (PTO-85B)	Issue_Fee_Payment.pdf	73510 9f8bc3bb49f5e9db5504bf4021312130a1 aede8	no	2
Warnings:					
Information:					
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Warnings:					
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/183,253	01/22/2013	8359501	NETL.046C1	7935

79141 7590 12/29/2012
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.

TO: Mail Stop 8 Director of the U.S. Patent & Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been
filed in the U.S. District Court NORTHERN CALIFORNIA on the following ☒ Patents or ☐ Trademarks:

DOCKET NO. CV 13-03901 DMR	DATE FILED 8/23/2013	U.S. DISTRICT COURT Northern District of California, 1301 Clay St. #400-S, Oakland, CA 94612
PLAINTIFF DIABLO TECHNOLOGIES INC		DEFENDANT NETLIST INC
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,001,434		***SEE ATTACHED COMPLAINT***
2 8,301,833		
3 8,359,501		
4 8,516,185		
5 8,516,187		

In the above—entitled case, the following patent(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK	
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK Richard W. Wieking	(BY) DEPUTY CLERK Clara Pierce	DATE August 27, 2013
---------------------------------	---------------------------------------	-----------------------------

Copy 1—Upon initiation of action, mail this copy to Commissioner Copy 3—Upon termination of action, mail this copy to Commissioner
Copy 2—Upon filing document adding patent(s), mail this copy to Commissioner Copy 4—Case file copy

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6 Facsimile: (650) 590-1900

7 Attorneys for Plaintiff
Diablo Technologies, Inc.
8

9 UNITED STATES DISTRICT COURT
10 NORTHERN DISTRICT OF CALIFORNIA
11 SAN FRANCISCO DIVISION

DMR

12 DIABLO TECHNOLOGIES, INC.,

13 Plaintiff,

14 v.

15 NETLIST, INC.,

16 Defendant.

CV Case No. 13 3901

COMPLAINT FOR DECLARATORY
JUDGMENT

DEMAND FOR JURY TRIAL

ORIGINAL
FILED

2013 AUG 23 A 9:23

RICHARD W. WILSON
CLERK, U.S. DISTRICT COURT
NORTHERN DISTRICT OF CALIFORNIA

1 For its Complaint against Defendant NETLIST, INC., ("NETLIST"), Plaintiff Diablo
2 Technologies, Inc. ("DIABLO") alleges as follows:

3 **FACTUAL BACKGROUND**

4 **DIABLO**

5 1. DIABLO is a corporation incorporated under the laws of Canada, with its
6 principal place of business at 80 Aberdeen Street, Suite 401, Ottawa, Ontario, K1S 5R5 Canada.

7 2. DIABLO, founded in 2003, delivers product solutions that enhance the
8 performance and capability of memory system designs. DIABLO developed the Memory
9 Channel Storage™ (MCST™) architecture for enterprise servers and storage systems.

10 3. DIABLO has been awarded "Most Innovative Flash Memory Enterprise Business
11 Application" in the Flash Memory Summit Best of Show Awards for enabling a new class of
12 ultra-low latency (ULL) SSDs and application accelerators. This innovative new architecture
13 radically transforms memory and storage for enterprise rack, blade and storage servers.

14
15 **NETLIST'S THREAT OF SUIT AGAINST DIABLO**

16 4. NETLIST purports to be incorporated under the laws of Delaware, with its
17 principal place of business at 51 Discovery, Suite 150, Irvine, California, 92618.

18 5. On August 22, 2013, NETLIST sent DIABLO a letter stating that "Netlist has
19 concluded that Smart Modular Technologies, Inc., Smart Storage Systems, Inc., Smart
20 Worldwide Holdings, Inc. and Diablo Technologies, Inc. (collectively "Defendants") are
21 directly and /or indirectly infringing the following United States patents at least through the
22 making, using, offering for sale and/or selling of the recently announced ULLtraDIMM product:

23 U.S. Patent No. 8,001,434, Memory Board With Self-Testing Capability

24 U.S. Patent No. 8,301,833, Non-Volatile Memory Module

25 U.S. Patent No. 8,359,501, Memory Board With Self-Testing Capability

26 U.S. Patent No. 8,516,185, System and Method Utilizing Distributed Byte-Wise Buffers
27 on a Memory Module

28 U.S. Patent No. 8,516,187, Data Transfer Scheme for Non-Volatile Memory Module"

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JURISDICTION AND VENUE

12 6. This Court has subject matter jurisdiction over this action pursuant to 28 U.S.C.
13 §§ 1331, 1338, 1367, 2201, and 2202.

14 7. This Court has personal jurisdiction over DIABLO. Among other things,
15 DIABLO transacts business in this jurisdiction.

16 8. This Court has personal jurisdiction over NETLIST. Among other things,
17 NETLIST transacts business in this jurisdiction and is a plaintiff in this District at least in the
18 pending action *Netlist, Inc. v. Google Inc.* 4-09-cv-05718 SBA (CAND).

19 9. Venue is proper in this District pursuant to 28 U.S.C. § 1391.

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INTRADISTRICT ASSIGNMENT

12 10. For purposes of intradistrict assignment pursuant to Local Rule 3-2(c), this
13 Intellectual Property Action is to be assigned on a district-wide basis.

COUNT I

(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,001,434)

17 11. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
18 the preceding paragraphs of this Complaint.

19 12. Upon information and belief, NETLIST is the owner of the '434 patent.

20 13. NETLIST is asserting the '434 patent against "ULLtraDIMM."

21 14. DIABLO, contends that it has not infringed and does not infringe, any valid and
22 enforceable claim of the '434 patent.

23 15. An actual and justiciable controversy has thus arisen between DIABLO and
24 NETLIST concerning the alleged infringement of the '434 patent.

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1 28. An actual and justiciable controversy has thus arisen between DIABLO and
2 NETLIST concerning the validity of the '833 patent.

3
4 **COUNT V**

5 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,359,501)**

6 29. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
7 the preceding paragraphs of this Complaint.

8 30. Upon information and belief, NETLIST is the owner of the '434 patent.

9 31. NETLIST is asserting the '501 patent against "ULLtraDIMM".

10 32. DIABLO contends that it has not infringed and does not infringe, any valid and
11 enforceable claim of the '501 patent.

12 33. An actual and justiciable controversy has thus arisen between DIABLO and
13 NETLIST concerning the alleged infringement of the '501 patent.

14
15 **COUNT VI**

16 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,359,501)**

17 34. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
18 the preceding paragraphs of this Complaint.

19 35. Upon information and belief, NETLIST contends that the '501 patent is valid.

20 36. DIABLO contends that the claims of the '501 patent are invalid under 35 U.S.C.
21 §§ 101, 102, 103, and/or 112.

22 37. An actual and justiciable controversy has thus arisen between DIABLO and
23 NETLIST concerning the validity of the '501 patent.

24
25 **COUNT VII**

26 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,516,185)**

27 38. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
28 the preceding paragraphs of this Complaint.

1 39. Upon information and belief, NETLIST is the owner of the '185 patent.

2 40. NETLIST is asserting the '185 patent against "ULLtraDIMM".

3 41. DIABLO, contends that it has not infringed and does not infringe, any valid and
4 enforceable claim of the '185 patent.

5 42. An actual and justiciable controversy has thus arisen between DIABLO and
6 NETLIST concerning the alleged infringement of the '185 patent.

7

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COUNT VIII

9 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,516,185)**

10 43. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
11 the preceding paragraphs of this Complaint.

12 44. Upon information and belief, NETLIST contends that the '185 patent is valid.

13 45. DIABLO contends that the claims of the '185 patent are invalid under 35 U.S.C.
14 §§ 101, 102, 103, and/or 112.

15 46. An actual and justiciable controversy has thus arisen between DIABLO and
16 NETLIST concerning the validity of the '185 patent.

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COUNT IX

19 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,516,187)**

20 47. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
21 the preceding paragraphs of this Complaint.

22 48. Upon information and belief, NETLIST is the owner of the '187 patent.

23 49. NETLIST is asserting the '187 patent against "ULLtraDIMM".

24 50. DIABLO, contends that it has not infringed and does not infringe, any valid and
25 enforceable claim of the '187 patent.

26 51. An actual and justiciable controversy has thus arisen between DIABLO and
27 NETLIST concerning the alleged infringement of the '187 patent.

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1 Respectfully submitted this 23rd day of August 2013.

2 DIABLO TECHNOLOGIES, INC.,

3 By: Sanjeet Dutta

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13 Telephone: (650) 590-0700

14 Facsimile: (650) 590-1900

15 Attorneys for Plaintiff

16 DIABLO TECHNOLOGIES, INC.

TO: Mail Stop 8
 Director of the U.S. Patent & Trademark Office
 P.O. Box 1450
 Alexandria, VA 22313-1450

**REPORT ON THE
 FILING OR DETERMINATION OF AN
 ACTION REGARDING A PATENT OR
 TRADEMARK**

In Compliance with 35 § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court Northern District of California on the following:
 (X) Patents or () Trademarks

DOCKET NO:
 13-cv-05889-YGR

DATE FILED:
 October 7, 2014

UNITED STATES DISTRICT COURT
 Ronald Dellums Federal Building
 1301 Clay Street
 Oakland, CA 94612

PLAINTIFF:
 Netlist, Inc

DEFENDANT:
 Smart Modular Technologies, Inc

PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1. <u>8,001,434</u>		See attached THIRD AMENDED COMPLAINT FOR PATENT INFRINGEMENT
2. <u>8,301,833</u>		
3. <u>8,359,501</u>		
4. <u>8,516,185</u>		
5. <u>8,516,187</u>		

In the above-entitled case, the following patent(s) have been included.

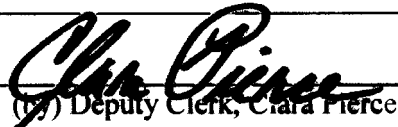
DATE INCLUDED INCLUDED BY:
 () Amendment () Answer () Cross Bill () Other Pleading

PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1. <u>7,881,150</u>		
2. <u>8,081,536</u>		
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In the above-entitled case, the following decision has been rendered or judgment issued:

DECISION/JUDGEMENT:

Richard W. Wieking, Clerk


 (by) Deputy Clerk, Clara Pierce

Copy 1 – Upon initiation of action, mail this copy to Commissioner
 Copy 2 – Upon filing document adding patent(s) mail this copy to Commissioner
 Copy 3 – Upon termination of action, mail this copy to the Commissioner
 Copy 4 – Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SANDISK CORPORATION,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-00971
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and GEORGIANNA
W. BRADEN, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

DECISION
Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

On June 18, 2014, Sandisk Corporation (“Petitioner”) filed a Petition (Paper 1, “Pet.”) to institute an *inter partes* review of claims 1–20 (the “challenged claims”) of U.S. Patent No. 8,359,501 B1 (Ex. 1001, “the ’501 patent”). 35 U.S.C. §§ 311–319. Netlist, Inc. (“Patent Owner”) timely filed a Preliminary Response (Paper 10, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may not be instituted “unless . . . there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” Upon consideration of the Petition and Patent Owner’s Preliminary Response, we determine Petitioner has established a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. We therefore institute an *inter partes* review as to: (1) claims 1–3, 5–8, and 11–20 on the ground that these claims are anticipated by Averbuj; (2) claims 1–5 and 14–16 on the ground that these claims would have been obvious over Huang; (3) claims 6–8, 11, and 17–20 on the ground that these claims would have been obvious over Huang and Averbuj; and (4) claims 9 and 10 on the ground that these claims would have been obvious over Huang, Averbuj, and JEDEC Standard. Our factual findings and conclusions at this stage of the proceeding are based on the evidentiary record developed thus far (prior to Patent Owner’s Response). This is not a final decision as to patentability of claims for which *inter partes* review is instituted. Our final decision will be based on the record as fully developed during trial.

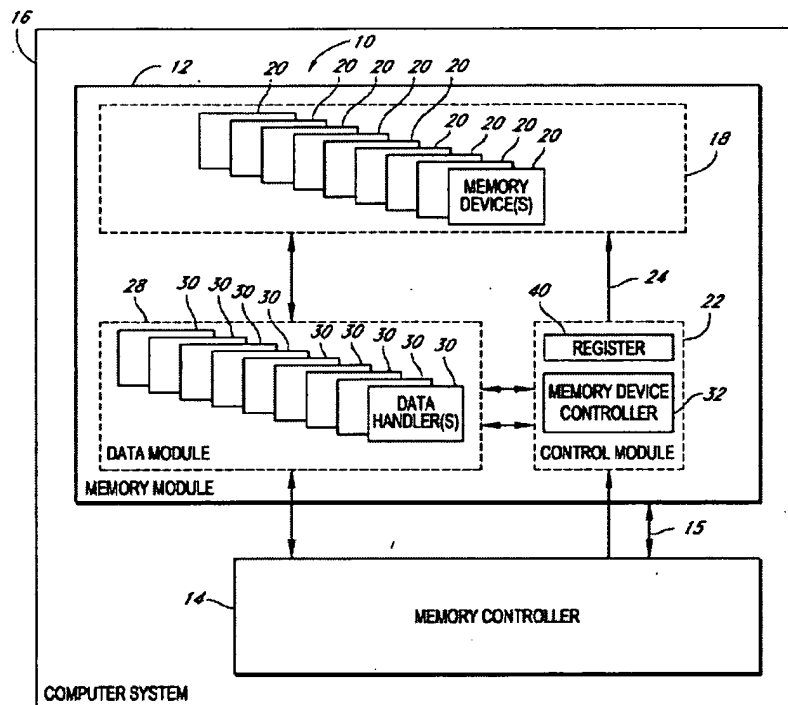
II. BACKGROUND

A. Related Matters

The '501 patent is also the subject of two infringement actions filed by Patent Owner, one of which names Petitioner as a defendant: *Netlist, Inc. v. Smart Modular Technologies, Inc.*, No. 4:13-cv-05889-YGR (Dec. 19, 2013, N.D. Cal.). SMART Modular Technologies, Inc. is a named defendant in both infringement actions, and filed two petitions for *inter partes* review of the '501 patent on August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01374 and IPR2014-01375 (PTAB). The '501 patent claims priority as a continuation of U.S. Patent No. 8,001,434 ("the '434 patent"). Petitioner has filed a petition for review of the '434 patent: *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (June 18, 2014, PTAB). On August 23, 2014, SMART Modular Technologies, Inc. also filed two petitions for *inter partes* review of the '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01372 and IPR2014-01373 (PTAB).

B. The '501 Patent (Ex. 1001)

The '501 patent relates to self-testing electronic memory modules. Ex. 1001, 1:28–29. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '501 patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10 includes a plurality of memory devices 18, e.g., ETT DRAM chips, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 3:34–36; 5:10–12.

Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:14–16, 42–46. “In various embodiments, the control module 22 includes . . . one or more application-specific integrated circuit[s]” *Id.* at 9:62–64. “[D]ata module 28 and/or [] control module 22 of certain embodiments are configured to test the plurality of memory devices

18 at the normal operating speed of [] memory devices 20.” *Id.* at 6:5–8.

“For example, [] data module 28 and/or [] control module 22 are configured to provide memory signals (e.g., data, address and control signals) according [to] the operating specification of [] memory devices 20.” *Id.* at 6:8–11.

“[E]ach of [] data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports” of the plurality of memory devices 18. *Id.* at 5:21–23. “In some embodiments, during testing, [] control module 22 generates address and control signals 24 associated with memory locations to be tested and [] data module 28 generates corresponding test data patterns and provides them to [] appropriate memory devices 20.” *Id.* at 5:48–52. “For example, [] data module 28 may receive a write command from [] control module 22 and provide data to be written to certain locations in [] memory devices 20 during a write operation.” *Id.* at 5:52–55.

C. Illustrative Claims

Of the challenged claims, claims 1 and 16 are independent, and are reproduced below:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - a control circuit configured to generate address and control signals, wherein the memory system is configured to test

the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

16. A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:

operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;

operating a control circuit to generate address and control signals; and

testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

D. Evidence Relied Upon

Petitioner's patentability challenges are based on the following references:

References	Patents/Printed Publications	Exhibit
Averbuj	U.S. Patent Application Publication No. 2005/0257109	1003
Huang	<i>An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers</i> , VLSI Design 2001, Fourteenth International Conference on VLSI Design (pp. 379–384), IEEE (2001)	1004
Tsern	U.S. Patent Application Publication No. 2007/0070669	1005

JEDEC Standard	JEDEC Standard, Fully Buffered DIMM (FBDIMM): DFX Design for Validation and Test, JESD82-28, (February 2008)	1006
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E. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of the '501 patent claims based on the following grounds:

References	Basis	Claims Challenged
Averbuj	§ 102(b)	1-3, 5-8, and 11-20
Averbuj and Tsern	§ 103(a)	1-8 and 11-20
Averbuj, Tsern, and JEDEC Standard	§ 103(a)	9 and 10
Huang and Tsern	§ 103(a)	1-5 and 14-16
Huang, Tsern, and Averbuj	§ 103(a)	6-8, 11-13, and 17-20
Huang, Tsern, Averbuj, and JEDEC Standard	§ 103(a)	9 and 10

In addition to these references, Petitioner relies on the Declaration of Donald Alpert, Ph.D. (Ex. 1007) under 37 C.F.R. § 1.68. Pet. 7.

III. CLAIM CONSTRUCTION

Claims of an unexpired patent are interpreted using the broadest reasonable construction in light of the specification of the patent. *See* Office Patent Trial Practice Guide, 77 Fed. Reg. 48756, 48766 (Aug. 14, 2012); 37 C.F.R. § 42.100(b). “Petitioner submits that, for the purposes of this review, each claim be construed in accordance with its plain and ordinary meaning under the required broadest reasonable interpretation.” Pet. 6-7. Petitioner

does not specifically state the broadest reasonable construction of any particular claim terms or phrases. *See id.*

Patent Owner contends the term “generate,” recited in each of independent claims 1 and 16, should be interpreted as meaning “produce.” Prelim. Resp. 24–25. In support of this construction, Patent Owner quotes column 6, lines 11–14 of the ’501 patent, which read: “In some embodiments, the control module 22 and the data module 28 *produce* memory addresses, control and/or data signals according to the JEDEC standard memory protocol” (emphasis added). Prelim. Resp. 24. Patent Owner also relies on two dictionary definitions which define “generate” as “produce.” *Id.* at 25 (citing Exs. 2002, 2003). Patent Owner argues the term “generate” should not be interpreted as meaning “to be the cause of.” *Id.*

As an initial matter, we note the description in the ’501 patent relied upon by Patent Owner relates to only “some embodiments,” and the claims do not recite that the data and control modules are configured to generate signals “according to the JEDEC standard memory protocol.” The specification states that the data module 28/data handlers 30 may be in communication with the control module 22 (Ex. 1001, 10:1–3), and that data signals and/or patterns of data signals generated from data generation element 54, present in certain embodiments of data handlers 30, “may be generated in a variety of ways” (*id.* at 10:53–54), including by programming from control module 22 (*id.* at 11:4–6). The specification also states that “data module 28 and/or [] *control module 22 are configured to provide* memory signals (e.g., *data*, address and control *signals*) according [to] the operating specification of [] memory devices 20.” *Id.* at 6:8–11 (emphasis added). Given such broad disclosure, Patent Owner has not convinced us

that the '501 patent specification supports its proposed narrow claim construction of “generate” as requiring, for example, that data “generated by” data handlers (*see* claims 1 and 16) is also “produced by” the data handlers.

Moreover, although both dictionaries cited by Patent Owner list “produce” as one definition of “generate” (Ex. 2002, 484; Ex. 2003, 230), Merriam Webster’s Collegiate Dictionary also defines generate as “to be the cause of” (Ex. 2002, 484). Patent Owner has not explained sufficiently why this latter definition of “generate” is inconsistent with the use of the term “generate” in the '501 patent specification and claims. *See* Prelim. Resp. 25 (“The term ‘generate’ should not be construed to mean, e.g., ‘to be the cause of.’”) (citing Ex. 2002).

Accordingly, for purposes of this decision, we interpret the claim term “generate” as meaning “produce” or “cause.” Thus, for example, as we interpret the claim term “generate,” “one or more data handlers . . . configured to generate data for writing to a corresponding one or more memory chips” as recited in claim 1, encompasses a data handler which writes programmable data patterns received from a control circuit. *See* Ex. 1001, 10:53–54; 11:4–6, discussed above.

IV. ANALYSIS

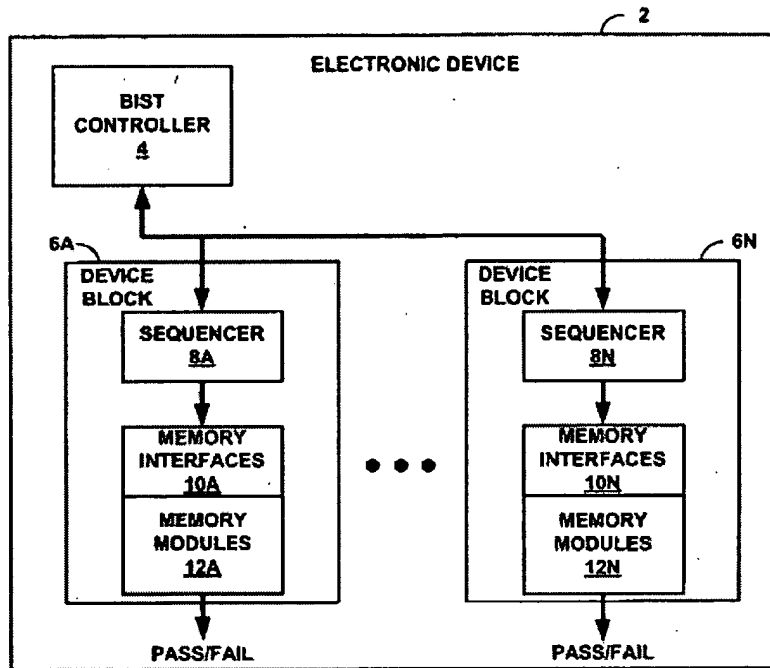
A. Anticipation of Claims 1–3, 5–8, and 11–20 by Averbuj

1. Averbuj

Averbuj (Ex. 1003)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the

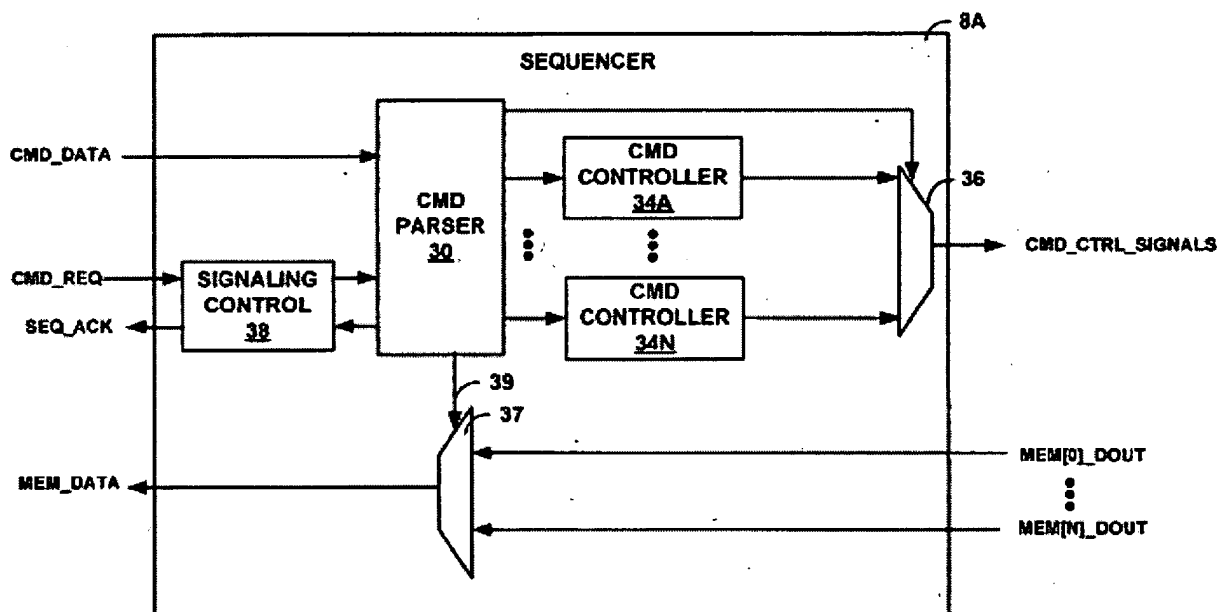
testing of one or more memory modules. Ex. 1003 ¶¶ 7–8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* ¶ 28. “[E]lectronic device 2 may be any device that incorporates memory modules, such as an embedded computing system; a computer, server, personal digital assistant (PDA), mobile computing device, mobile communication device, digital recording device, network appliance, mobile positioning device, and the like.” *Id.* ¶ 32. Averbuj states that “electronic devices . . . are [normally] constructed from many integrated circuit chips and many supporting components mounted on a circuit board.” *Id.* ¶ 5. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A

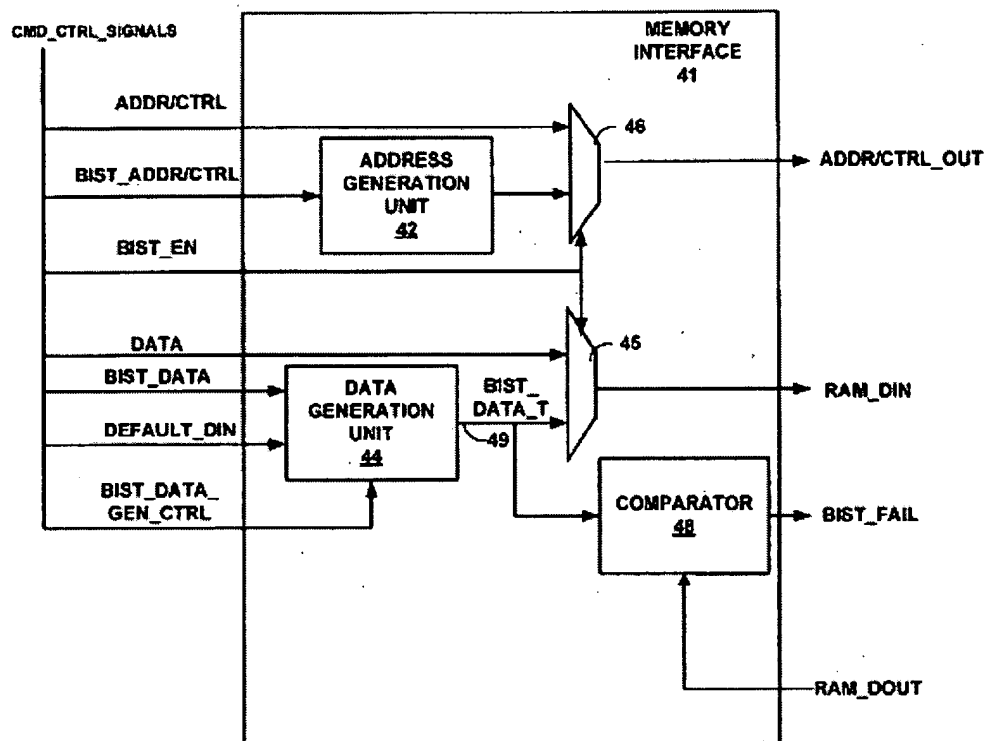
through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to sequencers 8 for application to device blocks 6. *Id.* ¶ 29.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* ¶ 22.



As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 and command controllers 34. *Id.* ¶¶ 41–42. Command parser 30 processes the commands from BIST controller 4 and invokes one of command controllers 34 which, in turn, issues command control signals, such as signals to provide a memory address and data, to receiving memory interfaces 10, 41. *Id.* ¶¶ 43, 51.

An exemplary memory interface is shown in Figure 6, below. *Id.* ¶ 23.



As shown in the Figure 6 embodiment, above, memory interface 41 includes a layer of multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals (BIST_ADDR/CTRL) and the test data provided by a respective higher level sequencer[, e.g., sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level sequencer [8A], and transforms the data and addresses provided by the sequencer as needed based on the particular

physical characteristics of the memory module” (*id.* ¶ 54) using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “[A]ddress generation unit 42 generates the address applied to [] memory module 12 based on the addressing requirements specified by sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* ¶ 51. “[D]ata generation unit 44 [] receives BIST_DATA signals and default data (DEFAULT_DIN) as provided by [] sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the sequencer and the specific physical characteristics of the corresponding memory module.” *Id.* ¶ 49. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.*

2. Analysis

Petitioner contends Averbuj describes “a memory system configured to be operatively coupled to a memory controller of a computer system,” as recited in the preambles of claims 1 and 16, because Averbuj states that under normal operating conditions, address/control and data signals are provided to memory modules 12 by a programmable processor. Pet. 8 (citing Ex. 1003 ¶ 48). Petitioner contends Averbuj’s sequencer 8A, memory interfaces 10, and memory modules 12 correspond, respectively, to the claimed “control circuit,” “data handlers,” and memory chips. *See* Pet. 14–17.

Patent Owner argues Averbuj fails to describe both a control circuit configured to generate address and control signals and data handlers configured to generate data as recited in each of independent claims 1 and 16. *See* Prelim. Resp. 28–30. Patent Owner argues one of ordinary skill in

the art would understand that either (1) Averbuj's sequencer 8A (the control circuit) generates the address, control, and data signals, or (2) Averbuj's sequencer 8A (the control circuit) generates the control signals and memory interfaces 10 (the data handlers) generate the address and data signals. *Id.* at 29–30.

Patent Owner's argument is not convincing in light of our interpretation (for purposes of this decision) of "generate" as meaning "produce" or "cause." *See* Section III, above. Averbuj describes sequencer 8A as including command controllers 34 which issue a sequence of one or more operations and sequentially drive the appropriate command control signals, such as signals to provide a memory address and data, to receiving memory interfaces 10. Pet. 12 (citing Ex. 1003, ¶ 43). This description supports Petitioner's contention that Averbuj describes a control circuit configured to generate/cause data. Averbuj describes an exemplary memory interface 41 as including address generation unit 42 and data generation unit 44 which transform the data and addresses provided by the sequencer and "generate" the address and data applied to the memory modules 12. Pet. 10–11 (citing Ex. 1003, ¶¶ 49, 51). The foregoing disclosure reasonably supports Petitioner's contention that Averbuj's memory interfaces 10 are data handlers configured to generate data. Patent Owner has not convinced us that because Averbuj's memory interfaces 10 may also generate address signals, they are not properly viewed as meeting the claim limitation of data handlers. *See* Prelim. Resp. 29. Although the independent claims require data handlers configured to either produce or cause data signals, Patent Owner has not explained why the claims should be narrowly construed as precluding the data handlers from being configured to also produce or cause

address signals.

On this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner identifies where Averbuj describes every element of the challenged claims. *See* Pet. 7–25. Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 1–3, 5–8, and 11–20 are anticipated by Averbuj.

B. Obviousness of Claims 1–8 and 1–20 over Averbuj and Tsern

As discussed in Section IV.A.2, above, Petitioner contends Averbuj anticipates claims 1–3, 5–8, and 11–20. Petitioner contends, in the alternative, that if this Board Panel determines Averbuj does not describe certain limitations recited in claims 1, 2, and 16, these limitations are described in Tsern (Ex. 1005) and

[t]he [person of ordinary skill] in the memory arts would [have been] motivated to combine Averbuj and Tsern for many reasons, such as to supplement the teachings of Averbuj relating to an efficient memory BIST architecture with Tsern’s teachings of efficient ways to incorporate a self-testing memory system within multiple integrated circuits on a printed circuit board.

Pet. 26–28. Petitioner also relies on Tsern for a teaching of the claim 4 limitation “wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” *Id.* at 29–30. Petitioner contends dependent claims 3, 5–8, 11–15, and 17–20 would have been obvious over Averbuj in view of Tsern. *Id.* at 28, 30. In other words, Petitioner has not asserted that claims 1–8 and 11–20 would have been obvious over Averbuj, alone. *See generally, id.* at 26–30.

In support of a motivation to combine Averbuj and Tsern, Petitioner relies on the Alpert Declaration. *See, e.g., id.* at 26 (citing Ex. 1007 ¶¶ 107, 110, and 112). Dr. Alpert's opinion that a person of ordinary skill in the art would have been motivated to combine Averbuj and Tsern, in turn, is either unsupported or relies on disclosure in the Detailed Description section of the '501 patent. *See* Ex. 1007 ¶¶ 110, 112, 115 (citing Ex. 1001, 3:32–37 and 7:43–66), ¶ 121. On this record, there is insufficient evidence that the cited portions of the '501 patent are a description of prior art. Moreover, Dr. Alpert's statements lack sufficient articulated reasoning with rational underpinning as to why an ordinary artisan, at the time of the invention, would have had a reason to combine the prior art elements in the manner required by the challenged claims.

On this record, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1–8 and 11–20 would have been obvious over Averbuj and Tsern.

C. Obviousness of Claims 1–5 and 14–16 over Huang and Tsern

1. Huang (Ex. 1004)

Huang describes a transparent built-in self-test (BIST) interface that is designed to perform testing of multiple embedded memory arrays in the normal mode. Ex. 1004, Abstract. Huang's transparent BIST system scheme includes a BIST controller, transparent test interface, memory arrays and a global multiple input shift register (MISR). *Id.* at 381, Fig. 2. As illustrated in Figure 2, the BIST controller generates Address, and control (Normal/BIST, shift clock, Read, and Write) signals. *Id.* “[A]ll memory modules and interface circuits receive the same test control signals from the transparent BIST controller.” *Id.* at 381, col. 2. The “test interface is

designed by adding several more multiplexers to the original input/output port of the memory array under test.” *Id.* at 380, col. 2. The test interface “implement[s] the functions of test pattern generation and test response evaluation.” *Id.* at 384, col. 2. According to Huang, “the test interface generate[s] test patterns *locally* to avoid test data routing from test controller to each memory array.” *Id.* at 381, col. 1. “[T]est patterns are generated by alternatively selecting Q or /Q as an input to each memory cell . . . , so the transition of $Q \rightarrow /Q$ and $/Q \rightarrow Q$ will be exercised in each memory cell. Thus, based on the memory content left by normal operations, new test patterns are generated by complementing each memory word several times.” *Id.*

2. Analysis

Petitioner contends Huang discloses all of the claim elements of independent claims 1 and 16. Pet. 39. Petitioner contends Huang’s BIST controller, transparent interfaces, and memory arrays correspond, respectively, to the claimed control circuit, data handlers, and memory chips. Pet. 31–36.

Patent Owner argues Huang’s data handlers (the transparent interfaces), “are not configured such that each is ‘operated independently’” as recited in claims 1 and 16, noting that Huang’s global MISR requires the concurrent testing of all memory modules. Prelim. Resp. 43–44. Patent Owner also argues Huang teaches away from performing any write operations to the memory arrays (memory chips), because one of Huang’s objectives is to keep the memory content unaltered and, therefore, the data handlers (transparent interfaces) are not configured to generate data for writing to the memory chips (memory arrays) as required by claims 1 and

16. *Id.* at 44–45.

Petitioner’s citation to Huang’s disclosure that the test interfaces generate test patterns locally to corresponding memory arrays, supports its contention that each of Huang’s test interfaces (data handlers) is independently operable from the other test interfaces and coupled to one or more corresponding memory arrays (memory chips) in the manner claimed. *See* Pet. 35 (citing Ex. 1004, 381, 1st column). Further, Petitioner points to Huang’s statement that “test patterns are generated by alternatively selecting Q or /Q as an input to each memory cell,” Pet. 34, which supports its contention that test data is written to the memory chips (memory arrays). Patent Owner has not explained how the claim language precludes a system which is capable of concurrently testing all memory arrays.

Patent Owner argues Huang does not describe a control circuit for generating address and control signals for testing the memory chips as recited in claims 1 and 16. Prelim. Resp. 46–47. We find Petitioner’s citation to Huang’s statement that the memory modules and interface circuits receive test control signals from the BIST controller, and citation to Figure 2 which illustrates “Normal/BIST” control selections, sufficient to support Petitioner’s contention that Huang’s controller is configured in the manner claimed. *See* Pet. 35–36.

On this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner identifies where Huang discloses or suggests every element of challenged claims 1–5 and 14–16. *See* Pet. 31–45. Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 1–5 and 14–16 would have been obvious over Huang.

As an alternative to its contention that Huang, alone, describes the limitations recited in claims 1 and 16, Petitioner asserts “[t]o the extent that Huang does not explicitly disclose that its memory system is ‘configured to be operatively coupled to a memory controller of a computer system,’ as recited in the preambles of claims 1 and 16,” and “[t]o the extent that Huang may not expressly disclose a plurality of memory ‘chips,’” these limitations are disclosed by Tsern. Pet. 32–33. Petitioner contends

[a] POSITA would have been motivated to combine the self-testing memory modules disclosed in Huang and Tsern because both relate to BIST systems for testing memory devices using circuitry that is located local to the memory devices. The POSITA in the memory arts would be motivated to combine Huang and Tsern for many reasons, such as to supplement the teachings of Huang relating to the desire to have test interfaces generate test patterns locally with Tsern’s teachings of efficient ways to incorporate a self-testing memory system within multiple integrated circuits on a printed circuit board. Further, the DIMM modules disclosed by Tsern are an efficient standard form factor for memory modules and would therefore yield a predictable result when combined with Huang.

Id. at 39. In support of a motivation to combine Huang and Tsern, Petitioner relies on Dr. Alpert’s opinion that the person of ordinary skill in the art would have found it obvious to combine the teachings of Huang and Tsern. *See, e.g.*, Pet. 39 (citing Ex. 1007 ¶¶ 129, 135, and 138). Petitioner’s challenge as to dependent claims 2–5, 14, and 15 based on the combination of Huang and Tsern relies on Petitioner’s contentions with respect to claims 1 and 16. *See* Pet. 39–45.

As discussed above in Section IV.B., Dr. Alpert’s opinions are based on disclosure in the ’501 patent Detailed Description, and lack sufficient articulated reasoning with rational underpinning as to why an ordinary

artisan at the time of the invention would have had a reason to combine the prior art elements in the manner required by the challenged claims.

Accordingly, on this record, Petitioner has not demonstrated a reasonable likelihood it would prevail in showing that claims 1–5 and 14–16 would have been obvious over Huang in view of Tsern.

D. Obviousness of Claims 6–8, 11–13, and 17–20 over Huang, Tsern, and Averbuj

In Section IV.C.2, we determined Petitioner identifies where Huang discloses or suggests every element of challenged independent claims 1 and 16, as well as dependent claims 2–5, 14, and 15. Petitioner contends the limitations in dependent claims 6–8, 11–13, and 17–20 are disclosed in Averbuj, and has identified where Averbuj discloses or suggests these limitations. Pet. 45–51.

In response to Petitioner’s challenge, Patent Owner relies on the same arguments made in response to Petitioner’s obviousness challenge based on Huang. *See* Prelim. Resp. 48–49. We are not persuaded by these arguments for the reasons discussed in Section IV.C.2, above.

Petitioner has explained the motivation to combine the teachings of Huang and Averbuj to achieve the inventions recited in claims 6–8, 11, and 17–20 such that on this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 6–8, 11, and 17–20 would have been obvious over Huang and Averbuj. *See id.* at 45–49. However, although Petitioner has identified where Averbuj discloses the limitations recited in claims 12 and 13, Petitioner fails to provide a statement of the motivation to modify Huang to

include the recited features. *See id.* at 49–51. Because Petitioner has not explained the motivation to combine Huang and Averbuj as to these two claims, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 12 and 13 would have been obvious over Huang and Averbuj. Likewise, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 6–8, 11–13, and 17–20 would have been obvious over Huang, Tsern, and Averbuj, because Petitioner has not explained sufficiently the motivation to combine Huang and Tsern for the reasons stated in Section IV.C.2, above.

E. Obviousness of Claims 9 and 10 over Averbuj, Tsern and JEDEC Standard, and over Huang, Tsern, Averbuj, and JEDEC Standard

Petitioner relies on JEDEC Standard (Ex. 1006) as evidence that at the time of the invention claimed in the '501 patent, it was known in the art to record failing data and corresponding addresses for BIST failures. Pet. 30, 51 (citing Ex. 1006 § 8.2.4.5). Petitioner contends it would have been obvious to have recorded failing data and corresponding addresses in Averbuj's and Huang's BIST systems to enable a failure analysis and determination of error locations. *Id.* at 31, 51. Patent Owner's arguments with respect to claims 9 and 10 are limited to its contention that JEDEC standard does not cure the deficiencies of Averbuj and/or Huang in combination with Tsern. Prelim. Resp. 39, 49–50.

On this record, Petitioner has identified where the limitations recited in claims 9 and 10 are disclosed or suggested in Huang, Averbuj, and JEDEC Standard, and sufficiently explained the motivation to combine these references. Accordingly, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner has demonstrated a

reasonable likelihood it would prevail on the ground that claims 9 and 10 would have been obvious over Huang, Averbuj, and JEDEC Standard.

Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 9 and 10 would have been obvious over Huang, Tsern, Averbuj, and JEDEC Standard, because Petitioner's motivation to combine Huang with Tsern is insufficient for the reasons explained in Section IV.C.2, above. As discussed in Sections IV.B., above, Petitioner has not asserted that the claims from which claims 9 and 10 depend would have been obvious over Averbuj, alone, but relies solely on the combination of Averbuj and Tsern. Because Petitioner's motivation to combine Averbuj with Tsern is insufficient for the reasons explained in Section IV.B., above, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 9 and 10 would have been obvious over Averbuj, Tsern, and JEDEC Standard.

V. CONCLUSION

Petitioner has demonstrated a reasonable likelihood that it would prevail on the grounds that claims 1–3, 5–8, and 11–20 are anticipated by Averbuj, claims 1–5 and 14–16 would have been obvious over Huang, claims 6–8, 11, and 17–20 would have been obvious over Huang and Averbuj, and claims 9 and 10 would have been obvious over Huang, Averbuj, and JEDEC Standard. Petitioner has not demonstrated a reasonable likelihood that it would prevail on the grounds that the challenged claims would have been obvious over Huang and/or Averbuj in view of Tsern, or Huang and/or Averbuj in view of Tsern and JEDEC Standard.

At this stage of the proceeding, the Board has not made a final determination as to the patentability of any challenged claim.

VI. ORDER

For the reasons given, it is

ORDERED that, pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '501 patent is instituted on the following grounds:

- 1) claims 1–3, 5–8, and 11–20 based on anticipation by Averbuj;
- 2) claims 1–5 and 14–16 based on obviousness over Huang;
- 3) claims 6–8, 11, and 17–20 based on obvious over Huang and Averbuj; and
- 4) claims 9 and 10 on the ground that these claims would have been obvious over Huang, Averbuj, and JEDEC Standard.

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '501 patent is hereby instituted commencing on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial.

FURTHER ORDERED that all other grounds raised in the Petition are denied, and no ground other than those specifically granted above is authorized for the *inter partes* review.

IPR2014-00971
Patent 8,359,501 B1

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01374
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and MATTHEW R.
CLEMENTS, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

DECISION
Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) to institute an *inter partes* review of claims 1–20 (the “challenged claims”) of U.S. Patent No. 8,359,501 B1 (Ex. 1001, “the ’501 patent”). 35 U.S.C. §§ 311–319. Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 10, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may not be instituted “unless . . . there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” Upon consideration of the Petition and Patent Owner’s Preliminary Response, we determine Petitioner has established a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. We therefore institute an *inter partes* review as to claims 1, 2, 4–6, 9–11, 14–17, and 20. Our factual findings and conclusions at this stage of the proceeding are based on the evidentiary record developed thus far (prior to Patent Owner’s Response). This is not a final decision as to patentability of claims for which *inter partes* review is instituted. Our final decision will be based on the record as fully developed during trial.

II. BACKGROUND

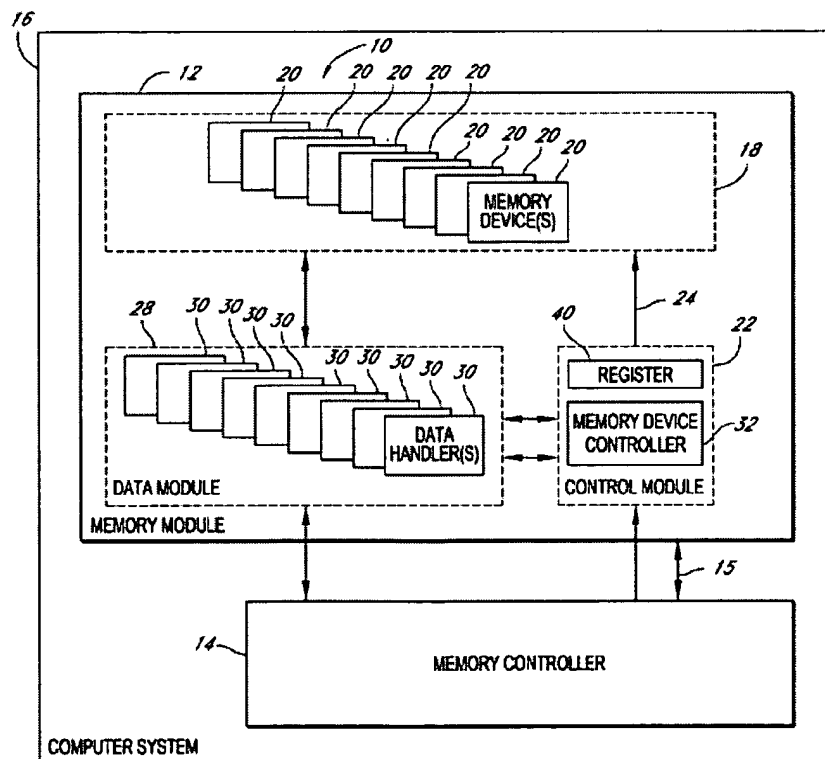
A. Related Matters

The ’501 patent is also the subject of *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB), wherein trial was instituted on claims 1–20. IPR2014-00970, Paper 12.. Contemporaneous with the instant Petition, Petitioner filed a second petition for *inter partes* review of the ’501 patent:

Smart Modular Technologies Inc. v. Netlist, Inc., Case IPR2014-01375 (PTAB). The '501 patent claims priority as a continuation of U.S. Patent No. 8,001,434 ("the '434 patent"). Also contemporaneous with the instant Petition, Petitioner filed two petitions for *inter partes* review of the '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01372 and IPR2014-01373 (PTAB).

B. The '501 Patent (Ex. 1001)

The '501 patent relates to self-testing electronic memory modules. Ex. 1001, 1:28–29. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '501 patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10

includes a plurality of memory devices 18, e.g., ETT DRAM chips, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 3:34–36; 5:10–12.

Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:14–16, 42–46. “In various embodiments, the control module 22 includes . . . one or more application-specific integrated circuit[s]” *Id.* at 9:62–64. “[D]ata module 28 and/or [] control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of [] memory devices 20.” *Id.* at 6:5–8. “For example, [] data module 28 and/or [] control module 22 are configured to provide memory signals (e.g., data, address and control signals) according [to] the operating specification of [] memory devices 20.” *Id.* at 6:8–11. “[E]ach of [] data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports” of the plurality of memory devices 18. *Id.* at 5:21–23. “In some embodiments, during testing, [] control module 22 generates address and control signals 24 associated with memory locations to be tested and [] data module 28 generates corresponding test data patterns and provides them to [] appropriate memory devices 20.” *Id.* at 5:48–52. “For example, [] data module 28 may receive a write command from [] control module 22 and provide data to be written to certain locations in [] memory devices 20 during a write operation.” *Id.* at 5:52–55.

C. Illustrative Claims

Of the challenged claims, claims 1 and 16 are independent, and are reproduced below:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.
16. A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:
 - operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;
 - operating a control circuit to generate address and control signals; and
 - testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

D. Evidence Relied Upon

Petitioner's patentability challenges are based on the following references:

Reference	Patent	Exhibit
Averbuj	US 7,392,442	1011
Best	US 8,233,303	1012
Pandey	US 6,934,205	1013
Rajan	US 7,379,316	1014

E. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of the '501 patent claims based on the following grounds:

References	Basis	Claims Challenged
Averbuj	§ 102(b)	1, 2, 4–6, 9–11, 14–17, and 20
Averbuj and Best	§ 103(a)	3
Averbuj, Best, and Pandey	§ 103(a)	7, 8, 12, 13, 18, and 19
Rajan and Averbuj	§ 103(a)	1–6, 9–11, 14–17, and 20 ¹
Rajan, Averbuj, and Pandey	§ 103(a)	7, 8, 12, 13, 18, and 19

¹Although the Petition lists claims 7, 8, 12, 13, 18, and 19 on page 7, elsewhere it lists and provides analysis for claims 1–6, 9–11, 14–17, and 20 (Pet. 8, 40–51).

In addition to these references, Petitioner relies on the Declaration of Dr. Nader Bagherzadeh (Ex. 1009) under 37 C.F.R. § 1.68. Pet. 6.

III. CLAIM CONSTRUCTION

Claims of an unexpired patent are interpreted using the broadest reasonable construction in light of the specification of the patent. *See* 37 C.F.R. § 42.100(b); Office Patent Trial Practice Guide, 77 Fed. Reg. 48756, 48766 (Aug. 14, 2012). Petitioner and Patent Owner offer respective constructions for five different claim terms. *See* Pet. 19–22; Prelim. Resp. 18–24. For purposes of this decision, we determine the only claim language requiring express construction is “generate,” recited in each of independent claims 1 and 16.

Petitioner contends the term “generate” should be interpreted as meaning “cause” or “initiate.” Pet. 20–21. In support of its proposed interpretation of “generate,” Petitioner relies on description in the ’501 patent which states that the data signals generated by data generation element 54 of data module 28 are “**programmable based on information received**” from memory controller 14 (Pet. 20 (quoting Ex. 1008, 11:4–6), and on *ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms* (4th Edition, 1988) (Ex. 1016), which defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters” (Ex. 1016, 411). Pet. 20–21. Petitioner’s proposed interpretation is consistent with our prior interpretation of the ’501 patent claim term “generate” (for purposes of instituting trial) as meaning “produce” or “cause.” *See* IPR2014-00971, Paper 12 at 9.

Patent Owner contends Petitioner's proposed definition of "generate" is based on an erroneous contention that programming and generating are synonymous. *See* Prelim. Resp. 22. Patent Owner argues, more specifically, that a programmable command transmitted to a generator "to cause" particular data to be produced does not itself comprise the particular data sought to be produced, and, therefore, does not "generate" the data. *Id.* As in IPR2014-00971, Paper 10 at 24, Patent Owner contends the claim term "generate" should be interpreted as meaning "produce." *Id.* at 20. In support of its proposed interpretation of "generate," Patent Owner relies on column 6, lines 11–14 of the '501 patent, which read: "In some embodiments, the control module 22 and the data module 28 **produce** memory addresses, control and/or data signals according to the JEDEC standard memory protocol." *Id.* at 20 (quoting Ex. 1001, 6:6–9 (emphasis added)). Patent Owner also relies on *Merriam-Webster's Collegiate Dictionary* (Ex. 2009) and *The New American Webster Handy College Dictionary* (Ex. 2010), each of which lists "produce" as one definition of "generate" (Ex. 2009, 484; Ex. 2010, 230). Prelim. Resp. 21.

As explained in our Decision to Institute in IPR2014-00971 (Paper 12 at 8), the description in the '501 patent relied upon by Patent Owner relates to only "some embodiments," and the claims do not recite that the data and control modules are configured to generate signals "according to the JEDEC standard memory protocol" As further explained in that prior decision, *Merriam Webster's Collegiate Dictionary* also defines generate as "to be the cause of" (IPR2014-00971, Paper 12, slip op. at 9). Ex. 2009, 484. Patent Owner argues the definition "to be the cause of" only applies in the context of "a situation, action, or state of mind." Prelim. Resp. 21. Patent Owner

contends “the context of the ’501 Patent is the physical process of generating electrical signals. Accordingly, the definition of ‘generate’ should be taken as ‘PRODUCE,’ since this is the ordinary meaning within the context of ‘a vital, chemical, or physical process.’” *Id.* at 21–22.

We agree with Patent Owner that the intrinsic evidence, as well as the dictionary definitions on this record, support an interpretation of the claim term “generate” as meaning, *inter alia*, “produce,” and, that a module “configured to generate” address and control signals or data, as claimed, encompasses a component which produces the exact signals or data used by the memory devices. On the record before us and for purposes of this decision, however, we are not convinced of error in our prior determination (*see* IPR2014-00971, Paper 12, slip op. at 9) that one of ordinary skill in the art would also understand the term “generate,” as used in the claims, to mean “cause.” Thus, for example, one of ordinary skill in the art would understand that a circuit “configured to generate” signals encompasses both a component that produces the exact signal used by a memory chip and a component which transmits a programmable command to that generator to cause production of the signal. Our interpretation of the term “generate” as meaning “cause” is supported by the ’501 patent (*see id.* at 8 (citing col. 10, lines 53–54 and column 11, lines 4–6)) and the dictionary definitions of record, including the additional definition of “generate” in *Merriam-Webster’s Collegiate Dictionary* as, in the context of a mathematical set or structure, “to define or originate . . . by the application of one or more rules or operations,” Ex. 2009, 484.

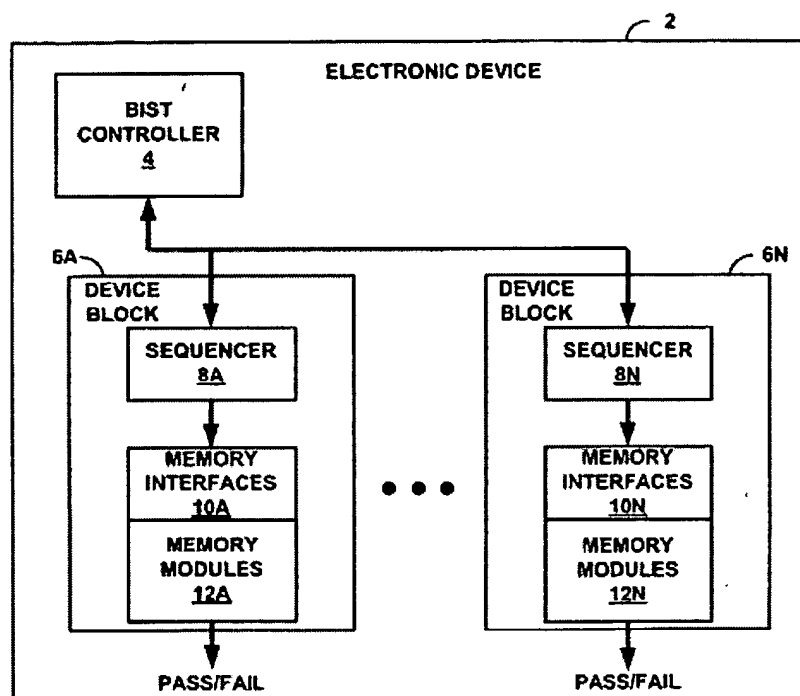
Accordingly, for purposes of this decision, we interpret the claim term “generate” as meaning “produce” or “cause.”

IV. ANALYSIS

A. Anticipation of Claims 1, 2, 4–6, 9–11, 14–17, and 20 by Averbuj

1. Averbuj (Ex. 1011)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1011, 1:64–65, 2:3–4. Figure 1 of Averbuj is reproduced below.



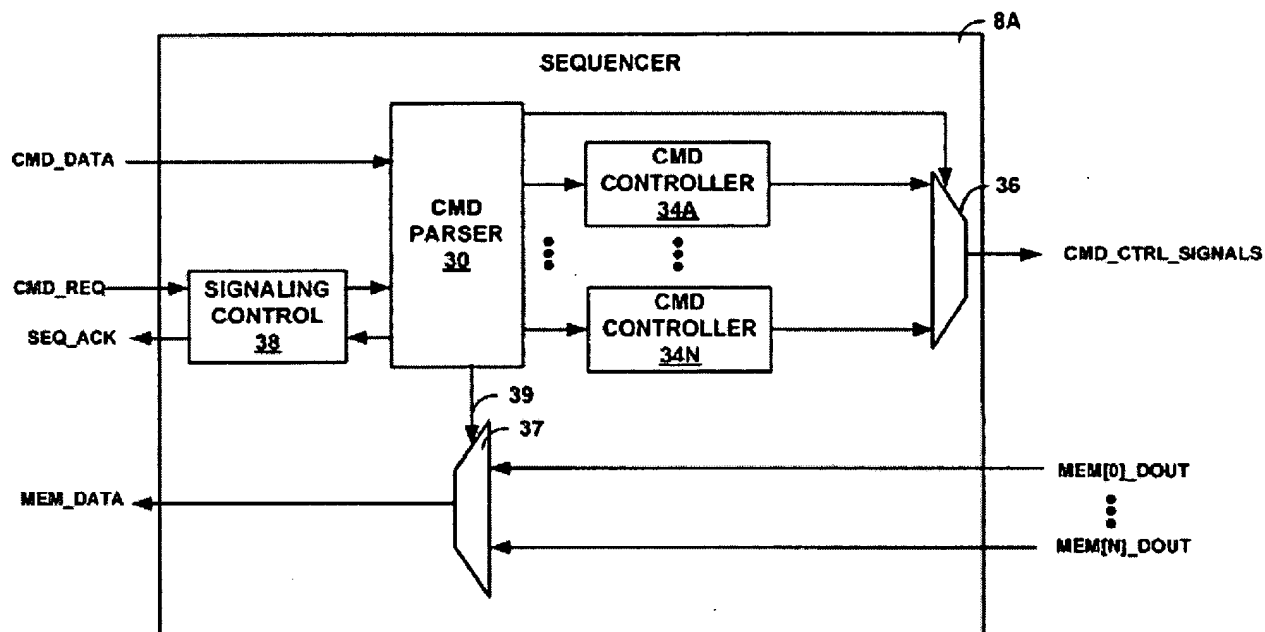
Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* at 3:62–64. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* at 3:64–67. “Each of device blocks 6 includes a sequencer 8, and a set of

one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* at 3:67–4:3.

BIST controller 4 includes algorithm controller 26 which is invoked either by a user or automatically upon power-up of electronic device 2. *Id.* at 5:12–16. Once invoked, algorithm controller 26 selects an algorithm stored within algorithm memory 20 of BIST controller 4. *Id.* at 5:16–19. The selected algorithm is applied to sequencers 8 for application to device blocks 6 as a stream of binary commands (CMD_DATA), each command defining a test within the overall algorithm. *Id.* at 4:4–7, 5:19–21, 44–46. In the embodiment illustrated in Figure 8, command 60 includes sequencer identifier 62, which identifies a sequencer to which command 60 is being issued, and payload 64. *Id.* at 9:17–21. Payload 64 includes operation code (OP CODE) 66 and a set of parameters 68. *Id.* at 9:30–31. “OP CODE 66 specifies a particular function to be performed by the receiving sequencers 8.” *Id.* at 9:32–34. “The format and meaning of parameters 68 is dependent upon the type of command, as specified by OP CODE 66.” *Id.* at 9:60–62. Figure 9A illustrates an exemplary embodiment wherein, for an OP CODE specifying a RESET command, which resets the sequencer(s) and selectively enables BIST mode, “the only relevant bit of parameters 68 is bit 28, which can be asserted or de-asserted to selectively enable and disable BIST mode.” *Id.* at 9:64–67. Another exemplary OP CODE is SET ADDRESS which “[s]ets a specific starting address as well as a maximum address limit for a test algorithm as applied to the memory modules.” *Id.* at 9, Table 1.

Upon assertion of the command request (CMD_REQ) interconnects, BIST controller 4 communicates a current command of the algorithm on the CMD_DATA interconnects to sequencers 8. *Id.* at 5:53–56. Figure 5,

below, illustrates an exemplary sequencer 8A. *Id.* at 3:46–47.

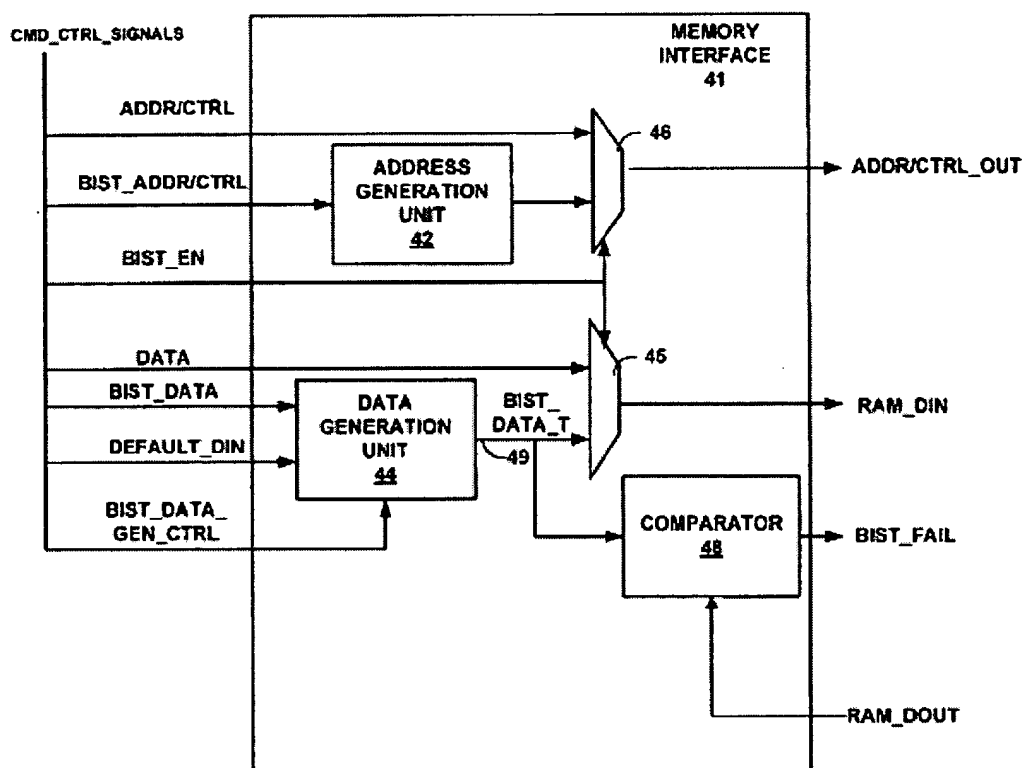


As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 that receives command data (CMD_DATA) from BIST controller 4. *Id.* at 6:24–26. Command parser 30 processes the commands from BIST controller 4 to identify a specified operation, e.g., by identifying an operational code (OP CODE) specified by the command. *Id.* at 6:26–29. Based on the specified operation, command parser 30 may extract one or more parameters from the command, and pass the extracted parameters to the command controller, (i.e., one of command controllers 34A–34N) that corresponds to the command. *Id.* at 6:30–37. The invoked command controller, in turn, issues a sequence of one or more operations to each memory interface 10, sequentially driving the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. *Id.* at 6:41–43. The command control signals may include signals

to provide memory address data to the receiving memory interfaces 10.

Id. at 6:46–48.

An exemplary memory interface is shown in Figure 6, below. *Id.* at 3:48–49.



As shown in the Figure 6 embodiment, above, memory interface 41 includes address generation unit 42 and data generation unit 44 that receive command control signals (**CMD_CTRL_SIGNALS**) from sequencer 8A. Address generation unit 42 receives address/control signals (**BIST_ADDR/CTRL** signals) and “generates the address applied to the memory module 12 based on the addressing requirements specified by sequencer 8A and the physical configuration of the rows and columns of the memory module. *Id.* at 7:63–67. “[D]ata generation unit 44 [] receives **BIST_DATA** signals and default

data (DEFAULT_DIN) as provided by [] sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the sequencer and the specific physical characteristics of the corresponding memory module 12.” *Id.* at 7:32–37. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.* at 7:38–40.

2. Analysis

Petitioner identifies the following components of Averbuj’s device as corresponding to the components recited in independent claims 1 and 16:

Challenged Claims 1 and 16		Averbuj’s Device	Pet.
A memory system configured to be operatively coupled to a memory controller of a computer system		Electronic device 2	23
A plurality of memory chips		Memory modules 12A-N	23
A control circuit configured to generate signals for testing the memory chips		BIST controller 4 generates OP CODE	29
a.	address signals	SET ADDRESS	24
b.	control signals	CMD_REQ	
One or more data handlers configured to generate data for testing a corresponding one or more memory chips		Sequencers 8A-N paired with their respective memory interfaces 10A-N, each of which pairs is a “data handler,” input RAM_DIN, generated by data generation unit 44, to the memory modules 12A-N	24–25

Patent Owner argues Petitioner has not identified sufficient support for its contention that Averbuj's BIST controller 4 generates address and control signals for testing memory chips as recited in the claims. *See* Prelim. Resp. 31–34. Patent Owner argues Averbuj's OP CODES (and the commands comprising OP CODES) define memory operations “without regard to the physical characteristics or time requirements of memory modules 12,” and, therefore, there is no evidence the OP CODES “could even be recognized by a memory chip, let alone test a memory chip.” *Id.* at 33 (emphasis omitted). Patent Owner further contends Averbuj's OP CODES are designed specifically to program a sequencer, and Petitioner has not shown that programming a sequencer is equivalent to testing memory chips or generating address and control signals. *Id.*

Patent Owner's arguments are based on a narrow construction of the term “generate” as requiring a control module configured to directly produce any address and control signals used to test the memory devices. As explained in Section III., above, for purposes of this decision, we interpret the claim term “generate” more broadly than proposed by Patent Owner as meaning “produce” *or* “cause.”

Averbuj discloses that the CMD_DATA produced by BIST controller 4 is a stream of binary commands, each of which identifies (1) a sequencer to which the command is being issued, (2) an operation (OP CODE) to be performed by the receiving sequencer (e.g., SET ADDRESS to set a starting address), and (3) a set of parameters. Ex. 1011, 9:17–21, 30–33, Table 1. In other words, upon assertion of a CMD_REQ, BIST controller 4 issues commands that “define a complete BIST algorithm” to an identified sequencer. Ex. 1011, 4:20–22. Although the commands are not sent

directly to memory devices 12A-N, but require interpretation by a sequencer and computational transformation by address generation unit 42 of a memory interface (*id.* at 4:18–19, 8:16–17), the above description in Averbuj supports Petitioner’s contention that the CMD_REQ and SET ADDRESS codes issued by BIST controller 4 cause the issuance of, i.e., “generate,” address and control signals.

Patent Owner also argues the position advanced by Petitioner is inconsistent, contending that if “Averbuj’s address and control ‘OP Codes’ are considered Address and Control Signals, then necessarily Averbuj’s data ‘OP Codes’ should be considered Data signals and Averbuj’s BIST controller 4 must necessarily generate BOTH the Data and Address/Control signals.” Prelim. Resp. 35.

Patent Owner’s argument is unpersuasive because it is not based on an interpretation of the claim term “generate” as encompassing components that both cause and produce the issuance of data signals. Under this interpretation of “generate,” applied for purposes of this decision, both Averbuj’s BIST controller 4 and memory interfaces 10 are properly viewed as generating data because BIST controller 4 causes the issuance of the exact data (RAM_DIN) produced by data generation unit 44 of memory interface 41 and applied to the inputs of memory devices 12A-N. Ex. 1011, 7:38–40.

On this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner identifies where Averbuj describes every element of challenged independent claims 1 and 16. Based on our review of the present record, we are persuaded likewise as to claims 2, 4–6, 9–11, and 14, 15, 17, and 20. *See* Pet. 22–34. Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that

claims 1, 2, 4–6, 9–11, 14–17, and 20 are anticipated by Averbuj.

B. Obviousness of Claims 1–20

Petitioner contends claim 3 would have been obvious over Averbuj in view of Best. Pet. 34–36. Petitioner further contends claims 1–6, 9–11, 14–17, and 20 would have been obvious over Rajan in view of Averbuj. *Id.* at 40–52. Petitioner contends the remaining claims 7, 8, 12, 13, 18, and 19 would have been obvious over Averbuj and Best in view of Pandey, and over Rajan and Averbuj in view of Pandey. *Id.* at 36–40, 52–56.

Petitioner contends “[a person of ordinary skill in the art] would [have been] motivated to combine Averbuj ‘442 and Best ‘303, because Best ‘303 teaches a multi-die memory device with a BIST interface die that may be disposed in place of conventional memory in memory modules, such as the module of Averbuj ‘442, which is similarly configured for BIST functions.” Pet. 35 (citations omitted). Petitioner similarly contends

[a person of ordinary skill in the art] would [have been] motivated to combine Rajan ‘316 and Averbuj ‘442, because Averbuj ‘442 provides a BIST architecture applicable to any electronic device, such as circuit boards with memory modules as disclosed in Rajan ‘316. . . . [A] POSITA would be motivated to implement the known BIST architecture and process in Averbuj ‘442 with the BIST-enabled circuit board-based memory module devices of Rajan ‘316, thereby replicating the testing capability of Averbuj ‘442 to Rajan ‘316.

Id. at 51 (citations omitted). In support of these contentions, Petitioner cites paragraphs 54 and 73, and Appendices C and E of the Declaration of Dr. Nader Bagherzadeh (Ex. 1009). *Id.* at 35–36, 51–52. Petitioner notes: “The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *Id.* at 36

(citations omitted).

As explained by the Federal Circuit, “[a]lthough predictability is a touchstone of obviousness, the ‘predictable result’ discussed in *KSR* refers not only to the expectation that prior art elements are capable of being physically combined, but also that the combination would have worked for its intended purpose.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 567 F.3d 1314, 1326 (Fed. Cir. 2009) (citing *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007)).

Petitioner contends the proposed combinations of Averbuj and Best, and Rajan and Averbuj would have been nothing more than combinations of familiar elements, but has not identified persuasive evidence which supports these contentions. In this regard, we acknowledge Petitioner relies on Dr. Bagherzadeh’s testimony that “Best ‘303 teaches a multi-die memory device with a BIST interface die that may be disposed in place of conventional memory in memory modules such as the module of Averbuj ‘442, which is similarly configured for BIST functions” (Ex. 1009 ¶ 54) and that “Averbuj ‘442 provides a BIST architecture applicable to any electronic device such as circuit boards with memory modules as disclosed in Rajan ‘316” (*id.* ¶ 97). However, Dr. Bagherzadeh has not identified sufficient support for these statements. For example, Dr. Bagherzadeh has not explained clearly why, or identified evidence that shows the invention described in the ’501 patent falls into a very predictable field, one of ordinary skill in the art would have possessed the requisite skills to make the proposed combinations, or one of ordinary skill in the art reasonably would have anticipated the combination to function in an ordinary and expected way. *See Rothman v. Target Corp.*, 556 F.3d 1310, 1319–20 (Fed. Cir.

2009). With respect to this latter point, although a teaching away supports a conclusion that a prior art combination would have produced an inoperative result, *see DePuy*, 567 F.3d at 1326, Petitioner has not identified case law to support Dr. Bagherzadeh's contention that the absence of a teaching away in Best and Averbuj evidences that the combination would be operative (Ex. 1009 ¶ 55). Petitioner's contentions regarding motivation to modify the references based on Pandey are likewise deficient. *See* Pet. 39–40, 55–56.

For the above-stated reasons, on this record, Petitioner has not demonstrated a reasonable likelihood it would prevail on the grounds that claim 3 would have been obvious over Averbuj in view of Best; claims 1–6, 9–11, 14–17, and 20 would have been obvious over Rajan in view of Averbuj; and claims 7, 8, 12, 13, 18, and 19 would have been obvious over Averbuj in view of Best and Pandey and over Rajan in view of Best and Pandey.

C. Patent Owner's Request to Deny the Petition as Redundant to the Petition in IPR2014-00971

Patent Owner contends we should deny Petitioner's request to institute an *inter partes* review because the Petition merely sets forth grounds redundant to those raised in *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971, Paper 1 (PTAB). *See* Prelim. Resp. 24-30. A similar argument was raised by Patent Owner in IPR2014-01372. For essentially the same reasons stated in our Decision to Institute in IPR2014-01372, we are not persuaded the Petition should be denied as to the ground of anticipation based on Averbuj. In particular, we note the Petition in the present case differs from that in IPR2014-00971 in that, *inter alia*, Petitioner relies on a different declarant, relies on different disclosure in Averbuj, and

raises a new challenge as to claims 4, 9, and 10 based on anticipation by Averbuj.

Accordingly, taking into account the burden on Netlist and the considerations set forth in 35 U.S.C. § 316(b), e.g., the efficient administration of the Office, institute trial as to the ground of anticipation based on Averbuj for the reasons discussed above in Section IV.A.2.

V. CONCLUSION

Petitioner has demonstrated a reasonable likelihood that it would prevail on the grounds that claims 1, 2, 4–6, 9–11, 14–17, and 20 are anticipated by Averbuj. Petitioner has not demonstrated a reasonable likelihood that it would prevail on its challenges under 35 U.S.C. §103(a).

At this stage of the proceeding, the Board has not made a final determination as to the patentability of any challenged claim.

VI. ORDER

For the reasons given, it is

ORDERED that, pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '501 patent is instituted as to claims 1, 2, 4–6, 9–11, 14–17, and 20 based on anticipation by Averbuj;

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '501 patent is hereby instituted commencing on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial; and

FURTHER ORDERED that all other grounds raised in the Petition are denied, and no ground other than those specifically granted above is authorized for the *inter partes* review.

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01375
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
MATTHEW R. CLEMENTS, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION
Denying Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) to institute an *inter partes* review of claims 1–20 (the “challenged claims”) of U.S. Patent No. 8,359,501 B1 (Ex. 1008, “the ’501 patent”). See 35 U.S.C. §§ 311–319. Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 10, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may not be instituted “unless . . . there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” Upon consideration of the Petition and Patent Owner’s Preliminary Response, we determine Petitioner has not established a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. We therefore decline to institute an *inter partes* review as to claims 1–20.

II. BACKGROUND

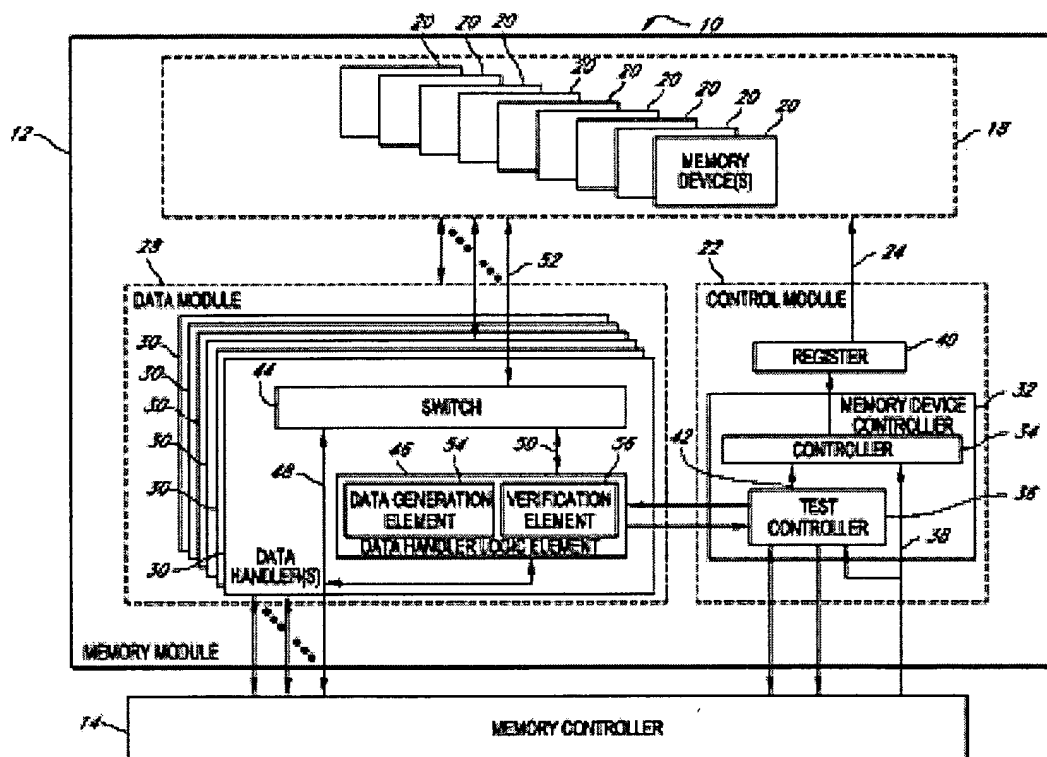
A. Related Matters

The ’501 patent is also the subject of *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB), wherein trial was instituted on claims 1–20. IPR2014-00971, Paper 12. Contemporaneous with the instant Petition, Petitioner also filed a second petition for *inter partes* review of the ’501 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01374 (PTAB). Also contemporaneous with the instant Petition, Petitioner filed two petitions for *inter partes* review of U.S. Patent No. 8,001,434 (“the ’434 patent”), to which the ’501 patent claims priority: *Smart Modular*

Technologies Inc. v. Netlist, Inc., Cases IPR2014-01372 and IPR2014-01373 (PTAB).

B. The '501 Patent (Ex. 1008)

The '501 patent relates to self-testing electronic memory modules. Ex. 1008, 1:26–27. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '501 patent, reproduced below.



As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:10–14. Memory module 10 further includes control module 22 and data module 28 comprising a plurality of independently operable data

handlers 30. *Id.* at 5:14–18. Control module 22 includes memory device controller 34 which receives address and control signals 38 from system memory controller 14 and address and control signals 42 from test controller 36. *Id.* at 9:40–45. Similarly, data module 28 includes switch 44, which selectively inputs to memory devices 18 either data signals 48 from memory controller 14 or data signals 50 from data handler logic element 46. *Id.* at 10:7–11.

During test mode, memory device controller 34 sends address and control signals 38 from test controller 36 to register 40 for input to address and control ports of memory devices 18. *Id.* at 9:40–50. “[T]est controller 36 updates each of [] data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.)” *Id.* at 14:7–9. Data generation element 54 of data handler logic element 46 is configured to generate data signals and/or patterns of data signals based on the information received from test controller 36, for writing to memory devices 18. *Id.* at 10:28–37, 15:25–27.

C. Illustrative Claim

Of the challenged claims, claims 1, 20, and 29 are independent.

Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - and

a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

D. Evidence Relied Upon

Petitioner's patentability challenges are based on the following references:

Reference	Patent/Printed Publication	Exhibit
Fleischman '320	US 6,321,320 B1	1011
JTAG 2001	IEEE JTAG Standard No. 1149.1-2001	1012
Adams '873	US 7,203,873 B1	1013
Huang	<i>An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers</i> , VLSI Design 2001, Fourteenth International Conference on VLSI Design (pp. 379–384), IEEE (2000)	1014

In addition to these references, Petitioner relies on the Declaration of Dr. Nader Bagherzadeh (Ex. 1009) under 37 C.F.R. § 1.68. Pet. 6.

E. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of the '501 patent claims based on the following grounds:

References	Basis	Claims Challenged
Fleischman '320 and JTAG 2001	§ 103(a)	1–20
Adams '873 and JTAG 2001	§ 103(a)	1–6, 9–11, 14–15, 16–17, and 20
Adams '873, JTAG 2001, and Huang	§ 103(a)	7–8, 12–13, and 18–19

III.CLAIM CONSTRUCTION

Claims of an unexpired patent are interpreted using the broadest reasonable construction in light of the specification of the patent. *See* 37 C.F.R. § 42.100(b); *see also In re Cuozzo Speed Techs. LLC*, No. 2014–1301, 2015 WL 448667 , at *6 (Fed. Cir. Feb. 04, 2015). Petitioner and Patent Owner offer respective constructions for seven different claim terms. *See* Pet. 18–21; Prelim. Resp. 26–32. For purposes of this decision, we determine the only claim language requiring express construction is the term “generate,” recited in each of independent claims 1 and 16.

Petitioner contends the term “generate” should be interpreted as meaning “cause” or “initiate.” Pet. 20. In support of its proposed interpretation of “generate,” Petitioner relies on a description in the '501 patent that states the data signals generated by data generation element 54 of data module 28 are programmable based on information received from memory controller 14 (Ex. 1008, 10:67–11:2), and on *IEEE 100 - STANDARD DICTIONARY OF ELECTRICAL AND ELECTRONIC TERMS*, (4th ed. 1988) (Ex.

1016), which defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters” (*id.* at 411). Pet 20. Petitioner’s proposed interpretation is consistent with our prior interpretation of the ’501 patent claim term “generate” (for purposes of instituting trial) as meaning “produce” *or* “cause.” See IPR2014-00971, Paper 12, slip op. at 9.

Patent Owner contends Petitioner’s proposed definition of “generate” is based on an erroneous contention that programming and generating are synonymous. See Prelim. Resp. 28. Patent Owner argues, more specifically, that a programmable command transmitted to a generator “to cause [data] generation” does not itself comprise the particular data sought to be produced, and, therefore, does not “generate” the data. *Id.* As in IPR2014-00971, (Paper 10, 24), Patent Owner contends the claim term “generate” should be interpreted as meaning only “produce.” *Id.* at 28. In support of its proposed interpretation of “generate,” Patent Owner relies on column 6, lines 6–9 of the ’501 patent, which read: “In some embodiments, the control module 22 and the data module 28 **produce** memory addresses, control and/or data signals according to the JEDEC standard memory protocol.” *Id.* at 28 (quoting Ex. 1008, 6:6–9) (emphasis added). Patent Owner also relies on *Merriam-Webster’s Collegiate Dictionary* (Ex. 2008) and *The New American Webster Handy College Dictionary* (Ex. 2009), each of which lists “produce” as one definition among several of “generate” (Ex. 2008, 484; Ex. 2009, 230). Prelim. Resp. 29.

As explained in our Decision to Institute in IPR2014-00971 (Paper 12, 8), the description in the ’501 patent relied upon by Patent Owner relates to only “some embodiments,” and the claims do not recite that the data and

control modules are configured to generate signals “according to the JEDEC standard memory protocol” (Prelim. Resp. 28). As further explained in that prior decision, *Merriam-Webster’s Collegiate Dictionary* also defines generate as “to be the cause of” (IPR2014-00971, Paper 12, slip op. at 9). Ex. 2008, 484. Patent Owner argues the definition “to be the cause of” applies only in the context of “a situation, action, or state of mind.” Prelim. Resp. 29. Patent Owner contends “the context of the ’501 Patent is the physical process of generating electrical signals. Accordingly, the definition of ‘generate’ should be taken as ‘PRODUCE,’ since this is the ordinary meaning within the context of ‘a vital, chemical, or physical process.’” *Id.* at 29.

We agree with Patent Owner that the intrinsic evidence, as well as the dictionary definitions on this record, support an interpretation of the claim term “generate” as meaning, *inter alia*, “produce,” and, that a module “configured to generate” address and control signals or data, as claimed, encompasses a component which produces the exact signals or data used by the memory devices. On the record before us, and for purposes of this decision, however, we are persuaded that one of ordinary skill in the art would also understand the term “generate,” as used in the claims, to mean “cause.” Thus, for example, one of ordinary skill in the art would understand that a module “configured to generate” signals or data encompasses both a component that produces the exact signal or data used by a memory device and a component which transmits a *programmable command* to that generator to cause production of the signal or data. Our interpretation of the term “generate” as including “cause” is supported by the ’501 patent (*see id.* at 9 (citing Ex. 1008, 10:49–50, 10:67–11:4)) and the

dictionary definitions of record, including the additional definition of “generate” in *Merriam-Webster’s Collegiate Dictionary* as, in the context of a mathematical set or structure, “to define or originate . . . by the application of one or more rules or operations” (Ex. 2008, 484). Accordingly, for purposes of this decision, we interpret the claim term “generate” as meaning “produce” or “cause.”

IV. ANALYSIS

A. Obviousness of Claims 1–20 over Fleischman ’320 and JTAG 2001

1. Fleischman ’320 (Ex. 1011)

Fleischman ’320 describes BIST architecture including multiple BIST engines. Ex. 1011, 3:29–31, 3:18–20, Figs. 3, 8. Fleischman ’320 also discloses that the BIST architecture includes BIST main control 110 and address generation unit 72 (together constituting a “control module”) and multiple data generation (DDG) units 94 (“plurality of data handlers”), each for testing the memory arrays. *Id.* at 6:5–8, 4:8–11. Further, the BIST architecture of Fleischman ’320 is configured to test memory arrays 1–N using address and control information issued by primary BIST main control 110 and address generator 72. *Id.* at 1:25–30, 8:1–4, 8:10–11.

2. JTAG 2001 (Ex. 1012)

JTAG 2001 discloses a general BIST architecture that is applicable to JTAG compliant circuitry, including printed circuit boards with memory devices for testing. Ex. 1012, 68, 128. JTAG 2001 also discloses the use of cyclic data, e.g., 1111X, 0XXX0, and other cyclic patterns for testing memory devices. *Id.* at 159.

3. Analysis

Petitioner contends claims 1–20 would have been obvious over Fleischman '320 in view of JTAG 2001. Pet. 21–36.

With respect to the motivation to combine the two references, Petitioner contends:

A POSITA would have been motivated to combine Fleischman '320 and JTAG 2001, because Fleischman '320 expressly discloses compliance with the JTAG standard. JTAG 2001 defines JTAG compliance, including the application of self-test to devices on a printed circuit board, and therefore defines an environment that is suitable for Fleischman '320. Accordingly, a POSITA would be motivated to adapt the BIST architecture and operations disclosed in Fleischman '320 to a PCB in accordance with the compliance requirements disclosed in JTAG 2001.

Id. at 36 (citations omitted). In support of these contentions, Petitioner cites paragraph 33 and Appendix B of the Declaration of Dr. Nader Bagherzadeh (Ex. 1009). *Id.* Petitioner notes: “The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *Id.* (citations omitted).

As explained by the Federal Circuit, “[a]lthough predictability is a touchstone of obviousness, the ‘predictable result’ discussed in *KSR* refers not only to the expectation that prior art elements are capable of being physically combined, but also that the combination would have worked for its intended purpose.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 567 F.3d 1314, 1326 (Fed. Cir. 2009) (citing *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007)).

We acknowledge Petitioner relies on Dr. Bagherzadeh’s testimony that is identical in substance to the text from the Petition reproduced above.

The specification of Fleischman '320 states, "BIST operation and extraction of test results are typically accomplished through IEEE Standard 1149.1 Joint Test Action Group (JTAG) boundary scan Test Access Port (TAP)." Ex. 1011, 2:61–63. However, this statement is a general description of the prior art, not the invention. Fleischman '320 describes embodiments using a JTAG TAP interface 122. Ex. 1011, 5:10–13, 9:47–55. In the '501 patent specification, the JTAG TAP interface is always connected to the BIST Main Control (BMC) 110.

Nonetheless, for the combination of Fleischman '320 and JTAG 2001, Petitioner proposes a different BIST architecture where the JTAG interface connects directly to the Memory Arrays in Fleischman '320. Pet. 26; Prelim. Resp. 38. Petitioner contends the proposed combination of Fleischman '320 and JTAG 2001 would have been nothing more than the combination of familiar elements, but has not identified persuasive evidence that supports these contentions. Pet. 26. Dr. Bagherzadeh also has not identified sufficient support for this combination. For example, Dr. Bagherzadeh has not explained clearly why, or identified evidence that shows that the invention described in the '501 patent falls into a very predictable field, or one of ordinary skill in the art would have possessed the requisite skills to make the proposed combination, or that one of ordinary skill in the art reasonably would have anticipated the combination to function in an ordinary and expected way. *See Rothman v. Target Corp.*, 556 F.3d 1310, 1319–20 (Fed. Cir. 2009). Given that the '501 specification teaches a different configuration of the JTAG TAP, Petitioner needed to explain why one of ordinary skill would have known to replace the existing configuration with the new configuration. *Compare Abbott Labs. v. Sandoz*,

Inc., 544 F.3d 1341, 1352 (Fed. Cir. 2008) (“[K]nowledge of the goal [such as meeting an FDA requirement] does not render its achievement obvious.”) with *Erico Int’ Corp. v. Vutec Corp.*, 516 F.3d 1350, 1356 (Fed. Cir. 2008) (a regulation that directs designers to a particular solution may render that solution obvious).

Dr. Bagherzadeh contends that the absence of a teaching away in Fleischman ’320 and JTAG 2001 is evidence that the combination would be operative. Ex. 1009 ¶ 35. Although a teaching away supports a conclusion that a prior art combination would have produced an inoperative result, *see DePuy*, 567 F.3d at 1326, Petitioner has not identified case law to support Dr. Bagherzadeh’s contention.

Accordingly, we are not persuaded that Petitioner has shown sufficiently a motivation to combine Fleischman ’320 and JTAG, and for the above-stated reasons, on this record, we determine Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1–20 would have been obvious over Fleischman ’320 and JTAG 2001.

B. Obviousness of Claims 1–6, 9–11, 14–15, 16–17, and 20 over Adams and JTAG 2001

1. Adams (Ex. 1013)

Adams ’873 discloses a BIST architecture that is applicable to any BIST environment, including environments compliant with the JTAG standard, and that is applicable for testing memory devices. Ex. 1013, 3:34–37, Fig. 2. Adams ’873 also discloses that the BIST architecture includes MBIST controller 210 (“control module”) and multiple collar circuitry 220a–c (“plurality of data handlers”), each for testing the corresponding

memory devices 225a-c. *Id.* at 2:6–12, 5:7–14, 5:58–6:5. Further, the BIST architecture of Adams ’873 is configured to test the memory devices 225a–c after being enabled by MBIST controller 210 and using data generated by collar circuitry 220a–c. *Id.* at 5:40–46, 5:58–6:5.

2. Analysis

Patent Owner argues Petitioner has not identified sufficient support for its contention that Adams ’873’s MBIST controller generates address and control signals for testing memory devices as recited in the claims. *See* Prelim. Resp. 43–46. Patent Owner argues the signals that enter Adams ’873’s memory 325 are instead all generated by the corresponding collars. *Id.* at 48.

Patent Owner’s arguments are based on a narrow construction of the term “generate” as requiring a control module configured to produce directly the actual address and control signals used to test the memory devices. As explained in Section III, above, for purposes of this decision, we interpret the claim term “generate” more broadly than proposed by Patent Owner as meaning “produce” *or* “cause.”

Adams ’873 discloses that the MBIST controller includes an “enable module 310a [that] is configured to provide enable signals for use in selecting, scheduling, and controlling BIST of collars 320 (and, therefore, the memory 325 in each collar).” Ex. 1013, 5:58–6:5; Pet. 37–38. In other words, MBIST controller 210 issues enable signals that communicate to an “enable element [that] is configured to selectively enable a collared memory for testing, e.g., one at a time.” Ex. 1013, 2:21–22; Pet. 37.

Although the commands are not sent directly to memory 325, but instead enable the collars (*id.*), the above description in Adams ’873

supports Petitioner's contention that the enable signals issued by MBIST controller 210 cause the issuance of, i.e., "generate," address and control signals. Nonetheless, Petitioner's ground fails—MBIST controller 210 is not "configured to generate" because, as set forth in Section III, above, we construe "configured to generate" signals or data as encompassing a component which transmits a *programmable command* to that generator to cause production of the signal or data. The enable signal in Adams pointed to by Petitioner is not programmable, i.e., it is not used by the collar to generate or program any specific signal. The enable signal simply "enables" the collar to enter test mode. Ex. 1013, 5:7–15.

For the above-stated reasons, on this record, we determine Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1–6, 9–11, 14–15, 16–17, and 20 would have been obvious over Adams '873 and JTAG 2001.

C. Obviousness of Claims 7–8, 12–13, and 18–19 over Adams, JTAG 2001, and Huang

Petitioner argues that claims 7–8, 12–13, and 18–19 would have been obvious over Adams '873, JTAG 2001, and Huang under 35 U.S.C. § 103(a). Pet. 48-53. Claims 7–8, 12–13, and 18–19 depend from independent claims 1 and 16, respectively. As discussed above, the combination of Adams '873 and JTAG 2001 fails to teach or suggest all of the elements of independent claims 1 and 16. Petitioner does not assert that Huang overcomes the aforementioned deficiency in Adams '873 and JTAG 2001. Thus, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood it would prevail with respect to claims 7–8, 12–13, and 18–19, on

the ground that these claims would have been obvious over Adams '873, JTAG 2001, and Huang.

V. CONCLUSION

The information presented does not show that there is a reasonable likelihood that Petitioner would prevail at trial with respect to at least one claim of the '501 patent, based on any grounds presented in the Petition. On this record, we deny the petition for *inter partes* review of claims 1–20.

VI. ORDER

Accordingly, it is

ORDERED that that the petition is DENIED as to all challenged claims, and no trial is instituted.

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01374
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
MATTHEW R. CLEMENTS, *Administrative Patent Judges*.

Opinion for the Board filed by *Administrative Patent Judge* GAUDETTE.

Opinion Dissenting filed by *Administrative Patent Judge* CLEMENTS.

GAUDETTE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. § 318(a) and 37 C.F.R. § 42.73

I. BACKGROUND

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) on August 23, 2014, requesting institution of an *inter partes* review of claims 1–20 (the “challenged claims”) of U.S. Patent No. 8,359,501 B1 (Ex. 1008, “the ’501 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 10). Based on these submissions, an *inter partes* review of claims 1, 2, 4–6, 9–11, 14–17, and 20 was instituted on March 10, 2015, pursuant to 35 U.S.C. § 314. Paper 12 (“Dec. on Inst.” or “Institution Decision”).

After institution, Patent Owner filed a Response (Paper 21, Corrected Response, “PO Resp.”), and Petitioner filed a Reply to the Patent Owner Response (Paper 26, “Reply”).

Petitioner filed a Motion to Exclude Evidence. Paper 25; *see also* Paper 36 (Patent Owner’s Opposition) and Paper 41 (Petitioner’s Reply). Patent Owner also filed a Motion to Exclude Evidence (Paper 33; *see also* Paper 38 (Petitioner’s Opposition) and Paper 40 (Patent Owner’s Reply)) and a Motion to Exclude Portions of Petitioner’s Reply (Paper 34; *see also* Paper 39 (Petitioner’s Opposition)).

Oral argument was held on November 17, 2015, and a transcript (Paper 42, “Tr.”) has been entered into the record.

The Board has jurisdiction under 35 U.S.C. § 6(c). This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73, addresses issues and arguments raised during trial.

For the reasons discussed below, we determine Petitioner has not met its burden to prove, by a preponderance of the evidence, that claims 1, 2, 4–

6, 9–11, 14–17, and 20 of the '501 patent are unpatentable. Petitioner's and Patent Owner's Motions to Exclude are dismissed.

A. Related Matters

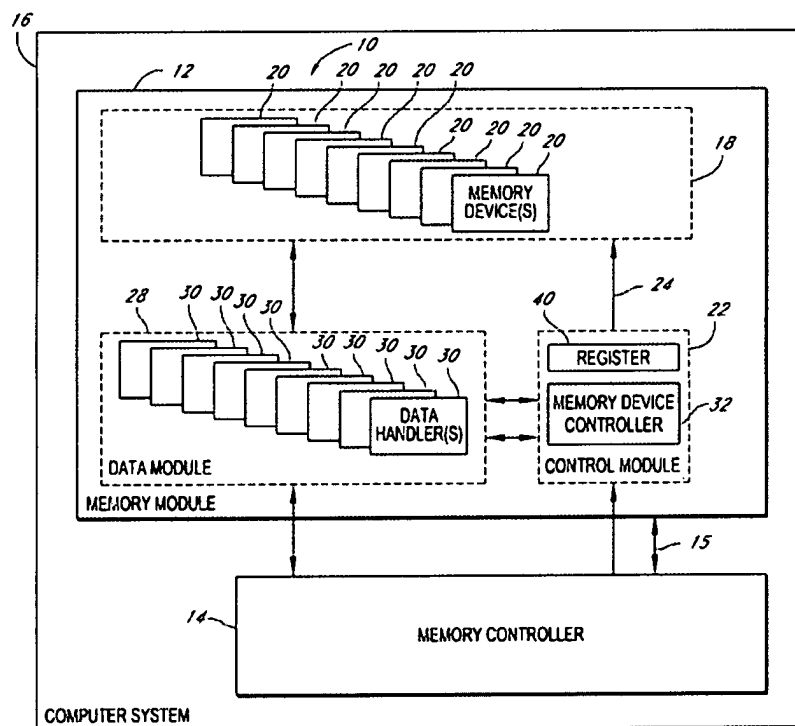
The '501 patent claims priority as a continuation of U.S. Patent Application No. 12/422,925, now U.S. Patent No. 8,001,434 ("the '434 patent"). Ex. 1008, 1. Petitioner also requested, and we instituted, an *inter partes* review of the '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01372 (PTAB March 10, 2015) ("IPR-1372"), Paper 13. We denied Petitioner's requests to institute *inter partes* reviews of the '434 patent and the '501 patent in two additional petitions, filed August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB March 13, 2015), Paper 16, and Case IPR2014-01375 (PTAB March 13, 2015), Paper 12, respectively. Oral argument in IPR2014-01372 was consolidated with the oral argument in the present *inter partes* review. A final decision in IPR2014-01372 is issued concurrently with this final decision.

The '501 patent also is the subject of IPR2014-00971 ("IPR-971"). We entered a final written decision in that case on December 14, 2015, concluding that claims 1–3, 5–8, and 11–20 are unpatentable on the following grounds: claims 1–3, 5–8, 11–14, and 16–20 under 35 U.S.C. § 102(b) based on Averbuj (U.S. Patent Application Publication No. 2005/0257109 A1, issued November 17, 2005); and claims 1–3, 5, and 14–16 under 35 U.S.C. § 103(a) based on Huang (*An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers*, Fourteenth International Conference on VLSI Design 379–384 (2001), IEEE). *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB

December 14, 2015), Paper 34 (“IPR-971 FWD”). Petitioner Sandisk Corporation has requested rehearing of our determination that it did not meet its burden to show unpatentability of claims 4, 9, and 10, and our determination that it failed to show unpatentability of claim 15 as anticipated by U.S. Patent Application Publication No. 2005/0257109 A1. *Id.*, Paper 35.

B. The '501 Patent (Ex. 1008)

The '501 patent relates to self-testing electronic memory modules. Ex. 1008, 1:28–29. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '501 patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10

includes a plurality of memory devices 18, e.g., ETT DRAM chips, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 3:34–36; 5:10–12.

Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:14–16, 42–46. “In various embodiments, the control module 22 includes . . . one or more application-specific integrated circuit[s]” *Id.* at 9:62–64. “[D]ata module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20.” *Id.* at 6:5–8. “For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according [to] the operating specification of the memory devices 20.” *Id.* at 6:8–11.

“In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20.” *Id.* at 5:48–52. “For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation.” *Id.* at 5:52–55.

A. Illustrative Claims

Of the challenged claims, claims 1 and 16 are independent, and are reproduced below:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

16. A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:
 - operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;
 - operating a control circuit to generate address and control signals; and
 - testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

C. The Instituted Ground of Unpatentability

We instituted the instant *inter partes* review of claims 1, 2, 4–6, 9–11, 14–17, and 20 based on anticipation under 35 U.S.C. § 102(b) by Averbuj, U.S. Patent No. 7,392,442 B2, issued June 24, 2008 (Ex. 1011, “Averbuj”). As indicated in Section I.A., above, in IPR-971, we held claims claims 1–3, 5–8, 11–14, and 16–20 of the ’501 patent unpatentable under 35 U.S.C. § 102(b) based on U.S. Patent Application Publication No. 2005/0257109 A1, the published application of the Averbuj patent on which the instituted ground of unpatentability in this proceeding is based.

D. Expert Testimony

Petitioner relies on the testimony of Dr. Nader Bagherzadeh in support of its patentability challenge. Pet. 1. Dr. Bagherzadeh executed a declaration (Ex. 1009, “the Bagherzadeh Declaration”) in support of the Petition. Dr. Bagherzadeh was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 2012.

Dr. Bagherzadeh testifies as follows: “In 1979 and 1987, respectively, [he] earned a master of science in electrical engineering and a doctorate degree in computer engineering from the University of Texas at Austin.” Ex. 1009 ¶ 4. He has been involved in design and development of digital systems for more than 30 years. *Id.* ¶ 6. Dr. Bagherzadeh joined the University of California, Irvine, in 1987, and has “been teaching, researching, and consulting regarding almost all aspects of memory design for high performance computer systems, including but not limited to DRAMs and SRAMs.” *Id.* ¶ 7. He has been employed as a professor in the department of Electrical Engineering and Computer Science at the

University of California, Irvine, since 2003. *Id.* ¶ 5. In 2000, Dr. Bagherzadeh became a cofounder of Morpho Technologies, a high tech company focused on the design and development of low power and high performance digital signal processors for mobile applications. *Id.* ¶ 7. He was involved in evaluating patents, technical reports and presentations related to memory chip designs, DSPs, and parallel processing algorithms for mobile platforms. *Id.*

Patent Owner relies on the testimony of Dr. Carl Sechen. PO Resp. 1. Dr. Sechen executed a declaration (Ex. 2019, “the Sechen Declaration”) in support of Patent Owner’s Response. Dr. Sechen was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 1031.

Dr. Sechen testifies as follows: he has an M.S. degree in Electrical Engineering from the Massachusetts Institute of Technology and “was awarded a Ph.D. in electrical engineering from the University of California at Berkeley in 1986.” Ex. 2019 ¶ 5. Dr. Sechen has been a Professor of Electrical Engineering for more than 28 years. *Id.* ¶ 2. During this time period, his research has focused on design and computer-aided design of digital integrated circuits, including the design of DRAM, and he has taught numerous students how to design DRAM memories. *Id.* Dr. Sechen has “also been involved in numerous research projects on VLSI design and memory design[, and has] taught numerous graduate researchers how to design digital integrated circuits, including memories.” *Id.* ¶ 4.

The parties do not dispute that Dr. Bagherzadeh and Dr. Sechen are qualified to testify as experts under FRE 702.

II. LEVEL OF ORDINARY SKILL IN THE ART

The level of ordinary skill in the art is relevant to claim construction and anticipation. *See Yorkey v. Diab*, 605 F.3d 1297, 1300 (Fed. Cir. 2010) (explaining that a determination of anticipation involves interpreting the claim language and then comparing the construed claim to a prior art reference); *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1259–60 (Fed. Cir. 2010) (“[C]laim language should be read in light of the specification as it would be interpreted by one of ordinary skill in the art.”).

Dr. Bagherzadeh and Dr. Sechen agree that the field of art is “memory module design with features for built-in self-test (BIST).” Ex. 2019 ¶ 21; *see also* Ex. 1009 ¶ 17.

Dr. Bagherzadeh testifies that

a person of ordinary skill in the art for the ‘501 patent in April 2008 would have a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering, and at least one year of experience relating to memory systems and BIST.

A person having a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering would have experience with digital circuit design, memories and computer architecture. An additional year of experience relating to memory systems and BIST would focus on those areas and allow one to recognize current issues with respect to the field.

Ex. 1009 ¶¶ 19–20; *see id.* ¶ 16 (“My understanding is that the earliest possible priority date of the ‘501 patent is April 14, 2008.”).

Dr. Sechen testifies that

one of ordinary skill in the art at the time of filing the application(s) for the ‘434 [sic, ‘501] Patent, who would be working on the design of memory devices and memory modules,

would have at least a Bachelor of Science degree in electrical engineering or computer engineering, and at least five years of industry experience designing memory devices and memory modules. Alternatively, one of ordinary skill in the art would have an M.S. degree in electrical engineering or computer engineering, and at least three years of industry experience designing memory devices and memory modules. Moreover, one of ordinary skill in the art would have a Ph.D. degree in electrical engineering or computer engineering, and have at least one year of industry experience designing memory devices and memory modules.

Ex. 2019 ¶ 21.

The difference in the opinions of the two experts is essentially that Dr. Sechen believes a person of ordinary skill in the art would have additional experience, education, or a combination of the two, beyond that proposed by Dr. Bagherzadeh. Both experts have been employed as professors since the mid 1980s. Dr. Bagherzadeh also has some experience working with engineers in the industry by virtue of his employment with AT&T Bell Labs from 1980–1984 (Ex. 1009 ¶ 6; *id.* Appx. A). Both experts were awarded bachelors and masters degrees in electrical engineering. Dr. Bagherzadeh was awarded a doctorate in computer engineering, while Dr. Sechen was awarded a doctorate in electrical engineering.

Based on a comparison of Dr. Bagherzadeh's and Dr. Sechen's declarations (Exs. 1009, 2019), we find Dr. Bagherzadeh has a broader range of educational and work experience, and, therefore, accord greater weight to his opinion as to the level of ordinary skill in the art. Dr. Bagherzadeh's opinion is consistent with our finding as to the level of ordinary skill in the art in our Final Written Decision in IPR-971. We, therefore, adopt our finding in IPR-971 as to the level of ordinary skill in the art:

[A] person of ordinary skill in the art at the time of the '501 patent would have a Bachelor's degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.

IPR-971 FWD, 10–11 (noting that our finding was “based on our review of the '501 patent and the types of problems and solutions described in the '501 patent and cited prior art”).

III. CLAIM CONSTRUCTION

In its Petition, Petitioner offers specific constructions for the claim terms “operatively coupled,” “ports,” “to generate,” “[operated/operating [. . .] independently,” and “cyclic data.” Pet. 19–22. We determined that, for purposes of our Institution Decision, only the term “generate” required express construction. Dec. on Inst. 7. We interpreted “generate” as meaning “produce” or “cause.” *Id.* at 9.

In its Response, Patent Owner disagrees with our interpretation of “generate” as meaning “cause.” PO Resp. 17–18. Patent Owner contends our “Institution Decision also provides a construction for ‘configured to generate,’” as recited in claim 1, and proposes its own construction for this phrase. *Id.* at 9 (citing Dec. on Inst. 9). In its Reply, Petitioner asserts that our interpretation of “generate” in the Institution Decision is correct (Reply 1), and offers a proposed construction for the term “configured” (*id.* at 9–10). Patent Owner’s and Petitioner’s proposed constructions for the disputed claim language are listed in the table below.

Claim Term/Phrase	Patent Owner's Proposed Construction	Petitioner's Proposed Construction
generate	produce (PO Resp. 17)	cause or initiate transmission (Pet. 20–21) produce or cause (Reply 1)
configured to generate	programmed to generate/produce (<i>see</i> PO Resp. 9)	arranged or prepared to produce or cause (Reply 9–10)

In support of their proposed constructions of the disputed claim language, the parties rely on language in the '501 patent and extrinsic evidence. *See generally*, Pet. 20–21; PO Resp. 8–33; Reply 1–17. Neither party relies on prosecution history disclaimer in support of its proposed constructions.

In an *inter partes* review, the Board interprets claim terms in an unexpired patent according to the broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b); *see also In re Cuozzo Speed Techs., LLC*, 793 F.3d 1268, 1278–80 (Fed. Cir. 2015) (“Congress implicitly approved the broadest reasonable interpretation standard in enacting the AIA,” and “the standard was properly adopted by PTO regulation.”), *cert. granted, sub nom. Cuozzo Speed Techs. LLC v. Lee*, 136 S. Ct. 890 (mem.) (2016). Under that standard, and absent any special definitions, we give claim terms their ordinary and customary meaning, as would be understood by one of ordinary skill in the art at the time of the invention. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007); *see also Trivascular, Inc. v. Samuels*, No. 2015-1631, 2016 WL 463539, at *3 (Fed. Cir. Feb. 5, 2016) (“Under a broadest reasonable interpretation, words of the claim must be given their plain meaning, unless such meaning is inconsistent with the specification and prosecution

history.”). Any special definitions for claim terms must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

“[T]he Board may rely on dictionaries ‘so long as the dictionary definition does not contradict any definition found in or ascertained by reading the patent document.’” *Belden Inc. v. Berk-Tek LLC*, 610 Fed. App’x 997, 1002 (Fed. Cir. 2015) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1322–23 (Fed. Cir. 2005)). Expert testimony is useful to explain terms of art, and the state of the art at any given time, but cannot be used to prove “the proper or legal construction of any instrument of writing.” *Teva Pharm. USA, Inc. v. Sandoz, Inc.*, 135 S. Ct. 831, 841 (2015) (internal citations omitted); *see also* 37 C.F.R. § 42.65(a) (“Expert testimony that does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.”).

Only terms which are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999). A court may revisit and alter its construction of claim terms as the record in a case develops. *See Pressure Prods. Med. Supplies, Inc. v. Greatbatch Ltd*, 599 F.3d 1308, 1316 (Fed. Cir. 2010).

A. Generate

The terms “generate” and “generated” are used in both independent claims 1 and 16. Petitioner contends “[t]he Institution Decision correctly found that the intrinsic and extrinsic record supported construing ‘generate’ as ‘cause’ or ‘produce.’” Reply 1. Patent Owner contends “‘generate’ is properly construed to mean ‘produce,’” but “a construction of generate that

includes ‘cause’ is unsupported by – and contradictory to – the intrinsic and extrinsic evidence, and fails to provide reasonable clarity.” PO Resp. 17–18. The same respective constructions of “generate” were advanced by Patent Owner and the petitioner, Sandisk Corporation, in IPR-971. *See* IPR-971 FWD, 23–25.

We provided an extensive analysis of the term “generate” in our Final Written Decision in IPR-971. *See id.* at 22–30. “[W]e agree[d] with Patent Owner that the broadest reasonable interpretation of the claim term ‘generate’ is ‘produce’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’” *Id.* at 29. We interpreted the claim language “address and control signals *generated* by the control circuit” (claims 1 and 16) as “encompassing signals . . . that originated in the control circuit . . . , including by transformation or modification of information and/or data received from another component. *Id.* We also stated that “[w]e [did] not interpret this language as encompassing signals and data received by the . . . control circuit from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the control circuit.” *Id.* at 29–30.

In support of its contention that “generate” also should be interpreted as meaning “cause,” Petitioner relies on extrinsic evidence that was not before us in IPR-971, i.e., the Bagherzadeh Declaration (Ex. 1009) ¶ 28, ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms, Fourth Edition, 1988 (Ex. 1016, “IEEE”), and *Webster’s II New College Dictionary*, 2001 (Ex. 1032, “Webster’s II”). *See* Pet. 20–21; Reply 1–9. We consider this evidence as it is relevant to whether the term “generate” has a particular meaning to one of ordinary skill in the art. *See Trivascular*,

2016 WL 463539, at *3 (“Construing individual words of a claim without considering the context in which those words appear is simply not ‘reasonable.’ Instead, it is the ‘use of the words in the context of the written description and customarily by those of skill in the relevant art that accurately reflects both the “ordinary” and “customary” meaning of the terms in the claims.’” (quoting *Ferguson Beauregard/Logic Controls, Div. of Dover Res., Inc. v. Mega Sys., LLC*, 350 F.3d 1327, 1338 (Fed. Cir. 2003))).

The IEEE defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters.” Ex. 1016, 5. Dr. Bagherzadeh cites the IEEE definition in support of his testimony that “[t]he broadest reasonable interpretation of the claim [term] ‘to generate’ in claims 1 and 16, is ‘to cause or initiate transmission.’” Ex. 1009 ¶ 28 (also citing Ex. 1008, 6:3–5, 10:67–11:2, 6:14–17). Dr. Bagherzadeh testifies that “[t]his interpretation is consistent with the specification and extrinsic evidence because they describe generating as initiating or transmitting, which is also the broadest interpretation of ‘generate.’” *Id.*

We are not persuaded by Dr. Bagherzadeh’s testimony because it is devoid of specific facts and analysis as to what led him to conclude that a person of ordinary skill in the art would interpret the claim term “generate” in the manner Petitioner proposes. *See id.* Dr. Bagherzadeh does not explain how the IEEE definition supports an interpretation of generate as “cause or initiate transmission,” terms that do not appear in the IEEE definition. *See id.* Nor does Dr. Bagherzadeh identify other extrinsic evidence which he believes to be consistent with an interpretation of

“generate” as meaning “cause or initiate transmission.” *See id.* Likewise, Dr. Bagherzadeh has not explained why he believes the ’501 patent “describe[s] generating as initiating or transmitting” (Ex. 1009 ¶ 28), as these terms do not appear in the cited portions of the ’501 patent. *See* Ex. 1008, 6:8–10, 11:4–11:6, 6:19–22.¹ Accordingly, Dr. Bagherzadeh’s testimony does not persuade us to accept Petitioner’s interpretation of “generate” as meaning “cause.”

Petitioner contends “Dr. Bagherzadeh’s use of the IEEE definition specifically illustrates that ‘generate’ includes ‘selection . . . from a set,’ which implies that ‘generate’ must not be limited to creating from previously undefined values, which may be a more traditional meaning, but holds little value in the intrinsic evidence in this proceeding.” Reply 6. Petitioner has not identified corresponding testimony by Dr. Bagherzadeh or other evidentiary support for this statement and, therefore, this argument is

¹ We understand Dr. Bagherzadeh’s citations to “6:3–5, 10:67–11:2, 6:14–17” to be a typographical mistake. In IPR-1372, Dr. Bagherzadeh cites to the same column and line numbers in the ’434 patent in support of an interpretation of the claim term “generate” as meaning “cause or initiate transmission.” *See* IPR-1372, Ex. 1009 ¶ 28. The disclosures at the cited column and line numbers differ in the ’434 and ’501 patents. Based on the citations to the ’501 patent relied upon by Petitioner (*see* Pet. 20–21), we presume Dr. Bagherzadeh intended to rely on column 6, lines 8–10 and 19–22, and column 11, lines 4–6 of the ’501 patent, which disclosure is identical to column 6, lines 3–5 and 14–17, and column 10, line 67–column 11, line 2 of the ’434 patent. *Compare* Ex. 1008, 6:8–10, 11:4–6, 6:19–22 *with* IPR-1372, Ex. 1008 (the ’434 patent), 6:3–5, 10:67–11:2, 6:14–17. This disclosure, like the disclosure in the ’501 patent cited in paragraph 28 of Dr. Bagherzadeh’s declaration, does not include the terms “initiating or transmitting.”

likewise unpersuasive. *See id.*

In support of its construction of “generate” as meaning “cause,” Petitioner also relies on a finding in our Institution Decision that such construction is supported by Merriam-Webster’s Collegiate Dictionary, wherein one definition of “generate” is “to be the cause of” (Ex. 2009, 3). Reply 2 (citing Dec. on Inst. 9 [sic, 8]). This argument is not persuasive because the definition in Merriam-Webster’s Collegiate Dictionary contradicts the definition of “generate” that we ascertained upon reading the ’501 patent as discussed in our Final Written Decision in IPR-971. *See Belden*, 610 Fed. App’x at 1002.

In sum, we are not persuaded that Petitioner’s extrinsic evidence supports a finding that a customary meaning of “generate” to one of ordinary skill in the art is “cause.”² We, therefore, determine the broadest reasonable construction of “generate” is “produce” for the reasons stated above and in our Final Written Decision in IPR-971. We adopt and incorporate by reference our analysis and construction of the term “generate” in our Final Written Decision in IPR-971 (IPR-971 FWD, 22–30).

B. Configured to

The term “configured to” is used in independent claim 1, but not in independent claim 16. In our Final Decision in IPR-971, we construed “configured to” as “designed to, adapted to, or arranged to.” *See* IPR-971 FWD, 18–22. Petitioner contends “configured” should be construed as

² Petitioner relies on *Webster’s II*, for a definition of “cause.” *See* Reply 4, 6–8 (citing Ex. 1032, 3). Having determined the broadest reasonable interpretation of “generate” does not encompass “cause,” we do not find this evidence relevant to our claim construction analysis.

“arranged” or “prepared.” Reply 9–10. Responsive to this Board panel’s questioning during the oral hearing, however, Petitioner agreed that “designed to, adapted to, or arranged to” was an acceptable definition for the claim term “configured to.” *See* Tr. 22:7–17.

Patent Owner contends the phrase “configured to generate” should be construed as “programmed to generate.” PO Resp. 9. Patent Owner does not propose a separate construction for the term “configured.” *See generally id.* at 9–17. In support of its proposed construction, Patent Owner relies on language in the ’501 patent and the testimony of its expert, Dr. Sechen. *See id.* Dr. Sechen testifies as to the meaning of “configured to generate,” but does not provide a separate discussion of the meaning of “configured to” outside the context of “configured to generate.” *See* Ex. 2019 ¶¶ 42–57.

As explained in our Final Decision in IPR-971, the term “configured to” is used in the ’501 patent in conjunction with numerous terms, such as “provide” and “selectively input,” and there is a presumption that the term “configured to” should carry the same meaning in each instance. *See* IPR-971 FWD, 18–19. The evidence relied on by Patent Owner in support of its proposed construction of the term “configured to” in the present *inter partes* review does not differ materially from the evidence relied upon by Patent Owner in IPR-971. *Compare* PO Resp. 9–17 *with* IPR-971, Paper 19, 14–19. Accordingly, we determine the broadest reasonable construction of “configured to” is “designed to, adapted to, or arranged to” for the reasons stated above and in our Final Written Decision in IPR-971. We adopt and incorporate by reference our analysis and construction of the term “configured to” in our Final Written Decision in IPR-971 (IPR-971 FWD, 18–22).

In summary, we construe the claim terms in controversy as follows:

Claim Term/Phrase	Interpretation
Generate	Produce
Configured to	Designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]
Configured to Generate	Designed, adapted, or arranged to produce

IV. ALLEGED ANTICIPATION OF CLAIMS 1, 2, 4–6, 9–11, 14–17, and 20 BY AVERBUJ

A. Principles of Law

To prevail in its challenges to the patentability of the claims, a petitioner must establish facts supporting its challenges by a preponderance of the evidence. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference.

Net MoneyIN, Inc. v. VeriSign, Inc., 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). While the elements must be arranged or combined in the same way as in the claim, “the reference need not satisfy an *ipsissimis verbis* test.” *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009).

A single prior art reference that discloses, either expressly or inherently, each limitation of a claim invalidates that claim by anticipation. Thus, a prior art reference without express reference to a claim limitation may nonetheless anticipate by inherency. “Under the principles of inherency, if the prior art necessarily functions in accordance with, or includes, the claims limitations, it anticipates.”

Perricone v. Medicis Pharm. Corp., 432 F.3d 1368, 1375–76 (Fed. Cir.

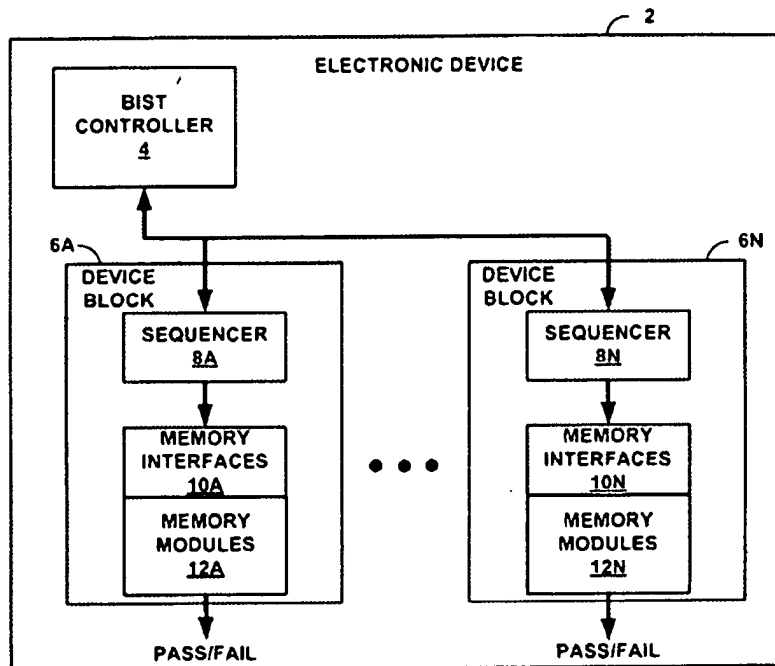
2005) (citations omitted). “In general, a limitation or the entire invention is inherent and in the public domain if it is the ‘natural result flowing from’ the explicit disclosure of the prior art.” *Schering Corp. v. Geneva Pharms., Inc.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003).

“[T]he Board is not bound by any findings made in its Institution Decision. . . . The Board is free to change its view of the merits after further development of the record, and *should do so* if convinced its initial inclinations were wrong.” *Trivascular*, 2016 WL 463539, at *9.

We analyze the instituted ground of unpatentability in accordance with the above-stated principles.

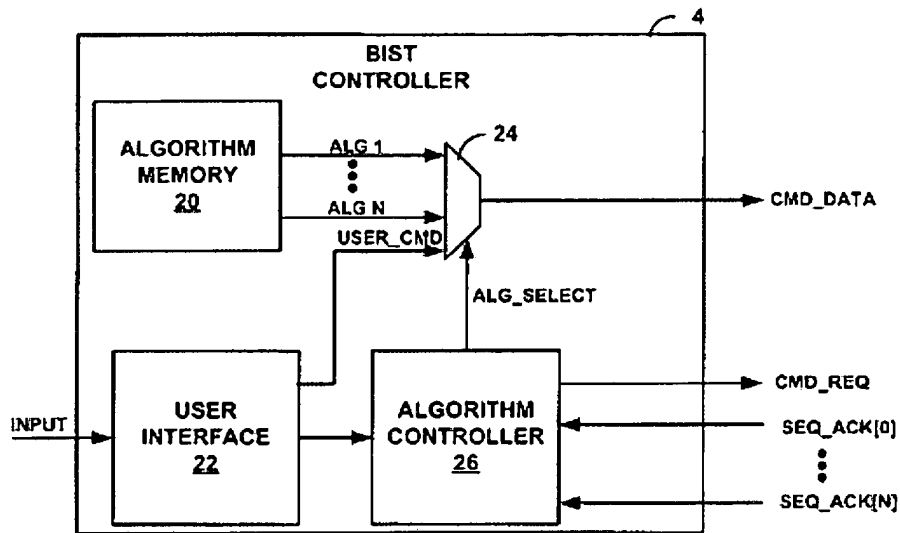
B. Averbuj (Ex. 1011)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1011, 1:64–65, 2:3–4. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* at 3:62–64. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* at 3:64–67. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* at 3:67–4:3.

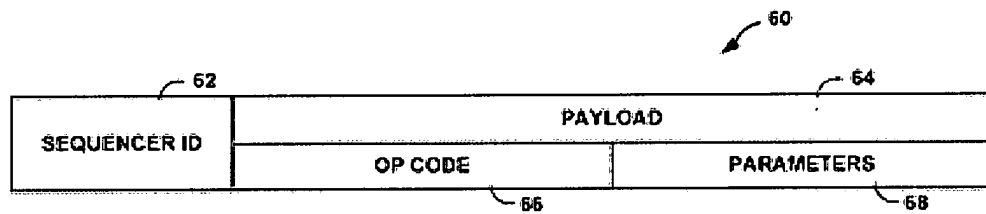
BIST controller 4 is illustrated in Figure 2, reproduced below.



Averbuj Figure 2, above, “is a block diagram illustrating an example embodiment of a BIST controller.” *Id.* at 3:39–40. BIST controller 4 includes algorithm controller 26 that can be invoked either by a user through user interface 22 or automatically upon power-up of electronic device 2. *Id.* at 5:12–16. “Once invoked, algorithm controller 26 provides an algorithm select signal (ALG_SELECT) to multiplexer 24 to select one of the algorithms stored within algorithm memory 20.” *Id.* at 5:16–19. Alternatively, user interface 22 may programmably receive algorithms via external input, and deliver the received algorithms to multiplexer 24. *Id.* at 5:41–43. The algorithms, whether stored in algorithm memory 20 or received via external input, have a similar form, i.e., “a sequence of binary commands in which each command defines a test within the overall algorithm.” *Id.* at 5:43–47. BIST controller 4 provides and communicates the selected algorithm to sequencers 8 for application to device blocks 6 as a stream of binary commands (CMD_DATA), “each command specif[ying] an operational code [(OP CODE)] and a set of parameters that define one or

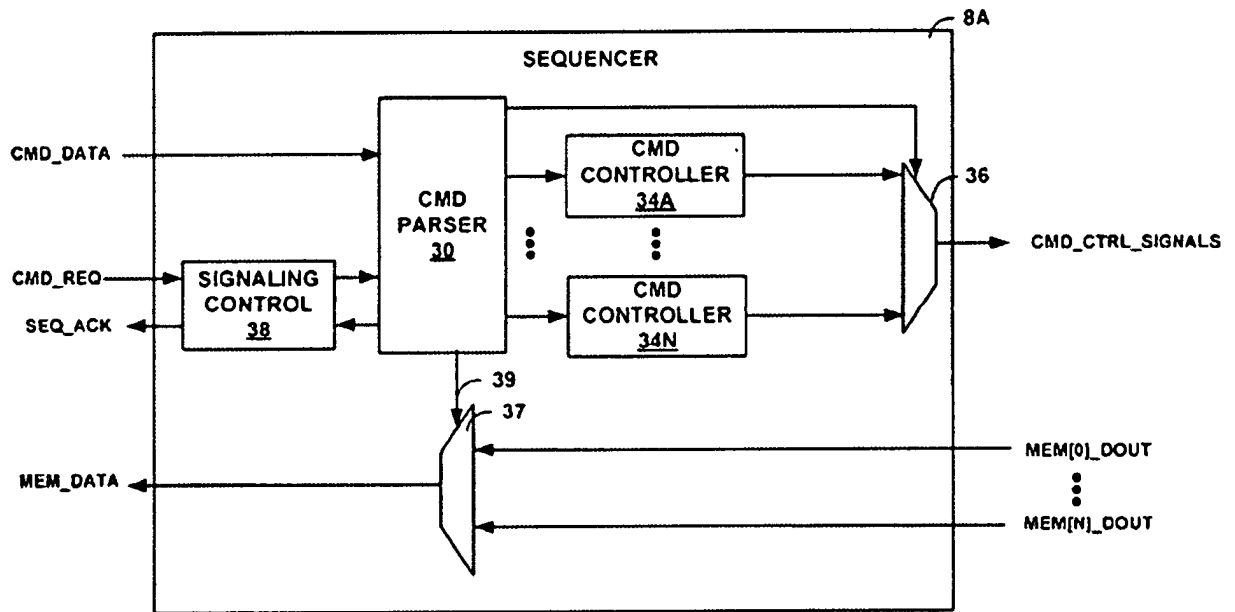
more memory operations without regard to the physical characteristics or timing requirements of memory modules 12.” *Id.* at 4:4–12, 5:19–21.

Figure 8, reproduced below, “is a block diagram illustrating an example data structure of a command issued by BIST controller 4.” *Id.* at 9:15–16.



In the embodiment shown in Figure 8, above, “command 60 includes a sequencer identifier (ID) 62, and a payload 64. Sequencer ID 62 identifies a sequencer, e.g., sequencer 8A, to which command 60 is being issued.” *Id.* at 9:17–20. “Payload 64 of command 60 carries binary data that defines the command itself. In particular, payload 64 includes an operational code (OP CODE) 66 and a set of parameters 68.” *Id.* at 9:29–31. In one embodiment, for example, “OP CODE 66 and parameters 68 comprise three bits and twenty-nine bits, respectively, to form a 32-bit command.” *Id.* at 9:58–60. “OP CODE 66 specifies a particular function to be performed by the receiving sequencers 8.” *Id.* at 9:32–33. An exemplary OP CODE is SET ADDRESS (OP CODE 100), which “[s]ets a specific starting address as well as a maximum address limit for a test algorithm as applied to the memory modules.” *Id.* at Table 1.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* at 3:46–47.



As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 that receives command data (CMD_DATA) from BIST controller 4. *Id.* at 6:24–26. Command parser 30 processes the commands from BIST controller 4 to identify a specified operation, e.g., by identifying an OP CODE specified by the command. *Id.* at 6:26–29. Based on the specified operation, command parser 30 may extract one or more parameters from the command, and pass the extracted parameters to a selected command controller (i.e., one of command controllers 34A–34N). *Id.* at 6:30–37. The invoked command controller, in turn, issues a sequence of one or more operations to each memory interface 10, sequentially driving the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. *Id.* at 6:41–43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with

the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* at 4:45–49.

C. Arguments

Independent claims 1 and 16 recite, respectively, “[a] memory system” and “[a] method of operating a memory system.” Both claims recite “a plurality of memory chips” and “a plurality of data handlers.” Petitioner contends the claimed “memory chips” read on Averbuj’s memory modules 12 and the claimed “data handlers” read on the combination of sequencers 8 and memory interfaces 10. Pet. 23. Independent claim 1 further recites “a control circuit *configured to generate address and control signals*, wherein the memory system is configured to test the one or more memory chips using the *address and control signals generated by the control circuit*.” Ex. 1008, claim 1 (emphasis added). Independent claim 16 recites steps of “operating a *control circuit to generate address and control signals*” and “testing the one or more memory locations of the one or more memory chips using the *address and control signals generated by the control circuit*.” *Id.*, claim 16 (emphasis added). With respect to the above-quoted limitations relating to generating/generated address and control signals, the Petition states:

the BIST controller 4 of Averbuj ‘442 represents circuitry of a control circuit that provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE* for testing the memory components 12A–N. Ex. 1011, Fig. 2 and Column 9: Table 1. The *SET ADDRESS* and *CMD_REQ* signals include address and control signals for the testing functions. Since the BIST controller 4 provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE*, a POSITA [(“person of ordinary skill in the art”)] would understand that the BIST controller 4 is programmable to provide address and

control signals as recited in part by the Challenged Claims. Ex. 1009, Bagherzadeh Decl., ¶ 33.

Pet. 24.

Patent Owner argues Averbuj's BIST controller 4 does not "generate" the OP CODE communicated to sequencers 8, and Petitioner has not met its burden to show that Averbuj's BIST controller 4 is "configured to generate" (claim 1) or can be "operat[ed] . . . to generate" (claim 16) address signals for testing the memory chips in the manner recited in claims 1 and 16 of the '501 patent. PO Resp. 45–46. As discussed in greater detail below, we find this argument persuasive, and, for this reason, conclude Petitioner failed to meet its burden to prove unpatentability of claims 1, 2, 4–6, 9–11, 14–17, and 20 of the '501 patent.³

D. Expert Testimony

In support of its argument that Averbuj's BIST controller 4 is not "configured to generate" (claim 1) and cannot be "operat[ed] . . . to generate" (claim 16) address signals, Patent Owner relies on paragraphs 133–141 of the Sechen Declaration. *See id.* at 45–49. Patent Owner also relies on Dr. Bagherzadeh's cross-examination testimony (*id.* at 46–47 (quoting Ex. 2012, 55:6–56:9)) regarding Averbuj Figure 2 (*see* Ex. 2012, 53:4–5; *see generally id.* at 53:4–56:9).

With reference to Averbuj Figure 2 (*see* Section IV.B., above, wherein this figure is reproduced and described), Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 are

³ Because we are persuaded by this argument that Petitioner has not met its burden to show unpatentability, we find it unnecessary to address the additional arguments advanced by Patent Owner in its Response.

generated by hand or by a tool (Ex. 2012, 55:20–56:9), and he identifies a compiler as a tool for generating such code (*id.* at 30:23–24, 41:1–4; *see id.* at 55:6–25).⁴ Dr. Bagherzadeh testifies that a compiler is not illustrated in Averbuj Figure 2, and that he did not recall any mention of the word “compiler” in Averbuj. *See* Ex. 2012, 54:8–55:5. Dr. Bagherzadeh further testifies that “99 percent of the designs or chips do not have a compiler on-board.” *Id.* at 54:21–22. Dr. Bagherzadeh testifies that Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and “the output [] reflect[s] what’s in the input, based on the selection.” *Id.* at 54:8–13. Dr. Bagherzadeh testifies that assuming “no other hardware components between . . . [CMD_DATA] and the input,” CMD_DATA is stored in the algorithm memory. *Id.* at 54:15–19.

⁴ Petitioner contends “[t]he discussion of a compiler, from PO’s deposition of [Dr. Bagherzadeh], was hypothetical and specific to an article coauthored by [Dr. Bagherzadeh].” Reply 22 (citing Ex. 2012, 15:2–19:3).⁴ We were unable to locate a discussion of a compiler or any reference to an article coauthored by Dr. Bagherzadeh in the cited portions of Dr. Bagherzadeh’s deposition transcript. Dr. Bagherzadeh’s first discussion of a compiler appears to be in connection with a use thereof in MorphoSys (*see* Ex. 2012, 30:12–20), a chip fabricated by Morpho Technologies (*id.* at 19:2–9), a company co-founded by Dr. Bagherzadeh (Ex. 1009 ¶ 7). An article relating to MorphoSys and co-authored by Dr. Bagherzadeh (*see* Ex. 2014, “MorphoSys: Case Study of A Reconfigurable Computing System Targeting Multimedia Applications”) is introduced on page 39 of the deposition transcript. *See* Ex. 2012, 39:19–22, cited in PO Resp. 25. In any event, Petitioner’s argument is not persuasive because we find the testimony on pages 54–56 of the Bagherzadeh transcript supports Patent Owner’s argument. *See* Section IV.E. *infra*.

Dr. Sechen testifies that

the Petition fails to identify every element of claims 1, 20, and 29 [sic, 1 and 16] for at least two reasons[:] (1) because the CMD_REQ and SET ADDRESS are not generated (“produced” or “caused”) by the purported control module (Averbuj’s BIST controller 4), and [(2)] because the OP CODE is not generated (“produced” or “caused”) by the alleged control module (Averbuj’s BIST controller 4).

Ex. 2019 ¶ 136. Dr. Sechen testifies, more specifically, that the Petition fails to explain how Averbuj’s BIST controller 4 “generate[s]/produce[s] OP CODE,” “because the Petition does not state that Averbuj’s BIST Controller includes a compiler, as described by Dr. Bagherzadeh.” *Id.* ¶ 138. Dr. Sechen testifies that “[b]ecause SET ADDRESS (OP CODE 100) is a predefined 3-bit sequence, . . . it is not ‘generated’ by Averbuj’s BIST Controller.” *Id.* ¶ 140. Dr. Sechen further testifies that “command 60, which carries the OP CODE [] (e.g. SET ADDRESS), . . . is not ‘generated’ by Averbuj’s BIST Controller,” but “is merely stored in memory and is output as part of ‘a stream of binary commands.’ (Ex. 1011 at 5:1–21.)” Ex. 2019 ¶ 141.

E. Analysis

After considering the parties’ arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses a control circuit that is “configured to generate” (claim 1) or can be “operat[ed] . . . to generate” (claim 16) address signals for testing the memory chips in the manner recited in claims 1 and 16 of the ’501 patent.

As noted in Section III, above, in its Petition, Petitioner proposes construing the claim term “generate” as “cause or initiate transmission” (Pet. 20–21). The Petition cites the following disclosure in the ’501 patent in

support of this construction: “the data module 28 and/or the control circuit 22 are configured to provide memory signals (e.g., data, address and control signals)’ Ex. 1008, ’501 patent, 6:8-10 (emphasis added).” Pet. 20. The Petition states that Averbuj teaches “a control circuit configured to generate” (claim 1) and “operating a control circuit to generate” (claim 16) address signals for testing the memory chips as recited in claims 1 and 16, because Averbuj’s “BIST controller 4 is programmable *to provide* address and control signals.” Pet. 24 (emphasis added); *see* PO Resp. 49–50.

For purposes of our Institution Decision, we interpreted “generate” as used in the ’501 patent claims as meaning “cause” or “produce.” *See* Dec. on Inst. 9. As indicated in Section III.A., above, subsequent to that decision and the oral hearing in the present *inter partes* review, we again considered the meaning of the claim term “generate” in our final decision in IPR-971, and determined that the broadest reasonable interpretation of the ’501 patent claim term “generate” is “produce,” and that “generate” does not mean “cause” or “cause to produce.” *See* IPR-971 FWD, 22–30. In Section III.A., above, we considered extrinsic evidence in support of Petitioner’s interpretation of “generate” as meaning “cause” that was not before us in IPR-971, but concluded the broadest reasonable interpretation of “generate” is “produce.” We, therefore, adopted and incorporated by reference in the present final decision our analysis and construction of the terms “generate” and “configured to generate” in our Final Written Decision in IPR-971. *See* Section III., above.

In construing the claim term “generate” in our final decision in IPR-971, we found that the terms “provide” and “generate” are used in the ’501 patent to describe different embodiments of the control circuit, and

determined the claim term “generate” did not mean “provide.” *See* IPR-971 FWD, 25–28. We also determined that the phrase “address and control signals generated by the control circuit” (claims 1 and 16) encompasses signals that originated in the control circuit, including by transformation or modification of information received from another component. IPR-971 FWD, 29. We further determined the broadest reasonable interpretation of this claim language does not encompass “signals . . . received by the control circuit . . . from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the control circuit.” *Id.* at 29–30. We stated that “generate,” as used in the ’501 patent, does not encompass the selection function of a multiplexer. *Id.* at 27.

Petitioner’s contention that Averbuj teaches “a control circuit configured to generate address . . . signals” (claim 1) and “operating a control circuit to generate address . . . signals” (claim 16) for testing memory chips, as recited in independent claims 1 and 16, is based on an interpretation of “generate” as meaning “cause.” Pet. 20, 24; *see also*, Reply 20–21 (asserting that Averbuj discloses the above-quoted limitations because “the BIST controller of Averbuj *causes* the CMD_REQ and SET ADDRESS signals, representing the claimed ‘address and controls signals’” (emphasis added)) (citing Dec. on Inst. 15–16). Petitioner has not explained sufficiently, however, how Averbuj’s BIST controller 4 is “configured to generate” (i.e., designed, adapted, or arranged to produce) or “operat[ed] . . . to generate” (i.e., operated to produce) address signals as recited in independent claims 1 and 16, respectively.⁵

⁵ During oral argument, Judge Clements asked Mr. Heafey, counsel for

Moreover, the evidence in this proceeding fails to support a finding that Averbuj's BIST controller 4 generates address signals for testing the memory devices in the manner claimed.

In Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS (*see* Ex. 1011, 4:4–12, 5:19–21, Table 1)) communicated to sequencers 8 are either received by user interface 22 via external input (*id.* at 5:41–43) or generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20 (*see* Ex. 2012, 30:23–24, 41:1–4, 54:21–22, 55:6–25, 55:20–56:9 (wherein Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 are generated by hand or by a tool, e.g., a compiler, and that “99 percent of the designs or chips do not have a compiler on-board”)). Multiplexer 24 *selects* one of the algorithms (i.e., CMD_DATA) and BIST controller 4 *provides* to sequencers 8 the same CMD_DATA (e.g., address signals) without transformation or modification. *See* Ex. 1011, 4:4–12, 5:16–21; Ex. 2012, 54:8–13 (“Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and the output [] reflect[s] what’s in the input, based on the selection.”). Based on our

Petitioner, the following question: “Assuming we adopt ‘to produce’ as the proper construction [of ‘to generate’], are [CMD]_REQ and SET ADDRESS produced by BIST controller 4?” Tr. 78:16–18. In response, Mr. Heafey stated: “[T]hose signals are definitely produced by BIST controller 4, yes, they are.” *Id.* at 78:19–20. Mr. Heafey did not elaborate, however, nor did he otherwise explain clearly during oral argument how Averbuj's BIST controller 4 is designed, adapted, or arranged to produce address signals for testing the memory devices.

interpretation of the term “generate,” as discussed above, an “address . . . signal[] generated by the control circuit,” as recited in claims 1 and 16, does not encompass Averbuj’s SET ADDRESS signal, because it is produced by a component external to BIST controller 4, and merely selected by multiplexer 24 and provided to sequencers 8 without transformation or modification by BIST controller 4.

In sum, after considering the parties’ arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses “a control circuit configured to generate address . . . signals” (claim 1) and “operating a control circuit to generate address . . . signals” (claim 16) for testing memory chips in the manner recited in independent claims 1 and 16. Petitioner does not correct this deficiency in its challenges as to dependent claims 2, 4–6, 9–11, 14, 15, 17, and 20. *See* Pet. 26–34; 35 U.S.C. § 112, ¶ 4 (“A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.”). Because Petitioner has not identified in Averbuj a teaching of each and every limitation recited in claims 1, 2, 4–6, 9–11, 14–17, and 20, Petitioner has not met its burden to show these claims are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj.

V. MOTIONS TO EXCLUDE

Both Patent Owner and Petitioner filed Motions to Exclude.

A. Petitioner’s Motion to Exclude

Petitioner moves to exclude Exhibit 2018 (the Deposition Transcript of Dr. Donald Alpert) and to strike Section IV(B)(5)(a) of Patent Owner’s Response, wherein Patent Owner quotes Dr. Alpert’s deposition testimony in

support of its proposed construction of the claim term “generate” (*see* PO Resp. 31–32). Paper 25, 2. Petitioner argues Dr. Alpert is Sandisk Corporation’s expert in IPR-971 and that Petitioner was not provided notice or otherwise invited to attend Dr. Alpert’s deposition. *Id.* at 3. Petitioner contends Dr. Alpert’s statements are not admissible as evidence in this proceeding “because they are hearsay and will unfairly prejudice Petitioner.” *Id.* at 2.

In construing the claim term “generate” in Section III.A., above, we adopted and incorporated by reference our analysis and construction of this term on pages 22–30 of our Final Written Decision in IPR-971. In our claim construction in IPR-971, we considered Dr. Alpert’s deposition and declaration testimony, but did not quote or cite to the testimony relied on by Patent Owner in Section IV(B)(5)(a) of its Patent Owner Response in the present proceeding. *Compare* IPR-971 FWD, 23–24 *with* PO Resp. 31–32. In IPR-971, we determined Dr. Alpert’s testimony that one of ordinary skill in the art would understand the claim term “generate” as meaning “cause” or “produce” was inconsistent with the ’501 patent’s implicit definition of this term as meaning only “produce.” *See* IPR-971 FWD, 27 n.4. We, therefore, did not rely on Dr. Alpert’s testimony in our construction of the claim term “generate” in IPR-971, or in the present case. *See id.*; *see also id.* at 32 (“Having considered the ordinary meaning of the term ‘generate’ in the context of both the claims and the ’501 patent as a whole, we agree with Patent Owner that the broadest reasonable interpretation of the claim term ‘generate’ is ‘produce,’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’”).

Because we did not rely on Exhibit 2018 or consider the argument in Section IV(B)(5)(a) of Patent Owner's Response in rendering our decision in this proceeding, we dismiss Petitioner's Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) as moot.

B. Patent Owner's Motions to Exclude

Patent Owner filed a Motion to Exclude "Exhibits 1026, 1029, 1030, and 1032 (or portions thereof)" and requests that we strike the corresponding citations to these exhibits in Petitioner's Reply. Paper 33, 2. We did not rely on these exhibits in rendering our decision. Accordingly, Patent Owner's Motion to Exclude these exhibits is dismissed as moot.

Patent Owner also filed a Motion to Exclude Portions of Petitioner's Reply, contending "Petitioner's Reply improperly presents new theories, never raised in its Petition or adopted by the Board's Institution Order, violating 37 CFR 42.22-23 and applicable law." Paper 34, 1. Patent Owner provides a table identifying the alleged new theories advanced by Petitioner in its Reply (*id.* at 2) and identifies by page and line number specific portions of sections II.A.3, II.B.1.a, II.B.5 & III.B.2 of the Reply that it contends should be excluded as advancing new theories (*see id.* at 2-8).

Our decision is not based on any of the alleged new theories identified by Patent Owner, and Patent Owner does not cite to any of the specific language in the Reply that Patent Owner contends should be excluded. Accordingly, Petitioner's Motion to Exclude portions of Petitioner's Reply is dismissed as moot.

VI. CONCLUSION

Petitioner has not shown by a preponderance of the evidence that claims 1, 2, 4–6, 9–11, 14–17, and 20 of the '501 patent are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2).

VII. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1, 2, 4–6, 9–11, 14–17, and 20 of U.S. Patent No. 8,359,501 B1 are not held unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2);

FURTHER ORDERED that Petitioner's Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) of Patent Owner's Response is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exhibits 1026, 1029, 1030, and 1032 to strike the corresponding citations to these exhibits in Petitioner's Reply is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exclude Portions of Petitioner's Reply is *dismissed*; and

FURTHER ORDERED that, because this is a final decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2014-01374
Patent 8,359,501 B1

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01374
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
MATTHEW R. CLEMENTS, *Administrative Patent Judges*.

CLEMENTS, *Administrative Patent Judge*, dissenting.

While I agree with the majority's construction of "generate" to mean "produce," I respectfully dissent from the majority's determination that "Petitioner has not explained sufficiently, however, how Averbuj's BIST controller 4 is 'configured to generate' (i.e., designed, adapted, or arranged to produce) or 'operat[ed] . . . to generate' (i.e., operated to produce) address signals as recited in independent claims 1 and 16, respectively." Majority Opinion ("Maj. Op.") 30–31. Although Petitioner's arguments are based on a construction of "generate" that includes cause, and we have construed

“generate” to mean “produce,” I believe the evidence cited by Petitioner nevertheless establishes sufficiently that BIST controller 4—identified by Petitioner as the recited “control circuit”—“produces” address signals.

The majority opinion finds that “[i]n Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS) communicated to sequencers 8 are either received by user interface 22 via external input or generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20.” Maj. Op. 31 (citations omitted). As support for its finding that the algorithms are generated by another component, the majority cites the cross-examination of Petitioner’s expert, Dr. Bagherzadeh, in which he identifies a person or tool as possible ways of generating the OP CODEs stored in algorithm memory 26:

Q Would a compiler be required to generate the op codes?

[Objection]

THE WITNESS: Which op codes?

BY MR. BRADLEY:

Q The op codes stored in algorithm memory?

A Required, no.

Q In what other way could code be generated for algorithm memory?

A By hand, as I explained to you.

Q Are there any other ways of generating the code?

[Objection]

THE WITNESS: Those codes that are residing in the memory, right, that's all that we're talking about?

BY MR. BRADLEY:

Q Right.

A So either you do it by hand or you have a tool do it.

Q Okay.

A There's no -- we're talking about code, not the generation of the -- or signals that are generated, just the bits in the code?

Q The generation of the op codes, yes.

A Yes. They have to be stored in the memory. Somebody did it by hand, wrote the code, or by a tool.

Q Okay.

A There's no other way around it.

Ex. 2012, 54:20–56:9.

The majority appears to understand this testimony by Dr. Bagherzadeh as a description of how BIST controller 4 works *in operation*. I understand this testimony, however, to explain only how the data stored in algorithm memory 26 could have been created and stored in algorithm memory 26 *in the first instance*. At no point in this testimony does Dr. Bagherzadeh suggest that a compiler plays any role during *operation* of BIST controller 4. Indeed, that would be inconsistent with Averbuj's explicit disclosure.

Averbuj explains that the *only* input received by BIST controller 4 is an invocation of some sort, whether an “external input, such as a control signal from an external testing apparatus” or an automatic invocation, and, as a result of that input, CMD_DATA, among other signals, is output:

User interface 22 invokes algorithm controller 26 in response to external input, such as a control signal from an external testing

apparatus. Alternatively, algorithm controller may be automatically invoked upon power-up of electronic device 2. Once invoked, algorithm controller 26 provides an algorithm select signal (ALG_SELECT) to multiplexer 24 to select one of the algorithms stored within algorithm memory 20. Once selected, a stream of binary commands that comprises the selected algorithm is applied to device blocks 6 as command data (CMD_DATA).

Ex. 1011, 5:12–21. Thus, in this embodiment, a control signal goes into BIST controller 4 and CMD_DATA, *inter alia*, comes out of BIST controller 4. In this embodiment, BIST controller 4 does not propagate signals through to its output because it does not receive CMD_DATA at user interface 22. For that reason, I find that BIST controller 4 “produces” CMD_DATA, including, *inter alia*, SET ADDRESS, which Petitioner identifies as the recited “address . . . signal[]” (Pet. 24).

The fact that the data output by BIST controller 4—i.e., CMD_DATA—is read from a memory within BIST controller 4 so that it can subsequently be output goes to *how* BIST controller 4 “produces” that data. It does not show that the data is “produced” by another component. Moreover, the fact that the data in algorithm memory 26 was first created by something other than BIST controller 4 before being stored in algorithm memory 26, e.g., at the time of manufacture, does not, in my view, preclude BIST controller 4 from “producing” that data during *operation* of the built-in self-test circuit. Finally, the fact that BIST controller 4 does not include a compiler is irrelevant, in my view, because BIST controller 4 does not need a compiler in order to “produce” address and control signals; when “invoked,” it “produces” them by retrieving them from its memory—i.e., algorithm memory 26—and outputting them.

The *only* scenario in which BIST controller 4 “receives address signals from another component . . . and merely outputs the same address signals,” as the majority opinion finds, is in an alternative mode of operation described at column 5, lines 41–49, where Averbuj discloses that “user interface 22 may programmably receive algorithms via external input” which are delivered directly to multiplexer 24. That, however, is not the mode of operation relied upon by Petitioner.

For all of the foregoing reasons, I disagree with the majority’s finding that “[i]n Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS) communicated to sequencers 8 are . . . generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20.” Maj. Op. 31.

Because I am persuaded that Petitioner has established sufficiently that BIST controller 4 “produces” the recited “address and control signals,” I would have analyzed the remainder of Patent Owner’s arguments.

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SANDISK CORPORATION,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-00971
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
GEORGIANNA W. BRADEN, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. § 318(a) and 37 C.F.R. § 42.73

I. BACKGROUND

Sandisk Corporation (“Petitioner”) filed a Petition (Paper 1, “Pet.”) on June 18, 2014, requesting institution of an *inter partes* review of claims 1–20 (the “challenged claims”) of U.S. Patent No. 8,359,501 B1 (Ex. 1001, “the ’501 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 10, “Prelim. Resp.”). Based on these submissions, an *inter partes* review of claims 1–20 was instituted on December 16, 2014, pursuant to 35 U.S.C. § 314. Paper 12 (“Dec. on Inst.” or “Institution Decision”).

After institution, Patent Owner filed a Response (Paper 19, Corrected Patent Owner Response, “PO Resp.”), and Petitioner filed a Reply to the Patent Owner Response (Paper 25, “Reply”).

Patent Owner also filed a Motion for Observation Regarding Cross-Examination of Dr. Donald Alpert. Paper 28 (“PO Mot.”). Petitioner filed a Response to the Motion (Paper 32, “Pet. Resp.”). Oral argument was held on July 27, 2015. A transcript of the oral hearing (“Tr.”) has been entered into the record. *See* Paper 33.

The Board has jurisdiction under 35 U.S.C. § 6(c). This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73.

For the reasons discussed below, we determine Petitioner has shown by a preponderance of the evidence that claims 1–3, 5–20 of the ’501 patent are unpatentable. Petitioner has not met its burden to show that claim 4 is unpatentable.

A. Related Proceedings

Petitioner also requested, and we instituted, an *inter partes* review of U.S. Patent No. 8,001,434 (“the ’434 patent”). The ’501 patent claims priority as a continuation of the ’434 patent: *Sandisk Corp. v. Netlist, Inc.*,

IPR2014-00971
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Case IPR2014-00970 (PTAB December 16, 2014). Oral argument in IPR2014-00970 was consolidated with the oral argument in the present *inter partes* review. A final decision in IPR2014-00970 is issued concurrently with this final decision.

Patent Owner informs us that the '501 patent is the subject of an infringement action filed by Patent Owner: *Netlist, Inc. v. Smart Modular Technologies, Inc.*, No. 4:13-cv-05889-YGR (N.D. Cal. Dec. 19, 2013). Paper 9, 2. SMART Modular Technologies, Inc., a named defendant, filed two petitions for *inter partes* review of the '501 patent on August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01374 and IPR2014-01375 (PTAB). *Id.* at 3. On August 23, 2014, SMART Modular Technologies, Inc. also filed two petitions for *inter partes* review of the related '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01372 and IPR2014-01373 (PTAB). *Id.* On March 10, 2015, we instituted *inter partes* reviews in IPR2014-01372 and IPR2014-01374. IPR2014-01372, Paper 13; IPR2014-01374, Paper 12. Oral arguments for IPR2014-01372 and IPR2014-01374 were consolidated, and held on November 17, 2015.

B. The '501 patent (Ex. 1001)

The '501 patent relates to self-testing electronic memory modules. Ex. 1001, 1:27–28. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '501 patent, reproduced below. *See id.* at 9:25–26.

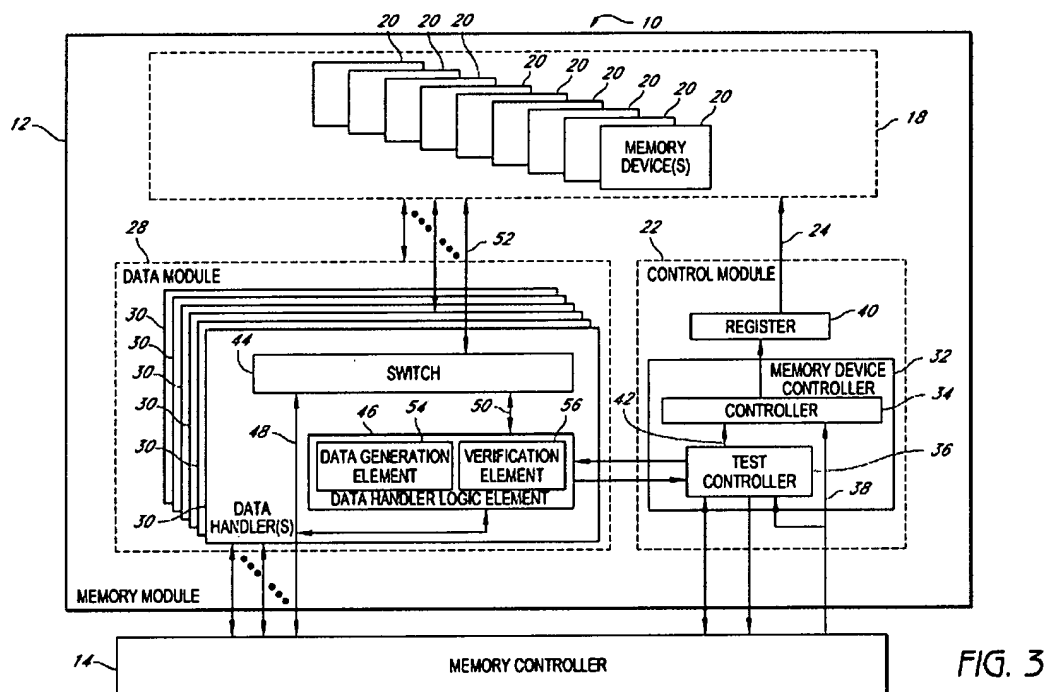


FIG. 3

As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:8–12. “[M]emory module 10 comprises a control module 22 [and] . . . a data module 28.” *Id.* at 5:12–15.

“[C]ontrol module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18.” *Id.* at 9:29–31. In the embodiment shown in Figure 3, control module 22 includes control mixer element 32 for controlling the address and control signals for the self-testing function. *Id.* at 9:35–37. Control mixer element 32 includes memory device controller 34 (e.g., a DRAM controller) and test controller 36. *Id.* at 9:33–35. “[T]est controller 36 controls the generation of the

address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28.” *Id.* at 9:53–56. “[M]emory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36.” *Id.* at 9:41–44.

Data module 28 comprises a plurality of data handlers 30 that “may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports” of memory devices 18. *Id.* at 5:14–16, 21–23. In the embodiment illustrated in Figure 3, each of data handlers 30 includes data handler logic element 46, comprising data generation element 54 and verification element 56. *Id.* at 10:31–34. “[D]ata generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports” of memory devices 18. *Id.* at 10:35–37. “The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information) the data handler logic element 46 receives from the control module 22.” *Id.* at 10:37–41.

C. Illustrative Claims

Independent claims 1 and 16 are illustrative of the claimed subject matter:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured

to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;

a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

16. A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:

operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;

operating a control circuit to generate address and control signals; and

testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

D. The Prior Art

The instituted grounds of unpatentability in this *inter partes* review are based on the following prior art:

Reference	Publication	Exhibit	Date
Averbuj	U.S. Patent Application Publication No. 2005/0257109 A1	1003	Nov. 17, 2005
Huang	<i>An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers</i> , Fourteenth International Conference on VLSI Design 379– 384 (2001), IEEE	1004	2001

Reference	Publication	Exhibit	Date
JEDEC Standard	JEDEC Standard, Fully Buffered DIMM (FBDIMM): DFX Design for Validation and Test, JESD82-28	1006	Feb. 2008

E. The Instituted Grounds of Unpatentability

We instituted the instant *inter partes* review of claims 1–20 on the following grounds:

Claims	Basis	Reference(s)
1–3, 5–8, and 11–20	§ 102(b)	Averbuj
1–5 and 14–16	§ 103(a)	Huang
6–8, 11, and 17–20	§ 103(a)	Huang and Averbuj
9 and 10	§ 103(a)	Huang and JEDEC Standard
9 and 10	§ 103(a)	Averbuj and JEDEC Standard

F. Expert Testimony

Petitioner relies on the testimony of Dr. Donald Alpert in support of its patentability challenges. Dr. Alpert executed a first declaration (Ex. 1007) in support of the Petition and a second declaration (Ex. 1009) in support of Petitioner’s Reply. Dr. Alpert was cross-examined on the subject matter of his first and second declarations, and transcripts of the testimony were filed as Exhibits 2013 and 2021, respectively.

Dr. Alpert testified that he “received an Electrical Engineering Ph.D. degree in 1984 from Stanford University.” Ex. 1007 ¶ 6. Dr. Alpert testified that he taught classes in computer architecture at Stanford, Tel Aviv, and Arizona State Universities (*id.*), and that he worked in the computer industry, i.e., designing microprocessors and components for Burroughs Corporation, Zilog, National Semiconductor, and Intel, at various times between 1976 and 1997 (*id.* ¶ 7). Dr. Alpert testified that his “own training and experience exceeds that of ordinary skill in the art.” *Id.* ¶ 21.

Patent Owner relies on the testimony of Dr. Carl Sechen. Dr. Sechen executed a declaration (Ex. 2012) in support of Patent Owner’s Response. Dr. Sechen was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 1010.

Dr. Sechen testified that he “was awarded a Ph.D. in electrical engineering from the University of California at Berkeley in 1986” (Ex. 2012 ¶ 6), and has been a Professor of Electrical Engineering for more than 28 years (*id.* ¶ 3). Dr. Sechen testified that during this time period, his research has focused on design and computer-aided design of digital integrated circuits, including the design of DRAM, and that he has taught numerous students how to design DRAM memories. *Id.* Dr. Sechen testified that he has “also been involved in numerous research projects on VLSI design and memory design[, and has] . . . taught numerous graduate researchers how to design digital integrated circuits, including memories.” *Id.* ¶ 5. Dr. Sechen testified that he is “at least a person of ordinary skill” in the art of memory module design. *Id.* ¶¶ 33–34.

To testify as an expert under Federal Rule of Evidence (FRE) 702, a person need not be a person of ordinary skill in the art, but rather “qualified

in the pertinent art.” *Sundance, Inc. v. DeMonte Fabricating Ltd.*, 550 F.3d 1356, 1363–64 (Fed. Cir. 2008); *see SEB S.A. v. Montgomery Ward & Co., Inc.*, 594 F.3d 1360, 1372–73 (Fed. Cir. 2010) (upholding a district court’s ruling to allow an expert to provide testimony at trial because the expert “had sufficient relevant technical expertise” and the expert’s “knowledge, skill, experience, training [and] education . . . [wa]s likely to assist the trier of fact to understand the evidence”); *Mytee Prods., Inc. v. Harris Research, Inc.*, 439 Fed. App’x 882, 886–87 (Fed. Cir. 2011) (non-precedential) (upholding admission of the testimony of an expert who “had experience relevant to the field of the invention,” despite admission that he was not a person of ordinary skill in the art). There is no dispute that Dr. Alpert and Dr. Sechen are qualified to testify as experts under FRE 702.

II. LEVEL OF ORDINARY SKILL IN THE ART

The level of ordinary skill in the art is relevant to claim construction and determinations of anticipation and obviousness. *See Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966) (explaining that obviousness is a question of law based on underlying factual findings including the level of ordinary skill in the pertinent art); *Yorkey v. Diab*, 605 F.3d 1297, 1300 (Fed. Cir. 2010) (explaining that a determination of anticipation involves, interpreting the claim language and then comparing the construed claim to a prior art reference); *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1259–60 (Fed. Cir. 2010) (“[C]laim language should be read in light of the specification as it would be interpreted by one of ordinary skill in the art.”).

Petitioner’s expert, Dr. Alpert, and Patent Owner’s expert, Dr. Sechen, disagree as to the level of ordinary skill in the art. Dr. Alpert testified that “a person of ordinary skill in the art for the ’501 patent would

have a bachelor's degree or the equivalent training or experience in electrical engineering and at least one year of experience relating to memory systems and BIST." Ex. 1007 ¶ 20. Dr. Sechen testified that

one of ordinary skill in the art at the time of filing the application(s) for the '501 patent, who would be working on the design of memory devices and memory modules, would have at least a Bachelor of Science degree in electrical engineering or computer engineering, and at least five years of industry experience designing memory devices and memory modules. Alternatively, one of ordinary skill in the art would have an M.S. degree in electrical engineering or computer engineering, and at least three years of industry experience designing memory devices and memory modules. Moreover, one of ordinary skill in the art would have a Ph.D. degree in electrical engineering or computer engineering, and have at least one year of industry experience designing memory devices and memory modules.

Ex. 2012 ¶ 33.

The difference in the opinions of the two experts is that Dr. Sechen believes a person of ordinary skill in the art would have an additional four years of experience, education, or a combination of the two, beyond that proposed by Dr. Alpert. *See* Ex. 1009 ¶ 7. According to Dr. Alpert, the difference in the proposed levels of ordinary skill in the art is not material to any dispute over claim construction or patentability in this *inter partes* review. *Id.*

We accord greater weight to Dr. Alpert's opinion as to the level of ordinary skill in the art because he has experience in both teaching students and working with engineers in the industry, while Dr. Sechen's experience is primarily in teaching. *Compare* Ex. 1007 ¶¶ 7, 21, *with* Ex. 2012 ¶¶ 3, 5. Moreover, based on our review of the '501 patent and the types of problems

and solutions described in the '501 patent and cited prior art, we agree with Dr. Alpert's opinion on the level of ordinary skill. Accordingly, we find that a person of ordinary skill in the art at the time of the '501 patent would have a Bachelor's degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.

III. CLAIM CONSTRUCTION

In its Petition, Petitioner requested that "each claim be construed in accordance with its plain and ordinary meaning under the required broadest reasonable interpretation" standard. Pet. 6–7. In its Preliminary Response, Patent Owner offered a specific construction for the claim term "generate." Prelim. Resp. 24–26. For purposes of our Institution Decision, we interpreted the claim term "generate" as meaning "produce" or "cause." Dec. on Inst. 9.

In its Response, Patent Owner disagrees with the Board's interpretation of "generate" (*see* PO Resp. 19), and further asserts that express construction of the claim language "configured to generate" (claim 1) is "necessary in view of the arguments made by Petitioner" (PO Resp. 14). Patent Owner also proposes a construction for "operated independently" (claim 1)/"operating . . . independently" (claim 16). PO Resp. 32.

In its Reply, Petitioner asserts that "'configured to,' 'operated independently,' and 'generate' are common words whose plain and ordinary meanings are well-known, even to a lay person" and, therefore, "the terms should be given their plain and ordinary meaning." Reply 1. Petitioner

contends Patent Owner's proposed constructions "import[] limitations from the specification into the claims, which is impermissible." *Id.* Petitioner also provides proposed constructions of the claim terms/phrases in dispute. *See generally* Reply 2–9.

Patent Owner's and Petitioner's proposed constructions for the disputed claim language are listed in the table below.

Claim Term/Phrase	Patent Owner's Proposed Construction	Petitioner's Proposed Construction
Operated independently/ Operating . . . independently	a plurality of data handlers configured to be <i>used to independently test/using</i> a plurality of data handlers <i>to independently test</i> (PO Resp. 32)	the operation of a data handler is not influenced or controlled by another data handler (Reply 4)
Generate	produce (PO Resp. 31)	produce or cause production of (Reply 8 n.2)
Configured to generate	programmed to produce (PO Resp. 20)	designed to, arranged to, or adapted to perform the function of producing or causing (Reply 7)

In an *inter partes* review, the Board interprets claim terms in an unexpired patent according to the broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b); *In re Cuozzo Speed Techs., LLC*, 793 F.3d 1268, 1278–79 (Fed. Cir. 2015). Under that standard, and absent any special definitions, we give claim terms their ordinary and customary meaning, as would be understood by one of ordinary skill in the art at the time of the invention. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definitions for claim terms must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir.

1994). “[E]xtrinsic evidence may be used only to assist in the proper understanding of the disputed limitation; it may not be used to vary, contradict, expand, or limit the claim language from how it is defined, even by implication, in the specification or file history.” *Tempo Lighting, Inc. v. Tivoli, LLC*, 742 F.3d 973, 977–78 (Fed. Cir. 2014) (internal quotations and citations omitted). Only terms which are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999). A court may revisit and alter its construction of claim terms as the record in a case develops. *See Pressure Prods. Med. Supplies, Inc. v. Greatbatch Ltd*, 599 F.3d 1308, 1322 (Fed. Cir. 2010).

Applying the foregoing principles of law, we review our construction of the claim term “generate,”¹ and also determine that the claim terms “configured to” and “operated independently”/“operating . . . independently” require construction.

A. Operated Independently/Operating . . . Independently

Independent claim 1 recites: “a plurality of data handlers configured to be operated independently from one other.” Ex. 1001, 16:42–43. Independent claim 16 recites “operating a plurality of data handlers independently from one another” *Id.* at 18:15–16.

In its Response, Patent Owner asserts that the above recitations in claims 1 and 16 mean that the data handlers are used to *independently test* a portion of one or more memory chips. PO Resp. 32. During the oral hearing, Patent Owner argued that “[o]perating independently” necessarily

¹ Our interpretation of the term “generate” also applies to other verb tenses used in the claims, i.e., “generating” and “generated.”

means that the use or operation of a data handler is not dependent on the use or operation of another data handler for testing. For example, to be ‘independently operable’ means that one data handler can be used – i.e., operated – in one mode while another data handler is operated or used in another mode.” Tr. 43:1–6. Petitioner contends “‘*operable independently*’ merely means that each data handler can operate without influence or control by any of the other data handlers.” Reply 3 (citing Ex. 1009 ¶ 15 (“This would encompass data handlers that receive common control signals from one controller, as long as the data handlers are not in communication with each other.”)).

In their papers, both Patent Owner (PO Resp. 32) and Petitioner (Reply 3) rely on column 8, lines 5–15 of the ’501 patent in support of their proposed constructions:²

Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28. *For example, each data handler 30 is configured to write to and/or read from the corresponding plurality of data ports of one or more of the memory devices 20 without being in communication [with] any of the other data handlers 30 or other data ports of the memory devices 20.* As such, each data handler 30 can be used to generally independently test a portion of the memory space of the memory module 10. For example, each data handler 30 may be used to independently test one memory device 20 of the memory module 10.

Ex. 1001, 8:5–15 (emphasis added). Petitioner quotes lines 1–11 in their entirety. Reply 3. Patent Owner omits the italicized language from its quotation. PO Resp. 32; *cf.* Tr. 43:16–18 (arguing that this language is

² Both parties also rely on the testimony of their experts and Petitioner relies on a dictionary definition. *See generally* PO Resp. 32; Reply 2–4.

immaterial to construction of the claim term “operable independently,” because it “relates to reading and writing to the data ports[, not] . . . to data generation, which is the claimed operation of the data handler”).

We begin our interpretation of the disputed terms with the language of the claims. *See Microsoft Corp. v. Proxyconn, Inc.*, 789 F.3d 1292, 1299 (Fed. Cir. 2015).

Independent claim 1 recites: “a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips.” Ex. 1001, 16:42–46. Claim 1 also recites “wherein the memory system is configured to test the one or more memory chips . . . using the data generated by the one or more data handlers. *Id.* at 16:48–51. Claim 1 thus requires that each of the plurality of data handlers is configured to be operated independently, but that only some of the data handlers (i.e., “one or more”) are further configured to generate data used for testing. Patent Owner’s proposed interpretation of “configured to be operated independently” (claim 1) as “configured to be used independently” (PO Resp. 32) narrows the meaning of this claim 1 phrase to encompass only the subset of data handlers that are also “configured to generate data” (claim 1). Patent Owner’s proposed interpretation of “configured to be operated independently” (claim 1) is, therefore, inconsistent with the use of this phrase in the context of claim 1 in its entirety.

Independent claim 16 recites “operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory

chips” and “testing the one or more memory locations of the one or more memory chips . . . using the data generated by the plurality of data handlers.” *Id.* at 18:15–18, 21–24. We apply a presumption that “operating . . . independently” in claim 16 has the same meaning as “operated independently” in claim 1. *See PODS, Inc. v. Porta Stor, Inc.*, 484 F.3d 1359, 1366 (Fed. Cir. 2007) (“We apply a ‘presumption that the same terms appearing in different portions of the claims should be given the same meaning unless it is clear from the specification and prosecution history that the terms have different meanings at different portions of the claims.’” (quoting *Fin Control Sys. Pty., Ltd. v. OAM, Inc.*, 265 F.3d 1311, 1318 (Fed. Cir. 2001))). In other words, unless usage of “operating . . . independently” in the context of the ’501 patent specification indicates otherwise, we presume one of ordinary skill in the art would not interpret the claim 16 phrase “operating a plurality of data handlers independently from one another” as limited to “*using* a plurality of data handlers *to independently test* a portion of one or more memory chips” (PO Resp. 32).

The term “operating” does not appear in the ’501 patent specification. The phrase “configured to be operated” appears only once in the ’501 patent specification as follows: “In certain embodiments, the memory module 10 is configured to be operated in a test mode.” *See* Ex. 1001, 5:64–65. In our interpretation of the similar ’434 patent claim term “operable independently” in our Final Decision in IPR2014-00970, we considered the disclosure in the related ’434 patent corresponding to the following ’501 patent disclosure:

Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports of one or more of the plurality of

memory devices 18. For example, each of the data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports. Each data handler 30 *is further configured to generate data for writing to the corresponding plurality of data ports.*

Ex. 1001, 5:16–23.

Because each of the data handlers 30 is operable independently of each of the other data handlers 30, the data handlers 30 are generally modular. As such, modifications in the configuration of the memory module 10 (e.g., changes in the bit-width of the memory bus, changes in the number of memory devices 20, etc.) may be less complicated to accommodate than in other types of self-testing memory modules 10. For example, where a new memory device 20 or set of memory devices 20 is added to the memory module 10, the change may be generally accommodated by adding a corresponding data handler 30. The change may be accommodated without having to implement a major reorganization of the memory module 10 or the self-testing logic of the memory module 10, for example.

Id. at 8:22–35. In our Final Decision in IPR2014-00970, we found that the above description of each data handler as “operable independently” and having the “further” feature of being “configured to generate data” (*see id.* at 5:16–23) supported Petitioner’s contention that the data generation function of the data handlers is an additional feature that is separate from the feature of independent operability. We found that based on the description quoted in column 8, lines 22–35, above, coupled with the discussion of the prior art problem of lack of flexibility in memory module self-test logic to accommodate changes to data width (*see id.* at 4:41–60, 8:24–29), one of ordinary skill in the art would understand the term “operable independently” to describe the data handlers as being discrete and functioning without being physically or logically connected to each other. In the present case, we find

similarly that the above-quoted language from the '501 patent specification is consistent with the plain meaning of the language in claim 1, i.e., that each of the plurality of data handlers is configured to be operated independently, but only some of the data handlers (i.e., “one or more”) are further configured to generate data used for testing. We agree with Petitioner, therefore, that when interpreted in the context of the '501 patent in its entirety, the claim term “operated independently” “merely means that each data handler can operate without influence or control by any of the other data handlers.” Reply 3. We find no basis in the '501 patent specification for applying a different construction to the claim 16 language “operating . . . independently.” Accordingly, we interpret “operated independently”/“operating . . . independently” in accordance with its plain and ordinary meaning: “the operation of a data handler is not influenced or controlled by another data handler” (Reply 4).

B. Configured to, and Generate

Independent claim 1 recites “a plurality of data handlers . . . *configured to generate* data for writing to a corresponding one or more memory chips” and “a control circuit *configured to generate* address and control signals.” Ex. 1001, 16:42–48. Independent claim 16 recites “operating a plurality of data handlers independently from one another to *generate* and transmit data” and “operating a control circuit to *generate* address and control signals.” *Id.* at 18:15–16, 19–20 (emphasis added).

1. Configured to

Prior to institution, neither party requested, nor did we deem necessary, construction of the term “configured to.” In its Response, Patent Owner now requests that we construe the term “configured to” in the context

of the claim phrase “configured to generate” (claim 1). *See* PO Resp. 14. As correctly observed by Petitioner, however, the term “configured to” is not used exclusively in conjunction with the term “generate” in claim 1, but is also used with the claim terms “test,” “receive,” “write,” “calculate,” “store,” “verify,” “be operatively coupled,” “be operated independently” and “selectively input.” *See* Reply 4. The term “configured to” is also used in the ’501 patent specification in conjunction with the terms “provide” (Ex. 1001, 6:6, 33), “operatively couple” (*id.* at 7:26), “read” (*id.* at 9–10), “receive” (*id.* at 11:11–12), “check” (*id.* at 11:14), “perform” (*id.* at 11:20–21), “report” (*id.* at 12:25), and “input” (*id.* at 14:5, 8). Presumptively, “configured to” should carry the same meaning, and “configured to generate,” “configured to test,” “configured to write,” etc. should have parallel meanings, differing only insofar as “test,” “write,” “generate,” etc., differ. *See Chamberlain Group, Inc. v. Lear Corp.*, 516 F.3d 1331, 1337 (Fed. Cir. 2008) (“‘[B]inary code’ and ‘trinary code’ should have parallel meanings, differing only insofar as ‘binary’ and ‘trinary’ differ in their relationships to the numbers 2 and 3.”).

Patent Owner does not offer an interpretation of the term “configured to” outside the context of the phrase “configured to generate” (*see* PO Resp. 14–19), but contends “‘configured to generate’ means ‘programmed to generate’ and enables the subsequent execution of a self-testing function (*e.g.*, data generation, writing and verification)” (*id.* at 18; *see also id.* at 15 (“‘Programmed’ is therefore synonymous with ‘configured.’”)).

Petitioner contends the claim term “configured to” should be construed consistently, whether used in conjunction with “generate” or with another function recited in the claims, and should be given its plain and

ordinary meaning of “designed to, adapted to, or arranged to.” Reply 4. In addition to relying on the testimony of its expert, Dr. Alpert, Petitioner cites two dictionary definitions in support of its proposed interpretation. Reply 5 (citing Ex. 1011, 186; Ex. 1012, 428).

Dr. Alpert, testified that, in the '501 patent, “‘program’ is one example of ‘configure,’ but not the only example.” Ex. 1009 ¶ 16. Dr. Alpert further testified “the '501 patent discloses that *configuring* can be accomplished in a wide variety of ways . . . , including those that do not require programming, for example using predefined values.” *See id.* ¶ 19 (quoting Ex. 1001, 6:19–22 (“The address sequences and/or the data patterns of certain embodiments may be programmable either through the I²C interface or they may be defaulted to pre-defined values.”); Ex. 1001, 10:67–11:4 (“[R]andom or pseudorandom data may be generated and written to the corresponding plurality of data ports. For example, a linear feedback shift register (LFSR) may be used in some embodiments. In addition, the data patterns may be programmable.”); *see also id.* at ¶ 20 (“The configuration of the data handlers can be fixed, hardwired.” (citing Ex. 1001, Figure 2, 8:5–19, 36–55)); Ex. 2021, 135:12–136:2 (wherein Dr. Alpert testified that a linear feedback shift register would not need to be programmed to generate data, but could be arranged or designed to generate data). In his second declaration, Dr. Alpert also testified that:

The '501 patent uses “configured” to describe hardwired arrangements of components that cannot be programmed for their configuration. Additionally, the '501 patent discloses various components included in the control module, which do not all require programming: The configuration of the verification circuitry can be fixed without any memory structure for storing a copy of the generated data, as disclosed at 11:28–

31. . . . [And,] control module 22 [can] include[] discrete logic [or] one or more application-specific integrated circuit (ASICs) . . . [b]oth . . . could be programmable or nonprogrammable as would have been understood by a person of ordinary skill.

Ex. 1009 ¶ 20 (emphasis omitted).

Patent Owner's expert, Dr. Sechen, testified that "the most conventional use of *configuration* refers to hardware programming" (Ex. 1010, 68:25–69:2 (emphasis added)), but that it could refer to software programming as well. *Id.* at 55:17–18. In his declaration, Dr. Sechen did not testify as to the meaning of "configured to" in conjunction with any term other than "generate." *See* Ex. 2012 ¶¶ 46–67. In response to questioning during cross-examination, however, Dr. Sechen testified that "configured to," as used in the context of the phrase "a printed circuit board configured to be operatively coupled to a memory controller of a computer system" (Ex. 1001, 2:25–27), "might mean designed [to]" because the printed circuit board has to be configured "such that it fits into the slot on the motherboard and is able to communicate successfully with the memory controller." Ex. 1010, 84:14–19. The '501 patent discloses that components of control module 22 and data module 28 may include discrete logic and application specific integrated circuits (ASICs) (*see* Ex. 1001, 9:62–64, 12:33–37), which, on cross-examination, Dr. Sechen conceded can be designed to operate without the need for hardware programming. Ex. 1010, 58:9–61:22.

Based on our analysis of the '501 patent claims and specification, and relying on the testimony of Drs. Alpert and Sechen regarding the technology,³ we find the '501 patent uses the term "configured to" to

³ "Although the patent file may often be sufficient to permit the judge to interpret the technical aspects of the patent properly, consultation of

describe components that can be programmed by a controller to perform a function, as well as components that can be hard-wired to perform a function and do not require programming. *See, e.g.*, Ex. 1001, 7:25–31 (“PCB 12 may include at least one connector (not shown) configured to operatively couple the memory module 10 to the memory controller 14 of the computer system 16. . . . For example, the memory module is electrically coupled, logically coupled, or both, with the memory controller 14”); *id.* at 8:50–53 (“[T]he data handlers 30 may be operatively coupled (e.g., electrically and/or logically coupled or connected) to the eight data ports 21 of one of the corresponding memory devices 20”). We do not agree with Patent Owner, therefore, that the broadest reasonable interpretation of “configured to” is “programmable to.” *Cf.* Ex. 1010, 89:3–5 (“‘Designed’ implies a certain hardwiredness, that that’s what it does, and ‘configured’ implies that maybe it only does that some of the time.”). Rather, we agree with Petitioner that one of ordinary skill in the art would interpret the claim term “configured to” in accordance with its plain and ordinary meaning, and would understand that this term encompasses components both programmable and hard-wired to perform the functions described in the ’501 patent. Accordingly, we interpret the term “configured to” as “designed, adapted, or arranged to.”

2. *Generate*

In addition to reciting “a plurality of data handlers . . . configured to

extrinsic evidence is particularly appropriate to ensure that his or her understanding of the technical aspects of the patent is not entirely at variance with the understanding of one skilled in the art.” *Pitney Bowes, Inc. v. Hewlett-Packard Co.*, 182 F.3d 1298, 1309 (Fed. Cir. 1999).

generate data” and “a control circuit configured to generate address and control signals,” claim 1 recites: “wherein the memory system is configured to test the one or more memory chips using the address and control signals *generated by* the control circuit and using the data *generated by* the one or more data handlers” (Ex. 1001, 16:48–51 (emphasis added)). Similarly, in addition to reciting “operating a plurality of data handlers independently from one another to *generate* and transmit data” and “operating a control circuit to *generate* address and control signals,” claim 16 recites a step of: “testing the one or more memory locations of the one or more memory chips using the address and control signals *generated by* the control circuit and using the data *generated by* the plurality of data handlers” (*id.* at 18:21–24 (emphasis added)).

For purposes of our Institution Decision, we interpreted the claim term “generate” as meaning “produce” or “cause.” *See* Dec. on Inst. 8–9. Our interpretation was based on findings that the plain and ordinary meaning of “generate,” as evidenced by Merriam-Webster’s Collegiate Dictionary, is “produce” or “be the cause of” (*id.* at 9 (citing Ex. 2002, 484)), and that interpreting “generate” to mean “cause” was not inconsistent with the ’501 patent specification, which broadly states that data can be generated by the data handlers in a variety of ways (*see id.* at 8–9 (citing Ex. 1001, 10:53–54)).

Petitioner contends the Board should maintain its construction of “generate” as “produce” or “cause,” because this construction “reflects the plain and ordinary meaning.” Reply 8–9. Petitioner maintains one of ordinary skill in the art “would appreciate that when substituted in the claims, the Board’s construction is ‘produce or cause production of

[data/address and control signals].” *Id.* at 8 n.2 (citing Ex. 1009 ¶ 29). During cross-examination, Dr. Alpert explained the difference between producing and causing to produce” as follows: “[P]roducing would actually be either creating or providing, in this case, some data and causing to produce would be having one device be the cause of or invoking another device to produce the data.” Ex. 2013, 43:5–11. Dr. Alpert testified that one of ordinary skill in the art would understand the description of the data handler being logically coupled to the memory devices at column 5, lines 20–23 of the ’501 patent as implying intermediary logical devices between the data handler and the memory device ports. *See id.* at 39:1–16. In other words, one of ordinary skill in the art would understand that the “physical system element” that produces the data used for testing in claim 1 “can be the data handler or it can be another device that the data handler causes to produce the data.” *Id.* at 32:4–6. During cross examination, Dr. Alpert identified “a buffer or a multiplex[e]r” as examples of devices that could “exist between the data handler and the memory device” to produce the data used for testing. *Id.* at 36:13–16. Dr. Alpert explained that a

buffer can take an input, information input and it can either produce a signal that contains that information or the complements of that information, or there could be several buffers that have potential to be in a third state, so-called tri-state buffers, such that the buffers can be used to select information that may be presented, that would be data that would be produced that would then be presented down to other circuits.

Id. at 36:24–37:8.

Patent Owner contends “‘generate’ is properly construed [only] as ‘produce.’” PO Resp. 31 (citing Ex. 2012 ¶¶ 68–96); *see also id.* at 20

(quoting Ex. 2012 ¶ 70 (“In the context of integrated circuit and printed circuit board design, *generate* or *produce* (which are synonyms in this context) are terms of art.”)). Patent Owner argues that interpreting “generate” as “cause,” or as “cause to produce,” as suggested by Petitioner, would render other claim language superfluous. *See id.* at 24–31. Specifically, Patent Owner contends that interpreting “generate” as “cause to produce” renders meaningless the claim limitations “the memory system is configured to test” (claim 1) and “testing” (claim 16) the memory chips “using the address and control signals generated by the control circuit and using the data generated by the . . . data handlers” (claim 1 and 16). *See id.* at 29–31; *see also id.* at 25 (“[T]he situation posited by the Board does not reflect that the data written by the data handler was ‘generated by the data handler’” and eliminates the claim requirements that data generated by the data handlers is used for testing.

“[W]here there are several common meanings for a claim term, the patent disclosure serves to point away from the improper meanings and toward the proper meaning.” *Renishaw PLC v. Marposs Societa' Per Azioni*, 158 F.3d 1243, 1250 (Fed. Cir. 2003). The '501 patent claims and specification use the term “generate” in conjunction with the memory module, control circuit/module, data module, and components thereof. The '501 patent further uses the term “generate” as a specific example of “produce,” with the term “produce” appearing only once in the '501 patent:

In some embodiments, the control module 22 and the data module 28 *produce* memory addresses, control and/or data signals according to the JEDEC standard memory protocol. In some embodiments, *for example*, the control module 22 and the data module 28 *generate* the memory interface signals with proper edge relationships based on the JEDEC standard.

Ex. 1001, 6:11–17 (emphasis added).

The '501 patent specification uses both the term “provide” and the term “generate” in describing the operation of data module 28:

[D]uring testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the *data module 28 generates corresponding test data patterns and provides them* to the appropriate memory devices 20. For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation.

Id. at 5:48–55 (emphasis added). The term “provide” is also used separately from, but in a similar manner to, the term “generate” to describe embodiments of control module 22 and data module 28 used for testing memory devices 18. *Compare id.* at 6:5–11 (“The data module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20. For example, the data module 28 and/or the control module 22 are *configured to provide* memory signals (e.g., data, address and control signals) according the operating specification of the memory devices 20” (emphasis added)) *with id.* at 9:29–31 (“The control module 22 can be *configured to generate* address and control signals 24 for testing the plurality of memory devices 18.” (emphasis added)) *and id.* at 10:31–37 (“Each of the data handlers 30 of certain embodiments further includes a data handler logic element 46. The data handler logic element 46 of certain embodiments comprises a data generation element 54 and a verification element 56. The data generation element 54 may be *configured to generate* data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports.” (emphasis added)).

In addition to using the term “provide” to describe embodiments of control module 22 and data module 28, the ’501 patent specification uses this term to describe the operation of switch 44: “The switch 44 may *provide* a bi-directional data multiplexer function” and is “configured to selectively input” data signals 48 from system memory controller 14 or data signals 50 from data handler logic element 46. *Id.* at 10:10–15 (emphasis added). The ’501 patent specification discloses that switch 44 may be positioned between data module 28 and memory devices 18 (*see id.* at 10:28–30), i.e., one of the hypothetical arrangements described by Dr. Alpert in which a data handler would cause an intermediary device (e.g., a multiplexer) to produce data. The ’501 patent specification does not use the term generate (or produce), however, to describe the configuration or operation of switch 44. The ’501 patent specification also describes an embodiment in which switch 44 is included in each of data handlers 30 (*id.* at 10:7–8), as well as an embodiment in which the function of selective inputting of data signals to memory devices 18 is not limited explicitly to a switch (*see, e.g., id.* at 6:1–4 (“[E]ach of the data handlers 30 write[s] the data generated by the data handler 30 to the corresponding plurality of data ports by selectively inputting data signals to the data ports of the plurality of memory devices 18.”)).

The above-noted usages of the terms “provide” and “generate” to describe different embodiments of control module 22 and data module 28, and to describe different functions of the data handlers, do not support Dr. Alpert’s opinion that the term “generate” is properly interpreted as

encompassing the selection function of a multiplexer.⁴ Use of the term “generate” to describe an embodiment in which the control module and data handlers “produce” memory addresses, control and/or data signals, however, does support Dr. Sechen’s opinion that “generate” and “produce” are synonymous.

In the absence of evidence to the contrary, there is a presumption that different claim terms have different meanings. *CAE Screenplates, Inc. v. Heinrich Fiedler GmbH & Co. KG*, 224 F.3d 1308, 1317 (Fed. Cir. 2000); *see also SEB S.A. v. Montgomery Ward & Co., Inc.*, 594 F.3d 1360, 1369 (Fed. Cir. 2010) (noting there is a presumption that meanings of claim terms do not overlap); *Merck & Co. v. Teva Pharms. USA, Inc.*, 395 F.3d 1364, 1372 (Fed. Cir. 2005) (“A claim construction that gives meaning to all the terms of the claim is preferred over one that does not do so.”).

The claim 1 recitation of “data handlers . . . configured to generate data for writing” and dependent claim 14 recitation of “the one or more circuits [sic, data handlers] . . . *further configured* to write data to the one or

⁴Dr. Alpert’s testimony, though consistent with Averbuj’s use of the term “generate” (*see* Ex. 1003 ¶ 56 (“[D]ata generation unit 44 generates the transformed data signal 49 (BIST_DATA_T) to fill the memory module 12 with the data dictated from the sequencer 8 without modification.”)), cannot be relied on to construe this claim term in a manner that varies from the implicit definition in the ’501 patent. *See Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1584–85 (Fed. Cir. 1996) ([P]rior art can often help to demonstrate how a disputed term is used by those skilled in the art. . . . [H]owever, reliance on such evidence is unnecessary, and indeed improper, when the disputed terms can be understood from a careful reading of the public record. Nor may it be used to vary claim terms from how they are defined, even implicitly, in the specification or file history.” (internal citation omitted)).

more memory chips by selectively inputting,” as well as the above disclosure in the ’501 patent specification, create a presumption that “generate” has a different meaning than “write” and/or “selectively input.” See also claim 1 (“a control circuit configured to generate address and control signals”) and dependent claim 15 (“the control circuit is *further configured* to selectively input to the address and control ports of the one or more memory chips either the address and control signals from the memory controller of the computer system or the address and control signals from the control circuit.” (emphasis added)). Likewise, the recitations of “operating a plurality of data handlers . . . to generate and transmit data” in independent claim 1, and “data transmitted to the one or more . . . memory chips” in dependent claims 17 and 20, create a presumption that “generate” has a different meaning than “transmit.” In other words, the ’501 patent specification and claims do not support an interpretation of “data generated” by a component as encompassing data selected, written, input, or transmitted by a first component to a second component, because such interpretation would result in an overlap in the meaning of “generate” and “write” and/or “selectively input” and/or “transmit.”

Having considered the ordinary meaning of the term “generate” in the context of both the claims and the ’501 patent as a whole, we agree with Patent Owner that the broadest reasonable interpretation of the claim term “generate” is “produce,” and that “generate” does not mean “cause” or “cause to produce.” We interpret the claim language “address and control signals *generated* by the control circuit” and “data *generated* by the . . . data handlers” (claims 1 and 16), as encompassing signals and data that originated in the control circuit and data handlers, including by

transformation or modification of information and/or data received from another component. *See e.g.* Ex. 1001, 10:37–41 (“The data signals and/or patterns of data signals may be *based on* information (e.g., programming or configuration information) the data handler logic element 46 receives from the control module 22.” (emphasis added)); *id.* at 10:61–63 (“The data may be generated *based [on]* previously written data (e.g., inverting each of the bits of a previously written data word)” (emphasis added)). We do not interpret this language as encompassing signals and data received by the control circuit and data handlers from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the control circuit and data handlers. *Compare, e.g.*, Ex. 1001, 9:54–57 (“In one embodiment, the memory device controller 34 *generates* the address and control signals for memory device (e.g., DRAM device) operations.” (emphasis added)), *with id.* at 9:40–53 (“In one embodiment, the memory device controller 34 *receives* signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) . . . [M]emory device controller 34 *may send* either the signals 38 from the system memory controller 14 or, alternatively, the signals 42 from the test controller 36, to the register 40.” (emphasis added)).

Thus, in summary, we construe “operated independently/operating independently,” “generate,” and “configured to” as follows:

Claim Term/Phrase	Interpretation
Operated independently/ Operating . . . independently	Operated, or operating, without influence or control by another

Claim Term/Phrase	Interpretation
Generate	produce
Configured to	designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]

IV. ANALYSIS

To prevail in its challenges to the patentability of the claims, a petitioner must establish facts supporting its challenges by a preponderance of the evidence. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference.

Net MoneyIN, Inc. v. VeriSign, Inc., 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). While the elements must be arranged or combined in the same way as in the claim, “the reference need not satisfy an *ipsissimis verbis* test.” *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009).

A single prior art reference that discloses, either expressly or inherently, each limitation of a claim invalidates that claim by anticipation. Thus, a prior art reference without express reference to a claim limitation may nonetheless anticipate by inherency. “Under the principles of inherency, if the prior art necessarily functions in accordance with, or includes, the claims limitations, it anticipates.”

Perricone v. Medicis Pharm. Corp., 432 F.3d 1368, 1375–76 (Fed. Cir. 2005) (citations omitted). “In general, a limitation or the entire invention is inherent and in the public domain if it is the ‘natural result flowing from’ the explicit disclosure of the prior art.” *Schering Corp. v. Geneva Pharms., Inc.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness, i.e., secondary considerations. *See Graham*, 383 U.S. at 17–18. An obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418.

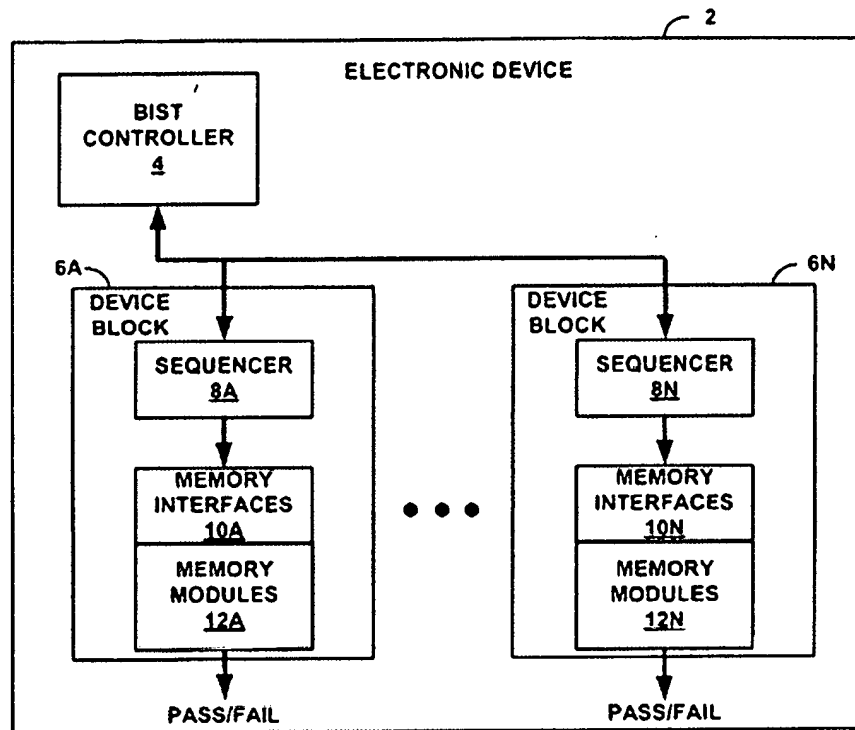
We analyze the instituted grounds of unpatentability in accordance with the above-stated principles.

A. Alleged unpatentability of claims 1–3, 5–8, and 11–20 under §102(b) based on Averbuj

1. Averbuj (Ex. 1003)

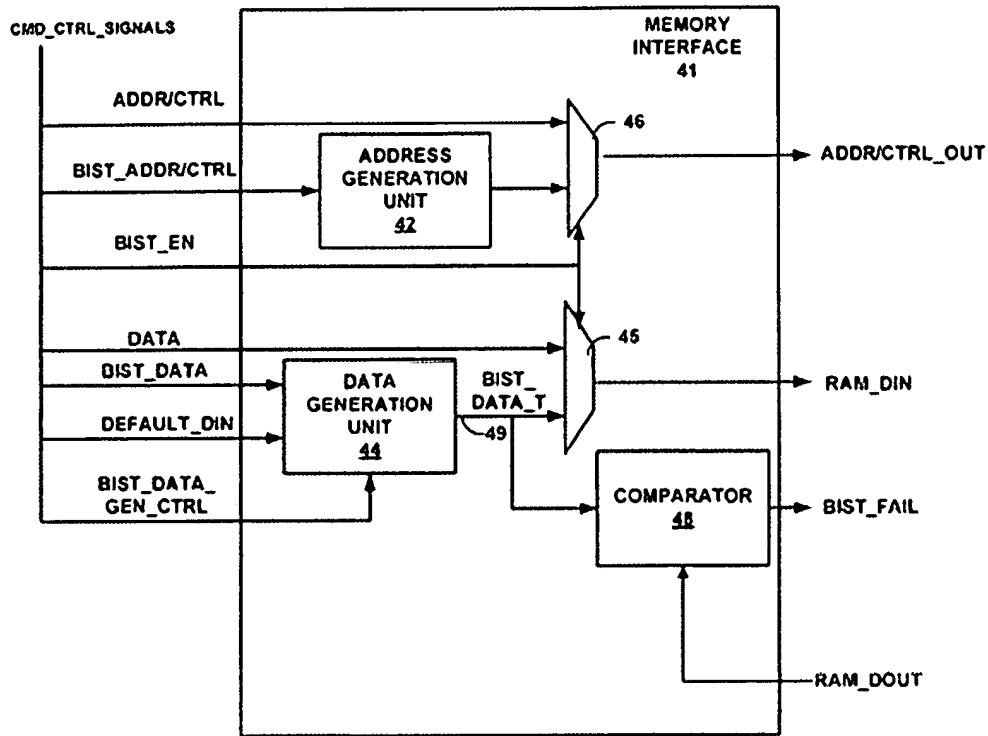
Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the

testing of one or more memory modules. Ex. 1003 ¶¶ 7–8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 . . . includ[ing] a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to sequencers 8 for application to device blocks 6. *Id.* ¶ 29. Sequencers 8, in turn, issue command control signals (CMD_CTRL_SIGNALS) to receiving memory interfaces 10. *Id.* ¶ 43.

Exemplary memory interface 41 is shown in Figure 6, below. *Id.* ¶ 23.



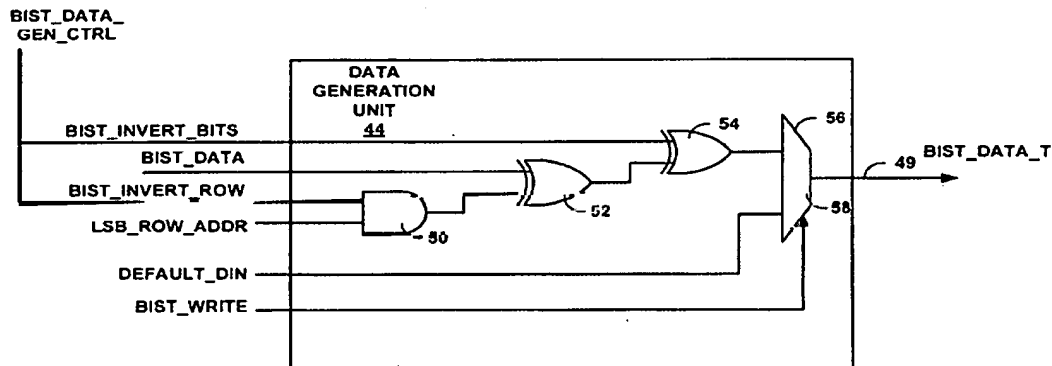
As shown in the Figure 6 embodiment, above, memory interface 41 includes a layer of multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals (BIST_ADDR/CTRL) and the test data provided by a respective higher-level sequencer [, e.g., sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level sequencer [8A], and transforms the data and addresses provided by the sequencer as needed based on the particular physical characteristics of the memory module” (*id.* ¶ 54) using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “For example,

sequencer 8 may request a checkerboard bit pattern that requires columns of data within the memory module to alternate between ones and zeros.

Different memory modules 12, however, may be arranged with different row and column configurations.” *Id.* ¶ 50. In this case, address generation unit 42 transforms the address provided by sequencer 8 to correspond to the physical configuration of the specified memory module before the address is applied to that memory module. *See id.* ¶¶ 51–52.

An example embodiment of data generation unit 44 is illustrated in Figure 7, reproduced below.



As shown in Figure 7, above, “data generation unit 44 receives a BIST_DATA signal, a default data signal (DEFAULT_DIN) provided by the sequencer 8[, and] . . . a number of control signals in the form of BIST_INVERT_BITS, BIST_INVERT_ROWS, LSB_ROW_ADDR, and BIST_WRITE.” *Id.* ¶ 55.

[S]equencer 8 asserts and deasserts the invert bits signal (BIST_INVERT_BITS), the invert rows signal (BIST_INVERT_ROWS), and the invert columns signal (not shown) to specify data patterns, such as solid, checkerboard, horizontal and vertical striped data patterns. If neither the invert bits nor the invert rows signals are asserted, then the BIST_DATA signal passes through XOR gates 52, 54 without

modification. As a result, data generation unit 44 generates the transformed data signal 49, (BIST _DATA_ T) to fill the memory module 12 with the data dictated from the sequencer 8 without modification.

Id. ¶ 56. By contrast, when the BIST_INVERT_BITS signal is asserted, data generation unit 44 automatically inverts the values dictated by the sequencer. *Id.* ¶ 57.

2. Analysis

Petitioner contends the claim 1 limitations (and corresponding claim 16 limitations) of “a control circuit configured generate address and control signals,” “data handlers [] configured to generate data,” and a “memory system [] configured to test [] one or memory chips” using these generated signals and data are met by Averbuj’s sequencers 8 and memory interfaces 10 that generate address/control signals and data, respectively, for testing memory modules 12. Pet. 12. Patent Owner argues: (1) Averbuj’s memory interfaces (the alleged data handlers) are not “configured to generate data,” as recited in claim 1; (2) Averbuj’s sequencer 8 (the alleged control circuit) is not “configured to generate” (claim 1), and does not “generate” (claim 16) the address and control signals used for testing the memory modules 12 (the alleged memory chips); (3) Averbuj’s memory interfaces 10 (the alleged data handlers) are not “configured to be operated independently” (claim 1) and cannot be “operating . . . independently” (claim 16); and (4) Averbuj’s sequencer (the alleged control circuit) is not “configured to selectively input” address and control signals to the memory chips, as recited in dependent claim 15. *See* PO Resp. 32–41. With the exception of the above-noted limitation in dependent claim 15, Patent Owner does not dispute directly Petitioner’s contentions with respect to the additional limitations

recited in dependent claims 2, 3, 5–8, 11–15, and 17–34, but contends these claims are not anticipated by Averbuj because Petitioner has failed to show Averbuj anticipates independent claims 1 and 16. *See id.* at 41.

a. *“one or more data handlers . . . configured to generate data” (claim 1) for testing memory chips*

Petitioner relies on paragraphs 43, 48, and 49 of Averbuj in support of its contention that Averbuj’s memory interfaces are data handlers that are configured to generate data for testing via data generation unit 44, described as “generat[ing] the exact data (RAM_DIN) applied to the memory inputs” of a memory module during testing. Pet. 15–16 (quoting Ex. 1003 ¶ 49).

Patent Owner argues Petitioner has not shown that Averbuj’s memory interfaces 10 (the alleged data handlers) are “configured to generate data” (claim 1) for testing memory modules 12 (the alleged memory chips), because “Petitioner has not shown that the data generation unit 44 in Averbuj is capable of being programmed.” PO Resp. 34 (“Averbuj’s control signals for data generation (BIST_DATA_GEN_CTRL) are not programmed – they are used to actively generate BIST_DATA_T. Without constant communication over the BIST_DATA_GEN_CTRL lines, BIST_DATA_T is not generated.”). Patent Owner’s argument is based on its interpretation of “configured to” as “programmed to.” *Id.* at 33.

In Section III.B.1, we determined the broadest reasonable interpretation of “configured to,” as used in the ’501 patent claims, is “designed, adapted, or arranged to,” and that one of ordinary skill in the art would understand that programmability is only one way that a component might be designed, adapted, or arranged to perform a function. In other words, lack of programmability is not determinative of whether data generation unit 44 is “configured to generate” data. We are persuaded

Petitioner has shown, by a preponderance of the evidence, that Averbuj's data generation unit 44 is capable of generating data for testing memory modules 12 and, therefore, is necessarily designed, adapted, or arranged to generate data in the manner recited in challenged claim 1.

- b. *"a control circuit configured to generate address and control signals" (claim 1)/"operating a control circuit to generate address and control signals" (claim 16) for testing memory chips*

Patent Owner argues Averbuj's sequencer 8 (the alleged control circuit) does not generate (i.e., produce) the address and control signals used to test memory modules 12 (the alleged memory chips) and, therefore, does not describe "a control circuit configured to generate address and control signals" or "operating a control circuit to generate address and control signals" (claim 16) for testing memory chips as recited in the challenged claim. PO Resp. 35. Patent Owner argues, more specifically, that in Averbuj, the address and control signals used to test memory modules 12 are produced by memory interfaces 10 (the alleged data handlers), not by sequencer 8 (the alleged control circuit). *See id.*

Petitioner references Averbuj paragraphs 37, 43, 48–53, 59–61, and 77–78 in support of its contention that Averbuj discloses a control circuit (sequencer 8) configured to generate address and control signals (CMD_CTRL_SIGNALS) used for testing memory chips (memory modules 12). Pet. 16–17. Petitioner contends "[t]he actual addressing information is produced by the [s]equencer, and the address generation unit 42 merely applies the received addressing information to a given physical memory configuration (i.e., row/column configuration)." Reply 12–13; *see also* Tr. 20:15–18 ("Averbuj teaches us a sequencer generates addresses, provides

them to the memory interface, which may pass them through or may translate them. . . . It generates address and control signals.”).

Averbuj discloses that “BIST controller 4 provides and communicates test algorithms to sequencers 8 for application to device blocks 6” (Ex. 1003 ¶ 29), and “[s]equencers 8 interpret” and “control the application of the test algorithms” to the memory modules (*id.* ¶ 30). *See also id.* ¶ 46 (“Sequencers 8 receive the generic BIST commands in accordance with the command protocol, and control the application of the commands by generating and issuing sequences of one or more memory operations for application to a set of respective memory modules 12.”). For example, in the exemplary embodiment illustrated in Figure 5, command parser 30 of sequencer 8A processes a command received from BIST controller 4, and extracts one or more parameters from the command. *Id.* ¶¶ 41–42. Command parser 30 passes the extracted parameters to command controller 34, which then issues a sequence of one or more operations to each memory interface 10. *Id.* ¶¶ 42–43.

In particular, . . . command controller[] 34 sequentially drives the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. The command control signals may include signals to provide a memory address and data to the receiving memory interfaces 10 and to direct the receiving memory interfaces to invert bits, perform read or write operations, invert rows, and the like.

Id. ¶ 43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* ¶ 31.

Averbuj further explains, in connection with the embodiment illustrated in Figure 6, that memory interface 41 “*transforms* the data and addresses provided by the sequencer *as needed* based on the particular physical characteristics of the memory module” (*id.* ¶ 54 (emphasis added)) using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). For example, if sequencer 8 asserts an invert rows signal and/or invert columns signal to specify a data pattern, data generation unit 44 inverts values dictated by the sequencer to generate transformed data signal 49 (BIST_DATA_T). *Id.* ¶ 57. If, however, none of the invert signals are asserted, then the BIST_DATA signal passes through XOR gates 52, 54 of data generation unit 44 without modification, and the BIST_DATA_T signal sent to memory module 12 is “the data dictated from the sequencer 8 *without modification.*” *Id.* ¶ 56 (emphasis added). Likewise, “in some BIST tests,” sequencer 8 may direct memory interface 41 to write BIST data in a row or column first fashion. *Id.* ¶ 51. In such case, address generation unit 42 must transform the address provided by sequencer 8 to conform to the particular row and column configuration of the memory module. *See id.* ¶¶ 51–52. “For example, a 256 bit memory module 12 may be organized as 128 rows and 2 columns, 32 rows by 8 columns, 16 rows by 16 columns, and the like.” *Id.* ¶ 50.

Based on the above description in Averbuj, we agree with Patent Owner that Averbuj describes memory interfaces (the alleged data handlers) that are configured to produce address and control signals, and that Averbuj describes producing address and control signals in certain circumstances, e.g., when Averbuj’s sequencer (the alleged control circuit) specifies writing data in a particular pattern that requires transformation to conform to a

particular configuration of a memory module. *See* PO Resp. 35. We also agree with Petitioner, however, that Averbuj's memory interfaces are further configured to apply to the memory modules the same address and control signals received from, and produced by, the sequencers for testing (*see* Reply 12–13), e.g., when the sequencer sends signals that can be applied, without modification or transformation (*see* Ex. 1003 ¶ 56), directly to a memory module irrespective of its physical configuration. Thus, Petitioner has shown, by a preponderance of the evidence, that Averbuj's sequencer meets the claim 1 limitation of “a control circuit configured to generate address and control signals” for testing memory chips, and that Averbuj describes “operating a control circuit [(i.e., a sequencer)] to generate address and control signals” used in testing memory chips as recited in claim 16 of the '501 patent.

- c. *“a plurality of data handlers configured to be operated independently from one [an]other” (claim 1)/ “operating a plurality of data handlers independently from one another” (claim 16)*

Petitioner contends Averbuj describes each memory interface 10A–C (the alleged data handlers) of Figure 6 as receiving control and address data from sequencer 8 and using this information to generate address and test data that is specific to its particular memory module 12. Pet. 11. Petitioner argues each memory interface operates independently from the other memory interfaces, because it “processes the sequential memory operations issued by the higher-level sequencer, and transforms the data and addresses provided by the sequencer as needed based on the particular physical characteristics of the memory module.” *Id.* (citing Ex. 1003 ¶ 54). In other words, each memory interface operates without being influenced or

controlled by another memory interface because “[a] given memory interface (*i.e.*, data handler) does not communicate with any of the other memory interfaces, or with any of the memory ports used by the other memory interfaces.” Reply 13 (citing Ex. 1003 ¶ 39).

Patent Owner argues Averbuj’s memory interfaces (the alleged data handlers) are not “configured to be operated independently” (claim 1) and cannot “operat[e] . . . independently” (claim 16) as recited in the challenged claims because they “are controlled by the same command control signals generated by the invoked command controller via a shared control line. Thus, as shown in Averbuj Fig. 4, the alleged plurality of data handlers cannot be controlled independently.” PO Resp. 38. In Section III.A., however, we determined the broadest reasonable interpretation of “operated independently” and “operating . . . independently” as recited in claim 1 and claim 16, respectively, is “operated, or operating, without influence or control by another.”

We find Petitioner has shown, by a preponderance of the evidence that Averbuj’s memory interfaces (the alleged data handlers) are “configured to be operated” (claim 1) and can perform the function of “operating” (claim 16) without influence or control by another data handler for the reasons stated in the Petition and above. Patent Owner’s argument is based on an overly narrow interpretation of “operated independently”/“operating . . . independently” as meaning that the data handlers are configured to be used, or are used “*to independently test* a portion of one or more memory chips” (*id.* at 38), and, therefore, does not persuade us that Petitioner failed to meet its burden of proof.

d. dependent claims 2, 3, 5–8, 11–15, and 17–20

Claim 15 depends from claim 1 and recites: “wherein the control circuit is further configured to selectively input to the address and control ports of the one or more memory chips either the address and control signals from the memory controller of the computer system or the address and control signals from the control circuit.” Ex. 1001, 18:5–10. Petitioner contends Averbuj’s multiplexer 46 meets these limitations. Pet. 24–25.

Patent Owner argues Averbuj’s sequencer 8 (the alleged control circuit) does not selectively input the BIST_ADDR/CTRL signal to memory modules 12 (the alleged memory chips). PO Resp. 40. Rather, sequencer 8 inputs the BIST_ADDR/CTRL signal to address generation unit 42, which is part of memory interface 41 (the alleged data handler), and thereafter to multiplexer 46, i.e., the alleged device that selectively inputs the address and control signals to memory modules 12. *Id.*

Averbuj describes sequencer 8 (the alleged control circuit) as providing a BIST enable signal that causes multiplexer 46 of memory interface 41 (the alleged data handler) to select the BIST address/control signals provided by the sequencer. *See* Ex. 1003 ¶ 48. Averbuj states that “[i]n this manner, the sequencer controls multiplexers 45, 46 of memory interface 41 to selectively isolate the respective memory module, thereby allowing BIST algorithms to be applied to that memory module.” *Id.*

We agree with Patent Owner that Petitioner has not explained adequately how Averbuj’s sequencer 8 (the alleged control circuit) is “configured to” (i.e., designed, adapted, or arranged to) “selectively input . . . address and control signals” (claim 15) to memory modules 12. Petitioner has not addressed Patent Owner’s argument that the device that

selectively inputs address and control signals to memory modules 12 is memory interface 41 (the alleged data handler), not sequencer 8 (the alleged control circuit). *See* Reply 14–15 (arguing “the Sequencer of Averbuj generates the address and control signals since it causes address generation unit 42 to produce the address and control signals received by the memory . . . Alternatively, the sequencer would also be understood to produce the address and control signals received by the memory,” but failing to explain how Averbuj’s sequencer is “configured to selectively input” (claim 15)).

Patent Owner also contends Petitioner has not met its burden to show dependent claims 2, 3, 5–8, 11–15, and 17–20 are anticipated by Averbuj, because Petitioner has failed to show independent claims 1 and 16, from which these claims depend, are anticipated. PO Resp. 41. We determined above that claims 1 and 16 are anticipated by Averbuj. With respect to claims 2, 3, 5–8, 11–14, and 17–20, we have reviewed the Petition and evidence cited therein, and determine Petitioner has met its burden to show these dependent claims are anticipated by Averbuj. *See* Pet. 17–24.

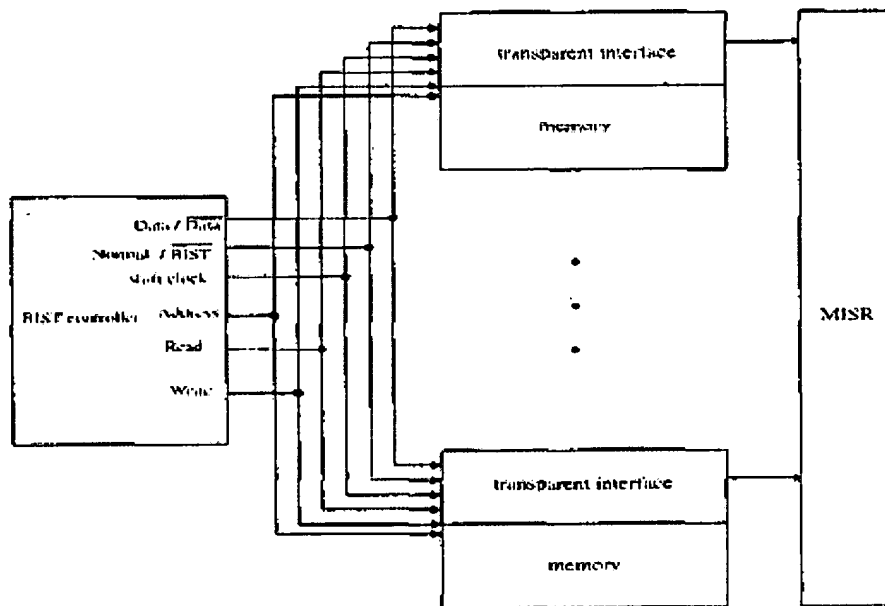
3. Conclusion

Petitioner has met its burden to show, by a preponderance of the evidence, that claims 1–3, 5–8, 11–14, and 16–20 are unpatentable under 35 U.S.C. §102(b) based on Averbuj. Patent Owner has persuaded us that Petitioner failed to meet its burden to establish unpatentability of claim 15 under 35 U.S.C. §102(b) based on Averbuj.

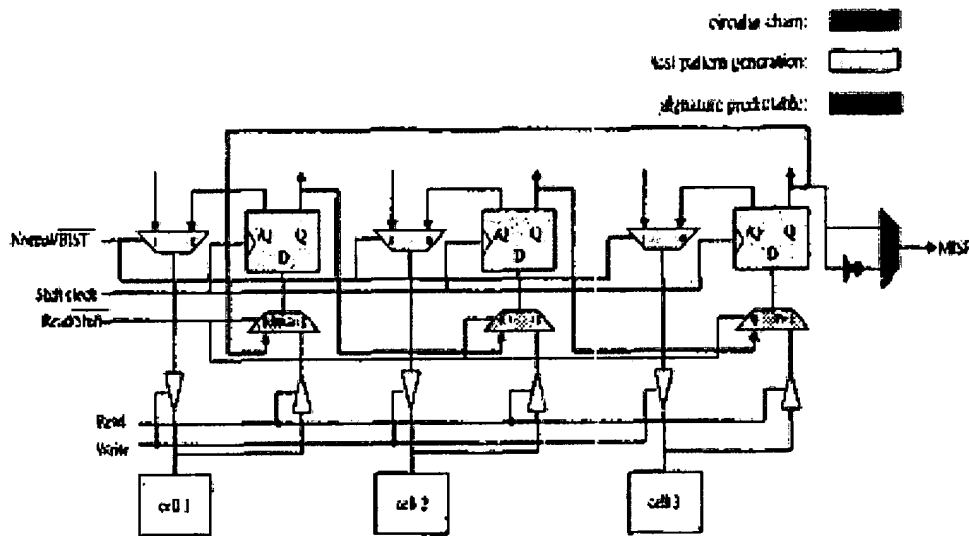
B. Alleged unpatentability of claims 1–5 and 14–16 under § 103(a) based on Huang

1. Huang (Ex. 1004)

Huang describes a transparent built-in self-test (TBIST) interface that is designed to perform testing of multiple embedded memory arrays in the normal mode. Ex. 1004, Abstract. Figure 2 of Huang is reproduced below:



As shown in Figure 2, above, Huang's transparent BIST system scheme includes a BIST controller, transparent test interface, memory arrays and a global multiple input shift register (MISR). *Id.* at 381, Fig. 2. As illustrated in Figure 2, the BIST controller generates Address, and control (Normal/BIST, shift clock, Read, and Write) signals. *Id.* "[A]ll memory modules and interface circuits receive the same test control signals from the transparent BIST controller." *Id.* at 381, col. 2. Figure 1 of Huang is reproduced below:



Huang's Figure 1, above, illustrates the test interface. The test interface "implement[s] the functions of test pattern generation and test response evaluation." *Id.* at 384, col. 2. According to Huang, "the test interface generate[s] test patterns *locally* to avoid test data routing from test controller to each memory array." *Id.* at 381, col. 1. "[T]est patterns are generated by alternatively selecting Q or /Q as an input to each memory cell . . . , so the transition of $Q \rightarrow /Q$ and $/Q \rightarrow Q$ will be exercised in each memory cell. Thus, based on the memory content left by normal operations, new test patterns are generated by complementing each memory word several times." *Id.*

2. Analysis

Petitioner contends Huang's BIST controller, test interfaces, and memory arrays teach or suggest a control circuit, data handlers and memory chips as recited in independent claims 1 and 16. Pet. 36–38. Patent Owner argues: (1) Huang's test interfaces (the alleged data handlers) are not "configured to generate data," and Huang's BIST controller (the alleged

control circuit) is not “configured to generate address and control signals,” as recited in claim 1; (2) Huang’s test interfaces (the alleged data handlers) are not “configured to be operated independently” (claim 1) and cannot be “operating . . . independently” (claim 16); and (3) Huang teaches away from “physically separate integrated circuit packages” as recited in dependent claim 4. PO Resp. 41–54. With the exception of the above-noted limitation in dependent claim 4, Patent Owner does not dispute directly Petitioner’s contentions with respect to the additional limitations recited in dependent claims 2–5, 14, and 15, but contends these claims are not unpatentable over Huang because Petitioner has failed to show independent claim 1, from which these claims depend, would have been obvious in view of Huang. *See id.* at 54.

- a. *“control circuit configured to generate address and control signals” and “one or more data handlers . . . configured to generate data” (claim 1)*

Petitioner contends Huang Figure 2 “illustrates the generation of address signals by the BIST controller and reception of those signals by the memory devices.” Reply 17; *see* Pet. 38–39. Petitioner asserts that Huang Figure 2 also illustrates the generation of “read” and “write,” i.e., control signals, by the BIST controller (the alleged control circuit), and Huang Figure 1 illustrates that these control signals are received by the memory arrays (the alleged memory chips). *Id.*; *see* Pet. 38–39. Petitioner contends Huang’s transparent interfaces (the alleged data handlers) are described as generating test patterns locally, and, therefore, it logically follows that they are “designed, arranged or adapted to perform the function of data generation.” Reply 16; *see* Pet. 37–38.

Patent Owner argues Huang's BIST controller and Huang's transparent interfaces are not "configured to generate address and control signals" and "data," respectively, as recited in claim 1, because they do not receive signals that can be used for programming. PO Resp. 41–43. Patent Owner's argument is based on an overly narrow interpretation of "configured to generate" as "programmed to generate." *See id.* at 42–43. In Section III.B.1, we determined the broadest reasonable interpretation of "configured to," as used in the '501 patent claims, is "designed, adapted, or arranged to," and that one of ordinary skill in the art would understand that programmability is only one way that a component might be designed, adapted, or arranged to perform a function.

We find that Petitioner has shown, by a preponderance of the evidence, that Huang's BIST controller and transparent interfaces teach or suggest the capability of generating address/control signals and data, respectively, for testing the memory arrays, and, therefore, are designed, adapted, or arranged to generate address/control signals and data in the manner recited in challenged claim 1.

- b. *"a plurality of data handlers configured to be operated independently from one [an]other" (claim 1)/"operating a plurality of data handlers independently from one another" (claim 16)*

Patent Owner argues Huang's transparent interfaces (the alleged data handlers) are not "configured to be operated independently" (claim 1) and cannot "operat[e] . . . independently" (claim 16) as recited in the challenged claims because the transparent interfaces "are (i) specified to receive the exact control signals, and (ii) designed to continuously shift the latched content and output multiple bit streams to the MISR in parallel, and thus are

not able to be used *independently* from each of the other data handlers of the plurality of data handlers in order to *independently* test a portion of one or more memory chips.” PO Resp. 43. Patent Owner’s arguments are based on an interpretation of “operable independently” as requiring that each data handler is able to be used to independently test a portion of the memory space. *See generally* PO Resp. 43–52. In Section III.A., however, we determined the broadest reasonable interpretation of “operable independently” is “capable of being put into use without influence or control by another.”

Petitioner argues that “[w]hen the proper construction of ‘operated independently’ is applied, it is apparent that each of Huang’s transparent memory interfaces (*i.e.*, data handlers) meets this claim limitation, since they each correspond to an associated memory device, and locally generate test data for the associated memory device.” Reply 18–19 (citing Pet. 35). Petitioner’s argument is persuasive, and Patent Owner has not explained sufficiently why each of Huang’s transparent memory interfaces is not capable of being put into use without influence or control by another transparent interface. *See generally*, PO Resp. 43–52; *see also* Reply 19 (citing Ex. 1010, 249:6–11 (wherein Dr. Sechen, when questioned about the operation of the transparent memory interfaces A and B (corresponding to the claimed data handlers), illustrated in Figure 4 of Huang, acknowledged that “when transparent memory interface A is applying a test pattern to memory module A,” it does not “affect how transparent memory interface B is applying its test pattern to memory module B”))). Therefore, Patent Owner has not refuted Petitioner’s *prima facie* showing that Huang teaches or suggests that its transparent memory interfaces are “configured to be

operated independently” (claim 1) and can “operat[e] . . . independently” (claim 16) as recited in the challenged claims.

c. dependent claims 2–5, 14 and 15

Claim 4 depends from claim 1 and recites: “wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” Ex. 1001, 16:61–65.

Patent Owner argues Petitioner’s challenge as to claim 4 is based on the combination of Huang and Tsern, and that Petitioner has not explained how Huang, alone, discloses or suggests two physically separate integrated circuit packages as claimed. PO Resp. 53. Patent Owner further relies on the following statement in Huang in support of its contention that Huang teaches away from a design in which individual data handlers are in separate integrated circuit packages: “Due to the improvement of VLSI technology, most systems can be condensed into a chip.” *Id.* at 53–54 (quoting Ex. 1004, 379).

We agree with Patent Owner that in its Petition, Petitioner does not identify clearly a teaching or suggestion in Huang, alone, of two physically separate integrated circuit packages. *See* Pet. 42 (“[A] POSITA would be motivated to combine Huang and Tsern . . . to supplement the teachings of Huang relating to the desire to have test interfaces generate test patterns locally with Tsern’s teachings of efficient ways to incorporate a self-testing memory system within multiple integrated circuits on a printed circuit board.”). Petitioner does not argue, until its Reply, that it would have been nothing more than a routine design choice to use physically separate

integrated circuit packages for individual transparent interfaces (the alleged data handlers). In support of its Reply, Petitioner relies exclusively on the second Alpert declaration, wherein Dr. Alpert explains, for the first time, that modification of Huang to achieve the limitations recited in claim 4 would have been within the knowledge and level of skill of the ordinary artisan. *Compare* Ex. 1007 ¶ 156 (testifying “[t]o the extent that this additional claim element may not be disclosed by Huang alone, it is disclosed by Tsern,” but failing to explain how the claim 4 limitations are disclosed or suggested by Huang), *with* Ex. 1009 ¶ 64 (“A person of ordinary skill in the art would have understood that an embedded computer system can be implemented as a single integrated circuit or as multiple integrated circuits (“chips”) and other components.”).

Rule 42.22(a)(2) provides that a petition must include “[a] full statement of the reasons for the relief requested, including a detailed explanation of the significance of the evidence including material facts, and the governing law, rules, and precedent.” 37 C.F.R. § 42.22(a)(2). Rule 42.23(b) provides that “[a] reply may only respond to arguments raised in the corresponding . . . patent owner response.” 37 C.F.R. § 42.23(b). Practice relating to replies is addressed in the Office Patent Trial Practice Guide, 77 Fed. Reg. 48,756, 48,767 (Aug. 14, 2012):

While replies can help crystalize issues for decision, a reply that raises a new issue or belatedly presents evidence will not be considered and may be returned. The Board will not attempt to sort proper from improper portions of the reply. Examples of indications that a new issue has been raised in a reply include new evidence necessary to make out a *prima facie* case for the patentability or unpatentability of an original or proposed substitute claim, and new evidence that could have been presented in a prior filing.

“Whether a reply contains arguments or evidence that are outside the scope of a proper reply under 37 C.F.R. § 42.23(b) is left to our determination.”
Vibrant Media Inc. v. General Electric Co., Case IPR2013-00170, slip op. at 31 (PTAB June 26, 2014) (Paper 56).

In our judgment, Petitioner’s arguments and evidence in the Reply are outside the scope of a proper reply because they do more than merely address Patent Owner’s argument that Huang teaches away; rather, they attempt to provide a new theory for modifying Huang that was not set forth clearly in the Petition. Consideration by the Board of these arguments in the Reply and evidence presented in Dr. Alpert’s second declaration in support thereof would be unfair to Patent Owner. We have, however, reviewed and considered the arguments in the Petition and evidence therein, and are in agreement with Patent Owner that Petitioner has not provided adequate explanation or evidence to support its contention that Huang discloses or suggests the limitations recited in claim 4. Accordingly, Petitioner has not met its burden to show, by a preponderance of the evidence, that claim 4 would have been unpatentable over Huang.

Patent Owner also contends Petitioner has not met its burden to show unpatentability of dependent claims 2–5, 14, and 15, because Petitioner has failed to show Huang discloses or suggests all of the limitations of independent claim 1, from which these claims depend. PO Resp. 54. With respect to claims 2, 3, 5, 14, and 15, we have reviewed the Petition and evidence cited therein, and determine Petitioner has met its burden to show these claims are unpatentable as obvious in view of Huang. *See* Pet. 39–41, 43–45.

3. Conclusion

Petitioner has met its burden to show that claims 1–3, 5, and 14–16 are unpatentable under 35 U.S.C. §103(a) based on Huang. Patent Owner has persuaded us that Petitioner failed to meet its burden to establish unpatentability of claim 4 under 35 U.S.C. §103(a) based on Huang.

C. Alleged unpatentability of claims 6–8, 11, and 17–20 under § 103(a) based on Huang and Averbuj

Claim 6 depends indirectly from claim 1 and claim 17 depends directly from claim 16. Claims 6 and 17 recite: “verifying that data received from the one or more memory locations of the one or more memory chips corresponds to the data” written/transmitted “to the one or more memory locations of the one or more memory chips.” Ex. 1001, 17:7–11, 18:29–32. Claims 7 and 8, and claims 18–20 include these limitations by virtue of their dependencies from claims 6 and 17, respectively. Claim 11 depends directly from claim 1 and recites that the data generated by the one or more data handlers comprises “*cyclic data*.” *Id.* at 17:33–35. Claim 20 includes a similar limitation. *Id.* at 18:43–45.

With respect to claims 6 and 17, Petitioner contends Averbuj’s comparator 48 corresponds to the claimed verification element. *See* Pet. 46 (citing Ex. 1003 ¶ 53; Figs. 6, 10). Petitioner contends one of ordinary skill in the art “would have been motivated to combine the self-testing memory modules disclosed in Huang . . . with the teachings of Averbuj because (among other reasons) adding this additional functionality within the memory interface furthers Huang’s desire to perform test data operations locally to avoid excess data routing.” Pet. 46 (citing Ex. 1007 ¶¶ 177, 181).

With respect to claims 11 and 20, Petitioner contends Averbuj discloses a data handler (i.e., data generation unit (44)) that “may be

instructed to generate different types of test patterns, including a checkerboard pattern (*i.e.*, cyclic data).” Pet. 49 (citing Ex. 1003 ¶ 49). Petitioner contends one of ordinary skill in the art “would have been motivated to combine the self-testing memory modules disclosed in Huang . . . with Averbuj’s disclosure of cyclic test patterns because this is a well-known technique commonly employed in BIST systems. *Id.* at 49 (citing Ex. 1007 ¶¶ 190).

Patent Owner contends Petitioner has not shown how Huang and Averbuj could be combined in the manner claimed. PO Resp. 54. Patent Owner relies on the declaration testimony of Dr. Sechen to establish Huang and Averbuj are incapable of being combined. *Id.* at 55–59 (citing Ex. 2012 ¶¶ 213, 216–217, 219–220, 223, 225–226).

Dr. Sechen testified that “the prior art elements of Averbuj (*i.e.*, cyclic data generation and data verification) that Petitioner cites are incapable of being combined with Huang” (Ex. 2012 ¶ 213) and “[e]ven if it were assumed that [the ordinary artisan] would have attempted to combine Huang and Averbuj, the combination would not have worked for its intended purpose” (*id.* ¶ 220). Dr. Sechen testified that “[t]he *cyclic data* generation in Averbuj is controlled by the sequencer, . . . [which] instructs the data generation unit 44 to generate different types of test patterns, including a checkerboard pattern (*i.e.*, the alleged *cyclic data*).” *Id.* ¶¶ 217–218. Dr. Sechen further testified that “Huang’s data generation is tied to the memory contents, and specific data patterns cannot be selected by Averbuj’s sequencer;” “Huang’s system cannot be instructed to generate different types of test patterns.” *Id.* ¶ 219. Dr. Sechen also testified as to why he believed

Averbuj's verification element could not be added to Huang. *See id.*

¶¶ 221–227.

In its Reply, Petitioner contends Patent Owner has “fail[ed] to consider the references as a whole,” and has “mistakenly argue[d] that applying specific components of Averbuj with the system of Huang would render the resulting system unfit for its intended purpose.” Reply 20–21. Petitioner explains that according to its proposed combination, Huang's Normal/BIST signal would be used to switch between the data outputs of Huang's testing scheme and Averbuj's testing scheme. *Id.* at 21. Petitioner relies on testimony in Dr. Alpert's second declaration in support of its arguments. *See id.* at 21–22 (citing Ex. 1009 ¶¶ 68–73).

Dr. Alpert testified that “Dr. Sechen's analysis fails to recognize the motivation for combining Huang and Averbuj because he erroneously conflates various types of testing that are used for different purposes at different times.” Ex. 1009 ¶ 68. Dr. Alpert goes on to provide detailed opinion testimony on the understanding of the ordinary artisan with respect to the teachings of Huang and Averbuj, and a detailed explanation of how Huang's system could be modified to include Averbuj's testing system and the motivation of the ordinary artisan to make such combination. *See id.* ¶¶ 69–72. According to Dr. Alpert, “[a] person of ordinary skill in the art would have seen the combination of Averbuj and Huang as choosing from a finite number of identified, predictable solutions, with a reasonable expectation of success.” *Id.* ¶ 72.

At oral argument, Patent Owner confirmed its understanding of Petitioner's challenge, as set forth in the Petition, as “starting with Huang and bringing in one element from Averbuj.” Tr. 59:24–25. Patent Owner

argued that “Petitioner’s reply changes the combination to comprise Averbuj’s entire system, not just the cyclic test generation portion.” *Id.* at 59:21–23. Patent Owner argued that the Petition includes only conclusory statements of a motivation to combine and fails to explain how Averbuj and Huang are combinable. *See id.* at 60:7–15. Patent Owner also argued Dr. Alpert’s second declaration and deposition present new opinions as to how Averbuj and Huang could be combined. *Id.* at 53:3–5. Petitioner contended the challenge, as set forth in the Petition, and Dr. Alpert’s first declaration in support thereof, were sufficient because combining Averbuj and Huang was trivial. *See id.* at 26:15–17.

Petitioner contended Dr. Alpert did not express new opinions in his second declaration and deposition, but was merely responding to Dr. Sechen’s testimony. *Id.* at 81:17–19; *see also, id.* at 82:20–83:3 (wherein Petitioner argued: “[I]n their response, Patent Owner took the position that . . . it was physically incapable to combine Averbuj and Huang. We’re entitled to respond to that and show that, in fact, they’re not correct, and we did it by saying, listen, we pointed to the multiplexers in the petition, it’s super, it’s simple to use those multiplexers to combine. . . . [I]t is not incapable. A person of skill in the art could have done that.”).

As discussed in Section IV.B.2.c., a petition must include “[a] full statement of the reasons for the relief requested, including a detailed explanation of the significance of the evidence including material facts, and the governing law, rules, and precedent” (37 C.F.R. § 42.22(a)(2)), and “[a] reply may only respond to arguments raised in the corresponding . . . patent owner response” (37 C.F.R. § 42.23(b)).

We agree with Patent Owner that the Reply does more than merely respond to or rebut arguments made in Patent Owner's Response; rather, it sets forth a theory that was not presented clearly in the Petition. Patent Owner's position is supported by the fact that Dr. Sechen, whose skill level is at least (and appears to exceed) that of the ordinary artisan, apparently was unable to envision how Huang and Averbuj could be combined. Further, Dr. Alpert's second declaration includes, at a minimum, opinions as to the understanding of the ordinary artisan with respect to the teachings of Huang and Averbuj that were not presented in his first declaration.

In our judgment, consideration by the Board of the above-noted arguments in the Reply and evidence presented in Dr. Alpert's second declaration in support thereof would be improper under 37 C.F.R. § 42.23, and unfair to Patent Owner. We have, however, reviewed and considered the arguments in the Petition and evidence therein, and are in agreement with Patent Owner that Petitioner has not provided adequate explanation or evidence to support its contention that one of ordinary skill in the art would have had a reasonable expectation of success in combining Huang and Averbuj. Accordingly, Petitioner has not met its burden to show, by a preponderance of the evidence, that claims 6–8, 11, and 17–20 would have been unpatentable over Huang and Averbuj.

D. Alleged unpatentability of claims 9 and 10 under 35 U.S.C. § 103(a) based on (1) Huang and JEDEC Standard, and (2) Averbuj and JEDEC Standard

JEDEC Standard “covers Design for Test, Design for Manufacturing and Design for Validation (‘DFx’) requirements and implementation guidelines for Fully Buffered DIMM technology.” Ex. 1006, 1. In its Petition, Petitioner relies on JEDEC Standard for a teaching that “failing

data and [the] corresponding address should be recorded in an error register.” Pet. 30, 51 (citing Ex. 1006 § 8.2.4.5). Petitioner argues it would have been obvious to combine the teachings of JEDEC Standard with the BIST architectures disclosed in Averbuj and Huang “for many reasons, including the well known benefit of recording memory errors in a BIST system to enable a failure analysis and determine where the errors are located.” *Id.* at 31, 52. Petitioner relies on Dr. Alpert’s first Declaration to support its contention that it would have been obvious to modify the respective memory systems of Averbuj and Huang to store addresses and data as taught by JEDEC Standard. Pet. 30–31, 51–52 (citing Ex. 1007 ¶¶ 123–127, 206–209). Dr. Alpert cites column 3, lines 46–49 of the ’501 patent in support of his testimony that “the ’501 patent acknowledges the relevancy of the JEDEC standard to self-testing memory modules.” Ex. 1007 ¶ 124; *see* Ex. 1001, 3:46–49 (“Certain embodiments described herein present a self-testing memory module that includes a control module and a data module which can generate memory addresses and data according to the JEDEC standard memory protocol.”).

In its Response, Patent Owner argues that claims 9 and 10 require a system configured to record “failures,” but in the system described in JEDEC Standard, testing stops after only one failure is recorded. Resp. 59–60. Patent Owner further argues claims 9 and 10 require an internal tester, but the tester in JEDEC Standard is external to the memory system being tested. *Id.* at 60. In support of these arguments, Patent Owner relies on the declaration testimony of its witness, Dr. Sechen (Exhibit 2012). *Id.* at 59–60 (citing Ex. 2012 ¶¶ 229–32).

Petitioner, in its Reply, argues “[t]he claims are not limited to recording failures during a single test. Instead, the cla[i]ms only require that the data and addresses associated with the failures be stored.” Reply 22. Petitioner directs us to the section of JEDEC Standard relied on its Petition, noting there is a subsection on “Multiple Failures” and a description of “a procedure to generate ‘a complete data log of all failures’ including the address and data associated with the failure.” *Id.* (citing Ex. 1006 § 8.2.4.5). In response to Patent Owner’s argument that the system in JEDEC Standard uses an external tester, Petitioner, relies on testimony in Dr. Alpert’s second Declaration (Ex. 1009). *See* Reply 22–23. Petitioner argues Dr. Sechen’s testimony is based on disclosure in JEDEC Standard relating to transparent mode testing (Ex. 1006 § 8.1), which differs from Memory BIST, described in the section of JEDEC Standard cited in the Petition (*id.* at § 8.2). *See id.*; Ex. 1009 ¶ 79 (asserting that the sections of JEDEC Standard relied on by Dr. Sechen, e.g., sections 5.1 and 5.5, relate to transparent mode). Dr. Alpert explains that “Transparent Mode is used during FB-DIMM manufacturing to detect defects in the memory devices, interface, and board assembly using data patterns supplied by an external tester. Memory BIST is used for testing a FB-DIMM in a system at power-on initialization without using test equipment external to the system.” Ex. 1009 ¶ 78. During the oral hearing, Patent Owner did not dispute these assertions. *See generally*, Paper 33 (transcript of the oral hearing).

We find that Dr. Sechen’s testimony related to JEDEC Standard’s description of testing in transparent mode and does not address adequately JEDEC Standard’s discussion of Memory BIST, i.e., the section of JEDEC Standard relied on by Petitioner in support of its patentability challenges as

to claims 9 and 10. We also agree with Petitioner that Patent Owner's arguments are not commensurate in scope with claims 9 and 10. We find Petitioner's contentions with respect to the teachings of JEDEC Standard and motivation for modifying the respective systems of Averbuj and Huang are supported by JEDEC Standard, and conclude Petitioner has met its burden to show, by a preponderance of the evidence that claims 9 and 10 are unpatentable over (1) Averbuj and JEDEC Standard, and (2) Huang and JEDEC Standard.

V. CONCLUSION

Petitioner has met its burden to show, by a preponderance of the evidence, unpatentability of: (1) claims 1–3, 5–8, 11–14, and 16–20 under 35 U.S.C. §102(b) based on Averbuj; (2) claims 1–3, 5, and 14–16 under 35 U.S.C. §103(a) based on Huang; (3) claims 9 and 10 under 35 U.S.C. § 103(a) based on Averbuj and JEDEC Standard; and (4) claims 9 and 10 under 35 U.S.C. § 103(a) based on Huang and JEDEC Standard.

Petitioner has failed to meet its burden to establish unpatentability of: (1) claim 15 under 35 U.S.C. §102(b) based on Averbuj; (2) claim 4 under 35 U.S.C. §103(a) based on Huang; and (3) claims 6–8, 11, and 17–20 under 35 U.S.C. §103(a) based on Huang and Averbuj.

VI. ORDER

For the reasons given, it is

ORDERED that claims 1–3 and 5–20 are unpatentable; and

FURTHER ORDERED that because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2014-00971
Patent 8,359,501 B1

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September 19, 2016

Attached is a copy of the Notice of Appeal for IPR 2014-01374, pursuant to 37 C.F.R. 1.302. The Notice of Appeal was timely filed at the United States Patent and Trademark Office and served on Patent Owner on September 8, 2016.

Sincerely,

/s Timothy H. Caine

Timothy H. Caine
Associate

ATTORNEY REFERENCE NO.: 21584-112005.4
IPR2014-01374

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE
BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES, INC.

Petitioner

v.

NETLIST, INC.

Patent Owner

Case IPR2014-01374

Patent 8,359,501

PETITIONER'S NOTICE OF APPEAL TO THE
U.S. COURT OF APPEALS FOR THE FEDERAL CIRCUIT

Via PRPS

Mail Stop PATENT BOARD
Patent Trial and Appeal Board
United States Patent and Trademark Office
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Pursuant to 35 U.S.C. Sections 141, 142, and 319, 37 C.F.R. Sections 90.2, 90.3, and 104.2; Rule 4(a)(3) of the Federal Rules of Appellate Procedure, and Rule 15 of the Federal Circuit Rules, the Petitioner appeals to the United States Court of Appeals for the Federal Circuit from the Final Written Decision (Paper No. 44), the Decision Denying Petitioner's Request for Rehearing (Paper No. 47), and from all underlying findings, orders, decisions, determinations, rulings, and opinions adverse to Petitioner. Petitioner indicates that the appeal includes at least the following issues:

1. the Board's judgment that claims 1, 2, 4-6, 9-11, 14-17 & 20 of U.S. Patent No. 8,359,501 are patentable; and
2. the Board's claim construction.

Petitioner reserves the right to challenge any findings, orders, decisions, determinations, rulings, and opinions adverse to Petitioner.

Dated: September 8, 2016

Respectfully submitted,

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Certificate Of Service In Compliance With 37 C.F.R. § 42.6(e)(4)

The undersigned certifies that a complete copy of this *PETITIONER'S MOTION TO EXCLUDE EVIDENCE* was served via email on Counsel for Patent Owner in this proceeding:

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Paper 44
Entered: March 9, 2016

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01374
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
MATTHEW R. CLEMENTS, *Administrative Patent Judges*.

Opinion for the Board filed by *Administrative Patent Judge* GAUDETTE.

Opinion Dissenting filed by *Administrative Patent Judge* CLEMENTS.

GAUDETTE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. § 318(a) and 37 C.F.R. § 42.73

IPR2014-01374
Patent 8,359,501 B1

I. BACKGROUND

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) on August 23, 2014, requesting institution of an *inter partes* review of claims 1–20 (the “challenged claims”) of U.S. Patent No. 8,359,501 B1 (Ex. 1008, “the ’501 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 10). Based on these submissions, an *inter partes* review of claims 1, 2, 4–6, 9–11, 14–17, and 20 was instituted on March 10, 2015, pursuant to 35 U.S.C. § 314. Paper 12 (“Dec. on Inst.” or “Institution Decision”).

After institution, Patent Owner filed a Response (Paper 21, Corrected Response, “PO Resp.”), and Petitioner filed a Reply to the Patent Owner Response (Paper 26, “Reply”).

Petitioner filed a Motion to Exclude Evidence. Paper 25; *see also* Paper 36 (Patent Owner’s Opposition) and Paper 41 (Petitioner’s Reply). Patent Owner also filed a Motion to Exclude Evidence (Paper 33; *see also* Paper 38 (Petitioner’s Opposition) and Paper 40 (Patent Owner’s Reply)) and a Motion to Exclude Portions of Petitioner’s Reply (Paper 34; *see also* Paper 39 (Petitioner’s Opposition)).

Oral argument was held on November 17, 2015, and a transcript (Paper 42, “Tr.”) has been entered into the record.

The Board has jurisdiction under 35 U.S.C. § 6(c). This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73, addresses issues and arguments raised during trial.

For the reasons discussed below, we determine Petitioner has not met its burden to prove, by a preponderance of the evidence, that claims 1, 2, 4–

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6, 9–11, 14–17, and 20 of the '501 patent are unpatentable. Petitioner's and Patent Owner's Motions to Exclude are dismissed.

A. Related Matters

The '501 patent claims priority as a continuation of U.S. Patent Application No. 12/422,925, now U.S. Patent No. 8,001,434 ("the '434 patent"). Ex. 1008, 1. Petitioner also requested, and we instituted, an *inter partes* review of the '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01372 (PTAB March 10, 2015) ("IPR-1372"), Paper 13. We denied Petitioner's requests to institute *inter partes* reviews of the '434 patent and the '501 patent in two additional petitions, filed August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB March 13, 2015), Paper 16, and Case IPR2014-01375 (PTAB March 13, 2015), Paper 12, respectively. Oral argument in IPR2014-01372 was consolidated with the oral argument in the present *inter partes* review. A final decision in IPR2014-01372 is issued concurrently with this final decision.

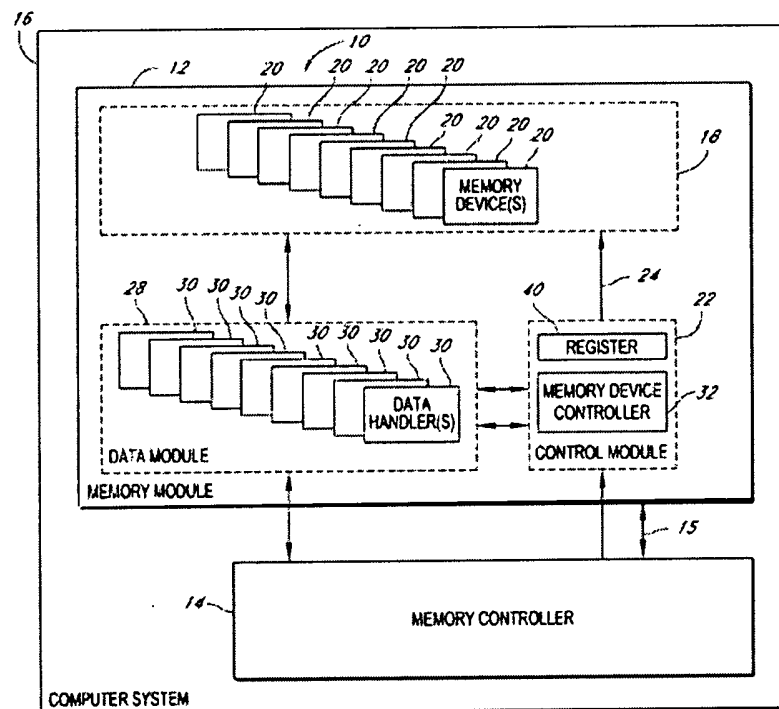
The '501 patent also is the subject of IPR2014-00971 ("IPR-971"). We entered a final written decision in that case on December 14, 2015, concluding that claims 1–3, 5–8, and 11–20 are unpatentable on the following grounds: claims 1–3, 5–8, 11–14, and 16–20 under 35 U.S.C. § 102(b) based on Averbuj (U.S. Patent Application Publication No. 2005/0257109 A1, issued November 17, 2005); and claims 1–3, 5, and 14–16 under 35 U.S.C. § 103(a) based on Huang (*An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers*, Fourteenth International Conference on VLSI Design 379–384 (2001), IEEE). *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB

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December 14, 2015), Paper 34 (“IPR-971 FWD”). Petitioner Sandisk Corporation has requested rehearing of our determination that it did not meet its burden to show unpatentability of claims 4, 9, and 10, and our determination that it failed to show unpatentability of claim 15 as anticipated by U.S. Patent Application Publication No. 2005/0257109 A1. *Id.*, Paper 35.

B. The '501 Patent (Ex. 1008)

The '501 patent relates to self-testing electronic memory modules. Ex. 1008, 1:28–29. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '501 patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10

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includes a plurality of memory devices 18, e.g., ETT DRAM chips, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 3:34–36; 5:10–12.

Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:14–16, 42–46. “In various embodiments, the control module 22 includes . . . one or more application-specific integrated circuit[s]” *Id.* at 9:62–64. “[D]ata module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20.” *Id.* at 6:5–8. “For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according [to] the operating specification of the memory devices 20.” *Id.* at 6:8–11.

“In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20.” *Id.* at 5:48–52. “For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation.” *Id.* at 5:52–55.

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A. Illustrative Claims

Of the challenged claims, claims 1 and 16 are independent, and are reproduced below:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:

a plurality of memory chips;

a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;

a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the one or more data handlers.

16. A method of operating a memory system configured to be operatively coupled to a memory controller of a computer system, the memory system having a plurality of memory chips, the method comprising:

operating a plurality of data handlers independently from one another to generate and transmit data to one or more memory locations of one or more memory chips of the plurality of memory chips;

operating a control circuit to generate address and control signals; and

testing the one or more memory locations of the one or more memory chips using the address and control signals generated by the control circuit and using the data generated by the plurality of data handlers.

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C. The Instituted Ground of Unpatentability

We instituted the instant *inter partes* review of claims 1, 2, 4–6, 9–11, 14–17, and 20 based on anticipation under 35 U.S.C. § 102(b) by Averbuj, U.S. Patent No. 7,392,442 B2, issued June 24, 2008 (Ex. 1011, “Averbuj”). As indicated in Section I.A., above, in IPR-971, we held claims claims 1–3, 5–8, 11–14, and 16–20 of the ’501 patent unpatentable under 35 U.S.C. § 102(b) based on U.S. Patent Application Publication No. 2005/0257109 A1, the published application of the Averbuj patent on which the instituted ground of unpatentability in this proceeding is based.

D. Expert Testimony

Petitioner relies on the testimony of Dr. Nader Bagherzadeh in support of its patentability challenge. Pet. 1. Dr. Bagherzadeh executed a declaration (Ex. 1009, “the Bagherzadeh Declaration”) in support of the Petition. Dr. Bagherzadeh was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 2012.

Dr. Bagherzadeh testifies as follows: “In 1979 and 1987, respectively, [he] earned a master of science in electrical engineering and a doctorate degree in computer engineering from the University of Texas at Austin.” Ex. 1009 ¶ 4. He has been involved in design and development of digital systems for more than 30 years. *Id.* ¶ 6. Dr. Bagherzadeh joined the University of California, Irvine, in 1987, and has “been teaching, researching, and consulting regarding almost all aspects of memory design for high performance computer systems, including but not limited to DRAMs and SRAMs.” *Id.* ¶ 7. He has been employed as a professor in the department of Electrical Engineering and Computer Science at the

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University of California, Irvine, since 2003. *Id.* ¶ 5. In 2000, Dr. Bagherzadeh became a cofounder of Morpho Technologies, a high tech company focused on the design and development of low power and high performance digital signal processors for mobile applications. *Id.* ¶ 7. He was involved in evaluating patents, technical reports and presentations related to memory chip designs, DSPs, and parallel processing algorithms for mobile platforms. *Id.*

Patent Owner relies on the testimony of Dr. Carl Sechen. PO Resp. 1. Dr. Sechen executed a declaration (Ex. 2019, “the Sechen Declaration”) in support of Patent Owner’s Response. Dr. Sechen was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 1031.

Dr. Sechen testifies as follows: he has an M.S. degree in Electrical Engineering from the Massachusetts Institute of Technology and “was awarded a Ph.D. in electrical engineering from the University of California at Berkeley in 1986.” Ex. 2019 ¶ 5. Dr. Sechen has been a Professor of Electrical Engineering for more than 28 years. *Id.* ¶ 2. During this time period, his research has focused on design and computer-aided design of digital integrated circuits, including the design of DRAM, and he has taught numerous students how to design DRAM memories. *Id.* Dr. Sechen has “also been involved in numerous research projects on VLSI design and memory design[, and has] taught numerous graduate researchers how to design digital integrated circuits, including memories.” *Id.* ¶ 4.

The parties do not dispute that Dr. Bagherzadeh and Dr. Sechen are qualified to testify as experts under FRE 702.

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II. LEVEL OF ORDINARY SKILL IN THE ART

The level of ordinary skill in the art is relevant to claim construction and anticipation. *See Yorkey v. Diab*, 605 F.3d 1297, 1300 (Fed. Cir. 2010) (explaining that a determination of anticipation involves interpreting the claim language and then comparing the construed claim to a prior art reference); *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1259–60 (Fed. Cir. 2010) (“[C]laim language should be read in light of the specification as it would be interpreted by one of ordinary skill in the art.”).

Dr. Bagherzadeh and Dr. Sechen agree that the field of art is “memory module design with features for built-in self-test (BIST).” Ex. 2019 ¶ 21; *see also* Ex. 1009 ¶ 17.

Dr. Bagherzadeh testifies that

a person of ordinary skill in the art for the ‘501 patent in April 2008 would have a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering, and at least one year of experience relating to memory systems and BIST.

A person having a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering would have experience with digital circuit design, memories and computer architecture. An additional year of experience relating to memory systems and BIST would focus on those areas and allow one to recognize current issues with respect to the field.

Ex. 1009 ¶¶ 19–20; *see id.* ¶ 16 (“My understanding is that the earliest possible priority date of the ‘501 patent is April 14, 2008.”).

Dr. Sechen testifies that

one of ordinary skill in the art at the time of filing the application(s) for the ‘434 [sic, ‘501] Patent, who would be working on the design of memory devices and memory modules,

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would have at least a Bachelor of Science degree in electrical engineering or computer engineering, and at least five years of industry experience designing memory devices and memory modules. Alternatively, one of ordinary skill in the art would have an M.S. degree in electrical engineering or computer engineering, and at least three years of industry experience designing memory devices and memory modules. Moreover, one of ordinary skill in the art would have a Ph.D. degree in electrical engineering or computer engineering, and have at least one year of industry experience designing memory devices and memory modules.

Ex. 2019 ¶ 21.

The difference in the opinions of the two experts is essentially that Dr. Sechen believes a person of ordinary skill in the art would have additional experience, education, or a combination of the two, beyond that proposed by Dr. Bagherzadeh. Both experts have been employed as professors since the mid 1980s. Dr. Bagherzadeh also has some experience working with engineers in the industry by virtue of his employment with AT&T Bell Labs from 1980–1984 (Ex. 1009 ¶ 6; *id.* Appx. A). Both experts were awarded bachelors and masters degrees in electrical engineering. Dr. Bagherzadeh was awarded a doctorate in computer engineering, while Dr. Sechen was awarded a doctorate in electrical engineering.

Based on a comparison of Dr. Bagherzadeh's and Dr. Sechen's declarations (Exs. 1009, 2019), we find Dr. Bagherzadeh has a broader range of educational and work experience, and, therefore, accord greater weight to his opinion as to the level of ordinary skill in the art. Dr. Bagherzadeh's opinion is consistent with our finding as to the level of ordinary skill in the art in our Final Written Decision in IPR-971. We, therefore, adopt our finding in IPR-971 as to the level of ordinary skill in the art:

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[A] person of ordinary skill in the art at the time of the '501 patent would have a Bachelor's degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.

IPR-971 FWD, 10–11 (noting that our finding was “based on our review of the '501 patent and the types of problems and solutions described in the '501 patent and cited prior art”).

III. CLAIM CONSTRUCTION

In its Petition, Petitioner offers specific constructions for the claim terms “operatively coupled,” “ports,” “to generate,” “[operated/operating [. . .] independently,” and “cyclic data.” Pet. 19–22. We determined that, for purposes of our Institution Decision, only the term “generate” required express construction. Dec. on Inst. 7. We interpreted “generate” as meaning “produce” or “cause.” *Id.* at 9.

In its Response, Patent Owner disagrees with our interpretation of “generate” as meaning “cause.” PO Resp. 17–18. Patent Owner contends our “Institution Decision also provides a construction for ‘configured to generate,’” as recited in claim 1, and proposes its own construction for this phrase. *Id.* at 9 (citing Dec. on Inst. 9). In its Reply, Petitioner asserts that our interpretation of “generate” in the Institution Decision is correct (Reply 1), and offers a proposed construction for the term “configured” (*id.* at 9–10). Patent Owner’s and Petitioner’s proposed constructions for the disputed claim language are listed in the table below.

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Claim Term/Phrase	Patent Owner's Proposed Construction	Petitioner's Proposed Construction
generate	produce (PO Resp. 17)	cause or initiate transmission (Pet. 20–21) produce or cause (Reply 1)
configured to generate	programmed to generate/produce (<i>see</i> PO Resp. 9)	arranged or prepared to produce or cause (Reply 9–10)

In support of their proposed constructions of the disputed claim language, the parties rely on language in the '501 patent and extrinsic evidence. *See generally*, Pet. 20–21; PO Resp. 8–33; Reply 1–17. Neither party relies on prosecution history disclaimer in support of its proposed constructions.

In an *inter partes* review, the Board interprets claim terms in an unexpired patent according to the broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b); *see also In re Cuozzo Speed Techs., LLC*, 793 F.3d 1268, 1278–80 (Fed. Cir. 2015) (“Congress implicitly approved the broadest reasonable interpretation standard in enacting the AIA,” and “the standard was properly adopted by PTO regulation.”), *cert. granted, sub nom. Cuozzo Speed Techs. LLC v. Lee*, 136 S. Ct. 890 (mem.) (2016). Under that standard, and absent any special definitions, we give claim terms their ordinary and customary meaning, as would be understood by one of ordinary skill in the art at the time of the invention. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007); *see also Trivascular, Inc. v. Samuels*, No. 2015-1631, 2016 WL 463539, at *3 (Fed. Cir. Feb. 5, 2016) (“Under a broadest reasonable interpretation, words of the claim must be given their plain meaning, unless such meaning is inconsistent with the specification and prosecution

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history.”). Any special definitions for claim terms must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

“[T]he Board may rely on dictionaries ‘so long as the dictionary definition does not contradict any definition found in or ascertained by reading the patent document.’” *Belden Inc. v. Berk-Tek LLC*, 610 Fed. App’x 997, 1002 (Fed. Cir. 2015) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1322–23 (Fed. Cir. 2005)). Expert testimony is useful to explain terms of art, and the state of the art at any given time, but cannot be used to prove “the proper or legal construction of any instrument of writing.” *Teva Pharm. USA, Inc. v. Sandoz, Inc.*, 135 S. Ct. 831, 841 (2015) (internal citations omitted); *see also* 37 C.F.R. § 42.65(a) (“Expert testimony that does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.”).

Only terms which are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999). A court may revisit and alter its construction of claim terms as the record in a case develops. *See Pressure Prods. Med. Supplies, Inc. v. Greatbatch Ltd*, 599 F.3d 1308, 1316 (Fed. Cir. 2010).

A. Generate

The terms “generate” and “generated” are used in both independent claims 1 and 16. Petitioner contends “[t]he Institution Decision correctly found that the intrinsic and extrinsic record supported construing ‘generate’ as ‘cause’ or ‘produce.’” Reply 1. Patent Owner contends “‘generate’ is properly construed to mean ‘produce,’” but “a construction of generate that

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includes ‘cause’ is unsupported by, – and contradictory to – the intrinsic and extrinsic evidence, and fails to provide reasonable clarity.” PO Resp. 17–18. The same respective constructions of “generate” were advanced by Patent Owner and the petitioner, Sandisk Corporation, in IPR-971. *See* IPR-971 FWD, 23–25.

We provided an extensive analysis of the term “generate” in our Final Written Decision in IPR-971. *See id.* at 22–30. “[W]e agree[d] with Patent Owner that the broadest reasonable interpretation of the claim term ‘generate’ is ‘produce’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’” *Id.* at 29. We interpreted the claim language “address and control signals *generated* by the control circuit” (claims 1 and 16) as “encompassing signals . . . that originated in the control circuit . . . , including by transformation or modification of information and/or data received from another component. *Id.* We also stated that “[w]e [did] not interpret this language as encompassing signals and data received by the . . . control circuit from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the control circuit.” *Id.* at 29–30.

In support of its contention that “generate” also should be interpreted as meaning “cause,” Petitioner relies on extrinsic evidence that was not before us in IPR-971, i.e., the Bagherzadeh Declaration (Ex. 1009) ¶ 28, ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms, Fourth Edition, 1988 (Ex. 1016, “IEEE”), and *Webster’s II New College Dictionary*, 2001 (Ex. 1032, “Webster’s II”). *See* Pet. 20–21; Reply 1–9. We consider this evidence as it is relevant to whether the term “generate” has a particular meaning to one of ordinary skill in the art. *See Trivascular*,

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2016 WL 463539, at *3 (“Construing individual words of a claim without considering the context in which those words appear is simply not ‘reasonable.’ Instead, it is the ‘use of the words in the context of the written description and customarily by those of skill in the relevant art that accurately reflects both the “ordinary” and “customary” meaning of the terms in the claims.” (quoting *Ferguson Beauregard/Logic Controls, Div. of Dover Res., Inc. v. Mega Sys., LLC*, 350 F.3d 1327, 1338 (Fed. Cir. 2003))).

The IEEE defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters.” Ex. 1016, 5. Dr. Bagherzadeh cites the IEEE definition in support of his testimony that “[t]he broadest reasonable interpretation of the claim [term] ‘to generate’ in claims 1 and 16, is ‘to cause or initiate transmission.’” Ex. 1009 ¶ 28 (also citing Ex. 1008, 6:3–5, 10:67–11:2, 6:14–17). Dr. Bagherzadeh testifies that “[t]his interpretation is consistent with the specification and extrinsic evidence because they describe generating as initiating or transmitting, which is also the broadest interpretation of ‘generate.’” *Id.*

We are not persuaded by Dr. Bagherzadeh’s testimony because it is devoid of specific facts and analysis as to what led him to conclude that a person of ordinary skill in the art would interpret the claim term “generate” in the manner Petitioner proposes. *See id.* Dr. Bagherzadeh does not explain how the IEEE definition supports an interpretation of generate as “cause or initiate transmission,” terms that do not appear in the IEEE definition. *See id.* Nor does Dr. Bagherzadeh identify other extrinsic evidence which he believes to be consistent with an interpretation of

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“generate” as meaning “cause or initiate transmission.” *See id.* Likewise, Dr. Bagherzadeh has not explained why he believes the ’501 patent “describe[s] generating as initiating or transmitting” (Ex. 1009 ¶ 28), as these terms do not appear in the cited portions of the ’501 patent. *See* Ex. 1008, 6:8–10, 11:4–11:6, 6:19–22.¹ Accordingly, Dr. Bagherzadeh’s testimony does not persuade us to accept Petitioner’s interpretation of “generate” as meaning “cause.”

Petitioner contends “Dr. Bagherzadeh’s use of the IEEE definition specifically illustrates that ‘generate’ includes ‘selection . . . from a set,’ which implies that ‘generate’ must not be limited to creating from previously undefined values, which may be a more traditional meaning, but holds little value in the intrinsic evidence in this proceeding.” Reply 6. Petitioner has not identified corresponding testimony by Dr. Bagherzadeh or other evidentiary support for this statement and, therefore, this argument is

¹ We understand Dr. Bagherzadeh’s citations to “6:3–5, 10:67–11:2, 6:14–17” to be a typographical mistake. In IPR-1372, Dr. Bagherzadeh cites to the same column and line numbers in the ’434 patent in support of an interpretation of the claim term “generate” as meaning “cause or initiate transmission.” *See* IPR-1372, Ex. 1009 ¶ 28. The disclosures at the cited column and line numbers differ in the ’434 and ’501 patents. Based on the citations to the ’501 patent relied upon by Petitioner (*see* Pet. 20–21), we presume Dr. Bagherzadeh intended to rely on column 6, lines 8–10 and 19–22, and column 11, lines 4–6 of the ’501 patent, which disclosure is identical to column 6, lines 3–5 and 14–17, and column 10, line 67–column 11, line 2 of the ’434 patent. *Compare* Ex. 1008, 6:8–10, 11:4–6, 6:19–22 with IPR-1372, Ex. 1008 (the ’434 patent), 6:3–5, 10:67–11:2, 6:14–17. This disclosure, like the disclosure in the ’501 patent cited in paragraph 28 of Dr. Bagherzadeh’s declaration, does not include the terms “initiating or transmitting.”

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likewise unpersuasive. *See id.*

In support of its construction of “generate” as meaning “cause,” Petitioner also relies on a finding in our Institution Decision that such construction is supported by Merriam-Webster’s Collegiate Dictionary, wherein one definition of “generate” is “to be the cause of” (Ex. 2009, 3). Reply 2 (citing Dec. on Inst. 9 [sic, 8]). This argument is not persuasive because the definition in Merriam-Webster’s Collegiate Dictionary contradicts the definition of “generate” that we ascertained upon reading the ’501 patent as discussed in our Final Written Decision in IPR-971. *See Belden*, 610 Fed. App’x at 1002.

In sum, we are not persuaded that Petitioner’s extrinsic evidence supports a finding that a customary meaning of “generate” to one of ordinary skill in the art is “cause.”² We, therefore, determine the broadest reasonable construction of “generate” is “produce” for the reasons stated above and in our Final Written Decision in IPR-971. We adopt and incorporate by reference our analysis and construction of the term “generate” in our Final Written Decision in IPR-971 (IPR-971 FWD, 22–30).

B. Configured to

The term “configured to” is used in independent claim 1, but not in independent claim 16. In our Final Decision in IPR-971, we construed “configured to” as “designed to, adapted to, or arranged to.” *See* IPR-971 FWD, 18–22. Petitioner contends “configured” should be construed as

² Petitioner relies on *Webster’s II*, for a definition of “cause.” *See* Reply 4, 6–8 (citing Ex. 1032, 3). Having determined the broadest reasonable interpretation of “generate” does not encompass “cause,” we do not find this evidence relevant to our claim construction analysis.

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“arranged” or “prepared.” Reply 9–10. Responsive to this Board panel’s questioning during the oral hearing, however, Petitioner agreed that “designed to, adapted to, or arranged to” was an acceptable definition for the claim term “configured to.” *See* Tr. 22:7–17.

Patent Owner contends the phrase “configured to generate” should be construed as “programmed to generate.” PO Resp. 9. Patent Owner does not propose a separate construction for the term “configured.” *See generally id.* at 9–17. In support of its proposed construction, Patent Owner relies on language in the ’501 patent and the testimony of its expert, Dr. Sechen. *See id.* Dr. Sechen testifies as to the meaning of “configured to generate,” but does not provide a separate discussion of the meaning of “configured to” outside the context of “configured to generate.” *See* Ex. 2019 ¶¶ 42–57.

As explained in our Final Decision in IPR-971, the term “configured to” is used in the ’501 patent in conjunction with numerous terms, such as “provide” and “selectively input,” and there is a presumption that the term “configured to” should carry the same meaning in each instance. *See* IPR-971 FWD, 18–19. The evidence relied on by Patent Owner in support of its proposed construction of the term “configured to” in the present *inter partes* review does not differ materially from the evidence relied upon by Patent Owner in IPR-971. *Compare* PO Resp. 9–17 with IPR-971, Paper 19, 14–19. Accordingly, we determine the broadest reasonable construction of “configured to” is “designed to, adapted to, or arranged to” for the reasons stated above and in our Final Written Decision in IPR-971. We adopt and incorporate by reference our analysis and construction of the term “configured to” in our Final Written Decision in IPR-971 (IPR-971 FWD, 18–22).

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In summary, we construe the claim terms in controversy as follows:

Claim Term/Phrase	Interpretation
Generate	Produce
Configured to	Designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]
Configured to Generate	Designed, adapted, or arranged to produce

IV. ALLEGED ANTICIPATION OF CLAIMS 1, 2, 4–6, 9–11, 14–17, and 20 BY AVERBUJ

A. Principles of Law

To prevail in its challenges to the patentability of the claims, a petitioner must establish facts supporting its challenges by a preponderance of the evidence. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference. *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). While the elements must be arranged or combined in the same way as in the claim, “the reference need not satisfy an *ipsissimis verbis* test.” *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009).

A single prior art reference that discloses, either expressly or inherently, each limitation of a claim invalidates that claim by anticipation. Thus, a prior art reference without express reference to a claim limitation may nonetheless anticipate by inherency. “Under the principles of inherency, if the prior art necessarily functions in accordance with, or includes, the claims limitations, it anticipates.”

Perricone v. Medicis Pharm. Corp., 432 F.3d 1368, 1375–76 (Fed. Cir.

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2005) (citations omitted). “In general, a limitation or the entire invention is inherent and in the public domain if it is the ‘natural result flowing from’ the explicit disclosure of the prior art.” *Schering Corp. v. Geneva Pharms., Inc.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003).

“[T]he Board is not bound by any findings made in its Institution Decision. . . . The Board is free to change its view of the merits after further development of the record, and *should do so* if convinced its initial inclinations were wrong.” *Trivascular*, 2016 WL 463539, at *9.

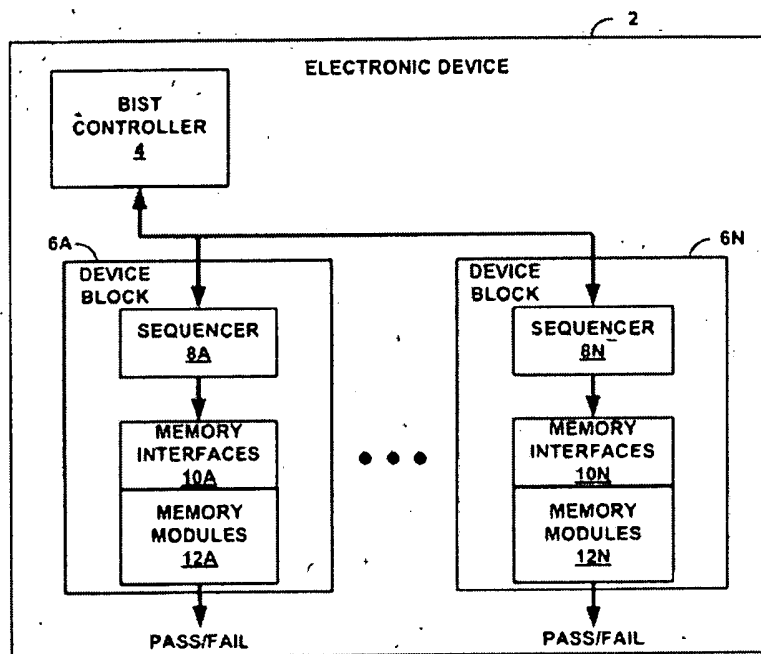
We analyze the instituted ground of unpatentability in accordance with the above-stated principles.

B. Averbuj (Ex. 1011)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1011, 1:64–65, 2:3–4. Figure 1 of Averbuj is reproduced below.

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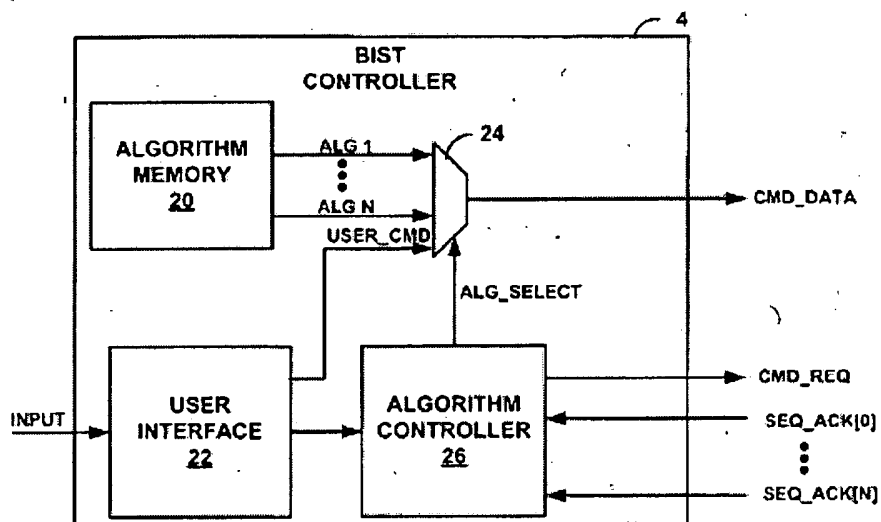


Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* at 3:62–64. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* at 3:64–67. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* at 3:67–4:3.

BIST controller 4 is illustrated in Figure 2, reproduced below.

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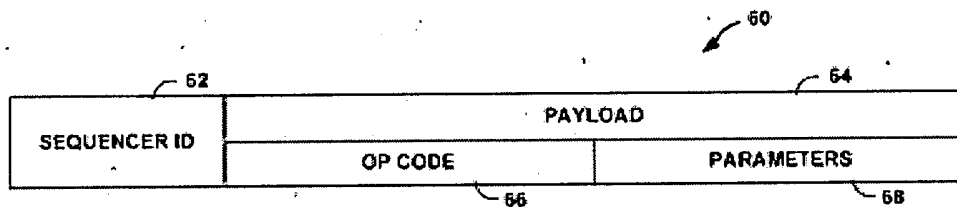


Averbuj Figure 2, above, “is a block diagram illustrating an example embodiment of a BIST controller.” *Id.* at 3:39–40. BIST controller 4 includes algorithm controller 26 that can be invoked either by a user through user interface 22 or automatically upon power-up of electronic device 2. *Id.* at 5:12–16. “Once invoked, algorithm controller 26 provides an algorithm select signal (ALG_SELECT) to multiplexer 24 to select one of the algorithms stored within algorithm memory 20.” *Id.* at 5:16–19. Alternatively, user interface 22 may programmably receive algorithms via external input, and deliver the received algorithms to multiplexer 24. *Id.* at 5:41–43. The algorithms, whether stored in algorithm memory 20 or received via external input, have a similar form, i.e., “a sequence of binary commands in which each command defines a test within the overall algorithm.” *Id.* at 5:43–47. BIST controller 4 provides and communicates the selected algorithm to sequencers 8 for application to device blocks 6 as a stream of binary commands (CMD_DATA), “each command specif[ying] an operational code [(OP CODE)] and a set of parameters that define one or

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more memory operations without regard to the physical characteristics or timing requirements of memory modules 12.” *Id.* at 4:4–12, 5:19–21.

Figure 8, reproduced below, “is a block diagram illustrating an example data structure of a command issued by BIST controller 4.” *Id.* at 9:15–16.

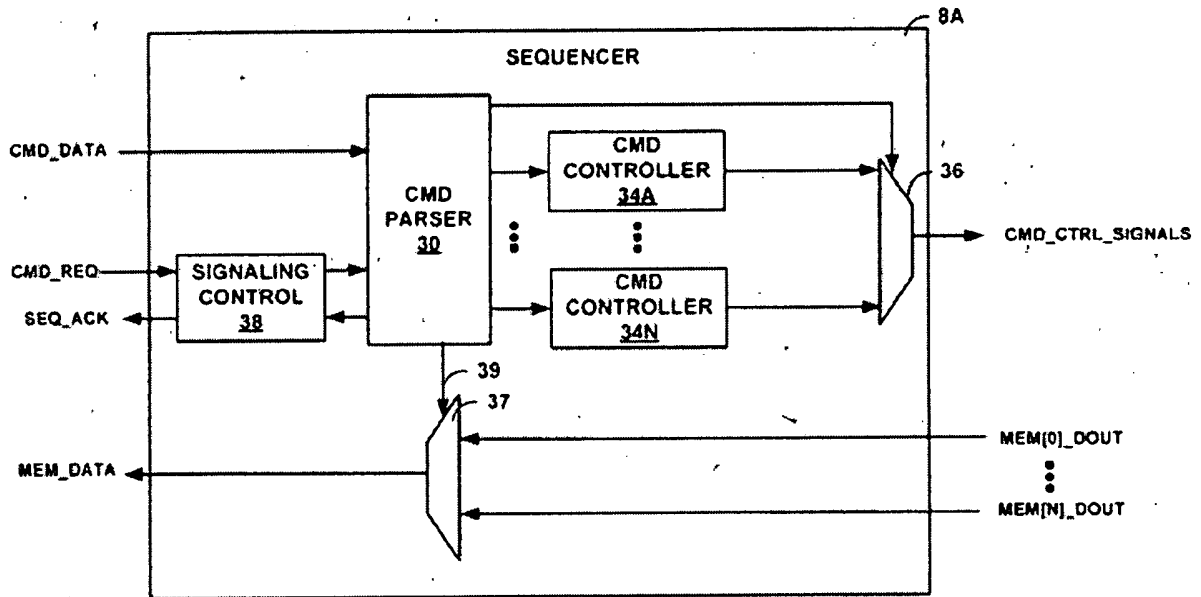


In the embodiment shown in Figure 8, above, “command 60 includes a sequencer identifier (ID) 62, and a payload 64. Sequencer ID 62 identifies a sequencer, e.g., sequencer 8A, to which command 60 is being issued.” *Id.* at 9:17–20. “Payload 64 of command 60 carries binary data that defines the command itself. In particular, payload 64 includes an operational code (OP CODE) 66 and a set of parameters 68.” *Id.* at 9:29–31. In one embodiment, for example, “OP CODE 66 and parameters 68 comprise three bits and twenty-nine bits, respectively, to form a 32-bit command.” *Id.* at 9:58–60. “OP CODE 66 specifies a particular function to be performed by the receiving sequencers 8.” *Id.* at 9:32–33. An exemplary OP CODE is SET ADDRESS (OP CODE 100), which “[s]ets a specific starting address as well as a maximum address limit for a test algorithm as applied to the memory modules.” *Id.* at Table 1.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* at 3:46–47.

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As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 that receives command data (CMD_DATA) from BIST controller 4. *Id.* at 6:24–26. Command parser 30 processes the commands from BIST controller 4 to identify a specified operation, e.g., by identifying an OP CODE specified by the command. *Id.* at 6:26–29. Based on the specified operation, command parser 30 may extract one or more parameters from the command, and pass the extracted parameters to a selected command controller (i.e., one of command controllers 34A–34N). *Id.* at 6:30–37. The invoked command controller, in turn, issues a sequence of one or more operations to each memory interface 10, sequentially driving the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. *Id.* at 6:41–43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with

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the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* at 4:45–49.

C. Arguments

Independent claims 1 and 16 recite, respectively, “[a] memory system” and “[a] method of operating a memory system.” Both claims recite “a plurality of memory chips” and “a plurality of data handlers.” Petitioner contends the claimed “memory chips” read on Averbuj’s memory modules 12 and the claimed “data handlers” read on the combination of sequencers 8 and memory interfaces 10. Pet. 23. Independent claim 1 further recites “a control circuit *configured to generate address and control signals*, wherein the memory system is configured to test the one or more memory chips using the *address and control signals generated by the control circuit*.” Ex. 1008, claim 1 (emphasis added). Independent claim 16 recites steps of “operating a *control circuit to generate address and control signals*” and “testing the one or more memory locations of the one or more memory chips using the *address and control signals generated by the control circuit*.” *Id.*, claim 16 (emphasis added). With respect to the above-quoted limitations relating to generating/generated address and control signals, the Petition states:

the BIST controller 4 of Averbuj ‘442 represents circuitry of a control circuit that provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE* for testing the memory components 12A–N. Ex. 1011, Fig. 2 and Column 9: Table 1. The *SET ADDRESS* and *CMD_REQ* signals include address and control signals for the testing functions. Since the BIST controller 4 provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE*, a POSITA [(“person of ordinary skill in the art”)] would understand that the BIST controller 4 is programmable to provide address and

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control signals as recited in part by the Challenged Claims. Ex. 1009, Bagherzadeh Decl., ¶ 33.

Pet. 24.

Patent Owner argues Averbuj's BIST controller 4 does not "generate" the OP CODE communicated to sequencers 8, and Petitioner has not met its burden to show that Averbuj's BIST controller 4 is "configured to generate" (claim 1) or can be "operat[ed] . . . to generate" (claim 16) address signals for testing the memory chips in the manner recited in claims 1 and 16 of the '501 patent. PO Resp. 45–46. As discussed in greater detail below, we find this argument persuasive, and, for this reason, conclude Petitioner failed to meet its burden to prove unpatentability of claims 1, 2, 4–6, 9–11, 14–17, and 20 of the '501 patent.³

D. Expert Testimony

In support of its argument that Averbuj's BIST controller 4 is not "configured to generate" (claim 1) and cannot be "operat[ed] . . . to generate" (claim 16) address signals, Patent Owner relies on paragraphs 133–141 of the Sechen Declaration. *See id.* at 45–49. Patent Owner also relies on Dr. Bagherzadeh's cross-examination testimony (*id.* at 46–47 (quoting Ex. 2012, 55:6–56:9)) regarding Averbuj Figure 2 (*see* Ex. 2012, 53:4–5; *see generally id.* at 53:4–56:9).

With reference to Averbuj Figure 2 (*see* Section IV.B., above, wherein this figure is reproduced and described), Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 are

³ Because we are persuaded by this argument that Petitioner has not met its burden to show unpatentability, we find it unnecessary to address the additional arguments advanced by Patent Owner in its Response.

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generated by hand or by a tool (Ex. 2012, 55:20–56:9), and he identifies a compiler as a tool for generating such code (*id.* at 30:23–24, 41:1–4; *see id.* at 55:6–25).⁴ Dr. Bagherzadeh testifies that a compiler is not illustrated in Averbuj Figure 2, and that he did not recall any mention of the word “compiler” in Averbuj. *See* Ex. 2012, 54:8–55:5. Dr. Bagherzadeh further testifies that “99 percent of the designs or chips do not have a compiler on-board.” *Id.* at 54:21–22. Dr. Bagherzadeh testifies that Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and “the output [] reflect[s] what’s in the input, based on the selection.” *Id.* at 54:8–13. Dr. Bagherzadeh testifies that assuming “no other hardware components between . . . [CMD_DATA] and the input,” CMD_DATA is stored in the algorithm memory. *Id.* at 54:15–19.

⁴ Petitioner contends “[t]he discussion of a compiler, from PO’s deposition of [Dr. Bagherzadeh], was hypothetical and specific to an article coauthored by [Dr. Bagherzadeh].” Reply 22 (citing Ex. 2012, 15:2–19:3).⁴ We were unable to locate a discussion of a compiler or any reference to an article coauthored by Dr. Bagherzadeh in the cited portions of Dr. Bagherzadeh’s deposition transcript. Dr. Bagherzadeh’s first discussion of a compiler appears to be in connection with a use thereof in MorphoSys (*see* Ex. 2012, 30:12–20), a chip fabricated by Morpho Technologies (*id.* at 19:2–9), a company co-founded by Dr. Bagherzadeh (Ex. 1009 ¶ 7). An article relating to MorphoSys and co-authored by Dr. Bagherzadeh (*see* Ex. 2014, “MorphoSys: Case Study of A Reconfigurable Computing System Targeting Multimedia Applications”) is introduced on page 39 of the deposition transcript. *See* Ex. 2012, 39:19–22, cited in PO Resp. 25. In any event, Petitioner’s argument is not persuasive because we find the testimony on pages 54–56 of the Bagherzadeh transcript supports Patent Owner’s argument. *See* Section IV.E. *infra*.

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Dr. Sechen testifies that

the Petition fails to identify every element of claims 1, 20, and 29 [sic, 1 and 16] for at least two reasons[:] (1) because the CMD_REQ and SET ADDRESS are not generated (“produced” or “caused”) by the purported control module (Averbuj’s BIST controller 4), and [(2)] because the OP CODE is not generated (“produced” or “caused”) by the alleged control module (Averbuj’s BIST controller 4).

Ex. 2019 ¶ 136. Dr. Sechen testifies, more specifically, that the Petition fails to explain how Averbuj’s BIST controller 4 “generate[s]/produce[s] OP CODE,” “because the Petition does not state that Averbuj’s BIST Controller includes a compiler, as described by Dr. Bagherzadeh.” *Id.* ¶ 138. Dr. Sechen testifies that “[b]ecause SET ADDRESS (OP CODE 100) is a predefined 3-bit sequence, . . . it is not ‘generated’ by Averbuj’s BIST Controller.” *Id.* ¶ 140. Dr. Sechen further testifies that “command 60, which carries the OP CODE [] (e.g. SET ADDRESS), . . . is not ‘generated’ by Averbuj’s BIST Controller,” but “is merely stored in memory and is output as part of ‘a stream of binary commands.’ (Ex. 1011 at 5:1–21.)” Ex. 2019 ¶ 141.

E. Analysis

After considering the parties’ arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses a control circuit that is “configured to generate” (claim 1) or can be “operat[ed] . . . to generate” (claim 16) address signals for testing the memory chips in the manner recited in claims 1 and 16 of the ’501 patent.

As noted in Section III, above, in its Petition, Petitioner proposes construing the claim term “generate” as “cause or initiate transmission” (Pet. 20–21). The Petition cites the following disclosure in the ’501 patent in

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support of this construction: “the data module 28 and/or the control circuit 22 are **configured to provide** memory signals (e.g., data, address and control signals)’ Ex. 1008, ’501 patent, 6:8-10 (emphasis added).” Pet. 20. The Petition states that Averbuj teaches “a control circuit configured to generate” (claim 1) and “operating a control circuit to generate” (claim 16) address signals for testing the memory chips as recited in claims 1 and 16, because Averbuj’s “BIST controller 4 is programmable *to provide* address and control signals.” Pet. 24 (emphasis added); *see* PO Resp. 49–50.

For purposes of our Institution Decision, we interpreted “generate” as used in the ’501 patent claims as meaning “cause” or “produce.” *See* Dec. on Inst. 9. As indicated in Section III.A., above, subsequent to that decision and the oral hearing in the present *inter partes* review, we again considered the meaning of the claim term “generate” in our final decision in IPR-971, and determined that the broadest reasonable interpretation of the ’501 patent claim term “generate” is “produce,” and that “generate” does not mean “cause” or “cause to produce.” *See* IPR-971 FWD, 22–30. In Section III.A., above, we considered extrinsic evidence in support of Petitioner’s interpretation of “generate” as meaning “cause” that was not before us in IPR-971, but concluded the broadest reasonable interpretation of “generate” is “produce.” We, therefore, adopted and incorporated by reference in the present final decision our analysis and construction of the terms “generate” and “configured to generate” in our Final Written Decision in IPR-971. *See* Section III., above.

In construing the claim term “generate” in our final decision in IPR-971, we found that the terms “provide” and “generate” are used in the ’501 patent to describe different embodiments of the control circuit, and

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determined the claim term “generate” did not mean “provide.” *See* IPR-971 FWD, 25–28. We also determined that the phrase “address and control signals generated by the control circuit” (claims 1 and 16) encompasses signals that originated in the control circuit, including by transformation or modification of information received from another component. IPR-971 FWD, 29. We further determined the broadest reasonable interpretation of this claim language does not encompass “signals . . . received by the control circuit . . . from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the control circuit.” *Id.* at 29–30. We stated that “generate,” as used in the ’501 patent, does not encompass the selection function of a multiplexer. *Id.* at 27.

Petitioner’s contention that Averbuj teaches “a control circuit configured to generate address . . . signals” (claim 1) and “operating a control circuit to generate address . . . signals” (claim 16) for testing memory chips, as recited in independent claims 1 and 16, is based on an interpretation of “generate” as meaning “cause.” Pet. 20, 24; *see also*, Reply 20–21 (asserting that Averbuj discloses the above-quoted limitations because “the BIST controller of Averbuj *causes* the CMD_REQ and SET ADDRESS signals, representing the claimed ‘address and controls signals’” (emphasis added)) (citing Dec. on Inst. 15–16). Petitioner has not explained sufficiently, however, how Averbuj’s BIST controller 4 is “configured to generate” (i.e., designed, adapted, or arranged to produce) or “operat[ed] . . . to generate” (i.e., operated to produce) address signals as recited in independent claims 1 and 16, respectively.⁵

⁵ During oral argument, Judge Clements asked Mr. Heafey, counsel for

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Moreover, the evidence in this proceeding fails to support a finding that Averbuj's BIST controller 4 generates address signals for testing the memory devices in the manner claimed.

In Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS (*see* Ex. 1011, 4:4–12, 5:19–21, Table 1)) communicated to sequencers 8 are either received by user interface 22 via external input (*id.* at 5:41–43) or generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20 (*see* Ex. 2012, 30:23–24, 41:1–4, 54:21–22, 55:6–25, 55:20–56:9 (wherein Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 are generated by hand or by a tool, e.g., a compiler, and that “99 percent of the designs or chips do not have a compiler on-board”)). Multiplexer 24 *selects* one of the algorithms (i.e., CMD_DATA) and BIST controller 4 *provides* to sequencers 8 the same CMD_DATA (e.g., address signals) without transformation or modification. *See* Ex. 1011, 4:4–12, 5:16–21; Ex. 2012, 54:8–13 (“Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and the output [] reflect[s] what’s in the input, based on the selection.”). Based on our

Petitioner, the following question: “Assuming we adopt ‘to produce’ as the proper construction [of ‘to generate’], are [CMD]_REQ and SET ADDRESS produced by BIST controller 4?” Tr. 78:16–18. In response, Mr. Heafey stated: “[T]hose signals are definitely produced by BIST controller 4, yes, they are.” *Id.* at 78:19–20. Mr. Heafey did not elaborate, however, nor did he otherwise explain clearly during oral argument how Averbuj's BIST controller 4 is designed, adapted, or arranged to produce address signals for testing the memory devices.

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interpretation of the term “generate,” as discussed above, an “address . . . signal[] generated by the control circuit,” as recited in claims 1 and 16, does not encompass Averbuj’s SET ADDRESS signal, because it is produced by a component external to BIST controller 4, and merely selected by multiplexer 24 and provided to sequencers 8 without transformation or modification by BIST controller 4.

In sum, after considering the parties’ arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses “a control circuit configured to generate address . . . signals” (claim 1) and “operating a control circuit to generate address . . . signals” (claim 16) for testing memory chips in the manner recited in independent claims 1 and 16. Petitioner does not correct this deficiency in its challenges as to dependent claims 2, 4–6, 9–11, 14, 15, 17, and 20. *See* Pet. 26–34; 35 U.S.C. § 112, ¶ 4 (“A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers.”). Because Petitioner has not identified in Averbuj a teaching of each and every limitation recited in claims 1, 2, 4–6, 9–11, 14–17, and 20, Petitioner has not met its burden to show these claims are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj.

V. MOTIONS TO EXCLUDE

Both Patent Owner and Petitioner filed Motions to Exclude.

A. Petitioner’s Motion to Exclude

Petitioner moves to exclude Exhibit 2018 (the Deposition Transcript of Dr. Donald Alpert) and to strike Section IV(B)(5)(a) of Patent Owner’s Response, wherein Patent Owner quotes Dr. Alpert’s deposition testimony in

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support of its proposed construction of the claim term “generate” (*see* PO Resp. 31–32). Paper 25, 2. Petitioner argues Dr. Alpert is Sandisk Corporation’s expert in IPR-971 and that Petitioner was not provided notice or otherwise invited to attend Dr. Alpert’s deposition. *Id.* at 3. Petitioner contends Dr. Alpert’s statements are not admissible as evidence in this proceeding “because they are hearsay and will unfairly prejudice Petitioner.” *Id.* at 2.

In construing the claim term “generate” in Section III.A., above, we adopted and incorporated by reference our analysis and construction of this term on pages 22–30 of our Final Written Decision in IPR-971. In our claim construction in IPR-971, we considered Dr. Alpert’s deposition and declaration testimony, but did not quote or cite to the testimony relied on by Patent Owner in Section IV(B)(5)(a) of its Patent Owner Response in the present proceeding. *Compare* IPR-971 FWD, 23–24 *with* PO Resp. 31–32. In IPR-971, we determined Dr. Alpert’s testimony that one of ordinary skill in the art would understand the claim term “generate” as meaning “cause” or “produce” was inconsistent with the ’501 patent’s implicit definition of this term as meaning only “produce.” *See* IPR-971 FWD, 27 n.4. We, therefore, did not rely on Dr. Alpert’s testimony in our construction of the claim term “generate” in IPR-971, or in the present case. *See id.*; *see also id.* at 32 (“Having considered the ordinary meaning of the term ‘generate’ in the context of both the claims and the ’501 patent as a whole, we agree with Patent Owner that the broadest reasonable interpretation of the claim term ‘generate’ is ‘produce,’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’”).

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Because we did not rely on Exhibit 2018 or consider the argument in Section IV(B)(5)(a) of Patent Owner's Response in rendering our decision in this proceeding, we dismiss Petitioner's Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) as moot.

B. Patent Owner's Motions to Exclude

Patent Owner filed a Motion to Exclude "Exhibits 1026, 1029, 1030, and 1032 (or portions thereof)" and requests that we strike the corresponding citations to these exhibits in Petitioner's Reply. Paper 33, 2. We did not rely on these exhibits in rendering our decision. Accordingly, Patent Owner's Motion to Exclude these exhibits is dismissed as moot.

Patent Owner also filed a Motion to Exclude Portions of Petitioner's Reply, contending "Petitioner's Reply improperly presents new theories, never raised in its Petition or adopted by the Board's Institution Order, violating 37 CFR 42.22-23 and applicable law." Paper 34, 1. Patent Owner provides a table identifying the alleged new theories advanced by Petitioner in its Reply (*id.* at 2) and identifies by page and line number specific portions of sections II.A.3, II.B.1.a, II.B.5 & III.B.2 of the Reply that it contends should be excluded as advancing new theories (*see id.* at 2-8).

Our decision is not based on any of the alleged new theories identified by Patent Owner, and Patent Owner does not cite to any of the specific language in the Reply that Patent Owner contends should be excluded. Accordingly, Petitioner's Motion to Exclude portions of Petitioner's Reply is dismissed as moot.

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VI. CONCLUSION

Petitioner has not shown by a preponderance of the evidence that claims 1, 2, 4–6, 9–11, 14–17, and 20 of the '501 patent are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2).

VII. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1, 2, 4–6, 9–11, 14–17, and 20 of U.S. Patent No. 8,359,501 B1 are not held unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2);

FURTHER ORDERED that Petitioner's Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) of Patent Owner's Response is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exhibits 1026, 1029, 1030, and 1032 to strike the corresponding citations to these exhibits in Petitioner's Reply is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exclude Portions of Petitioner's Reply is *dismissed*; and

FURTHER ORDERED that, because this is a final decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01374
Patent 8,359,501 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
MATTHEW R. CLEMENTS, *Administrative Patent Judges*.

CLEMENTS, *Administrative Patent Judge*, dissenting.

While I agree with the majority’s construction of “generate” to mean “produce,” I respectfully dissent from the majority’s determination that “Petitioner has not explained sufficiently, however, how Averbuj’s BIST controller 4 is ‘configured to generate’ (i.e., designed, adapted, or arranged to produce) or ‘operat[ed] . . . to generate’ (i.e., operated to produce) address signals as recited in independent claims 1 and 16, respectively.” Majority Opinion (“Maj. Op.”) 30–31. Although Petitioner’s arguments are based on a construction of “generate” that includes cause, and we have construed

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“generate” to mean “produce,” I believe the evidence cited by Petitioner nevertheless establishes sufficiently that BIST controller 4—identified by Petitioner as the recited “control circuit”—“produces” address signals.

The majority opinion finds that “[i]n Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS) communicated to sequencers 8 are either received by user interface 22 via external input or generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20.” Maj. Op. 31 (citations omitted). As support for its finding that the algorithms are generated by another component, the majority cites the cross-examination of Petitioner’s expert, Dr. Bagherzadeh, in which he identifies a person or tool as possible ways of generating the OP CODEs stored in algorithm memory 26:

Q Would a compiler be required to generate the op codes?

[Objection]

THE WITNESS: Which op codes?

BY MR. BRADLEY:

Q The op codes stored in algorithm memory?

A Required, no.

Q In what other way could code be generated for algorithm memory?

A By hand, as I explained to you.

Q Are there any other ways of generating the code?

[Objection]

THE WITNESS: Those codes that are residing in the memory, right, that's all that we're talking about?

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BY MR. BRADLEY:

Q Right.

A So either you do it by hand or you have a tool do it.

Q Okay.

A There's no -- we're talking about code, not the generation of the -- or signals that are generated, just the bits in the code?

Q The generation of the op codes, yes.

A Yes. They have to be stored in the memory. Somebody did it by hand, wrote the code, or by a tool.

Q Okay.

A There's no other way around it.

Ex. 2012, 54:20–56:9.

The majority appears to understand this testimony by Dr. Bagherzadeh as a description of how BIST controller 4 works *in operation*. I understand this testimony, however, to explain only how the data stored in algorithm memory 26 could have been created and stored in algorithm memory 26 *in the first instance*. At no point in this testimony does Dr. Bagherzadeh suggest that a compiler plays any role during *operation* of BIST controller 4. Indeed, that would be inconsistent with Averbuj's explicit disclosure.

Averbuj explains that the *only* input received by BIST controller 4 is an invocation of some sort, whether an “external input, such as a control signal from an external testing apparatus” or an automatic invocation, and, as a result of that input, CMD_DATA, among other signals, is output:

User interface 22 invokes algorithm controller 26 in response to external input, such as a control signal from an external testing

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apparatus. Alternatively, algorithm controller may be automatically invoked upon power-up of electronic device 2. Once invoked, algorithm controller 26 provides an algorithm-select signal (ALG_SELECT) to multiplexer 24 to select one of the algorithms stored within algorithm memory 20. Once selected, a stream of binary commands that comprises the selected algorithm is applied to device blocks 6 as command data (CMD_DATA).

Ex. 1011, 5:12–21. Thus, in this embodiment, a control signal goes into BIST controller 4 and CMD_DATA, *inter alia*, comes out of BIST controller 4. In this embodiment, BIST controller 4 does not propagate signals through to its output because it does not receive CMD_DATA at user interface 22. For that reason, I find that BIST controller 4 “produces” CMD_DATA, including, *inter alia*, SET ADDRESS, which Petitioner identifies as the recited “address . . . signal[]” (Pet. 24).

The fact that the data output by BIST controller 4—i.e., CMD_DATA—is read from a memory within BIST controller 4 so that it can subsequently be output goes to *how* BIST controller 4 “produces” that data. It does not show that the data is “produced” by another component. Moreover, the fact that the data in algorithm memory 26 was first created by something other than BIST controller 4 before being stored in algorithm memory 26, e.g., at the time of manufacture, does not, in my view, preclude BIST controller 4 from “producing” that data during *operation* of the built-in self-test circuit. Finally, the fact that BIST controller 4 does not include a compiler is irrelevant, in my view, because BIST controller 4 does not need a compiler in order to “produce” address and control signals; when “invoked,” it “produces” them by retrieving them from its memory—i.e., algorithm memory 26—and outputting them.

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The *only* scenario in which BIST controller 4 “receives address signals from another component . . . and merely outputs the same address signals,” as the majority opinion finds, is in an alternative mode of operation described at column 5, lines 41–49, where Averbuj discloses that “user interface 22 may programmably receive algorithms via external input” which are delivered directly to multiplexer 24. That, however, is not the mode of operation relied upon by Petitioner.

For all of the foregoing reasons, I disagree with the majority’s finding that “[i]n Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS) communicated to sequencers 8 are . . . generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20.” Maj. Op. 31.

Because I am persuaded that Petitioner has established sufficiently that BIST controller 4 “produces” the recited “address and control signals,” I would have analyzed the remainder of Patent Owner’s arguments.

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO. 8:16-cv-01605	DATE FILED 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,756,364	June 17, 2014	Netlist, Inc.
2 8,516,185	August 20, 2013	Netlist, Inc.
3 8,001,434	August 16, 2011	Netlist, Inc.
4 8,359,501	January 22, 2013	Netlist, Inc.
5 8,689,064	April 1, 2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK	
1			
2			
3			
4			
5			

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and
SK HYNIX MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00562
Patent 8,359,501 B1

Before BRYAN F. MOORE, MATHEW CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION
Instituting *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

SK hynix Inc., SK hynix America Inc., and SK hynix memory solutions Inc., (collectively “Petitioner”) filed a Petition requesting an *inter partes* review of claims 1 and 4 of U.S. Patent No. 8,359,501 B1 (Ex. 1001, “the ’501 Patent”). Paper 1 (“Pet.”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response. Paper 6 (“Prelim. Resp.”).

We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may be authorized only if “the information presented in the petition . . . and any [preliminary] response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a). Pursuant to 35 U.S.C. § 314, we determine that Petitioner has demonstrated a reasonable likelihood that it would prevail with respect to claims 1 and 4 of the ’501 Patent and, thus, institute an *inter partes* review as to the challenged claims.

A. Related Proceedings

Petitioner recites a list of District Court proceedings related to this *inter partes* review. Pet. 2. This *inter partes* review challenges the same patent at issue in the decision entered in IPR2014-00971 (the 971 IPR). *See Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB Apr. 27, 2016) (Paper 37) (“971 FWD”). In the 971 IPR, as to the claims at issue here, claim 1 was held unpatentable as anticipated by Averbuj. 971 FWD 60. That decision is on appeal to Federal Circuit. In the 971 IPR, claim 4 was also challenged as obvious over Averbuj and Tsern, but that challenge was not instituted. *Sandisk*, IPR2014-00970 (PTAB Dec. 16, 2014) (Paper 12, 15–16).

The ’501 Patent was also challenged in *Smart Modular Techns Inc. v. Netlist, Inc.*, Case IPR2014-01374 (the 1374 IPR). In the 1374 IPR, as to the claims at issue here, we did not hold claims 1 and 4 to be unpatentable as anticipated by Averbuj. *Smart Modular Techns Inc. v. Netlist, Inc.*, Case IPR2014-01374 (PTAB Mar. 9, 2016) (Paper 44, 35). The decision in the 1374 IPR was not appealed to Federal Circuit.

The '501 Patent was also challenged in IPR2014-01375 in which institution was denied as to all challenges on the merits. *Smart Modular Techs Inc. v. Netlist, Inc.*, Case IPR2014-01375 (PTAB Mar. 13, 2015) (Paper 12). Neither Averbuj nor Tsern was asserted in the 1375 IPR.

We have discretion to institute, under 35 U.S.C. § 325(d), when considering “the same or substantially the same prior art or arguments previously . . . presented to the Office” and under 35 U.S.C. § 314(a) (authorizing institution of an inter partes review under particular circumstances, but not requiring institution under any circumstances). Considering the particular circumstances of this case, we address the merits of the Petition and do not exercise our discretion under 35 U.S.C. § 325(d) and/or 35 U.S.C. § 314(a). We consider the following factors relevant to exercising our discretion not to institute:

(1) whether the same petitioner previously filed a petition directed to the same claims of the same patent;

(2) whether the petitioner reasonably could have raised, in its earlier petition, the challenge now asserted;

(3) the elapsed time between the petitioner’s current challenge and earlier proceedings, and the degree to which petitioner’s challenge, or the timing thereof, has been unfairly informed by information gleaned, by word or deed, from an earlier proceeding; and

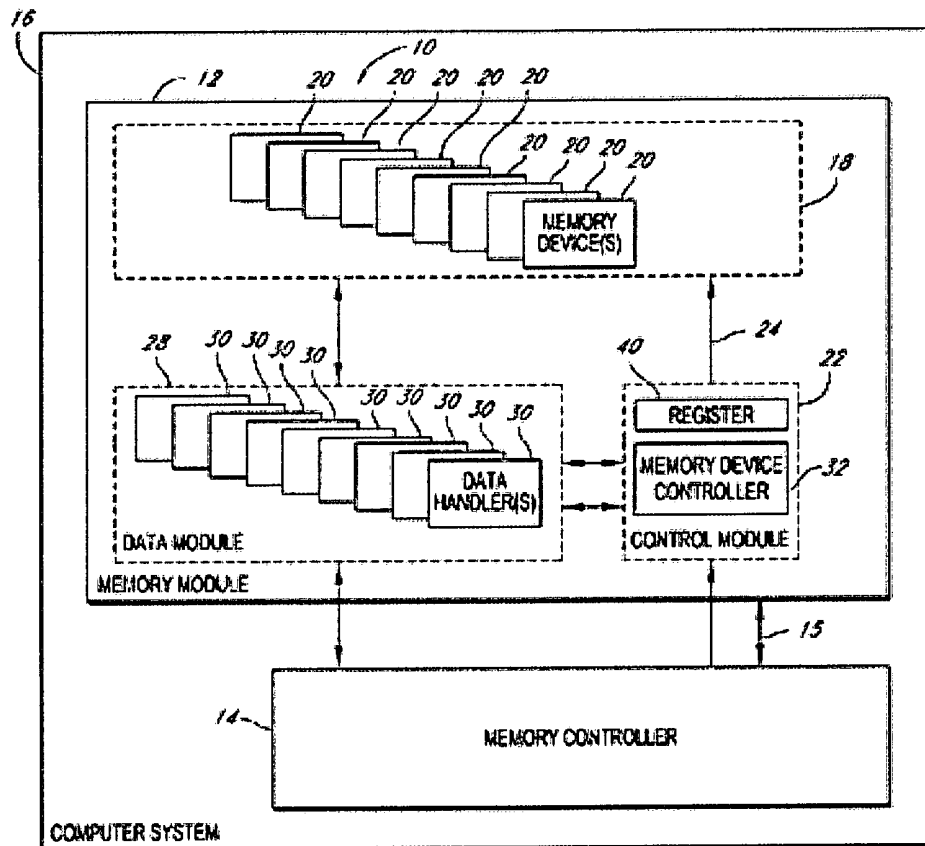
(4) whether petitioner identifies any mitigating factors that justify another challenge to a patent.¹

¹ These factors are a more generalized statement of factors analyzed by other panels of the Board. *See, e.g., NVIDIA Corp. v. Samsung Elec. Co.*, Case IPR2016-00134 (PTAB May 4, 2016) (Paper 9); *Xactware Solutions, Inc. v.*

Applying these factors, and weighing the equities between Petitioner and Patent Owner, we do not exercise our discretion not to institute review for the following reasons. As noted above, Petitioner was not a party to any of the prior proceedings. The first and second factor therefore weigh against a discretionary denial. As to the third factor, we acknowledge that the third petition was filed almost three years after the first petition. However, as a mitigating factor, Petitioner in this challenge was not sued until some significant amount of time after the Petitioner in the 971 IPR was sued. Thus, the fourth factor weighs against a discretionary denial. On balance, we determine that the particular facts in this case weigh in favor of considering the Petition on the merits.

B. The '501 Patent (Ex. 1001)

The '501 Patent relates to self-testing electronic memory modules. Ex. 1001, 1:28–29. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '501 Patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10 includes a plurality of memory devices 18, e.g., ETT DRAM chips, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 3:34–36, 5:10–12. Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:14–16, 42–46. “In various embodiments, the control module 22 includes .

... one or more application-specific integrated circuit[s].” *Id.* at 9:62–64.
“[D]ata module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20.” *Id.* at 6:5–8. “For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according [to] the operating specification of the memory devices 20.” *Id.* at 6:8–11. “In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20.” *Id.* at 5:48–52. “For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation.” *Id.* at 5:52–55.

C. Illustrative Claim

Of the challenged claims, claim 1 is independent. Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address

and control signals generated by the control circuit and using the data generated by the one or more data handlers.

2. Prior Art Relied Upon

Petitioner relies upon the following prior art references:

Reference	Patent/Printed Publication Number	Exhibit Number
Averbuj	U.S. Patent Application Publication No. 2005/0257109 A1	Ex. 1005
Tsern	U.S. Patent Application Publication No. 2007/0070669 A1	Ex. 1006

3. The Asserted Grounds

Petitioner asserts that the challenged claims are unpatentable based on the following grounds:

Reference[s]	Basis	Claims Challenged
Averbuj	§ 102	1 and 4
Averbuj	§ 103	1 and 4
Averbuj and Tsern	§ 103	4

II. ANALYSIS

A. Claim Construction

In an *inter partes* review, claim terms in an unexpired patent are given their broadest reasonable construction in light of the specification of the patent in which they appear. *See* 37 C.F.R. § 42.100(b). Under the broadest reasonable construction standard, claim terms are given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art in the context of the entire disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definition for a claim term

must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

1. *generate, operable independently, and configured to*

Petitioner and Patent Owner requested construction of several claim terms to be generally consistent with how those terms were previously construed in the 970 IPR. IPR2014-00971, Paper 12, 33; *see also* IPR2014-00971, Paper 12, 10–33 (analyzing those claims); Pet. 8–9; Prelim. Resp. 20–22. Patent Owner agrees with the proposed constructions, but also asserts that our analysis regarding the constructions should be considered to be a part of the construction. Prelim. Resp. 20–22. We have considered the claim term constructions and analysis, and discern no reason to alter those constructions used in the 971 IPR. The following chart indicates constructions we adopt for the purpose of this decision.

Claim Term(s)	Claims Challenged
<i>Generate</i>	Produce
<i>Operable independently</i>	Operable without influence or control by another
<i>Configured to</i>	Designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]

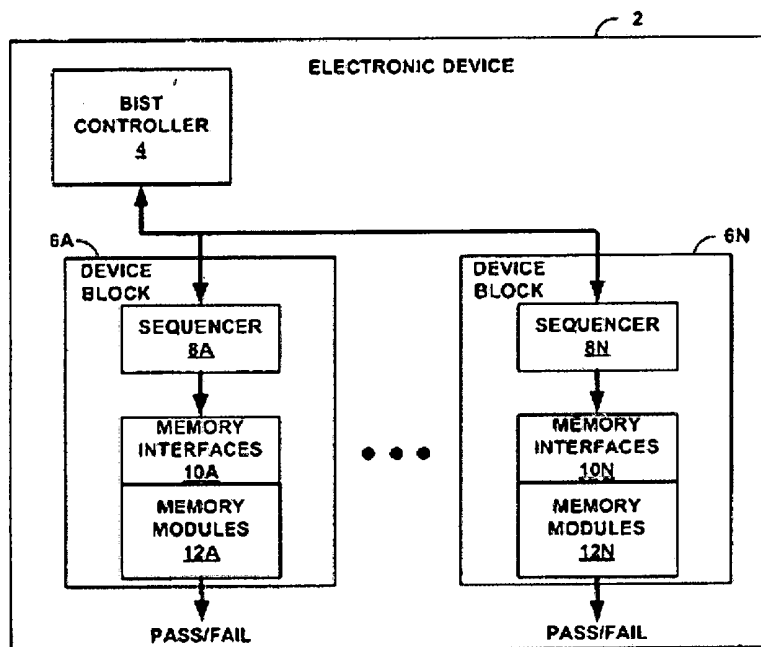
2. “*memory controller*”

Patent Owner and Petitioner appear to agree the term “memory controller” should mean “a device, or a portion of a device that manages the flow of data to and from the memory.” Ex. 1003 ¶ 88 (cited at Pet. 21); Prelim. Resp. 21, 22, 31. Patent Owner leaves out the reference to “a portion of a device” but does not explain why and does not appear to make

an argument based on that language. Thus, for the purpose of this Decision we construe the term “memory controller” to mean “a device, or a portion of a device, that manages the flow of data to and from the memory.”

B. Claims 1 and 4 — Anticipation by Averbuj (Ex. 1005)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1005 ¶¶ 7–8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* ¶ 28. “[E]lectronic device 2 may be any device that incorporates memory modules, such as an embedded computing system, a computer, server, personal digital assistant (PDA), mobile computing device, mobile communication device, digital recording device, network appliance, mobile positioning device, and the like.” *Id.* ¶ 32. Averbuj states

that “electronic devices . . . are [normally] constructed from many integrated circuit chips and many supporting components mounted on a circuit board.” *Id.* ¶ 5. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to sequencers 8 for application to device blocks 6. *Id.* ¶ 29.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* ¶ 22.

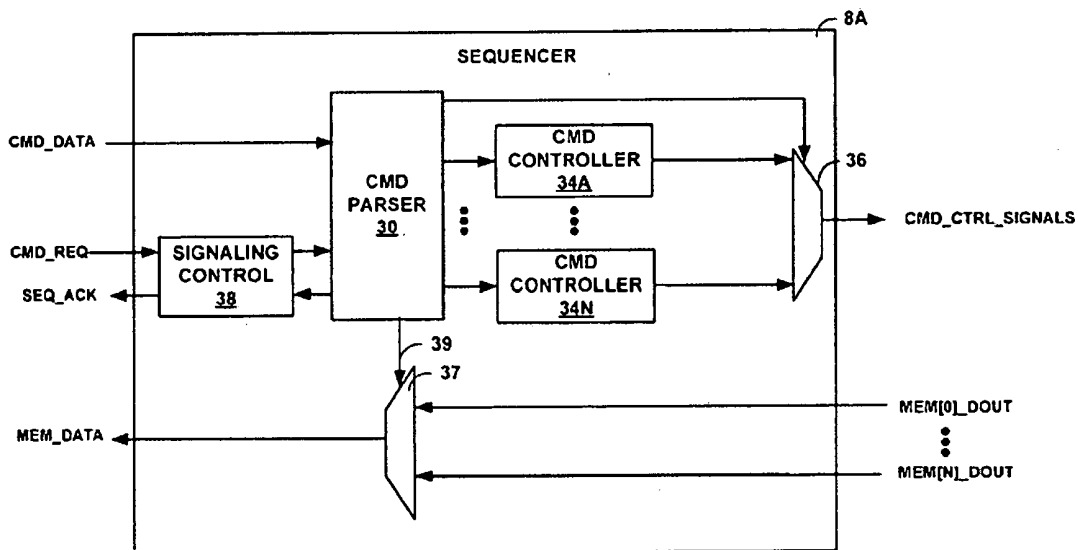


FIG. 5

As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 and command controllers 34. *Id.* ¶¶ 41–42. Command parser 30 processes the commands from BIST controller 4 and invokes one of command controllers 34, which, in turn, issues command control signals,

such as signals to provide a memory address and data, to receiving memory interfaces 10, 41. *Id.* ¶¶ 43, 51. An exemplary memory interface is shown in Figure 6, below. *Id.* ¶ 23.

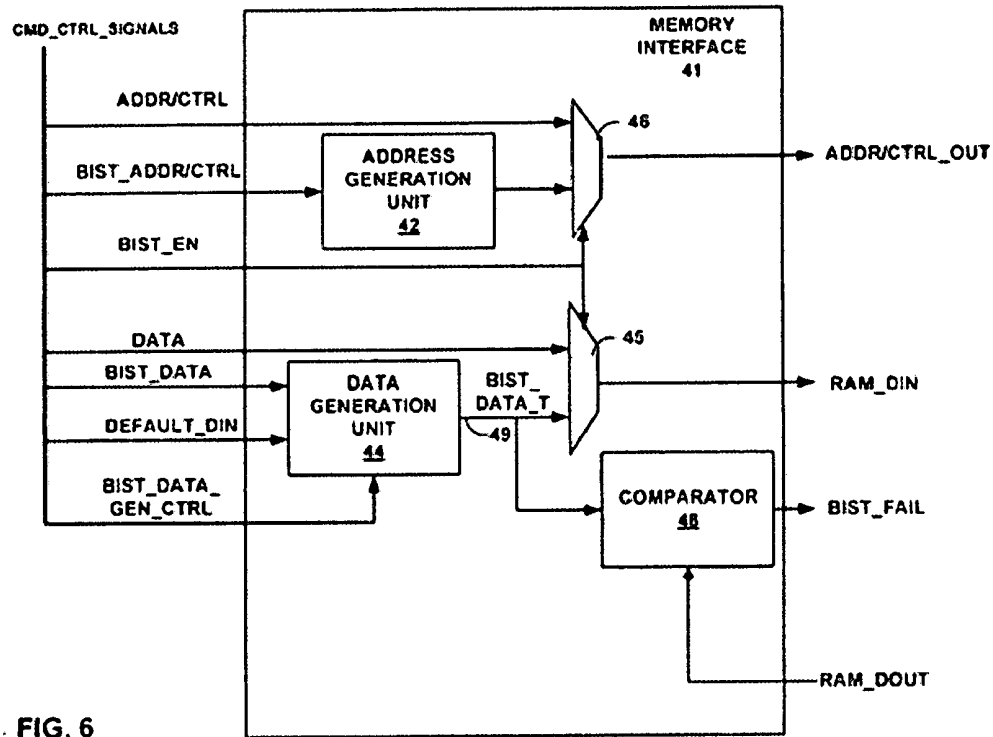


FIG. 6

As shown in the Figure 6 embodiment, above, memory interface 41 includes a layer of multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals

(BIST_ADDR/CTRL) and the test data provided by a respective higher level sequencer[, e.g., sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level sequencer [8A], and transforms the data and addresses provided by the sequencer as needed based on the particular physical characteristics of the memory module” (*id.* ¶ 54) using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “[A]ddress generation unit 42 generates the address applied to [] memory module 12 based on the addressing requirements specified by sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* ¶ 51. “[D]ata generation unit 44 [] receives BIST_DATA signals and default data (DEFAULT_DIN) as provided by [] sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the sequencer and the specific physical characteristics of the corresponding memory module.” *Id.* ¶ 49. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.*

2. *Analysis*

a. *memory controller*

Petitioner contends Averbuj’s electronic device (2) includes a plurality of device blocks (6), each of which includes a sequencer (8), a set of memory interfaces (10) interfacing with a set of corresponding memory modules (12). Pet. 20. Petitioner relies on Averbuj’s a BIST controller (4) that operates as a centralized “memory controller” during test mode by sending test commands (CMD_DATA and CMD_REQ) to the device blocks (“memory system”) of the electronic device, (“a computer system”) and

monitoring the status of the memory testing operation (e.g., SEQ_ACK). *Id.* at 20–21.

Petitioner further contends the preamble does not actually require that “a memory system” be “operatively coupled to a memory controller.” Pet. 21. It is alleged that all that is required is that “a memory system” be “designed to, adapted to, or arranged to [be] or be capable of [being]” “operatively coupled to a memory controller.” *Id.* (citing IPR2014-00971, Paper 37 at 31.) We agree. Thus, the fact that Averbuj’s electronic device is capable of receiving address, control, and data signals from an external device demonstrates that it is capable of being operatively coupled to a device that manages the data flow of a memory by issuing address, control, and data signals. *Id.* at 21–22 (citing Ex. 1008, 497; Ex. 1009, Fig. 1–2, Ex. 1003 ¶ 91).

Patent Owner asserts Averbuj “does not disclose that its BIST controller or programmable processor manages the flow of data to and from the memory.” Prelim. Resp. 31. Patent Owner argues that Averbuj’s programmable processor does not manage the flow of data from the memory. However, Patent Owner does not explain why the control signal is not related to the flow of data in and out of the memory as suggested by Petitioner. Thus, we are not persuaded by Patent Owner’s argument.

b. memory chips

Petitioner asserts “[e]ach memory module (12) disclosed in Averbuj is a ‘memory chip’ because Averbuj equates ‘memory chips’ with the various types of memory (e.g., DRAM, Flash memory) that are disclosed as comprising the memory modules (12) of his electronic device.” Pet. 22. As to this and all other limitations, Petitioner relies on the same aspects of

Averbuj relied on in the petition, and accepted by the Board, in the final written decision of the IPR 971. IPR2014-0971, Paper 1 (Petition), 8–17; Paper 37 (Final Written Decision), 31–44. In this decision, we have considered the fact that these aspects of Averbuj were determined to anticipate claim 1 in the previous final written decision by the Board. Petitioner also relies on several alternative bases of anticipation by Averbuj for various limitations for claim 1. Pet. 26–28, 32–34. At this juncture, we do not rely on those alternative bases in this Decision, but we do not exclude them from the trial.

Patent Owner argues Averbuj does not describe “a plurality of memory chips,” as recited in claim 1. Prelim. Resp. 32–33. Patent Owner argues that due to the small size of the memory modules of Averbuj, they cannot be “chips.” *Id.* Patent Owner does not construe the term “chip” nor provide evidence that the size of a memory module defines whether it can be considered a “chip.” Thus, we are not persuaded by Patent Owner’s argument.

c. control signal

As to the claimed “control signal,” Petitioner asserts “the output signal (CMD_CTRL_SIGNALS) includes test address and control signals (BIST_ADDR/CTRL), which are selected and applied to the memory module (12) during self-test mode. Ex. 1005 at Fig. 6, ¶¶ [0043-0044]; Ex. 1003 at ¶¶ 109-110.” Pet. 27. Patent Owner further argues Averbuj fails to describe a “memory system is configured to test the one or more memory chips using . . . control signals generated by the control circuit,” as recited in claim 1. *See* Prelim. Resp. 33–37. Patent Owner argues the ADDR/CTRL_OUT is an address signal. *Id.* at 36–38. As supporting

evidence, Patent Owner relies on the Averbuj provisional application (Ex. 2001) as allegedly showing only one signal. *Id.* Patent Owner does not explain why the claimed address and control signal must be described as two signals beyond the name ADDR/CTRL_OUT. Contrary to Patent Owner's argument, however, the '501 Patent describes, similar to Averbuj, single signal line 24 or single line 38 respectively as "address and control signals" and does not provide disclosures that label in a meaningful way two distinct "address" and "control" signals. *See generally*, Ex. 1001. Additionally, at this stage of the proceeding, we decline to impose a limited view of disclosure of Averbuj based on an embodiment of the Avebuj provisional application. Thus, we are not persuaded by Patent Owner's argument.

d. control circuit

Petitioner asserts "the sequencers (8) disclosed in Averbuj constitute "a control circuit" as claimed" Pet. 26. Patent Owner asserts that Averbuj's sequencer cannot be the claimed "control circuit" because it does not input address and control signals to the "memory chips." Prelim. Resp. 38. Specifically, Patent Owner asserts that Petitioner incorrectly relies on a statement from the 971 IPR final written decision for this proposition. *Id.* (citing Pet. 26–27). Patent Owner asserts the quoted statement is incorrect because "[p]aragraph 56 of Averbuj is only directed to the data generation—not address and control signals." Prelim. Resp. 39. Therefore, according to Patent Owner, Petitioner has not shown that the alleged "control module" of Averbuj "generates" the address and control signals, as opposed to data signals, used to test the memory devices. Prelim. Resp. 38–40.

To provide context, in the 971 IPR's institution decision the Board construed the term "a control module configured to generate address and

control signals for testing the memory devices” as requiring the control module to “produce” or “cause [to be produced]” address and control signals. IPR2014-00971, Paper 12, 8–9. However, in the 971 IPR’s final written decision, the Board construed that term as limited to a control module that produces address and control signals. IPR2014-00971, Paper 37, 22–30. Thus, the issue in the 971 IPR was whether Averbuj sent address and control signals that were produced by the sequencer and delivered unchanged to the memory devices rather than a signal that was transformed in some way after it leaves the sequencer but before it is used in testing the memory devices.

For further context, we discuss in detail below Petitioner’s contention as to “control circuit.” Petitioner contends that Averbuj discloses a control circuit (sequencer 8) configured to generate address and control signals (CMD_CTRL_SIGNALS) used for testing memory devices (memory modules 12). Pet. 26–27. Averbuj discloses that “BIST controller 4 provides and communicates test algorithms to sequencers 8 for application to device blocks 6” (Ex. 1005 ¶ 29), and “[s]equencers 8 interpret” and “control the application of the test algorithms” to the memory modules (*id.* ¶ 30). *See also id.* ¶ 46 (“Sequencers 8 receive the generic BIST commands in accordance with the command protocol, and control the application of the commands by generating and issuing sequences of one or more memory operations for application to a set of respective memory modules 12.”).

For example, in the exemplary embodiment illustrated in Figure 5, command parser 30 of sequencer 8A processes a command received from BIST controller 4, and extracts one or more parameters from the command. *Id.* ¶¶ 41–42. Command parser 30 passes the extracted parameters to

command controller 34, which then issues a sequence of one or more operations to each memory interface 10. *Id.* ¶¶ 42–43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* ¶ 31.

In the 971 IPR, the Board relied on disclosure other than paragraph 56 to show the control module limitation was met. The 970 IPR’s final written decision states:

Averbuj further explains, in connection with the embodiment illustrated in Figure 6, that memory interface 41 “*transforms* the data and addresses provided by the sequencer *as needed* based on the particular physical characteristics of the memory module” (*id.* ¶ 54 (emphasis added)) using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51).

971 IPR Paper 37, 40. In other words, paragraph 54 of Averbuj suggests that data *and addresses* may be passed through the memory interface without being transformed or being transformed based on the necessity for transformation. The final written decision cited a specific example in which a data signal, BIST_DATA_T is sent to the memory module explicitly “without modification.” *Id.* (citing IPR 971, Ex 1003 (Averbuj) ¶ 56). Even if Patent Owner is correct that the description in paragraph 56 of Averbuj is limited to data signals sent by the data generation unit, the statement in paragraph 54 of Averbuj sufficiently shows that address signals can also be sent to the memory module without being transformed by memory interface 41. Thus, we are not persuaded by Patent Owner’s argument on this issue.

Petitioner also presents alternative theories on how Averbuj meets the “control module” limitation of claims 1. Pet. 32–34. Patent Owner responds

to those theories. Prelim. Resp. 50–52. Although we do not limit the trial to Petitioner’s sequencer theory, we need not discuss Petitioner’s other theories for the purpose of institution because the sequencer theory is sufficient for institution.

For the reasons above, Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 is anticipated by Averbuj.

g. Claim 4

Claim 4 recites “[t]he memory system of claim 1, wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” Petitioner asserts that

Averbuj discloses that BIST units are “commonly incorporated into memory chips and other integrated circuits.” Ex. 1005 at ¶ [0003]. Because the memory interfaces are part of a BIST unit, Averbuj also discloses the memory interface (10, 41) is commonly incorporated into (memory or other) chips, or “integrated circuits.” Ex. 1012 at 98, 277; Ex. 1003 at ¶ 126. Averbuj further discloses that the device block (6) and the components contained therein, such as the sequencer (8) and the memory modules (12), are “distributed” and “located throughout an electronic device,” i.e., physically separated. *See e.g.*, Ex. 1005 at ¶¶ [0013], [0015]; Ex. 1003 at ¶ 127. Since memory interfaces (10, 41) are also components of each device block (6), at a minimum, the different sets of memory interfaces (10, 41) belonging to different device blocks (6) must also be physically separated. Ex. 1003 at ¶ 127. Averbuj also discloses that the various integrated circuits or chips of an electronic device are “mounted on a circuit board,” indicating that the integrated circuits and chips comprising the electronic device are packaged integrated circuits or chips. Ex. 1005 at ¶ [0005]; Ex. 1016 at 1:15-27; Ex. 1003 at ¶ 128. Therefore, Averbuj discloses that the memory interfaces (each containing a “data handler”) of different

device blocks (which comprise a “memory system”) must be distributed across “at least two physically separate integrated circuit packages.” Ex. 1003 at ¶ 129.

Pet. 41. In other words, Petitioner asserts that because the sequencer can be distributed throughout an electronic device, the memory system comprises two separate chip packages. We disagree. Claim 4 requires that the memory module be a “two physically separate integrated circuit packages.”

Petitioner has not shown sufficient support for its broad reading that the memory system as claimed would be distributed across the device as Petitioner suggests. Thus, Petitioner has not shown a reasonable likelihood of success as to anticipation of claim 4 by Averbuj.

C. Claims 1 and 4—Obviousness over Averbuj (Ex. 1005)

Petitioner argues that claims 1 and 4 are unpatentable over Averbuj under 35 U.S.C. § 103. Pet. 31–38. As to claim 1, Petitioner includes an obviousness ground in case we determine that Averbuj does not disclose a “memory system configured to be operatively coupled to a memory controller of a computer system” and “memory chip.” *Id.* at 31–34. We determine above that both of these claim limitations are disclosed by Averbuj. Inasmuch as “anticipation is the epitome of obviousness” (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)), we determine that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj alone.²

As to claim 4, we determine above that Petitioner has not shown sufficiently that Averbuj anticipates those claims. Therefore, we discuss

² There is no evidence of objective indicia of non-obviousness yet of record.

below Petitioner's contention that dependent claim 4 would have been obvious over Averbuj.

Petitioner's argument regarding obviousness of claims 4 over Averbuj alone are based on Petitioner's declarant's testimony that it was known to house data handlers in physically separate integrated circuit packages. Pet. 34–38 (citing Ex. 1003 ¶¶ 143–161)³. Petitioner argues that to do so would “have been the use of a known technique for its known purpose and would have achieved only predictable results, such as the protection of integrated circuits by the packaging.” Pet. 35. Petitioner asserts that memory modules at the time of the invention were composed of memory chips mounted on different portions of a memory board with BIST units in each memory chip. *Id.*

Given that memory chips mounted on different portions of a memory board with BIST units in each memory chip were known, according to Petitioner, Petitioner asserts that reducing redundancy would have been a motivation to modify Averbuj. Specifically, Petitioner asserts

[a] skilled artisan that seeks to reduce the redundant BIST circuitry of conventional memory chips and BIST units would have been motivated to modify the conventional design in such a way as to avoid unnecessary modifications to the conventional

³ Petitioner's arguments regarding obviousness of claims 4 over Averbuj alone are based in part on disclosures in Tsern. Pet. 34–35 (citing Ex. 1006 ¶¶ 29, 52–53, Fig. 9A). Petitioner's arguments based on the disclosure of Tsern rely on structures in Tsern to modify Averbuj rather than only Tsern's general teachings to one of ordinary skill in the art and, thus, those arguments are properly considered below in Petitioner's ground based upon the combination of Averbuj and Tsern. As such we do not consider them here.

design. Ex. 1003 ¶ 151. Specifically, a skilled artisan would have understood that the BIST controller and sequencer portions of conventional BIST units would need to be carved out from each memory chip and combined to reduce redundant circuitry. Ex. 1003 ¶ 151. However, a skilled artisan would also have understood and recognized that the memory interface portions need not be excised and implemented as a separate chip since doing so would not reduce much, if any, circuitry. Ex. 1003 ¶ 151. Hence, a skilled artisan would have been motivated to avoid unnecessary design changes by keeping the original design with respect to the memory interface portion—i.e., retain the memory interfaces (each of which includes a “*data handler*”) within each memory chip that was separately packaged and mounted on a printed circuit board of a memory module (“*physically separate integrated circuit packages*.”). Ex. 1003 ¶ 152.

Pet. 53–54. We disagree.

As to Petitioner’s arguments supported by its declarant, we agree with Patent Owner and find those to employ hindsight to piece together the device as claimed. Prelim. Resp. 46. The only evidence in the record of separating the data handlers and BIST controllers as claimed presented by Petitioner, other than the long complicated explanation of its declarant or the misplaced reliance on the structures of Tsern under this challenge, is in the ’501 Patent itself. “The inventor’s own path itself never leads to a conclusion of obviousness; that is hindsight. What matters is the path that the person of ordinary skill in the art would have followed, as evidenced by the pertinent prior art.” *Otsuka Pharm. Co., v. Sandoz, Inc.*, 678 F.3d 1280, 1296 (Fed. Cir. 2012). Thus, absent impermissible hindsight reconstruction, we are not persuaded that Petitioner has provided a sufficient rationale for modifying Averbuj as proposed in the Petition. Moreover, Petitioner declarant’s testimony lacks sufficient articulated reasoning with rational underpinning as to why an ordinary artisan, at the time of the invention,

would have had a reason to modify the prior art in the manner required by the challenged claims. *See* Ex. 1003 ¶¶ 141–161. Thus, Petitioner has not shown sufficiently that Averbuj discloses the physically separate integrated circuit packages limitation for the reasons discussed above.

D. Claim 4 — Obviousness over Averbuj (Ex. 1005) and Tsern (Ex. 1006)

Petitioner argues that claim 4 is unpatentable over Averbuj and Tsern under 35 U.S.C. § 103. Pet. 38–42.

1. Tsern

Figure 1 of Tsern is reproduced below.

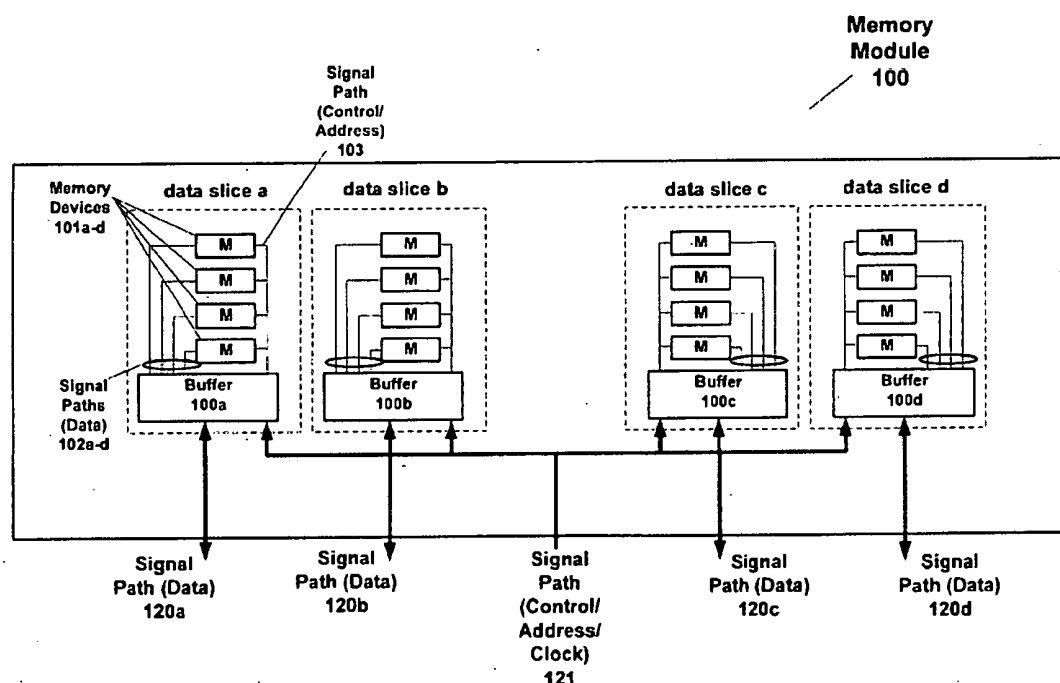


Fig. 1

Figure 1 of Tsern, above, illustrates an embodiment of a memory module with memory devices (101a–d) and corresponding buffer devices (100a–d). Figure 18 of Tsern is reproduced below.

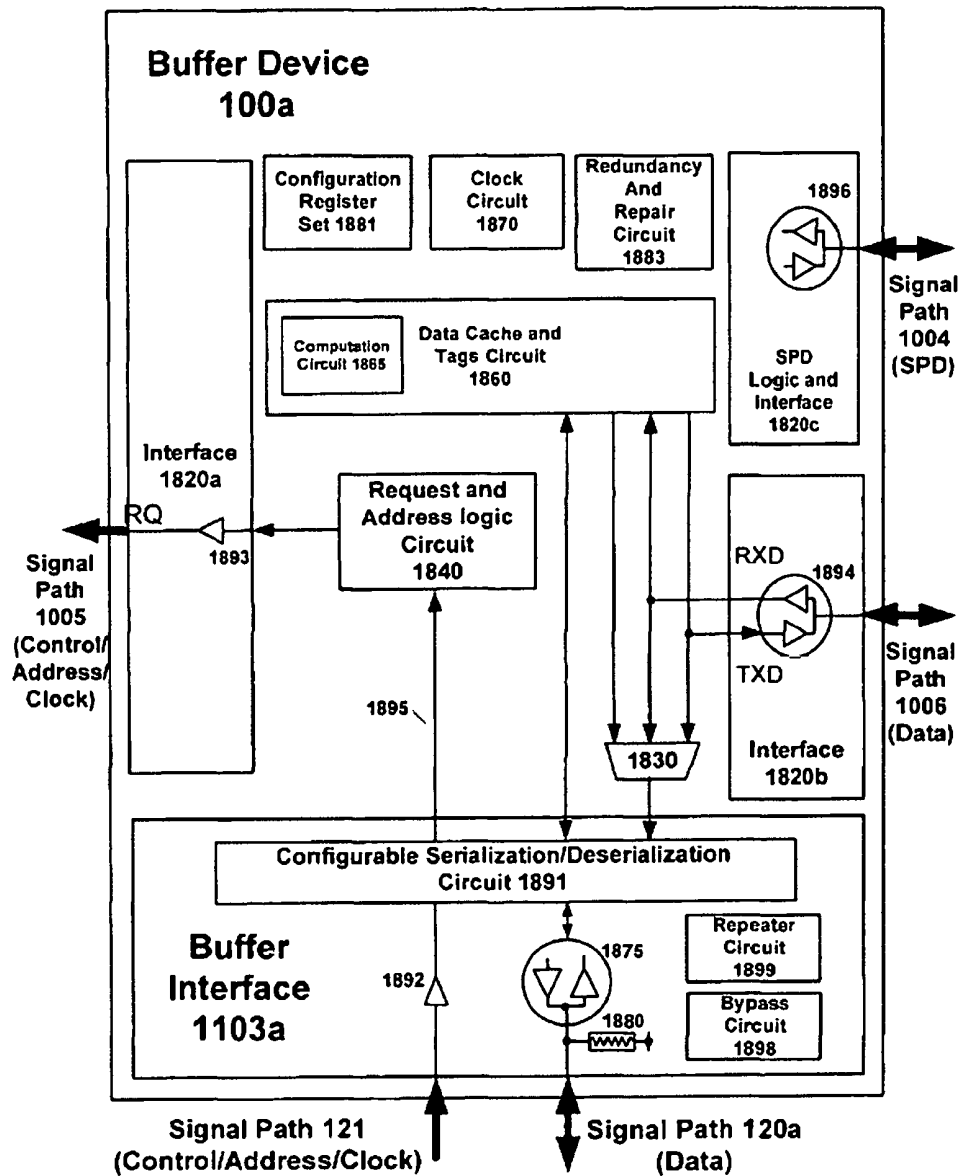


Fig. 18

As shown in Figure 18, above, Tsern discloses that, in one embodiment, each buffer device (100a) includes a redundancy and repair unit (1883) which tests and repairs the corresponding memory device(s). Ex. 1006, Fig. 18, ¶ 77. Each such buffer device (100a) includes data, address, and control interfaces (1820a and 1820b) that can be programmed

or configured to support memory modules with different number, size, width, and type of memory devices. *Id.*, Fig. 18, ¶ 99.

Figure 1 of Tsern shows that each buffer device (100a–d) communicates address, control, and data signals with its corresponding memory device(s) (101a–d) without being in communication with any of the other buffer devices (100a–d) or other data ports of the other memory devices. Ex. 1006, Fig. 1.

2. *Analysis*

As to claim 1, Petitioner does not includes an obviousness ground with Tsern. As discussed above, we determine that claim 1 is anticipated by Averbuj. Inasmuch as “anticipation is the epitome of obviousness” (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)), we determine that Petitioner has also established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj alone. We are not persuaded, however, that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 4 is anticipated by, or would have been obvious over, Averbuj. Thus, we discuss below the combination of Averbuj and Tsern as it relates to claim 4.

Claim 4 recites that “[t]he memory system of claim 1, wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” Petitioner relies on Figure 9 of Tsern, reproduced below.

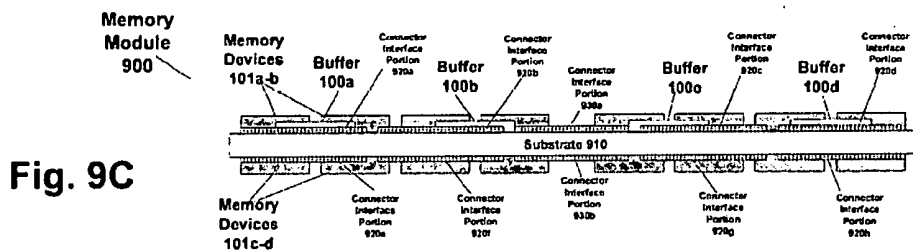
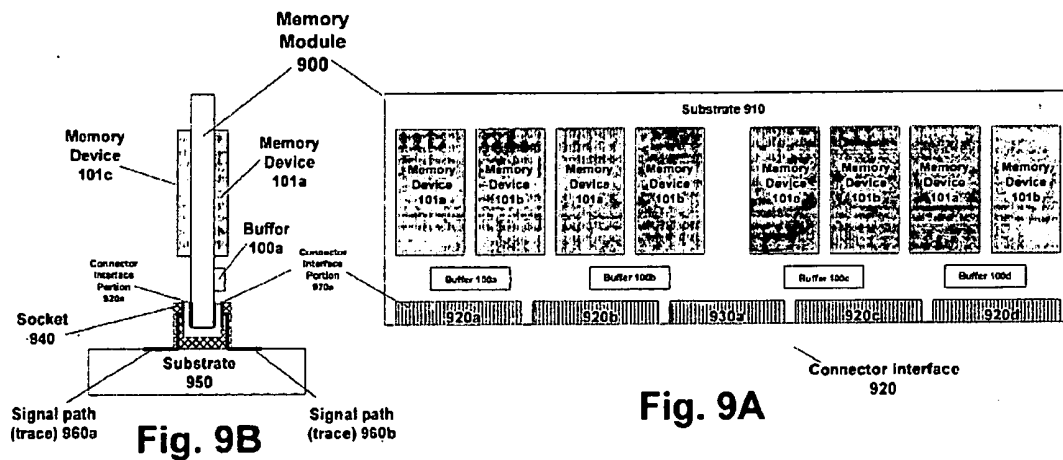


Figure 9 of Tsern, above, shows an implementation of a memory module in which each of the four buffer devices (100a–d) is implemented as a physically separate integrated circuit package and mounted on different portions of the printed circuit board that houses the memory module. Ex. 1006, Figs. 9A–C.

For purposes of this decision and based on Petitioner’s reliance on specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it also would have been obvious to modify the teachings of Averbuj with Tsern’s teachings, to include physically separate integrated chip packages. Pet. 60–62. For example, Petitioner asserts:

It would have been obvious to include the “separate integrated circuit packages” configuration of Tsern, including the distribution of self-test circuitry, in the system of Averbuj for several reasons. First, to do so would have been merely the arrangement of old elements with each performing the same function it had been known to perform and yielding no more than one would expect from such an arrangement. Tsern demonstrates that, before the priority date of the ’501 Patent, it was within the average skill of the art to include a plurality of data handler circuits in physically separate integrated circuit packages, on a printed circuit board and that such circuits would operate as expected. Ex. 1003 at ¶ 167.

A skilled artisan would have been further motivated to implement the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj, particularly for the memory interfaces of Averbuj, in order to reduce the busing area a centralized approach would require. *See, e.g.*, Ex. 1017 at 3:45-53; Ex. 1003 at ¶ 168.

A skilled artisan also would also have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj so that the self-test circuitry could be conveniently placed in the same packaging as buffer circuitry used to access and isolate different portions of the memory array. By the priority date of the ’501 Patent it was known that separate buffers, in separate packages, for different portions of the memory array could advantageously reduce the load experienced by the memory controllers and improve the memory timing. Ex. 1018 at 1:5-23, Fig. 3; Ex. 1019 at 1:32-2:3, Fig. 5. A skilled artisan would therefore have been motivated to place the sequencers and the memory interfaces of Averbuj within such buffer components in order to reduce the load on the memory controller and also improve memory timing. Ex. 1003 at ¶ 169.

A skilled artisan also would have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj so that failing sequencers and memory interface components could be easily replaced, by

simply replacing a standard package on the printed circuit board, without the necessity to replace other self-test circuitry in other packages that was operating normally. Ex. 1003 at ¶ 170.

A skilled artisan also would have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj in order to place each data handler at positions on the PCB close and perhaps closest to its associated memory. Doing so would (i) simplify the wiring plan on the circuit board (Ex. 1021 at 6); (ii) reduce EMI interference and improve signal integrity (Ex. 1010 at 23-24, 207); and (iii) limit the capacitive load on those lines and permit faster and more efficient data transmission (Ex. 1020 at 2:27–3:6). See *above* at § V.C.1; Ex. 1003 at ¶ 171.

Id.

Patent Owner argues that Petitioner has not shown that Tsern’s buffers (the alleged data handlers in Tsern) “generate” data as required by the claims. Prelim. Resp. 48. However, Petitioner relies on the teachings of both Averbuj (sequencer) and Tsern (buffer) in combination, so we are not persuaded by those arguments.

Patent Owner also argues that “the Petition fails to explain why a POSITA designing a circuit for an embedded memory BIST architecture (such as Averbuj) would even consider a ‘separate integrated circuit packages’ configuration, as it would increase the size of the circuit by several orders of magnitude, contrary to the teaching of Averbuj.” Prelim. Resp. 49. In considering obviousness, there is no requirement that the product of a combination of teachings must be optimal or better than other known structures or techniques. See *In re Fulton*, 391 F.3d 1195, 1200 (Fed.Cir.2004) (“[C]ase law does not require that a particular combination must be the preferred, or the most desirable, combination described in the

prior art in order to provide the motivation for the current invention.”).

Thus, we are not persuaded by this argument.

Patent Owner asserts that Petitioner’s combination would be inoperative. Prelim Resp. 49–50. We find this argument is conclusory and relies on an expectation that Averbuj and Tsern must be bodily incorporated. Patent Owner also argues in conclusory fashion that the motivation to combine is “illogical and unsupported,” that there would be no benefit from the combination, that Averbuj and Tsern are from “different fields of art,” that Averbuj teaches away from the combinations, and that modifying Averbuj would change its principle of operation. *Id.* 51–53. We are not persuaded by those conclusory arguments.

Patent Owner also asserts “[e]ven if Averbuj’s sequencer and memory interface were capable of being physically combined into Tsern’s buffer, . . . the combination would not have worked for its intended purpose. Averbuj’s sequencers require generalized command data, and the Petition fails to show how this could be compatible with the standards-based interface protocol in Tsern.” *Id.* 52–53. Patent Owner relies only on attorney argument and thus we are not persuaded.

Accordingly, upon review of Petitioner’s analysis and supporting evidence, we determine that Petitioner has demonstrated that there is a reasonable likelihood that it would prevail in showing that claims 1 and 4 are unpatentable as obvious over the combination of Averbuj and Tsern.

III. CONCLUSION

For the foregoing reasons, we determine that the information presented establishes a reasonable likelihood that Petitioner would prevail in showing that claims 1 and 4 of the ’501 Patent are unpatentable. At this

preliminary stage, we have not made a final determination with respect to the patentability of the challenged claims or any underlying factual and legal issues.

IV. ORDER

Accordingly, it is:

ORDERED that pursuant to 35 U.S.C. § 314(a), an *inter partes* review is hereby instituted as to

1. claim 1 of the '501 Patent under 35 U.S.C. § 102(b) as anticipated by Averbuj;
2. claim 1 of the '501 Patent under 35 U.S.C. § 103(a) as obvious over Averbuj; and
3. claim 4 of the '501 Patent under 35 U.S.C. 103(a) as obvious over Averbuj and Tsern;

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial, which commences on the entry date of this decision; and

FURTHER ORDERED that the trial is limited to the ground identified immediately above, and no other ground is authorized.

IPR2017-00562
Patent 8,359,501 B1

PETITIONER:

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UNITED STATES COURT OF APPEALS FOR THE FEDERAL CIRCUIT

**NOTICE OF ENTRY OF
JUDGMENT WITHOUT OPINION**

JUDGMENT ENTERED: 11/14/2017

The judgment of the court in your case was entered today pursuant to Rule 36. This Court affirmed the judgment or decision that was appealed. None of the relief sought in the appeal was granted. No opinion accompanied the judgment. The mandate will be issued in due course.

Information is also provided about petitions for rehearing and suggestions for rehearing en banc. The questions and answers are those frequently asked and answered by the Clerk's Office.

Costs are taxed against the appellant in favor of the appellee under Rule 39. The party entitled to costs is provided a bill of costs form and an instruction sheet with this notice.

The parties are encouraged to stipulate to the costs. A bill of costs will be presumed correct in the absence of a timely filed objection.

Costs are payable to the party awarded costs. If costs are awarded to the government, they should be paid to the Treasurer of the United States. Where costs are awarded against the government, payment should be made to the person(s) designated under the governing statutes, the court's orders, and the parties' written settlement agreements. In cases between private parties, payment should be made to counsel for the party awarded costs or, if the party is not represented by counsel, to the party pro se. Payment of costs should not be sent to the court. Costs should be paid promptly.

If the court also imposed monetary sanctions, they are payable to the opposing party unless the court's opinion provides otherwise. Sanctions should be paid in the same way as costs.

Regarding exhibits and visual aids: Your attention is directed to FRAP 34(g) which states that the clerk may destroy or dispose of the exhibits if counsel does not reclaim them within a reasonable time after the clerk gives notice to remove them. (The clerk deems a reasonable time to be 15 days from the date the final mandate is issued.)

FOR THE COURT

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court

cc: Wayne Bradley
Michael Francis Heafey
Scott P McBride
Gregory Schodde
Anup M. Shah
Ronald Spuhler
Thomas J. Wimbiscus

16-2666, -2667: Smart Modular Technologies v. Netlist, Inc.
United States Patent and Trademark Office, Case Nos. IPR2014-01372, IPR2014-01374

NOTE: This disposition is nonprecedential.

**United States Court of Appeals
for the Federal Circuit**

SMART MODULAR TECHNOLOGIES, INC.,
Appellant

v.

NETLIST, INC.,
Appellee

2016-2666, 2016-2667

Appeals from the United States Patent and Trade-
mark Office, Patent Trial and Appeal Board in Nos.
IPR2014-01372, IPR2014-01374.

JUDGMENT

MICHAEL FRANCIS HEAFEY, Troutman Sanders LLP,
San Francisco, CA, argued for appellant. Also represent-
ed by ANUP M. SHAH, Charlotte, NC.

THOMAS J. WIMBISCUS, McAndrews, Held & Malloy,
Ltd., Chicago, IL, argued for appellee. Also represented
by WAYNE BRADLEY, SCOTT P. MCBRIDE, GREGORY
SCHODDE, RONALD SPUHLER.

THIS CAUSE having been heard and considered, it is

ORDERED and ADJUDGED:

PER CURIAM (PROST, *Chief Judge*, WALLACH and
HUGHES, *Circuit Judges*).

AFFIRMED. See Fed. Cir. R. 36.

ENTERED BY ORDER OF THE COURT

November 14, 2017

Date

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court

UNITED STATES COURT OF APPEALS **FOR THE FEDERAL CIRCUIT**

Questions and Answers

Petitions for Rehearing (Fed. Cir. R. 40) **and** **Petitions for Hearing or Rehearing En Banc (Fed. Cir. R. 35)**

Q. When is a petition for rehearing appropriate?

A. Petitions for panel rehearing are rarely successful because they most often fail to articulate sufficient grounds upon which to grant them. For example, a petition for panel rehearing should not be used to reargue issues already briefed and orally argued; if a party failed to persuade the court on an issue in the first instance, a petition for panel rehearing should not be used as an attempt to get a second "bite at the apple." This is especially so when the court has entered a judgment of affirmance without opinion under Fed. Cir. R. 36. Such dispositions are entered if the court determines the judgment of the trial court is based on findings that are not clearly erroneous, the evidence supporting the jury verdict is sufficient, the record supports the trial court's ruling, the decision of the administrative agency warrants affirmance under the appropriate standard of review, or the judgment or decision is without an error of law.

Q. When is a petition for hearing or rehearing en banc appropriate?

A. En banc decisions are extraordinary occurrences. To properly answer the question, one must first understand the responsibility of a three-judge merits panel of the court. The panel is charged with deciding individual appeals according to the law of the circuit as established in the court's precedential opinions. While each merits panel is empowered to enter precedential opinions, the ultimate duty of the court en banc is to set forth the law of the Federal Circuit, which merit panels are obliged to follow.

Thus, as a usual prerequisite, a merits panel of the court must have entered a precedential opinion in support of its judgment for a suggestion for rehearing en banc to be appropriate. In addition, the party seeking rehearing en banc must show that either the merits panel has failed to follow identifiable decisions of the U.S. Supreme Court or

Federal Circuit precedential opinions or that the merits panel has followed circuit precedent, which the party seeks to have overruled by the court en banc.

Q. How frequently are petitions for rehearing granted by merits panels or petitions for rehearing en banc accepted by the court?

A. The data regarding petitions for rehearing since 1982 shows that merits panels granted some relief in only three percent of the more than 1900 petitions filed. The relief granted usually involved only minor corrections of factual misstatements, rarely resulting in a change of outcome in the decision.

En banc petitions were accepted less frequently, in only 16 of more than 1100 requests. Historically, the court itself initiated en banc review in more than half (21 of 37) of the very few appeals decided en banc since 1982. This sua sponte, en banc review is a by-product of the court's practice of circulating every precedential panel decision to all the judges of the Federal Circuit before it is published. No count is kept of sua sponte, en banc polls that fail to carry enough judges, but one of the reasons that virtually all of the more than 1100 petitions made by the parties since 1982 have been declined is that the court itself has already implicitly approved the precedential opinions before they are filed by the merits panel.

Q. Is it necessary to have filed either of these petitions before filing a petition for certiorari in the U.S. Supreme Court?

A. No. All that is needed is a final judgment of the Court of Appeals. As a matter of interest, very few petitions for certiorari from Federal Circuit decisions are granted. Since 1982, the U.S. Supreme Court has granted certiorari in only 31 appeals heard in the Federal Circuit. Almost 1000 petitions for certiorari have been filed in that period.

UNITED STATES COURT OF APPEALS FOR THE FEDERAL CIRCUIT

INFORMATION SHEET

FILING A PETITION FOR A WRIT OF CERTIORARI

There is no automatic right of appeal to the Supreme Court of the United States from judgments of the Federal Circuit. You must file a petition for a writ of certiorari which the Supreme Court will grant only when there are compelling reasons. (See Rule 10 of the Rules of the Supreme Court of the United States, hereinafter called Rules.)

Time. The petition must be filed in the Supreme Court of the United States within 90 days of the entry of judgment in this Court or within 90 days of the denial of a timely petition for rehearing. The judgment is entered on the day the Federal Circuit issues a final decision in your case. [The time does not run from the issuance of the mandate, which has no effect on the right to petition.] (See Rule 13 of the Rules.)

Fees. Either the \$300 docketing fee or a motion for leave to proceed in forma pauperis with an affidavit in support thereof must accompany the petition. (See Rules 38 and 39.)

Authorized Filer. The petition must be filed by a member of the bar of the Supreme Court of the United States or by the petitioner representing himself or herself.

Format of a Petition. The Rules are very specific about the order of the required information and should be consulted before you start drafting your petition. (See Rule 14.) Rules 33 and 34 should be consulted regarding type size and font, paper size, paper weight, margins, page limits, cover, etc.

Number of Copies. Forty copies of a petition must be filed unless the petitioner is proceeding in forma pauperis, in which case an original and ten copies of the petition for writ of certiorari and of the motion for leave to proceed in forma pauperis. (See Rule 12.)

Where to File. You must file your documents at the Supreme Court.

Clerk
Supreme Court of the United States
1 First Street, NE
Washington, DC 20543
(202) 479-3000

No documents are filed at the Federal Circuit and the Federal Circuit provides no information to the Supreme Court unless the Supreme Court asks for the information.

Access to the Rules. The current rules can be found in Title 28 of the United States Code Annotated and other legal publications available in many public libraries.

UNITED STATES COURT OF APPEALS FOR THE FEDERAL CIRCUIT

16-2666, 16-2667

SMART MODULAR TECHNOLOGIES, INC.,

Appellant

v.

NETLIST, INC.,

Appellee

Appeals from the United States Patent and Trademark Office in nos. IPR2014-01372, IPR2014-01374

Administrative Patent Judge Linda M. Gaudette, Administrative Patent

Judge Matthew R. Clements, Administrative Patent Judge Bryan F. Moore, Administrative Patent

Judge Peter P. Chen

MANDATE

In accordance with the judgment of this Court, entered November 14, 2017, and pursuant to Rule 41(a) of the Federal Rules of Appellate Procedure, the formal mandate is hereby issued.

FOR THE COURT

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and
SK HYNIX MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00562
Patent 8,359,501 B1

Before BRYAN F. MOORE, MATTHEW R. CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. 318(a)

I. INTRODUCTION

SK hynix Inc., SK hynix America Inc., and SK hynix memory solutions Inc., (collectively “Petitioner”) filed a Petition (Paper 1, “Pet.”) pursuant to 35 U.S.C. §§ 311–319 to institute an *inter partes* review of claims 1 and 4 (“the challenged claims”) of U.S. Patent No. 8,359,501 B1 (“the ’501 Patent,” Ex. 1001). The Petition is supported by the Declaration of Pinaki Mazumder, Ph.D. (“Mazumder Declaration,” “Mazumder Dec.,” Ex. 1003). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (“Prelim. Resp.,” Paper 6).

On July 7, 2017, we instituted an *inter partes* review of claims 1 and 4 of the ’501 Patent, but did not institute on all grounds. Paper 7, 7, 29 (“Inst. Dec.”). Patent Owner filed a Response. Paper 14 (“PO Resp.”). Patent Owner Response is supported by the Declaration of R. Jacob Baker, Ph.D. (“Baker Declaration,” “Baker Dec.,” Ex. 2010). Petitioner filed a Reply. Paper 18 (“Reply”).

On February 28, 2018, Petitioner and Patent Owner filed motions to exclude. Paper 21 (“Pet. Mot. to Excl.”); Paper 23 (“PO Mot. to Excl.”). Petitioner and Patent Owner filed responses to their respective motions to exclude. Paper 25 (“Pet. Mot. to Excl. Resp.”); Paper 26 (“PO Mot. to Excl. Resp.”). Petitioner and Patent Owner filed replies to those respective responses. Paper 29 (“Pet. Mot. to Excl. Reply”); Paper 30 (“PO Mot. to Excl. Reply”). Patent Owner also filed a Motion to Strike directed to portions of Petitioner’s Reply, with Petitioner filing an Opposition. Paper 32 (“Mot. to Strike”); Paper 33 (“Opp. Mot. to Strike”).

On March 13, 2018, we entered a *sua sponte* Order to Show Cause why the *inter partes* review should not be terminated as to claim 1 because it

has been finally adjudicated as unpatentable. Paper 24 (“Show Cause Order”). Petitioner and Patent Owner responded to the order. Paper 27 (“Pet. Show Cause Resp.”); Paper 28 (“PO Show Cause Resp.”).

An oral hearing was held on April 6, 2018. Paper 34 (“Tr.”).

On April 24, 2018, the Supreme Court held that a decision on institution under 35 U.S.C. § 314 may not institute on less than all claims challenged in the petition. 138 S. Ct. 1348 (2018). We modified our Institution Decision to institute trial on all of the challenged claims and all of the grounds presented in the Petition. Paper 35 (“SAS Order”). We invited the parties to request briefing regarding the newly added grounds and no request was made. *Id.*

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is issued pursuant to 35 U.S.C. § 318(a). For the reasons that follow, we determine that Petitioner has shown by a preponderance of the evidence that claim 4 is unpatentable. Based on collateral estoppel, we also terminate the trial pursuant to 37 C.F.R. § 42.72 as to claim 1.

A. Related Proceedings

Petitioner recites a list of District Court proceedings related to this *inter partes* review. Pet. 2. This *inter partes* review challenges the same patent at issue in the decision entered in IPR2014-00971 (the ’971 IPR). *See Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB Dec. 16, 2014) (Paper 12). In the ’971 IPR, as to the claims at issue here, claim 1 was held unpatentable as anticipated by Averbuj in a final written decision. *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB Apr. 27, 2016) (Paper 37). That decision was appealed to Federal Circuit, the decision was affirmed, and the mandate issued on December 21, 2017. *See* Exs. 1029,

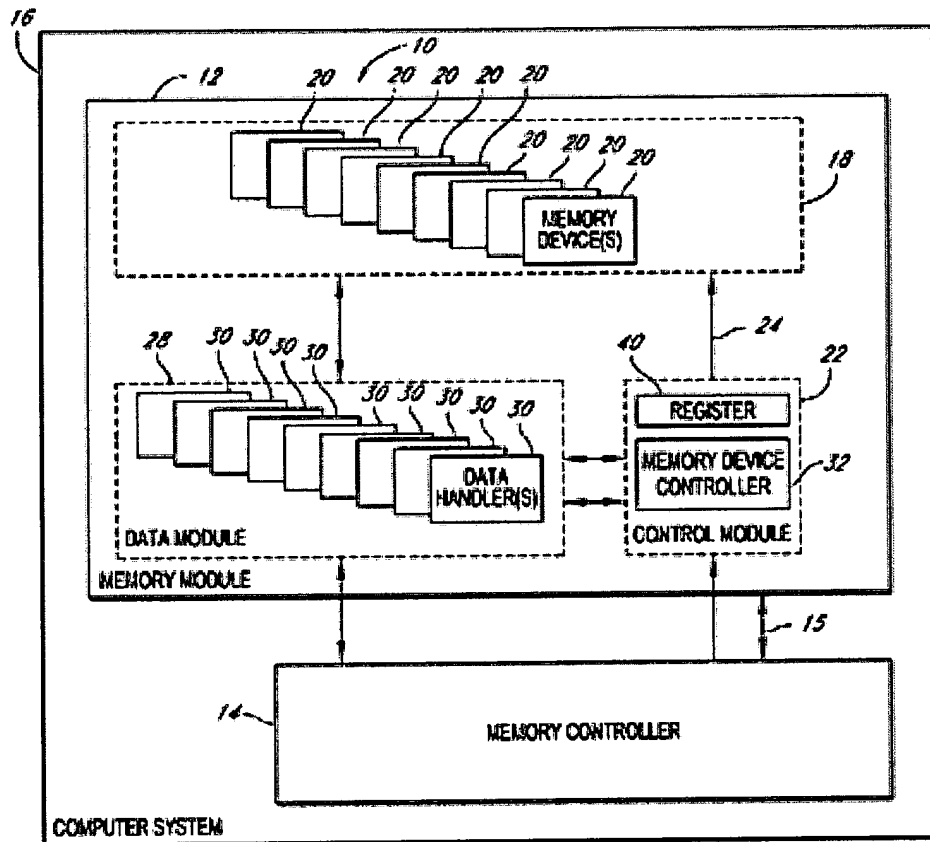
1034. In the '971 IPR, claims 1 and 4 were also challenged as obvious over Averbuj and Tsern, but that challenge was not instituted. *Sandisk*, IPR2014-00971 (Paper 12, 15–16, 23).

The '501 Patent was also challenged in IPR2014-01374 (the '1374 IPR). In the 1374 IPR, as to the claims at issue here, claims 1 and 4 were held not unpatentable as anticipated by Averbuj (under a different theory of anticipation than the '971 IPR) in a final written decision. *Smart Modular Techs. Inc. v. Netlist, Inc.*, Case IPR2014-01374 (PTAB March 6, 2016) (Paper 44). The decision in the '1374 IPR was not appealed to Federal Circuit.

The '501 Patent was also challenged in IPR2014-01375 (the '1375 IPR), in which institution was denied as to all challenges on the merits. *Smart Modular Techs. Inc. v. Netlist, Inc.*, Case IPR2014-01375 (PTAB Mar. 13, 2015) (Paper 12). Neither Averbuj nor Tsern was asserted in the '1375 IPR.

B. The '501 Patent

The '501 Patent relates to self-testing electronic memory modules. Ex. 1001, 1:28–29. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '501 Patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:6–8. Memory module 10 includes a plurality of memory devices 18, e.g., ETT DRAM chips, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 3:34–36, 5:10–12. Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:14–16, 42–46. “In various embodiments, the control module 22 includes .

. . one or more application-specific integrated circuit[s].” *Id.* at 9:62–64.
“[D]ata module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20.” *Id.* at 6:5–8. “For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according [to] the operating specification of the memory devices 20.” *Id.* at 6:8–11. “In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20.” *Id.* at 5:48–52. “For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation.” *Id.* at 5:52–55.

C. Illustrative Claim

Of the challenged claims, claim 1 is independent. Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A memory system configured to be operatively coupled to a memory controller of a computer system, the memory system comprising:
 - a plurality of memory chips;
 - a plurality of data handlers configured to be operated independently from one other, wherein one or more data handlers of the plurality of data handlers are configured to generate data for writing to a corresponding one or more memory chips of the plurality of memory chips;
 - a control circuit configured to generate address and control signals, wherein the memory system is configured to test the one or more memory chips using the address and control signals

generated by the control circuit and using the data generated by the one or more data handlers.

Ex. 1001, 16:38–51.

D. Instituted Grounds of Unpatentability

We instituted trial on the following grounds (Inst. Dec. 29; SAS Order 2):

Reference[s]	Basis	Claims Challenged
Averbuj ¹	§ 102	1 and 4
Averbuj	§ 103	1 and 4
Averbuj and Tsern ²	§ 103	4

II. ANALYSIS

A. Collateral Estoppel

The parties raised an issue arising from challenges to the subject patent in related *inter partes* review proceedings brought by a different petitioner, namely the '971 IPR. In the '971 IPR, the Board held that claim 1 is unpatentable over art that is the same as the art considered in this proceeding as against a different petitioner (Sandisk) than this proceeding. *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (PTAB Apr. 27, 2016) (Paper 37). The Federal Circuit affirmed that decision. *Netlist, Inc. v. SanDisk LLC*, 702 F. App'x 986 (Fed. Cir. 2017) (per curiam). Accordingly, claim 1, which is involved in this proceeding, has been held

¹ U.S. Patent Application Publication No. 2005/0257109 (Ex. 1005, "Averbuj")

² U.S. Patent Application Publication No. 2007/0070669 (Ex. 1006, "Tsern")

unpatentable. This raises the issue of whether we should terminate the proceeding as to claim 1 based on collateral estoppel.

“It is well established that collateral estoppel, also known as issue preclusion, applies in the administrative context.” *MaxLinear, Inc. v. CF CRESPE LLC*, 880 F.3d 1373, 1376 (Fed. Cir. 2018) (citations omitted); *Webpower, Inc.*, Case IPR2016-01239, slip. op. at 26–28 (PTAB Dec. 26, 2017) (Paper 21). A patentee is estopped from asserting the validity of a patent that has been declared invalid in a prior suit against a different defendant, unless patentee demonstrates that he did not have full and fair opportunity, procedurally, substantively, and evidentially, to litigate the validity of his patent in the prior suit. *MaxLinear, Inc.*, 880 F.3d at 1377 (citing *Blonder-Tongue Labs., Inc. v. Univ. of Illinois Found.*, 402 U.S. 313, 91 S. Ct. 1501, 28 L. Ed. 2d 788 (1971) (finding collateral estoppel applies to a patentee who had a full and fair opportunity to litigate the validity of a patent in a prior federal case.)) Thus, as to a different petitioner, when the prior decisions finding claims unpatentable have “subsequently been affirmed by [the Federal Circuit] those prior decisions, having been affirmed by [the Federal Circuit], are binding in this proceeding, as a matter of collateral estoppel.” *Id.* at 1376.

In response to our Order to Show Cause why the proceeding should not be terminated as to claim 1, Patent Owner argued that “[t]his IPR should be terminated as to claim 1” to avoid wasting the resources of the Board. PO Show Cause Resp. 1. Petitioner argued “[c]ollateral estoppel only binds Patent Owner, not Petitioners, and Petitioners, in addition to advancing the same mapping of Averbuj to claim 1 as did SanDisk in IPR2014-0971, also present alternative mappings as well.” Pet. Show Cause Br. 1. Petitioner

further argues “[g]iven that the Board may need to consider whether Petitioners’ alternative mappings read on the limitations of claim 1 when addressing the patentability of claims 4, there may be little efficiency gained by terminating this IPR as to claim 1.” *Id.* at 2. We recognize that Petitioner presents alternative mappings as to claim 1. Nevertheless, as explained below, we find that claims 4 is unpatentable under the mappings used in the ’971 IPR. Thus, given that we find that collateral estoppel applies, judicial efficiency is advanced by declining to reassess the patentability of claim 1 under alternative mappings.

Patent Owner asserts that this *inter partes* review relies on obviousness based on the combination of Averbuj and Tsern rather than the ground under which claim 1 was found unpatentable in the ’971 IPR, i.e. anticipation by Averbuj. Thus, according to Patent Owner, “Petitioners must also provide evidence of a reason why a POSITA would have modified/combined the limitations of claim 1 (as allegedly shown in Averbuj) with the limitations of dependent claims 4 (as allegedly shown in Tsern), to create the claimed inventions.” PO Show Cause Br. 1. We agree that Petitioner must show a motivation to combine Tsern with Averbuj, but this does not mean that Petitioner must prove again that Averbuj meets the limitations of claim 1; it means only that Petitioner need show a rationale for combining Tsern with those aspects of Averbuj that the Board has already determined meet the limitations of claim 1.

Patent Owner asserts that “collateral estoppel does not apply to the application of the claim 1 limitations in assessing dependent claim 4.” *Id.* at 2. Patent Owner further asserts “[a] party seeking to apply collateral estoppel based on a prior action must show that (1) the identical issue was

actually litigated; (2) the issue was actually decided in a final decision on the merits; (3) the issue was necessary to the final decision; and (4) the party being estopped was adequately represented in the prior action [i.e. “the *United Access* test”].” *Id.* at 2 (citing *United Access Techs., LLC v. CenturyTel Broadband Services LLC*, 778 F.3d 1327, 1331 (Fed. Cir. 2015)). We agree that this is the proper test to use to assess the applicability of collateral estoppel. Nevertheless, we note *United Access* applies collateral estoppel in a different context and suggests that issue preclusion can be applied in an obviousness challenge, even if the final decision on the original issue was considered under an anticipation challenge. In particular, in *United Access* it was noted that “[i]ssue preclusion [traditionally called “collateral estoppel”³] applies to issues litigated in prior disputes . . . [and, thus, i]t thus makes no difference that the earlier no-damages-due-to-competition ruling arose in a case under the Truth in Negotiation Act and that this dispute arises under the False Claims Act and the common law.” *United Access*, 778 F.3d. at 726. The “issue” of whether Averbuj teaches the limitations of claim 1 is identical to the issue that was litigated in the ‘971 IPR which found that Averbuj anticipates the limitations of claim 1. The question becomes how does the fact that the limitations of claim 1 rely on an obviousness combination with Tsern effect the analysis. To that point, we recognize that “though anticipation is the epitome of obviousness, [they] are separate and distinct concepts.” *Jones v. Hardy*, 727 F.2d 1524, 1529 (Fed.Cir.1984). It is that difference that is important here.

³ *Mother’s Rest., Inc. v. Mama’s Pizza, Inc.*, 723 F.2d 1566, 1569 (Fed. Cir. 1983)

The Federal Circuit has noted that obviousness is different because: more than one reference can be used in obviousness; secondary considerations must be analyzed in obviousness; inherency in anticipation is different than obviousness (*Cohesive Techs., Inc. v. Waters Corp.*, 543 F.3d 1351, 1364 (Fed. Cir. 2008)); and single reference obviousness requires a showing of motivation to modify a reference when a party argues that such a modification must be made to meet the limitations of a claim (*Monsanto Tech. LLC v. E.I. DuPont de Nemours & Co.*, 878 F.3d 1336, 1346 (Fed. Cir. 2018)). Here, however, Petitioner relies on Averbuj alone, not on the combination of Averbuj and Tsern (or any modification of Averbuj), to meet any limitation of claim 1, Petitioner (here and in the '971 IPR) did not rely on inherency to show any limitations of claim 1, and Patent Owner has not argued secondary considerations. *See generally*, Pet.; PO Resp.; Ex. 1028 ('971 IPR FWD).

Thus, the relevant legal issue, in this essentially single reference obviousness situation as to claim 1, is the same as in anticipation, i.e. does the reference teach the limitations of the claim. In fact, anticipation requires a higher level of teaching requiring that references teach limitations of a claim explicitly or inherently. Under obviousness, the reference need only render a limitation obvious to one of skill in the art at the time of the invention. Thus, the adage “anticipation is the epitome of obviousness.” *Jones*, 727 F.2d at 1529. The Board is allowed to rely on findings regarding whether a reference anticipates a limitation in its obviousness analysis.⁴ *See*

⁴ In fact, the Board relied on this principle in the Decision to Institute. Dec. to Inst. 19 (“[W]e determine that claim 1 is anticipated by Averbuj.

Wasica Fin. GmbH v. Cont'l Auto. Sys., Inc., 853 F.3d 1272, 1278 (Fed. Cir. 2017) (stating approvingly in a footnote that the “Board noted that Oselin rendered [the challenged claims] obvious by virtue of its anticipation of them.”) (citing *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983)).

Also, as to the remaining prongs of the *United Access* test, the issue of whether Averbuj teaches the limitations of claim 1 was actually decided in a final decision on the merits, the issue was necessary to the final decision because it went to the ultimate issue of patentability, and there is no allegation that Patent Owner was not adequately represented in the prior action. Thus, for the reasons above, claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel and the issue of whether the limitations of claim 1 are met by Averbuj by obviousness over Averbuj and Tsern also is established under the doctrine of collateral estoppel. Thus, we terminate this proceeding as to claim 1.

Our reviewing court has recently ordered the Board, on remand, to assess whether claims depending from a claim that has been finally adjudicated as unpatentable should not also be found unpatentable for the same reasons as the independent claim. *MaxLinear, Inc.*, 880 F.3d at 1377–1378. We apply this analysis to claim 4. Claim 4 has not been finally adjudicated as unpatentable in any previous case. Because the differences between the unadjudicated patent claim 4 and the adjudicated patent claim 1

Inasmuch as ‘anticipation is the epitome of obviousness’ (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)), we determine that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj alone.”).

materially alter the question of unpatentability, collateral estoppel does not apply. *C.f. In re Arunachalam*, 709 F. App'x 699, 702 (Fed. Cir. 2017) (unpublished) (finding collateral estoppel applies to an adjudicated claim that is not “materially different” than the adjudicated claims); *accord MaxLinear, Inc.*, 880 F.3d at 1377–1378. In this case, there are material differences between claim 4 and claim 1. For example, claim 4, recites “the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” Separate ICs is not contemplated in claim 1. It is still true, however, as Petitioner asserts, “Patent Owner is precluded from relitigating here whether Averbuj discloses all the limitations of claim 1, as arranged in the claim, including for purposes of determining the patentability of claim 4.” Pet. Show Cause Resp. 3. Thus, we do not terminate as to claim 4 under collateral estoppel.

B. Relevant Law

1. Obviousness Law

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and, (4) where in evidence, so-called secondary

considerations, including commercial success, long-felt but unsolved needs, failure of others, and unexpected results.⁵ *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966) (“the *Graham* factors”).

2. *Level of Skill Law*

The level of ordinary skill in the art usually is evidenced by the references themselves. See *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978). For an obviousness analysis, prior art references must be “considered together with the knowledge of one of ordinary skill in the pertinent art.” *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994) (quoting *In re Samour*, 571 F.2d 559, 562 (CCPA 1978)). Moreover, “it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). That is because an obviousness analysis “need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; see also *In re Translogic Tech., Inc.*, 504 F.3d at 1259.

3. *Claim Interpretation Law*

In an *inter partes* review, claim terms in an unexpired patent are given their broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b). Under this standard, we

⁵ Patent Owner does not put forth evidence it alleges tend to show secondary considerations of non-obviousness in its Response.

interpret claim terms using “the broadest reasonable meaning of the words in their ordinary usage as they would be understood by one of ordinary skill in the art, taking into account whatever enlightenment by way of definitions or otherwise that may be afforded by the written description contained in the applicant’s specification.” *In re Morris*, 127 F.3d 1048, 1054 (Fed. Cir. 1997).

We presume that claim terms have their ordinary and customary meaning. *See Trivascular, Inc. v. Samuels*, 812 F.3d 1056, 1062 (Fed. Cir. 2016) (“Under a broadest reasonable interpretation, words of the claim must be given their plain meaning, unless such meaning is inconsistent with the specification and prosecution history.”); *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007) (“The ordinary and customary meaning is the meaning that the term would have to a person of ordinary skill in the art in question.” (internal citation and quotation marks omitted)). A patentee, however, may rebut this presumption by acting as his or her own lexicographer, providing a definition of the term in the specification with “reasonable clarity, deliberateness, and precision.” *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994). Only those terms that are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999); *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017).

C. Level of Skill Analysis

Petitioner asserts “[a] a person of ordinary skill in the art at the time of the ‘501 Patent would have been someone with “a Bachelor’s degree in electrical engineering, computer engineering, or in a related field and at least

one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST” Pet. 6 (citing, *inter alia*, Ex. 1003 ¶ 51). Patent Owner asserts through its declarant:

[A Person of ordinary skill in the art at the time of the invention would have been] working on the design of memory devices, would have at least a Bachelor of Science degree in Electrical and/or Computer Engineering, and at least five years of industry experience designing memory devices and controllers. Alternatively, one of ordinary skill in the art would have a Master of Science degree in Electrical and/or Computer Engineering, and at least three years of industry experience designing memory devices and controllers. Also alternatively, one of ordinary skill in the art would have a Doctorate degree in Electrical and/or Computer Engineering, and have at least one year of industry experience designing memory devices and controllers.

I agree with Petitioner’s expert that the memory industry requires a POSITA with industry experience.

Ex. 2010 ¶¶ 32–33.

Neither party has argued that the difference between the parties’ articulated level of skill should change the analysis as to patentability. Therefore, we adopt Petitioner’s articulation of the level of skill, above, and acknowledge that, commensurate with that articulation of the level of skill, the level of ordinary skill in the art is also reflected by the prior art of record. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978).

D. Claim Construction Analysis

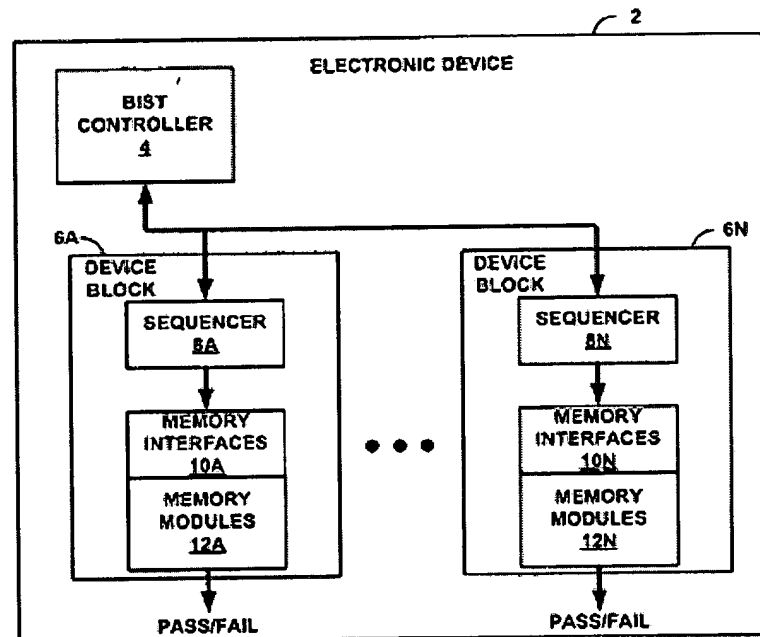
Claim 4 does not recite claim limitations whose constructions are in controversy. Patent Owner disputes only the construction for the term “memory module,” recited in claim 1. PO Resp. 16–19. The “memory controller” is not argued as a reason that Averbuj or Tsern do not meet the limitations of claim 4. *Id.* at 52–56. Thus, we do not further construe any terms including memory controller in this Final Written Decision.

E. Claims 1 and 4 — Anticipation by Averbuj (Ex. 1005) (Ground 1)

Petitioner argues that claims 1 and 4 are unpatentable over Averbuj under 35 U.S.C. § 102. Pet. 19–31. We refer to this as “Ground 1.”

1. Averbuj (Ex. 1005)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1005 ¶¶ 7–8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* ¶ 28. “[E]lectronic device 2 may be any device that incorporates memory modules, such as an embedded computing system, a computer, server, personal digital assistant (PDA), mobile computing device, mobile communication device, digital recording device, network appliance, mobile positioning device, and the like.” *Id.* ¶ 32. Averbuj states that “electronic devices . . . are [normally] constructed from many integrated circuit chips and many supporting components mounted on a circuit board.” *Id.* ¶ 5. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device blocks 6 includes a Sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to Sequencers 8 for application to device blocks 6. *Id.* ¶ 29.

Figure 5, below, illustrates an exemplary Sequencer 8A. *Id.* ¶ 22

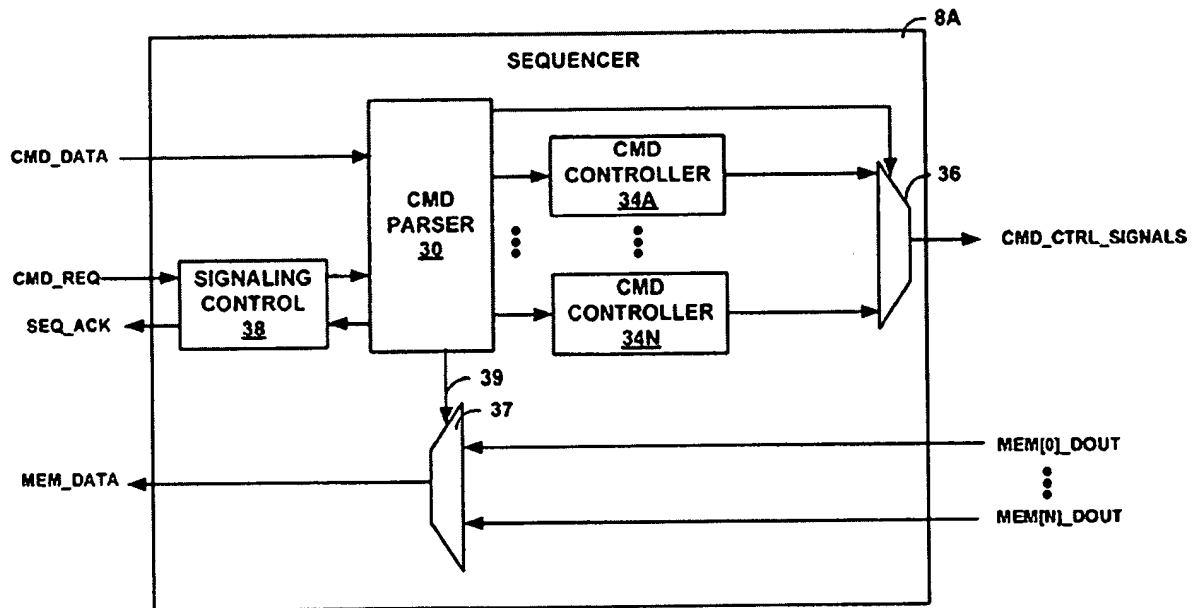


FIG. 5

As shown in Figure 5, above, exemplary Sequencer 8A includes command parser 30 and command controllers 34. *Id.* ¶¶ 41–42. Command parser 30 processes the commands from BIST controller 4 and invokes one of command controllers 34, which, in turn, issues command control signals, such as signals to provide a memory address and data, to receiving memory interfaces 10, 41. *Id.* ¶¶ 43, 51. An exemplary memory interface is shown in Figure 6, below. *Id.* ¶ 23.

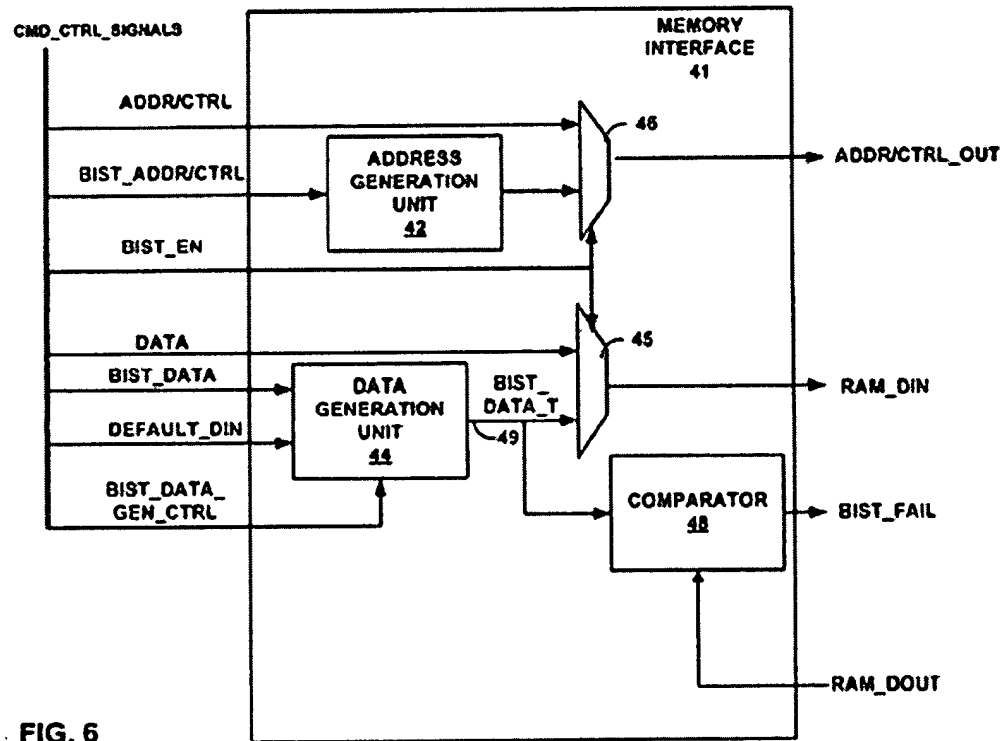


FIG. 6

As shown in the Figure 6 embodiment, above, memory interface 41 includes multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals (BIST_ADDR/CTRL) and the test data provided by a respective higher level Sequencer[, e.g., Sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level Sequencer [8A], and transforms the data and

addresses provided by the Sequencer as needed based on the particular physical characteristics of the memory module” (*id.* ¶ 54), using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “[A]ddress generation unit 42 generates the address applied to [] memory module 12 based on the addressing requirements specified by Sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* ¶ 51. “[D]ata generation unit 44 [] receives BIST_DATA signals and default data (DEFAULT_DIN) as provided by [] Sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the Sequencer and the specific physical characteristics of the corresponding memory module.” *Id.* ¶ 49. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.*

2. *Analysis*

a. *Claim 1*

Claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. *See supra* Section II. A.

b. *Claim 4*

Claim 4 recites “[t]he memory system of claim 1, wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” Petitioner asserts that

Averbuj discloses that BIST units are “commonly incorporated into memory chips and other integrated circuits.” Ex. 1005 at ¶ [0003]. Because the memory interfaces are part of a BIST unit, Averbuj also discloses the memory interface (10,

41) is commonly incorporated into (memory or other) chips, or “integrated circuits.” Ex. 1012 at 98, 277; Ex. 1003 at ¶ 126. Averbuj further discloses that the device block (6) and the components contained therein, such as the sequencer (8) and the memory modules (12), are “distributed” and “located throughout an electronic device,” i.e., physically separated. *See e.g.*, Ex. 1005 at ¶¶ [0013], [0015]; Ex. 1003 at ¶ 127. Since memory interfaces (10, 41) are also components of each device block (6), at a minimum, the different sets of memory interfaces (10, 41) belonging to different device blocks (6) must also be physically separated. Ex. 1003 at ¶ 127. Averbuj also discloses that the various integrated circuits or chips of an electronic device are “mounted on a circuit board,” indicating that the integrated circuits and chips comprising the electronic device are packaged integrated circuits or chips. Ex. 1005 at ¶ [0005]; Ex. 1016 at 1:15-27; Ex. 1003 at ¶ 128. Therefore, Averbuj discloses that the memory interfaces (each containing a “data handler”) of different device blocks (which comprise a “memory system”) must be distributed across “at least two physically separate integrated circuit packages.” Ex. 1003 at ¶ 129.

Pet. 30–31.

In other words, Petitioner asserts that because the sequencer can be distributed throughout an electronic device, the memory system comprises two separate chip packages. We disagree. Claim 4 requires that the memory module comprise “two physically separate integrated circuit packages.” Petitioner has not shown sufficient support for its broad reading that the memory system as claimed would be distributed across the device as Petitioner suggests. Thus, Petitioner has not established a preponderance of the evidence as to anticipation of claim 4 by Averbuj.

F. Claims 1 and 4 — Obviousness over Averbuj (Ex. 1005) (Ground 2)

Petitioner argues that claims 1 and 4 are unpatentable over Averbuj under 35 U.S.C. § 103. Pet. 31–38. We refer to this as “Ground 2.” As to

claim 1, Petitioner includes an obviousness ground in case we determine that Averbuj does not disclose a “memory system configured to be operatively coupled to a memory controller of a computer system” and “memory chip.” *Id.* at 31–34. As explained above, claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. As also explained above, we determine that claim 1 can also be considered obvious over Averbuj.

As to claim 4, we explained above that we are not persuaded that Petitioner has shown sufficiently that Averbuj anticipates that claim. In the Institution Decision, we found Petitioner failed to establish a reasonable likelihood of prevailing in its assertion that claim 4 would have been obvious over Averbuj. Inst. Dec. 19–22. Therefore, we discuss the obviousness challenge to dependent claim 4 below under the preponderance of the evidence standard.

Petitioner’s argument regarding obviousness of claim 4 over Averbuj alone is based on Petitioner’s declarant’s testimony that it was known to house data handlers in physically separate integrated circuit packages. Pet. 34–38 (citing Ex. 1003 ¶¶ 143–161)⁶. Petitioner argues that to do so would “have been the use of a known technique for its known purpose and would

⁶ Petitioner’s arguments regarding obviousness of claim 4 over Averbuj alone are based in part on disclosures in Tsern. Pet. 34–35 (citing Ex. 1006 ¶¶ 29, 52–53, Fig. 9A). Petitioner’s arguments based on the disclosure of Tsern rely on structures in Tsern to modify Averbuj rather than only Tsern’s general teachings to one of ordinary skill in the art and, thus, those arguments are properly considered below in Petitioner’s ground based upon the combination of Averbuj and Tsern. As such we do not consider them here.

have achieved only predictable results, such as the protection of integrated circuits by the packaging.” Pet. 35. Petitioner asserts that memory modules at the time of the invention were composed of memory chips mounted on different portions of a memory board with BIST units in each memory chip. *Id.*

Given that memory chips mounted on different portions of a memory board with BIST units in each memory chip were known, according to Petitioner, Petitioner asserts that reducing redundancy would have been a motivation to modify Averbuj. Specifically, Petitioner asserts

[a] skilled artisan that seeks to reduce the redundant BIST circuitry of conventional memory chips and BIST units would have been motivated to modify the conventional design in such a way as to avoid unnecessary modifications to the conventional design. Ex. 1003 ¶ 151. Specifically, a skilled artisan would have understood that the BIST controller and sequencer portions of conventional BIST units would need to be carved out from each memory chip and combined to reduce redundant circuitry. Ex. 1003 ¶ 151. However, a skilled artisan would also have understood and recognized that the memory interface portions need not be excised and implemented as a separate chip since doing so would not reduce much, if any, circuitry. Ex. 1003 ¶ 151. Hence, a skilled artisan would have been motivated to avoid unnecessary design changes by keeping the original design with respect to the memory interface portion—i.e., retain the memory interfaces (each of which includes a “*data handler*”) within each memory chip that was separately packaged and mounted on a printed circuit board of a memory module (“*physically separate integrated circuit packages*.”). Ex. 1003 ¶ 152.

Pet. 53–54.

Patent Owner's Preliminary Response⁷ argued that "modifying Averbuj as the Petitioner proposes would eliminate the 'built-in' aspect of Averbuj's memory BIST (built-in self-test). The proposed modification would therefore be contrary to Averbuj's principle of operation." *Id.* at 46. Patent Owner's Preliminary Response arguments assume that Averbuj would not be modified based on the single chip structure of Averbuj. In this Final Written Decision, we do not need to, and do not rely on this argument. Here, as in the Decision to Institute (Inst. Dec. 25), we focus on the motivation given by Petitioner to modify Averbuj.

Patent Owner also argued, in its Preliminary Response, that "the skilled artisan that the Petition imagines is one who has Averbuj and uses hindsight to turn Averbuj into the '501 patent. According to the Petition, this particularly skilled artisan is looking to implement Averbuj's memory BIST circuitry as a memory module." Prelim. Resp. 46.

"There must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness." *Innogenetics, N.V. v. Abbott Labs.*, 512 F.3d 1363, 1373 (Fed. Cir. 2008) (internal quotations omitted) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)). For example, in *ActiveVideo Networks, Inc. v. Verizon*, our reviewing court

⁷ Patent Owner chose not to request additional briefing to address this ground and arguments in the Preliminary Response are generally waived (Paper 8, 2–3), however, we acknowledge these arguments due to the unusual circumstance created by the Supreme Court's decision in *SAS* and to explain the background in the context of Patent Owners' Motion to Strike.

held that the following expert testimony on the motivation to combine was deficient as a matter of law:

[Y]ou wanted to build something better. You wanted a system that was more efficient, cheaper, or you wanted a system that had more features, makes it more attractive to your customers, because by combining these two things you could do something new that hadn't been able to do before.

Active Video Networks, Inc. v. Verizon Commc'ns, Inc., 694 F.3d 1312, 1328 (Fed. Cir. 2012). The court noted that because the expert “failed to explain how specific references could be combined, which combination(s) of elements in specific references would yield a predictable result, or how any specific combination would operate or read on the asserted claims,” the expert’s testimony was conclusory and bore no relation to any specific combination of prior art elements. *Id.* at 1327. Although the declarant’s testimony here goes further than in *Active Video*, it fills in gaps about how the combination would be made with only conclusory testimony that appears to be tainted with hindsight.

As to other of Petitioner’s arguments supported by its declarant, we find those to employ hindsight to piece together the device as claimed. The expert provides a host of rationales for separating the components of Averbuj into separate ICs including structures of Tsern (Ex. 1003 ¶ 143) and other secondary references. We acknowledge that Tsern contains structures that suggest the claimed configuration of the data handlers, but in this ground Petitioner does not assert Tsern as a reference, rather simply as evidence of the knowledge of one of skill in the art. Pet. 34–38. Relying on knowledge in the art does not obviate the need to prove a reference is prior art as of the critical date and show a motivation to combine existing

components in the manner claimed free of hindsight motivation. This amounts to an argument that the individual elements of the claim were known and that combining them would have yielded the claimed invention, but Petitioner provides insufficient explanation why an ordinarily skilled artisan would have combined those elements in the claimed manner without the explicit reliance on the structures of Tsern. *See KSR*, 550 U.S. at 418 (“[A] patent composed of several elements is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.”); *Metalcraft of Mayville, Inc. v. The Toro Co.*, 848 F.3d 1358, 1366 (Fed. Cir. 2017) (“it is insufficient to simply conclude the combination would have been obvious without identifying any reason why a person of skill in the art would have made the combination”) (citation omitted).

For the above reasons, we find the evidence and argument presented on the ground based on Averbuj alone is insufficient to show by a preponderance of the evidence that claim 4 would have been obvious.

Petitioner has not shown sufficiently that Averbuj teaches or suggests the physically separate ICs limitation for the reasons discussed above. Thus, Petitioner has failed to establish, by a preponderance of the evidence, its assertion that claim 4 would have been obvious over Averbuj.

G. Claim 4 — Obviousness over Averbuj (Ex. 1005) and Tsern (Ex. 1006) (Ground 3)

Petitioner argues that claim 4 is unpatentable over Averbuj and Tsern under 35 U.S.C. § 103. Pet. 38–42. We refer to this as “Ground 3.” Claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. Upon consideration of the full record of evidence and arguments, we determine that Petitioner has shown by a preponderance of the evidence that claim 4 is unpatentable under 35 U.S.C. § 103(a) as obvious over Averbuj and Tsern.

1. *Tsern*

Figure 1 of Tsern is reproduced below.

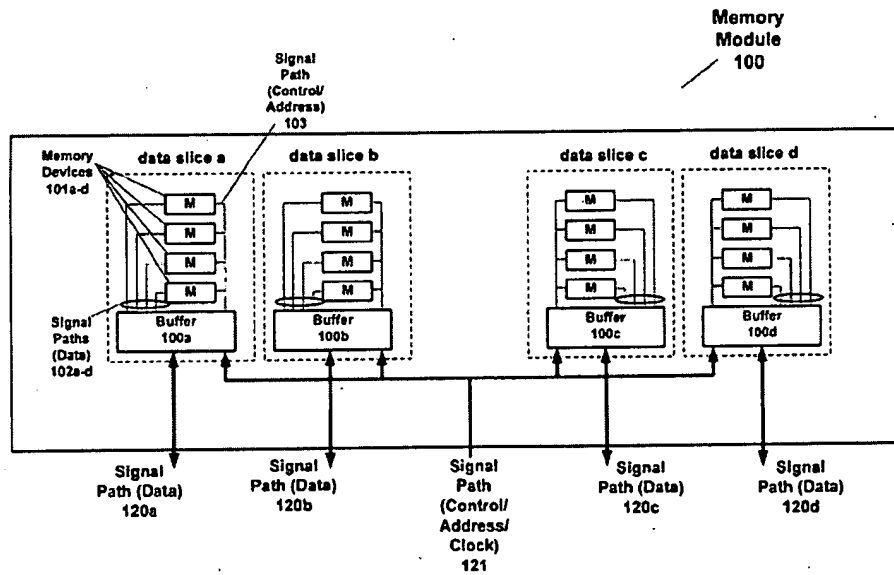


Fig. 1

Figure 1 of Tsern, above, illustrates an embodiment of a memory module with memory devices (101a-d) and corresponding buffer devices (100a-d). Figure 18 of Tsern is reproduced below.

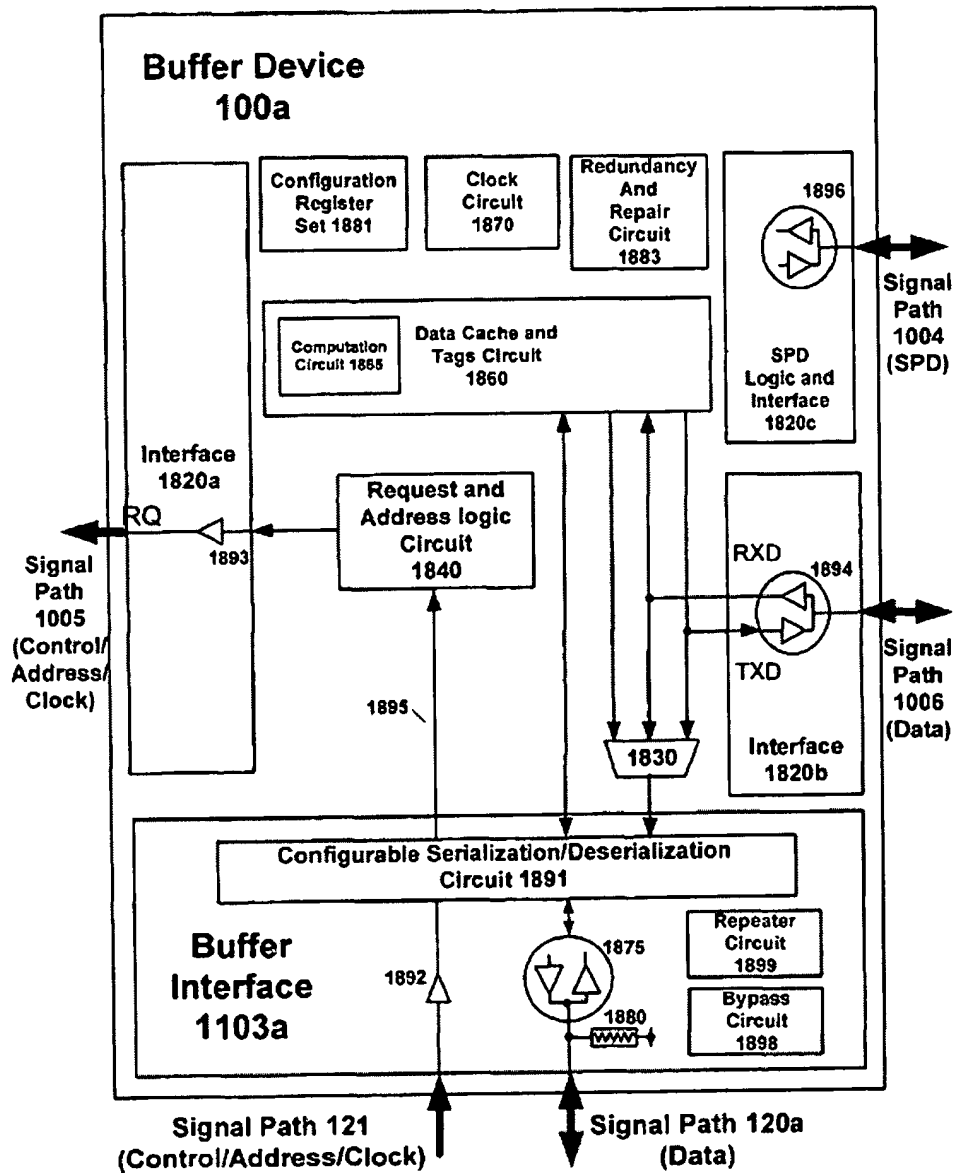


Fig. 18

As shown in Figure 18, above, Tsern discloses that, in one embodiment, each buffer device (100a) includes a redundancy and repair unit (1883), which tests and repairs the corresponding memory device(s). Ex. 1006, Fig. 18, ¶ 77. Each such buffer device (100a) includes data, address, and control interfaces (1820a and 1820b) that can be programmed

or configured to support memory modules with different number, size, width, and type of memory devices. *Id.* ¶ 99, Fig. 18.

Figure 1 of Tsern shows that each buffer device (100a–d) communicates address, control, and data signals with its corresponding memory device(s) (101a–d) without being in communication with any of the other buffer devices (100a–d) or other data ports of the other memory devices. Ex. 1006, Fig. 1.

2. *Analysis*

As discussed above, claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel, and we determined that claim 1 is also rendered obvious by Averbuj.

As to claim 4, we did not find that those claims are anticipated by, or would have been obvious over, Averbuj. Thus, below we discuss the combination of Averbuj and Tsern as it relates to claim 4.

Claim 4 recites that “[t]he memory system of claim 1, wherein the memory system comprises at least two physically separate integrated circuit packages, wherein each of the at least two physically separate integrated circuit packages comprises at least one data handler of the plurality of data handlers.” Petitioner relies on Figure 9 of Tsern, reproduced below.

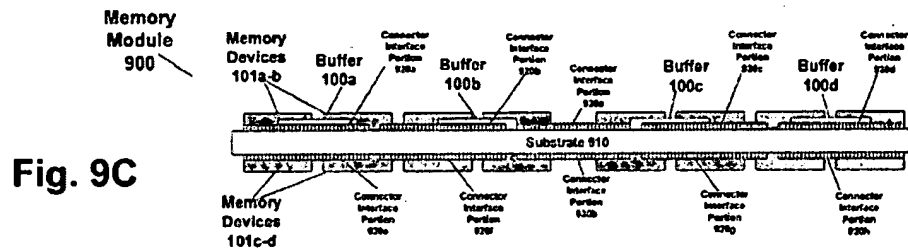
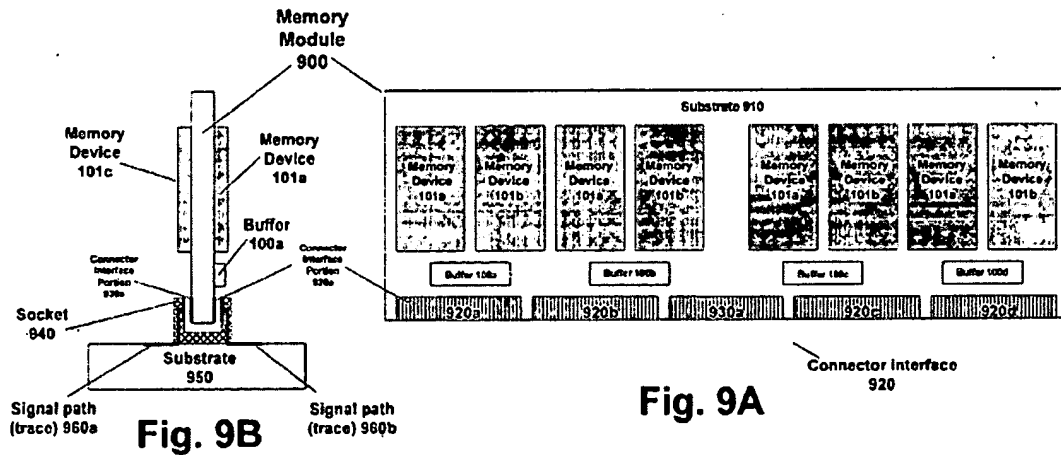


Figure 9 of Tsern, above, shows an implementation of a memory module in which each of the four buffer devices (100a–d) is implemented as a physically separate integrated circuit package and mounted on different portions of the printed circuit board that houses the memory module. Ex. 1006, Figs. 9A–C.

Based on Petitioner’s arguments on specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern’s teachings to include physically separate integrated chip packages. Pet. 40–42. For example, Petitioner asserts:

It would have been obvious to include the “separate integrated circuit packages” configuration of Tsern, including the distribution of self-test circuitry, in the system of Averbuj for several reasons. First, to do so would have been merely the arrangement of old elements with each performing the same function it had been known to perform and yielding no more than one would expect from such an arrangement. Tsern demonstrates that, before the priority date of the ’501 Patent, it was within the average skill of the art to include a plurality of data handler circuits in physically separate integrated circuit packages, on a printed circuit board and that such circuits would operate as expected. Ex. 1003 at ¶ 167.

A skilled artisan would have been further motivated to implement the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj, particularly for the memory interfaces of Averbuj, in order to reduce the busing area a centralized approach would require. See, e.g., Ex. 1017 at 3:45-53; Ex. 1003 at ¶ 168.

A skilled artisan also would also have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj so that the self-test circuitry could be conveniently placed in the same packaging as buffer circuitry used to access and isolate different portions of the memory array. By the priority date of the ’501 Patent it was known that separate buffers, in separate packages, for different portions of the memory array could advantageously reduce the load experienced by the memory controllers and improve the memory timing. Ex. 1018 at 1:5-23, Fig. 3; Ex. 1019 at 1:32-2:3, Fig. 5. A skilled artisan would therefore have been motivated to place the sequencers and the memory interfaces of Averbuj within such buffer components in order to reduce the load on the memory controller and also improve memory timing. Ex. 1003 at ¶ 169.

A skilled artisan also would have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj so that failing sequencers and memory interface components could be easily replaced, by

simply replacing a standard package on the printed circuit board, without the necessity to replace other self-test circuitry in other packages that was operating normally. Ex. 1003 at ¶ 170.

A skilled artisan also would have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj in order to place each data handler at positions on the PCB close and perhaps closest to its associated memory. Doing so would (i) simplify the wiring plan on the circuit board (Ex. 1021 at 6); (ii) reduce EMI interference and improve signal integrity (Ex. 1010 at 23-24, 207); and (iii) limit the capacitive load on those lines and permit faster and more efficient data transmission (Ex. 1020 at 2:27–3:6). See above at § V.C.1; Ex. 1003 at ¶ 171.

Id.

We also find that these findings regarding claim 4 are consistent with the findings of the '971 IPR that Averbuj meets the limitations of claim 1 such that those limitations, as they apply to claim 4, are met by collateral estoppel in this *inter partes* review.

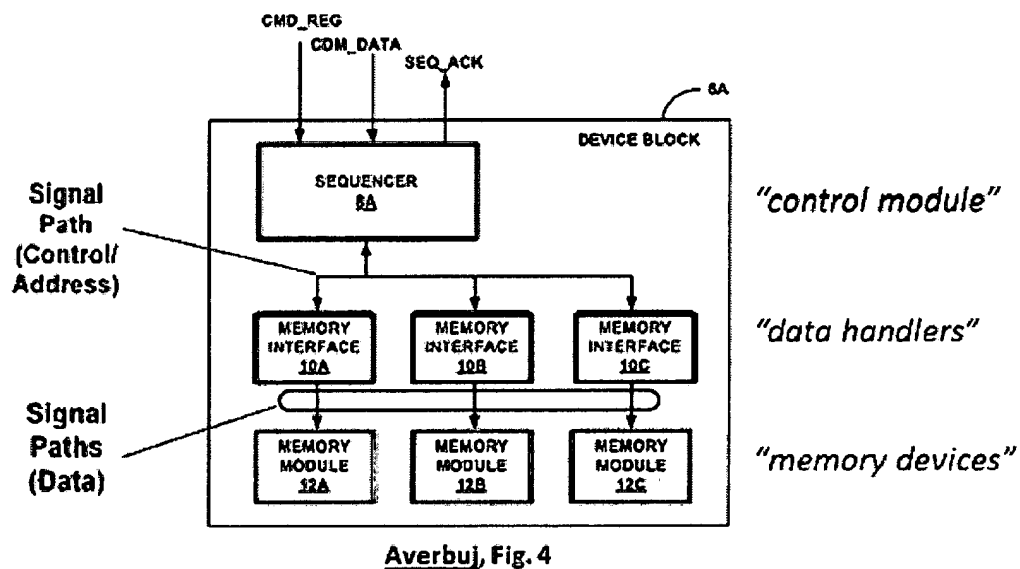
a. Patent Owner Response –Mapping of claim 1

Patent Owner’s Response to the Petition was directed to an alternative mapping in which the address generation unit is mapped to the “control circuit.” See generally PO Resp. Thus, the arguments in the Patent Owner Response are not relevant to any issue we seek to resolve in this Final Written Decision. This is because Patent Owner contends that Petitioner did not assert in the Petition that the limitations of claim 4 were met under the same mapping of Averbuj to claim 1 as was used in the '971 IPR. Mot. To Strike, 2–5.

To summarize the argument, the Petition presents three mappings that are directed to which component of Averbuj is mapped to the “control circuit” recited in claim 1. “Mapping 1,” which is the same mapping used in

the '971 IPR, maps Averbuj's Sequencer to the "control circuit." Pet. 26–27. "Mapping 2," which was not at issue in the '971 IPR, maps the address generation unit to the "control circuit." Pet. 27–28. "Mapping 3," which was also not used in the '971 IPR, maps both the Sequencer and the address generation unit to the "control circuit." Pet. 28–29.

To illustrate Patent Owner's argument, we include below several annotated figures provided by Patent Owner. Below is Patent Owner's annotated version of Figure 4 of Averbuj alleged to represent Mapping 1. PO Show Cause Resp. 3.

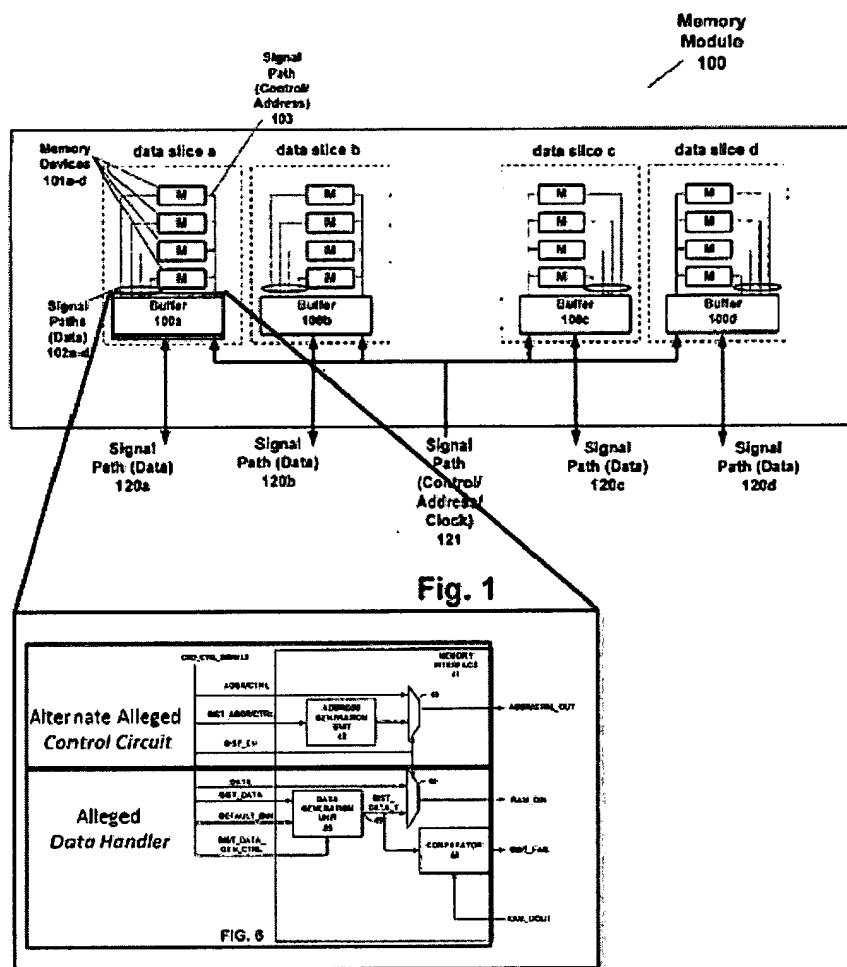


MAPPING 1

The annotated Figure 4, above, shows Mapping 1 of claim 1 limitations to Averbuj, the mapping used in the '971 IPR, in which the Sequencer alone is

the “control circuit.”⁸ PO Show Cause Resp. 3; Pet. 26–27 (Petitioner’s contention that the sequencer alone meet the “control circuit” limitation).

Below are Patent Owner's edited and annotated versions of Figures 1 and 6 of Averbuj. PO Resp. 55.



MAPPING 2

⁸ Unlike other Patent Owner annotated figures used in this Final Written Decision, this annotated figure does not show how each element would be placed on a separate IC as required by claim 4, however, in that case each element in the colored box would represent a different IC.

The annotated Figures 1 and 6, above, represents Mapping 2 in which the Address Generation Unit of Averbuj is mapped to the “control circuit.”

Patent Owner does not include an annotated figure for Mapping 3, but it would be similar to Mapping 2, with the Sequencer included in the red box containing the Address Generation Unit. *See* Pet. 27–28.

As noted above, in Ground 3 discussing the combination of Averbuj and Tsern, the Petition states “A skilled artisan would have been further motivated to include the ‘separate components’ configuration of Tsern in the system of Averbuj, *particularly for the Memory Interfaces of Averbuj*, in order to reduce the busing area a centralized approach would require.” Pet. 28–29 (emphasis added) (citations omitted). This statement does not mention where the “control circuit” is located or what component is mapped to the “control circuit.” Petitioner asserts that claim 4 is directed to separating the “data handlers” from each other and is, at best, agnostic as to whether the “control circuit” is in the same IC as the data handlers. Tr. 12:14–19. We find these facts are sufficient to find that Petitioner did not limit its contentions regarding the combination of Averbuj and Tsern to Mapping 2 or Mapping 3.

Nevertheless, Patent Owner presents a further argument that claim 4 requires a one-to-many relationship between the “control circuit” and the “data handlers” such that Petitioner’s theory under Ground 3 cannot meet the limitations of claim 1. We discuss that below.

b. Patent Owner Motion to Strike–One-to-Many

Patent Owner, in the Motion to Strike, asserts that because claim 1 recites that “wherein the memory system is configured to test the one or more memory chips using the address and control signals generated by the

control circuit and using the data generated by the one or more data handlers” there must be a one-to-many relationship between the “control circuit” and the “data handlers.” Mot. to Strike 1–2. We disagree. The claim explicitly recites using signals from a “control circuit” and from a plurality of data handlers, but no explicit required connection between the plurality of “data handlers” and the “control circuit” is recited such that a one-to-many relationship is required. We also do not find any one-to-many relationship limitation was imported into the claims from the Specification by the Patentee. For that reason, as explained above, Petitioner has set forth a sufficient contention that the combination of Averbuj and Tsern would render obvious putting the “data handlers” in physically separate components.

Nevertheless, we will analyze this issue assuming that a one-to-many relationship is required such that having one “control circuit” in a chip with only one “data handler” would not meet the limitations of claim 1, which are incorporated into claim 4. This exercise may be slightly difficult to follow because it appears that Petitioner did not anticipate this one-to-many argument and simply focused on showing that the “data handlers” each could be put in separate components regardless of the mapping of the “control circuit.” An additional difficulty is that this issue is separate and distinct from the mappings issue presented above.

i. Alleged Separate Component Theories

We begin by explaining Petitioner’s theories regarding where the “control circuit” and “data handlers” are mapped in the grounds presented in the Petition. In the Petition, as to anticipation by Averbuj, Petitioner argued that

Averbuj further discloses that the device block (6) and the components contained therein, such as the sequencer (8) and the memory modules (12), are ‘distributed’ and ‘located throughout an electronic device,’ i.e., physically separated. *See e.g.*, Ex. 1005 ¶¶ 13, 15; Ex. 1003 ¶ 127. Because memory interfaces (10, 41) are also components of each device block (6), at a minimum, the different sets of memory interfaces (10, 41) belonging to different device blocks (6) must also be physically separated. Ex. 1003 at ¶ 127

Pet. 30. Petitioner further argued that “Averbuj also discloses that the various integrated circuits or chips of an electronic device are ‘mounted on a circuit board,’ indicating that the integrated circuits and chips comprising the electronic device are packaged integrated circuits or chips.” *Id.* Thus, Petitioner argued the memory interfaces 10A, 10B and 10C were physically separate. We disagreed that this contention, as presented in the Petition, met the reasonable likelihood standard for institution, stating “[c]laim 4 requires that the memory module be [] ‘two physically separate integrated circuit packages.’ Petitioner has not shown sufficient support for its broad reading that the memory system as claimed would be distributed across the device as Petitioner suggests.” Inst. Dec. 19. Thus, we found that Averbuj did not show sufficiently that memory interfaces 10A, 10B, and 10C were physically separate ICs under Ground 1 as to anticipation by Averbuj.

As to Ground 2, Petitioner argues that separating components of the BIST architecture into physically separate ICs would have been “the use of a known technique for its known purpose and would have achieved only predictable results, such as the protection of integrated circuits by the packaging.” Pet. 35. Petitioner asserts that memory modules at the time of the invention were composed of memory chips mounted on different portions of a memory board with BIST units in each memory chip. *Id.* at

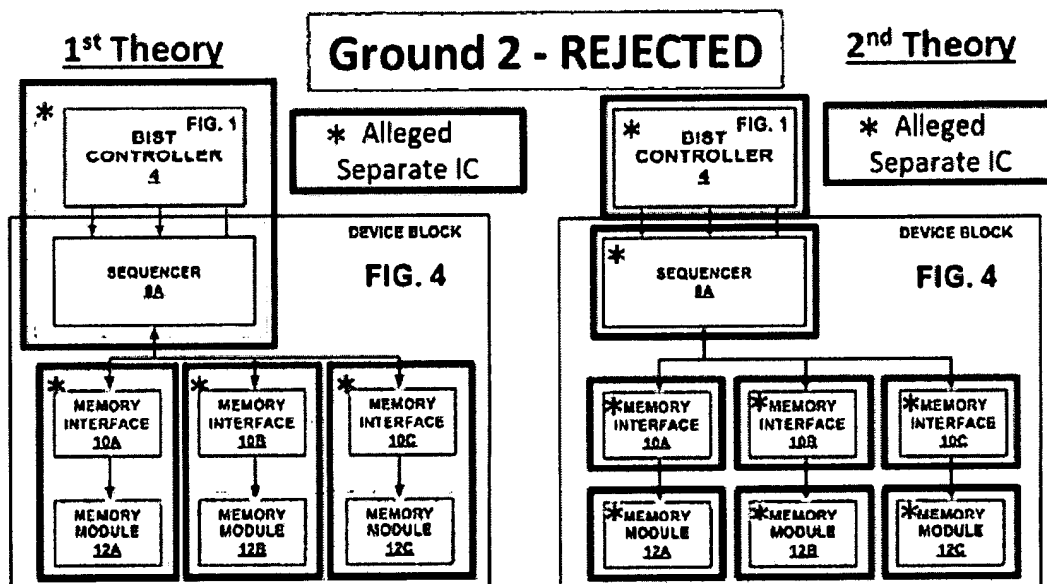
35–37. Given its assertion that memory chips mounted on different portions of a memory board with BIST units in each memory chip were known, Petitioner asserts that reducing redundancy is a motivation to modify Averbuj. *Id.*

Specifically, Petitioner presents a first alternative asserting a skilled artisan would have understood that the BIST controller and sequencer portions of conventional BIST units would need to be removed from each memory chip and combined in order to reduce redundant circuitry . . . [but] a skilled artisan would also have understood and recognized that the memory interface portions need not be excised and implemented as a separate chip since doing so would not reduce much, if any, circuitry [thus remaining in a chip with the memory devices].

Pet. 36.

Later, also as to Ground 2, Petitioner presents a second alternative asserting “a skilled artisan, particularly a skilled artisan looking to implement Averbuj’s BIST circuitry in a memory module that does not have a BIST circuitry, would have been motivated to incorporate each component of Averbuj’s BIST circuitry—i.e., BIST controller, sequencers, and memory interfaces—into separate integrated circuit chips/packages” Pet. 37.

Reproduced below are Patent Owner’s annotated versions of Figure 4 of Averbuj representing these two alternate theories as to Ground 2 which is obviousness based on Averbuj alone.

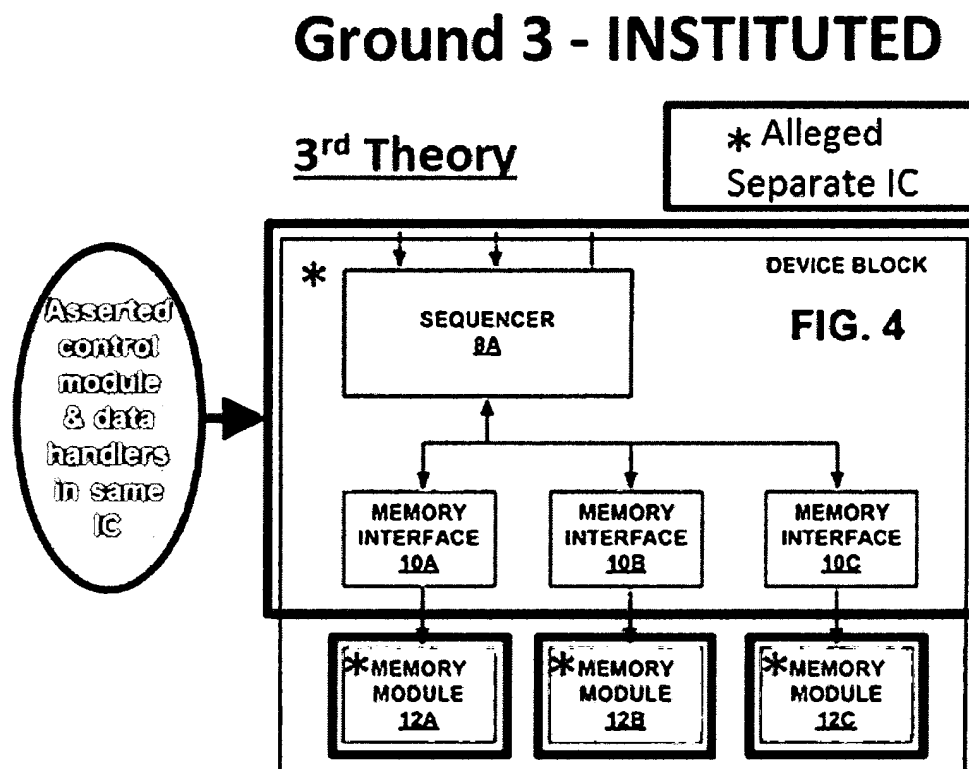


Patent Owner's annotated versions of Figure 4, above, accurately shows two alternate theories which are associated with Ground 2 based on obviousness over Averbuj alone. Mot. to Strike 3. These theories do not specify which component is the "control circuit." Pet. 35–37. In our Decision to Institute, we rejected these theories as relying on hindsight stating:

As to Petitioner's arguments supported by its declarant, we agree with Patent Owner and find those to employ hindsight to piece together the device as claimed. Prelim. Resp. 46. The only evidence in the record of separating the data handlers and BIST controllers as claimed presented by Petitioner, other than the long complicated explanation of its declarant or the misplaced reliance on the structures of Tsern under this challenge, is in the '501 Patent itself.

Inst. Dec. 21. Thus, we found only that there was insufficient motivation shown, absent a combination with Tsern, to break up Averbuj's single chip architecture into an architecture in which the "data handlers" are in physically separate ICs.

Patent Owner represents what it considers Petitioner's Ground 3 theory in the annotated Figure 4 of Averbuj below. Mot. To Strike, 3.



As illustrated in the annotated Figure 4 of Averbuj, above, Patent Owner contends Petitioner is limited to a theory in which the Sequencer (the alleged “control circuit”) and the memory interfaces (the alleged data handlers) are in the same IC. Mot. To Strike, 3.

- ii. *Patent Owner Asserts Petitioner's Ground 3 Theory does not include Sequencer Alone as the “Control Circuit”*

With that background, Patent Owner's argument focuses on Ground 3 and the following statement in the Petition:

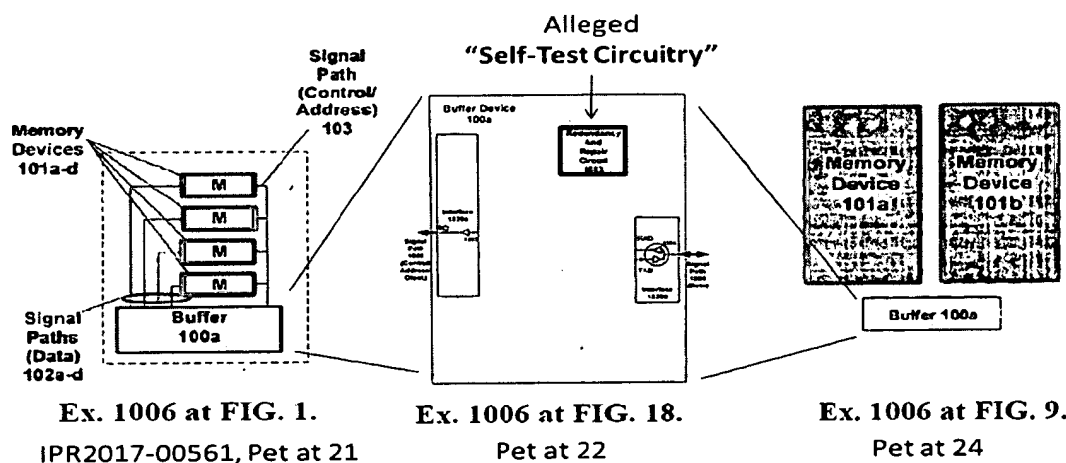
A skilled artisan also would also have been motivated to include the “separate integrated circuit packages” configuration of Tsern in the system of Averbuj so that *the self-test circuitry could be*

conveniently placed in the same packaging as buffer circuitry used to access and isolate different portions of the memory array. By the priority date of the '501 Patent it was known that separate buffers, in separate packages, for different portions of the memory array could advantageously reduce the load experienced by the memory controllers and improve the memory timing. Ex. 1018 at 1:5-23, Fig. 3; Ex. 1019 at 1:32-2:3, Fig. 5. A skilled artisan would therefore have been motivated to *place the sequencers and the memory interfaces of Averbuj within such buffer components* in order to reduce the load on the memory controller and also improve memory timing. Ex. 1003 ¶ 169.

Pet. 40–41 (emphasis added). Based on this statement, Patent Owner asserts Petitioner's theory under Ground 3 regarding the combination of Averbuj and Tsern is limited to a configuration in which the Sequencer and the memory interfaces are in the same chip such that the plurality of "data handlers" (memory interfaces) are not physically separate from the "control circuit." Mot. To Strike, 3–4. We note this paragraph does not mention the "control circuit" at all, but we will address that later. According to Patent Owner, this configuration would violate the so called "one-to-many" relationship required by claim 1. *Id.*

Petitioner responds that the statement that "the self-test circuitry could be conveniently placed in the same packaging as buffer circuitry" in the Petition was meant to express "that the components can be placed in the same type of packaging, not a single package." Mot. To Strike Resp. 5. We disagree. A reasonable reading of this statement is that a single buffer circuitry package would include "the self-test circuitry." Petitioner also responds that the statement "the Sequencers and the Memory Interfaces of Averbuj within such buffer components" in the Petition means "that each of the Sequencers (plural) and each of the Memory Interfaces can be placed in

a “buffer component” (i.e., “separate component”), not all in one.” *Id.* We disagree. A reasonable reading of this sentence is that the Sequencers and Memory Interfaces would both be in the buffer component of Tsern. This is supported by Patent Owner’s annotated figures, reproduced below, illustrating the Petition’s summary of Tsern.



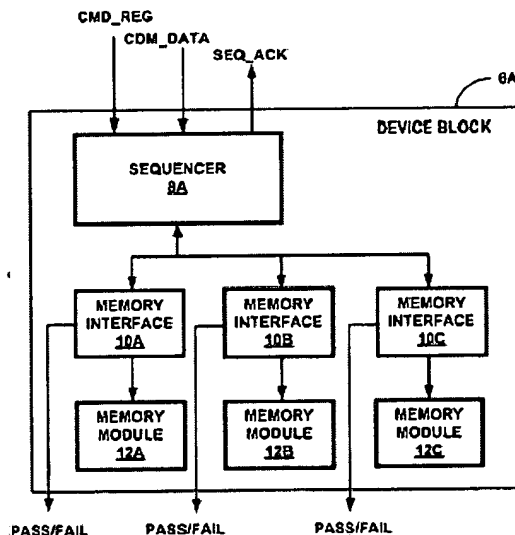
PO Show Cause Resp. 4. The annotated figures illustrate Patent Owner’s assertion that the buffer that Petitioner refers to as the “separate component” contains the “self-test circuitry” which contains the alleged “control circuit” (Sequencer) and the memory interface which is the alleged data handler. *Id.*

Nevertheless, Petitioner also responds that it had presented an argument that the “data handler” alone would be placed in separate chips. Resp. Mot. To Strike. 1–4. Petitioner ties a citation to the following statement from the Ground 2 Section of the Petition regarding obviousness over Averbuj alone to its argument regarding Averbuj in combination with Tsern. Reply 15 (citing Pet. at 34-35):

It was known in the art to include circuit components of a memory system, including those components responsible for

handling the transfer of data to and from a memory device, in physically separate integrated circuit packages, and to do so would have been well within the average skill in the art. See, e.g., Ex. 1006 at Fig. 9A; ¶¶ [0052-0053] (explaining that the depicted separate “buffer devices” are in one embodiment “housed in separate packages”); ¶ [0029] (noting that as used therein “an integrated circuit buffer device is also referred to as a buffer or buffer device.”); Ex. 1003 at ¶ 143.

Pet. 34 (emphasis added). Petitioner’s declarant states in the cited portion of his testimony that “the idea to implement the components of a memory module, such as the components responsible for handling the transfer of data to and from a memory device, in physically separate integrated circuit packages had already been proposed and adopted by the priority date of the ‘501 Patent.” Ex. 1003 ¶ 143 (citing Ex. 1006, Fig. 9A, ¶¶ 52–53, 29). Reproduced below is an annotated version of Figure 4 of Averbuj from Petitioner’s declarant’s testimony. *Id.* ¶ 153.



Annotated Figure 4 above shows each memory interface in a separate IC. Ex. 1003 ¶ 153. This testimony is cited at page 37 of the Petition. Pet.

37. Thus, we find that Petitioner presented a contention that Averbuj, as modified in light of the structures of Tsern, teaches a Sequencer in a separate component from its associated memory interfaces. We acknowledge that petitioners routinely rely on earlier anticipation or single reference obviousness grounds to support and build up later combination grounds. *See Silicon Labs., Inc. v. Cresta Tech. Corp.*, Case IPR2015-00626, slip op. at 38 (PTAB Aug. 11, 2016) (Paper 65) (“Patent Owner characterizes the ground of unpatentability as if Petitioner relies only on Grumman for a ‘signal processor’ as claimed. Such is not the case. The ground of unpatentability against claim 24 must be understood in the context that it builds upon the ground against claim 1.”).

Given that context, the Petition states later in the Ground 3 Section dealing with the combination of Averbuj and Tsern that

Tsern further discloses that his buffer devices may be separate integrated circuit devices that can be housed in separate packages. Ex. 1006 at ¶ [0031] (“Likewise in an embodiment, an integrated circuit buffer device is distinguished from a buffer die in that a buffer die is a monolithic integrated circuit formed from semiconductor materials and performs at least one or more buffer functions described herein, whereas an integrated circuit buffer device is a buffer die having at least some form of packaging or interface that allows communication with the buffer die.”) (emphasis added.); *see also id.* at ¶ [0053] (“In an embodiment, each memory device and buffer device are housed in separate packages.”); ¶ [0043] (“packages used to house buffer devices 100a and 100d”); Ex. 1003 at ¶ 165.

Pet. 39. In the quoted portion above, the Petition cites the same section of Tsern, i.e. paragraph 53, as cited in support of Petitioner’s contention that Averbuj alone rendered claim 2 obvious. The Petition also recites “[a] skilled artisan would have been further motivated to implement the ‘separate

integrated circuit packages' configuration of Tsern in the system of Averbuj, particularly for the memory interfaces of Averbuj, in order to reduce the busing area a centralized approach would require.." *Id.* at 40 (citing Ex. 1017 at 3:45–53; Ex. 1003 ¶ 168.) Thus, as Petitioner points out in its response to the Motion to Strike, there are several statements in the Ground 3 portion of the Petition that can apply to Mapping 1, which was asserted in the '971 IPR, and which places the Sequencer itself (as the alleged "control circuit") in a separate IC from the "data handlers" thus satisfying the alleged one-to-many requirement of the claims. Mot. To Strike Resp., 2–5.

Patent Owner, in its Motion to Strike, argues that the contentions under this ground are either not sufficiently tied to Petitioner's theory under Ground 3 or that they were specifically rejected as hindsight in the Institution Decision. Mot. to Strike. 2–5. We note Petitioner cannot expect the Board to read through the entire record, searching for relevant evidence that might support Petitioner's arguments. *Cf., DeSilva v. DiLeonardi*, 181 F.3d 865, 866–67 (7th Cir. 1999)("[B]rief must make all arguments accessible to the judges, rather than ask them to play archaeologist with the record."). Nevertheless, the reference to Tsern and its structures under this ground is plainly relevant to Ground 3 based on Averbuj and Tsern. We decline to limit Petitioner's Ground 3 based on obviousness over Averbuj and Tsern to only the arguments made specifically in the section of the Petition labeled Ground 3.

We find that, based on all the parts of the Petition cited above, the Petition—without any *ex post facto* reconstruction by the Board—can be reasonably read to present two theories based on Averbuj and Tsern—one in which each portion of the device is in a separate component (or IC) and

possibly one in which the Sequencer and memory interfaces are together in the same component. *Cf. In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1380–81 (Fed. Cir. 2016) (rejecting an argument that the Board properly “ma[de] an obviousness argument on behalf of [petitioner]” that “could have been included in a properly drafted petition,” because “petitioner ... bears the burden of proof”). As noted above, we rely on collateral estoppel to show the limitations of claim 1. Thus, for the reasons above, Petitioner has shown sufficiently in the Petition its theory that each portion of the device, including the Sequencer and each memory interface, would be in separate components mounted on a printed circuit board, as recited by claim 4 even under Patent Owner’s “one-to-many” construction of claims 1 and 4. Finally, as explained above, Petitioner has shown sufficient motivation to combine Averbuj and Tsern.

For the reasons above, we are persuaded by and adopt Petitioner’s contentions as to the limitations of claim 4. Accordingly, upon review of the full record before us, we determine that Petitioner has demonstrated by a preponderance of the evidence that claim 4 is unpatentable over the combination of Averbuj and Tsern.

H. Patent Owner’s Motion to Strike

Patent Owner moves to strike pages 13–16 of Petitioners’ Reply and to exclude any argument based thereon, as violating Patent Owner’s due process and procedural rights. Mot. to Strike 1. Pages 13–16 of Petitioner’s Reply are directed to Ground 3 of the Petition which challenges claims 1 and 4 as rendered obvious by Averbuj and Tsern. In those pages, Petitioner argues that Patent Owner ignored Petitioner’s alternative theory that the Sequencer alone is the “control circuit.” Reply 13–16. As explained above

(Sections II. G. 2. b.), we do not find that Petitioner's Reply introduces a new theory. *See* Mot. to Strike Resp. 1–5. Thus, Patent Owner's Motion to Strike is denied.

I. Petitioner's Motion to Exclude

Petitioner moves to exclude Patent Owner's Exhibits 2005, 2006, 2012. Pet. Mot. to Excl. 1–2. Exhibits 2005 and 2012 relate to whether the BIST unit is in a single embedded chip and whether Averbuj's Sequencer is the "control circuit." This issue is moot because the limitations of claim 1 have been shown to be met by Averbuj by collateral estoppel. Thus, we deny the Petitioner's Motion to Exclude as to Exhibits 2005 and 2012.

Patent Owner does not oppose the motion to exclude Exhibit 2006. Pet. Mot. to Excl. Resp. 6. Thus, Petitioner's Motion to Exclude Exhibit 2006 is granted.

J. Patent Owner's Motion to Exclude

Patent Owner summarily moves to exclude paragraphs 1–45, 55–63, 71, 73, 80–82, 93, 99, 105, 114, 121, 125, 131, 138, 142, 157, 162 and 166 of Exhibit 1003, along with the entirety of Exhibit 1022, because those paragraphs and that exhibit were submitted with, but were not cited in, the Petition. PO Mot. to Excl. 1.

Petitioner explains why each of the paragraph/exhibits are referred to indirectly; why they are relevant to a foreseeable issue even though they are not explicitly cited; and why Petitioner's reliance on these exhibits is not unfairly prejudicial to Patent Owner. PO Mot. to Excl. Reply 1–3. Patent Owner does not respond to those specific arguments except as to Exhibit 1022. PO Mot. to Excl. Resp. 2–3.

The paragraphs cited in Exhibit 1003 are general background of the expert and how he conducted his analysis and summaries of analysis that is cited in the Petition. Additionally, as above in relation to Petitioner's Motion to Exclude, many of the relevant paragraphs are moot due to the collateral estoppel issue.

The Board acts as both the gatekeeper of evidence and as the weigher of evidence. Rather than excluding evidence that is allegedly confusing, misleading, untimely, and/or irrelevant, we will simply not rely on it or give it little weight, as appropriate, in our analysis. Similar to a district court in a bench trial, the Board, sitting as a non-jury tribunal with administrative expertise, is well positioned to determine and assign appropriate weight to evidence presented, including giving it no weight. *See, e.g., Donnelly Garment Co. v. NLRB*, 123 F.2d 215, 224 (8th Cir. 1941) ("One who is capable of ruling accurately upon the admissibility of evidence is equally capable of sifting it accurately after it has been received"). Thus, in this *inter partes* review, the better course is to have a complete record of the evidence to facilitate public access as well as appellate review. Thus, as to the relevant paragraphs of Exhibit 1003, Patent Owner's Motion to Exclude is denied.

As to Exhibit 1022, this is a reference relied on by Dr. Mazumder to show the state of the art. PO Mot. to Excl. Reply 3. Patent Owner argues that this is an improper incorporation by reference. PO Mot. to Excl. Reply. 2–3 (citing 37 C.F.R. § 42.6(a)(3) (prohibiting incorporation by reference); *Oxford Nanopore Tech's Inc. v. University of Washington*, Case IPR2014-00512, slip op. (PTAB Sept. 14, 2014) (Paper 12); *Biodelivery Sciences Intern., Inc. v. Monosol RX, LLC*, Case IPR2015-00167, slip op.

(PTAB.May 20, 2015) (Paper 6); *Conopco, Inc. v. Procter & Gamble Co.*, Case IPR2013-00510, slip op. at 8 (PTAB Feb. 12, 2014) (Paper 9) (“We decline to consider information presented in a supporting declaration, but not discussed in a petition, because, among other reasons, doing so would encourage the use of declarations to circumvent the page limits that apply to petitions.”)). 37 C.F.R. § 42.6(a)(3) prohibits incorporating arguments into the Petition by reference to other documents, and 37 C.F.R. § 42.104(b)(5) gives us discretion to “exclude or give no weight to . . . evidence” where the Petition “fail[s] to state its relevance.” *See* 37 C.F.R. § 42.6(a)(3) (“Arguments must not be incorporated by reference from one document into another document.”); 37 C.F.R. § 42.104(b)(5) (“The Board may exclude or give no weight to the evidence where a party has failed to state its relevance or to identify specific portions of the evidence that support the challenge.”). Nevertheless, we are encouraged to consider all evidence of record that is probative of the background knowledge of a person of ordinary skill in the relevant technology. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362–63 (Fed. Cir. 2013). There is a balance between making sure the Petition is complete, as a stand-alone document, in stating everything necessary to meet the elements of establishing a preponderance of evidence of unpatentability, with the need for citation to, and reliance on, evidence to support each argument. Here, the balance tips in favor of Petitioner. The testimony regarding Exhibit 1022 is relevant to why Averbuj would be modified to the configuration in claim 4 in which the components are on separate ICs. This argument is fully explored in the Petition despite the lack of specific citation to Exhibit 1022. *See* Pet. 34–38 (citing, *inter alia*, Ex. 1003 ¶¶ 148–149 (citing Exhibit 1022)). We do not find that Dr. Mazumder’s reference to this

Exhibit introduces argument that was not contained in the Petition or introduces material that was either excessive or material that the relevance of which is not explained. Thus, we deny Patent Owner's Motion to Exclude as to Exhibit 1022.

III. CONCLUSION

For the foregoing reasons, we determine claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. Thus, pursuant to 37 C.F.R. § 42.72, and the doctrine of collateral estoppel, this trial is terminated as to claim 1. Additionally, Petitioner has demonstrated, by a preponderance of the evidence, that claim 4 of the '501 Patent is unpatentable as obvious in view of Averbuj and Tsern.

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claim 4 of the '501 Patent has been shown to be unpatentable;

FURTHER ORDERED, pursuant to 37 C.F.R. § 42.72 and the doctrine of collateral estoppel, that trial is terminated as to claim 1; and

FURTHER ORDERED Patent Owner's Motion to Strike is *denied*; Patent Owner's Motion to Exclude is *denied*; Petitioner's Motion to Exclude is *granted* as to Exhibit 2006 and *denied* as to Exhibit 2005 and 2012.

This is a final decision. Parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2017-00562
Patent 8,359,501 B1

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(12) **INTER PARTES REVIEW CERTIFICATE** (1075th)

United States Patent
Lee et al.

(10) **Number:** **US 8,359,501 K1**
(45) **Certificate Issued:** **Sep. 28, 2018**

(54) **MEMORY BOARD WITH SELF-TESTING
CAPABILITY**

(76) **Inventors: Hyun Lee; Jayesh R. Bhakta;
Soonju Choi**

Trial Numbers:

IPR2014-00971 filed Jun. 18, 2014
IPR2014-01374 filed Aug. 23, 2014

Inter Partes Review Certificate for:

Patent No.: **8,359,501**
Issued: **Jan. 22, 2013**
Appl. No.: **13/183,253**
Filed: **Jul. 14, 2011**

The results of IPR2014-00971 and IPR2014-01374 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,359,501 K1
Trial No. IPR2014-00971
Certificate Issued Sep. 28, 2018

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AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claim **4** is found patentable.

5

Claims **1-3** and **5-20** are cancelled.

* * * * *

NOTE: This disposition is nonprecedential.

**United States Court of Appeals
for the Federal Circuit**

NETLIST, INC.,
Appellant

v.

**SK HYNIX INC., SK HYNIX AMERICA INC., SK
HYNIX MEMORY SOLUTIONS INC.,**
Appellees

2018-2357, 2018-2358

Appeals from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in Nos. IPR2017-
00561, IPR2017-00562.

JUDGMENT

THOMAS J. WIMBISCUS, McAndrews, Held & Malloy,
Ltd., Chicago, IL, argued for appellant. Also represented
by WAYNE BRADLEY, RONALD SPUHLER.

JOSEPH A. MICALLEF, Sidley Austin LLP, Washington,
DC, argued for appellees. Also represented by CARTER
GLASGOW PHILLIPS, RYAN C. MORRIS, BRIAN R. NESTER.

THIS CAUSE having been heard and considered, it is

ORDERED and ADJUDGED:

PER CURIAM (PROST, *Chief Judge*, NEWMAN and
MOORE, *Circuit Judges*).

AFFIRMED. See Fed. Cir. R. 36.

ENTERED BY ORDER OF THE COURT

December 12, 2019
Date

/s/ Peter R. Marksteiner
Peter R. Marksteiner
Clerk of Court

**United States Court of Appeals
for the Federal Circuit**

NETLIST, INC.,
Appellant

v.

**SK HYNIX INC., SK HYNIX AMERICA INC., SK
HYNIX MEMORY SOLUTIONS INC.,**
Appellees

2018-2357, 2018-2358

Appeals from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in Nos. IPR2017-
00561, IPR2017-00562.

MANDATE

In accordance with the judgment of this Court, entered
December 12, 2019, and pursuant to Rule 41 of the Federal
Rules of Appellate Procedure, the formal mandate is
hereby issued.

FOR THE COURT

January 21, 2020

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court

(12) **INTER PARTES REVIEW CERTIFICATE** (1952nd)

United States Patent
Lee et al.

(10) **Number:** **US 8,359,501 K2**
(45) **Certificate Issued:** **Mar. 1, 2021**

(54) **MEMORY BOARD WITH SELF-TESTING
CAPABILITY**

(76) **Inventors: Hyun Lee; Jayesh R. Bhakta;
Soonju Choi**

Trial Number:

IPR2017-00562 filed Jan. 3, 2017

Inter Partes Review Certificate for:

Patent No.: **8,359,501**
Issued: **Jan. 22, 2013**
Appl. No.: **13/183,253**
Filed: **Jul. 14, 2011**

The results of IPR2017-00562 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,359,501 K2
Trial No. IPR2017-00562
Certificate Issued Mar. 1, 2021

1

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AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claim 4 is cancelled.

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* * * * *

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.)

<u>DOCKET NO.</u> 8:16-cv-01605 JLS	<u>DATE FILED</u> 8/31/2016	<u>U.S. DISTRICT COURT</u> Central District of California, Santa Ana Division
<u>PLAINTIFF</u> NETLIST, INC.		<u>DEFENDANT</u> SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
<u>PATENT OR TRADEMARK NO.</u>	<u>DATE OF PATENT OR TRADEMARK</u>	<u>HOLDER OF PATENT OR TRADEMARK</u>
1 8,756,364	June 17, 2014	Netlist, Inc.
2 8,516,185	August 20, 2013	Netlist, Inc.
3 8,001,434	August 16, 2011	Netlist, Inc.
4 8,359,501	January 22, 2013	Netlist, Inc.
5 8,689,064	April 1, 2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

<u>DATE INCLUDED</u>	<u>INCLUDED BY</u> ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
<u>PATENT OR TRADEMARK NO.</u>	<u>DATE OF PATENT OR TRADEMARK</u>	<u>HOLDER OF PATENT OR TRADEMARK</u>	
1			
2			
3			
4			
5			

In the above—entitled case, the following decision has been rendered or judgement issued:

<u>DECISION/JUDGEMENT</u> "SEE ATTACHED COPY OF ORDER DISMISSING CASE"

<u>CLERK</u> Kiry Gray	<u>(BY) DEPUTY CLERK</u> /s/ J. Lam	<u>DATE</u> 04/05/2021
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Copy 3—Upon termination of action,

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.)

<u>DOCKET NO.</u> 8:16-cv-01605	<u>DATE FILED</u> 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,489,837	July 16, 2013	Netlist, Inc.
2		
3		
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5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

<u>DATE INCLUDED</u>	<u>INCLUDED BY</u> ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT "SEE ATTACHED COPY OF ORDER DISMISSING CASE"

CLERK Kiry Gray	(BY) DEPUTY CLERK /s/ J. Lam	DATE 04/05/2021
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UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
13/183,253	07/14/2011	411122

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167

UTILITY APPLICATION	Attorney Docket No.: NETL.046A
	First Named Inventor: Hyun Lee
	Title: MEMORY BOARD WITH SELF-TESTING CAPABILITY
Direct all correspondence to Customer No.: 20995	
Date: April 13, 2009 Page 1	

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

The following enclosures are transmitted herewith to be filed in the patent application of:

Inventor(s):

1. Hyun Lee
2. Jayesh R. Bhakta
3. Soonju Choi

APPLICATION:

- (X) Specification in 30 pages.
(X) Drawings in 5 sheets.

CONTINUITY INFORMATION:

Application	Relationship	Parent App. No.	Filing Date	Status
This Application	Non-provisional of	61/044801	04/14/08	Pending
This Application	Non-provisional of	61/044825	04/14/08	Pending
This Application	Non-provisional of	61/044839	04/14/08	Pending

Reference to prior domestic applications is made in the:

- (X) Specification.
(X) Enclosed Application Data Sheet (ADS).

FILING FEES:

FEE CALCULATION				
FEE TYPE		FEE CODE	CALCULATION	TOTAL
Basic Utility	1.16(a)(1)	1011 (\$330)		\$330
Search Fee	1.16(k)	1111 (\$540)		\$540
Examination Fee	1.16(o)	1311 (\$220)		\$220
Excess Claims > 20	35 - 20 = 15	1202 (\$52)	15 x 52 =	\$780
			TOTAL FEE DUE	\$1870

UTILITY APPLICATION

Attorney Docket No.: NETL.046A

First Named Inventor: Hyun Lee

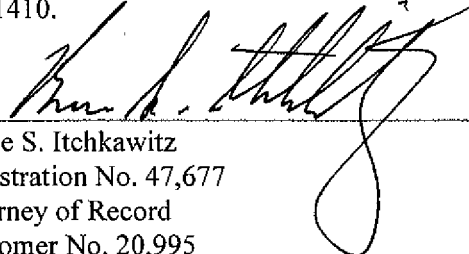
Title: MEMORY BOARD WITH SELF-TESTING CAPABILITY

Direct all correspondence to Customer No.: 20995

Date: April 13, 2009

Page 2

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment to Account No. 11-1410.

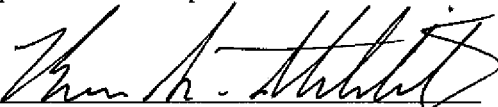


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

**NONPUBLICATION REQUEST
UNDER
35 U.S.C. § 122(b)(2)(B)(i)**

I hereby certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral agreement, that requires publication at eighteen months after filing.

I hereby request that the attached application not be published under 35 USC 122(b).



Bruce S. Itchkawitz

April 13, 2009

This request is being signed in compliance with 37 CFR 1.33(b) and submitted with the application **upon filing**.

Applicant understands that this nonpublication request may be rescinded at any time. If applicant rescinds a request that an application not be published under 35 USC 122(b), the application will be scheduled for publication at eighteen months from the earliest claimed filing date for which a benefit is claimed.

Applicant also understands that if applicant subsequently files an application directed to the invention disclosed in the attached application in another country, or under a multilateral international agreement, that requires publication of applications eighteen months after filing, the applicant **must** notify the United States Patent and Trademark Office of such filing within forty-five (45) days after the date of the filing of such foreign or international application. **Failure to do so will result in abandonment of this application (35 USC 122(b)(2)(B)(iii)).**

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046A
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Applicant Information:

Applicant 1					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence	US
Citizenship under 37 CFR 1.41(b)		US			
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Address 2					
City	Ladera Ranch	State/Province	CA		
Postal Code	92694	Country	US		
Applicant 2					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jayesh	R.	Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cerritos	State/Province	CA	Country of Residence	US
Citizenship under 37 CFR 1.41(b)		US			
Mailing Address of Applicant:					
Address 1		12220 Rose Street			
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country	US		
Applicant 3					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Soonju		Choi		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence	US

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046A
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

Citizenship under 37 CFR 1.41(b)		KR	
Mailing Address of Applicant:			
Address 1	195 Woodbury		
Address 2			
City	Irvine	State/Province	CA
Postal Code	92620	Country	US
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.			Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
Attorney Docket Number	NETL.046A	Small Entity Status Claimed ☐	
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	5	Suggested Figure for Publication (if any)	1

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☒ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046A
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61/044801	2008-04-14
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61/044825	2008-04-14
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61/044839	2008-04-14

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

Remove			
Application Number	Country	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

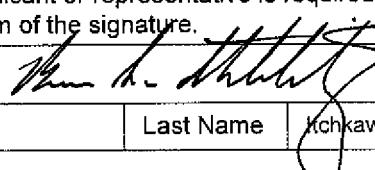
Assignee 1	
If the Assignee is an Organization check here.	☒
Organization Name	NETLIST, INC.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046A
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

Mailing Address Information:			
Address 1		51 Discovery	
Address 2		Ste 150	
City		Irvine	State/Province CA
Country	US	Postal Code	92618
Phone Number		949-435-0025	Fax Number 949-435-0031
Email Address		hlee@netlist.com	
Additional Assignee Data may be generated within this form by selecting the Add button.			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.			
Signature			Date (YYYY-MM-DD) 2009-04-13
First Name	Bruce	Last Name	Kichkowitz
Registration Number		47677	

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

MEMORY BOARD WITH SELF-TESTING CAPABILITY**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application claims the benefit of priority from U.S. Provisional Application No. 61/044,801, filed April 14, 2008, from U.S. Provisional Application No. No. 61/044,825, filed April 14, 2008, and from U.S. Provisional Application No. 61/044,839, filed April 14, 2008, each of which are incorporated in their entirety by reference herein. This application is related to U.S. Patent Application No. _____, filed on even date herewith and entitled "Self-Adjusting Damper", and to U.S. Application No. _____, filed on even data herewith and entitled "Circuit Providing Load Isolation and Noise Reduction", both of which are incorporated in their entirety by reference herein.

BACKGROUNDField

[0002] The present invention relates to self-testing electronic modules and, more particularly, to self-testing electronic memory modules.

Description of the Related Art

[0003] The failure of memory components in an electronic system may result in the loss of valid data. Therefore, it is important to ensure proper memory operation in an electronic system. Memory integrated circuits ("memory chips") often go through a series of tests at various stages of system manufacture. Once memory chips are deployed in a system, they also generally go through a system level memory test each time the system is booted. In addition, memory chips may undergo a parity checking process during normal system operation.

[0004] There are typically at least three test phases which memories undergo during system manufacture. Each phase generally tests for memory defects and for the correct operation of the input/output interface. The first test phase is typically conducted by the memory chip manufacturer and generally involves checking for bit failures, correct memory access speed, etc. The second test phase is typically done by memory module manufacturers and generally involves testing the signal quality, the noise susceptibility, and the operational

speed of the memory module as a single unit. The second test phase may also include checking for bit failures in individual memory chips. The third phase is usually carried out by the system manufacturer. During the third phase, the interaction of the memory subsystem with other components in the system is tested. During the third phase, the individual memory module operation is also tested again and the memory array is checked for defects. Because of the significant amount of testing that memories undergo during the manufacturing process, there is generally substantial test cost and test time associated with ensuring the proper memory operation. This test cost and test time translate into an increase in system cost and a decrease in system performance.

[0005] There are a number of memory test methodologies that employ either external test hardware, embedded self-test logic ("MBIST"), or both. However, the usefulness of these test methodologies is limited due to the high cost and other limitations associated with them. For instance, external test hardware such as automatic test equipment ("ATE") is very expensive. Moreover, the development time and cost associated with implementing MBIST is relatively high. These costs and limitations are especially significant when testing dynamic random access memory ("DRAM"). For example, technological developments, such as increases in DRAM speed, may require manufacturers to upgrade ATE machines relatively frequently. In addition, MBIST in DRAM chips generally cannot be fully utilized for system level testing of memory boards.

[0006] Because of the increasing cost, complexity, and time involved with fully testing DRAM chips, DRAM manufacturers often provide "effectively tested" ("ETT") DRAM chips to memory module manufacturers at a lower price rather than providing fully tested DRAM chips. Memory module manufacturers often prefer the ETT DRAM chips mainly due to their greater availability. Memory module manufacturers who receive ETT DRAM chips then have to assume a part of the responsibility of validating the DRAM chips, adding to the complexity of the memory module test process.

SUMMARY

[0007] In certain embodiments, a self-testing memory module is provided comprising a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module may further comprise a plurality of

memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprises data, address, and control ports. In certain embodiments, the memory module includes a control module configured to generate address and control signals for testing the memory devices and a data module comprising a plurality of data handlers. Each data handler in certain embodiments is operable independently from each of the other data handlers of the plurality of data handlers. Each data handler may be operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports.

[0008] A self-testing memory module is provided in certain embodiments comprising a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module may further comprise a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprises data, address, and control ports in some embodiments. The memory module of certain embodiments includes a control module configured to generate address and control signals for testing the memory devices. In certain embodiments, the memory module comprises a data module comprising at least one data handler operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate cyclic data for writing to the corresponding plurality of data ports.

[0009] In certain embodiments, a method of self-testing a memory module includes providing a self-testing memory module. The memory module comprises a printed circuit board configured to be operatively coupled to a memory controller of a computer system. The memory module further comprises a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices comprising data, address, and control ports. In certain embodiments, the memory module comprises a control module configured to generate address and control signals for testing the memory devices. The memory module may further comprise a data module comprising a plurality of data handlers. Each data handler may be operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports. The method of certain embodiments further includes generating, by each of the data handlers, data for writing to the corresponding plurality of data ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein.

[0011] Figure 2 is a block diagram of an example self-testing memory module including eight memory devices and eight data handlers in accordance with certain embodiments described herein.

[0012] Figure 3 is a block diagram of an example data handler module and an example control module in accordance with certain embodiments described herein.

[0013] Figure 4 is a flowchart of an example method of self-testing a memory module in accordance with certain embodiments described herein.

[0014] Figure 5 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

[0015] Figure 6 is a flow diagram illustrating the self-testing operation of an example memory module in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0016] Certain embodiments described herein present a self-testing electronic system, such as for example, a self-testing electronic memory module. Some embodiments described herein present a self-testing memory module that is populated with ETT DRAM chips. Certain embodiments described herein present a self-testing registered dual in-line memory module ("RDIMM"). Some embodiments described herein present a self-testing RDIMM that does not require any additional pins other than the pins on the standard JEDEC RDIMM connector. For example, the memory module may utilize the address and control signals (e.g., address and control signals generated by the memory module to test the memory module) along with a test signal to enable and execute a self testing function.

[0017] Certain embodiments described herein present a self-testing memory module that includes a control module and a data module which can generate memory addresses and data according to the JEDEC standard memory protocol.

[0018] Some embodiments described herein presents a self-testing memory module that can be configured through an I²C interface and that allows test results to be read out through the I²C interface. Certain embodiments described herein present a self-testing

memory module that allows a test function to be configured, controlled, and/or executed without substantial system memory controller involvement. Some embodiments present a self-testing memory module that can be tested without any external test equipment. For example, certain embodiments present a self-testing memory module that can be tested without any system driven test procedure. Various embodiments described herein present a self-testing memory module that can be self-tested at a target system speed. Certain embodiments described herein present a self-testing memory module that can generate data (DQ) and data-strobe (DQS) signals with wave characteristics that resemble the wave characteristics of DQ and DQS signals from a system memory controller.

[0019] MBIST is commonly used to test memories embedded in application specific integrated circuits ("ASICs") or system on chip integrated circuits ("SoCs") such as, for example, advanced memory buffers ("AMBs"). The MBIST implementation usually includes three distinct functional blocks: the address/control generator, the data generator/checker (sometimes referred to as a signature analyzer), and the test interface controller/register ("JTAG"). The test instructions and test patterns are generally loaded to the address/control generator and the data generator/checker through the JTAG interface. The test results are generally read out through the JTAG interface.

[0020] In memory module applications, there are obstacles associated with using MBIST. One of these obstacles is the large number of signals used to interface the three MBIST functional blocks. The inter-block timing constraints can present another obstacle. In addition, the ability to control the test is limited. For example, synchronizing the address/or control signals with the data signal can be difficult. The relatively large amount of information to be gathered and stored while testing the memory presents yet another obstacle.

[0021] Unlike on an ASIC, in which the three MBIST functional blocks are in a single chip, the MBIST functional blocks on a memory module generally would be segregated into multiple chips on the memory module due to physical and electrical limitations and requirements. This makes implementing MBIST on a memory module difficult because, while there is virtually no limit on the number of available interface signals among the three MBIST functional blocks in an ASIC, the memory module can support only a limited amount of interface signals between memory chips. In addition, because the MBIST

functional blocks are spread out to multiple chips on a memory module, the inter-block signal delay is generally much longer on a memory module than on an ASIC. It is also generally not feasible to implement cross-checking logic that operates the three MBIST functional blocks in lock-operational block. This is due to the limitation on the number of interface signals and to the relatively long inter-block signal delay time on a memory module versus on an ASIC.

[0022] In some cases such as where a memory module includes an AMB, the self-test logic (MBIST) implemented in the AMB includes command and address generation logic in addition to a data generator and checker. Each of these functional blocks may be implemented on a single physical AMB device (e.g., a single integrated circuit package). However, because all of the self-test command, address and data signals are combined in one physical area of the memory module, a memory module (e.g., DIMM) level routing problem can occur, making it difficult to route the self-test signals on the memory module and resulting in performance degradation and/or implementation difficulty. In addition, the data width of self-test logic of the memory module will be limited to the data width of the AMB (e.g., to the data width of the self-test logic implemented on the AMB, to the number of available ports on the AMB, etc.). As such, implementing memory module test logic on an AMB is not flexible (e.g., to changes in the data width of the memory module) and generally supports only memory modules having certain predetermined, fixed data widths.

[0023] Finally, in most cases, since an ASIC cannot be repaired, the ASIC MBIST is generally only capable of detecting and reporting the pass/fail status of memory tests. For memory module test results, on the other hand, it is generally advantageous to include both addresses of the memory locations where failures occur and the data patterns that were read back from the failed memory locations. This type of reporting can help to facilitate the repair of the memory module by, for example, allowing for the identification and replacement of failed components.

[0024] Figure 1 is a block diagram of an example self-testing memory module 10 in accordance with certain embodiments described herein. The memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board (PCB) 12, each memory device 20 of the plurality of memory

devices 18 comprising data, address, and control ports. The memory module 10 comprises a control module 22 configured to generate address and control signals 24 for testing the memory devices 18. The memory module 10 also includes a data module 28 comprising a plurality of data handlers 30. Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports of one or more of the plurality of memory devices 18. For example, each of the data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports. Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. The memory module 10 may further include an I²C interface 15 in certain embodiments.

[0025] As described more fully below, in certain embodiments the data module 28 generates test data patterns to write to the plurality of memory devices 18 of the memory module 12 and checks the data patterns read or received back from the plurality of memory devices 18 for agreement with corresponding data patterns that are expected to be read back from the plurality of memory devices 18. For example, in one embodiment, the data module 28 generates cyclic patterns to write to the plurality of memory devices 18. In some embodiments, the data module 18 also isolates the data path from the system board of the computer system 16 to the plurality of memory devices 18 while the memory module 10 is not accessed by the computer system 16. For example, the data module 28 may isolate the data path from the system board to the plurality of memory devices 18 when the memory module 10 is in a self-testing mode. The control module 22 may include, for example, a dual input register (e.g., the memory device controller 32 described more fully below) for registering address and control signals coming from either self-testing logic (e.g., from the test controller 36 described more fully below) or from the memory controller 14 on the system board. In some embodiments, during testing, the control module 22 generates address and control signals 24 associated with memory locations to be tested and the data module 28 generates corresponding test data patterns and provides them to the appropriate memory devices 20. For example, the data module 28 may receive a write command from the control module 22 and provide data to be written to certain locations in the memory devices 20 during a write operation. The data module 28 may then receive a read command to read back

the data from those locations and check the read data for agreement with the expected data. If there is a mismatch between the read data and the expected data, the data module 28 may, for example, store the failure information (e.g., the failed data word) and inform the control module 22 about the failure. The control module may save the address of the memory location where the failure occurred.

[0026] In certain embodiments, the memory module 10 is configured to be operated in a test mode in which the control module 22 selectively inputs the address and control signals to the address and control ports of the plurality of memory devices 18. Moreover, in the test mode, each of the data handlers 30 write the data generated by the data handler 30 to the corresponding plurality of data ports by selectively inputting data signals to the data ports of the plurality of memory devices 18. The data module 28 and/or the control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of the memory devices 20. For example, the data module 28 and/or the control module 22 are configured to provide memory signals (e.g., data, address and control signals) according the operating specification of the memory devices 20. In some embodiments, the control module 22 and the data module 28 produce memory addresses, control and/or data signals according to the JEDEC standard memory protocol. In some embodiments, for example, the control module 22 and the data module 28 generate the memory interface signals with proper edge relationships based on the JEDEC standard. In certain embodiments, the test speed, for example, may be defined by the speed of the clock (e.g., the system clock). The address sequences and/or the data patterns of certain embodiments may be programmable either through the I²C interface or they may be defaulted to pre-defined values.

[0027] In certain embodiments, data module 28 is configured to generate data signals with programmable slew rates and/or with variable peak values. In one embodiment, for example, the data module 28 is also able to generate data (DQ) and data-strobe (DQS) signals with programmable slew rates and programmable peak values so that the characteristics of the signals generated by the data module 28 generally correspond to the characteristics DQ and DQS signals generated by the system memory controller 14.

[0028] In some embodiments, the data module 28 and/or the control module 22 are configured to test the plurality of memory devices 18 under non-normal conditions. For example, the data module 28 and/or control module 22 may be configured to provide signals having frequencies which are higher or lower than the normal operating frequencies of the memory devices 20.

[0029] In certain embodiments, the memory module 10 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, or 8-GB. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bytes, 8 bytes, 16 bytes, 32 bytes, or 32 bits, 64 bits, 128 bits, 256 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 12 has an industry-standard form factor. For example, the PCB 12 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 12 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 12 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0030] In certain embodiments, the plurality of memory devices 18 of the memory module 10 may be arranged as ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 20 of the plurality of memory devices 18 that can be accessed by the memory controller 14 to access the full bit-width of the memory bus of the memory module 10. For example, a memory module 10 in which each rank of the memory module is 64 bits wide is described as having an "x 64" organization. Similarly, a memory module 10 having 72-bit-wide ranks is described as having an "x 72" organization. The number of memory devices 20 and corresponding memory capacity of a memory module 10 can be increased by increasing the

number of memory devices 20 per rank or by increasing the number of ranks. For example, a memory module with four ranks with each rank having N 512-MB memory devices 20 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 20 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices 20. During operation, the ranks of a memory module 10 may be selected or activated by control signals that are received from a component of the system (e.g., a system memory controller 14 or a local memory controller of the memory module 10). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 10 comprises only one rank of memory devices 20.

[0031] As discussed, the PCB 12 may include at least one connector (not shown) configured to operatively couple the memory module 10 to the memory controller 14 of the computer system 16. The computer system 16 may include a host computer system. For example, the memory module is electrically coupled, logically coupled, or both, with the memory controller 14. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 14 may comprise a disk controller of the computer system 16, for example. The memory controller 14 may be mounted on a system board of the host computer 16. The connector can comprise a plurality of edge connections which fit into a corresponding slot connector of the host system 16. The connector of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 10 and the host system 16. For example, the connector can comprise a standard DDR2, DDR3, and other future generation edge connectors. Additionally, in certain embodiments, more than one memory module 10 is coupled to the host system 16.

[0032] The plurality of memory devices 18 on the PCB 12 may include one or more volatile memory components. For example, the plurality of memory devices 18 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 20. Types of DRAM devices 20 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM

(SDRAM). The memory devices 18 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 20 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 20 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP). The plurality of memory devices 18 may further include one or more non-volatile memory devices 20, such as, for example, flash memories. The plurality of memory devices 18 of certain embodiments may include both volatile and non-volatile memory devices 20. For example, the plurality of memory devices 18 may include one or more of DRAM, SRAM, and/or flash memory devices in some embodiments.

[0033] Each data handler 30 is operable independently from each of the other data handlers 30 of the plurality of data handlers 28. For example, each data handler 30 is configured to write to and/or read from the corresponding plurality of data ports of one or more of the memory devices 20 without being in communication any of the other data handlers 30 or other data ports of the memory devices 20. As such, each data handler 30 can be used to generally independently test a portion of the memory space of the memory module 10. For example, each data handler 30 may be used to independently test one memory device 20 of the memory module 10. In such a configuration, the corresponding plurality of data ports of each data handler 30 may comprise each data port of the corresponding memory device 20. In other embodiments, each data handler 30 may be used to test a segment of one memory device 20, more than one memory device 20, segments more than one memory device 20, or any combination or sub-combination thereof. Because each of the data handlers 30 is operable independently of each of the other data handlers 30, the data handlers 30 are generally modular. As such, modifications in the configuration of the memory module 10 (e.g., changes in the bit-width of the memory bus, changes in the number of memory devices 20, etc.) may be less complicated to accommodate than in other types of self-testing memory modules 10. For example, where a new memory device 20 or set of memory devices 20 is added to the memory module 10, the change may be generally accommodated by adding a corresponding data handler 30. The change may be accommodated without having to

implement a major reorganization of the memory module 10 or the self-testing logic of the memory module 10, for example.

[0034] Each data handler 30 is further configured to generate data for writing to the corresponding plurality of data ports. Figure 2 is a block diagram of an example self-testing memory module 10 including eight memory devices 20 (e.g., memory devices 40a-40h) and a data module 28 comprising eight data handlers 30 (e.g., data handlers 30a-30h) in accordance with certain embodiments described herein. Each of the memory devices 20 includes an eight bit output data word and eight corresponding data ports. In addition, the system memory bus 50 between the memory controller 14 and the example memory module 10 is 64 bits wide and each of the data handlers 30 receives an eight bit segment of the system memory bus 50. Each of the data handlers 30 is operatively coupled to a corresponding plurality of data ports 21 of a corresponding one of the memory devices 20. As such, the data handlers 30 may be operatively coupled (e.g., electrically and/or logically coupled or connected) to the eight data ports 21 of one of the corresponding memory devices 20. For example, the data handler 30a may be operatively coupled to the eight data ports 21 of the memory device 40a of Figure 2.

[0035] The configuration shown in Figure 2 is for the purposes of illustration and is not intended to be limiting. For example, while the example memory module 10 of Figure 2 includes an equal number of memory devices 20 and data handlers 30, other configurations are possible. In some configurations there are more memory devices 20 than data handlers 30 or vice versa. Moreover, the one or more data handlers 30 may be operatively coupled to a subset of the data ports 21 of one the memory devices 20 instead of all of the data ports 21 of one of the memory devices 20. In other embodiments, one or more data handlers 30 may be operatively coupled to a subset or all of the data ports 21 of more than one of the memory devices 20. For example, in one embodiment, each of the data handlers 30 are operatively coupled to all of the data ports of two memory devices 20.

[0036] In certain embodiments, the plurality of data handlers 28 comprises at least two physically separate components mounted on the PCB 12. For example, the plurality of data handlers 28 may include at least two physically separate integrated circuit packages. The physically separate integrated circuit packages are mounted on different portions of the

PCB 12 in some embodiments. For example, each of the eight data handlers 30a-30h shown in Figure 2 may include physically separate integrated circuit packages mounted on different portions of the PCB 12. While eight data handlers 30 are shown in Figure 2, other numbers of data handlers 30 are possible including fewer or more than eight.

[0037] In certain embodiments, each of the plurality of data handlers 30 is positioned on the PCB 12 proximate to the corresponding plurality of data ports. For example, each data handler 30 of certain embodiments is positioned closer to the corresponding plurality of data ports 21 than the data handler 30 is to the other data ports 21 of the plurality of memory devices 18. For example, the data handler 30a is positioned closer to the corresponding plurality of data ports 21 of the memory device 40a than to the other data ports 21 of the other memory devices 40b-40h.

[0038] Figure 3 is a block diagram an example data module 28 and an example control module 22 in accordance with certain embodiments described herein. The control module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18. In some embodiments, the control module 22 includes a control mixer element 32. The control mixer element 32 may include a memory device controller 34 (e.g., a DRAM controller) and a test controller 36. In certain embodiments, the control mixer element 32 generally controls the address and the control signals for the self-testing function.

[0039] In certain embodiments, the memory device controller 34 generally pre-processes address and control information before it sends the information to a register 40. In one embodiment, the memory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36. The control module 22 of certain embodiments is configured to selectively input to the address and control ports of the plurality memory devices 18 either the address and control signals 38 from the system memory controller 14 or the address and control signals 42 from the control module 22 (e.g., from the test controller 36). For example, the memory device controller 34 may send either the signals 38 from the system memory controller 14 or, alternatively, the signals 42 from the test controller 36, to the register 40 depending on whether the memory module 10 is in normal (non-test) mode or

in a test mode, respectively. In one embodiment, the memory device controller 34 generates the address and control signals for memory device (e.g., DRAM device) operations. The test controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28. The control module 22 may be implemented in the control register of the memory module 10 in certain embodiments. In various embodiments, the control module 22 includes discrete logic, one or more application-specific integrated circuit (ASICs), one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic device (CPLDs).

[0040] The data module 28 and the subcomponents thereof (e.g., the data handlers 30) may be in communication with one or more of the memory devices 20, the control module 22, and the memory controller 14. In certain embodiments, the data module 28 comprises a plurality of data handlers 30. In other embodiments the data module 28 includes at least one data handler 30. Each of the data handlers 30 of certain embodiments comprises a switch 44. For example, the switch 44 may include a data multiplexer/demultiplexer ("data mux/demux"). The switch 44 may provide a bi-directional data multiplexer function. In certain embodiments, the switch 44 is configured to selectively input to the corresponding plurality of data ports either data signals 48 from the system memory controller 14 or data signals 50 from the data handler logic element 46. The switch 44 of certain embodiments may further be configured to receive data signals 52 (e.g., during a read operation) from the plurality of memory devices 18 and to propagate the data signals 52 to the data handler logic element 46 and/or the memory controller 14. In some embodiments, for example, the switch 44 selectively inputs the data signals 48 to be written to the plurality of memory devices 18 from the system memory controller 14 when the memory module 10 is a normal (non-test mode) mode and, alternatively, inputs the data signals 50 from the data handler logic element 46 during a test mode. While the switch 44 is shown as being included in the data handler 30 in the example of Figure 3, other configurations are possible. For example, in other embodiments the switch 44 may be logically and/or physically separated from the data handler module 28 and/or the data handlers 30.

[0041] Each of the data handlers 30 of certain embodiments further includes a data handler logic element 46. The data handler logic element 46 of certain embodiments comprises a data generation element 54 and a verification element 56. The data generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports, for example. The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information) the data handler logic element 46 receives from the control module 22, for example. The data may be cyclic data in some embodiments or non-cyclic data in other embodiments. For example, the cyclic data may comprise at least one predetermined pattern of data which repeats or is cycled two or more times. In various embodiments, the data comprises one or more incrementing patterns or decrementing patterns, for example. In other embodiments, the data comprises a pattern which alternates each bit on successive memory writes. For example, a memory write comprising one or more hexadecimal "A" characters (each corresponding to a four-bit binary word of "1010") may be followed by a memory write comprising one or more hexadecimal "5" characters (each corresponding to a four-bit binary word of "0101"). The data may be generated in a variety of ways. In one embodiment, the data is generated based on a current write address value. For example, in one example configuration, on a first write cycle, hexadecimal "A's" are generated and written to even address locations and hexadecimal "5's" are generated and written to odd address locations, and on a second write cycle, "5's" are written to even addresses and "A's" are written to odd addresses, and this pattern repeats in subsequent cycles. The data may be generated based previously written data (e.g., inverting each of the bits of a previously written data word) in some embodiments. In general, any manner of generating a cyclic or otherwise deterministic data pattern may be compatible with embodiments described herein. In other embodiments, random or pseudorandom data may be generated and written to the corresponding plurality of data ports. For example, a linear feedback shift register (LFSR) may be used in some embodiments. In addition, the data patterns may be programmable. For example, the data patterns may be programmable based on information received by the data generation element 54 from the memory controller 14 (e.g., through the control module 22), or from the control module 22.

[0042] The plurality of data handlers 30 are further configured to read data from the corresponding plurality of data ports. For example, the verification element 56 may be configured to receive data from the corresponding plurality of data ports (e.g., through the switch 44 during a test mode). The verification element 56 may further be configured to check for failures in the operation of the plurality memory devices 18 by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler 30 and written to the corresponding plurality of data ports.

[0043] In certain embodiments, the verification element 56 is configured to perform the verification without storing a copy of the data written to the corresponding plurality of data ports or accessing a stored copy beyond the data read from the plurality of memory devices 18. For example, the verification element 56 does not store or access a copy of the data that is written to the corresponding plurality of data ports except for the data stored and read back from the plurality of memory devices 18. As such, the memory module 10 of certain embodiments advantageously does not require separate memory for storing duplicate copies of test data that is written to the plurality of memory devices 18 for later comparison. For example, the verification element 56 may calculate comparison data and may compares the comparison data to the data read from the corresponding plurality of data ports. In certain embodiments, the comparison data comprises data which expresses the data or values expected to be received from the plurality of memory devices 18 if the write, store, and read processes of the data using the data module 28 and the plurality of memory devices 18 are performed correctly or as expected. The calculation may be performed simultaneously or substantially simultaneously with receiving the data read from the corresponding plurality of data ports in certain embodiments. In other embodiments, the calculation is performed either before or after receiving the data. Other configurations are possible. For example, in one embodiment, the verification element 56 does store a separate copy of the data written to the corresponding plurality of data ports upon writing the data and compares the separate copy to the read data received from the plurality of memory devices 18.

[0044] The verification element 56 of certain embodiments calculates the comparison data based on the cyclic data. For example, in one embodiment, the verification

element 56 calculates the comparison data in substantially the same manner that the data generation element 54 generates the data as described above (e.g., based on a current write address, using an LFSR, etc.). As such, the comparison data of certain embodiments is substantially a repeat of the data written. In one example embodiment, the data handler 30 is configured (e.g., is programmed by the control module 22) to write an alternating series of "A's" and "5's" to the data ports of the corresponding plurality of data ports as described herein. For example, the data handler 30 may be configured to write one or more "A's" to the first address location of an N-word memory device 20 including the corresponding plurality of data ports. The data handler 30 may then write one or more "5's" to the second address location, one or more "A's" to the third address location and so on until the data handler 30 has written to all N memory locations. The verification element 56 of the example embodiment then calculates the comparison data based on the cyclic data written to the corresponding plurality of data ports. For example, the verification element 56 calculates a comparison word including one or more "A's", "5's," and "A's," respectively, to be compared to the data read from the first, second, and third address locations based on the cyclic data (e.g., based on the known cycle of the data). In some embodiments, the comparison data is calculated based on a current read address.

[0045] In certain embodiments, data associated with failures in the operation of the plurality of memory devices 18 are stored in the data module 28. For example, data read from the corresponding plurality of data ports which do not correspond to (e.g., match) the comparison data calculated by the verification element 56 may be stored in the data module. Moreover, in some embodiments, memory addresses associated with the failures in the operation of the plurality of memory devices 18 are stored in the control module 22. For example, the data handler 30 may communicate data failures (e.g., when data read from the corresponding plurality of data ports does not correspond to calculated comparison data) to the control module 22 which may then store the addresses corresponding to the data failure. In certain embodiments, the memory module 12 is configured to report failures (e.g., the failed data, the address corresponding to the memory location of the failed data, and/or expected data) via the I²C interface 15 to the memory controller 14. In addition, in certain embodiments the test controller 36 and/or the data handlers 30 may be updated through the

I²C interface 15 with new data patterns and/or with alternative memory access sequences to conduct AC tests (e.g., tests of the power, current, I/O speed, etc.).

[0046] In various embodiments, the components of the data module 28 (e.g., the switch 44, the data handlers 30, the data handler logic element 46, the data generation element 54, and/or verification element 56) may include discrete logic, one or more application-specific integrated circuits (ASICs) one or more microprocessors, one or more field-programmable gate arrays (FPGAs), or one or more computer-programmable logic devices (CPLDs). Additionally, one or more of the various functional blocks (e.g., the switch 44) of the data module 28 of Figure 3 may not be included. In some embodiments, additional functional blocks may be included. Moreover, some of the functional blocks are described as separate functional blocks for illustration purposes and may comprise one physical component. For example, in one embodiment, each of the data handlers 30 and the corresponding switch 44, data generation element 54, and verification element 56 comprise one physical component (e.g., are included in one integrated circuit package). In another embodiment, the data module 28 comprises one physical component.

[0047] Referring again to Figure 1, a self-testing memory module 10 of certain embodiments comprises a printed circuit board (PCB) 12 and is configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 further includes a plurality of memory devices 18 on the printed circuit board 12 where each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. The memory module 10 further comprises a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. In certain embodiments, the memory module 10 further comprises a data module 28 comprising at least one data handler 30 and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices 20. The data handler 30 is configured to generate cyclic data for writing to the corresponding plurality of data ports. In certain embodiments, there may be one data handler 30, for example. In other embodiments, there may be more than one data handler 30. The elements of the memory module 10 may be compatible with any of the embodiments described herein.

[0048] Figure 4 is a flowchart of an example method 70 of self-testing a memory module 10 in accordance with certain embodiments described herein. While the method 70 is described herein by reference to the memory module 10, other memory modules, electronic systems or subsystems, and/or circuits are also compatible with the embodiments described herein. The method 70 of certain embodiments comprises providing a self-testing memory module 10 at operational block 72. The memory module 10 may comprise a printed circuit board (PCB) 12 configured to be operatively coupled to a memory controller 14 of a computer system 16. The memory module 10 may further comprise a plurality of memory devices 18 on the printed circuit board 12. Each memory device 20 of the plurality of memory devices 18 may comprise data, address, and control ports. The memory module 10 may comprise a control module 22 configured to generate address and control signals for testing the plurality of memory devices 18. The memory module 10 may further comprise a data module 28 comprising a plurality of data handlers 30. In certain embodiments, the data module 28 comprises at least one data handler 30. Each data handler 30 of certain embodiments is operable independently from each of the other data handlers 30 of the plurality of data handlers 28 and is operatively coupled to a corresponding plurality of the data ports. At operational block 74, the method 70 further comprises generating, by each of the data handlers 30, data for writing to the corresponding plurality of data ports.

[0049] Figure 5 is a flow diagram 100 illustrating self-testing operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. At operational block 110 the memory module 10 enters an idle state after power up or, in some embodiments, when the memory board of the computer system 16 is connected to a mother board or to a test board. In one embodiment, at operational block 110, the contents of the register 40 are undefined and the controllers (e.g., the test controller 36 and the memory device controller 34) are in unknown state. After power up, the memory module 10 is reset (e.g., a reset command is executed) at operational block 111. At operational block 111, the logic of the data handler 30 (e.g., data generation element 54 and/or verification element 56) and the logic of the control mixer 32 are set to default values and/or states. For example, the memory module 10 may default to a non-test mode (e.g., normal operational mode) and the switches 44 of the data handlers 30 and the memory device controller 34 are

not configured in a test mode. At operational block 112, the test mode is configured. For example, the computer system 16 (e.g., through the memory controller 14) may configure the test mode. The configuration may be through the I²C interface 15, for example, and may include configuring the test controller 36 for test mode. For example, configuring the test mode may include switching (e.g., by configuring the test controller 36) the mode of the memory module 10 from a normal operation mode to the test mode.

[0050] At operational block 113, the test mode is initiated. Initiating the test mode may comprise configuring a particular test case (e.g., particular test data or data patterns, particular write and/or read address sequences, etc.). The initiation may be achieved by activating the test controller 36 to initiate the test mode. For example, at operational block 113, the data handlers 30 and the control mixer 32 are readied (e.g., via a tester) beginning the self-testing of the memory module 10. For example, the tester may comprise an ATM tester, a server, a specialized tester. The tester may ready the data handlers 30 and the control mixer through the I²C interface 15 or through the control signals 38, for example. For example, the data handlers 30 are configured to input data generated by the data handlers 30 to the corresponding plurality of data ports of the plurality of memory devices 18 and the control mixer 32 is configured to input address and control signals from the test controller 36 to the plurality of memory devices 18. At operational block 113 the test controller 36 updates each of the data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.).

[0051] In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal ("Par-in") is un-asserted (e.g., set to a "low" value), it is asserted (e.g., set to a "high" value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. The memory module 10 executes the test mode (e.g., writes and reads test data patterns) at operational block 114. The memory module 10 continues executing the test mode at operational block 114 until the self-test is complete or until the memory module 10 detects a certain number of memory failures such that a failure count exceeds a preset number.

[0052] If the failure count exceeds the preset number, the memory module 10 enters operational block 115 and "errors out" of the test mode. If the self-test is completed at

operational block 114 and the failure count does not exceed the preset number, the memory module 10 enters operational block 115. At operational block 115, the memory module 10 sends out a test completion indication signal (e.g., through the test controller 36 to the memory controller 14). At operational block 116, the fail status of the test may be read out through, for example, the I²C interface 15. For example, one or more address values associated with failed memory locations, and/or the data read from those locations, and/or expected data may be read. In other cases, where the test fails, the memory module 10 may report the failure to the memory controller 14 through any available signal (e.g., bidirectional data or data strobe signal) between the memory module 10 and the memory controller 14.

[0053] In some embodiments, the memory module 10 is generally interruptible. For example, the memory module 10 may exit the self-test and return to operational block 110 if the signal going into the parity-in pin is, for example, de-asserted or removed at operational block 155. In other embodiments, the interrupt signal or condition may be different and may not be a parity-in signal. For example, in one embodiment, the memory module 10 may exit the self-test and return to operational block 110 if a timer having a pre-determined count expires. When the self-testing is interrupted, the memory module 10 will return to operational block 110 and the bit failure information in the control mixer 32 and the data handler 30 becomes invalid.

[0054] Figure 6 is a flow diagram 200 illustrating the operation of an example self-testing memory module 10 in accordance with certain embodiments described herein. For example, one or more of the operational blocks of the flow diagram 200 of Figure 6 may correspond to one or more of the operational blocks of Figure 5. At operational block 210, the memory module 10 may be generally inactive. For example, the testing logic (e.g., the control module 22 and/or the data module 28) may be generally inactive and the memory module 10 may default to a functional (e.g., non-test) mode. In certain embodiments, the memory module 10 executes one or more tasks which configure the self-test mode of the self-testing memory module using an I²C interface 15. For example, the memory module 200 may configure the test controller 36 through the I²C interface 15 (e.g., as described with respect to Figure 5). In other embodiments, the test mode is configured using the control signals 38. For

example, the test mode may be updated at operational block 211 in certain embodiments. In certain embodiments, updating the test mode comprises programming the test controller 36 with test configuration information such as test data patterns, test data sequences, test failure conditions, etc. After updating the test mode at operational block 211, the test controller 36 may then be activated through the I²C interface 15 (e.g., as described with respect to Figure 5) in certain embodiments. Alternatively, as shown, the test controller 36 may be activated through the I²C interface 15 without first updating the test mode.

[0055] At operational block 212, the memory module test mode is ready for execution. In one embodiment, the input signal on a pin (e.g., a parity-in pin) of the memory module 10 is asserted or toggled (e.g., by the memory controller 14). For example, if the parity-in signal is low, it is asserted (e.g., set to a “high” value) and held in the asserted state. Alternatively, if the parity-in signal is already in the asserted state, it may be toggled and then held in the asserted state. At operational block 113, a set of status registers (e.g., one or more registers of the control module 22 or the data module 28) are cleared.

[0056] The memory module 200 generates address and data signals for testing the plurality of memory devices 18 at operational block 214. For example, the data generation elements 54 of the data handlers 30 may generate the data for writing to the plurality of memory devices 18 as described herein. In addition, the control module 22 may generate address and control signals for testing the plurality of memory devices 18 as described herein. At operational block 215, the memory module 10 performs a burst write function. For example, in certain embodiments, the memory module 10 writes multiple memory locations in the memory module 10 that are to be tested at operational block 215 in a relatively short amount of time (or in a burst). In certain embodiments, the memory module 10 can perform multiple write bursts at operational block 215. Next, at operational block 216, the memory module 10 performs a read & compare function. For example, in certain embodiments, the memory module 10 reads back certain memory locations at operational block 216 that have been written at operational block 215 and compares the values with certain expected data. For example, a verification element 56 of each of the data handlers 30 may calculate the expected data and/or compare the values as described herein. In some embodiments, the memory module 200 reads from multiple memory locations in the memory module 10 in a relatively

short amount of time (or in a burst). In certain embodiments, the memory module 10 can execute multiple read bursts at operational block 216 as it compares the read data with expected data. In some embodiments, the memory module 200 (e.g., the data handlers 30 of the memory module 200) stores any failed read data and the associated expected data at operational block 217. In some embodiments, the memory module sends one or more failure indications to the memory controller 14 over one or more available signals between memory module 10 and the memory controller 14. For example, the I²C interface 15 or one or more bi-directional data pins (e.g., one or more of the data pins 48) may be used.

[0057] The memory module 200 stores the addresses of the memory locations associated with the failed read data using the test controller 32 at operational block 217. In other embodiments the failed read data, expected data, and the addresses of the memory locations associated with the failed read data may be stored using the test controller 32. For example, in one embodiment, the test controller 32 stores the failed read data, expected data, and the addresses of the memory locations associated with the failed read data at operational block 217. In some embodiments, the test controller 32 is generally inaccessible through the I²C interface 15 during self-test operation. For example, the test controller is inaccessible through the I²C interface 15 until a test failure occurs at operational block 217 or until the test completes at operational block 219. In some embodiments, when the test is complete, the test results can be read out of the memory module 10 (e.g., through the I²C interface 15). In some cases, the test can be interrupted using the I²C interface 15 or using the parity-in signal (e.g., by de-asserting the parity-in signal).

[0058] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects

or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports.
2. The memory module of Claim 1, wherein the plurality of data handlers comprise at least two physically separate components mounted on the printed circuit board.
3. The memory module of Claim 2, wherein the plurality of data handlers comprise at least two physically separate integrated circuit packages.
4. The memory module of Claim 3, wherein the physically separate integrated circuit packages are mounted on different portions of the printed circuit board.
5. The memory module of Claim 2, wherein each of the plurality of data handlers is positioned on the printed circuit board proximate to the corresponding plurality of data ports.
6. The memory module of Claim 5, wherein each of the plurality of data handlers is positioned closer to the corresponding plurality of data ports than to the other data ports of the plurality of memory devices.
7. The memory module of Claim 6, wherein each of the data handlers is further configured to read from the corresponding plurality of data ports and further comprises a verification element for checking for failures in the operation of the memory devices by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler and written to the corresponding plurality of data ports.

8. The memory module of Claim 7, wherein the verification element is configured to verify that the data read from the corresponding plurality of data ports corresponds to the data generated by the data handler and written to the corresponding plurality of data ports without storing a copy of the data written to the corresponding plurality of data ports.

9. The memory module of Claim 7, wherein the verification element calculates comparison data and compares the comparison data to the data read from the corresponding plurality of data ports.

10. The memory module of Claim 9, wherein the data signals written to the corresponding plurality of data ports comprises cyclic data.

11. The memory module of Claim 10, wherein the verification element calculates the comparison data based on the cyclic data.

12. The memory module of Claim 10, wherein the verification element calculates the comparison data based on a current read address.

13. The memory module of Claim 7, wherein data associated with failures in the operation of the memory devices is stored in the data module and wherein memory addresses associated with failures in the operation of the memory devices are stored in the control module.

14. The memory module of Claim 1, each of the plurality of data handlers further configured to write data to the memory devices by selectively inputting to the corresponding plurality of data ports either data signals from the system memory controller or data signals from the data handler.

15. The memory module of Claim 14, wherein each of the plurality of data handlers comprises a switch which performs the selective inputting to the corresponding plurality of data ports.

16. The memory module of Claim 15, wherein the switch comprises a data multiplexer-demultiplexer.

17. The memory module of Claim 1, the control module further configured to selectively input to the address and control ports of the memory devices either the address

and control signals from the memory controller or the address and control signals from the control module.

18. The memory module of Claim 17, wherein the control module further comprises a DRAM controller which performs the selective inputting of the address and control ports.

19. The memory module of Claim 1, wherein the memory module is configured to be operated in a test mode in which the control module selectively inputs to the address and control ports of the memory devices the address and control signals from the control module and wherein each of the data handlers write the data generated by the data handler to the corresponding plurality of data ports by selectively inputting to the data ports of the memory devices data signals from the data handler.

20. A self-testing memory module, comprising:

- a printed circuit board configured to be operatively coupled to a memory controller of a computer system;

- a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;

- a control module configured to generate address and control signals for testing the memory devices; and

- a data module comprising at least one data handler operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate cyclic data for writing to the corresponding plurality of data ports.

21. The memory module of Claim 20, wherein each of the data handlers is further configured to read data from the corresponding plurality of data ports and further comprises a verification element for checking for failures in the operation of the memory devices by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler and written to the corresponding plurality of data ports.

22. The memory module of Claim 21, wherein the verification element is configured to verify that the data read from the corresponding plurality of data ports

corresponds the data generated by the data handler and written to the corresponding plurality of data ports without storing a copy of the data written to the memory devices.

23. The memory module of Claim 22, wherein the verification element calculates comparison data and compares the comparison data to the data read from the corresponding plurality of data ports.

24. The memory module of Claim 23, wherein the verification element calculates the comparison data based on the cyclic data.

25. The memory module of Claim 23, wherein the verification element calculates the comparison data based on a current read address.

26. The memory module of Claim 21, wherein data associated with failures in the operation of the memory devices is stored in the data module and wherein memory addresses associated with failures in the operation of the memory devices are stored in the control module.

27. The memory module of Claim 20, wherein each of the at least one data handlers is operable independently from each of the other data handlers.

28. The memory module of Claim 20, wherein the at least one data handler is positioned on the printed circuit board proximate to the corresponding plurality of data ports.

29. A method of self-testing a memory module, comprising:

providing a self-testing memory module, comprising:

a printed circuit board configured to be operatively coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;

a control module configured to generate address and control signals for testing the memory devices; and

a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports; and

generating, by each of the data handlers, data for writing to the corresponding plurality of data ports.

30. The method of Claim 29, further comprising:

reading, by each of the data handlers, data from the corresponding plurality of data ports; and

checking for failures in the operation of the memory devices by each of the plurality of data handlers by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler and written to the corresponding plurality of data ports.

31. The method of Claim 30, further comprising verifying that the data read from the corresponding plurality of data ports corresponds to the data generated by the data handler and written to the corresponding plurality of data ports without storing a copy of the data written to the corresponding plurality of data ports.

32. The method of Claim 31, further comprising:

calculating comparison data; and

comparing the comparison data to the data read from the corresponding plurality of data ports.

33. The method of Claim 32, wherein the data signals written to the corresponding plurality of data ports comprises cyclic data and calculating comprises calculating the data based on the cyclic data.

34. The method of Claim 32, wherein the data signals written to the corresponding plurality of data ports comprises cyclic data calculating comprises calculating the data based on a current read address.

35. The method of Claim 30, further comprising operating the memory module in a test mode in which the control module selectively inputs to the address and control ports of the memory devices the address and control signals from the control module and wherein each of the data handlers write the data generated by the data handler to the corresponding plurality of data ports by selectively inputting to the data ports of the memory devices data signals from the data handler.

MEMORY BOARD WITH SELF-TESTING CAPABILITY

ABSTRACT

A self-testing memory module includes a printed circuit board configured to be operatively coupled to a memory controller of a computer system and includes a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports. The memory module also includes a control module configured to generate address and control signals for testing the memory devices. The memory module includes a data module comprising a plurality of data handlers. Each data handler is operable independently from each of the other data handlers of the plurality of data handlers. Each data handler is operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and is configured to generate data for writing to the corresponding plurality of data ports.

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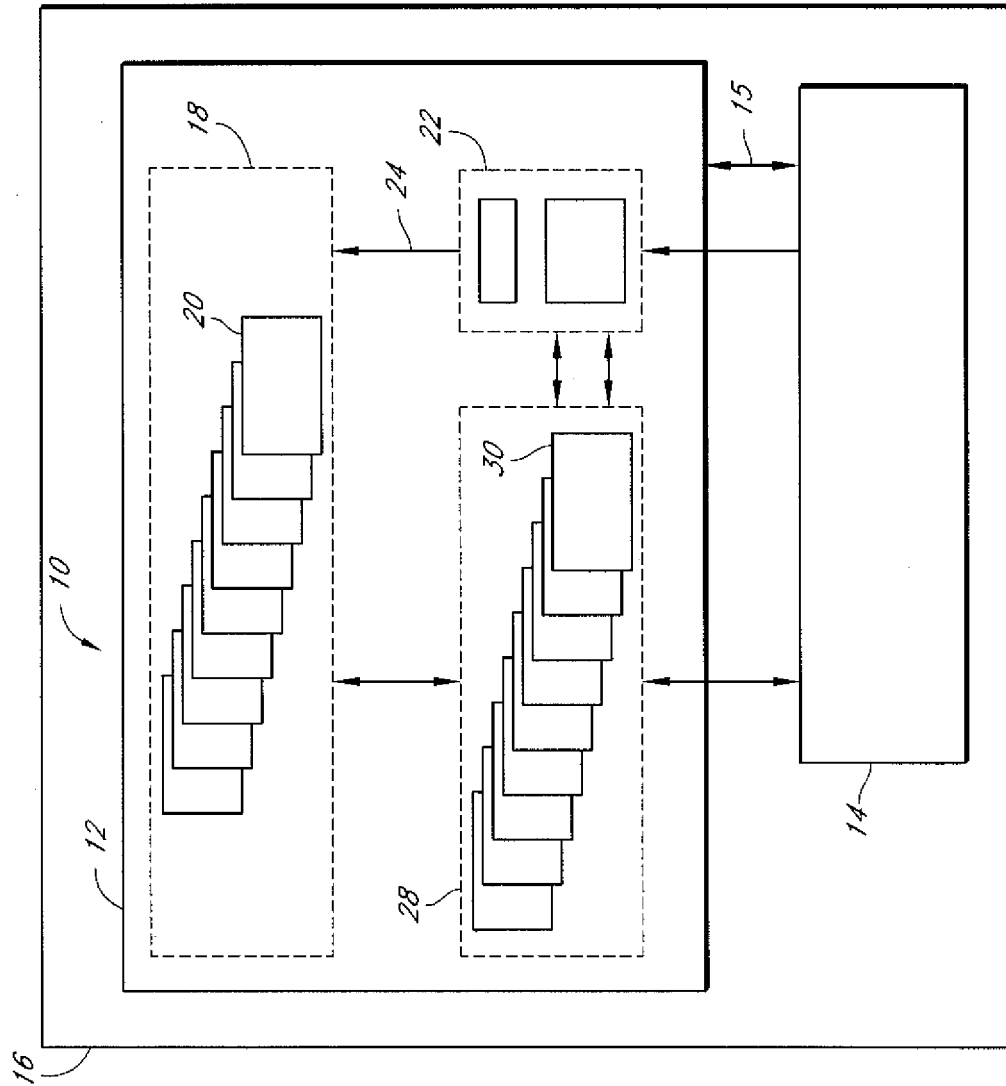


FIG. 1

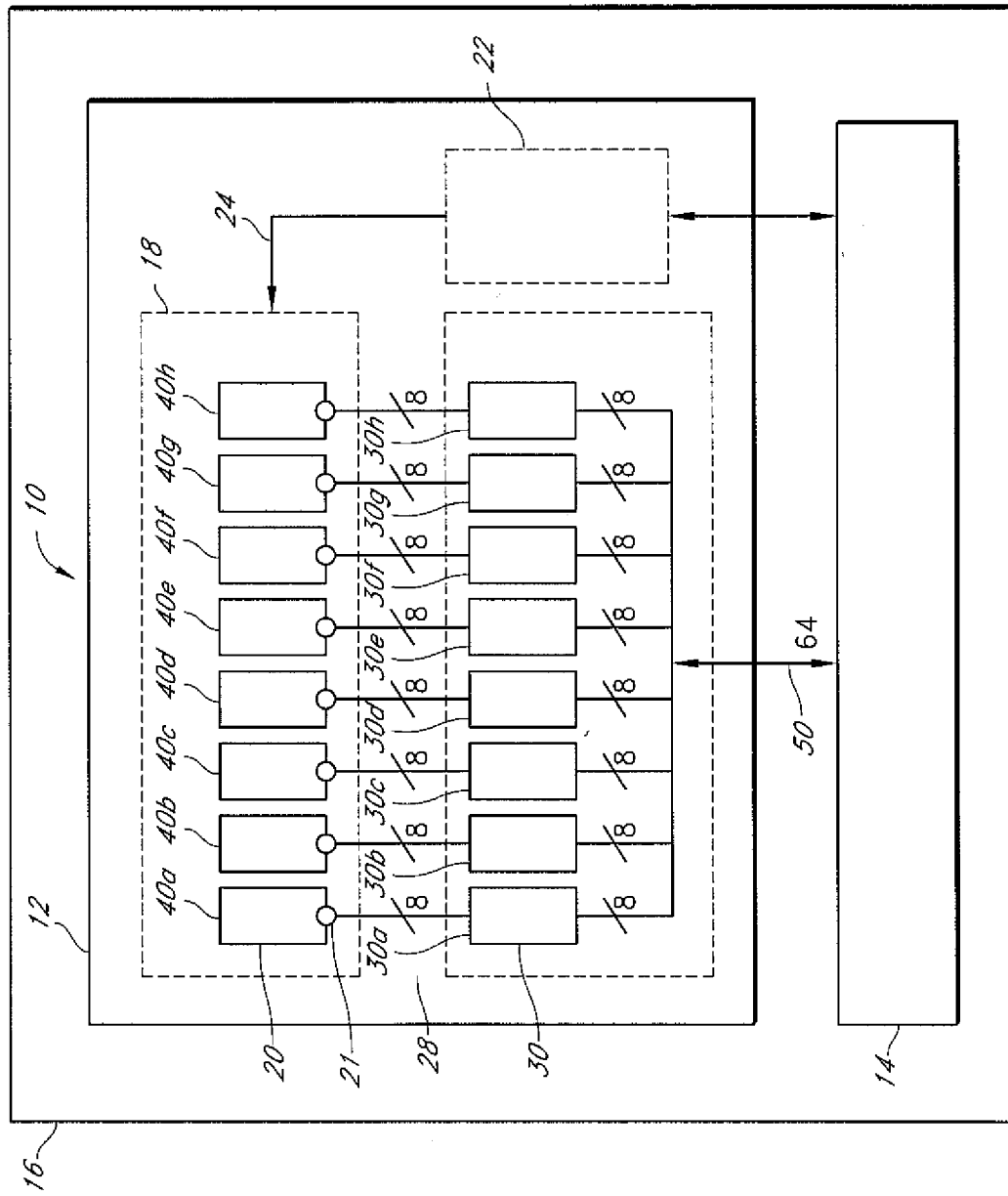


FIG. 2

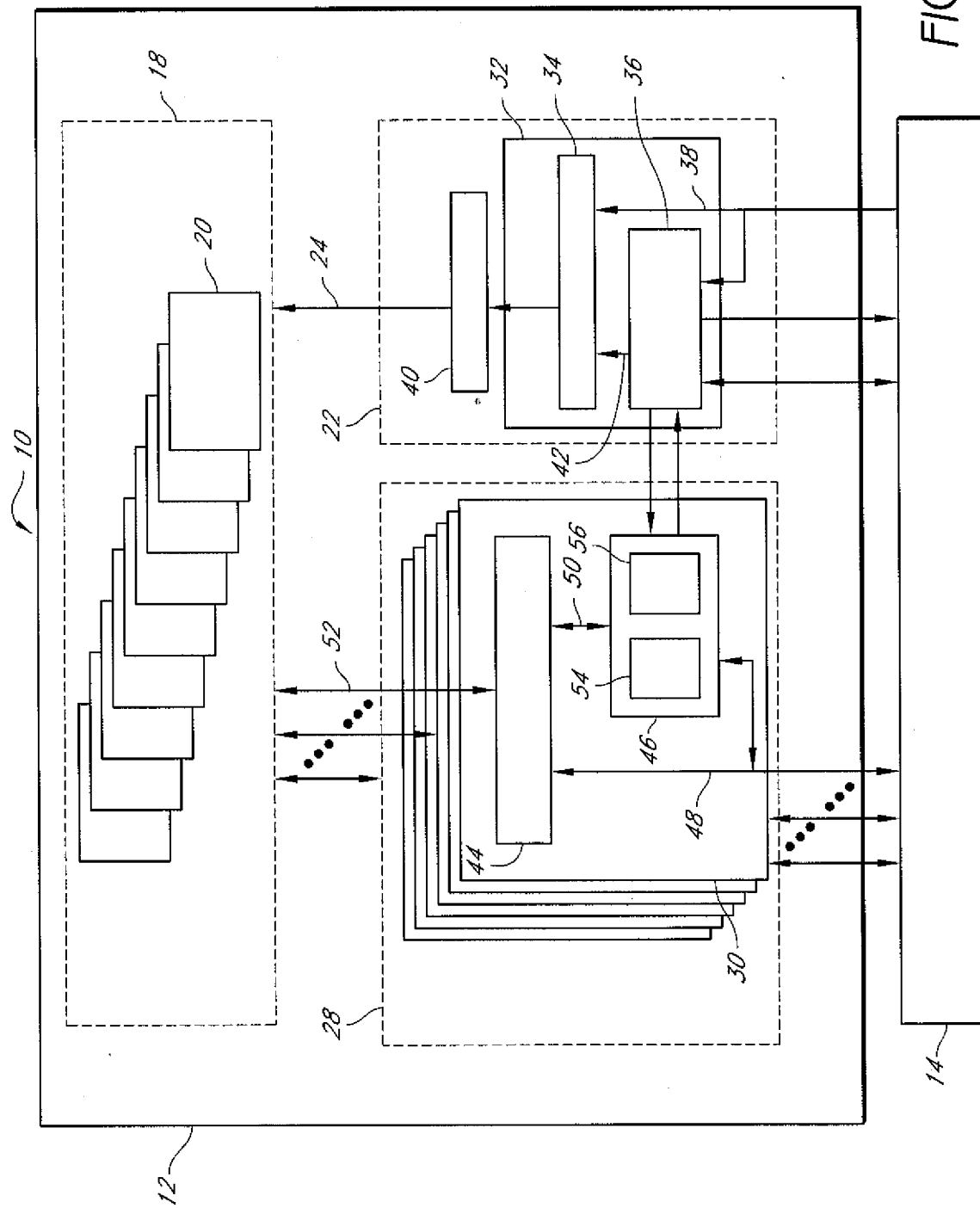


FIG. 3

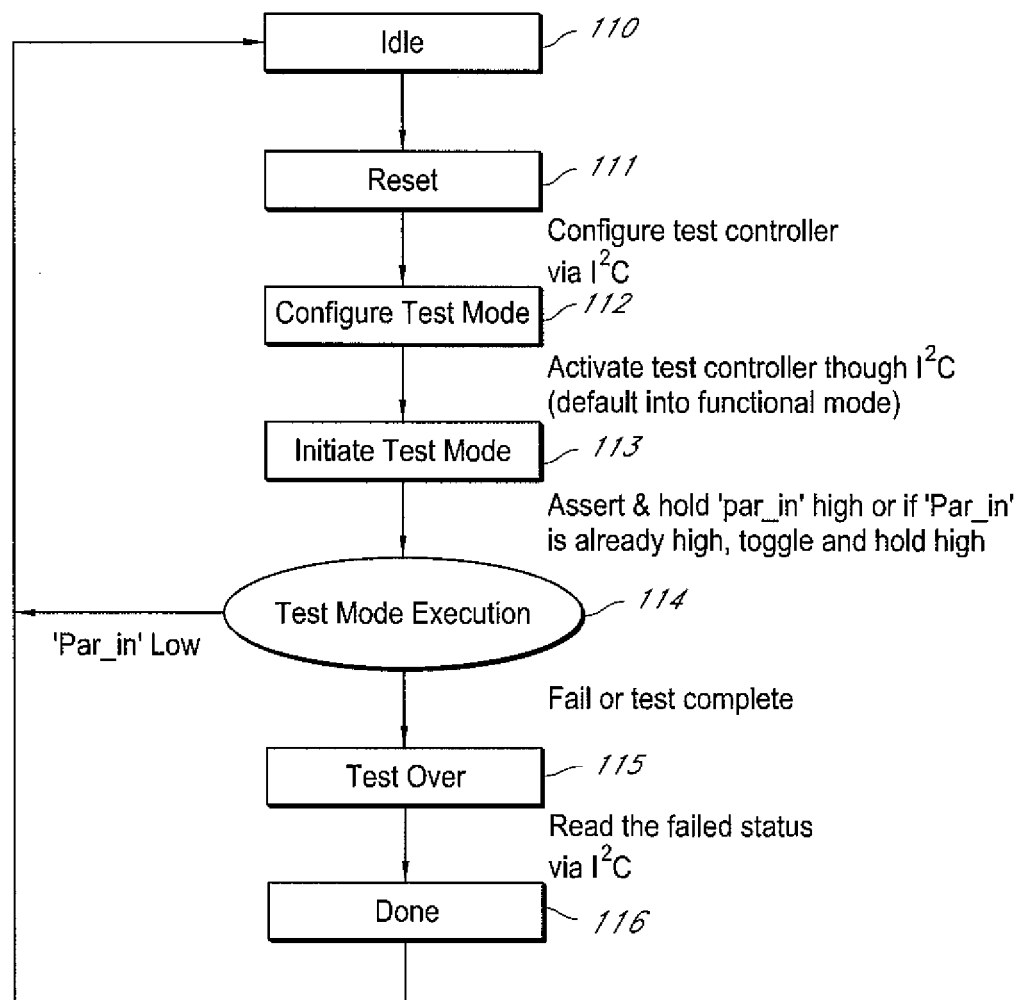
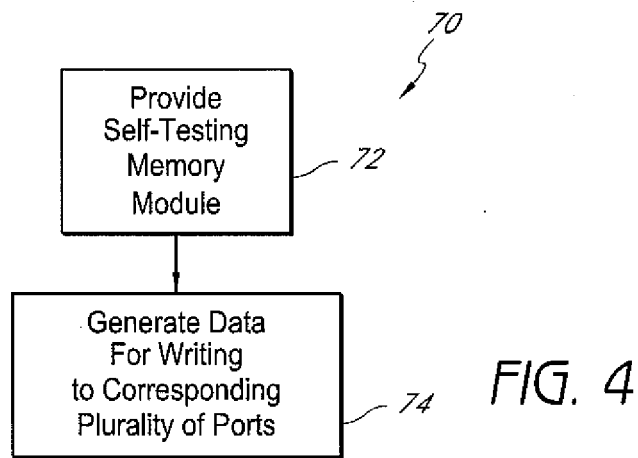
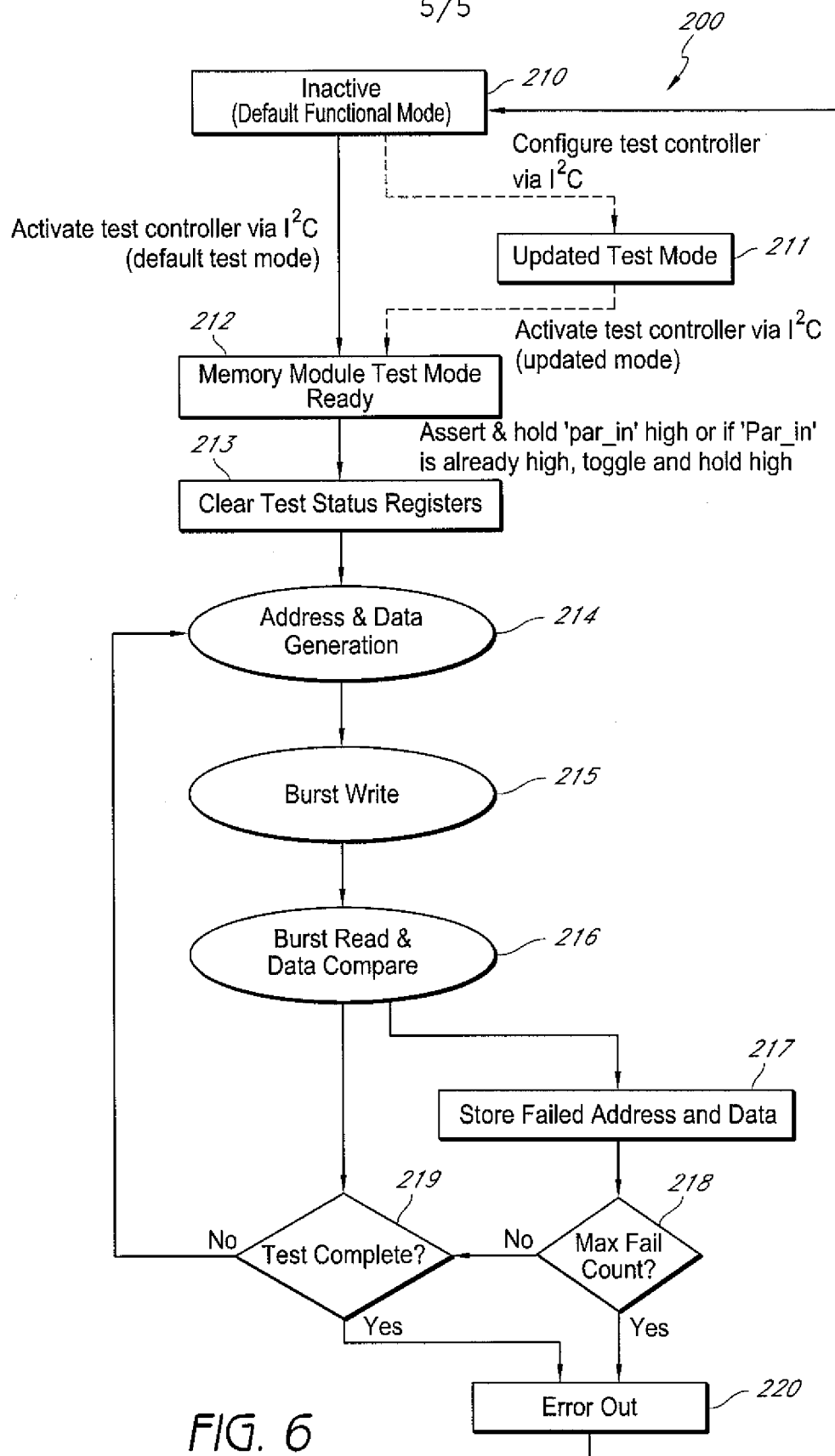


FIG. 5



Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Bruce S. Itchkawitz/Jodi Marasia

Attorney Docket Number:

NETL.046A

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	330	330
Utility Search Fee	1111	1	540	540
Utility Examination Fee	1311	1	220	220

Pages:

Claims:

Claims in excess of 20	1202	15	52	780
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Miscellaneous-Filing:

Petition:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1870

Electronic Acknowledgement Receipt

EFS ID:	5146487
Application Number:	12422925
International Application Number:	
Confirmation Number:	7596
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	45338
Filer:	Bruce S. Itchkawitz/angela chavez
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046A
Receipt Date:	13-APR-2009
Filing Date:	
Time Stamp:	20:37:42
Application Type:	Utility under 35 USC 111(a)

Payment information:

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Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1870
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1	Transmittal Letter	Transmittal_NETL046A.pdf	64252 e1a2265803abd52e84ce7177057cf6e2626cb10	no	2
Warnings:					
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2	Application Data Sheet	ADS_NETL046A.pdf	204429 c9b100a8d9c5faee5c3f08c3b64e07e8436a4d93	no	5
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3		Specification_and_Drawings- NETL046A.pdf	1323836 553581aa53104a787e42035eb52b1a3558993c90	yes	35
	Multipart Description/PDF files in .zip description				
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	Specification		1	24	
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4	Fee Worksheet (PTO-06)	fee-info.pdf	36995 824ff89209e59b45d082d2719db8a52f9ea8d79f	no	2
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Filing Date: 04/13/2009

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875						Application or Docket Number 12/422,925				
APPLICATION AS FILED – PART I										
(Column 1)		(Column 2)		SMALL ENTITY		OR	LARGE ENTITY			
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)				
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A		N/A	330				
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A		N/A	540				
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A		N/A	220				
TOTAL CLAIMS (37 CFR 1.16(i))	35	15	X\$ 26		2X\$52	780				
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	0	X\$110		1X\$220	0				
APPLICATION SIZE FEE (37 CFR 1.16(s))			If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$270 (\$135 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR							
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))			195		390					
			TOTAL		TOTAL	1870				
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APPLICATION AS AMENDED – PART II										
(Column 1)		(Column 2)		(Column 3)		SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)	ADDITIONAL FEE (\$)			
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	Application Size Fee (37 CFR 1.16(s))									
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/422,925	04/13/2009	2829	1870	NETL.046A	35	3

CONFIRMATION NO. 7596

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FILING RECEIPT



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Date Mailed: 05/05/2009

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Applicant(s)

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Soonju Choi, Irvine, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications

If Required, Foreign Filing License Granted: 04/24/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/422,925**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

Preliminary Class

324

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/422,925	04/13/2009	Hyun Lee	NETL.046A

CONFIRMATION NO. 7596

FORMALITIES LETTER



OC000000035665316

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 05/05/2009

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The oath or declaration is missing.

A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.

Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of \$130 for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is \$130 for a non-small entity

- \$130 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/twyatt/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2829
	Examiner	Unknown
SHEET 1 OF 4	Attorney Docket No.	NETL.046A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
	11	4,910,597	03-20-1990	Harada et al.	
	12	4,942,556	07-17-1990	Sasaki et al.	
	13	4,987,321	01-22-1991	Toohy	
	14	5,138,619	08-11-1992	Fasang et al.	
	15	5,173,906	12-22-1992	Dreibelbis et al.	
	16	5,222,066	06-22-1993	Grula et al.	
	17	5,241,503	08-31-1993	Cheng	
	18	5,304,856	04-19-1994	Rainal	
	19	5,337,254	08-09-1994	Knee et al.	
	20	5,359,235	10-25-1994	Coyle et al.	
	21	5,394,037	02-28-1995	Josephson et al.	
	22	5,430,335	07-04-1995	Tanoi	
	23	5,525,917	06-11-1996	Wong et al.	
	24	5,841,296	11-24-1998	Churcher et al.	
	25	5,914,543	06-22-1999	Scherpenberg, et al.	
	26	6,000,048	12-07-1999	Krishna et al.	
	27	6,044,481	03-28-2000	Kornachuk et al.	
	28	6,070,217	05-30-2000	Connolly et al.	
	29	6,169,696	01-02-2001	Bissey	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2829
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 2 OF 4	Attorney Docket No.	NETL.046A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	30	6,194,959	02-27-2001	Kamoshida et al.	
	31	6,216,240	04-10-2001	Won et al.	
	32	6,467,056	10-15-2002	Satou et al.	
	33	6,560,740	05-06-2003	Zuraski, Jr. et al.	
	34	6,614,702	09-02-2003	Lee	
	35	6,721,150	04-13-2004	Guerrero, Jr.	
	36	6,812,869	11-02-2004	Rahman et al.	
	37	6,829,728	12-07-2004	Cheng et al.	
	38	6,918,072	07-12-2005	Cowels et al.	
	39	6,928,024	08-09-2005	Pfeiffer et al.	
	40	6,928,593	08-09-2005	Halbert et al.	
	41	6,930,509	08-16-2005	Banik	
	42	6,934,900	08-23-2005	Cheng et al.	
	43	7,036,064	04-25-2006	Kebichi et al.	
	44	7,053,470	05-30-2006	Sellers et al.	
	45	7,062,696	06-13-2006	Barry et al.	
	46	7,184,915	02-27-2007	Hansquine et al.	
	47	7,190,210	03-13-2007	Azrai et al.	
	48	7,203,873	04-10-2007	Adams	
	49	7,253,652	08-07-2007	Azimi et al.	
	50	7,284,166	10-16-2007	Zappa et al.	
	51	2001/0048342	12-06-2001	Yoshida et al.	
	52	2002/0000847	01-03-2002	Taguchi	
	53	2002/0131535	09-19-2002	Huber	
	54	2002/0140523	10-03-2002	Park et al.	
	55	2003/0098742	05-29-2003	Nakagawa et al.	
	56	2003/0126346	07-03-2003	Kuo	
	57	2003/0197797	10-23-2003	Segura	
	58	2003/0218491	11-27-2003	Nagasue	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2829
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 3 OF 4	Attorney Docket No.	NETL.046A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	59	2004/0155702	08-12-2004	Danielsson	
	60	2004/0199843	10-07-2004	Hansquine et al.	
	61	2005/0093620	05-05-2005	Ho et al.	
	62	2005/0127989	06-16-2005	Miyagi	
	63	2005/0289423	12-29-2005	Yabuta	
	64	2006/0082383	04-20-2006	Choi	
	65	2006/0107156	05-18-2006	Lee et al.	
	66	2006/0140015	06-26-2006	Kasamsetty	
	67	2006/0144015	07-06-2006	Cash et al.	
	68	2006/0147217	07-06-2006	Hahin et al.	
	69	2006/0192653	08-31-2006	Atkinson et al.	
	70	2006/0242458	10-26-2006	Feldman et al.	
	71	2006/0262586	11-23-2006	Solomon et al.	
	72	2006/0271748	11-30-2006	Jain et al.	
	73	2007/0030814	02-08-2007	Shin et al.	
	74	2007/0058471	03-15-2007	Rajan et al.	
	75	2007/0079199	04-05-2007	Chorn et al.	
	76	2007/0109707	05-17-2007	Honda	
	77	2007/0152743	07-05-2007	Keeth et al.	
	78	2007/0204075	08-30-2007	Rajan et al.	
	79	2007/0223296	09-27-2007	Miller et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2829
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 4 OF 4	Attorney Docket No.	NETL.046A

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
	80	DER-CHANG ET AL. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, April 2002, Volume 21, Issue 4, Pages 449-465	
	81	BAYARD "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
	82	MUTOH et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, November 2-6, 2004, Volume 1, pages 963-968	
	83	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

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Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

Electronic Acknowledgement Receipt

EFS ID:	5671145
Application Number:	12422925
International Application Number:	
Confirmation Number:	7596
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Valerie Jones
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046A
Receipt Date:	09-JUL-2009
Filing Date:	13-APR-2009
Time Stamp:	15:47:15
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		IDS_NETL046A.pdf	282593 810d066f169d72429b851fe9ac5670a0c5c74bee	yes	5

	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	1	
	Information Disclosure Statement (IDS) Filed (SB/08)		2	5	
Warnings:					
Information:					
2	NPL Documents	Der-Chang_2002.PDF	351911	no	17
			9d99ac67a7fedfd832c8a72c41db496210e7045a		
Warnings:					
Information:					
3	NPL Documents	Bayard_1999.PDF	426723	no	9
			2378ecec191f2e9ec715ed2b1ed73de94916c07c		
Warnings:					
Information:					
4	NPL Documents	Mutoh_2004.pdf	529276	no	6
			65af400f07b5c6424cab138fa7ad7fb2b39f946e		
Warnings:					
Information:					
5	NPL Documents	Sekiguchi_1995.pdf	620932	no	6
			cea3ee0409d761ddb879a98ffb38bd9d344b9163		
Warnings:					
Information:					
Total Files Size (in bytes):			2211435		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT

Applicant : Hyun Lee et al.
App. No. : 12/422,925
Filed : April 13, 2009
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Examiner : Unknown
Art Unit : 2829
Conf No. : 7596

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a PTO/SB/08 Equivalent listing 83 references to be considered by the Examiner. Also enclosed are 4 non-patent literature as listed on the Information Disclosure Statement.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 11-1410.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: _____

7/9/09

By: _____

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Please Direct All Correspondence to Customer Number 20995

RESPONSE TO FORMALITIES NOTICE

Applicant	: Lee et al.
App. No	: 12/422,925
Filed	: April 13, 2009
For	: MEMORY BOARD WITH SELF- TESTING CAPABILITY
Art Unit	: 2829

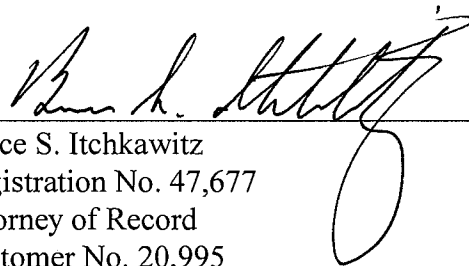
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without a Declaration and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following.

- (X) A Declaration in 5 pages.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.



Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

Doc Code: OATH

Document Description: Oath or declaration filed

PTO/SB/01 (10-08)
Approved for use through 06/30/2010. OMB 0651-0032
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

DECLARATION FOR UTILITY OR DESIGN PATENT APPLICATION (37 CFR 1.63)	Attorney Docket Number	NETL.046A		
	First Named Inventor	Hyun Lee		
	COMPLETE IF KNOWN			
	Application Number	12/422,925		
	Filing Date	April 13, 2009		
	Art Unit	Unknown		
☐ Declaration Submitted With Initial Filing	OR	☒ Declaration Submitted after Initial Filing (surcharge (37 CFR 1.16 (f)) required)	Examiner Name	Unknown

I hereby declare that: (1) Each inventor's residence, mailing address, and citizenship are as stated below next to their name; and (2) I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is claimed and for which a patent is sought on the invention entitled:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

(Title of the Invention)

the application of which

☐ is attached hereto

OR

☒ was filed on (MM/DD/YYYY) April 13, 2009 as United States Application Number or PCT International

Application Number 12/422,925 and was amended on (MM/DD/YYYY) (if applicable).

I hereby state that I have reviewed and understand the contents of the above identified application, including the claims, as amended by any amendment specifically referred to above.

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56, including for continuation-in-part applications, material information which became available between the filing date of the prior application and the national or PCT international filing date of the continuation-in-part application.

Authorization To Permit Access To Application by Participating Offices

☐ If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), and any other intellectual property offices in which a foreign application claiming priority to the above-identified application is filed access to the above-identified patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, or other intellectual property office in which a foreign application claiming priority to the above-identified application is filed to have access to the application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the application-as-filed with respect to: 1) the above-identified application, 2) any foreign application to which the above-identified application claims priority under 35 USC 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the above-identified US application, and 3) any U.S. application from which benefit is sought in the above-identified application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing the Authorization to Permit Access to Application by Participating Offices.

[Page 1 of 3]

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

DECLARATION — Utility or Design Patent Application

Claim of Foreign Priority Benefits

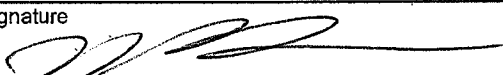
I hereby claim foreign priority benefits under 35 U.S.C. 119(a)-(d) or (f), or 365(b) of any foreign application(s) for patent, inventor's or plant breeder's rights certificate(s), or 365(a) of any PCT international application which designated at least one country other than the United States of America, listed below and have also identified below, by checking the box, any foreign application for patent, inventor's or plant breeder's rights certificate(s), or any PCT international application having a filing date before that of the application on which priority is claimed.

Prior Foreign Application Number(s)	Country	Foreign Filing Date (MM/DD/YYYY)	Priority Not Claimed	Certified Copy Attached?	
				YES	NO
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐

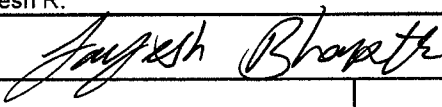
☐ Additional foreign application numbers are listed on a supplemental priority data sheet PTO/SB/02B attached hereto.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

DECLARATION — Utility or Design Patent Application

Direct all correspondence to:	☒	The address associated with Customer Number:	20995	OR	☐	Correspondence address below
Name						
Address						
City State						ZIP
Country T		elephone		Email		
WARNING:						
Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form P TO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213 (a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available. Petitioner/applicant is advised that documents which form the record of a patent application (such as the PTO/SB/01) are placed into the Privacy Act system of records DEPARTMENT OF COMMERCE, COMMERCE-PAT-7, System name: <i>Patent Application Files</i>. Documents not retained in an application file (such as the PTO-2038) are placed into the Privacy Act system of COMMERCE/PAT-TM-10, System name: <i>Deposit Accounts and Electronic Funds Transfer Profiles</i>. I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under 18 U.S.C. 1001 and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.						
NAME OF SOLE OR FIRST INVENTOR:				☐ A petition has been filed for this unsigned inventor		
Given Name (first and middle [if any])				Family Name or Surname		
Hyun				Lee		
Inventor's Signature					Date	
					10/1/09	
Residence: City	State	Country	Citizenship			
Ladera Ranch	CA	US	US			
Mailing Address						
21 Thalia						
City State	Zip		Country			
Ladera Ranch	92694		US			
☒ Additional inventors or a legal representative are being named on the ¹ supplemental sheet(s) PTO/SB/02A or 02LR attached hereto.						

DECLARATION	ADDITIONAL INVENTOR(S) Supplemental Sheet
Page <u>1</u> of <u>1</u>	

Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Jayesh R.		Bhakta	
Inventor's Signature 		Date <u>10-1-09</u>	
Residence: City Cerritos	State CA	Country US	Citizenship US
Mailing Address 12220 Rose Street			
City Cerritos	State CA	Zip 90703	Country US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Soonju		Choi	
Inventor's Signature 		Date <u>10/1/2009</u>	
Residence: City Irvine	State CA	Country US	Citizenship KR
Mailing Address 195 Woodbury			
City Irvine	State CA	Zip 92620	Country US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Inventor's Signature		Date	
Residence: City	State	Country	Citizenship
Mailing Address			
City	State	Zip	Country

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786-9199) and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	12422925			
Filing Date:	13-Apr-2009			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Bruce S. Itchkawitz/Jessica Noda			
Attorney Docket Number:	NETL.046A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late filing fee for oath or declaration	1051	1	130	130
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 3 months with \$0 paid	1253	1	1110	1110
Miscellaneous:				
Total in USD (\$)				1240

Electronic Acknowledgement Receipt

EFS ID:	6189140
Application Number:	12422925
International Application Number:	
Confirmation Number:	7596
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Valerie Jones
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046A
Receipt Date:	01-OCT-2009
Filing Date:	13-APR-2009
Time Stamp:	19:26:29
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1240
RAM confirmation Number	5024
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		mparts.pdf	319881 508dc0d6f2d7499ff2091844f7f73571e289dc8b	yes	6
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Applicant Response to Pre-Exam Formalities Notice		1	1	
	Oath or Declaration filed		2	6	
Warnings:					
Information:					
2	Fee Worksheet (PTO-875)	fee-info.pdf	32460 50ab98ba80701b10f973a77db4ffc5df40a5abe1	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			352341		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/422,925	04/13/2009	2829	2000	NETL.046A	35	3

CONFIRMATION NO. 7596

UPDATED FILING RECEIPT



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 10/14/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/044,801 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,839 04/14/2008

Foreign Applications

If Required, Foreign Filing License Granted: 04/24/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/422,925**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

Preliminary Class

324

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER**Title 35, United States Code, Section 184****Title 37, Code of Federal Regulations, 5.11 & 5.15****GRANTED**

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as

set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).



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UNITED STATES DEPARTMENT OF COMMERCE
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/422,925

04/13/2009

Hyun Lee

NETL.046A

7596

20995

7590

12/08/2010

KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

EXAMINER

CHUNG, PHUNG M

ART UNIT

PAPER NUMBER

2117

NOTIFICATION DATE

DELIVERY MODE

12/08/2010

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jcartee@kmob.com
efiling@kmob.com
eOAPilot@kmob.com

Office Action Summary	Application No. 12/422,925	Applicant(s) LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTH(S) OR THIRTY (30) DAYS, WHICHEVER IS LONGER, FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☐ Responsive to communication(s) filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims

- 4) ☒ Claim(s) 1-35 is/are pending in the application.
- 4a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 5) ☐ Claim(s) ____ is/are allowed.
- 6) ☒ Claim(s) 1-35 is/are rejected.
- 7) ☐ Claim(s) ____ is/are objected to.
- 8) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

Application Papers

- 9) ☒ The specification is objected to by the Examiner.
- 10) ☒ The drawing(s) filed on 13 April 2009 is/are: a) ☐ accepted or b) ☒ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).
- 11) ☐ The oath or declaration is objected to by the Examiner. Note the attached Office Action or form PTO-152.

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some * c) ☐ None of:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

* See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|---|---|
| 1) ☒ Notice of References Cited (PTO-892) | 4) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☐ Notice of Draftperson's Patent Drawing Review (PTO-948) | 5) ☐ Notice of Informal Patent Application |
| 3) ☒ Information Disclosure Statement(s) (PTO/SB/08)
Paper No(s)/Mail Date <u>7/9/09</u> . | 6) ☐ Other: ____. |

Specification

1. Paragraph (0001) lines 5 and 7, U.S. Patent Application No. are missing.

Applicant is requested to fill out the missing No. and/or update the status of these applications.

Drawings

2. New corrected drawings in compliance with 37 CFR 1.121(d) are required in this application because all rectangular boxes of Figures 1-3 should have numbers and elements to indicate which numbers and elements the boxes represent. Applicant is advised to employ the services of a competent patent draftsman outside the Office, as the U.S. Patent and Trademark Office no longer prepares new drawings. The corrected drawings are required in reply to the Office action to avoid abandonment of the application. The requirement for corrected drawings will not be held in abeyance.

INFORMATION ON HOW TO EFFECT DRAWING CHANGES

Replacement Drawing Sheets

Drawing changes must be made by presenting replacement sheets which incorporate the desired changes and which comply with 37 CFR 1.84. An explanation of the changes made must be presented either in the drawing amendments section, or remarks, section of the amendment paper. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either "Replacement Sheet" or "New Sheet" pursuant to 37 CFR 1.121(d). A replacement sheet must include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of the amended drawing(s) must not be labeled as "amended." If the changes to the drawing figure(s) are not accepted by the examiner, applicant will be notified of any required corrective action in the next Office action. No further drawing submission will be required, unless applicant is notified.

Identifying indicia, if provided, should include the title of the invention, inventor's name, and application number, or docket number (if any) if an application number has not

Art Unit: 2117

been assigned to the application. If this information is provided, it must be placed on the front of each sheet and within the top margin.

Annotated Drawing Sheets

A marked-up copy of any amended drawing figure, including annotations indicating the changes made, may be submitted or required by the examiner. The annotated drawing sheet(s) must be clearly labeled as "Annotated Sheet" and must be presented in the amendment or remarks section that explains the change(s) to the drawings.

Timing of Corrections

Applicant is required to submit acceptable corrected drawings within the time period set in the Office action. See 37 CFR 1.85(a). Failure to take corrective action within the set period will result in ABANDONMENT of the application.

If corrected drawings are required in a Notice of Allowability (PTOL-37), the new drawings **MUST** be filed within the **THREE MONTH** shortened statutory period set for reply in the "Notice of Allowability." Extensions of time may **NOT** be obtained under the provisions of 37 CFR 1.136 for filing the corrected drawings after the mailing of a Notice of Allowability.

Claim Rejections - 35 USC § 112

3. The following is a quotation of the second paragraph of 35 U.S.C. 112:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

4. Claims 1-35 are rejected under 35 U.S.C. 112, second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention.

As per claim 1, line 1, the preamble of the claim is "a self-testing memory module" however, there isn't any testing means, testing devices or testing circuit recites in the body of the claim for self-testing and/or how it doing the self-test is unclear.

Appropriate correction and/or clarification are required.

As per claims 2-19 and 21-28, line 1, "The memory module" should be changed to - - The self-testing memory module - - .

As per claim 20, , line 1, the preamble of the claim is "a self-testing memory module" however, there isn't any tester, test device or test circuit recites in the body of the claim for self-testing and/or how it doing the self-test is unclear;

Lines 8-9, "a data module comprising at least one data handler operatively coupled to a corresponding plurality of data ports" is not clear how it's coupled to the corresponding plurality of data ports when there is only one data handler? Appropriate correction and/or clarification are required.

As per claim 21, line 1, "wherein each of the data handlers" does not have a clear antecedent basis. appropriate correction is required.

As per claim 29, lines 1-2, "A method of self-testing a memory module, comprising:

providing a self-testing memory module, comprising:" It is not clear which part of the claim is the preamble and which part of the claim is the body of the claim. It is also not clear what comprising what?

Lines 1-15, "A method of self-testing a memory module, comprising: providing a self-testing memory module, comprising:...data ports." The claim as a whole is not clear as to what it meant. There is only one step of providing the self-testing memory module, but how it does the self-test and what is the test result are unclear. Appropriate correction and/or clarification are required.

Art Unit: 2117

As per claim 30, line 6, "the data handler" should be changed to - - the data handlers - -. Appropriate correction is required.

As per claim 31, line 2, "the data handler" should be changed to - - the data handlers - -. Appropriate correction is required.

As per claims 32-34, these claims are also rejected because they dependent upon the rejected base claim.

As per claim 35, lines 4 and 6, "the data handler" should be changed to - - the data handlers - -. Appropriate correction is required.

5. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Phung M. Chung whose telephone number is (571)272-3818. The examiner can normally be reached on Monday to Friday.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Beausoliel, Robert can be reached on 571-272-3645. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Art Unit: 2117

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Phung M. Chung/
Primary Examiner
Art Unit 2117

Notice of References Cited	Application/Control No. 12/422,925	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner Phung M. Chung	Art Unit 2117	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-5,051,997	09-1991	Sakashita et al.	714/732
*	B	US-5,033,048	07-1991	Pierce et al.	714/719
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 12422925	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner Phung M Chung	Art Unit 2117

SEARCHED			
Class	Subclass	Date	Examiner
714	718,719,736,819,733,734,30	12/2/10	pmc
365	200,201	12/2/10	pmc

SEARCH NOTES		
Search Notes	Date	Examiner

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner

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<i>Index of Claims</i> 	Application/Control No. 12422925	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner Phung M Chung	Art Unit 2117

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	12/02/2010								
	1	v								
	2	v								
	3	v								
	4	v								
	5	v								
	6	v								
	7	v								
	8	v								
	9	v								
	10	v								
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	29	v								
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	31	v								
	32	v								
	33	v								
	34	v								
	35	v								



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BIB DATA SHEET

CONFIRMATION NO. 7596

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.		
12/422,925	04/13/2009	324	2117	NETL.046A		
RULE						
APPLICANTS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Soonju Choi, Irvine, CA;						
** CONTINUING DATA ***** This appln claims benefit of 61/044,801 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,839 04/14/2008						
** FOREIGN APPLICATIONS *****						
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 04/24/2009						
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and Acknowledged <u>/PHUNG M. CHUNG/</u> Examiner's Signature		☐ Met after Allowance Initials	STATE OR COUNTRY	SHEETS DRAWINGS	TOTAL CLAIMS	INDEPENDENT CLAIMS
			CA	5	35	3
ADDRESS						
KNOBBE MARTENS OLSON & BEAR LLP 2040 MAIN STREET FOURTEENTH FLOOR IRVINE, CA 92614 UNITED STATES						
TITLE						
MEMORY BOARD WITH SELF-TESTING CAPABILITY						
FILING FEE RECEIVED 2000	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees			
			☐ 1.16 Fees (Filing)			
			☐ 1.17 Fees (Processing Ext. of time)			
			☐ 1.18 Fees (Issue)			
			☐ Other _____			
☐ Credit						

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	1	(self-testing memory module) or (built-in self test memory module) or (in-circuit testing memory module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:30
L2	9	(self-testing memory module) or (built-in self test memory module) or (in-circuit testing memory module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:31
L3	360581	printed circuit board	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:31
L4	427450	memory device\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:31

L5	122011	memory controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:31
L6	618473	computer system	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:31
L7	109200	control module	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:34
L8	169	(control module) same (address signal\$1 and control signal\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:34
L9	9	data module same (data handler\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:35
L10	7	I2 and I3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:35

L11	1	I10 and I4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:35
L12	1	I2 and I4 and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:36
L13	6	I10 and I5	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:40
L16	12569	built-in selft test or built-in test or in-circuit test or (BIST)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:41
L17	6	I13 and I16	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:41
L18	84144	address signal\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:42

L19	960820	control signal\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:42
L20	5635131	data	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:42
L22	6	l17 and (l18 or l19 or l20)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:43
L23	8476	(714/733,734,718,719,736,735,819).ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:44
L24	1	l22 and l23	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:44

L25	29	(("3,660,675") or ("3,757,235") or ("4,305,091") or ("4,586,168") or ("4,701,845") or ("4,752,741") or ("4,782,487") or ("4,837,743") or ("4,885,799") or ("4,903,266") or ("4,910,597") or ("4,942,556") or ("4,987,321") or ("5,138,619") or ("5,173,906") or ("5,222,066") or ("5,241,503") or ("5,304,856") or ("5,337,254") or ("5,359,235") or ("5,394,037") or ("5,430,335") or ("5,525,917") or ("5,841,296") or ("5,914,543") or ("6,000,048") or ("6,044,481") or ("6,070,217") or ("6,169,696")).PN.	US-PGPUB; USPAT; JPO	OR	OFF	2010/11/29 14:47
L28	21	(("20040155702") or ("20040199843") or ("20050093620") or ("20050127989") or ("20050289423") or ("20060082383") or ("20060107156") or ("20060140015") or ("20060144015") or ("20060147217") or ("20060192653") or ("20060242458") or ("20060262586") or ("20060271748") or ("20070030814") or ("20070058471") or ("20070079199") or ("20070109707") or ("20070152743") or ("20070204075") or ("20070223296")).PN.	US-PGPUB; USPAT; JPO	OR	OFF	2010/11/29 14:48
L29	29	(("6,194,959") or ("6,216,240") or ("6,467,056") or ("6,560,740") or ("6,614,702") or ("6,721,150") or ("6,812,869") or ("6,829,728") or ("6,918,072") or ("6,928,024") or ("6,928,593") or ("6,930,509") or ("6,934,900") or ("7,036,064") or ("7,053,470") or ("7,062,696") or ("7,184,915") or ("7,190,210") or ("7,203,873") or ("7,253,652") or ("7,284,166") or ("20010048342") or ("20020000847") or ("20020131535") or ("20020140523") or ("20030098742") or ("20030126346") or ("20030197797") or ("20030218491")).PN.	US-PGPUB; USPAT; JPO	OR	OFF	2010/11/29 14:49

L30	2	I2 and I4	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2010/11/29 14:59
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11/ 29/ 10 3:02:44 PM

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2629 2117
	Examiner	Unknown Phung M. Chung
SHEET 1 OF 4	Attorney Docket No.	NETL.046A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
/PC/	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
	11	4,910,597	03-20-1990	Harada et al.	
	12	4,942,556	07-17-1990	Sasaki et el.	
	13	4,987,321	01-22-1991	Toohey	
	14	5,138,619	08-11-1992	Fasang et al.	
	15	5,173,906	12-22-1992	Dreibelbis et al.	
	16	5,222,066	06-22-1993	Grula et al.	
	17	5,241,503	08-31-1993	Cheng	
	18	5,304,856	04-19-1994	Rainal	
	19	5,337,254	08-09-1994	Knee et al.	
	20	5,359,235	10-25-1994	Coyle et al.	
	21	5,394,037	02-28-1995	Josephson et al.	
	22	5,430,335	07-04-1995	Tanoi	
	23	5,525,917	06-11-1996	Wong et al.	
	24	5,841,296	11-24-1998	Churcher et al.	
	25	5,914,543	06-22-1999	Scherpenberg, et al.	
	26	6,000,048	12-07-1999	Krishna et al.	
	27	6,044,481	03-28-2000	Kornachuk et al.	
	28	6,070,217	05-30-2000	Connolly et al.	
/PC/	29	6,169,696	01-02-2001	Bissey	

Examiner Signature	/Phung Chung/	Date Considered	11/29/2010
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***Examiner:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2009 2117
	Examiner	Unknown Phung Chung
SHEET 2 OF 4	Attorney Docket No.	NETL.046A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
/PC/	30	6,194,959	02-27-2001	Kamoshida et al.	
	31	6,216,240	04-10-2001	Won et al.	
	32	6,467,056	10-15-2002	Satou et al.	
	33	6,560,740	05-06-2003	Zuraski, Jr. et al.	
	34	6,614,702	09-02-2003	Lee	
	35	6,721,150	04-13-2004	Guerrero, Jr.	
	36	6,812,869	11-02-2004	Rahman et al.	
	37	6,829,728	12-07-2004	Cheng et al.	
	38	6,918,072	07-12-2005	Cowels et al.	
	39	6,928,024	08-09-2005	Pfeiffer et al.	
	40	6,928,593	08-09-2005	Halbert et al.	
	41	6,930,509	08-16-2005	Banik	
	42	6,934,900	08-23-2005	Cheng et al.	
	43	7,036,064	04-25-2006	Kebichi et al.	
	44	7,053,470	05-30-2006	Sellers et al.	
	45	7,062,696	06-13-2006	Barry et al.	
	46	7,184,915	02-27-2007	Hansquine et al.	
	47	7,190,210	03-13-2007	Azrai et al.	
	48	7,203,873	04-10-2007	Adams	
	49	7,253,652	08-07-2007	Azimi et al.	
	50	7,284,166	10-16-2007	Zappa et al.	
	51	2001/0048342	12-06-2001	Yoshida et al.	
	52	2002/0000847	01-03-2002	Taguchi	
	53	2002/0131535	09-19-2002	Huber	
	54	2002/0140523	10-03-2002	Park et al.	
	55	2003/0098742	05-29-2003	Nakagawa et al.	
	56	2003/0126346	07-03-2003	Kuo	
	57	2003/0197797	10-23-2003	Segura	
/PC/	58	2003/0218491	11-27-2003	Nagasue	

Examiner Signature	/Phung Chung/	Date Considered	11/29/2010
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2000 2117
	Examiner	Unknown Phung Chung
SHEET 3 OF 4	Attorney Docket No.	NETL.046A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
/PC/	59	2004/0155702	08-12-2004	Danielsson	
	60	2004/0199843	10-07-2004	Hansquaine et al.	
	61	2005/0093620	05-05-2005	Ho et al.	
	62	2005/0127989	06-16-2005	Miyagi	
	63	2005/0289423	12-29-2005	Yabuta	
	64	2006/0082383	04-20-2006	Choi	
	65	2006/0107156	05-18-2006	Lee et al.	
	66	2006/0140015	06-26-2006	Kasamsetty	
	67	2006/0144015	07-06-2006	Cash et al.	
	68	2006/0147217	07-06-2006	Hahin et al.	
	69	2006/0192653	08-31-2006	Atkinson et al.	
	70	2006/0242458	10-26-2006	Feldman et al.	
	71	2006/0262586	11-23-2006	Solomon et al.	
	72	2006/0271748	11-30-2006	Jain et al.	
	73	2007/0030814	02-08-2007	Shin et al.	
	74	2007/0058471	03-15-2007	Rajan et al.	
	75	2007/0079199	04-05-2007	Chorn et al.	
	76	2007/0109707	05-17-2007	Honda	
	77	2007/0152743	07-05-2007	Keeth et al.	
	78	2007/0204075	08-30-2007	Rajan et al.	
/PC/	79	2007/0223296	09-27-2007	Miller et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

NON PATENT LITERATURE DOCUMENTS			
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Examiner Signature	/Phung Chung/	Date Considered	11/29/2010
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i> SHEET 4 OF 4	Application No.	12/422,925
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2029 2117
	Examiner	Unknown Phung Chung
	Attorney Docket No.	NETL.046A

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
/PC/	80	DER-CHANG ET AL. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, April 2002, Volume 21, Issue 4, Pages 449-465	
/PC/	81	BAYARD "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
/PC/	82	MUTOH et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, November 2-6, 2004, Volume 1, pages 963-968	
/PC/	83	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

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061909

Examiner Signature	/Phung Chung/	Date Considered	11/29/2010
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Lee, et al.
App. No. : 12/422925
Filed : April 13, 2009
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Examiner : Phung M. Chung
Art Unit : 2117
Conf No. : 7596

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March 3, 2011

(Date)

/Sean Ambrosius/

Sean Ambrosius, Reg. No. 65,290

AMENDMENT AND RESPONSE TO
NON-FINAL OFFICE ACTION DATED DECEMBER 8, 2010

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

In response to the December 8, 2010 Office Action ("Office Action"), please amend the above-identified application as follows:

Amendments to the Specification begin on page 2 of this paper.

Amendments to the Drawings begin on page 3 of this paper.

Amendments to the Claims are reflected in the listing of claims which begins on page 4 of this paper.

Remarks begin on page 10 of this paper.

Application No.: 12/422925
Filing Date: April 13, 2009

AMENDMENTS TO THE SPECIFICATION

Amendments to the specification are provided below in highlighted form. Insertions appear as underlined text (e.g., insertions) while deletions appear as strikethrough text or double-bracketed text (e.g., ~~deletions~~ or [[deletions]]).

On page 1, please amend paragraph number [0001] as follows:

[0001] The present application claims the benefit of priority from U.S. Provisional Application No. 61/044,801, filed April 14, 2008, from U.S. Provisional Application No. [[No.]] 61/044,825, filed April 14, 2008, and from U.S. Provisional Application No. 61/044,839, filed April 14, 2008, each of which are incorporated in their entirety by reference herein. This application is related to U.S. Patent Application No. 12/422,912 [[_____]], filed on even date herewith and entitled “Self-Adjusting Damper”, and to U.S. Application No. 12/422,853 [[_____]], filed on even data herewith and entitled “Circuit Providing Load Isolation and Noise Reduction”, both of which are incorporated in their entirety by reference herein.

Application No.: 12/422925
Filing Date: April 13, 2009

AMENDMENTS TO THE DRAWINGS

Applicants are submitting herewith three replacement sheets in which Figures 1-3 have been amended to include various labels as shown on the annotated version of the three replacement sheets included in the attached Appendix. Applicants submit that the amended drawings include no new matter.

AMENDMENTS TO THE CLAIMS

The listing of claims replaces all prior versions and listings of claims. Only those claims being amended herein show their changes in highlighted form, where insertions appear as underlined text (e.g., insertions) while deletions appear as strikethrough text or double-bracketed text (e.g., ~~deletions~~ or ~~[[deletions]]~~).

1. **(Currently Amended)** A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.
2. **(Currently Amended)** The self-testing memory module of Claim 1, wherein the plurality of data handlers comprise at least two physically separate components mounted on the printed circuit board.
3. **(Currently Amended)** The self-testing memory module of Claim 2, wherein the plurality of data handlers comprise at least two physically separate integrated circuit packages.
4. **(Currently Amended)** The self-testing memory module of Claim 3, wherein the physically separate integrated circuit packages are mounted on different portions of the printed circuit board.

5. **(Currently Amended)** The self-testing memory module of Claim 2, wherein each of the plurality of data handlers is positioned on the printed circuit board proximate to the corresponding plurality of data ports.

6. **(Currently Amended)** The self-testing memory module of Claim 5, wherein each of the plurality of data handlers is positioned closer to the corresponding plurality of data ports than to the other data ports of the plurality of memory devices.

7. **(Currently Amended)** The self-testing memory module of Claim 6, wherein each of the data handlers is further configured to read from the corresponding plurality of data ports and further comprises a verification element for checking for failures in the operation of the memory devices by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler for writing and ~~written~~ to the corresponding plurality of data ports.

8. **(Currently Amended)** The self-testing memory module of Claim 7, wherein the verification element is configured to verify that the data read from the corresponding plurality of data ports corresponds to the data generated by the data handler for writing and ~~written~~ to the corresponding plurality of data ports without storing a copy of the data generated by the data handler for writing ~~written~~ to the corresponding plurality of data ports.

9. **(Currently Amended)** The self-testing memory module of Claim 7, wherein the verification element calculates comparison data and compares the comparison data to the data read from the corresponding plurality of data ports.

10. **(Currently Amended)** The self-testing memory module of Claim 9, wherein the data for writing ~~signals written~~ to the corresponding plurality of data ports comprises cyclic data.

11. **(Currently Amended)** The self-testing memory module of Claim 10, wherein the verification element calculates the comparison data based on the cyclic data.

12. **(Currently Amended)** The self-testing memory module of Claim 10, wherein the verification element calculates the comparison data based on a current read address.

13. **(Currently Amended)** The self-testing memory module of Claim 7, wherein data associated with failures in the operation of the memory devices is stored in the data module and wherein memory addresses associated with failures in the operation of the memory devices are stored in the control module.

14. **(Currently Amended)** The self-testing memory module of Claim 1, each of the plurality of data handlers further configured to write data to the memory devices by selectively inputting to the corresponding plurality of data ports either data signals from the system memory controller or data signals from the data handler.

15. **(Currently Amended)** The self-testing memory module of Claim 14, wherein each of the plurality of data handlers comprises a switch which performs the selective inputting to the corresponding plurality of data ports.

16. **(Currently Amended)** The self-testing memory module of Claim 15, wherein the switch comprises a data multiplexer-demultiplexer.

17. **(Currently Amended)** The self-testing memory module of Claim 1, the control module further configured to selectively input to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the control module.

18. **(Currently Amended)** The self-testing memory module of Claim 17, wherein the control module further comprises a DRAM controller which performs the selective inputting of the address and control ports.

19. **(Currently Amended)** The self-testing memory module of Claim 1, wherein the memory module is configured to be operated in a test mode in which the control module selectively inputs to the address and control ports of the memory devices the address and control signals from the control module and wherein each of the data handlers write the data generated by the data handler to the corresponding plurality of data ports by selectively inputting to the data ports of the memory devices data signals from the data handler.

20. **(Currently Amended)** A self-testing memory module, comprising:

a printed circuit board configured to be operatively coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and

a circuit comprising:

a control module configured to generate address and control signals for testing the memory devices; and

a data module comprising at least one data handler operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate cyclic data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the cyclic data generated by the at least one data handler.

21. **(Currently Amended)** The self-testing memory module of Claim 20, wherein each of the at least one data handlers handler is further configured to read data from the corresponding plurality of data ports and further comprises a verification element for checking for failures in the operation of the memory devices by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler for writing and ~~written~~ to the corresponding plurality of data ports.

22. **(Currently Amended)** The self-testing memory module of Claim 21, wherein the verification element is configured to verify that the data read from the corresponding plurality of data ports corresponds the data generated by the data handler for writing and ~~written~~ to the corresponding plurality of data ports without storing a copy of the data written to the memory devices.

23. **(Currently Amended)** The self-testing memory module of Claim 22, wherein the verification element calculates comparison data and compares the comparison data to the data read from the corresponding plurality of data ports.

24. **(Currently Amended)** The self-testing memory module of Claim 23, wherein the verification element calculates the comparison data based on the cyclic data.

25. **(Currently Amended)** The self-testing memory module of Claim 23, wherein the verification element calculates the comparison data based on a current read address.

26. **(Currently Amended)** The self-testing memory module of Claim 21, wherein data associated with failures in the operation of the memory devices is stored in the data module and wherein memory addresses associated with failures in the operation of the memory devices are stored in the control module.

27. **(Currently Amended)** The self-testing memory module of Claim 20, wherein there are at least two data handlers and each of the at least one data handlers is operable independently from each of the other data handlers.

28. **(Currently Amended)** The self-testing memory module of Claim 20, wherein the at least one data handler is positioned on the printed circuit board proximate to the corresponding plurality of data ports.

29. **(Currently Amended)** A method of self-testing a memory module, comprising:
providing a self-testing memory module, the self-testing memory module comprising:

a printed circuit board configured to be operatively coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and

a circuit comprising:

a control module configured to generate address and control signals for testing the memory devices; and

a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports; and

generating, by the control module, address and control signals for testing the memory devices;

generating, by each of the data handlers, data for writing to the corresponding plurality of data ports; and

using the address and control signals and the generated data to test the memory devices.

30. **(Currently Amended)** The method of Claim 29, further comprising:
reading, by each of the plurality of data handlers, data from the corresponding plurality of data ports; and

checking for failures in the operation of the memory devices by each of the plurality of data handlers by verifying that data read from the corresponding plurality of data ports corresponds to the generated data for writing ~~generated by the data handler and written~~ to the corresponding plurality of data ports.

31. **(Currently Amended)** The method of Claim 30, further comprising verifying that the data read from the corresponding plurality of data ports corresponds to the generated data for writing ~~generated by the data handler and written~~ to the corresponding plurality of data ports without storing a copy of the data written to the corresponding plurality of data ports.

32. **(Original)** The method of Claim 31, further comprising:
calculating comparison data; and
comparing the comparison data to the data read from the corresponding plurality of data ports.

33. **(Original)** The method of Claim 32, wherein the data signals written to the corresponding plurality of data ports comprises cyclic data and calculating comprises calculating the data based on the cyclic data.

34. **(Original)** The method of Claim 32, wherein the data signals written to the corresponding plurality of data ports comprises cyclic data calculating comprises calculating the data based on a current read address.

35. **(Currently Amended)** The method of Claim 30, further comprising operating the memory module in a test mode in which the control module selectively inputs to the address and control ports of the memory devices the address and control signals from the control module and wherein each of the data handlers write the generated data ~~generated by the data handler~~ to the corresponding plurality of data ports by selectively inputting to the data ports of the memory devices data signals representative of the generated data ~~from the data handler~~.

REMARKS

Applicants thank the Examiner for their examination of the present application. By way of summary, Claims 1-35 were pending in this application. In the present amendment, Applicants have amended the specification and supplied replacement drawings for Figures 1-3. Additionally, Applicants have amended Claims 1-31 and 35, and Claims 32-34 remain as originally filed. Accordingly, Claims 1-35 remain pending for consideration.

Response to Comments Regarding the Specification

In the Office Action, the Examiner requests that paragraph [0001] of the specification be updated to recite the U.S. application numbers of certain related applications. Applicants have complied with this request by amending paragraph [0001] of the specification, as indicated above.

Response to the Objection to the Drawings

In the Office Action, the Examiner indicates that correction of the drawings is required, stating that “all rectangular boxes of Figures 1-3 should have numbers and elements to indicate which numbers and elements the boxes represent.” While Applicants do not necessarily agree with the Examiner’s statement, Applicants have amended Figures 1-3 to advance prosecution. Accordingly, please replace Figures 1-3 with the “Replacement Sheets” submitted herewith. Annotated versions of these replacement sheets are included in the Appendix of this response for the Examiner’s convenience.

Rejection of Claims 1-35 Under 35 U.S.C. § 112, ¶ 2

Claim 1-35 are rejected under 35 U.S.C. § 112, ¶ 2, as allegedly being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention.

Claim 1

In the Office Action, the Examiner indicates that “the preamble of the claim is ‘a self-testing memory module,’ however there isn’t any testing means, testing devices or testing circuit recites in the body of the claim for self-testing and/or how it doing the self-test is unclear.” While Applicants traverse the rejection of Claim 1, Applicants have amended Claim 1 in view of the Examiner’s comments to advance prosecution. Applicants respectfully submit that amended Claim 1 satisfies 35 U.S.C. § 112, ¶ 2 and therefore respectfully request that the rejection of Claim 1 be withdrawn.

Claims 2-19

In the Office Action, the Examiner indicates that for each of Claims 2-19, “The memory module” should be changed to “The self-testing memory module.” To advance prosecution, Applicants have amended each of Claims 2-19 accordingly. Additionally, while each of Claims 2-19 depends from rejected independent Claim 1, as indicated above, Applicants submit that amended independent Claim 1 satisfies 35 U.S.C. § 112, ¶ 2. Thus, Applicants respectfully request that the rejections of dependent Claims 2-19 be withdrawn.

Claim 20

In the Office Action, the Examiner indicates that “the preamble of the claim is ‘a self-testing memory module,’ however there isn’t any tester, test device or test circuit recites in the body of the claim for self-testing and/or how it doing the self-test is unclear.” While Applicants traverse the Examiner’s statements, Applicants have amended Claim 20 in view of the Examiner’s comments to advance prosecution.

Additionally, in the Office Action the Examiner indicates that lines 8-9 of Claim 20 recite “a data module comprising at least one data handler operatively coupled to a corresponding plurality of data ports,” (emphasis in the Office Action) but that it “is not clear how it’s coupled to the corresponding plurality of data ports when there is only one data handler?” In response to the Examiner’s request for clarification, Applicants refer to paragraph [0034] and Figure 2 of the present application. Paragraph [0034] recites, in part, that “Figure 2 is a block diagram of an example self-testing memory module 10 including eight memory devices 20 (e.g., memory devices 40a-40h) and a data module 28 comprising eight data handlers 30 (e.g., data handlers 30a-30h) in accordance with certain embodiments described herein. Each of the memory devices 20 includes an eight bit output data word and eight corresponding data ports.” Thus, in the example embodiment described in paragraph [0034] and shown in Figure 2, each data handler 30 is coupled to “the eight corresponding data ports” of the respective memory device 20. Thus, “at least one data handler [is] operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices,” as recited by Claim 20.

Thus, for the foregoing reasons, Applicants respectfully submit that amended Claim 20 satisfies 35 U.S.C. § 112, ¶ 2 and respectfully request that the rejection of amended Claim 20 be withdrawn.

Claim 21

In the Office Action, the Examiner indicates that “wherein each of the data handlers” (emphasis in Office Action) in Claim 21 does not have clear antecedent basis. Applicants have amended Claim 21 in response to the Examiner’s comments. Additionally, while Claim 21 depends from rejected independent Claim 20, as indicated above, Applicants submit that amended independent Claim 20 satisfies 35 U.S.C. § 112, ¶ 2. Thus, Applicants respectfully request that the rejection of amended dependent Claim 21 be withdrawn.

Claims 22-28

Each of Claims 22-28 depend from rejected independent Claim 20, and Claims 22-25 additionally depend from rejected dependent Claim 21. Since the Examiner has not stated a separate basis for rejecting Claims 22-28 under 35 U.S.C. § 112, ¶ 2, Applicants assume that Claims 22-28 stand rejected as being dependent upon a rejected base claim. If this is not the case, Applicants respectfully request clarification regarding the rejections of Claims 22-28. Otherwise, as indicated above, Applicants submit that both amended independent Claim 20 and amended dependent Claim 21 as well as Claims 22-28, satisfy 35 U.S.C. § 112, ¶ 2. Thus, Applicants respectfully request that the rejections of Claims 22-28 be withdrawn.

Claim 29

In the Office Action, the Examiner indicates that “[i]t is not clear which part of the claim is the preamble and which part of the claim is the body of the claim. It is also not clear what comprising what?” The Examiner further indicates that “[t]he claim as a whole is not clear as to what it meant. There is only one step of providing the self-testing memory module, but how it does the self-test and what is the test result are unclear.”

Applicants respectfully traverse the rejection of Claim 29, and specifically would like to point out that Claim 29 as originally filed included “generating...data for writing to the corresponding plurality of data ports” in addition to “providing a self-testing memory module,” contrary to the statement made in the Office Action. However, in the interest of advancing prosecution, Applicants have amended Claim 29 in view of the Examiner’s comments. Applicants respectfully submit that amended Claim 29 satisfies 35 U.S.C. § 112, ¶ 2, and respectfully request that the rejection of Claim 29 be withdrawn.

Claims 30, 31, and 35

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In the Office Action, the Examiner indicates that “the data handler” of Claims 30, 31, and 35 should be changed to “the data handlers.” To advance prosecution, Applicants have amended Claims 30, 31, and 35. Moreover, while Claims 30, 31, and 35 depend from rejected independent Claim 29, as indicated above, Applicants submit that amended independent Claim 29 satisfies 35 U.S.C. § 112, ¶ 2. Thus, Applicants respectfully request that the rejections of dependent Claims 30, 31, and 35 be withdrawn.

Claims 32-34

Each of Claims 32-34 depend from rejected independent Claim 29 and were rejected because they depend upon a rejected base claim. As indicated above, Applicants submit that amended independent Claim 29 satisfies 35 U.S.C. § 112, ¶ 2. Thus, Applicants respectfully request that the rejections of Claims 32-34 be withdrawn.

No Disclaimers or Disavowals

Although the present communication may include alterations to the application or claims, or characterizations of claim scope or referenced art, Applicants are not conceding in this application that previously pending claims are not patentable over the cited references. Rather, any alterations or characterizations are being made to facilitate expeditious prosecution of this application. Applicants reserve the right to pursue at a later date any previously pending or other broader or narrower claims that capture any subject matter supported by the present disclosure, including subject matter found to be specifically disclaimed herein or by any prior prosecution. Accordingly, reviewers of this or any parent, child or related prosecution history shall not reasonably infer that Applicants have made any disclaimers or disavowals of any subject matter supported by the present application.

Summary

For the foregoing reasons, Applicants submit that Claims 1-35 are in condition for allowance, and respectfully request such action.

Application No.: 12/422925
Filing Date: April 13, 2009

Please charge any additional fees, including any fees for additional extension of time, or credit overpayment to Deposit Account No. 11-1410.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

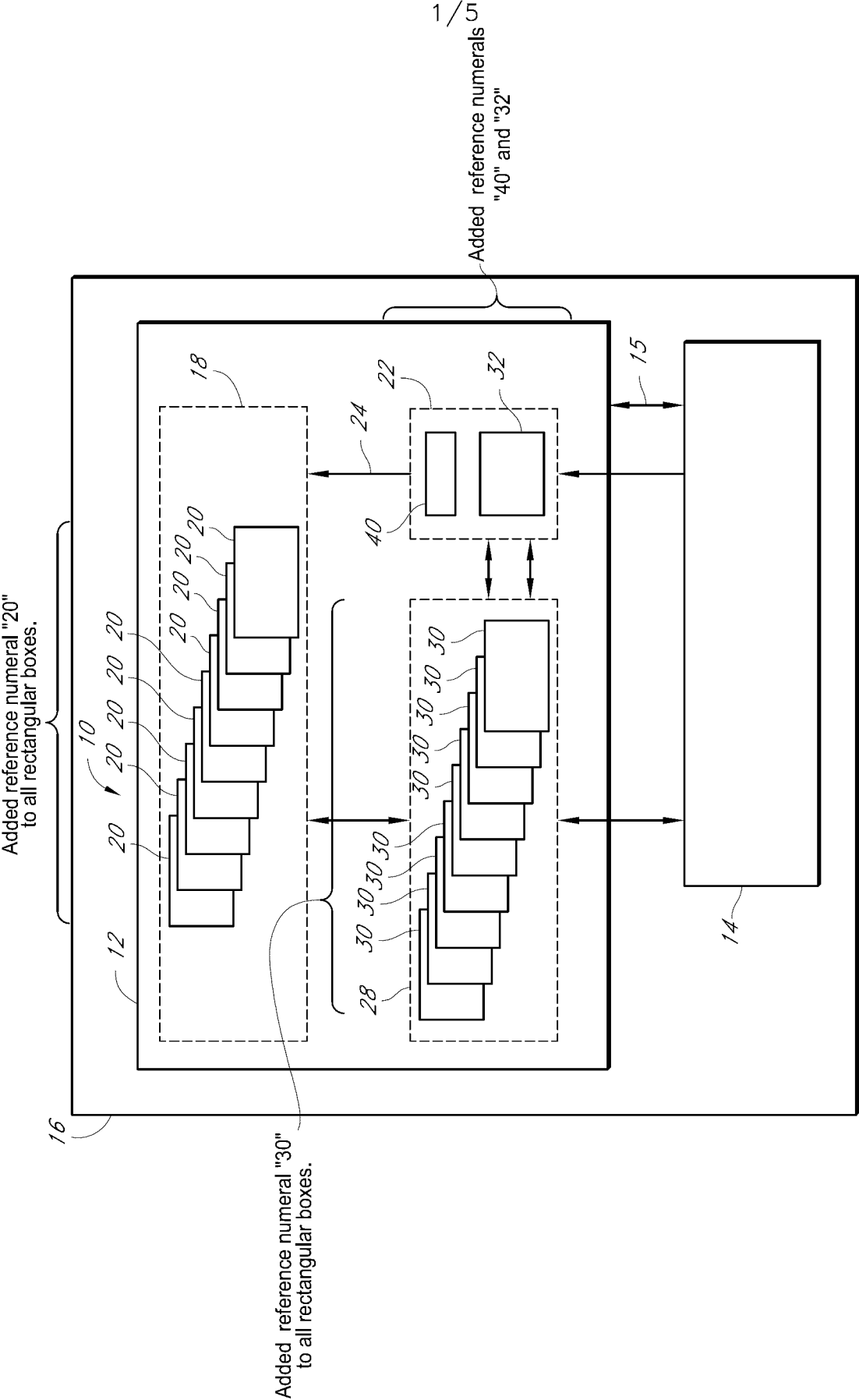
Dated: March 3, 2011

By: /Sean Ambrosius/
Sean Ambrosius
Registration No. 65,290
Attorney of Record
Customer No. 20995
(949) 760-0404

Application No.: 12/422925
Filing Date: April 13, 2009

APPENDIX

10748028



1 / 5
Added reference numerals
"40" and "32"

FIG. 1

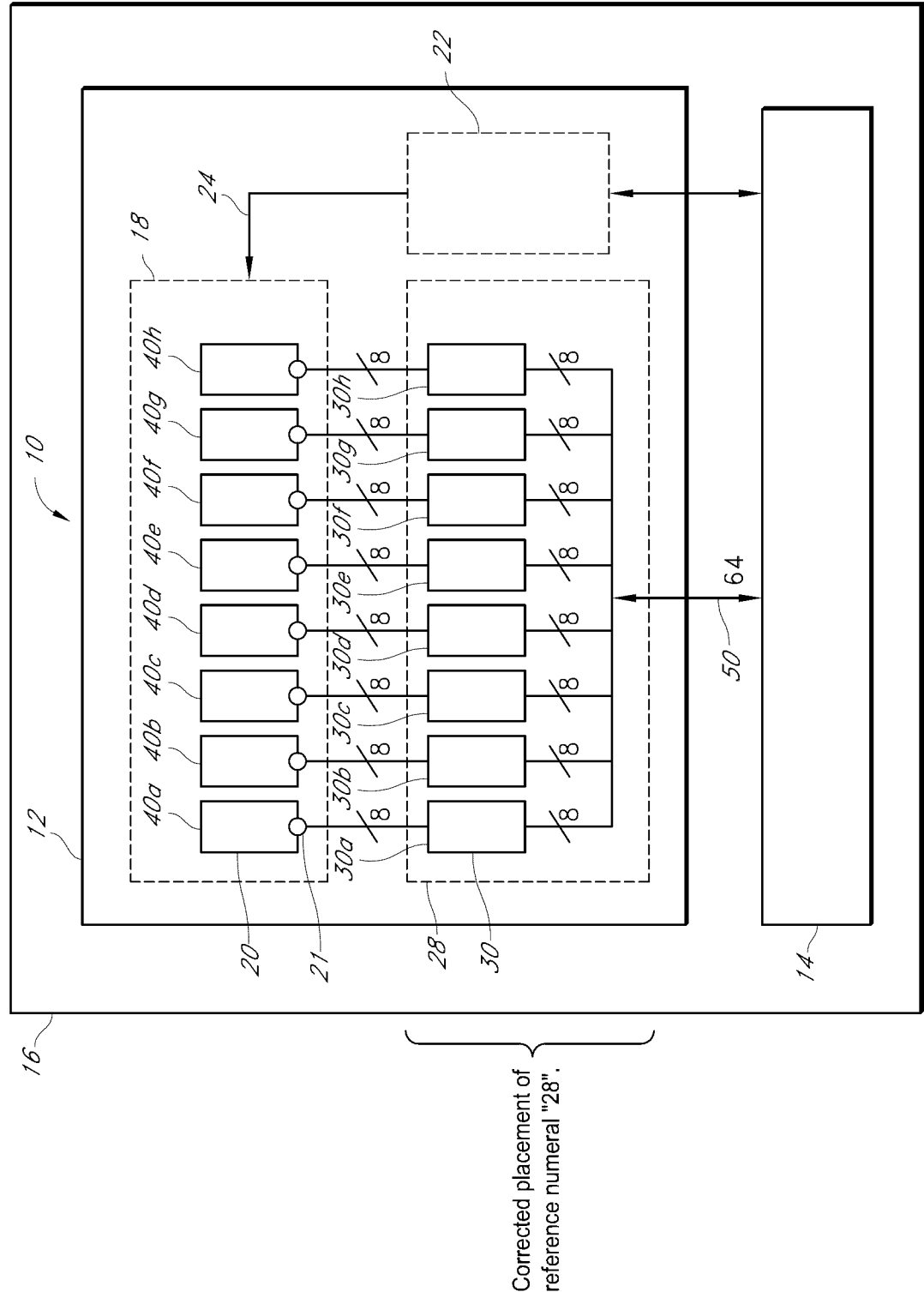


FIG. 2



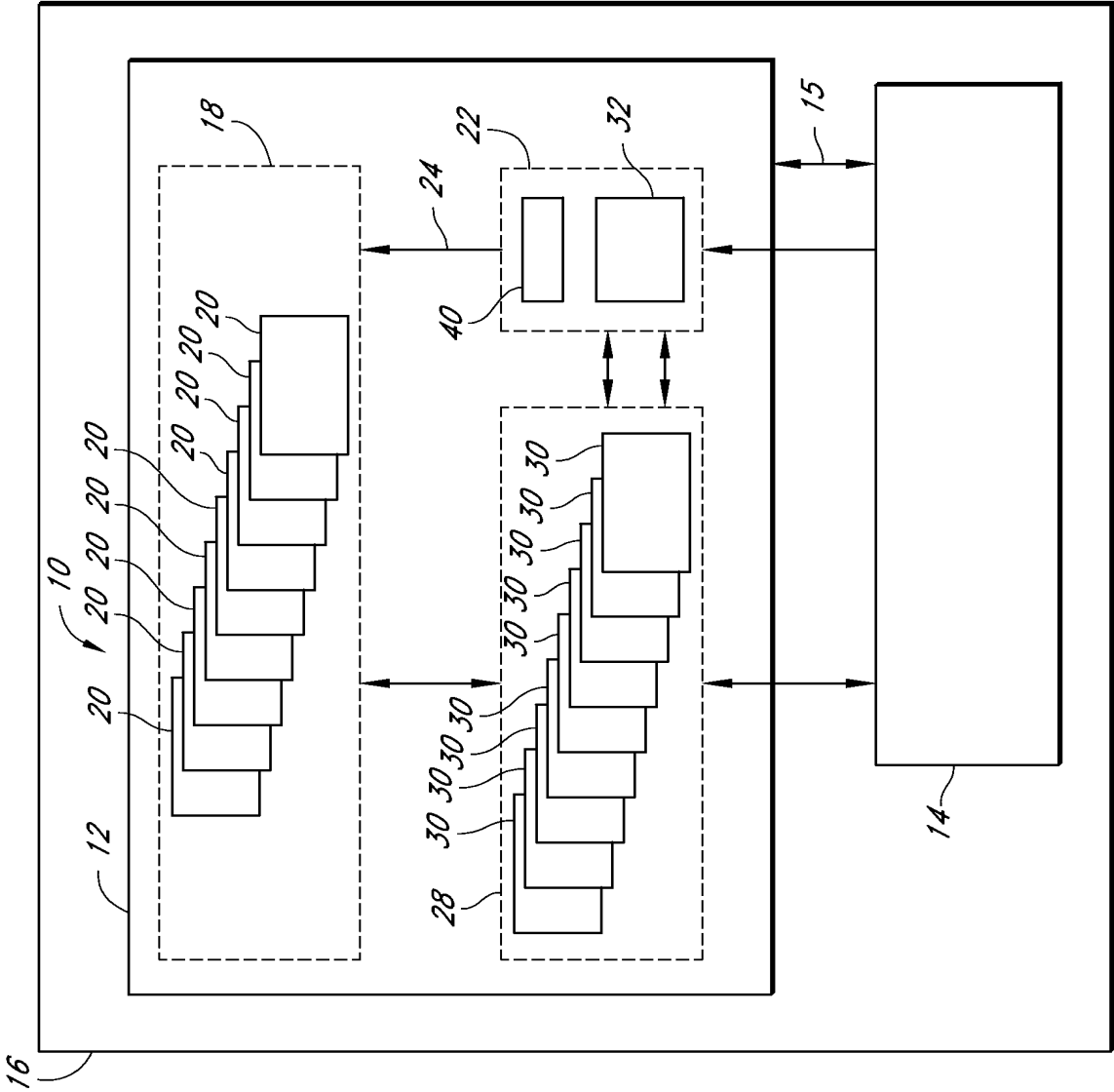


FIG. 1

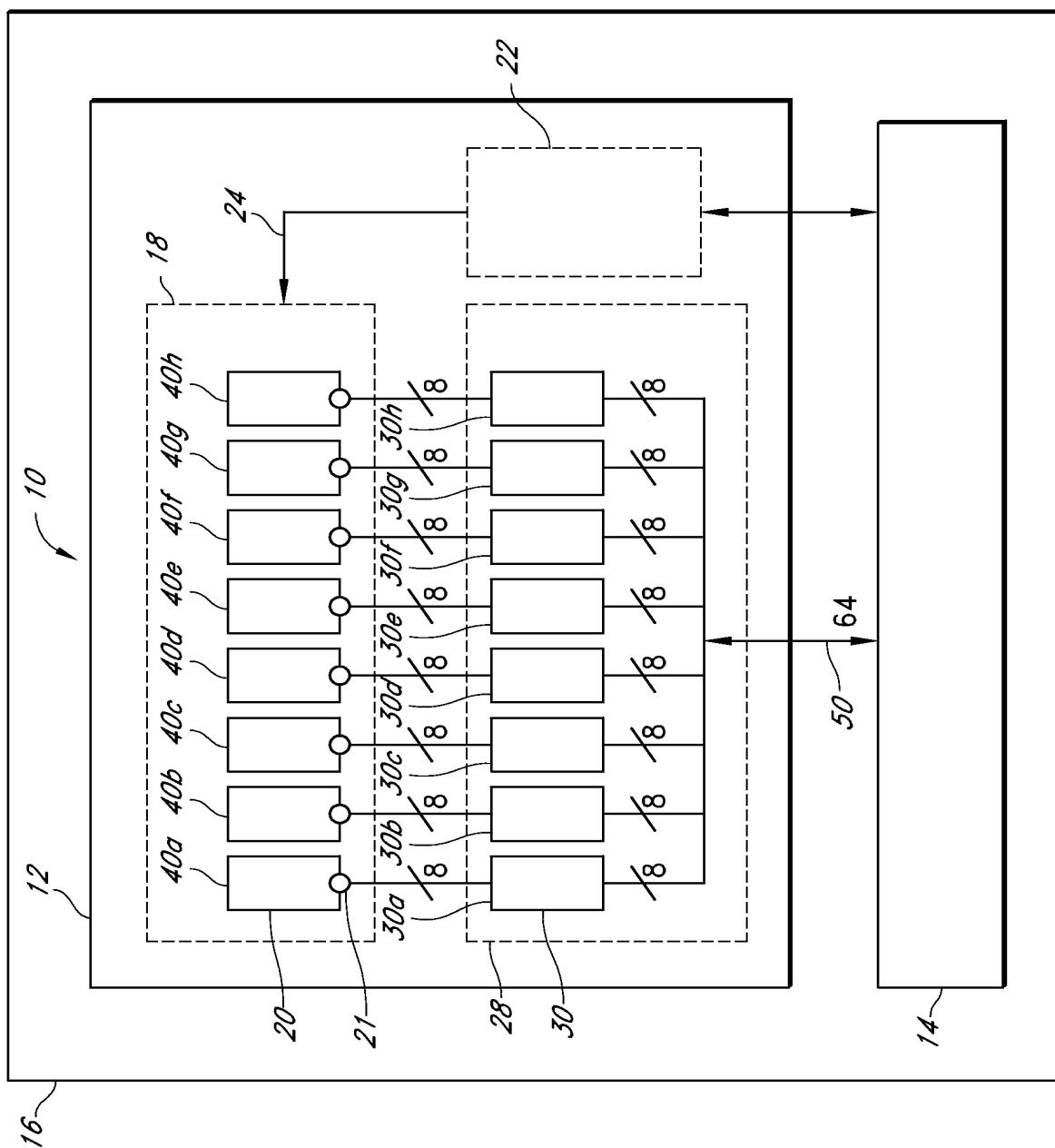
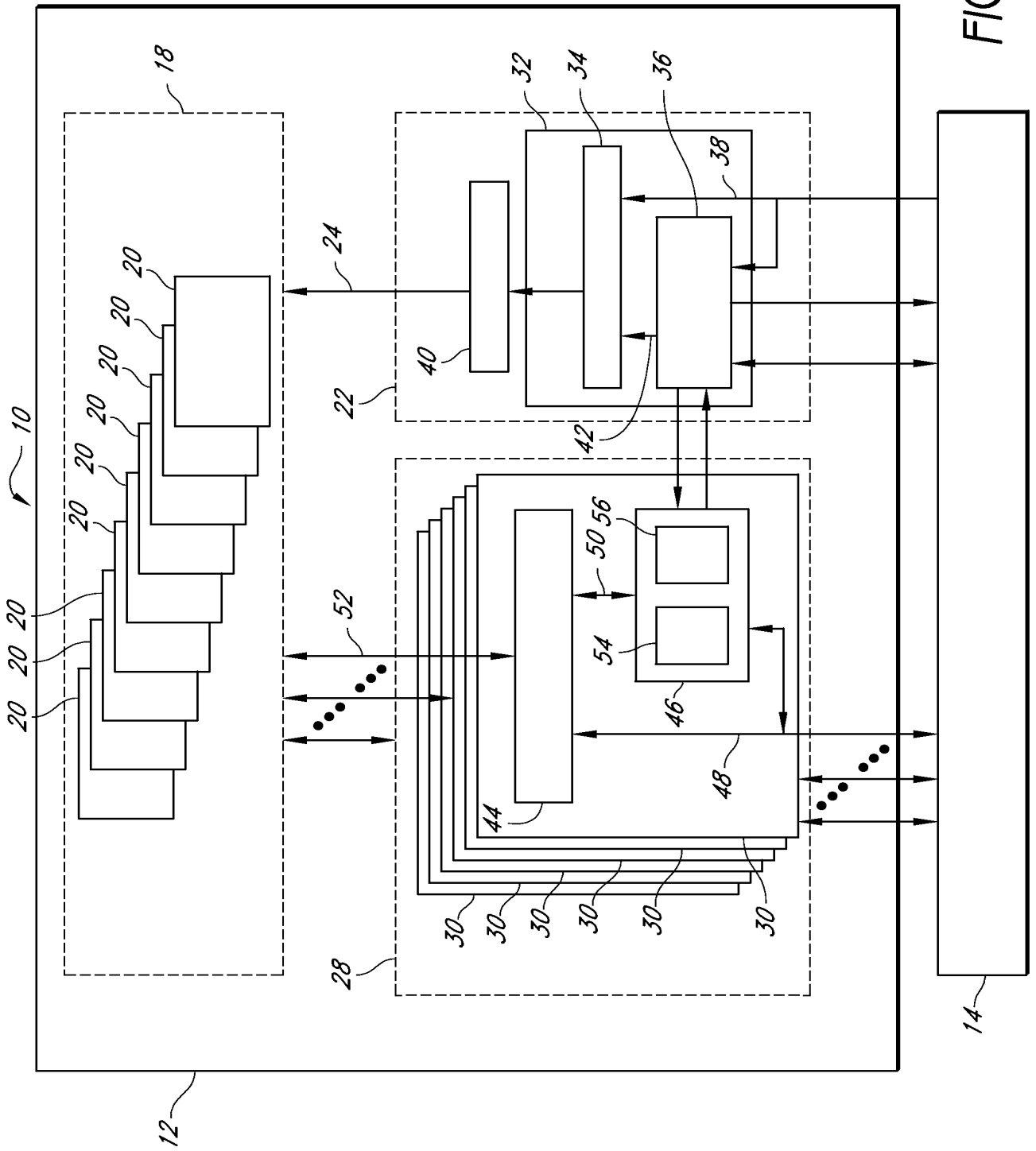
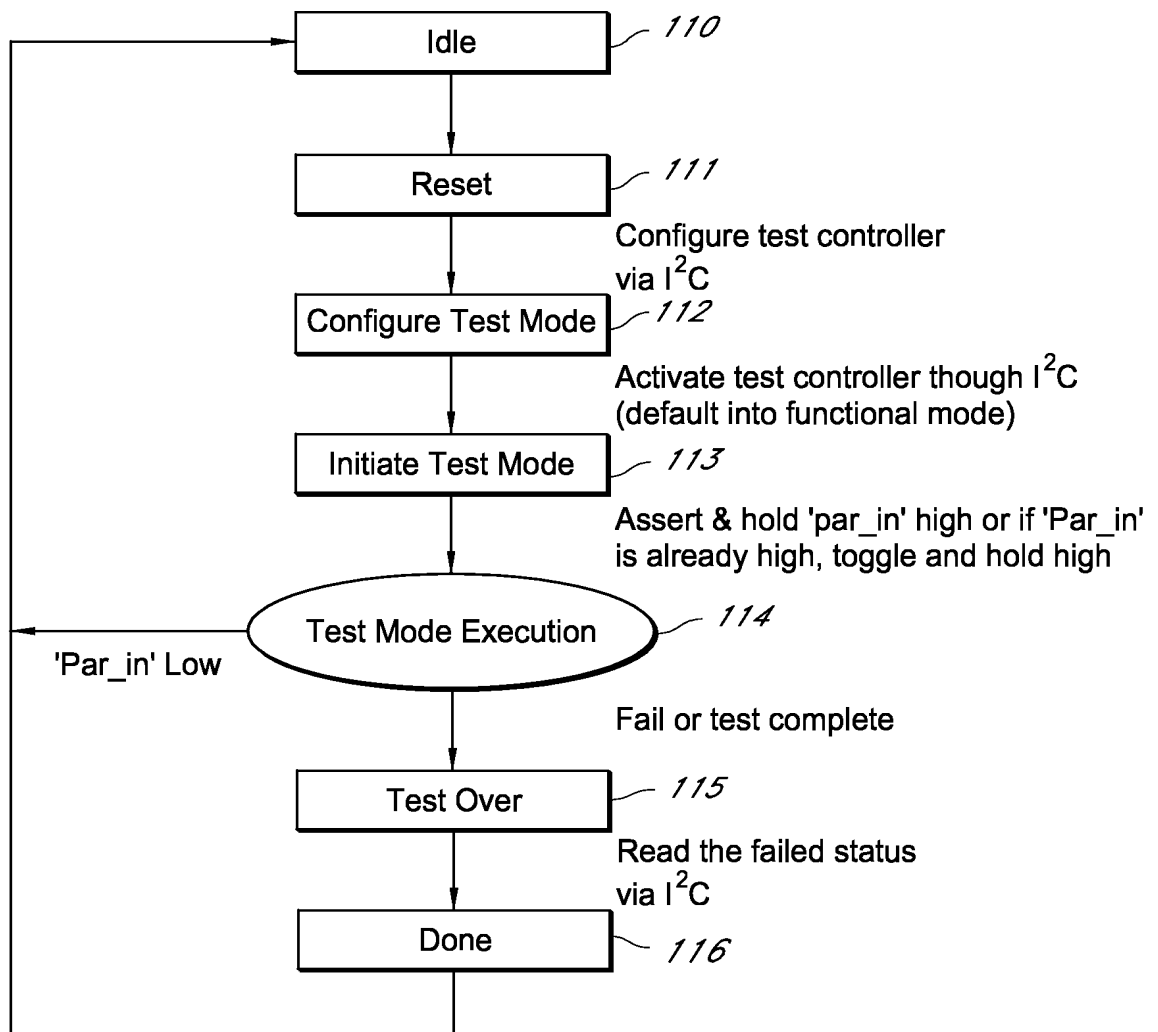
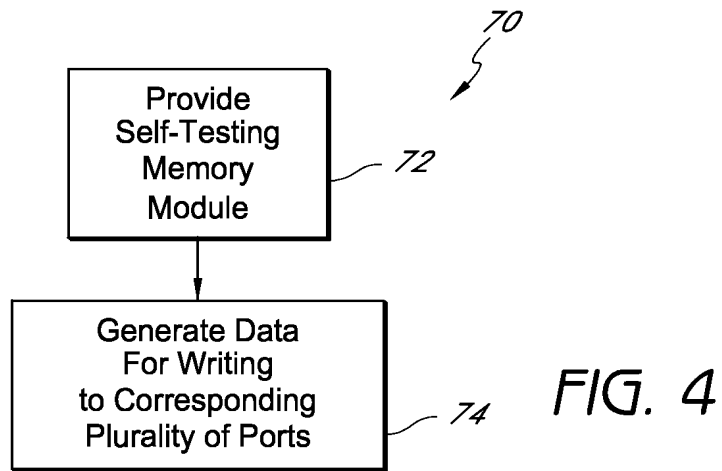


FIG. 2





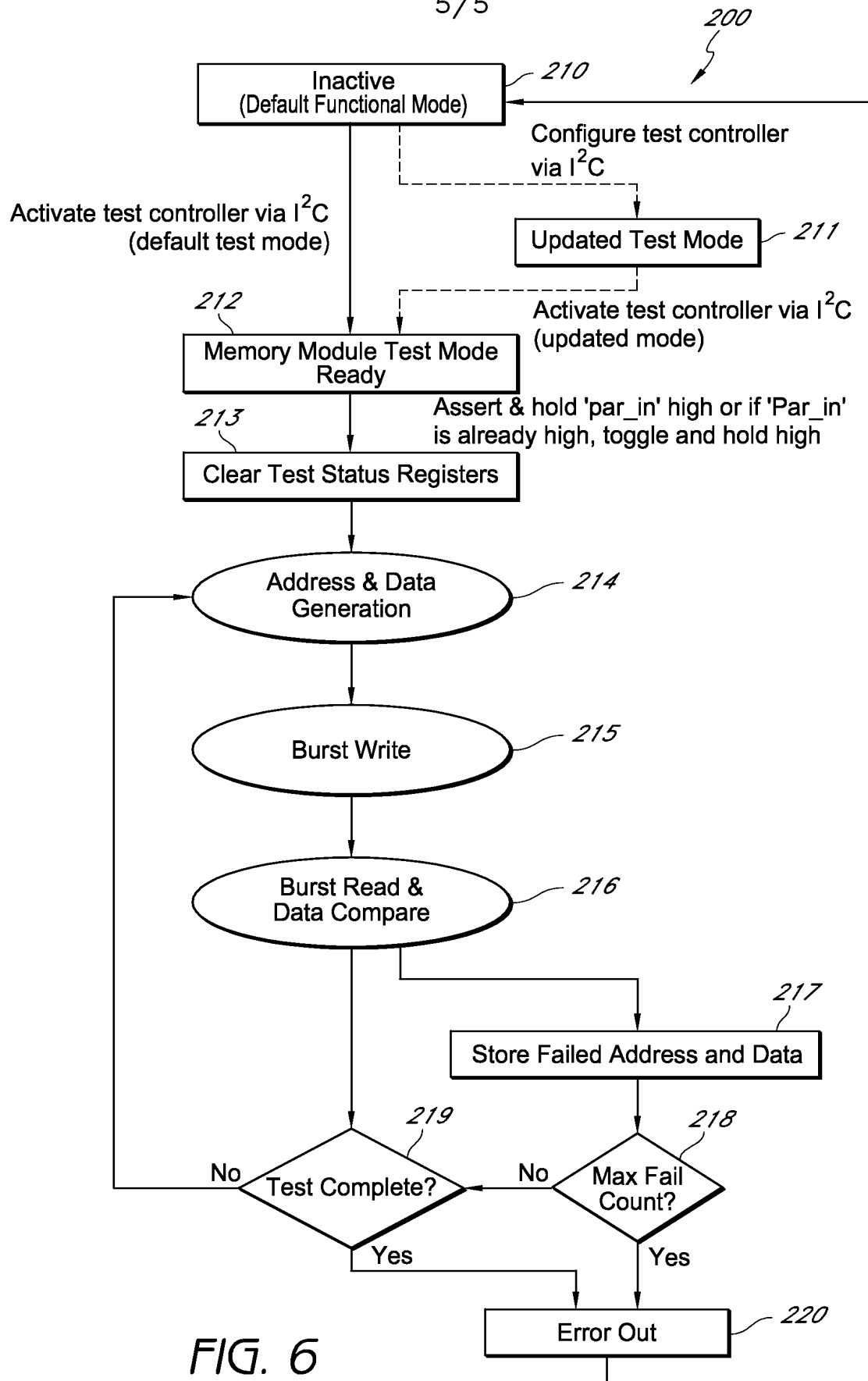


FIG. 6

Electronic Acknowledgement Receipt

EFS ID:	9582526
Application Number:	12422925
International Application Number:	
Confirmation Number:	7596
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Sean Patrick Ambrosius/Quyen Lieu
Filer Authorized By:	Sean Patrick Ambrosius
Attorney Docket Number:	NETL.046A
Receipt Date:	03-MAR-2011
Filing Date:	13-APR-2009
Time Stamp:	19:33:56
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	Trans_NETL_046A.pdf	31414 f111f676d3ba7af5fdb4fb44e0ccce34e01688f9	no	1

Warnings:

Information:

2		Amend_NETL_046A.pdf	607550 c7de5a445cc0e177e7a38d3e241ef8328f9e3ce	yes	18
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	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Specification		2	2	
	Drawings-only black and white line drawings		3	3	
	Claims		4	9	
	Applicant Arguments/Remarks Made in an Amendment		10	18	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	Replacement_NETL_046A.pdf	243360 8922af14ad2bbcc9393bab38d4daf8ff3418bc97	no	5
Warnings:					
Information:					
Total Files Size (in bytes):			882324		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

Please Direct All Correspondence to Customer Number **20995**

AMENDMENT / RESPONSE TRANSMITTAL

Applicant : Lee, et al.

App. No : 12/422925

Filed : April 13, 2009

For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY

Examiner : Chung, Phung M.

Art Unit : 2117

CERTIFICATE OF EFS WEB
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March 3, 2011

(Date)

/Sean Ambrosius/

Sean Ambrosius, Reg. No. 65,290

Mail Stop Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

Transmitted herewith for filing in the above-identified application are the following enclosures:

- (X) Amendment after Non-Final Office Action in 18 pages
- (X) with Appendix including 3 pages of Annotated Drawings Sheets
- (X) Replacement Drawing Sheets in 5 pages

Please charge any additional fees, including any fees for additional extension of time, or credit overpayment to Deposit Account No. 11-1410.

/Sean Ambrosius/

Sean Ambrosius

Registration No. 65,290

Attorney of Record

Customer No. 20,995

(949) 760-0404

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/422,925		Filing Date 04/13/2009		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	03/03/2011	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 35	Minus	** 35	= 0	X \$	=		X \$52=	0
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	X \$	=		X \$220=	0
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	0
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=		X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=		X \$	=
☐ Application Size Fee (37 CFR 1.16(s))										
☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/DEBORAH POLLARD/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

20995 7590 04/04/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

EXAMINER

CHUNG, PHUNG M

ART UNIT

PAPER NUMBER

2117

DATE MAILED: 04/04/2011

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/422,925

04/13/2009

Hyun Lee

NETL.046A

7596

TITLE OF INVENTION: MEMORY BOARD WITH SELF-TESTING CAPABILITY

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1510	\$0	\$0	\$1510	07/05/2011

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

20995 7590 04/04/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/422,925 04/13/2009 Hyun Lee NETL.046A 7596

TITLE OF INVENTION: MEMORY BOARD WITH SELF-TESTING CAPABILITY

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional NO \$1510 \$0 \$0 \$1510 07/05/2011

EXAMINER	ART UNIT	CLASS-SUBCLASS
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CHUNG, PHUNG M 2117 714-719000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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20995 7590 04/04/2011
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IRVINE, CA 92614

EXAMINER

CHUNG, PHUNG M

ART UNIT	PAPER NUMBER
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2117

DATE MAILED: 04/04/2011

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 178 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 178 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	12/422,925	LEE ET AL.	
	Examiner	Art Unit	
	Phung M. Chung	2117	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment dated on 3/3/11.
2. ☒ The allowed claim(s) is/are 1-35.
3. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. ____.
3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).
- * Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

4. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
5. ☒ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
- (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
- 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
- (b) ☒ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| <p>1. ☐ Notice of References Cited (PTO-892)</p> <p>2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948)</p> <p>3. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____</p> <p>4. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material</p> | <p>5. ☐ Notice of Informal Patent Application</p> <p>6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____.</p> <p>7. ☒ Examiner's Amendment/Comment</p> <p>8. ☒ Examiner's Statement of Reasons for Allowance</p> <p>9. ☒ Other <u>the attached examiner's amendment to the drawing.</u></p> |
|---|--|

/Phung M. Chung/
Primary Examiner
Art Unit: 2117

EXAMINER'S COMMENT

Drawings

1. The drawings were received on 3/3/11. These drawings are not approved because there isn't any element in the rectangular boxes as mentioned in paragraph 2 of the office action dated on 12/8/10. For a suggestion, for example, see the attached examiner's amendment to the drawing.

Allowable Subject Matter

2. The following is an examiner's statement of reasons for allowance: Claims 1-35 are allowable over the art of record. This is because the art of record does not teach or disclose a self-testing memory module as recited in claims 1, 20 and 29, and including:

a printed circuit board configured to be operatively coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and

a circuit comprising:

a control module configured to generate address and control signals for testing the memory devices; and

a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the

Art Unit: 2117

memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

//Phung M. Chung//
Primary Examiner, Art Unit 2117

Search Notes 	Application/Control No. 12422925	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner Phung M Chung	Art Unit 2117

SEARCHED			
Class	Subclass	Date	Examiner
714	718,719,736,733,734,30	3/24/11	pmc
365	230.03,189.011,201	3/24/11	pmc

SEARCH NOTES		
Search Notes	Date	Examiner
Interference searched.	3/24/11	pmc

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
714	719,733	3/23/11	pmc

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L20	12638	(self-testing memory module) or (built-in self test memory module) or (in-circuit testing memory module) or (BIST) or (in-circuit test) or (built-in self test) or (built in self-test) or (built-in self tester) or (built-in self-tester)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:36
L21	127038	memory controller	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:36
L22	613179	(memory device\$1 or memory array or memory cell\$1 or memory bank\$1 or memory block\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:37
L23	425246	printed circuit board or (PCB)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:37
L24	16143	data module or (data handler\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:37

L25	2500	(write-in same read out) and (compar\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:38
L26	55031	comparison data or calculated data or computed data	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:39
L27	608	compar\$3 and ((predetermined data or expected data or computed data or calculated data or known data) same (read out data or response data))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:40
L28	406	(control module) and (address signal\$1 same control signal\$1)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:40
L29	19154	(714/718,719,736,733,734,30;365/230.03,189.011,201).ccs.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:41
L30	5016	I20 and I22	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:41
L31	892	I21 and I30	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:41

L32	259	I31 and I23	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:42
L33	13	I32 and I24	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:42
L37	10	I29 and I33	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:43
L38	642684	computer system\$1	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:43
L39	10	I37 and I38	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	ADJ	ON	2011/03/24 22:43

3/ 24/ 11 10:45:06 PM

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EAST Search History**EAST Search History (I nterference)**

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L40	0	(memory devices AND memory controller AND test controller AND data handler AND memory BIST AND data module AND control module AND printed circuit board AND data port AND address port AND control port AND computer system AND address signal AND control signal AND data). clm.	US- PGPUB; USPAT; UPAD	ADJ	ON	2011/03/24 22:49

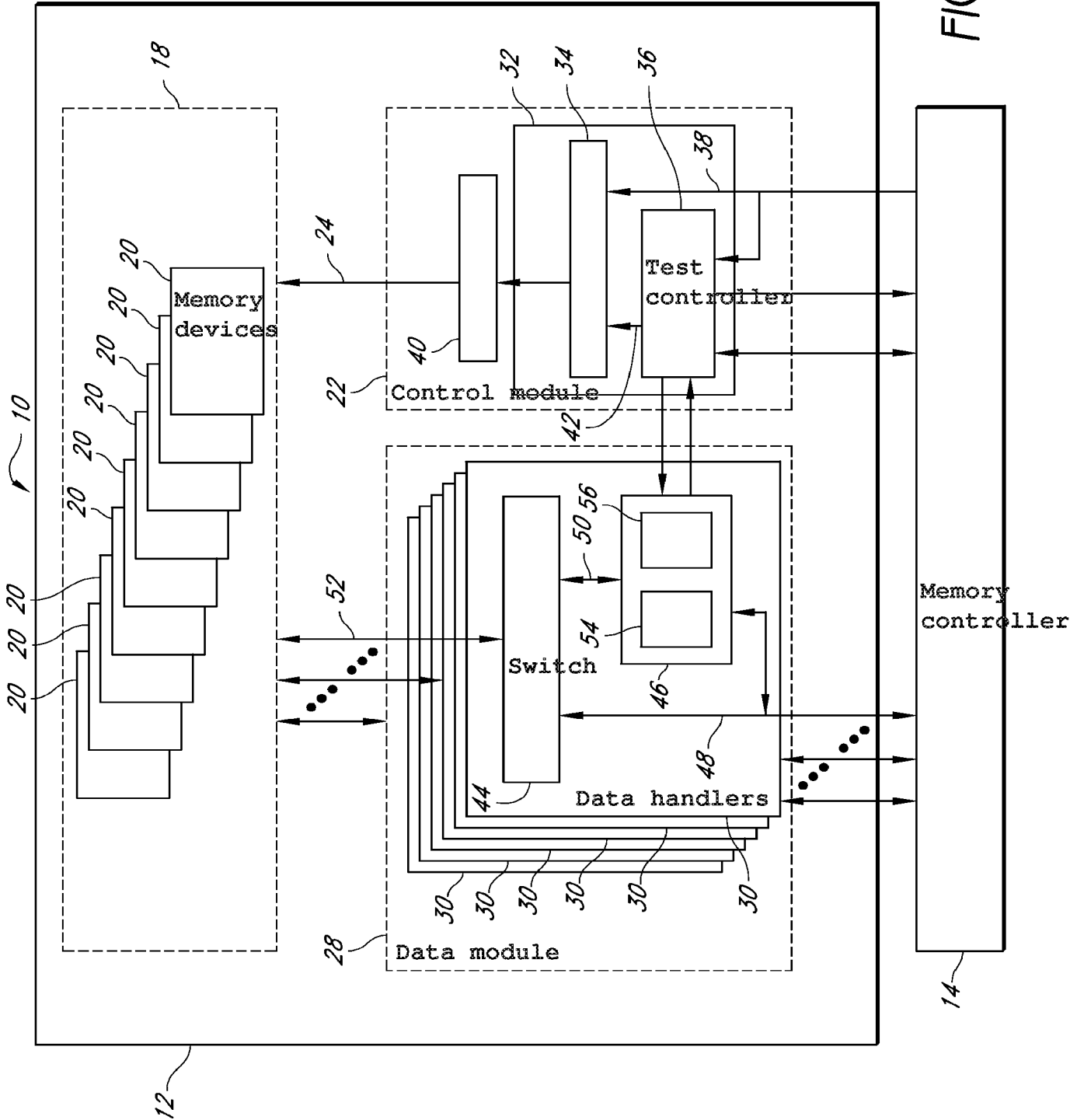
3/ 24/ 11 10:50:04 PM**C:\ Documents and Settings\ pchung\ My Documents\ EAST\ Workspaces
\ 12422925.w sp**

(for example)

Replacement Sheet

3/5

The attached examiner's
amendment to the
drawing.
/PMC/
3/24/11



PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

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IRVINE, CA 92614

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/422,925	04/13/2009	Hyun Lee	NETL.046A	7596

TITLE OF INVENTION: MEMORY BOARD WITH SELF-TESTING CAPABILITY

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EXAMINER	ART UNIT	CLASS-SUBCLASS
CHUNG, PHUNG M	2117	714-719000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

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2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1. Knobbe Martens
2. Olson & Bear LLP
3. _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

NETLIST, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 11-1410 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____

Date July 1, 2011

Typed or printed name Jerom D. Kesler

Registration No. 57,046

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

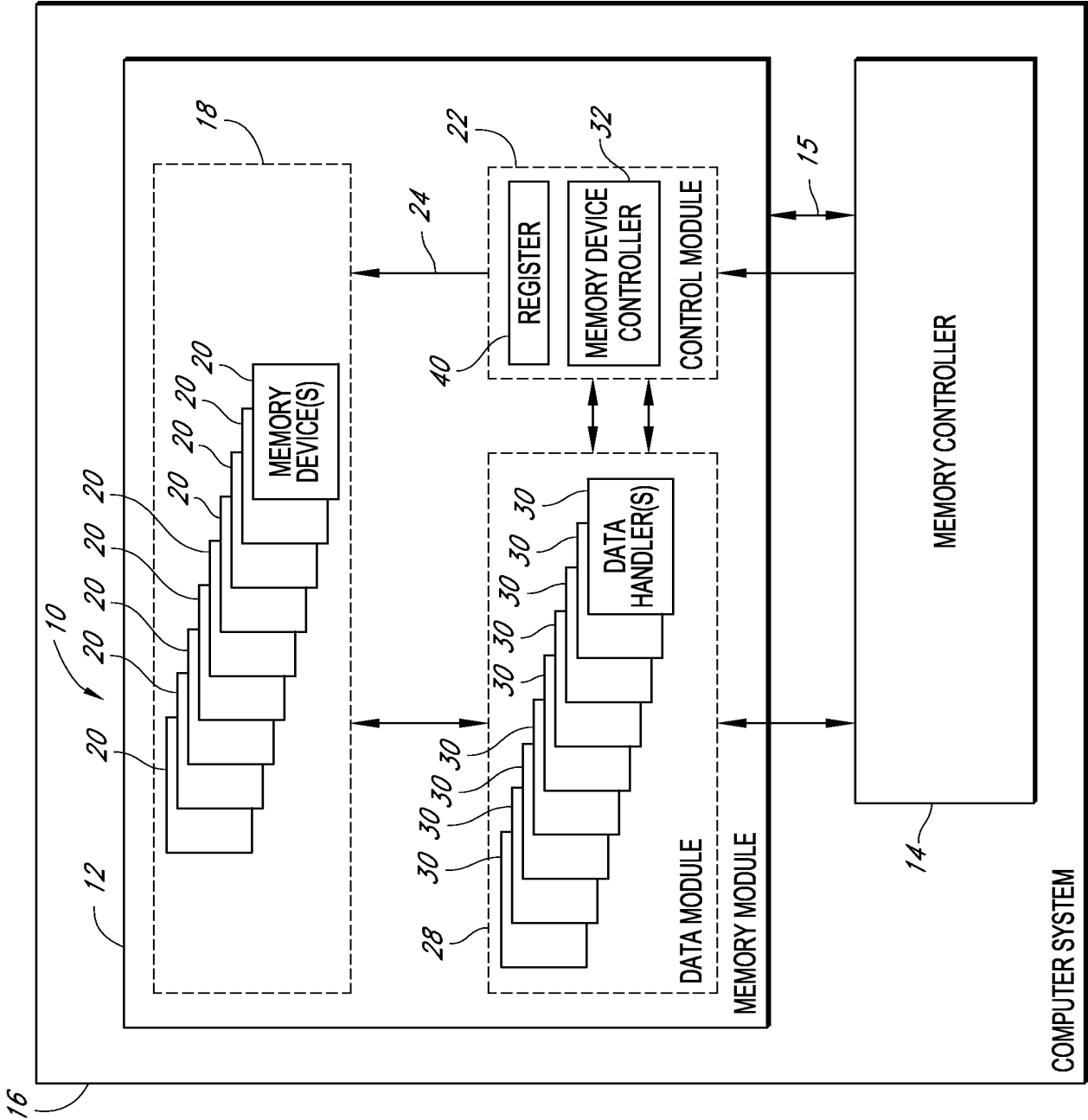


FIG. 1

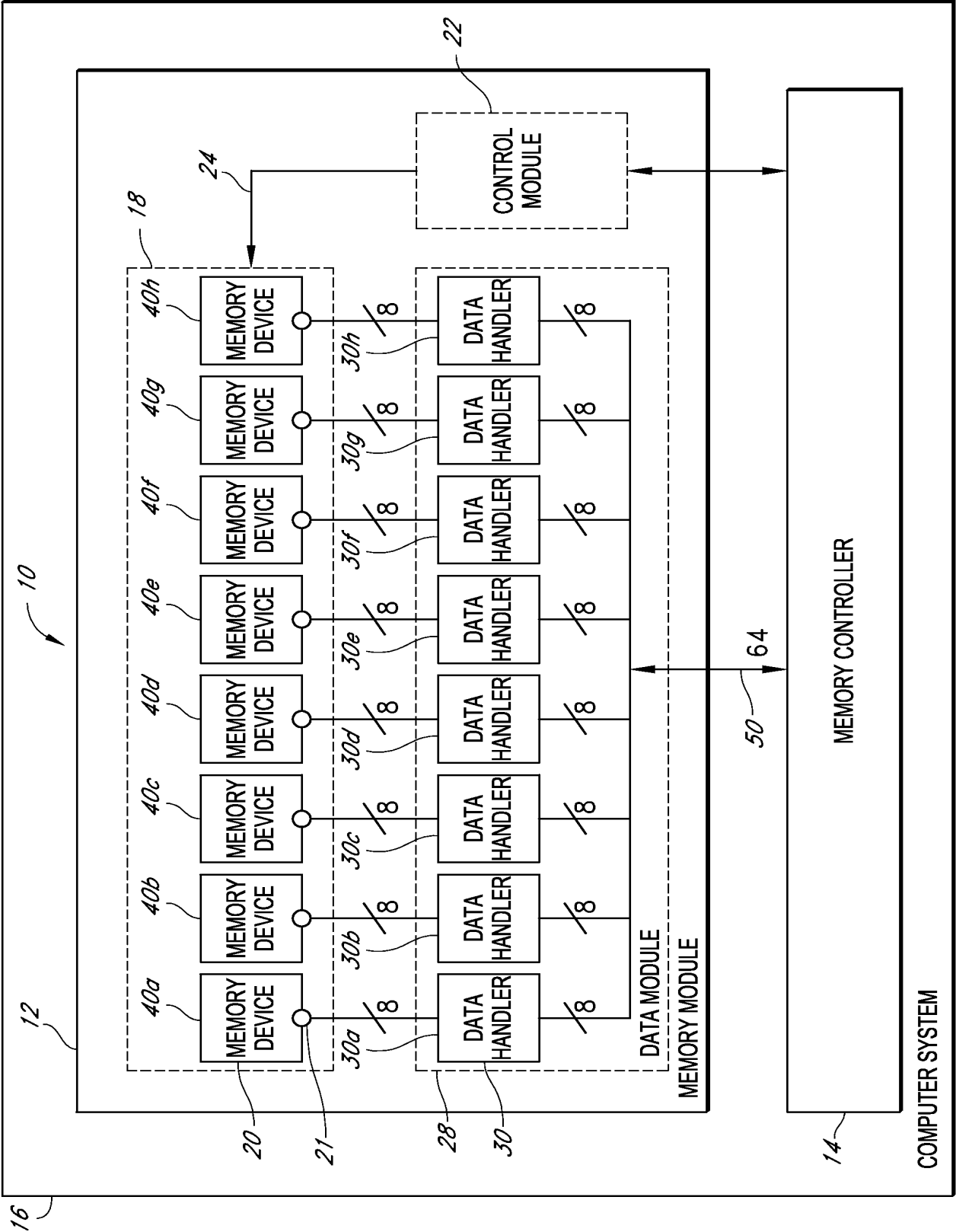


FIG. 2

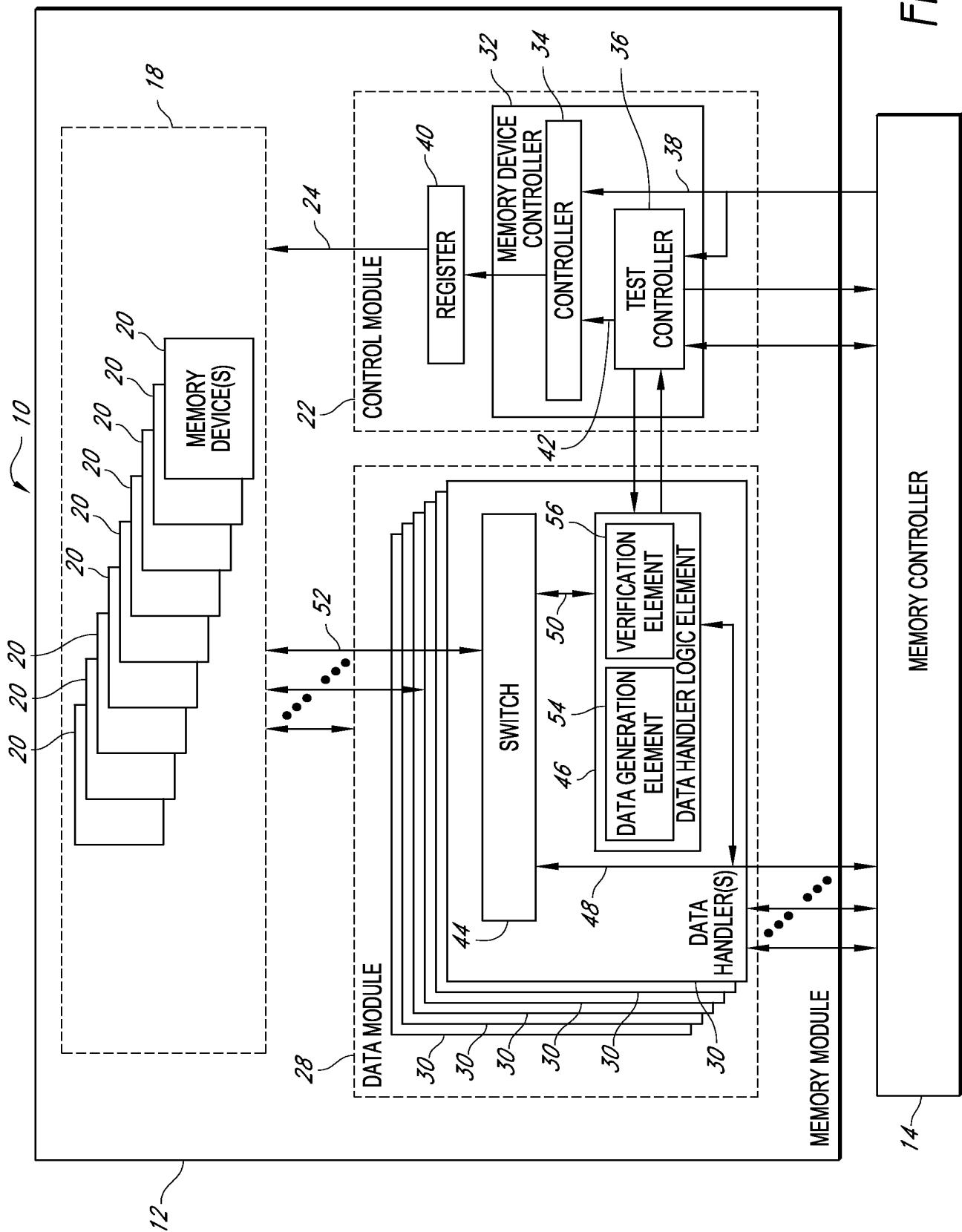


FIG. 3

4/5

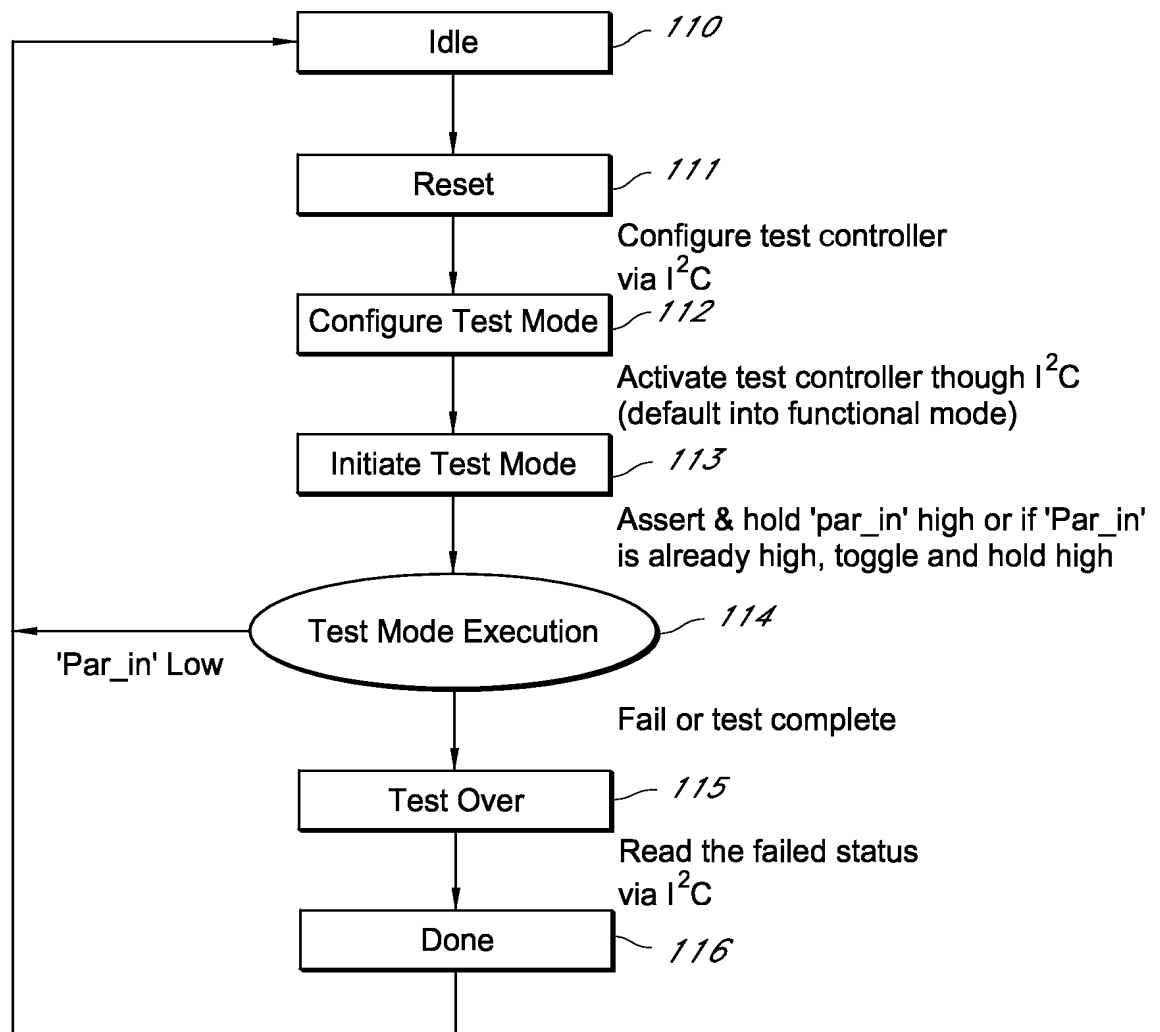
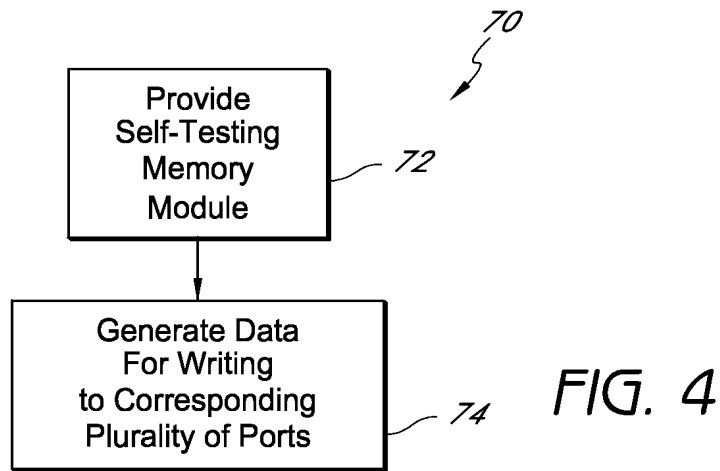
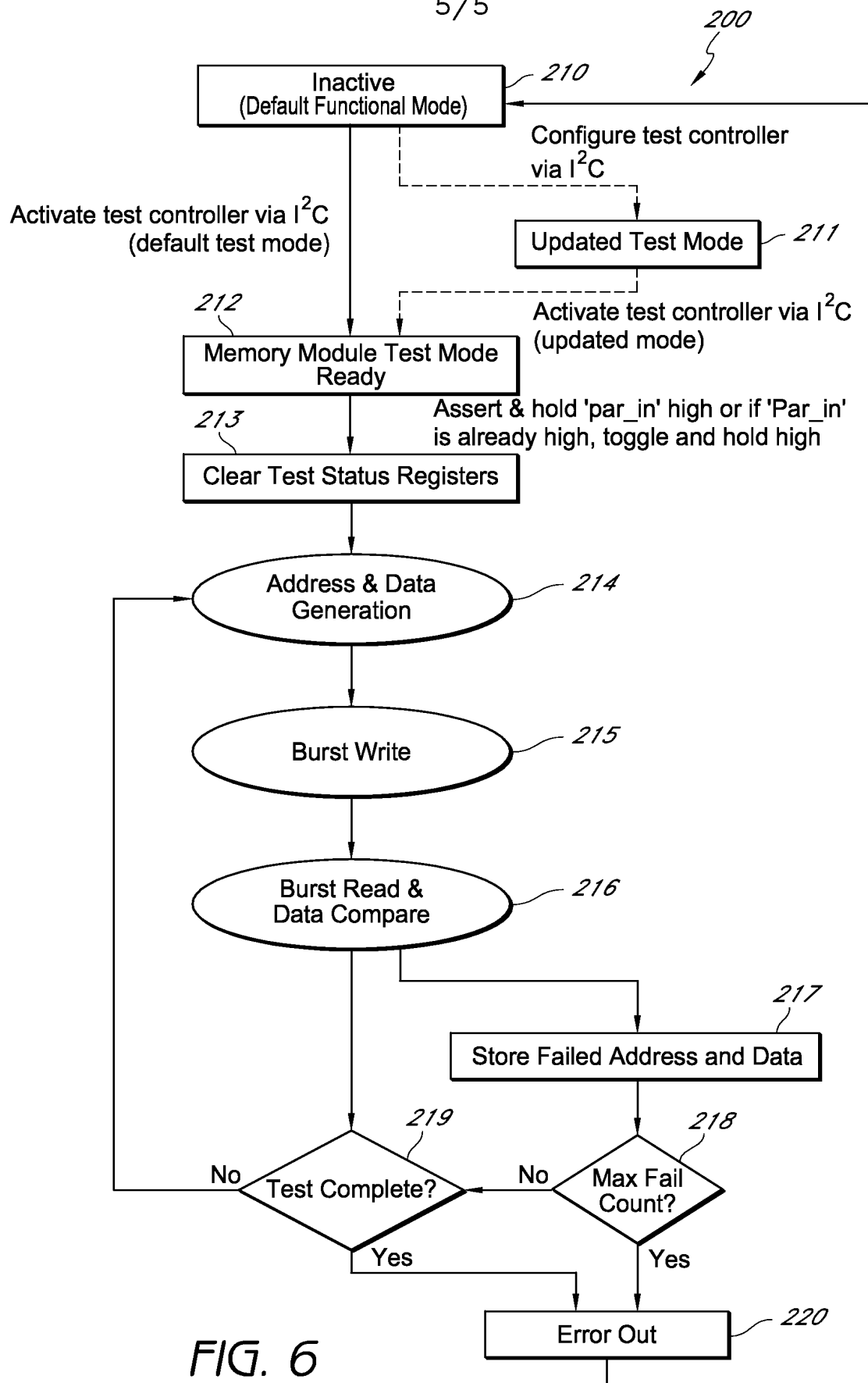


FIG. 5

5/5



IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Hyun Lee, et al.
App. No : 12/422,925
Filed : April 13, 2009
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Examiner : Chung, Phung M.
Art Unit : 7596
Conf No. : 2117

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July 1, 2011

(Date)

/Jarom Kesler/

Jarom D. Kesler, Reg. No. 57,046

COMMENTS ON STATEMENT OF REASONS FOR ALLOWANCE**Mail Stop Issue Fee**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Applicants respectfully disagree with the Examiner's Statement of Reasons for Allowance to the extent that the features recited by the Examiner are not present in all of the claims. To the extent that there is any implication that the patentability of the claims rests on the recitation of a single feature, Applicants respectfully disagree with the Examiner's Statement because it is the combination of features that makes the claims patentable. Accordingly, Applicants submit that the claims of the present application are allowable because each of the claims recites a combination of features that are not taught or suggested by the prior art.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: July 1, 2011

By: /Jarom Kesler/
Jarom D. Kesler
Registration No. 57,046
Attorney of Record
Customer No. 20,995
(949) 760-0404

Please Direct All Correspondence to Customer Number **20995**

ISSUE FEE TRANSMITTAL LETTER

Applicant : Hyun Lee, et al.
App. No : 12/422925
Filed : April 13, 2009
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Art Unit : 2117
Class/Sub-Class : 714-719000
Examiner : Chung, Phung M.

**CERTIFICATE OF EFS WEB
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July 1, 2011

(Date)

/Jarom Kesler/

Jarom D. Kesler, Reg. No. 57,046

MAIL STOP ISSUE FEE

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing is the Issue Fee for the above-identified application:

- (X) Form PTOL-85.
- (X) Five (5) sheets of Replacement Drawings.
- (X) Comments on Statement of Reasons for Allowance in 1 page.
- (X) The amount of **\$1510** will be paid via EFS Web.

The Commissioner is hereby authorized to charge any additional fees which may be required, or credit any overpayment, to Account No. 11-1410.

/Jarom Kesler/

Jarom D. Kesler

Registration No. 57,046

Attorney of Record

Customer No. 20,995

Electronic Patent Application Fee Transmittal

Application Number:	12422925			
Filing Date:	13-Apr-2009			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jarom D. Kesler/Nadin Hamoui			
Attorney Docket Number:	NETL.046A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1510	1510
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				1510

Electronic Acknowledgement Receipt

EFS ID:	10440802
Application Number:	12422925
International Application Number:	
Confirmation Number:	7596
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Jarom D. Kesler/ThuyQuyen Nguyen
Filer Authorized By:	Jarom D. Kesler
Attorney Docket Number:	NETL.046A
Receipt Date:	01-JUL-2011
Filing Date:	13-APR-2009
Time Stamp:	17:44:47
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Deposit Account
Payment was successfully received in RAM	\$ 1510
RAM confirmation Number	3996
Deposit Account	111410
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	Issue.pdf	87055 e7d79ea858f030e7b2e82de3fb5160c9d1b679e6	no	1
Warnings:					
Information:					
2	Drawings-only black and white line drawings	Drawings-046A.pdf	829694 8ea75d2d38dbc427848d36ac8c40287f3f022eae	no	5
Warnings:					
Information:					
3	Miscellaneous Incoming Letter	comments.pdf	38623 f16e40091f30866e4b92eb1c26f42d99d129106a	no	1
Warnings:					
Information:					
4	Transmittal Letter	transmittal.pdf	29996 4041a732c27a7b799c4fd6a446a03861d18ced5	no	1
Warnings:					
Information:					
5	Fee Worksheet (SB06)	fee-info.pdf	30199 2ffb6c197e825fb7009050e5b6cec2cd1b2c17c4	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			1015567		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/422,925	08/16/2011	8001434	NETL.046A	7596

20995 7590 07/27/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 131 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;



IFJ

PTO/SB/81 (01-09)

Approved for use through 11/30/2011. OMB 0651-0035
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	12/422925
	Filing Date	13-Apr-2009
	First Named Inventor	Lee, Hyun
	Title	MEMORY BOARD WITH SELF-TESTING CAPABILITY
	Art Unit	2117
	Examiner Name	CHUNG, PHUNG M
	Attorney Docket Number	NETL.046A

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

79141

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

I am the:

☒ Applicant/Inventor.

OR

☐ Assignee of record of the entire interest. See 37 CFR 3.71.
Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____

SIGNATURE of Applicant or Assignee of Record

Signature		Date	August 24, 2012
Name	Hyun Lee	Telephone	949-435-0025
Title and Company	VP and CTO		

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.

☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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APPLICATION NUMBER	FILING DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/422,925	04/13/2009	Hyun Lee	NETL.046A

CONFIRMATION NO. 7596



20995
 KNOBBE MARTENS OLSON & BEAR LLP
 2040 MAIN STREET
 FOURTEENTH FLOOR
 IRVINE, CA 92614

Cc: THE LAW OFFICE OF JAMIE ZHENG, PH.D ESQ.
 P.O. BOX 60573
 PALO ALTO, CA 94306

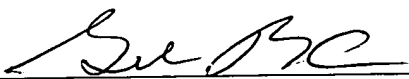
Date Mailed: 09/12/2012

DENIAL OF REQUEST FOR POWER OF ATTORNEY

The request for Power of Attorney filed 08/28/2012 is acknowledged. However, the request cannot be granted at this time for the reason stated below.

- ☐ The Power of Attorney you provided did not comply with the new Power of Attorney rules that became effective on June 25, 2004. See 37 CFR 1.32.
- ☐ The revocation is not signed by the applicant, the assignee of the entire interest, or one particular principal attorney having the authority to revoke.
- ☐ The Power of Attorney is from an assignee and the Certificate required by 37 CFR 3.73(b) has not been received.
- ☐ The person signing for the assignee has omitted their empowerment to sign on behalf of the assignee.
- ☐ The inventor(s) is without authority to appoint attorneys since the assignee has intervened as provided by 37 CFR 3.71.
- ☒ The signature(s) of Jayesh R. Bhakta, Soonju Choi, a co-inventor in this application, has been omitted. The Power of Attorney will be entered upon receipt of confirmation signed by said co-inventor(s).
- ☐ The person(s) appointed in the Power of Attorney is not registered to practice before the U.S. Patent and Trademark Office.

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/422,925	04/13/2009	Hyun Lee	NETL.046A

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

CONFIRMATION NO. 7596
MISCELLANEOUS NOTICE



Date Mailed: 09/13/2012

A communication which cannot be delivered in electronic form has been mailed to the applicant.

**REVOCATION
AND
GENERAL POWER OF ATTORNEY**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

The undersigned is an empowered representative of the Assignee and hereby appoints the registrant, Jamie J. Zheng, Customer No. 79141, as attorney to represent the Assignee before the United States Patent and Trademark Office (USPTO) in connection with the patent assigned to the Assignee according to the USPTO assignment records or assignment documents supplied with the accompanying Statement Under 37 CFR § 3.73(b). This appointment is to be to the exclusion of the inventor(s) and his attorney(s) in accordance with the provisions of 37 CFR § 3.71.

Submission of this paper in connection with any matter of the below named assignee, together with a statement under 37 CFR 3.73(b), shall serve to revoke any previous powers of attorneys in that matter.

A Statement under 37 CFR § 3.73(b) is attached setting forth a full chain of title for the subject patent owned by the Assignee named below.

Please recognize or change the correspondence address for the patent identified in the attached Statement to Customer No. 79141.

By:

Date: October 1, 2012

Name: Gail Sasaki

Assignee: Netlist Inc.

Title: Vice President and CFO

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 8,001,434Filed/Issue Date: 16-Aug-2011

Titled:

MEMORY BOARD WITH SELF-TESTING CAPABILITYNetlist, Inc., a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)
the patent application/patent identified above, by virtue of either:

A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 025230, Frame 0797, or for which a copy therefore is attached.

OR

B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (*i.e.*, a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

Signature

Gail Sasaki

Printed or Typed Name

October 1, 2012

Date

VP and CFO

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

Electronic Acknowledgement Receipt

EFS ID:	13925589
Application Number:	12422925
International Application Number:	
Confirmation Number:	7596
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.046A
Receipt Date:	05-OCT-2012
Filing Date:	13-APR-2009
Time Stamp:	19:00:01
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	poa8001434.pdf	232953 43deabd56c834f02680974dc39dd945b3490a43b	no	2

Warnings:

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Information:

Total Files Size (in bytes):

232953

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/422,925	04/13/2009	Hyun Lee	NETL.046A

79141
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

CONFIRMATION NO. 7596
POA ACCEPTANCE LETTER



Date Mailed: 10/25/2012

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 10/05/2012.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/zabaha/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

TO: Mail Stop 8 Director of the U.S. Patent & Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been
filed in the U.S. District Court NORTHERN CALIFORNIA on the following ☒ Patents or ☐ Trademarks:

DOCKET NO. CV 13-03901 DMR	DATE FILED 8/23/2013	U.S. DISTRICT COURT Northern District of California, 1301 Clay St. #400-S, Oakland, CA 94612
PLAINTIFF DIABLO TECHNOLOGIES INC		DEFENDANT NETLIST INC
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,001,434		***SEE ATTACHED COMPLAINT***
2 8,301,833		
3 8,359,501		
4 8,516,185		
5 8,516,187		

In the above—entitled case, the following patent(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK	
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In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK Richard W. Wieking	(BY) DEPUTY CLERK Clara Pierce	DATE August 27, 2013
---------------------------------	---------------------------------------	-----------------------------

Copy 1—Upon initiation of action, mail this copy to Commissioner Copy 3—Upon termination of action, mail this copy to Commissioner
Copy 2—Upon filing document adding patent(s), mail this copy to Commissioner Copy 4—Case file copy

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8

9 UNITED STATES DISTRICT COURT
10 NORTHERN DISTRICT OF CALIFORNIA
11 SAN FRANCISCO DIVISION

DMR

12 DIABLO TECHNOLOGIES, INC.,

13 Plaintiff,

14 v.

15 NETLIST, INC.,

16 Defendant.

CV Case No. 13 3901

COMPLAINT FOR DECLARATORY
JUDGMENT

DEMAND FOR JURY TRIAL

ORIGINAL
FILED

2013 AUG 23 A 9:23

RICHARD W. WILSON
CLERK, U.S. DISTRICT COURT
NORTHERN DISTRICT OF CALIFORNIA

1 For its Complaint against Defendant NETLIST, INC., ("NETLIST"), Plaintiff Diablo
2 Technologies, Inc. ("DIABLO") alleges as follows:

3 **FACTUAL BACKGROUND**

4 **DIABLO**

5 1. DIABLO is a corporation incorporated under the laws of Canada, with its
6 principal place of business at 80 Aberdeen Street, Suite 401, Ottawa, Ontario, K1S 5R5 Canada.

7 2. DIABLO, founded in 2003, delivers product solutions that enhance the
8 performance and capability of memory system designs. DIABLO developed the Memory
9 Channel Storage™ (MCST™) architecture for enterprise servers and storage systems.

10 3. DIABLO has been awarded "Most Innovative Flash Memory Enterprise Business
11 Application" in the Flash Memory Summit Best of Show Awards for enabling a new class of
12 ultra-low latency (ULL) SSDs and application accelerators. This innovative new architecture
13 radically transforms memory and storage for enterprise rack, blade and storage servers.

14
15 **NETLIST'S THREAT OF SUIT AGAINST DIABLO**

16 4. NETLIST purports to be incorporated under the laws of Delaware, with its
17 principal place of business at 51 Discovery, Suite 150, Irvine, California, 92618.

18 5. On August 22, 2013, NETLIST sent DIABLO a letter stating that "Netlist has
19 concluded that Smart Modular Technologies, Inc., Smart Storage Systems, Inc., Smart
20 Worldwide Holdings, Inc. and Diablo Technologies, Inc. (collectively "Defendants") are
21 directly and /or indirectly infringing the following United States patents at least through the
22 making, using, offering for sale and/or selling of the recently announced ULLtraDIMM product:

23 U.S. Patent No. 8,001,434, Memory Board With Self-Testing Capability

24 U.S. Patent No. 8,301,833, Non-Volatile Memory Module

25 U.S. Patent No. 8,359,501, Memory Board With Self-Testing Capability

26 U.S. Patent No. 8,516,185, System and Method Utilizing Distributed Byte-Wise Buffers
27 on a Memory Module

28 U.S. Patent No. 8,516,187, Data Transfer Scheme for Non-Volatile Memory Module"

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JURISDICTION AND VENUE

6. This Court has subject matter jurisdiction over this action pursuant to 28 U.S.C. §§ 1331, 1338, 1367, 2201, and 2202.

7. This Court has personal jurisdiction over DIABLO. Among other things, DIABLO transacts business in this jurisdiction.

8. This Court has personal jurisdiction over NETLIST. Among other things, NETLIST transacts business in this jurisdiction and is a plaintiff in this District at least in the pending action *Netlist, Inc. v. Google Inc.* 4-09-cv-05718 SBA (CAND).

9. Venue is proper in this District pursuant to 28 U.S.C. § 1391.

INTRADISTRICT ASSIGNMENT

10. For purposes of intradistrict assignment pursuant to Local Rule 3-2(c), this Intellectual Property Action is to be assigned on a district-wide basis.

COUNT I

(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,001,434)

11. DIABLO incorporates by reference, as if fully set forth herein, the allegations of the preceding paragraphs of this Complaint.

12. Upon information and belief, NETLIST is the owner of the '434 patent.

13. NETLIST is asserting the '434 patent against "ULLtraDIMM."

14. DIABLO, contends that it has not infringed and does not infringe, any valid and enforceable claim of the '434 patent.

15. An actual and justiciable controversy has thus arisen between DIABLO and NETLIST concerning the alleged infringement of the '434 patent.

1 **COUNT II**

2 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,001,434)**

3 16. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
4 the preceding paragraphs of this Complaint.

5 17. Upon information and belief, NETLIST contends that the '434 patent is valid.

6 18. DIABLO contends that the claims of the '434 patent are invalid under 35 U.S.C.
7 §§ 101, 102, 103, and/or 112.

8 19. An actual and justiciable controversy has thus arisen between DIABLO and
9 NETLIST concerning the validity of the '434 patent.

10
11 **COUNT III**

12 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,301,833)**

13 20. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
14 the preceding paragraphs of this Complaint.

15 21. Upon information and belief, NETLIST is the owner of the '833 patent.

16 22. NETLIST is asserting the '833 patent against "ULLtraDIMM".

17 23. DIABLO, contends that it has not infringed and does not infringe, any valid and
18 enforceable claim of the '833 patent.

19 24. An actual and justiciable controversy has thus arisen between DIABLO and
20 NETLIST concerning the alleged infringement of the '833 patent.

21
22 **COUNT IV**

23 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,301,833)**

24 25. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
25 the preceding paragraphs of this Complaint.

26 26. Upon information and belief, NETLIST contends that the '833 patent is valid.

27 27. DIABLO contends that the claims of the '833 patent are invalid under 35 U.S.C.
28 §§ 101, 102, 103, and/or 112.

1 28. An actual and justiciable controversy has thus arisen between DIABLO and
2 NETLIST concerning the validity of the '833 patent.

3
4 **COUNT V**

5 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,359,501)**

6 29. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
7 the preceding paragraphs of this Complaint.

8 30. Upon information and belief, NETLIST is the owner of the '434 patent.

9 31. NETLIST is asserting the '501 patent against "ULLtraDIMM".

10 32. DIABLO contends that it has not infringed and does not infringe, any valid and
11 enforceable claim of the '501 patent.

12 33. An actual and justiciable controversy has thus arisen between DIABLO and
13 NETLIST concerning the alleged infringement of the '501 patent.

14
15 **COUNT VI**

16 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,359,501)**

17 34. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
18 the preceding paragraphs of this Complaint.

19 35. Upon information and belief, NETLIST contends that the '501 patent is valid.

20 36. DIABLO contends that the claims of the '501 patent are invalid under 35 U.S.C.
21 §§ 101, 102, 103, and/or 112.

22 37. An actual and justiciable controversy has thus arisen between DIABLO and
23 NETLIST concerning the validity of the '501 patent.

24
25 **COUNT VII**

26 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,516,185)**

27 38. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
28 the preceding paragraphs of this Complaint.

1 39. Upon information and belief, NETLIST is the owner of the '185 patent.

2 40. NETLIST is asserting the '185 patent against "ULLtraDIMM".

3 41. DIABLO, contends that it has not infringed and does not infringe, any valid and
4 enforceable claim of the '185 patent.

5 42. An actual and justiciable controversy has thus arisen between DIABLO and
6 NETLIST concerning the alleged infringement of the '185 patent.

7

8

COUNT VIII

9 **(Declaratory Judgment of Invalidity of U.S. Patent No. 8,516,185)**

10 43. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
11 the preceding paragraphs of this Complaint.

12 44. Upon information and belief, NETLIST contends that the '185 patent is valid.

13 45. DIABLO contends that the claims of the '185 patent are invalid under 35 U.S.C.
14 §§ 101, 102, 103, and/or 112.

15 46. An actual and justiciable controversy has thus arisen between DIABLO and
16 NETLIST concerning the validity of the '185 patent.

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18

COUNT IX

19 **(Declaratory Judgment of Non-Infringement of U.S. Patent No. 8,516,187)**

20 47. DIABLO incorporates by reference, as if fully set forth herein, the allegations of
21 the preceding paragraphs of this Complaint.

22 48. Upon information and belief, NETLIST is the owner of the '187 patent.

23 49. NETLIST is asserting the '187 patent against "ULLtraDIMM".

24 50. DIABLO, contends that it has not infringed and does not infringe, any valid and
25 enforceable claim of the '187 patent.

26 51. An actual and justiciable controversy has thus arisen between DIABLO and
27 NETLIST concerning the alleged infringement of the '187 patent.

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1 Respectfully submitted this 23rd day of August 2013.

2 DIABLO TECHNOLOGIES, INC.,

3 By: Sanjeet Dutta

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13 Telephone: (650) 590-0700

14 Facsimile: (650) 590-1900

15 Attorneys for Plaintiff

16 DIABLO TECHNOLOGIES, INC.

TO: Mail Stop 8
 Director of the U.S. Patent & Trademark Office
 P.O. Box 1450
 Alexandria, VA 22313-1450

**REPORT ON THE
 FILING OR DETERMINATION OF AN
 ACTION REGARDING A PATENT OR
 TRADEMARK**

In Compliance with 35 § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court Northern District of California on the following:
 (X) Patents or () Trademarks

DOCKET NO:
 13-cv-05889-YGR

DATE FILED:
 October 7, 2014

UNITED STATES DISTRICT COURT
 Ronald Dellums Federal Building
 1301 Clay Street
 Oakland, CA 94612

PLAINTIFF:
 Netlist, Inc

DEFENDANT:
 Smart Modular Technologies, Inc

PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1. <u>8,001,434</u>		See attached THIRD AMENDED COMPLAINT FOR PATENT INFRINGEMENT
2. <u>8,301,833</u>		
3. <u>8,359,501</u>		
4. <u>8,516,185</u>		
5. <u>8,516,187</u>		

In the above-entitled case, the following patent(s) have been included.

DATE INCLUDED INCLUDED BY:
 () Amendment () Answer () Cross Bill () Other Pleading

PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1. <u>7,881,150</u>		
2. <u>8,081,536</u>		
3.		
4.		
5.		

In the above-entitled case, the following decision has been rendered or judgment issued:

DECISION/JUDGEMENT:

Richard W. Wieking, Clerk


 (by) Deputy Clerk, Clara Pierce

Copy 1 – Upon initiation of action, mail this copy to Commissioner
 Copy 2 – Upon filing document adding patent(s) mail this copy to Commissioner
 Copy 3 – Upon termination of action, mail this copy to the Commissioner
 Copy 4 – Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SANDISK CORPORATION,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-00970
Patent 8,001,434 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
PETER P. CHEN, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

DECISION
Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

On June 18, 2014, Sandisk Corporation (“Petitioner”) filed a Petition (Paper 1, “Pet.”) to institute an *inter partes* review of claims 1–35 (the “challenged claims”) of U.S. Patent No. 8,001,434 B1 (Ex. 1001, “the ’434 patent”). 35 U.S.C. §§ 311–319. Netlist, Inc. (“Patent Owner”) timely filed a Preliminary Response (Paper 10, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may not be instituted “unless . . . there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” Upon consideration of the Petition and Patent Owner’s Preliminary Response, we determine Petitioner has established a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. We therefore institute an *inter partes* review as to: (1) claims 1, 14, 15, 17–25, 27, and 29–35 on the ground that these claims are anticipated by Averbuj; (2) claim 28 on the ground that this claim would have been obvious over Averbuj; (3) claims 1, 2, 5, 6, 14, 15, 19, and 29 on the ground that these claims would have been obvious over Huang; (4) claims 7–12, 17, 18, 20–25, 27, 28, and 30–35 on the ground that these claims would have been obvious over Huang and Averbuj; and (5) claims 13 and 26 on the ground that these claims would have been obvious over Huang, Averbuj, and JEDEC Standard. We decline to institute *inter partes* review as to claims 3, 4, and 16. Our factual findings and conclusions at this stage of the proceeding are based on the evidentiary record developed thus far (prior to Patent Owner’s Response). This is not a final decision as to patentability of claims for which *inter partes* review is instituted. Our final

decision will be based on the record as fully developed during trial.

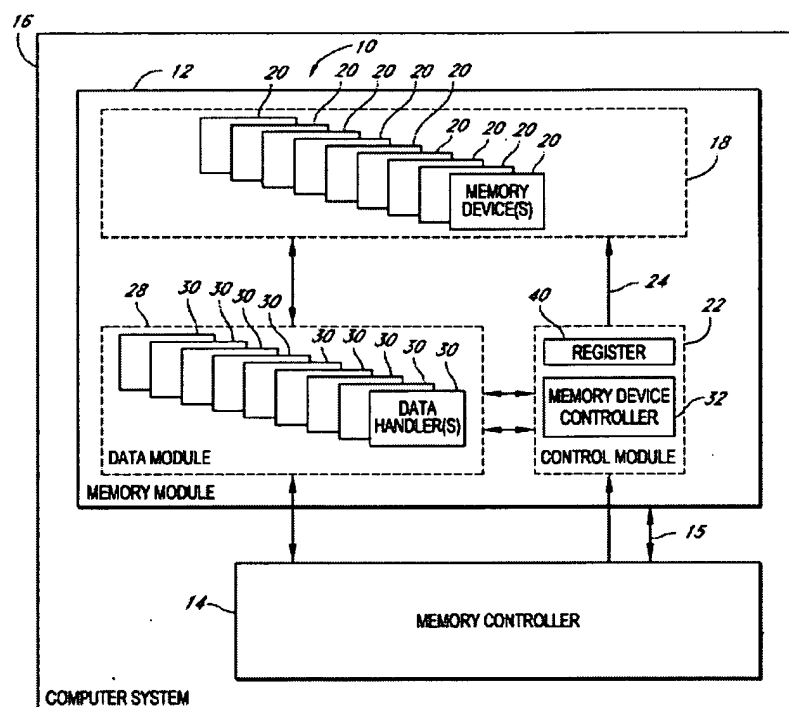
II. BACKGROUND

A. Related Matters

The '434 patent is also the subject of two infringement actions filed by Patent Owner, one of which names Petitioner as a defendant: *Netlist, Inc. v. Smart Modular Technologies, Inc.*, No. 4:13-cv-05889-YGR (Dec. 19, 2013, N.D. Cal.). SMART Modular Technologies, Inc. is a named defendant in both infringement actions, and filed two petitions for *inter partes* review of the '434 patent on August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01372 and IPR2014-01373 (PTAB). Petitioner has filed a petition for review of U.S. Patent No. 8,359,501 ("the '501 patent"), which claims priority as a continuation of the '434 patent: *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00971 (June 18, 2014, PTAB). On August 23, 2014, SMART Modular Technologies, Inc. also filed two petitions for *inter partes* review of the '501 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01374 and IPR2014-01375 (PTAB).

B. The '434 Patent (Ex. 1001)

The '434 patent relates to self-testing electronic memory modules. Ex. 1001, 1:23—24. A block diagram of an exemplary self-testing memory module is shown in Figure 1 of the '434 patent, reproduced below.



As illustrated in Figure 1, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:1–3. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:3–7.

Memory module 10 further includes data module 28 comprising a plurality of independently operable data handlers 30, and a control module 22 that may include a dual input register for registering address and control signals coming from either self-testing logic or from memory controller 14 on the system board. *Id.* at 5:9–11, 37–42. “[D]ata module 28 and/or [] control module 22 of certain embodiments are configured to test the plurality of memory devices 18 at the normal operating speed of [] memory devices 20.” *Id.* at 5:67–6:3. “For example, [] data module 28 and/or [] control module 22 are configured to provide memory signals (e.g., data, address and

control signals) according [to] the operating specification of [] memory devices 20.” *Id.* at 6:3–6. “[E]ach of [] data handlers 30 may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports” of the plurality of memory devices 18. *Id.* at 5:15–18. “In some embodiments, during testing, [] control module 22 generates address and control signals 24 associated with memory locations to be tested and [] data module 28 generates corresponding test data patterns and provides them to [] appropriate memory devices 20.” *Id.* at 5:43–47. “For example, [] data module 28 may receive a write command from [] control module 22 and provide data to be written to certain locations in [] memory devices 20 during a write operation.” *Id.* at 5:47–50.

C. Illustrative Claim

Of the challenged claims, claims 1, 20, and 29 are independent.

Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to

generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

D. Evidence Relied Upon

Petitioner's patentability challenges are based on the following references:

References	Patents/Printed Publications	Exhibit
Averbuj	U.S. Patent Application Publication No. 2005/0257109	1003
Huang	<i>An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers</i> , VLSI Design 2001, Fourteenth International Conference on VLSI Design (pp. 379–384), IEEE (2001)	1004
Tsern	U.S. Patent Application Publication No. 2007/0070669	1005
JEDEC Standard	JEDEC Standard, Fully Buffered DIMM (FBDIMM): DFX Design for Validation and Test, JESD82–28, (February 2008)	1006

In addition to these references, Petitioner relies on the Declaration of Donald Alpert, Ph.D. (Ex. 1007) under 37 C.F.R. § 1.68. Pet. 7.

E. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of the '434 patent claims based on the following grounds:

References	Basis	Claims Challenged
Averbuj	§ 102(b)	1, 14, 15, 17–25, 27, and 29–35
Averbuj and Tsern	§ 103(a)	1–12, 14–25, and 27–35
Averbuj, Tsern, and JEDEC Standard	§ 103(a)	13 and 26
Huang and Tsern	§ 103(a)	1–6, 14–16, 19, and 29
Huang, Tsern, and Averbuj	§ 103(a)	7–12, 17, 18, 20–25, 27, 28, and 30–35
Huang, Tsern, Averbuj, and JEDEC Standard	§ 103(a)	13 and 26

III. CLAIM CONSTRUCTION

Claims of an unexpired patent are interpreted using the broadest reasonable construction in light of the specification of the patent. *See* Office Patent Trial Practice Guide, 77 Fed. Reg. 48756, 48766 (Aug. 14, 2012); 37 C.F.R. § 42.100(b). “Petitioner submits that, for the purposes of this review, each claim be construed in accordance with its plain and ordinary meaning under the required broadest reasonable interpretation.” Pet. 6–7. Petitioner does not specifically state the broadest reasonable construction of any particular claim terms or phrases. *See id.* Patent Owner, however, offers specific constructions for the following claim language, recited in each of independent claims 1, 20, and 29: “memory module,” “self-testing memory module,” and “generate.” Prelim. Resp. 22–27. For purposes of this decision, we determine the only claim language requiring express construction is “self-testing memory module” and “generate.”

A. Self-testing Memory Module

Patent Owner contends “self-testing memory module” is properly

construed as a “memory module that is able to test one or more elements of the memory module without the use of a testing device external to the memory module.” Prelim. Resp. 25 (emphasis omitted). In support of this claim construction, Patent Owner relies on column 3, lines 52–54 of the ’434 patent, which read: “[s]ome embodiments present a self-testing memory module that can be tested without any external test equipment.” Patent Owner has not directed us to, nor do we find in the specification or claims, any basis for limiting the claims to encompass only those particular embodiments wherein the memory module is capable of being tested without the use of external test equipment. In this regard, we note the term “self-testing memory module” is used to describe other embodiments of the invention which are not explicitly described as capable of being tested without the use of external test equipment. *See e.g.*, Ex. 1001, 3:46–52. Accordingly, for purposes of this decision, we interpret the claim language “self-testing memory module” as encompassing memory modules that can be tested with internal and/or external test equipment.

B. Generate

Patent Owner contends the claim term “generate” should be interpreted as meaning “produce.” Prelim. Resp. 26–27. Patent Owner argues, more specifically, that claim phrases such as “a control module configured to generate address and control signals” (Ex. 1001, claims 1, 20, 29), should be construed as meaning that the control module itself produces the address and control signals, but should not be construed as encompassing a control module which merely sends a command to another device or component to generate address and control signals. Prelim. Resp. 26–27. In support of this construction, Patent Owner cites column 6, lines 6–9 of the

'434 patent, which read: "In some embodiments, the control module 22 and the data module 28 *produce* memory addresses, control and/or data signals according to the JEDEC standard memory protocol." *Id.* at 26 (quoting Ex. 1001, 6:6–9 (emphasis added)). Patent Owner also relies on two dictionary definitions of "generate." *Id.* (citing Exs. 2002, 2003).

As an initial matter, we note the description in the '434 patent relied upon by Patent Owner relates to only "some embodiments," and the claims do not recite that the data and control modules are configured to generate signals "according to the JEDEC standard memory protocol." The specification states that the data module 28/data handlers 30 may be in communication with the control module 22 (Ex. 1001, 9:64–66), and that data signals and/or patterns of data signals generated from data generation element 54, present in certain embodiments of data handlers 30, "may be generated in a variety of ways" (*id.* at 10:49–50), including by programming from control module 22 (*id.* at 10:67–11:4). The specification also states that "data module 28 and/or [] *control module 22 are configured to provide* memory signals (e.g., *data*, address and control *signals*) according [to] the operating specification of [] memory devices 20." *Id.* at 5:67–6:6 (emphasis added). Given such broad disclosure, Patent Owner has not convinced us that the '434 patent specification supports its proposed narrow claim construction of "generate" as requiring, for example, that data "generated by" a data handler (*see* claims 1 and 20) is also "produced by" the data handler.

Moreover, although both dictionaries cited by Patent Owner list "produce" as one definition of "generate" (Ex. 2002, 484; Ex. 2003, 230), Merriam Webster's Collegiate Dictionary also defines generate as "to be the

cause of” (Ex. 2002, 484). Patent Owner has not explained sufficiently why this latter definition of “generate” is inconsistent with the use of the term “generate” in the ’434 patent specification and claims. *See* Prelim. Resp. 26 (“The term ‘generate’ should not be construed to mean, e.g., ‘to be the cause of.’”) (citing Ex. 2002).

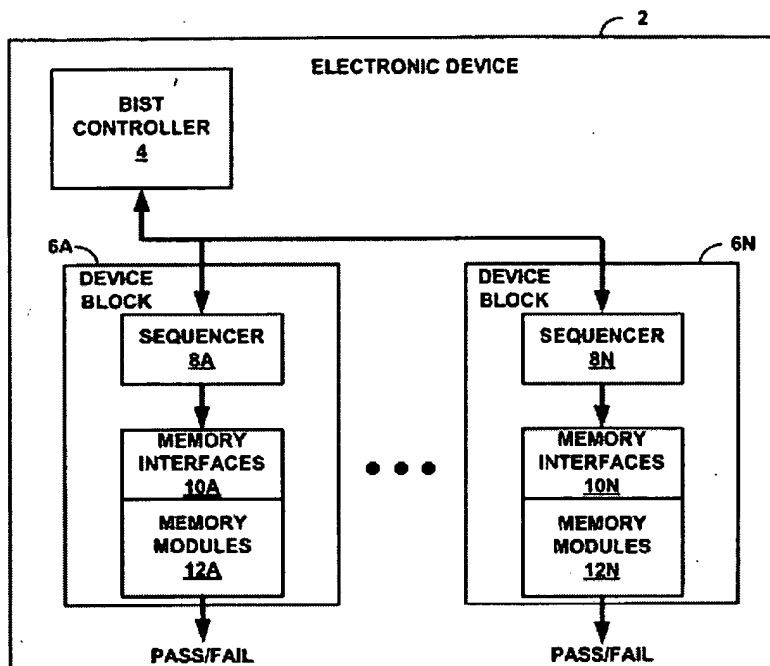
Accordingly, for purposes of this decision, we interpret the claim term “generate” as meaning “produce” or “cause.” Thus, for example, as we interpret the claim term “generate,” “a data module configured to generate data for writing to the corresponding plurality of data ports,” as recited in claim 1, encompasses a data module/data handler which writes programmable data patterns received from a control module. *See* Ex. 1001, 10:49–50; 10:67–11:4, discussed above.

IV. ANALYSIS

A. Anticipation of Claims 1, 14, 15, 17–25, 27, and 29–35 by Averbuj

1. Averbuj (Ex. 1003)

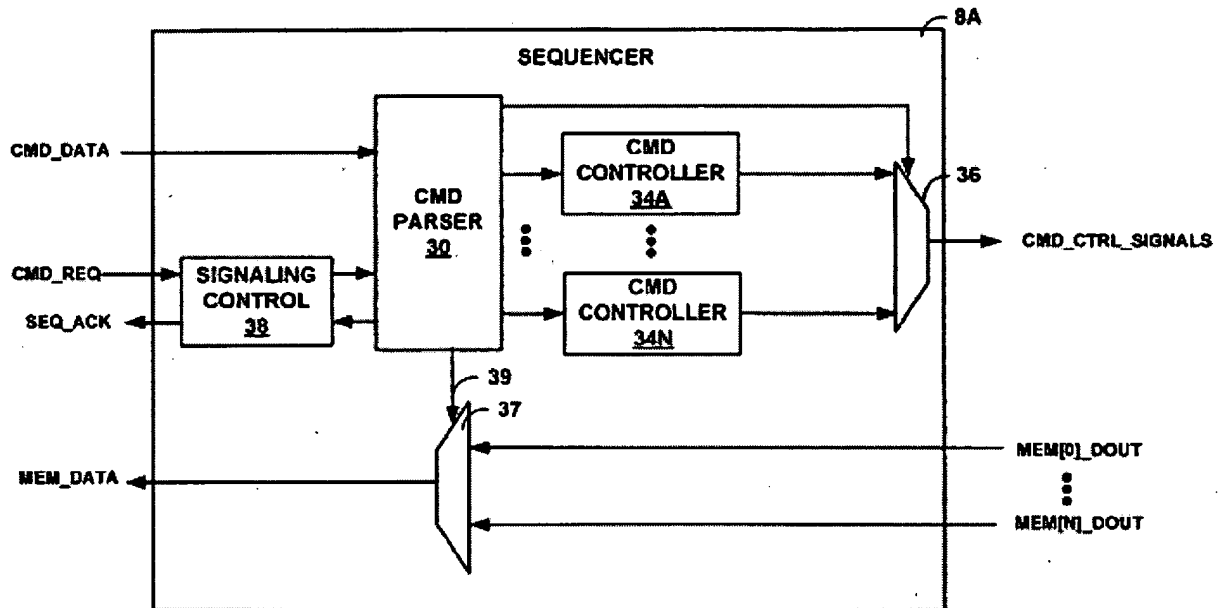
Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1003 ¶¶ 7-8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* ¶ 28. “[E]lectronic device 2 may be any device that incorporates memory modules, such as an embedded computing system, a computer, server, personal digital assistant (PDA), mobile computing device, mobile communication device, digital recording device, network appliance, mobile positioning device, and the like.” *Id.* ¶ 32. Averbuj states that “electronic devices . . . are [normally] constructed from many integrated circuit chips and many supporting components mounted on a circuit board.” *Id.* ¶ 5. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces

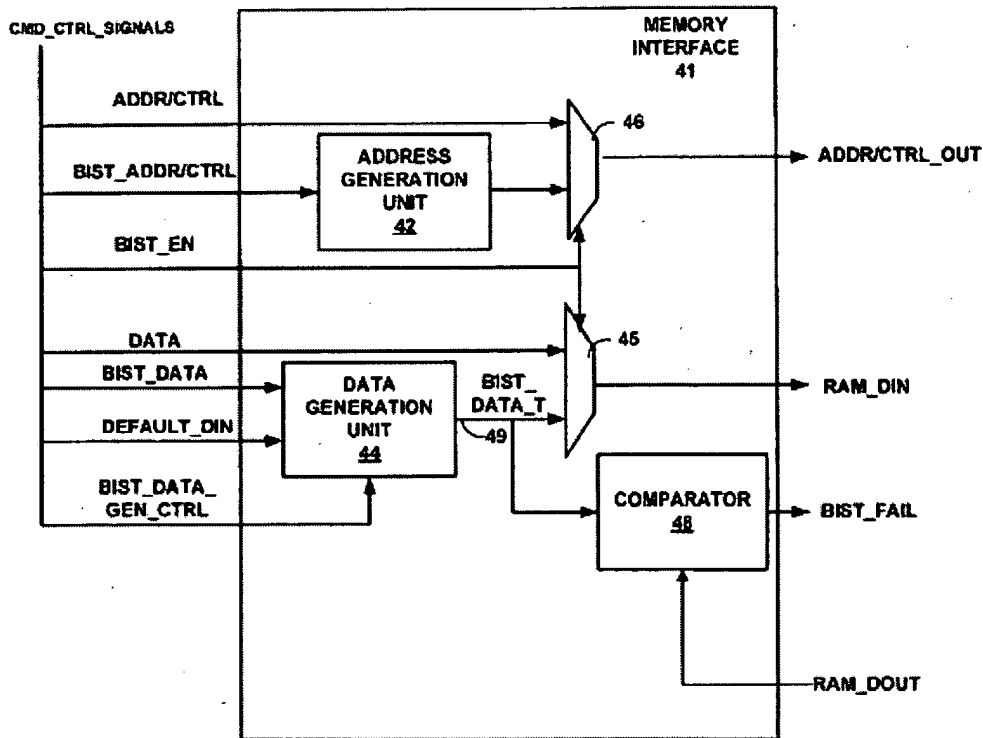
10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to sequencers 8 for application to device blocks 6. *Id.* ¶ 29.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* ¶ 22.



As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 and command controllers 34. *Id.* ¶¶ 41-42. Command parser 30 processes the commands from BIST controller 4 and invokes one of command controllers 34 which, in turn, issues command control signals, such as signals to provide a memory address and data, to receiving memory interfaces 10, 41. *Id.* ¶¶ 43, 51.

An exemplary memory interface is shown in Figure 6, below. *Id.* ¶ 23.



As shown in the Figure 6 embodiment, above, memory interface 41 includes a layer of multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals (BIST_ADDR/CTRL) and the test data provided by a respective higher level sequencer[, e.g., sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level sequencer [8A], and transforms the data and addresses provided by the sequencer as needed based on the particular physical characteristics of the memory module” (*id.* ¶ 54) using address

generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “[A]ddress generation unit 42 generates the address applied to [] memory module 12 based on the addressing requirements specified by sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* ¶ 51. “[D]ata generation unit 44 [] receives BIST_DATA signals and default data (DEFAULT_DIN) as provided by [] sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the sequencer and the specific physical characteristics of the corresponding memory module.” *Id.* ¶ 49. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.*

2. Analysis

Petitioner contends each of Averbuj’s device blocks 6A-6N is “a ‘self-testing memory module,’ as recited in the preambles of claims 1 and 20, that performs ‘a method of self-testing a memory module,’ as recited in the preamble of claim 29.” Pet. 8. Petitioner relies on the statement in Averbuj ¶ 5 that BIST units are normally constructed from integrated circuit chips mounted on a circuit board, to establish that device blocks 6 comprise a printed circuit board. Pet. 8. Petitioner further contends device blocks 6 are configured to be operatively coupled to a memory controller of a computer system, because Averbuj states that under normal operating conditions, address/control and data signals are provided by a programmable processor. *Id.* at 9 (citing Ex. 1003 ¶ 48). Petitioner contends Averbuj’s sequencer 8A, memory interfaces 10, 41, and memory modules 12 correspond, respectively, to the claimed “control module,” “data handlers,” and memory devices. *See* Pet. 16–18.

Patent Owner contends Averbuj's memory modules 12 are not "self-testing" as recited in independent claims 1, 20, and 29, because the components used in providing the self-testing, e.g. sequencer 8A and memory interfaces 10, 41, are external to memory modules 12. Prelim. Resp. 28. Patent Owner's argument is not convincing in light of our interpretation (for purposes of this Decision) of "self-testing memory module" as encompassing memory modules that can be tested with internal and/or external test equipment (*see* Section III.A., above).

Patent Owner argues Averbuj does not describe "a printed circuit board configured to be operatively coupled to a memory controller of a computer system." Prelim. Resp. 31 (quoting Ex. 1001, claims 1, 20, 29). Patent Owner argues Averbuj ¶ 5, relied on by Petitioner for a teaching of this limitation (*see* Pet. 15), is a description of prior art devices, and that Petitioner fails to give any explanation of how Averbuj's electronic device 2 could be "operatively coupled to a memory controller of a computer system." Prelim. Resp. 32 (quoting Ex. 1001, claims 1, 20, 29).

Patent Owner's argument is not convincing because Averbuj ¶ 32, also relied on by Petitioner for a teaching of this limitation (*see* Pet. 9), indicates electronic device 2 can be any number of conventional electronic devices, and Averbuj ¶ 5 describes electronic devices, in general, as constructed from integrated circuit chips and supporting components mounted on a circuit board. *See* Ex. 1003 ¶¶ 5, 32. Further, Averbuj ¶ 48 explicitly indicates multiplexers 45, 46 of memory interface 41, which are part of electronic device 2, are capable of receiving signals from a programmable processor. *See* Pet. 9; Ex. 1003 ¶ 48. Patent Owner has not explained sufficiently why the foregoing description in Averbuj does not

support Petitioner's contention that Averbuj's electronic device 2 can be "operatively coupled to a memory controller of a computer system," as recited in claims 1, 20, and 29.

Patent Owner further argues Averbuj fails to describe both a control module configured to generate address and control signals and a data handler configured to generate data as recited in each of independent claims 1, 20, and 29. *See* Prelim. Resp. 33–40. Patent Owner argues one of ordinary skill in the art would understand that either (1) Averbuj's sequencer 8A generates the address, control, and data signals, or (2) Averbuj's sequencer 8A generates the control signals and memory interface 41 generates the address and data signals. *Id.* at 39–40.

Patent Owner's argument is not convincing in light of our interpretation (for purposes of this decision) of "generate" as meaning "produce" or "cause." *See* Section III.B., above. Averbuj describes sequencer 8A as including command controllers 34 which issue a sequence of one or more operations and sequentially drive the appropriate command control signals, such as signals to provide a memory address and data, to receiving memory interfaces 10, 41. Pet. 10 (citing Ex. 1003, ¶ 43); *see* Ex. 1003, ¶ 51. This description supports Petitioner's contention that Averbuj describes a control module configured to generate/cause data. Averbuj describes memory interface 41 as including address generation unit 42 and data generation unit 44 which transform the data and addresses provided by the sequencer and "generate" the address and data applied to the memory module 12. Pet. 12–13 (citing Ex. 1003, ¶ 49); *see* Ex. 1003 ¶¶ 51, 54. The foregoing disclosure reasonably supports Petitioner's contention that Averbuj's memory interfaces 10, 41 are data handlers configured to

generate data. Patent Owner has not convinced us that because Averbuj's memory interfaces 10, 41 may also generate address signals, they are not properly viewed as meeting the claim limitation of data handlers. Although the independent claims require data handlers configured to either produce or cause data signals, Patent Owner has not explained why the claims should be narrowly construed as precluding the data handlers from being configured to also produce or cause address signals.

On this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner identifies where Averbuj describes every element of the challenged claims. *See* Pet. 7–24. Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 1, 14, 15, 17–25, 27, and 29–35 are anticipated by Averbuj.

B. Obviousness of Claims 1–12, 14–25, and 27–35 over Averbuj and Tsern

1. Claims 1, 14, 15, 17–25, 27, and 29–35

As discussed in Section IV.A.2., above, Petitioner contends Averbuj anticipates claims 1, 14, 15, 17–25, 27, and 29–35. Petitioner contends, in the alternative, that if this Board Panel determines Averbuj does not describe certain limitations recited in claims 1, 14, 15, 17–25, 27, and 29–35, these limitations are described in Tsern (Ex. 1005) and

[t]he [person of ordinary skill] in the memory arts would [have been] motivated to combine Averbuj and Tsern for many reasons, such as to supplement the teachings of Averbuj relating to an efficient memory BIST architecture with Tsern's teachings of efficient ways to incorporate a self-testing memory system within multiple integrated circuits on a printed circuit board.

Pet. 27; *see id.* at 33. Petitioner has not asserted that claims 1, 14, 15, 17–

25, 27, and 29–35 would have been obvious over Averbuj, alone. *See generally, id.* at 24–27 and 33.

In support of a motivation to combine Averbuj and Tsern, Petitioner relies on the Alpert Declaration. *See, e.g., id.* at 27 (citing Ex. 1007 ¶¶ 157, 159). Dr. Alpert’s opinion that a person of ordinary skill in the art would have been motivated to combine Averbuj and Tsern, in turn, is either unsupported or relies on disclosure in the Detailed Description section of the ’434 patent. *See* Ex. 1007 ¶ 157 (citing Ex. 1001, 3:32–37 and 7:21–66), ¶ 159 (citing Ex. 1001, 3:32–37 and 7:43–66), ¶ 186. On this record, there is insufficient evidence that the cited portions of the ’434 patent are a description of prior art. Moreover, Dr. Alpert’s statements lack sufficient articulated reasoning with rational underpinning as to why an ordinary artisan, at the time of the invention, would have had a reason to combine the prior art elements in the manner required by the challenged claims.

On this record, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1, 14, 15, 17–25, and 27–35 would have been obvious over Averbuj and Tsern.

2. Claims 2–12 and 16

In support of its obviousness challenge as to dependent claims 2–12 and 16, Petitioner relies on the same motivation to combine Averbuj and Tsern relied on in support of its obviousness challenge as to independent claim 1. Petitioner has not asserted that claims 2–12 and 16 would have been obvious over Averbuj, alone. *See generally, Pet.* 24–33.

Petitioner’s motivation to combine is insufficient for the reasons stated above. Therefore, on this record, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 2–12 and 16

would have been obvious over Averbuj and Tsern.

3. Claim 28

Claim 28 depends from independent claim 20 and recites: “wherein the at least one data handler is positioned on the printed circuit board proximate to the corresponding plurality of data ports.” As discussed in Section IV.A.2, above, Petitioner has identified where Averbuj describes every element of claim 20. Petitioner contends Averbuj discloses the limitation recited in claim 28, and has likewise identified where Averbuj describes this limitation. Pet. 29 (citing Ex. 1003, Fig. 4). Patent Owner has not raised any arguments specific to claim 28. *See* Prelim. Resp. 47.

Petitioner’s alternative contention is that Tsern discloses the limitation recited in claim 28. Pet. 29–30. However, Petitioner has not explained sufficiently the motivation to combine Averbuj and Tsern for the reasons stated above.

On this record, Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claim 28 would have been obvious over Averbuj. Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1–12, 14–25, and 27–35 would have been obvious over Averbuj and Tsern.

C. Obviousness of Claims 1–6, 14–16, 19, and 29 over Huang and Tsern

1. Huang (Ex. 1004)

Huang describes a transparent built-in self-test (BIST) interface that is designed to perform testing of multiple embedded memory arrays in the normal mode. Ex. 1004, Abstract. Huang’s transparent BIST system scheme includes a BIST controller, transparent test interface, memory arrays

and a global multiple input shift register (MISR). *Id.* at 381, Fig. 2. As illustrated in Figure 2, the BIST controller generates Address, and control (Normal/BIST, shift clock, Read, and Write) signals. *Id.* “[A]ll memory modules and interface circuits receive the same test control signals from the transparent BIST controller.” *Id.* at 381, col. 2. The test interface “implement[s] the functions of test pattern generation and test response evaluation.” *Id.* at 384, col. 2. According to Huang, “the test interface generate[s] test patterns *locally* to avoid test data routing from test controller to each memory array.” *Id.* at 381, col. 1. “[T]est patterns are generated by alternatively selecting Q or /Q as an input to each memory cell . . . , so the transition of Q→/Q and /Q→Q will be exercised in each memory cell. Thus, based on the memory content left by normal operations, new test patterns are generated by complementing each memory word several times.” *Id.*

2. Analysis

Petitioner contends Huang teaches all of the claim elements of independent claims 1 and 29. Pet. 39; *see* Ex. 1007 ¶ 196 (stating that Huang’s Figure 2 transparent BIST system is a “memory module” as claimed). Petitioner contends Huang’s BIST controller, transparent interfaces, and memory arrays correspond, respectively, to the claimed control module, data handlers, and memory devices. Pet. 41–42 (citing Ex. 1007 ¶¶ 200–201, 207–208, and 209–210).

Patent Owner argues Huang does not disclose a self-testing memory module as claimed because test response evaluation is performed external to the memory modules. *Id.* at 50. As explained in Section III.A., above, for purposes of this Decision, we do not interpret the term “self-testing” as

precluding test response evaluation performed external to the memory modules, e.g., by Huang's MISR. Therefore, this argument is not persuasive.

Patent Owner argues Huang's data handlers (the transparent interfaces), are not "operable independently from each of the other data handlers," as recited in claims 1 and 29, because all memory arrays are tested concurrently. Prelim. Resp. 51–52. Patent Owner also argues Huang teaches away from performing any write operations to the memory arrays and, therefore, the data handlers (transparent interfaces) are not configured to generate data for writing to the memory devices (memory arrays). *Id.* at 52–53.

Patent Owner's arguments are not convincing because Petitioner's citation to Huang's disclosure that the test interfaces generate test patterns locally to corresponding memory arrays, supports its contention that each of Huang's test interfaces (data handlers) is independently operable from the other test interfaces and coupled to one or more corresponding memory arrays (memory devices) in the manner claimed. *See* Pet. 37–38 (citing Ex. 1004, 381, 1st column). Further, Petitioner points to Huang's statement that "test patterns are generated by alternatively selecting Q or /Q as an input to each memory cell," *id.* at 37, which supports its contention that test data is written to the memory devices (memory arrays). Patent Owner has not explained how the claim language precludes a system which is capable of concurrently testing all memory arrays.

Patent Owner argues Huang does not describe "a control module configured to generate address and control signals for testing the memory devices" as recited in claims 1 and 29. Prelim. Resp. 54–55. However, for

purposes of this decision, Petitioner's citation to Huang's statement that the memory modules and interface circuits receive test control signals from the BIST controller, and citation to Figure 2 which illustrates "Normal/BIST" control selections, is sufficient to support Petitioner's contention that Huang's BIST controller is configured in the manner claimed. *See* Pet. 36.

On this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner identifies where Huang discloses or suggests every element of challenged claims 1, 2, 5, 6, 14, 15, 19, and 29. *See* Pet. 7–24. Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 1, 2, 5, 6, 14, 15, 19, and 29 would have been obvious over Huang.

With respect to claims 3, 4, and 16, Petitioner relies exclusively on the combination of Huang and Tsern, and has not asserted that Huang, alone, discloses or suggests the limitations recited in these claims. Moreover, as an alternative to its contention that Huang, alone, describes the limitations recited in claims 1, 2, 5, 6, 14, 15, 19, and 29, Petitioner asserts

to the extent [] Huang does not specifically disclose "a printed circuit board configured to be operatively coupled to a memory controller of a computer system," a plurality of memory devices "on the printed circuit board," or "each memory device of the plurality of memory devices comprising data, address, and control ports," these limitations are trivial implementation details and are obvious in view of Tsern.

Pet. 39. Petitioner contends Tsern describes a memory module that may communicate with a memory controller, and includes a printed circuit board and a memory device having data, address, and control ports. *Id.* at 40–41 (citing Ex. 1005 ¶¶ 52, 58). Petitioner contends

[a] POSITA would have been motivated to combine the self-testing memory modules disclosed in Huang and Tsern because both relate to BIST systems for testing memory devices using circuitry that is located local to the memory devices. The POSITA in the memory arts would be motivated to combine Huang and Tsern for many reasons, such as to supplement the teachings of Huang relating to the desire to have test interfaces generate test patterns locally with Tsern's teachings of efficient ways to incorporate a self-testing memory system within multiple integrated circuits on a printed circuit board. Further, the DIMM modules disclosed by Tsern are an efficient standard form factor for memory modules and would therefore yield a predictable result when combined with Huang.

Id. at 43. In support of a motivation to combine Huang and Tsern, Petitioner relies on Dr. Alpert's opinion that the person of ordinary skill in the art would have found it obvious to combine the teachings of Huang and Tsern. *See, e.g., id.* (citing Ex. 1007, ¶¶ 193, 199, and 206). Dr. Alpert's opinion that a person of ordinary skill in the art would have been motivated to combine Huang and Tsern, in turn, is either unsupported or relies on disclosure in the '434 patent Detailed Description. *See, e.g.,* Ex. 1007 ¶¶ 193, 199, and 206. As discussed above in Section IV.B.1., there is no evidence on this record that the cited portions of the '434 patent are a description of prior art. Moreover, Dr. Alpert's statements lack sufficient articulated reasoning with rational underpinning as to why an ordinary artisan at the time of the invention would have had a reason to combine the prior art elements in the manner required by the challenged claims. Accordingly, on this record, Petitioner has not demonstrated a reasonable likelihood it would prevail in showing that claims 1–6, 14–16, 19, and 29 would have been obvious over Huang in view of Tsern.

*D. Obviousness of Claims 7–12, 17, 18, 20–25, 27, 28, and 30–35
over Huang, Tsern, and Averbuj*

In Section IV.C.2, we determined Petitioner identifies where Huang discloses or suggests every element of challenged independent claims 1 and 29. Independent claim 20 is similar to claims 1 and 29, but additionally includes a limitation that the data handler is “configured to generate cyclic data for writing to the corresponding plurality of data ports.” Pet. 50 (quoting claim 20). Petitioner contends this limitation is disclosed by Averbuj. *Id.* at 50–51 (citing Ex. 1003 ¶ 49). In support of its obviousness challenge as to independent claim 20, Petitioner relies on Dr. Alpert’s opinion that both Huang and Averbuj “relate to capabilities of self-testing memory modules that include various memory widths,” and that “a person having ordinary skill in the art would [have found] it advantageous to augment the teachings of Huang with those of Averbuj in order to improve test coverage, for example by using pattern-based tests during manufacture or initial system power-up when the memory content does not need to be preserved.” Ex. 1007 ¶¶ 281–282; *see* Pet. 51. Petitioner has identified where Averbuj discloses or suggests the limitations recited in dependent claims 7–12, 17, 18, 21–25, 27, 28, and 30–35, and explained the motivation to combine the teachings of Huang and Averbuj. *See* Pet. 51–57.

In response to this challenge, Patent Owner relies on the same arguments made in response to Petitioner’s anticipation challenge based on Averbuj and obviousness challenges based on either Huang or Averbuj. *See* Prelim. Resp. 57–59. We are not persuaded by these arguments for the reasons discussed in Sections IV.A.2 and IV.C.2, above.

On this record, we are persuaded, based on the discussion above and

the further arguments in the Petition, that Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 7–12, 17, 18, 20–25, 27, 28, and 30–35 would have been obvious over Huang and Averbuj. However, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that these claims would have been obvious over Huang, Tsern, and Averbuj, because Petitioner has not explained sufficiently the motivation to combine Huang and Tsern for the reasons stated in Section IV.C.2., above.

E. Obviousness of claims 13 and 26 over Averbuj, Tsern and JEDEC Standard, and over Huang, Tsern, Averbuj, and JEDEC Standard

Petitioner relies on JEDEC Standard (Ex. 1006) as evidence that at the time of the invention claimed in the '434 patent, it was known in the art to record failing data and corresponding addresses for BIST failures. Pet. 33–34, 57–58 (citing Ex. 1006 § 8.2.4.5). Petitioner contends it would have been obvious to have recorded failing data and corresponding addresses in Averbuj's and Huang's BIST systems to enable a failure analysis and determination of error locations. *Id.* at 34, 58. Patent Owner's arguments with respect to claims 13 and 26 are limited to its contention that JEDEC standard does not cure the deficiencies of Averbuj and/or Huang in combination with Tsern. Prelim. Resp. 47, 59–60.

On this record, Petitioner has identified where the limitations recited in claims 13 and 26 are disclosed or suggested in Huang, Averbuj, and JEDEC Standard, and sufficiently explained the motivation to combine these references. Accordingly, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 13 and 26

would have been obvious over Huang, Averbuj, and JEDEC Standard.

Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 13 and 26 would have been obvious over Huang, Tsern, Averbuj, and JEDEC Standard, because Petitioner's motivation to combine Huang with Tsern is insufficient for the reasons explained in Section IV.C.2., above. As discussed in Sections IV.B.1 & 2, above, Petitioner has not asserted that the claims from which claims 13 and 26 depend would have been obvious over Averbuj, alone, but relies solely on the combination of Averbuj and Tsern. Because Petitioner's motivation to combine Averbuj with Tsern is insufficient for the reasons explained in Sections IV.B.1. & 2, above, Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 13 and 26 would have been obvious over Averbuj, Tsern, and JEDEC Standard.

V. CONCLUSION

Petitioner has demonstrated a reasonable likelihood that it would prevail on the grounds that claims 1, 14, 15, 17–25, 27, and 29–35 are anticipated by Averbuj, claim 28 would have been obvious over Averbuj, claims 1, 2, 5, 6, 14, 15, 19, and 29 would have been obvious over Huang, claims 7–12, 17, 18, 20–25, 27, 28, and 30–35 would have been obvious over Huang and Averbuj, and claims 13 and 26 would have been obvious over Huang, Averbuj, and JEDEC Standard. Petitioner has not demonstrated a reasonable likelihood that it would prevail on the grounds that the challenged claims would have been obvious over Huang and/or Averbuj in view of Tsern, or Huang and/or Averbuj in view of Tsern and JEDEC

Standard. Petitioner has not demonstrated a reasonable likelihood that it would prevail on any ground as to claims 3, 4, and 16.

At this stage of the proceeding, the Board has not made a final determination as to the patentability of any challenged claim.

VI. ORDER

For the reasons given, it is

ORDERED that, pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '434 patent is instituted on the following grounds:

- 1) claims 1, 14, 15, 17–25, 27, and 29–35 based on anticipation by Averbuj;
- 2) claim 28 based on obviousness over Averbuj;
- 3) claims 1, 2, 5, 6, 14, 15, 19, and 29 based on obviousness over Huang;
- 4) claims 7–12, 17, 18, 20–25, 27, 28, and 30–35 based on obviousness over Huang and Averbuj; and
- 5) claims 13 and 26 based on obviousness over Huang, Averbuj, and JEDEC Standard.

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '434 patent is hereby instituted commencing on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial.

FURTHER ORDERED that all other grounds raised in the Petition are denied, and no ground other than those specifically granted above is authorized for the *inter partes* review.

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Docket No.: NETL046A

Customer No. 79141

ASSERTION OF ENTITLEMENT TO SMALL ENTITY STATUS UNDER 37 CFR 1.27

Applicant	: Lee et al
App. No.	: 12/422,925
Pat. No.	: 8,001,434
Issued	: August 16, 2011
For	: MEMORY BOARD WITH SELF- TESTING CAPABILITY
Docket No.	: NETL046A

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

The Assignee, Netlist, Inc., of all rights and title to the above-identified patent is a small entity concern as defined in 37 C.F.R. 1.27(a). Assertion of entitlement to small entity status under 37 C.F.R. 1.27(c) is hereby made for this patent.

Applicants acknowledge the duty to file notification of a loss of entitlement to small entity status in the patent prior to paying, or at the time of paying, any maintenance fee due after the date on which status as a small entity is no longer appropriate.

Respectfully submitted,

Date: January 29, 2015

Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01372
Patent 8,001,434 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
PETER P. CHEN, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

DECISION
Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) to institute an *inter partes* review of claims 1–35 (the “challenged claims”) of U.S. Patent No. 8,001,434 B1 (Ex. 1008, “the ’434 patent”). 35 U.S.C. §§ 311–319. Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 11, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may not be instituted “unless . . . there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” Upon consideration of the Petition and Patent Owner’s Preliminary Response, we determine Petitioner has established a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. We, therefore, institute an *inter partes* review as to claims 1–4, 14–20, 27, and 29. Our factual findings and conclusions at this stage of the proceeding are based on the evidentiary record developed thus far (prior to Patent Owner’s Response). This is not a final decision as to patentability of claims for which *inter partes* review is instituted. Our final decision will be based on the record as fully developed during trial.

II. BACKGROUND

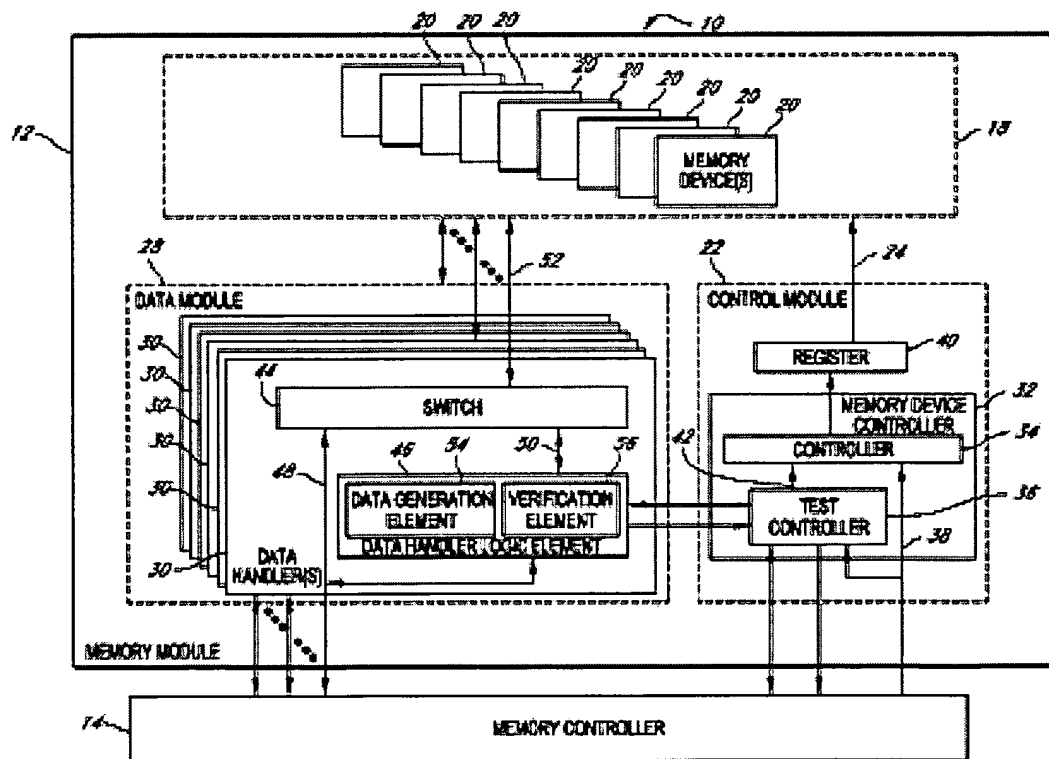
A. Related Matters

The ’434 patent is also the subject of *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB), wherein trial was instituted on claims 1, 2, 5–15, and 17–35. IPR2014-00970, Paper 12. Contemporaneous with the instant Petition, Petitioner also filed a second petition for *inter partes* review

of the '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB). Also contemporaneous with the instant Petition, Petitioner filed two petitions for *inter partes* review of U.S. Patent No. 8,359,501 ("the '501 patent"), which claims priority as a continuation of the '434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01374 and IPR2014-01375 (PTAB).

B. The '434 Patent (Ex. 1001)

The '434 patent relates to self-testing electronic memory modules. Ex. 1001, 1:23–24. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '434 patent, reproduced below.



As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:1–3. Memory module 10

includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:3–7. Memory module 10 further includes control module 22 and data module 28 comprising a plurality of independently operable data handlers 30. *Id.* at 5:7–11. Control module 22 includes memory device controller 34, which receives address and control signals 38 from system memory controller 14 and address and control signals 42 from test controller 36. *Id.* at 9:40–45. Similarly, data module 28 includes switch 44, which selectively inputs to memory devices 18 either data signals 48 from memory controller 14 or data signals 50 from data handler logic element 46. *Id.* at 10:7–11.

During test mode, memory device controller 34 sends address and control signals 38 from test controller 36 to register 40 for input to address and control ports of memory devices 18. *Id.* at 9:40–50. “[T]est controller 36 updates each of [] data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.)” *Id.* 14:7–9. Data generation element 54 of data handler logic element 46 is configured to generate data signals and/or patterns of data signals based on the information received from test controller 36, for writing to memory devices 18. *Id.* at 10:28–37, 15:25–27.

C. Illustrative Claim

Of the challenged claims, claims 1, 20, and 29 are independent. Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
a printed circuit board configured to be operatively coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and

a circuit comprising:

a control module configured to generate address and control signals for testing the memory devices; and

a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

D. Evidence Relied Upon

Petitioner's patentability challenges are based on the following references:

Reference	Patent	Exhibit
Averbuj	US 7,392,442	1011
Best	US 8,233,303	1012
Pandey	US 6,934,205	1013
Rajan	US 7,379,316	1014

In addition to these references, Petitioner relies on the Declaration of Dr. Nader Bagherzadeh (Ex. 1009) under 37 C.F.R. § 1.68. Pet. 6.

E. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of the '434 patent claims based on the following grounds:

References	Basis	Claims Challenged
Averbuj	§ 102(b)	1–4, 14–20, 27, and 29
Averbuj and Best	§ 103(a)	5–7, 13, 21, 26, 28, 30, and 35
Averbuj, Best, and Pandey	§ 103(a)	8–12, 22–25, and 31–34
Rajan and Averbuj	§ 103(a)	1–7, 13–21, 26–30, and 35
Rajan, Averbuj, and Pandey	§ 103(a)	8–12, 22–25, and 31–34

III. CLAIM CONSTRUCTION

Claims of an unexpired patent are interpreted using the broadest reasonable construction in light of the specification of the patent. *See* 37 C.F.R. § 42.100(b); Office Patent Trial Practice Guide, 77 Fed. Reg. 48756, 48766 (Aug. 14, 2012). Petitioner and Patent Owner offer respective constructions for seven different claim terms. *See* Pet. 23–28; Prelim. Resp. 18–26. For purposes of this decision, we determine the only claim language requiring express construction is the term “generate,” recited in each of independent claims 1, 20, and 29.

Petitioner contends the term “generate” should be interpreted as meaning “cause” or “initiate.” Pet. 24–25. In support of its proposed interpretation of “generate,” Petitioner relies on description in the '434 patent which states that the data signals generated by data generation element 54 of data module 28 are programmable based on information

received from memory controller 14 (Ex. 1008, 10:67–11:2), and on *ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms* (4th Edition, 1988) (Ex. 1016), which defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters” (Ex. 1016, 411). Pet 24–25. Petitioner’s proposed interpretation is consistent with our prior interpretation of the ’434 patent claim term “generate” (for purposes of instituting trial) as meaning “produce” or “cause.” See IPR2014-00970, Paper 12, slip op. at 10.

Patent Owner contends Petitioner’s proposed definition of “generate” is based on an erroneous contention that programming and generating are synonymous. See Prelim. Resp. 22. Patent Owner argues, more specifically, that a programmable command transmitted to a generator “to cause [data] generation” does not itself comprise the particular data sought to be produced, and, therefore, does not “generate” the data. *Id.* As in IPR2014-00970, Paper 10 at 25, Patent Owner contends the claim term “generate” should be interpreted as meaning “produce.” *Id.* at 20. In support of its proposed interpretation of “generate,” Patent Owner relies on column 6, lines 6–9 of the ’434 patent, which read: “In some embodiments, the control module 22 and the data module 28 **produce** memory addresses, control and/or data signals according to the JEDEC standard memory protocol.” *Id.* at 20–21 (quoting Ex. 1001, 6:6–9) (emphasis added). Patent Owner also relies on *Merriam-Webster’s Collegiate Dictionary* (Ex. 2009) and *The New American Webster Handy College Dictionary* (Ex. 2010), each of which lists “produce” as one definition of “generate” (Ex. 2009, 484; Ex. 2010, 230). Prelim. Resp. 21.

As explained in our Decision to Institute in IPR2014-00970 (Paper 12 at 9), the description in the '434 patent relied upon by Patent Owner relates to only "some embodiments," and the claims do not recite that the data and control modules are configured to generate signals "according to the JEDEC standard memory protocol" (Prelim. Resp. 20–21). As further explained in that prior decision, *Merriam-Webster's Collegiate Dictionary* also defines generate as "to be the cause of" (IPR2014-00970, Paper 12, slip op. at 9–10). Ex. 2009, 484. Patent Owner argues the definition "to be the cause of" only applies in the context of "a situation, action, or state of mind." Prelim. Resp. 21–22. Patent Owner contends "the context of the '434 Patent is the physical process of generating electrical signals. Accordingly, the definition of 'generate' should be taken as 'PRODUCE,' since this is the ordinary meaning within the context of 'a vital, chemical, or physical process.'" *Id.* at 22.

We agree with Patent Owner that the intrinsic evidence, as well as the dictionary definitions on this record, support an interpretation of the claim term "generate" as meaning, *inter alia*, "produce," and, that a module "configured to generate" address and control signals or data, as claimed, encompasses a component which produces the exact signals or data used by the memory devices. On the record before us and for purposes of this decision, we, however, are not convinced of error in our prior determination (*see* IPR2014-00970, Paper 12, slip op. at 10) that one of ordinary skill in the art would also understand the term "generate," as used in the claims, to mean "cause." Thus, for example, one of ordinary skill in the art would understand that a module "configured to generate" signals or data encompasses both a component that produces the exact signal or data used

by a memory device and a component which transmits a programmable command to that generator to cause production of the signal or data. Our interpretation of the term “generate” as meaning “cause” is supported by the ’434 patent (*see id.* at 9 (citing col. 10, lines 49–50 and column 10, line 67 to column 11, line 4)) and the dictionary definitions of record, including the additional definition of “generate” in *Merriam-Webster’s Collegiate Dictionary* as, in the context of a mathematical set or structure, “to define or originate . . . by the application of one or more rules or operations,” Ex. 2009, 484.

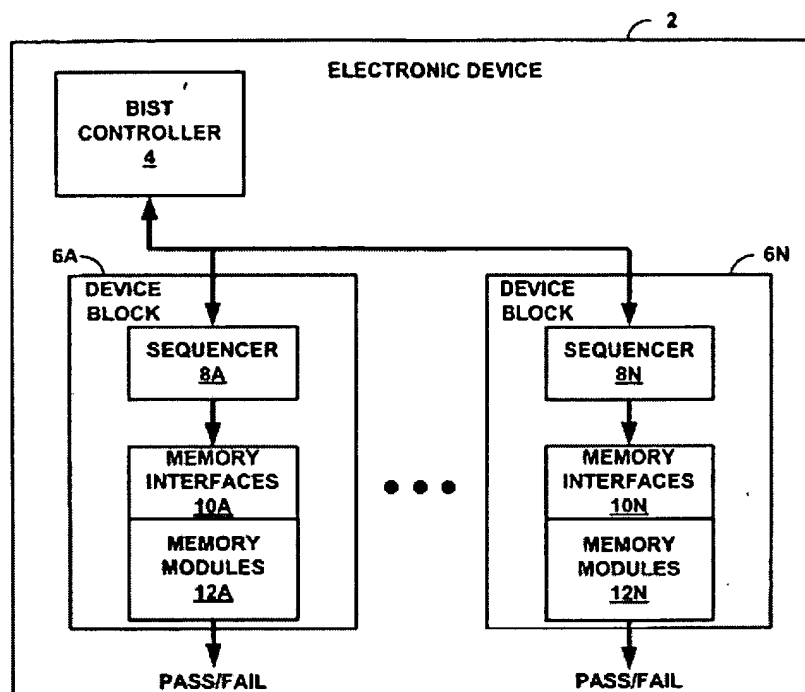
Accordingly, for purposes of this decision, we interpret the claim term “generate” as meaning “produce” or “cause.”

IV. ANALYSIS

A. Anticipation of Claims 1–4, 14–20, 27, and 29 by Averbuj

1. Averbuj (Ex. 1011)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1011, 1:64–65, 2:3–4. Figure 1 of Averbuj is reproduced below.

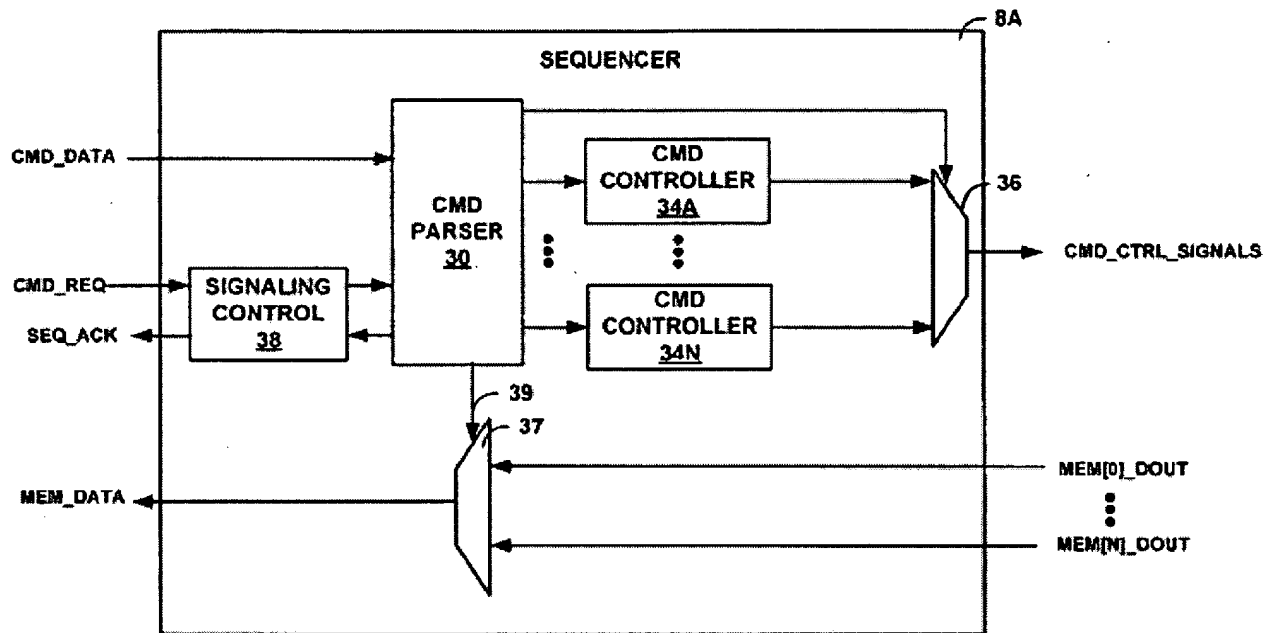


Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* at 3:62–64. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* at 3:64–67. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* at 3:67–4:3.

BIST controller 4 includes algorithm controller 26 which is invoked either by a user or automatically upon power-up of electronic device 2. *Id.* at 5:12–16. Once invoked, algorithm controller 26 selects an algorithm stored within algorithm memory 20 of BIST controller 4. *Id.* at 5:16–19. The selected algorithm is applied to sequencers 8 for application to device

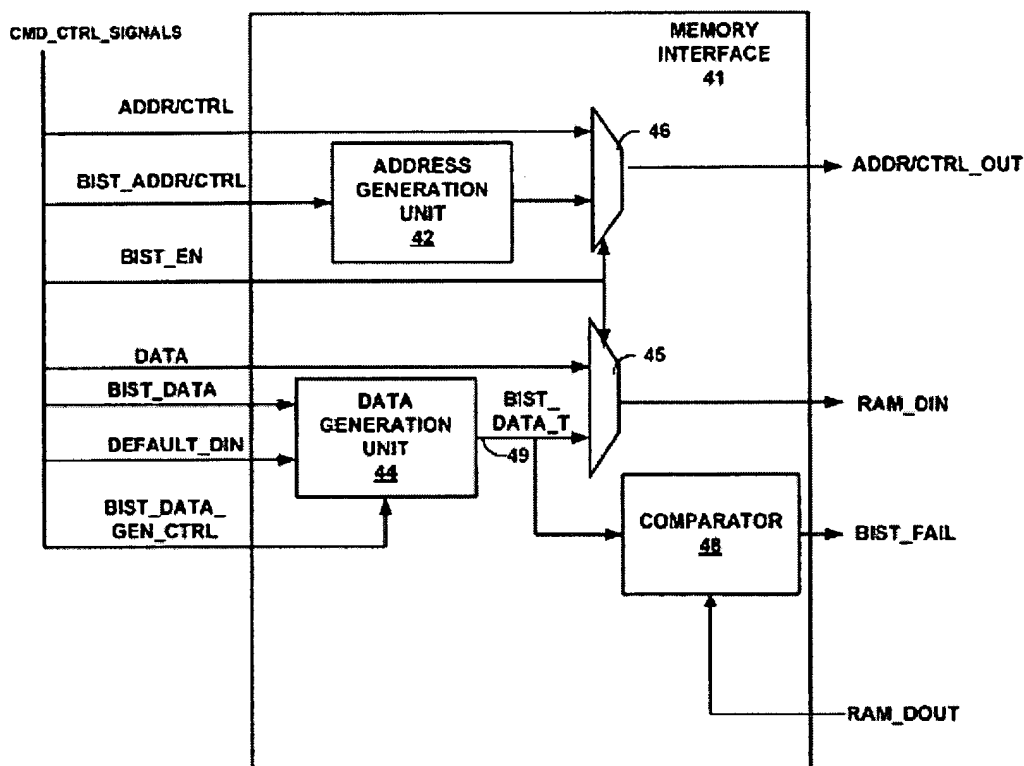
blocks 6 as a stream of binary commands (CMD_DATA), each command defining a test within the overall algorithm. *Id.* at 4:4–7, 5:19–21, 44–46. In the embodiment illustrated in Figure 8, command 60 includes sequencer identifier 62, which identifies a sequencer to which command 60 is being issued, and payload 64. *Id.* at 9:17–21. Payload 64 includes operation code (OP CODE) 66 and a set of parameters 68. *Id.* at 9:30–31. “OP CODE 66 specifies a particular function to be performed by the receiving sequencers 8.” *Id.* at 9:32–34. “The format and meaning of parameters 68 is dependent upon the type of command, as specified by OP CODE 66.” *Id.* at 9:60–62. Figure 9A illustrates an exemplary embodiment wherein, for an OP CODE specifying a RESET command, which resets the sequencer(s) and selectively enables BIST mode, “the only relevant bit of parameters 68 is bit 28, which can be asserted or de-asserted to selectively enable and disable BIST mode.” *Id.* at 9:64–67. Another exemplary OP CODE is SET ADDRESS which “[s]ets a specific starting address as well as a maximum address limit for a test algorithm as applied to the memory modules.” *Id.* at 9, Table 1.

Upon assertion of the command request (CMD_REQ) interconnects, BIST controller 4 communicates a current command of the algorithm on the CMD_DATA interconnects to sequencers 8. *Id.* at 5:53–56. Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* at 3:46–47.



As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 that receives command data (CMD_DATA) from BIST controller 4. *Id.* at 6:24–26. Command parser 30 processes the commands from BIST controller 4 to identify a specified operation, e.g., by identifying an operational code (OP CODE) specified by the command. *Id.* at 6:26–29. Based on the specified operation, command parser 30 may extract one or more parameters from the command, and pass the extracted parameters to the command controller, (i.e., one of command controllers 34A–34N) that corresponds to the command. *Id.* at 6:30–37. The invoked command controller, in turn, issues a sequence of one or more operations to each memory interface 10, sequentially driving the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. *Id.* at 6:41–43. The command control signals may include signals to provide memory address data to the receiving memory interfaces 10. *Id.* at 6:46–48.

An exemplary memory interface is shown in Figure 6, below. *Id.* at 3:48–49.



As shown in the Figure 6 embodiment, above, memory interface 41 includes address generation unit 42 and data generation unit 44 that receive command control signals (CMD_CTRL_SIGNALS) from sequencer 8A. Address generation unit 42 receives address/control signals (BIST_ADDR/CTRL signals) and “generates the address applied to the memory module 12 based on the addressing requirements specified by sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* at 7:63–67. “[D]ata generation unit 44 [] receives BIST_DATA signals and default data (DEFAULT_DIN) as provided by [] sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals

(BIST_DATA_GEN_CTRL) provided by the sequencer and the specific physical characteristics of the corresponding memory module 12.” *Id.* at 7:32–37. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.* at 7:38–40.

2. Analysis

Petitioner identifies the following components of Averbuj’s device as corresponding to the components recited in independent claims 1, 20, and 29:

Challenged Claims 1, 20, and 29		Averbuj’s Device	Pet.
A self-testing memory module comprising a printed circuit board		Electronic device 2	28
A plurality of memory devices		Memory modules 12A-N	29
A control module configured to generate signals for testing the memory devices		BIST controller 4 generates OP CODE	29
a.	address signals	SET ADDRESS	29
b.	control signals	CMD_REQ	
A data module comprising			
a.	a plurality of data handlers	Sequencers 8A-N paired with their respective corresponding memory interfaces 10A-N, each of which pairs is a “data handler”	30
b.	the data handlers generating data	Each memory interface 10 includes data generation unit 44 for generating RAM_DIN	31

Patent Owner argues Petitioner has not identified sufficient support for its contention that Averbuj's BIST controller 4 generates address and control signals for testing memory devices as recited in the claims. *See* Prelim. Resp. 32–37. Patent Owner argues the signals that enter Averbuj's memory modules 12A–N all are generated by the corresponding memory interfaces 10A–N. *Id.* at 15. Patent Owner argues Averbuj's OP CODES (and the commands comprising OP CODES) define memory operations “without regard to the physical characteristics or time requirements of memory modules 12,” and, therefore, there is no evidence the OP CODES “could even be recognized by a memory chip, let alone test a memory chip.” *Id.* at 34–35 (emphasis omitted). Patent Owner further contends Averbuj's OP CODES are designed specifically to program a sequencer, and Petitioner has not shown that programming a sequencer is equivalent to testing memory chips or generating address and control signals. *Id.* at 35.

Patent Owner's arguments are based on a narrow construction of the term “generate” as requiring a control module configured to produce directly any address and control signals used to test the memory devices. As explained in Section III., above, for purposes of this decision, we interpret the claim term “generate” more broadly than proposed by Patent Owner as meaning “produce” or “cause.”

Averbuj discloses that the CMD_DATA produced by BIST controller 4 is a stream of binary commands, each of which identifies (1) a sequencer to which the command is being issued, (2) an operation (OP CODE) to be performed by the receiving sequencer (e.g., SET ADDRESS to set a starting address), and (3) a set of parameters. Ex. 1011, 9:17–21, 30–33, Table 1. In other words, upon assertion of a CMD_REQ, BIST controller 4 issues

commands that “define a complete BIST algorithm” to an identified sequencer. Ex. 1011, 4:20–22. Although the commands are not sent directly to memory devices 12A–N, but require interpretation by a sequencer and computational transformation by address generation unit 42 of a memory interface (*id.* at 4:18–19, 8:16–17), the above description in Averbuj supports Petitioner’s contention that the CMD_REQ and SET ADDRESS codes issued by BIST controller 4 cause the issuance of, i.e., “generate,” address and control signals.

Patent Owner also argues the position advanced by Petitioner is inconsistent, contending that if “Averbuj’s address and control ‘OP Codes’ are considered Address and Control Signals, then necessarily Averbuj’s data ‘OP Codes’ should be considered Data signals and Averbuj’s BIST controller 4 must necessarily generate BOTH the Data and Address/Control signals.” Prelim. Resp. 37.

Patent Owner’s argument is unpersuasive because it is not based on an interpretation of the claim term “generate” as encompassing components that both cause and produce the issuance of data signals. Under this interpretation of “generate,” applied for purposes of this decision, both Averbuj’s BIST controller 4 and memory interfaces are properly viewed as generating data because BIST controller 4 causes the issuance of the exact data (RAM_DIN) produced by data generation unit 44 of memory interface 41 and applied to the inputs of memory devices 12A–N. Ex. 1011, 7:38–40.

On this record, we are persuaded, based on the discussion above and the further arguments in the Petition, that Petitioner identifies where Averbuj describes every element of challenged independent claims 1, 20, and 29. Based on our review of the present record, we are persuaded likewise as to

claims 2–4, 14–19, and 27. *See* Pet. 28–39. Petitioner has demonstrated a reasonable likelihood it would prevail on the ground that claims 1–4, 14–20, 27, and 29 are anticipated by Averbuj.

B. Obviousness of Claims 1–35

Petitioner contends claims 5–7, 13, 21, 26, 28, 30, and 35 would have been obvious over Averbuj in view of Best. Pet. 39–44. Petitioner further contends claims 1–7, 13–21, 26–30, and 35 would have been obvious over Rajan in view of Averbuj. *Id.* at 48–57. Petitioner contends the remaining claims 8–12, 22–25, and 31–34 would have been obvious over Averbuj and Best in view of Pandey, and over Rajan and Averbuj in view of Pandey. *Id.* at 44–48, 58–59.

Petitioner contends “[a person of ordinary skill in the art] would [have been] motivated to combine Averbuj ‘442 and Best ‘303, because Best ‘303 teaches a multi-die memory device with a BIST interface die that may be disposed in place of conventional memory in memory modules, such as the module of Averbuj ‘442, which is similarly configured for BIST functions.” Pet. 44 (citations omitted). Petitioner similarly contends

[a person of ordinary skill in the art] would [have been] motivated to combine Rajan ‘316 and Averbuj ‘442, because Averbuj ‘442 provides a BIST architecture applicable to any electronic device, such as circuit boards with memory modules as disclosed in Rajan ‘316. . . . [A] POSITA would be motivated to implement the known BIST architecture and process in Averbuj ‘442 with the BIST-enabled circuit board-based memory module devices of Rajan ‘316, thereby replicating the testing capability of Averbuj ‘442 to Rajan ‘316.

Id. at 57 (citations omitted). In support of these contentions, Petitioner cites paragraphs 61, 63, and 97, and Appendix E of the Declaration of Dr. Nader

Bagherzadeh (Ex. 1009). *Id.* at 44, 57. Petitioner notes: “The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *Id.* at 44 (citations omitted).

As explained by the Federal Circuit, “[a]lthough predictability is a touchstone of obviousness, the ‘predictable result’ discussed in *KSR* refers not only to the expectation that prior art elements are capable of being physically combined, but also that the combination would have worked for its intended purpose.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 567 F.3d 1314, 1326 (Fed. Cir. 2009) (citing *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007)).

Petitioner contends the proposed combinations of Averbuj and Best, and Rajan and Averbuj would have been nothing more than combinations of familiar elements, but has not identified persuasive evidence which supports these contentions. In this regard, we acknowledge Petitioner relies on Dr. Bagherzadeh’s testimony that “Best ‘303 teaches a multi-die memory device with a BIST interface die that may be disposed in place of conventional memory in memory modules such as the module of Averbuj ‘442, which is similarly configured for BIST functions” (Ex. 1009 ¶ 61) and that “Averbuj ‘442 provides a BIST architecture applicable to any electronic device such as circuit boards with memory modules as disclosed in Rajan ‘316” (*id.* ¶ 97). However, Dr. Bagherzadeh has not identified sufficient support for these statements. For example, Dr. Bagherzadeh has not explained clearly why, or identified evidence that shows, the invention described in the ’434 patent falls into a very predictable field, one of ordinary skill in the art would have possessed the requisite skills to make the

proposed combinations, or one of ordinary skill in the art reasonably would have anticipated the combination to function in an ordinary and expected way. *See Rothman v. Target Corp.*, 556 F.3d 1310, 1319–20 (Fed. Cir. 2009). With respect to this latter point, although a teaching away supports a conclusion that a prior art combination would have produced an inoperative result, *see DePuy*, 567 F.3d at 1326, Petitioner has not identified case law to support Dr. Bagherzadeh’s contention that the absence of a teaching away in Best and Averbuj evidences that the combination would be operative (Ex. 1009 ¶ 61). Petitioner’s contentions regarding motivation to modify the references based on Pandey are likewise deficient. *See* Pet. 48, 59.

For the above-stated reasons, on this record, Petitioner has not demonstrated a reasonable likelihood it would prevail on the grounds that claims 5–7, 13, 21, 26, 28, 30, and 35 would have been obvious over Averbuj in view of Best; claims 1–7, 13–21, 26–30, and 35 would have been obvious over Rajan in view of Averbuj; and claims 8–12, 22–25, and 31–34 would have been obvious over Averbuj in view of Best and Pandey, and over Rajan in view of Best and Pandey.

C. Unpatentability of Claims 5–13 and 28 Under 35 U.S.C. §112, Second Paragraph, as Indefinite

In addition to the above five grounds, Petitioner advances a sixth ground of unpatentability in which it contends that if we determine its proposed interpretation of “proximate” in claims 5 and 28 is overly broad, then “claim 5, its corresponding dependent claims 6–13, and claim 28 are unpatentable under section 112 as indefinite.” Pet. 59. We will not consider this challenge as to claims 5–13 and 28 because it exceeds the scope of *inter partes* review. *See* 35 U.S.C. § 311(b)(2013) (“A petitioner in an inter

partes review may request to cancel as unpatentable 1 or more claims of a patent only on a ground that could be raised under section 102 or 103 and only on the basis of prior art consisting of patents or printed publications.”).

D. Patent Owner’s Request to Deny the Petition as Redundant to the Petition in IPR2014-00970

Patent Owner contends we should deny Petitioner’s request to institute an *inter partes* review because the Petition merely sets forth grounds redundant to those raised in *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970, Paper 1 (PTAB). *See* Prelim. Resp. 26–31. Patent Owner cites several Board decisions in support of this argument. *See id.* (citing, e.g., *Unified Patents, Inc. v. Personalweb Techs.*, Case IPR2014-00702, Paper 13 (PTAB July 24, 2014)).

The Petition in the present case differs from that in IPR2014-00970 in that, *inter alia*, Petitioner relies on a different declarant, relies on different disclosure in Averbuj¹, and raises a new challenge as to claims 2–4 and 16 based on anticipation by Averbuj. Taking into account the burden on Netlist and the considerations set forth in 35 U.S.C. § 316(b), e.g., the efficient administration of the Office, we exercise our discretion and institute trial as to the ground of anticipation based on Averbuj for the reasons discussed above in Section IV.A.2. *See Rackspace US, Inc. v. Personalweb Techs, LLC*, Case IPR2014-00057, Paper 9 (PTAB Oct. 28, 2014) (instituting trial

¹ IPR2014-00970 relies on US 2005/0257109, which is the published application of the Averbuj patent relied on in the instant Petition. Prelim. Resp. 28. Neither Petitioner nor Patent Owner has identified any substantive differences in the disclosures of the Averbuj published application and the ’434 patent.

as to substantially the same ground on which trial was instituted in another pending proceeding).

V. CONCLUSION

Petitioner has demonstrated a reasonable likelihood that it would prevail on the grounds that claims 1–4, 14–20, 27, and 29 are anticipated by Averbuj. Petitioner has not demonstrated a reasonable likelihood that it would prevail on its challenges under 35 U.S.C. §103(a). We cannot consider the challenge to claims 5–13 and 28 as indefinite under 35 U.S.C. §112, second paragraph, because it exceeds the scope of *inter partes* review.

At this stage of the proceeding, the Board has not made a final determination as to the patentability of any challenged claim.

VI. ORDER

For the reasons given, it is

ORDERED that, pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '434 patent is instituted as to claims 1–4, 14–20, 27, and 29 based on anticipation by Averbuj;

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(a), *inter partes* review of the '434 patent is hereby instituted commencing on the entry date of this Order, and pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial; and

FURTHER ORDERED that all other grounds raised in the Petition are denied, and no ground other than that specifically granted above is authorized for the *inter partes* review.

IPR2014-01372
Patent 8,001,434 B1

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01373
Patent 8,001,434 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
PETER P. CHEN, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION
Denying Institution of *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 10, Corrected Petition (“Pet.”)) to institute an *inter partes* review of claims 1–35 (the “challenged claims”) of U.S. Patent No. 8,001,434 B1 (Ex. 1008, “the ’434 patent”). *See* 35 U.S.C. §§ 311–319. Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 14, “Prelim. Resp.”). We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may not be instituted “unless . . . there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” Upon consideration of the Petition and Patent Owner’s Preliminary Response, we determine Petitioner has not established a reasonable likelihood that it would prevail in showing the unpatentability of at least one of the challenged claims. We therefore decline to institute an *inter partes* review as to claims 1–35.

II. BACKGROUND

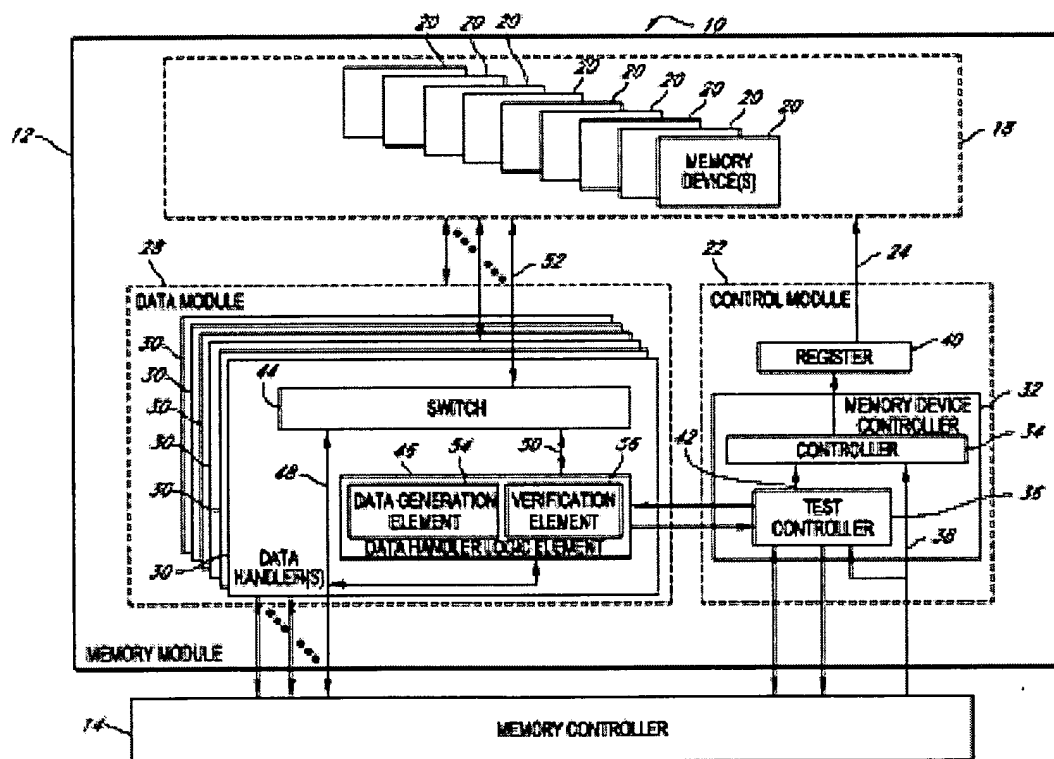
A. Related Matters

The ’434 patent is also the subject of *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB), wherein trial was instituted on claims 1, 2, 5–15, and 17–35. IPR2014-00970, Paper 12. Contemporaneous with the instant Petition, Petitioner also filed a second petition for *inter partes* review of the ’434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01372 (PTAB). Also contemporaneous with the instant Petition, Petitioner filed two petitions for *inter partes* review of U.S. Patent No. 8,359,501 (“the ’501 patent”), which claims priority as a continuation of the

'434 patent: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Cases IPR2014-01374 and IPR2014-01375 (PTAB).

B. The '434 Patent (Ex. 1008)

The '434 patent relates to self-testing electronic memory modules. Ex. 1008, 1:23–24. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '434 patent, reproduced below.



As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:1–3. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:3–7. Memory module 10 further includes control module 22 and data module 28 comprising a plurality of independently operable data

handlers 30. *Id.* at 5:7–11. Control module 22 includes memory device controller 34 which receives address and control signals 38 from system memory controller 14 and address and control signals 42 from test controller 36. *Id.* at 9:40–45. Similarly, data module 28 includes switch 44, which selectively inputs to memory devices 18 either data signals 48 from memory controller 14 or data signals 50 from data handler logic element 46. *Id.* at 10:7–11.

During test mode, memory device controller 34 sends address and control signals 38 from test controller 36 to register 40 for input to address and control ports of memory devices 18. *Id.* at 9:40–50. “[T]est controller 36 updates each of [] data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.)” *Id.* at 14:7–9. Data generation element 54 of data handler logic element 46 is configured to generate data signals and/or patterns of data signals based on the information received from test controller 36, for writing to memory devices 18. *Id.* at 10:28–37, 15:25–27.

C. Illustrative Claim

Of the challenged claims, claims 1, 20, and 29 are independent. Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and

a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

D. Evidence Relied Upon

Petitioner's patentability challenges are based on the following references:

Reference	Patent/Printed Publication	Exhibit
Fleischman '320	US 6,321,320 B1	1011
JTAG 2001	IEEE JTAG Standard No. 1149.1-2001	1012
Adams '873	US 7,203,873 B1	1013
Huang	<i>An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers</i> , VLSI Design 2001, Fourteenth International Conference on VLSI Design (pp. 379–384), IEEE (2000)	1014

In addition to these references, Petitioner relies on the Declaration of Dr. Nader Bagherzadeh (Ex. 1009) under 37 C.F.R. § 1.68. Pet. 6.

E. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of the '434 patent claims based

on the following grounds:

References	Basis	Claims Challenged
Fleischman '320 and JTAG 2001	§ 103(a)	1–35
Adams '873 and JTAG 2001	§ 103(a)	1–7, 13–21, 26–30, and 35
Adams '873, JTAG 2001, and Huang	§ 103(a)	8–12, 22–25, and 31–34

III.CLAIM CONSTRUCTION

Claims of an unexpired patent are interpreted using the broadest reasonable construction in light of the specification of the patent. *See* 37 C.F.R. § 42.100(b); *see also In re Cuozzo Speed Techs.*, No. 2014–1301, 2015 WL 448667, at *6 (Fed. Cir. Feb. 04, 2015). Petitioner and Patent Owner offer respective constructions for seven different claim terms. *See* Pet. 22–27; Prelim. Resp. 18–26. For purposes of this decision, we determine the only claim language requiring express construction is the term “generate,” recited in each of independent claims 1, 20, and 29.

Petitioner contends the term “generate” should be interpreted as meaning “cause” or “initiate.” Pet. 24–25. In support of its proposed interpretation of “generate,” Petitioner relies on a description in the '434 patent that states the data signals generated by data generation element 54 of data module 28 are programmable based on information received from memory controller 14 (Ex. 1008, 10:67–11:2), and on *ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms*, (4th ed. 1988) (Ex. 1016), which defines “generate (computing systems)” as “[t]o produce a program by selection of subsets from a set of skeletal coding under the

control of parameters” (*id.* at 411). Pet 24–25. Petitioner’s proposed interpretation is consistent with our prior interpretation of the ’434 patent claim term “generate” (for purposes of instituting trial) as meaning “produce” or “cause.” See IPR2014-00970, Paper 12, slip op. at 10.

Patent Owner contends Petitioner’s proposed definition of “generate” is based on an erroneous contention that programming and generating are synonymous. See Prelim. Resp. 30. Patent Owner argues, more specifically, that a programmable command transmitted to a generator “to cause [data] generation” does not itself comprise the particular data sought to be produced, and, therefore, does not “generate” the data. *Id.* As in IPR2014-00970 (Paper 10, 25), Patent Owner contends the claim term “generate” should be interpreted as meaning only “produce.” *Id.* at 30. In support of its proposed interpretation of “generate,” Patent Owner relies on column 6, lines 6–9, of the ’434 patent, which read: “In some embodiments, the control module 22 and the data module 28 **produce** memory addresses, control and/or data signals according to the JEDEC standard memory protocol.” *Id.* at 28 (quoting Ex. 1008, 6:6–9) (emphasis added). Patent Owner also relies on *Merriam Webster’s Collegiate Dictionary* (Ex. 2008) and *The New American Webster Handy College Dictionary* (Ex. 2009), each of which lists “produce” as one definition among several of “generate” (Ex. 2008, 484; Ex. 2009, 230). Prelim. Resp. 29.

As explained in our Decision to Institute in IPR2014-00970 (Paper 12, 9), the description in the ’434 patent relied upon by Patent Owner relates to only “some embodiments,” and the claims do not recite that the data and control modules are configured to generate signals “according to the JEDEC standard memory protocol” (Prelim. Resp. 28). As further explained in that

prior decision, *Merriam Webster's Collegiate Dictionary* also defines generate as “to be the cause of” (IPR2014-00970, Paper 12, slip op. at 9–10). Ex. 2009, 484. Patent Owner argues the definition “to be the cause of” only applies in the context of “a situation, action, or state of mind.” Prelim. Resp. 29. Patent Owner contends “the context of the '434 Patent is the physical process of generating electrical signals. Accordingly, the definition of ‘generate’ should be taken as ‘PRODUCE,’ since this is the ordinary meaning within the context of ‘a vital, chemical, or physical process.’” *Id.* at 29–30.

We agree with Patent Owner that the intrinsic evidence, as well as the dictionary definitions on this record, support an interpretation of the claim term “generate” as meaning, *inter alia*, “produce,” and, that a module “configured to generate” address and control signals or data, as claimed, encompasses a component which produces the exact signals or data used by the memory devices. On the record before us, and for purposes of this decision, however, we are persuaded that one of ordinary skill in the art would also understand the term “generate,” as used in the claims, to mean “cause.” Thus, for example, one of ordinary skill in the art would understand that a module “configured to generate” signals or data encompasses both a component that produces the exact signal or data used by a memory device and a component which transmits a *programmable command* to that generator to cause production of the signal or data. Our interpretation of the term “generate” as including “cause” is supported by the '434 patent (*see id.* at 24 (citing Ex. 1008, 10:67–11:4)) and the dictionary definitions of record, including the additional definition of “generate” in *Merriam Webster's Collegiate Dictionary* as, in the context of a

mathematical set or structure, “to define or originate . . . by the application of one or more rules or operations” (Ex. 2008, 484).

Accordingly, for purposes of this decision, we interpret the claim term “generate” as meaning “produce” or “cause.”

IV. ANALYSIS

A. Obviousness of Claims 1–35 over Fleischman ’320 and JTAG 2001

1. Fleischman ’320 (Ex. 1011)

Fleischman ’320 describes BIST architecture including multiple BIST engines. Ex. 1011, 3:29–31, 3:18–20, Figs. 3, 8. Fleischman ’320 also discloses that the BIST architecture includes a BIST main control 110 and address generation unit 72 (together constituting a “control module”) and multiple data generation (DDG) units 94 (“plurality of data handlers”), each for testing the memory arrays. *Id.* at 6:5–8, 4:8–11. Further, the BIST architecture of Fleischman ’320 is configured to test the memory arrays 1–N using address and control information issued by the primary BIST main control 110 and the address generator 72. *Id.* at 1:25–30, 8:1–4, 8:10–11.

2. JTAG 2001 (Ex. 1012)

JTAG 2001 discloses a general BIST architecture that is applicable to JTAG compliant circuitry, including printed circuit boards with memory devices for testing. Ex. 1012, 68, 128. JTAG 2001 also discloses the use of cyclic data, e.g., 1111X, 0XXX0, and other cyclic patterns for testing memory devices. *Id.* at 159.

3. Analysis

Petitioner contends claims 1–35 would have been obvious over Fleischman ’320 in view of JTAG 2001. Pet. 27–43.

With respect to the motivation to combine the two references,
Petitioner contends:

A POSITA would have been motivated to combine Fleischman '320 and JTAG 2001, because Fleischman '320 expressly discloses compliance with the JTAG standard. JTAG 2001 defines JTAG compliance, including the application of self-test to devices on a printed circuit board, and therefore defines an environment that is suitable for Fleischman '320. Accordingly, a POSITA would be motivated to adapt the BIST architecture and operations disclosed in Fleischman '320 to a PCB in accordance with the compliance requirements disclosed in JTAG 2001.

Id. at 43–44 (citations omitted). In support of these contentions, Petitioner cites paragraph 34 and Appendix B of the Declaration of Dr. Nader Bagherzadeh (Ex. 1009). *Id.* Petitioner notes: “The combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results.” *Id.* (citations omitted).

As explained by the Federal Circuit, “[a]lthough predictability is a touchstone of obviousness, the ‘predictable result’ discussed in *KSR* refers not only to the expectation that prior art elements are capable of being physically combined, but also that the combination would have worked for its intended purpose.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 567 F.3d 1314, 1326 (Fed. Cir. 2009) (citing *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398 (2007)).

We acknowledge Petitioner relies on Dr. Bagherzadeh’s testimony that is identical in substance to the text from the Petition reproduced above. The specification of the Fleischman '320 states, “BIST operation and extraction of test results are typically accomplished through IEEE Standard 1149.1 Joint Test Action Group (JTAG) boundary scan Test Access Port

(TAP).” Ex. 1011, 2:61–63. However, this statement is a general description of the prior art, not the invention. Fleischman ’320 describes embodiments using a JTAG TAP interface 122. Ex. 1011, 5:10–13, 9:47–55. In the ’434 patent specification, the JTAG TAP interface is always connected to the BIST Main Control (BMC) 110.

Nonetheless, for the combination of Fleischman ’320 and JTAG 2001, Petitioner proposes a different BIST architecture where the JTAG interface connects directly to the Memory Arrays in Fleischman ’320. Pet. 29–30. Prelim. Resp. 40. Petitioner contends the proposed combination of Fleischman ’320 and JTAG 2001 would have been nothing more than the combination of familiar elements, but has not identified persuasive evidence that supports these contentions. Pet. 43. Dr. Bagherzadeh also has not identified sufficient support for this combination. For example, Dr. Bagherzadeh has not explained clearly why, or identified evidence that shows, the invention described in the ’434 patent falls into a very predictable field, or that one of ordinary skill in the art would have possessed the requisite skills to make the proposed combination, or that one of ordinary skill in the art reasonably would have anticipated the combination to function in an ordinary and expected way. *See Rothman v. Target Corp.*, 556 F.3d 1310, 1319–20 (Fed. Cir. 2009). Given that the ’434 specification teaches a different configuration of the JTAG TAP, Petitioner needed to explain why one of ordinary skill would know to replace the existing configuration with the new configuration. *Compare Abbott Labs. v. Sandoz, Inc.*, 544 F.3d 1341, 1352 (Fed. Cir. 2008) (“[K]nowledge of the goal [such as meeting an FDA requirement] does not render its achievement obvious.”) *with Erico Int’l Corp. v. Vutec Corp.*, 516 F.3d 1350, 1356 (Fed. Cir. 2008)

(a regulation that directs designers to a particular solution may render that solution obvious).

Dr. Bagherzadeh contends that the absence of a teaching away in Fleischman '320 and JTAG 2001 evidences that the combination would be operative. Ex. 1009 ¶ 35. Although a teaching away supports a conclusion that a prior art combination would have produced an inoperative result, see *DePuy*, 567 F.3d at 1326, Petitioner has not identified case law to support Dr. Bagherzadeh's contention.

Accordingly, we are not persuaded that Petitioner has shown sufficiently a motivation to combine Fleischman '320 and JTAG, and for the above-stated reasons, on this record, we determine Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1–35 would have been obvious over Fleischman '320 and JTAG 2001.

B. Obviousness of Claims 1–7, 13–21, 26–30, and 35 over Adams and JTAG 2001

1. Adams (Ex. 1013)

Adams '873 discloses a BIST architecture that is applicable to any BIST environment, including environments compliant with the JTAG standard, and that is applicable for testing memory devices. Ex. 1013, 3:34–37, Fig. 2. Adams '873 also discloses that the BIST architecture includes an MBIST controller 210 (“control module”) and multiple collar circuitry 220a–c (“plurality of data handlers”), each for testing the corresponding memory devices 225a–c. *Id.* at 2:6–12, 5:7–14, 5:58–6:5. Further, the BIST architecture of Adams '873 is configured to test the memory devices 225a–c

after being enabled by the MBIST controller 210, and using data generated by the collar circuitry 220a–c. *Id.* at 5:40–46, 5:58–6:5.

2. Analysis

Patent Owner argues Petitioner has not identified sufficient support for its contention that Adams ’873’s MBIST controller generates address and control signals for testing memory devices as recited in the claims. *See* Prelim. Resp. 45–49. Patent Owner argues the signals that enter Adams ’873’s memory 325 are instead all generated by the corresponding collars. *Id.* at 48.

Patent Owner’s arguments are based on a narrow construction of the term “generate” as requiring a control module configured to produce directly the actual address and control signals used to test the memory devices. As explained in Section III above, for purposes of this decision, we interpret the claim term “generate” more broadly than proposed by Patent Owner as meaning “produce” or “cause.”

Adams ’873 discloses that MBIST controller includes an “enable module 310a [that] is configured to provide enable signals for use in selecting, scheduling, and controlling BIST of collars 320 (and, therefore, the memory 325 in each collar)”. Ex. 1013, 5:58–6:5; Pet. 46. In other words, MBIST controller 210 issues enable signals that communicate to an “enable element [that] is configured to selectively enable a collared memory for testing, e.g., one at a time.” Ex. 1013, 2:21–22; Pet. 46.

Although the commands are not sent directly to memory 325, but instead enable the collars (*id.*), the above description in Adams ’873 supports Petitioner’s contention that the enable signals issued by MBIST controller 210 cause the issuance of, i.e., “generate,” address and control

signals. Nonetheless, Petitioner's ground fails—MBIST controller 210 is not “configured to generate” because, as set forth in Section III above, we construe “configured to generate” signals or data as encompassing a component which transmits a *programmable command* to that generator to cause production of the signal or data. The enable signal in Adams pointed to by Petitioner is not programmable, i.e., it is not used by the collar to generate or program any specific signal. The enable signal simply “enables” the collar to enter test mode. Ex. 1013, 5:7–15.

For the above-stated reasons, on this record, we determine Petitioner has not demonstrated a reasonable likelihood it would prevail on the ground that claims 1–7, 13–21, 26–30, and 35 would have been obvious over Adams '873 and JTAG 2001.

C. Obviousness of Claims 8–12, 22–25, and 31–34 over Adams, JTAG 2001, and Huang

Petitioner argues that claims 8–12, 22–25, and 31–34 would have been obvious over Adams '873, JTAG 2001, and Huang under 35 U.S.C. § 103(a). Pet. 55–59. Claims 8–12, 22–25, and 31–34 depend from independent claims 1, 20, and 29, respectively. As discussed above, the combination of Adams '873 and JTAG 2001 fails to teach or suggest all of the elements of independent claims 1, 20, and 29. Petitioner does not assert that Huang overcomes the aforementioned deficiency in Adams '873 and JTAG 2001. Thus, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has not demonstrated that there is a reasonable likelihood it would prevail with respect to claims 8–12, 22–25, and 31–34, on the ground that these claims would have been obvious over Adams '873, JTAG 2001, and Huang.

*D. Unpatentability of Claims 5–13 and 28 under 35 U.S.C. §112,
Second Paragraph, as Indefinite*

In addition to the above three grounds, Petitioner advances a fourth ground of unpatentability in which it contends that if we determine its proposed interpretation of “proximate” in claims 5 and 28 is overly broad, then “claim 5, its corresponding dependent claims 6–13, and claim 28 are unpatentable . . . under section 112 as indefinite.” Pet. 27. We will not consider this challenge as to claims 5–13 and 28 because it exceeds the scope of *inter partes* review. See 35 U.S.C. § 311(b)(2013) (“A petitioner in an inter partes review may request to cancel as unpatentable 1 or more claims of a patent only on a ground that could be raised under section 102 or 103 and only on the basis of prior art consisting of patents or printed publications.”).

V. CONCLUSION

The information presented does not show that there is a reasonable likelihood that Petitioner would prevail at trial with respect to at least one claim of the ’434 patent, based on any ground presented in the Petition. On this record, we deny the petition for *inter partes* review of claims 1–35. We cannot consider the challenge to claims 5–13 and 28 as indefinite under 35 U.S.C. §112, second paragraph, because it exceeds the scope of *inter partes* review.

VI. ORDER

Accordingly, it is

ORDERED that that the petition is *denied* as to all challenged claims, and no trial is instituted.

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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01372
Patent 8,001,434 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
PETER P. CHEN, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. § 318(a) and 37 C.F.R. § 42.73

I. BACKGROUND

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) on August 23, 2014, requesting institution of an *inter partes* review of claims 1–35 (the “challenged claims”) of U.S. Patent No. 8,001,434 B1 (Ex. 1008, “the ’434 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 11). Based on these submissions, an *inter partes* review of claims 1–4, 14–20, 27, and 29 was instituted on March 10, 2015, pursuant to 35 U.S.C. § 314. Paper 13 (“Dec. on Inst.” or “Institution Decision”).

After institution, Patent Owner filed a Response (Paper 22, “PO Resp.”), and Petitioner filed a Reply to the Patent Owner Response (Paper 27, “Reply”).

Petitioner filed a Motion to Exclude Evidence. Paper 26; *see also* Paper 37 (Patent Owner’s Opposition) and Paper 42 (Petitioner’s Reply). Patent Owner also filed a Motion to Exclude Evidence (Paper 34; *see also* Paper 39 (Petitioner’s Opposition) and Paper 41 (Patent Owner’s Reply)) and a Motion to Exclude Portions of Petitioner’s Reply (Paper 35; *see also* Paper 40 (Petitioner’s Opposition)).

Oral argument was held on November 17, 2015, and a transcript (Paper 43, “Tr.”) has been entered into the record.

The Board has jurisdiction under 35 U.S.C. § 6(c). This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73, addresses issues and arguments raised during trial.

For the reasons discussed below, we determine Petitioner has not met its burden to prove, by a preponderance of the evidence, that claims 1–4, 14–

20, 27, and 29 of the '434 patent are unpatentable. Petitioner's and Patent Owner's Motions to Exclude are dismissed.

A. Related Matters

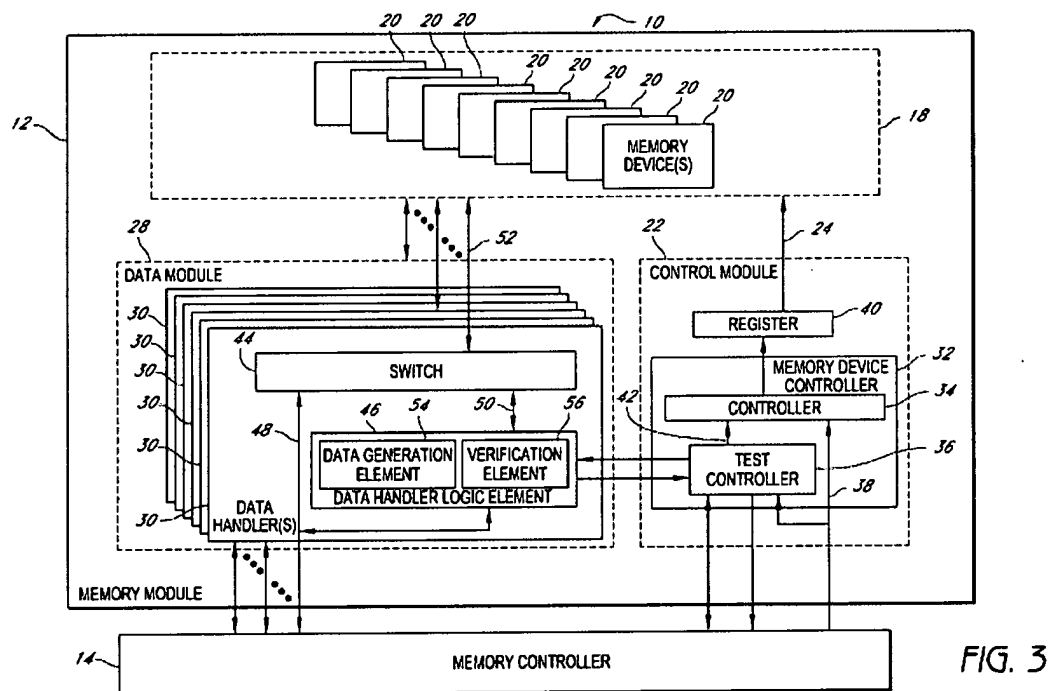
Petitioner also requested, and we instituted, an *inter partes* review of U.S. Patent No. 8,359,501 ("the '501 patent"), which claims priority as a continuation of U.S. Patent Application No. 12/422,925 (now the '434 patent): *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01374 (PTAB March 10, 2015), Paper 12. We denied Petitioner's requests to institute *inter partes* reviews of the '434 patent and the '501 patent in two additional petitions, filed August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB March 13, 2015), Paper 16, and Case IPR2014-01375 (PTAB March 13, 2015), Paper 12, respectively. Oral argument in IPR2014-01374 was consolidated with the oral argument in the present *inter partes* review. A final decision in IPR2014-01374 is issued concurrently with this final decision.

The '434 patent also is the subject of IPR2014-00970 ("IPR-970"). We entered a final written decision in that case on December 14, 2015, concluding that claims 1, 14, 15, 19–25, and 27–34 are unpatentable on the following grounds: claims 1, 14, 15, 20–25, 27, and 29–34 under 35 U.S.C. § 102(b) based on Averbuj (U.S. Patent Application Publication No. 2005/0257109 A1, issued November 17, 2005); claim 28 under 35 U.S.C. § 103(a) based on Averbuj; and claims 1, 14, 15, 19, and 29 under 35 U.S.C. § 103(a) based on Huang (*An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers*, Fourteenth International Conference on VLSI Design 379–384 (2001), IEEE). *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB December 14, 2015), Paper 32

(“IPR-970 FWD”). Petitioner Sandisk Corporation has requested rehearing of our determination that it did not meet its burden to show unpatentability of claims 2, 5, 6, 17, 18, and 35. *Id.*, Paper 33.

B. The '434 Patent (Ex. 1008)

The '434 patent relates to self-testing electronic memory modules. Ex. 1008, 1:23–24. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '434 patent, reproduced below. *See id.* at 9:23–25.



As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:1–3. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports.

Id. at 5:3–7. “[M]emory module 10 comprises a control module 22 [and] . . . a data module 28.” *Id.* at 5:7–10.

“[C]ontrol module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18.” *Id.* at 9:25–27. In the embodiment shown in Figure 3, control module 22 includes control mixer element 32 for controlling the address and control signals for the self-testing function. *Id.* at 9:31–33. Control mixer element 32 includes memory device controller 34 (e.g., a DRAM controller) and test controller 36. *Id.* at 9:29–31. “[T]est controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28.” *Id.* at 9:53–56. “[M]emory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36.” *Id.* at 9:37–40.

Data module 28 comprises a plurality of data handlers 30 that “may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports” of memory devices 18. *Id.* at 5:10–11, 16–18. In the embodiment illustrated in Figure 3, each of data handlers 30 includes data handler logic element 46, comprising data generation element 54 and verification element 56. *Id.* at 10:27–30. “[D]ata generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports” of memory devices 18. *Id.* at 10:31–33. “The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information)

the data handler logic element 46 receives from the control module 22.” *Id.* at 10:33–37.

C. Illustrative Claim

Of the challenged claims, claims 1, 20, and 29 are independent.

Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

D. The Instituted Ground of Unpatentability

We instituted the instant *inter partes* review of claims 1–4, 14–20, 27, and 29 based on anticipation under 35 U.S.C. § 102(b) by Averbuj, U.S. Patent No. 7,392,442 B2, issued June 24, 2008 (Ex. 1011, “Averbuj”). As indicated in Section I.A., above, in IPR-970, we held claims 1, 14, 15, 20–

25, 27, and 29–34 of the '434 patent unpatentable under 35 U.S.C. § 102(b) and claim 28 unpatentable under 35 U.S.C. § 103(a) based on U.S. Patent Application Publication No. 2005/0257109 A1, the published application of the Averbuj patent on which the instituted ground of unpatentability in this proceeding is based.

E. Expert Testimony

Petitioner relies on the testimony of Dr. Nader Bagherzadeh in support of its patentability challenge. Pet. 1. Dr. Bagherzadeh executed a declaration (Ex. 1009, “the Bagherzadeh Declaration”) in support of the Petition. Dr. Bagherzadeh was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 2012.

Dr. Bagherzadeh testifies as follows: “In 1979 and 1987, respectively, [he] earned a master of science in electrical engineering and a doctorate degree in computer engineering from the University of Texas at Austin.” Ex. 1009 ¶ 4. He has been involved in design and development of digital systems for more than 30 years. *Id.* ¶ 6. Dr. Bagherzadeh joined the University of California, Irvine, in 1987, and has “been teaching, researching, and consulting regarding almost all aspects of memory design for high performance computer systems, including but not limited to DRAMs and SRAMs.” *Id.* ¶ 7. He has been employed as a professor in the department of Electrical Engineering and Computer Science at the University of California, Irvine, since 2003. *Id.* ¶ 5. In 2000, Dr. Bagherzadeh became a cofounder of Morpho Technologies, a high tech company focused on the design and development of low power and high performance digital signal processors for mobile applications. *Id.* ¶ 7. He

was involved in evaluating patents, technical reports and presentations related to memory chip designs, DSPs, and parallel processing algorithms for mobile platforms. *Id.*

Patent Owner relies on the testimony of Dr. Carl Sechen. PO Resp. 1. Dr. Sechen executed a declaration (Ex. 2019, “the Sechen Declaration”) in support of Patent Owner’s Response. Dr. Sechen was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 1031.

Dr. Sechen testifies as follows: he has an M.S. degree in Electrical Engineering from the Massachusetts Institute of Technology and “was awarded a Ph.D. in electrical engineering from the University of California at Berkeley in 1986.” Ex. 2019 ¶ 5. Dr. Sechen has been a Professor of Electrical Engineering for 29 years. *Id.* ¶ 2. During this time period, his research has focused on design and computer-aided design of digital integrated circuits, including the design of DRAM, and he has taught numerous students how to design DRAM memories. *Id.* Dr. Sechen has “also been involved in numerous research projects on VLSI design and memory design[, and has] taught numerous graduate researchers how to design digital integrated circuits, including memories.” *Id.* ¶ 4.

The parties do not dispute that Dr. Bagherzadeh and Dr. Sechen are qualified to testify as experts under FRE 702.

II. LEVEL OF ORDINARY SKILL IN THE ART

The level of ordinary skill in the art is relevant to claim construction and anticipation. *See Yorkey v. Diab*, 605 F.3d 1297, 1300 (Fed. Cir. 2010) (explaining that a determination of anticipation involves interpreting the

claim language and then comparing the construed claim to a prior art reference); *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1259–60 (Fed. Cir. 2010) (“[C]laim language should be read in light of the specification as it would be interpreted by one of ordinary skill in the art.”).

Dr. Bagherzadeh and Dr. Sechen agree that the field of art is “memory module design with features for built-in self-test (BIST).” Ex. 2019 ¶ 21; see also Ex. 1009 ¶ 17.

Dr. Bagherzadeh testifies that

a person of ordinary skill in the art for the ‘434 patent in April 2008 would have a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering, and at least one year of experience relating to memory systems and BIST.

A person having a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering would have experience with digital circuit design, memories and computer architecture. An additional year of experience relating to memory systems and BIST would focus on those areas and allow one to recognize current issues with respect to the field.

Ex. 1009 ¶¶ 19–20; *see id.* ¶ 16 (“My understanding is that the earliest possible priority date of the ‘434 patent is April 14, 2008.”).

Dr. Sechen testifies that

one of ordinary skill in the art at the time of filing the application(s) for the ‘434 Patent, who would be working on the design of memory devices and memory modules, would have at least a Bachelor of Science degree in electrical engineering or computer engineering, and at least five years of industry experience designing memory devices and memory modules. Alternatively, one of ordinary skill in the art would have an M.S. degree in electrical engineering or computer engineering, and at least three years of industry experience designing

memory devices and memory modules. Moreover, one of ordinary skill in the art would have a Ph.D. degree in electrical engineering or computer engineering, and have at least one year of industry experience designing memory devices and memory modules.

Ex. 2019 ¶ 21.

The difference in the opinions of the two experts is essentially that Dr. Sechen believes a person of ordinary skill in the art would have additional experience, education, or a combination of the two, beyond that proposed by Dr. Bagherzadeh. Both experts have been employed as professors since the mid 1980s. Dr. Bagherzadeh also has some experience working with engineers in the industry by virtue of his employment with AT&T Bell Labs from 1980–1984 (Ex. 1009 ¶ 6; *id.* Appx. A). Both experts were awarded bachelors and masters degrees in electrical engineering. Dr. Bagherzadeh was awarded a doctorate in computer engineering, while Dr. Sechen was awarded a doctorate in electrical engineering.

Based on a comparison of Dr. Bagherzadeh's and Dr. Sechen's declarations (Exs. 1009, 2019), we find Dr. Bagherzadeh has a broader range of educational and work experience, and, therefore, accord greater weight to his opinion as to the level of ordinary skill in the art. Dr. Bagherzadeh's opinion is consistent with our finding as to the level of ordinary skill in the art in our Final Written Decision in IPR-970. We, therefore, adopt our finding in IPR-970 as to the level of ordinary skill in the art:

[A] person of ordinary skill in the art at the time of the '434 patent would have a Bachelor's degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.

IPR-970 FWD, 10–11 (noting that our finding was “based on our review of the ’434 patent and the types of problems and solutions described in the ’434 patent and cited prior art”).

III. CLAIM CONSTRUCTION

In its Petition, Petitioner offers specific constructions for the claim terms “operatively coupled,” “ports,” “to generate”/“generating,” “data module,” “operable independently,” “cyclic data,” and “proximate.” Pet. 23–28. We determined that, for purposes of our Institution Decision, only the term “generate” required express construction. Dec. on Inst. 6. We interpreted “generate” as meaning “produce” or “cause.” *Id.* at 9.

In its Response, Patent Owner disagrees with our interpretation of “generate” as meaning “cause.” PO Resp. 18. Patent Owner contends our “Institution Decision also provides a construction for ‘configured to generate,’” and proposes its own construction for this phrase. *Id.* at 9 (citing Dec. on Inst. 8–9). In its Reply, Petitioner asserts that our interpretation of “generate” in the Institution Decision is correct (Reply 1), and offers a proposed construction for the term “configured” (*id.* at 9). Patent Owner’s and Petitioner’s proposed constructions for the disputed claim language are listed in the table below.

Claim Term/Phrase	Patent Owner’s Proposed Construction	Petitioner’s Proposed Construction
generate	produce (PO Resp. 18)	cause or initiate transmission (Pet. 24–25) produce or cause (Reply 1)
configured to generate	programmed to generate/produce (<i>see</i> PO Resp. 9)	arranged or prepared to produce or cause (Reply 9–10)

In support of their proposed constructions of the disputed claim language, the parties rely on language in the '434 patent and extrinsic evidence. *See generally*, Pet. 24–25; PO Resp. 8–36; Reply 1–17. Neither party relies on prosecution history disclaimer in support of its proposed constructions.

In an *inter partes* review, the Board interprets claim terms in an unexpired patent according to the broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b); *see also In re Cuozzo Speed Techs., LLC*, 793 F.3d 1268, 1278–80 (Fed. Cir. 2015) (“Congress implicitly approved the broadest reasonable interpretation standard in enacting the AIA,” and “the standard was properly adopted by PTO regulation.”), *cert. granted, sub nom. Cuozzo Speed Techs. LLC v. Lee*, 136 S. Ct. 890 (mem.) (2016). Under that standard, and absent any special definitions, we give claim terms their ordinary and customary meaning, as would be understood by one of ordinary skill in the art at the time of the invention. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007); *see also Trivascular, Inc. v. Samuels*, No. 2015-1631, 2016 WL 463539, at *3 (Fed. Cir. Feb. 5, 2016) (“Under a broadest reasonable interpretation, words of the claim must be given their plain meaning, unless such meaning is inconsistent with the specification and prosecution history.”). Any special definitions for claim terms must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

“[T]he Board may rely on dictionaries ‘so long as the dictionary definition does not contradict any definition found in or ascertained by reading the patent document.’” *Belden Inc. v. Berk-Tek LLC*, 610 Fed.

App'x 997, 1002 (Fed. Cir. 2015) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1322–23 (Fed. Cir. 2005)). Expert testimony is useful to explain terms of art, and the state of the art at any given time, but cannot be used to prove “the proper or legal construction of any instrument of writing.” *Teva Pharm. USA, Inc. v. Sandoz, Inc.*, 135 S. Ct. 831, 841 (2015) (internal citations omitted); *see also* 37 C.F.R. § 42.65(a) (“Expert testimony that does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.”).

Only terms which are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999). A court may revisit and alter its construction of claim terms as the record in a case develops. *See Pressure Prods. Med. Supplies, Inc. v. Greatbatch Ltd*, 599 F.3d 1308, 1316 (Fed. Cir. 2010).

A. Generate

Petitioner contends “[t]he Institution Decision correctly found that the intrinsic and extrinsic record supported construing ‘generate’ as ‘cause’ or ‘produce.’” Reply 1. Patent Owner contends “‘generate’ is properly construed to mean ‘produce,’” but “a construction of generate that includes ‘cause’ is unsupported by – and contradictory to – the intrinsic and extrinsic evidence, and fails to provide reasonable clarity.” PO Resp. 18. The same respective constructions of “generate” were advanced by Patent Owner and the petitioner, Sandisk Corporation, in IPR-970. *See* IPR-970 FWD, 25–27.

We provided an extensive analysis of the term “generate” in our Final Written Decision in IPR-970. *See id.* at 24–33. “[W]e agree[d] with Patent Owner that the broadest reasonable interpretation of the claim term

‘generate’ is ‘produce’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’” *Id.* at 32.

We interpret[ed] the claim language “address and control signals *generated* by the control module and the data *generated* by the plurality of data handlers” (claim 1), and the corresponding limitations in claims 20 and 29, as encompassing signals and data that originated in these modules, including by transformation or modification of information and/or data received from another component.

Id. We also stated that “[w]e [did] not interpret this language as encompassing signals and data received by the data and control modules from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the data and control modules.” *Id.*

In support of its contention that “generate” also should be interpreted as meaning “cause,” Petitioner relies on extrinsic evidence that was not before us in IPR-970, i.e., the Bagherzadeh Declaration (Ex. 1009) ¶ 28, ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms, Fourth Edition, 1988 (Ex. 1016, “IEEE”), and *Webster’s II New College Dictionary*, 2001 (Ex. 1032, “Webster’s II”). *See* Pet. 24–25; Reply 1–9. We consider this evidence as it is relevant to whether the term “generate” has a particular meaning to one of ordinary skill in the art. *See Trivascular*, 2016 WL 463539, at *3 (“Construing individual words of a claim without considering the context in which those words appear is simply not ‘reasonable.’ Instead, it is the ‘use of the words in the context of the written description and customarily by those of skill in the relevant art that accurately reflects both the “ordinary” and “customary” meaning of the terms in the claims.” (quoting *Ferguson Beauregard/Logic Controls, Div.*

of Dover Res., Inc. v. Mega Sys., LLC, 350 F.3d 1327, 1338 (Fed. Cir. 2003))).

The IEEE defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters.” Ex. 1016, 5. Dr. Bagherzadeh cites the IEEE definition in support of his testimony that “[t]he broadest reasonable interpretation of the claim [terms] ‘to generate’ and ‘generating’ in claims 1, 20, and 29, is ‘to cause or initiate transmission’ and ‘causing or initiating.’” Ex. 1009 ¶ 28 (also citing Ex. 1008, 6:3–5, 10:67–11:2, 6:14–17). Dr. Bagherzadeh testifies that “[t]his interpretation is consistent with the specification and extrinsic evidence because they describe generating as initiating or transmitting, which is also the broadest interpretation of ‘generate.’” *Id.*

We are not persuaded by Dr. Bagherzadeh’s testimony because it is devoid of specific facts and analysis as to what led him to conclude that a person of ordinary skill in the art would interpret the claim term “generate” in the manner Petitioner proposes. *See id.* Dr. Bagherzadeh does not explain how the IEEE definition supports an interpretation of generate as “cause or initiate transmission,” terms that do not appear in the IEEE definition. *See id.* Nor does Dr. Bagherzadeh identify other extrinsic evidence which he believes to be consistent with an interpretation of “generate” as meaning “cause or initiate transmission.” *See id.* Likewise, Dr. Bagherzadeh has not explained why he believes the ’434 patent “describe[s] generating as initiating or transmitting” (Ex. 1009 ¶ 28), as these terms do not appear in the cited portions of the ’434 patent. *See* Ex. 1008, 6:3–5, 10:67–11:2, 6:14–17. Accordingly, Dr. Bagherzadeh’s

testimony does not persuade us to accept Petitioner's interpretation of "generate" as meaning "cause."

Petitioner contends "Dr. Bagherzadeh's use of the IEEE definition specifically illustrates that 'generate' includes 'selection . . . from a set,' which implies that 'generate' must not be limited to creating from previously undefined values, which may be a more traditional meaning, but holds little value in the intrinsic evidence in this proceeding." Reply 6. Petitioner has not identified corresponding testimony by Dr. Bagherzadeh or other evidentiary support for this statement and, therefore, this argument is likewise unpersuasive. *See id.*

In support of its construction of "generate" as meaning "cause," Petitioner also relies on a finding in our Institution Decision that such construction is supported by Merriam-Webster's Collegiate Dictionary, wherein one definition of "generate" is "to be the cause of" (Ex. 2009, 3). Reply 2 (citing Dec. on Inst. 9 [sic, 8]). This argument is not persuasive because the definition in Merriam-Webster's Collegiate Dictionary contradicts the definition of "generate" that we ascertained upon reading the '434 patent as discussed in our Final Written Decision in IPR-970. *See Belden*, 610 Fed. App'x at 1002.

In sum, we are not persuaded that Petitioner's extrinsic evidence supports a finding that a customary meaning of "generate" to one of ordinary skill in the art is "cause."¹ We, therefore, determine the broadest reasonable

¹ Petitioner relies on *Webster's II*, for a definition of "cause." *See* Reply 4, 6–8 (citing Ex. 1032, 3). Having determined the broadest reasonable interpretation of "generate" does not encompass "cause," we do not find this evidence relevant to our claim construction analysis.

construction of “generate” is “produce” for the reasons stated above and in our Final Written Decision in IPR-970. We adopt and incorporate by reference our analysis and construction of the term “generate” in our Final Written Decision in IPR-970 (IPR-970 FWD, 24–33).

B. Configured to

In our Final Decision in IPR-970, we construed “configured to” as “designed to, adapted to, or arranged to.” *See* IPR-970 FWD, 20–24. Petitioner contends “configured” should be construed as “arranged” or “prepared.” Reply 9. Responsive to this Board panel’s questioning during the oral hearing, however, Petitioner agreed that “designed to, adapted to, or arranged to” was an acceptable definition for the claim term “configured to.” *See* Tr. 22:7–17.

Patent Owner contends the phrase “configured to generate” should be construed as “programmed to generate.” PO Resp. 9. Patent Owner does not propose a separate construction for the term “configured.” *See generally id.* at 9–18. In support of its proposed construction, Patent Owner relies on language in the ’434 patent and the testimony of its expert, Dr. Sechen. *See id.* Dr. Sechen testifies as to the meaning of “configured to generate,” but does not provide a separate discussion of the meaning of “configured to” outside the context of “configured to generate.” *See* Ex. 2019 ¶¶ 42–57.

As explained in our Final Decision in IPR-970, the term “configured to” is used in the ’434 patent in conjunction with numerous terms, such as “provide” and “selectively input,” and there is a presumption that the term “configured to” should carry the same meaning in each instance. *See* IPR-970 FWD, 20–21. The evidence relied on by Patent Owner in support of its proposed construction of the term “configured to” in the present *inter partes*

review does not differ materially from the evidence relied upon by Patent Owner in IPR-970. *Compare* PO Resp. 9–18 with IPR-970, Paper 16, 12–17. Accordingly, we determine the broadest reasonable construction of “configured to” is “designed to, adapted to, or arranged to” for the reasons stated above and in our Final Written Decision in IPR-970. We adopt and incorporate by reference our analysis and construction of the term “configured to” in our Final Written Decision in IPR-970 (IPR-970 FWD, 20–24).

In summary, we construe the claim terms in controversy as follows:

Claim Term/Phrase	Interpretation
Generate	Produce
Configured to	Designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]
Configured to Generate	Designed, adapted, or arranged to produce

IV. ALLEGED ANTICIPATION OF CLAIMS 1–4, 14–20, 27, AND 29 BY AVERBUJ

A. Principles of Law

To prevail in its challenges to the patentability of the claims, a petitioner must establish facts supporting its challenges by a preponderance of the evidence. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference.

Net MoneyIN, Inc. v. VeriSign, Inc., 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). While the elements must be arranged or combined in the same way

as in the claim, “the reference need not satisfy an *ipsissimis verbis* test.” *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009).

A single prior art reference that discloses, either expressly or inherently, each limitation of a claim invalidates that claim by anticipation. Thus, a prior art reference without express reference to a claim limitation may nonetheless anticipate by inherency. “Under the principles of inherency, if the prior art necessarily functions in accordance with, or includes, the claims limitations, it anticipates.”

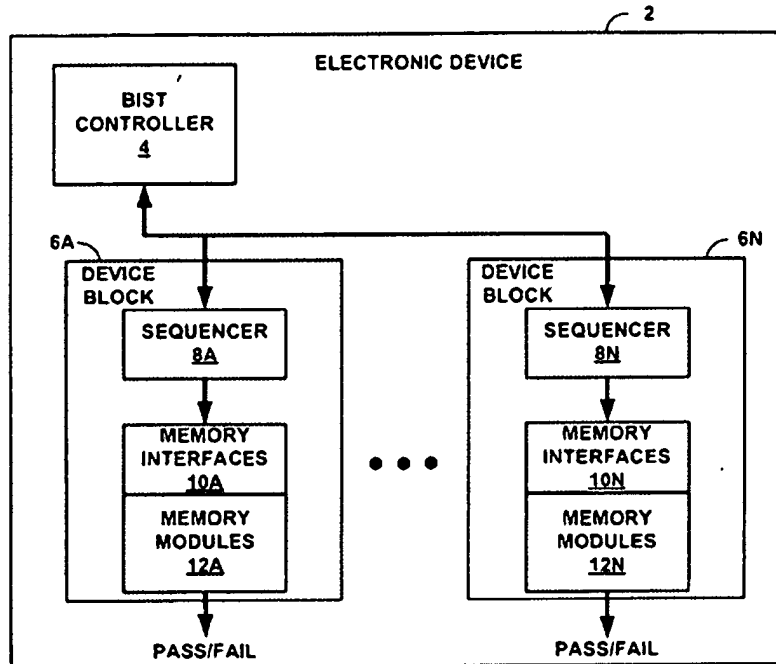
Perricone v. Medicis Pharm. Corp., 432 F.3d 1368, 1375–76 (Fed. Cir. 2005) (citations omitted). “In general, a limitation or the entire invention is inherent and in the public domain if it is the ‘natural result flowing from’ the explicit disclosure of the prior art.” *Schering Corp. v. Geneva Pharms., Inc.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003).

“[T]he Board is not bound by any findings made in its Institution Decision. . . . The Board is free to change its view of the merits after further development of the record, and *should do so* if convinced its initial inclinations were wrong.” *Trivascular*, 2016 WL 463539, at *9.

We analyze the instituted ground of unpatentability in accordance with the above-stated principles.

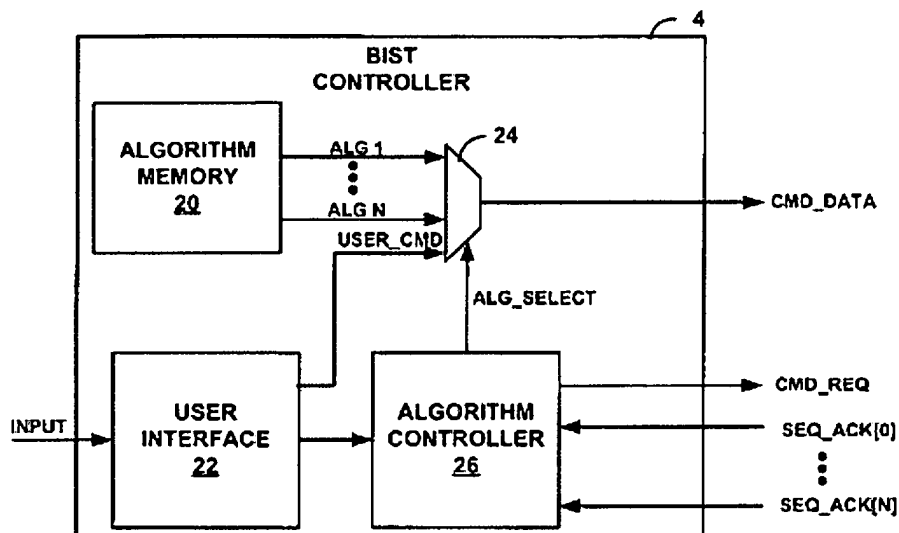
B. Averbuj (Ex. 1011)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1011, 1:64–65, 2:3–4. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* at 3:62–64. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* at 3:64–67. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* at 3:67–4:3.

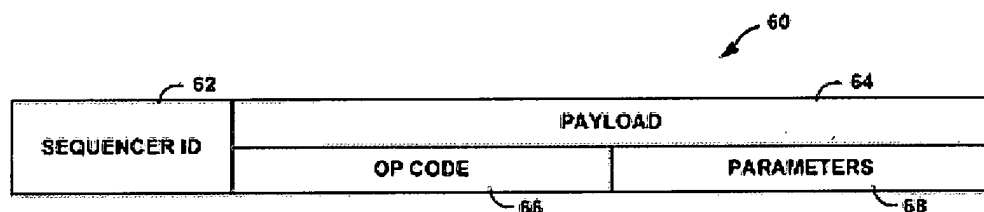
BIST controller 4 is illustrated in Figure 2, reproduced below.



Averbuj Figure 2, above, “is a block diagram illustrating an example embodiment of a BIST controller.” *Id.* at 3:39–40. BIST controller 4 includes algorithm controller 26 that can be invoked either by a user through user interface 22 or automatically upon power-up of electronic device 2. *Id.* at 5:12–16. “Once invoked, algorithm controller 26 provides an algorithm select signal (ALG_SELECT) to multiplexer 24 to select one of the algorithms stored within algorithm memory 20.” *Id.* at 5:16–19. Alternatively, user interface 22 may programmably receive algorithms via external input, and deliver the received algorithms to multiplexer 24. *Id.* at 5:41–43. The algorithms, whether stored in algorithm memory 20 or received via external input, have a similar form, i.e., “a sequence of binary commands in which each command defines a test within the overall algorithm.” *Id.* at 5:43–47. BIST controller 4 provides and communicates the selected algorithm to sequencers 8 for application to device blocks 6 as a stream of binary commands (CMD_DATA), “each command specif[ying] an operational code [(OP CODE)] and a set of parameters that define one or

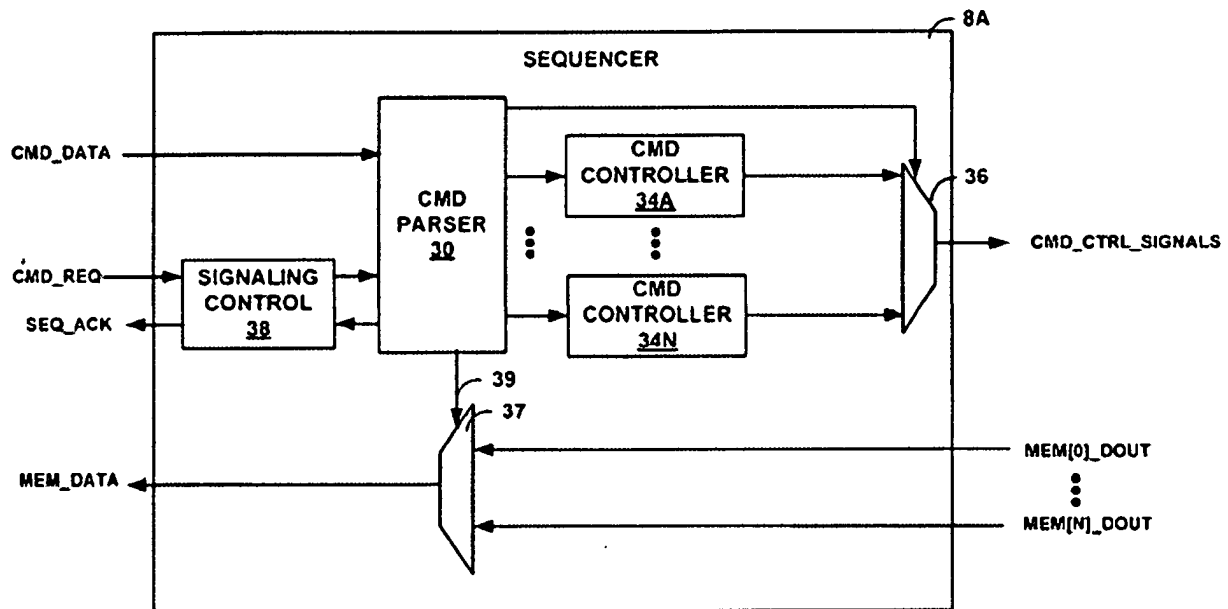
more memory operations without regard to the physical characteristics or timing requirements of memory modules 12.” *Id.* at 4:4–12, 5:19–21.

Figure 8, reproduced below, “is a block diagram illustrating an example data structure of a command issued by BIST controller 4.” *Id.* at 9:15–16.



In the embodiment shown in Figure 8, above, “command 60 includes a sequencer identifier (ID) 62, and a payload 64. Sequencer ID 62 identifies a sequencer, e.g., sequencer 8A, to which command 60 is being issued.” *Id.* at 9:17–20. “Payload 64 of command 60 carries binary data that defines the command itself. In particular, payload 64 includes an operational code (OP CODE) 66 and a set of parameters 68.” *Id.* at 9:29–31. In one embodiment, for example, “OP CODE 66 and parameters 68 comprise three bits and twenty-nine bits, respectively, to form a 32-bit command.” *Id.* at 9:58–60. “OP CODE 66 specifies a particular function to be performed by the receiving sequencers 8.” *Id.* at 9:32–33. An exemplary OP CODE is SET ADDRESS (OP CODE 100), which “[s]ets a specific starting address as well as a maximum address limit for a test algorithm as applied to the memory modules.” *Id.* at Table 1.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* at 3:46–47.



As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 that receives command data (CMD_DATA) from BIST controller 4. *Id.* at 6:24–26. Command parser 30 processes the commands from BIST controller 4 to identify a specified operation, e.g., by identifying an OP CODE specified by the command. *Id.* at 6:26–29. Based on the specified operation, command parser 30 may extract one or more parameters from the command, and pass the extracted parameters to a selected command controller (i.e., one of command controllers 34A–34N). *Id.* at 6:30–37. The invoked command controller, in turn, issues a sequence of one or more operations to each memory interface 10, sequentially driving the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. *Id.* at 6:41–43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with

the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* at 4:45–49.

C. Arguments

Independent claims 1, 20, and 29 recite “a plurality of memory devices” and “a data module.” Petitioner contends the claimed “memory devices” read on Averbuj’s memory modules 12 and the claimed “data module” reads on the combination of sequencers 8 and memory interfaces 10. Pet. 28. Independent claims 1, 20, and 29 further recite “a control module configured to generate address and control signals for testing the memory devices.” *See also* claims 1 and 20 (reciting a circuit configured to test the memory devices using the address and control signals generated by the control module); claim 29 (reciting a step of “generating, by the control module, address and control signals for testing the memory devices”). With respect to this limitation, the Petition states:

the BIST controller 4 of Averbuj ‘442 represents circuitry of a control module that provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE* for testing the memory components 12A–N. Ex. 1011, Fig. 2 and Column 9: Table 1. The *SET ADDRESS* and *CMD_REQ* signals include address and control signals for the testing functions. Since the BIST controller 4 provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE*, a person of ordinary skill in the art (“POSITA”) would understand that the BIST controller 4 is programmable to provide address and control signals as recited in part by the Challenged Claims. Ex. 1009, Bagherzadeh Decl., ¶ 35.

Pet. 29.

Patent Owner argues Averbuj’s BIST controller 4 does not “generate” the *OP CODE* communicated to sequencers 8, and Petitioner has not met its burden to show that Averbuj’s BIST controller 4 is “configured to generate

address . . . signals for testing the memory devices” as recited in claims 1, 20, and 29 of the ’434 patent. PO Resp. 48–49. As discussed in greater detail below, we find this argument persuasive, and, for this reason, conclude Petitioner failed to meet its burden to prove unpatentability of claims 1–4, 14–20, 27, and 29 of the ’434 patent.²

D. Expert Testimony

In support of its argument that Averbuj’s BIST controller 4 is not configured to generate address signals as claimed in claims 1, 20, and 29, Patent Owner relies on paragraphs 133–141 of the Sechen Declaration. *Id.* at 48–51. Patent Owner also relies on Dr. Bagherzadeh’s cross-examination testimony (*id.* at 49–50 (quoting Ex. 2012, 55:6–56:9)) regarding Averbuj Figure 2 (*see* Ex. 2012, 53:4–5; *see generally id.* at 53:4–56:9).

With reference to Averbuj Figure 2 (*see* Section IV.B., above, wherein this figure is reproduced and described), Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 is generated by hand or by a tool (Ex. 2012, 55:20–56:9), and he identifies a compiler as a tool for generating such code (*id.* at 30:23–24; 41:1–4; *see id.* at 55:6–25).³ Dr. Bagherzadeh testifies that a compiler is not illustrated in

² Because we are persuaded by this argument that Petitioner has not met its burden to show unpatentability, we find it unnecessary to address the additional arguments advanced by Patent Owner in its Response.

³ Petitioner contends “[t]he discussion of a compiler, from PO’s deposition of [Dr. Bagherzadeh], was hypothetical and specific to an article coauthored by [Dr. Bagherzadeh].” Reply 22 (citing Ex. 2012, 15:2–19:3).³ We were unable to locate a discussion of a compiler or any reference to an article coauthored by Dr. Bagherzadeh in the cited portions of Dr. Bagherzadeh’s deposition transcript. Dr. Bagherzadeh’s first discussion of a compiler appears to be in connection with a use thereof in MorphoSys (*see* Ex. 2012,

Averbuj Figure 2, and that he did not recall any mention of the word “compiler” in Averbuj. *See* Ex. 2012, 54:8–55:5. Dr. Bagherzadeh further testifies that “99 percent of the designs or chips do not have a compiler on-board.” *Id.* at 54:21–22. Dr. Bagherzadeh testifies that Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and “the output [] reflect[s] what’s in the input, based on the selection.” *Id.* at 54:8–13. Dr. Bagherzadeh testifies that assuming “no other hardware components between . . . [CMD_DATA] and the input,” CMD_DATA is stored in the algorithm memory. *Id.* at 54:15–19.

Dr. Sechen testifies that

the Petition fails to identify every element of claims 1, 20, and 29 for at least two reasons[:] (1) because the CMD_REQ and SET ADDRESS are not generated (“produced” or “caused”) by the purported control module (Averbuj’s BIST controller 4), and [(2)] because the OP CODE is not generated (“produced” or “caused”) by the alleged control module (Averbuj’s BIST controller 4).

30:12–20), a chip fabricated by Morpho Technologies (*id.* at 19:2–9), a company co-founded by Dr. Bagherzadeh (Ex. 1009 ¶ 7). An article relating to MorphoSys and co-authored by Dr. Bagherzadeh (*see* Ex. 2014, “MorphoSys: Case Study of A Reconfigurable Computing System Targeting Multimedia Applications”) is introduced on page 39 of the deposition transcript. *See* Ex. 2012, 39:19–22, cited in PO Resp. 27. In any event, Petitioner’s argument is not persuasive because we find the testimony on pages 54–56 of the Bagherzadeh transcript supports Patent Owner’s argument. *See* Section IV.E. *infra*.

Ex. 2019 ¶ 136. Dr. Sechen testifies, more specifically, that the Petition fails to explain how Averbuj's BIST controller 4 "generate[s]/produce[s] OP CODE," "because the Petition does not state that Averbuj's BIST Controller includes a compiler, as described by Dr. Bagherzadeh." *Id.* ¶ 138. Dr. Sechen testifies that "[b]ecause SET ADDRESS (OP CODE 100) is a predefined 3-bit sequence, . . . it is not 'generated' by Averbuj's BIST Controller. Rather, it is stored in Algorithm Memory 20 of Fig. 2." *Id.* ¶ 140. Dr. Sechen further testifies that "command 60, which carries the OP CODE . . . (e.g. . . . the SET ADDRESS opcode), . . . is not 'generated' by Averbuj's BIST Controller," but "is merely stored in memory (Algorithm Memory 20 in Fig. 2) and is output as part of 'a stream of binary commands.' (Ex. 1011 at 5:1–21.)" Ex. 2019 ¶ 141.

E. Analysis

After considering the parties' arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses "a control module configured to generate address . . . signals for testing the memory devices" as recited in independent claims 1, 20, and 29.

As noted in Section III, above, in its Petition, Petitioner proposes construing the claim term "generate" as "cause or initiate transmission" (Pet. 24). The Petition cites the following disclosure in the '434 patent in support of this construction: "the data module 28 and/or the control module 22 are **configured to provide** memory signals (e.g., data, address and control signals)" Ex. 1008, '434 patent, 6:3-5 (emphasis added)." Pet. 24. The Petition states that Averbuj teaches "a control module configured to generate address and control signals for testing the memory devices,"

because Averbuj's "BIST controller 4 is programmable *to provide* address and control signals." Pet. 29 (emphasis added); *see* PO Resp. 53.

For purposes of our Institution Decision, we interpreted "generate" as used in the '434 patent claims as meaning "cause" or "produce." *See* Dec. on Inst. 9. As indicated in Section III.A., above, subsequent to that decision and the oral hearing in the present *inter partes* review, we again considered the meaning of the claim term "generate" in our final decision in IPR-970, and determined that the broadest reasonable interpretation of the '434 patent claim term "generate" is "produce," and that "generate" does not mean "cause" or "cause to produce." *See* IPR-970 FWD, 24–33. In Section III.A., above, we considered extrinsic evidence in support of Petitioner's interpretation of "generate" as meaning "cause" that was not before us in IPR-970, but concluded the broadest reasonable interpretation of "generate" is "produce." We, therefore, adopted and incorporated by reference in the present final decision our analysis and construction of the terms "generate" and "configured to generate" in our Final Written Decision in IPR-970. *See* Section III., above.

In construing the claim term "generate" in our final decision in IPR-970, we found that the terms "provide" and "generate" are used in the '434 patent to describe different embodiments of the control module, and determined the claim term "generate" did not mean "provide." *See* IPR-970 FWD, 28–30. We also determined that the phrases "address and control signals generated by the control module" (claims 1, 20) and "generating, by the control module, address and control signals" (claim 29) encompass signals that originated in the control module, including by transformation or modification of information received from another component. IPR-970

FWD, 32. We further determined the broadest reasonable interpretation of this claim language does not encompass “signals received by the control module from another component, that are merely provided, propagated, sent, or input to the memory devices, without transformation or modification by the control module.” *Id.* We stated that “generate,” as used in the ’434 patent, does not encompass the selection function of a multiplexer. *Id.* at 29–30.

Petitioner’s contention that Averbuj teaches “a control module configured to generate address . . . signals for testing the memory devices,” as recited in independent claims 1, 20, and 29, is based on an interpretation of “generate” as meaning “cause.” Pet. 24, 29; *see also*, Reply 20–21 (asserting that Averbuj’s BIST controller 4 is “configured to generate address and control signals” as claimed in the ’434 patent because “the BIST controller of Averbuj *causes* the CMD_REQ and SET ADDRESS signals, representing the claimed ‘address and controls signals’” (emphasis added)) (citing Dec. on Inst. 15–16). Petitioner has not explained sufficiently, however, how Averbuj’s BIST controller 4 is designed, adapted, or arranged to produce address signals for testing the memory devices.⁴

⁴ During oral argument, Mr. Heafey, counsel for Petitioner, was asked the following question: “Assuming we adopt ‘to produce’ as the proper construction [of ‘to generate’], are [CMD]_REQ and SET ADDRESS produced by BIST controller 4?” Tr. 78:16–18. In response, Mr. Heafey stated: “[T]hose signals are definitely produced by BIST controller 4, yes, they are.” *Id.* at 78:19–20. Mr. Heafey did not elaborate, however, nor did he otherwise explain clearly during oral argument how Averbuj’s BIST controller 4 is designed, adapted, or arranged to produce address signals for testing the memory devices.

Moreover, the evidence in this proceeding fails to support a finding that Averbuj's BIST controller 4 is configured to generate address signals for testing the memory devices in the manner claimed.

In Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS (*see* Ex. 1011, 4:4–12, 5:19–21, Table 1)) communicated to sequencers 8 are either received by user interface 22 via external input (*id.* at 5:41–43) or generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20 (*see* Ex. 2012, 30:23–24, 41:1–4, 54:21–22, 55:6–25, 55:20–56:9 (wherein Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 are generated by hand or by a tool, e.g., a compiler, and that “99 percent of the designs or chips do not have a compiler on-board”)). Multiplexer 24 *selects* one of the algorithms (i.e., CMD_DATA) and BIST controller 4 *provides* to sequencers 8 the same CMD_DATA (e.g., address signals) without transformation or modification. *See* Ex. 1011, 4:4–12, 5:16–21; Ex. 2012, 54:8–13 (“Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and the output [] reflect[s] what’s in the input, based on the selection.”). Based on our interpretation of the terms “configured to” and “generate,” as discussed above, the claim recitation “a control module configured to generate address . . . signals for testing the memory devices” (claims 1, 20, and 29) does not read on Averbuj's BIST controller 4, because the SET ADDRESS signal output by BIST controller 4 is produced by a component external to BIST controller 4, and merely selected by multiplexer 24 and provided to sequencers 8 without transformation or modification by BIST controller 4.

In sum, after considering the parties' arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses "a control module configured to generate address . . . signals for testing the memory devices" as recited in independent claims 1, 20, and 29. Petitioner does not correct this deficiency in its challenges as to dependent claims 2–4, 14–19, and 27. *See* Pet. 32–39; 35 U.S.C. § 112, ¶ 4 ("A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers."). Because Petitioner has not identified in Averbuj a teaching of each and every limitation recited in claims 1–4, 14–20, 27, and 29, Petitioner has not met its burden to show these claims are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj.

V. MOTIONS TO EXCLUDE

Both Patent Owner and Petitioner filed Motions to Exclude.

A. Petitioner's Motion to Exclude

Petitioner moves to exclude Exhibit 2018 (the Deposition Transcript of Dr. Donald Alpert) and to strike Section IV(B)(5)(a) of Patent Owner's Response, wherein Patent Owner quotes Dr. Alpert's deposition testimony in support of its proposed construction of the claim term "generate" (*see* PO Resp. 34). Paper 26, 2. Petitioner argues Dr. Alpert is Sandisk Corporation's expert in IPR-970 and that Petitioner was not provided notice or otherwise invited to attend Dr. Alpert's deposition. *Id.* at 3. Petitioner contends Dr. Alpert's statements are not admissible as evidence in this proceeding "because they are hearsay and will unfairly prejudice Petitioner." *Id.* at 2.

In construing the claim term “generate” in Section III.A., above, we adopted and incorporated by reference our analysis and construction of this term on pages 24–33 of our Final Written Decision in IPR-970. In our claim construction in IPR-970, we considered Dr. Alpert’s deposition and declaration testimony, but did not quote or cite to the testimony relied on by Patent Owner in Section IV(B)(5)(a) of its Patent Owner Response in the present proceeding. *Compare* IPR-970 FWD, 25–26 *with* PO Resp. 34. In IPR-970, we determined Dr. Alpert’s testimony that one of ordinary skill in the art would understand the claim term “generate” as meaning “cause” or “produce” was inconsistent with the ’434 patent’s implicit definition of this term as meaning only “produce.” *See* IPR-970 FWD, 30 n.4. We, therefore, did not rely on Dr. Alpert’s testimony in our construction of the claim term “generate” in IPR-970, or in the present case. *See id.*; *see also id.* at 32 (“Having considered the ordinary meaning of the term ‘generate’ in the context of both the claims and the ’434 patent as a whole, we agree with Patent Owner that the broadest reasonable interpretation of the claim term ‘generate’ is ‘produce,’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’”).

Because we did not rely on Exhibit 2018 or consider the argument in Section IV(B)(5)(a) of Patent Owner’s Response in rendering our decision in this proceeding, we dismiss Petitioner’s Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) as moot.

B. Patent Owner’s Motions to Exclude

Patent Owner filed a Motion to Exclude “Exhibits 1026, 1029, 1030, and 1032 (or portions thereof)” and requests that we strike the corresponding citations to these exhibits in Petitioner’s Reply. Paper 34, 2. We did not

rely on these exhibits in rendering our decision. Accordingly, Patent Owner's Motion to Exclude these exhibits is dismissed as moot.

Patent Owner also filed a Motion to Exclude Portions of Petitioner's Reply, contending "Petitioner's Reply improperly presents new theories, never raised in its Petition or adopted by the Board's Institution Order, violating 37 CFR 42.22-23 and applicable law." Paper 35, 1. Patent Owner provides a table identifying the alleged new theories advanced by Petitioner in its Reply (*id.* at 2) and identifies by page and line number specific portions of sections II.A.3, II.B.1.a, II.B.5 & III.B.2 of the Reply that it contends should be excluded as advancing new theories (*see id.* at 2–8).

Our decision is not based on any of the alleged new theories identified by Patent Owner, and Patent Owner does not cite to any of the specific language in the Reply that Patent Owner contends should be excluded. Accordingly, Petitioner's Motion to Exclude portions of Petitioner's Reply is dismissed as moot.

VI. CONCLUSION

Petitioner has not shown by a preponderance of the evidence that claims 1–4, 14–20, 27, and 29 of the '434 patent are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2).

VII. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–4, 14–20, 27, and 29 of U.S. Patent No. 8,001,434 B1 are not held unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2);

FURTHER ORDERED that Petitioner's Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) of Patent Owner's Response is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exhibits 1026, 1029, 1030, and 1032 to strike the corresponding citations to these exhibits in Petitioner's Reply is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exclude Portions of Petitioner's Reply is *dismissed*; and

FURTHER ORDERED that, because this is a final decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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Patent 8,001,434 B1

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September 19, 2016

Attached is a copy of the Notice of Appeal for IPR 2014-01372, pursuant to 37 C.F.R. 1.302. The Notice of Appeal was timely filed at the United States Patent and Trademark Office and served on Patent Owner on September 8, 2016.

Sincerely,

/s/ Timothy H. Caine

Timothy H. Caine
Associate

ATTORNEY REFERENCE NO.: 21584-112005.4
IPR2014-01372

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE
BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES, INC.

Petitioner

v.

NETLIST, INC.

Patent Owner

Case IPR2014-01372

Patent 8,001,434

**PETITIONER'S NOTICE OF APPEAL TO THE
U.S. COURT OF APPEALS FOR THE FEDERAL CIRCUIT**

Via PRPS

Mail Stop PATENT BOARD
Patent Trial and Appeal Board
United States Patent and Trademark Office
P.O. Box 1450
Alexandria, VA 22313-1450

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Pursuant to 35 U.S.C. Sections 141, 142, and 319, 37 C.F.R. Sections 90.2, 90.3, and 104.2; Rule 4(a)(3) of the Federal Rules of Appellate Procedure, and Rule 15 of the Federal Circuit Rules, the Petitioner appeals to the United States Court of Appeals for the Federal Circuit from the Final Written Decision (Paper No. 45), the Decision Denying Petitioner's Request for Rehearing (Paper No. 48), and from all underlying findings, orders, decisions, determinations, rulings, and opinions adverse to Petitioner. Petitioner indicates that the appeal includes at least the following issues:

1. the Board's judgment that claims 1-4, 14-20, 27 & 29 of U.S. Patent No. 8,001,434 are patentable; and
2. the Board's claim construction.

Petitioner reserves the right to challenge any findings, orders, decisions, determinations, rulings, and opinions adverse to Petitioner.

Dated: September 8, 2016

Respectfully submitted,

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Certificate Of Service In Compliance With 37 C.F.R. § 42.6(e)(4)

The undersigned certifies that a complete copy of this *PETITIONER'S MOTION TO EXCLUDE EVIDENCE* was served via email on Counsel for Patent Owner in this proceeding:

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Paper 45
Entered: March 9, 2016

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SMART MODULAR TECHNOLOGIES INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2014-01372
Patent 8,001,434 B1

Before LINDA M. GAUDETTE, BRYAN F. MOORE, and
PETER P. CHEN, *Administrative Patent Judges*.

GAUDETTE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. § 318(a) and 37 C.F.R. § 42.73

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Patent 8,001,434 B1

I. BACKGROUND

Smart Modular Technologies Inc. (“Petitioner”) filed a Petition (Paper 7, Corrected Petition (“Pet.”)) on August 23, 2014, requesting institution of an *inter partes* review of claims 1–35 (the “challenged claims”) of U.S. Patent No. 8,001,434 B1 (Ex. 1008, “the ’434 patent”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (Paper 11). Based on these submissions, an *inter partes* review of claims 1–4, 14–20, 27, and 29 was instituted on March 10, 2015, pursuant to 35 U.S.C. § 314. Paper 13 (“Dec. on Inst.” or “Institution Decision”).

After institution, Patent Owner filed a Response (Paper 22, “PO Resp.”), and Petitioner filed a Reply to the Patent Owner Response (Paper 27, “Reply”).

Petitioner filed a Motion to Exclude Evidence. Paper 26; *see also* Paper 37 (Patent Owner’s Opposition) and Paper 42 (Petitioner’s Reply). Patent Owner also filed a Motion to Exclude Evidence (Paper 34; *see also* Paper 39 (Petitioner’s Opposition) and Paper 41 (Patent Owner’s Reply)) and a Motion to Exclude Portions of Petitioner’s Reply (Paper 35; *see also* Paper 40 (Petitioner’s Opposition)).

Oral argument was held on November 17, 2015, and a transcript (Paper 43, “Tr.”) has been entered into the record.

The Board has jurisdiction under 35 U.S.C. § 6(c). This Final Written Decision, issued pursuant to 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73, addresses issues and arguments raised during trial.

For the reasons discussed below, we determine Petitioner has not met its burden to prove, by a preponderance of the evidence, that claims 1–4, 14–

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20, 27, and 29 of the '434 patent are unpatentable. Petitioner's and Patent Owner's Motions to Exclude are dismissed.

A. Related Matters

Petitioner also requested, and we instituted, an *inter partes* review of U.S. Patent No. 8,359,501 ("the '501 patent"), which claims priority as a continuation of U.S. Patent Application No. 12/422,925 (now the '434 patent): *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01374 (PTAB March 10, 2015), Paper 12. We denied Petitioner's requests to institute *inter partes* reviews of the '434 patent and the '501 patent in two additional petitions, filed August 23, 2014: *Smart Modular Technologies Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB March 13, 2015), Paper 16, and Case IPR2014-01375 (PTAB March 13, 2015), Paper 12, respectively. Oral argument in IPR2014-01374 was consolidated with the oral argument in the present *inter partes* review. A final decision in IPR2014-01374 is issued concurrently with this final decision.

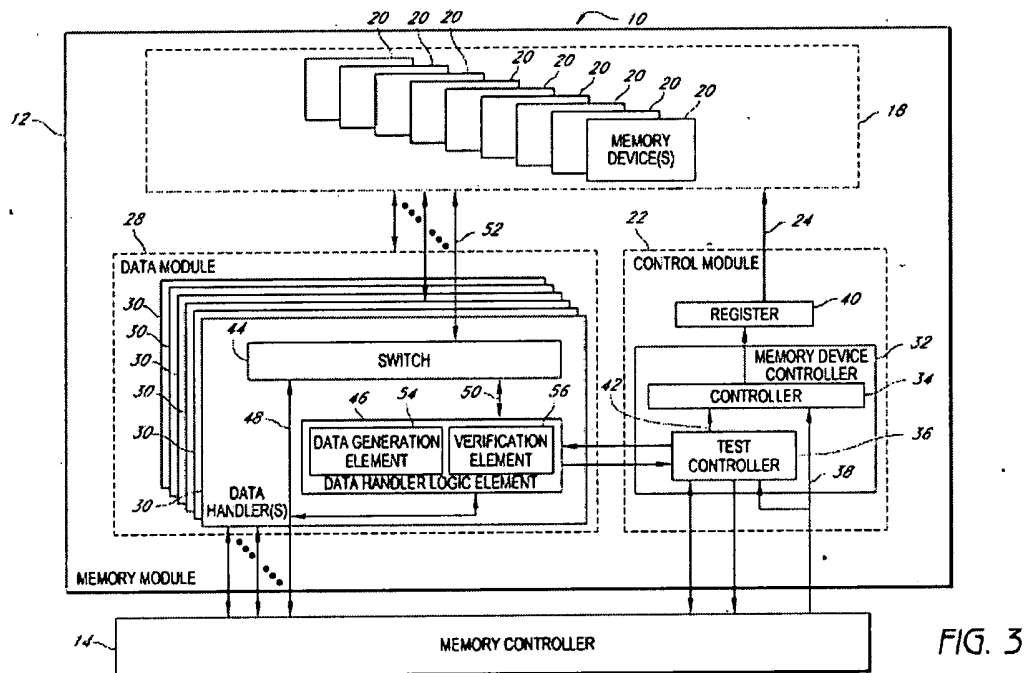
The '434 patent also is the subject of IPR2014-00970 ("IPR-970"). We entered a final written decision in that case on December 14, 2015, concluding that claims 1, 14, 15, 19–25, and 27–34 are unpatentable on the following grounds: claims 1, 14, 15, 20–25, 27, and 29–34 under 35 U.S.C. § 102(b) based on Averbuj (U.S. Patent Application Publication No. 2005/0257109 A1, issued November 17, 2005); claim 28 under 35 U.S.C. § 103(a) based on Averbuj; and claims 1, 14, 15, 19, and 29 under 35 U.S.C. § 103(a) based on Huang (*An Efficient Parallel Transparent BIST Method for Multiple Embedded Memory Buffers*, Fourteenth International Conference on VLSI Design 379–384 (2001), IEEE). *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB December 14, 2015), Paper 32

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("IPR-970 FWD"). Petitioner Sandisk Corporation has requested rehearing of our determination that it did not meet its burden to show unpatentability of claims 2, 5, 6, 17, 18, and 35. *Id.*, Paper 33.

B: The '434 Patent (Ex. 1008)

The '434 patent relates to self-testing electronic memory modules. Ex. 1008, 1:23–24. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '434 patent, reproduced below. *See id.* at 9:23–25.



As illustrated in Figure 3, above, "memory module 10 includes a printed circuit board 12 configured to be operatively coupled to a memory controller 14 of a computer system 16." *Id.* at 5:1–3. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports.

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Id. at 5:3–7. “[M]emory module 10 comprises a control module 22 [and] . . . a data module 28.” *Id.* at 5:7–10.

“[C]ontrol module 22 can be configured to generate address and control signals 24 for testing the plurality of memory devices 18.” *Id.* at 9:25–27. In the embodiment shown in Figure 3, control module 22 includes control mixer element 32 for controlling the address and control signals for the self-testing function. *Id.* at 9:31–33. Control mixer element 32 includes memory device controller 34 (e.g., a DRAM controller) and test controller 36. *Id.* at 9:29–31. “[T]est controller 36 controls the generation of the address and control signal sequences to be used during the self-testing operation of the memory module 10 and also communicates with the data module 28.” *Id.* at 9:53–56. “[M]emory device controller 34 receives signals 38 (e.g., address and control signals) from the system memory controller 14 and signals 42 (e.g., address and controls signals) from the test controller 36.” *Id.* at 9:37–40.

Data module 28 comprises a plurality of data handlers 30 that “may be operatively coupled to (e.g., logically and/or electrically coupled to) the corresponding plurality of data ports” of memory devices 18. *Id.* at 5:10–11, 16–18. In the embodiment illustrated in Figure 3, each of data handlers 30 includes data handler logic element 46, comprising data generation element 54 and verification element 56. *Id.* at 10:27–30. “[D]ata generation element 54 may be configured to generate data signals (e.g., patterns of data signals) for writing to the corresponding plurality of data ports” of memory devices 18. *Id.* at 10:31–33. “The data signals and/or patterns of data signals may be based on information (e.g., programming or configuration information)

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the data handler logic element 46 receives from the control module 22.” *Id.* at 10:33–37.

C. Illustrative Claim

Of the challenged claims, claims 1, 20, and 29 are independent.

Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

D. The Instituted Ground of Unpatentability

We instituted the instant *inter partes* review of claims 1–4, 14–20, 27, and 29 based on anticipation under 35 U.S.C. § 102(b) by Averbuj, U.S. Patent No. 7,392,442 B2, issued June 24, 2008 (Ex. 1011, “Averbuj”). As indicated in Section I.A., above, in IPR-970, we held claims 1, 14, 15, 20–

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25, 27, and 29–34 of the '434 patent unpatentable under 35 U.S.C. § 102(b) and claim 28 unpatentable under 35 U.S.C. § 103(a) based on U.S. Patent Application Publication No. 2005/0257109 A1, the published application of the Averbuj patent on which the instituted ground of unpatentability in this proceeding is based.

E. Expert Testimony

Petitioner relies on the testimony of Dr. Nader Bagherzadeh in support of its patentability challenge. Pet. 1. Dr. Bagherzadeh executed a declaration (Ex. 1009, “the Bagherzadeh Declaration”) in support of the Petition. Dr. Bagherzadeh was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 2012.

Dr. Bagherzadeh testifies as follows: “In 1979 and 1987, respectively, [he] earned a master of science in electrical engineering and a doctorate degree in computer engineering from the University of Texas at Austin.” Ex. 1009 ¶ 4. He has been involved in design and development of digital systems for more than 30 years. *Id.* ¶ 6. Dr. Bagherzadeh joined the University of California, Irvine, in 1987, and has “been teaching, researching, and consulting regarding almost all aspects of memory design for high performance computer systems, including but not limited to DRAMs and SRAMs.” *Id.* ¶ 7. He has been employed as a professor in the department of Electrical Engineering and Computer Science at the University of California, Irvine, since 2003. *Id.* ¶ 5. In 2000, Dr. Bagherzadeh became a cofounder of Morpho Technologies, a high tech company focused on the design and development of low power and high performance digital signal processors for mobile applications. *Id.* ¶ 7. He

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was involved in evaluating patents, technical reports and presentations related to memory chip designs, DSPs, and parallel processing algorithms for mobile platforms. *Id.*

Patent Owner relies on the testimony of Dr. Carl Sechen. PO Resp. 1. Dr. Sechen executed a declaration (Ex. 2019, “the Sechen Declaration”) in support of Patent Owner’s Response. Dr. Sechen was cross-examined on the subject matter of his declaration, and a transcript of the testimony was filed as Exhibit 1031.

Dr. Sechen testifies as follows: he has an M.S. degree in Electrical Engineering from the Massachusetts Institute of Technology and “was awarded a Ph.D. in electrical engineering from the University of California at Berkeley in 1986.” Ex. 2019 ¶ 5. Dr. Sechen has been a Professor of Electrical Engineering for 29 years. *Id.* ¶ 2. During this time period, his research has focused on design and computer-aided design of digital integrated circuits, including the design of DRAM, and he has taught numerous students how to design DRAM memories. *Id.* Dr. Sechen has “also been involved in numerous research projects on VLSI design and memory design[, and has] taught numerous graduate researchers how to design digital integrated circuits, including memories.” *Id.* ¶ 4.

The parties do not dispute that Dr. Bagherzadeh and Dr. Sechen are qualified to testify as experts under FRE 702.

II. LEVEL OF ORDINARY SKILL IN THE ART

The level of ordinary skill in the art is relevant to claim construction and anticipation. *See Yorkey v. Diab*, 605 F.3d 1297, 1300 (Fed. Cir. 2010) (explaining that a determination of anticipation involves interpreting the

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claim language and then comparing the construed claim to a prior art reference); *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1259–60 (Fed. Cir. 2010) (“[C]laim language should be read in light of the specification as it would be interpreted by one of ordinary skill in the art.”).

Dr. Bagherzadeh and Dr. Sechen agree that the field of art is “memory module design with features for built-in self-test (BIST).” Ex. 2019 ¶ 21; see also Ex. 1009 ¶ 17.

Dr. Bagherzadeh testifies that

a person of ordinary skill in the art for the ‘434 patent in April 2008 would have a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering, and at least one year of experience relating to memory systems and BIST.

A person having a bachelor’s degree or the equivalent training or experience in electrical engineering or computer engineering would have experience with digital circuit design, memories and computer architecture. An additional year of experience relating to memory systems and BIST would focus on those areas and allow one to recognize current issues with respect to the field.

Ex. 1009 ¶¶ 19–20; *see id.* ¶ 16 (“My understanding is that the earliest possible priority date of the ‘434 patent is April 14, 2008.”).

Dr. Sechen testifies that

one of ordinary skill in the art at the time of filing the application(s) for the ‘434 Patent, who would be working on the design of memory devices and memory modules, would have at least a Bachelor of Science degree in electrical engineering or computer engineering, and at least five years of industry experience designing memory devices and memory modules. Alternatively, one of ordinary skill in the art would have an M.S. degree in electrical engineering or computer engineering, and at least three years of industry experience designing

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memory devices and memory modules. Moreover, one of ordinary skill in the art would have a Ph.D. degree in electrical engineering or computer engineering, and have at least one year of industry experience designing memory devices and memory modules.

Ex. 2019 ¶ 21.

The difference in the opinions of the two experts is essentially that Dr. Sechen believes a person of ordinary skill in the art would have additional experience, education, or a combination of the two, beyond that proposed by Dr. Bagherzadeh. Both experts have been employed as professors since the mid 1980s. Dr. Bagherzadeh also has some experience working with engineers in the industry by virtue of his employment with AT&T Bell Labs from 1980–1984 (Ex. 1009 ¶ 6; *id.* Appx. A). Both experts were awarded bachelors and masters degrees in electrical engineering. Dr. Bagherzadeh was awarded a doctorate in computer engineering, while Dr. Sechen was awarded a doctorate in electrical engineering.

Based on a comparison of Dr. Bagherzadeh's and Dr. Sechen's declarations (Exs. 1009, 2019), we find Dr. Bagherzadeh has a broader range of educational and work experience, and, therefore, accord greater weight to his opinion as to the level of ordinary skill in the art. Dr. Bagherzadeh's opinion is consistent with our finding as to the level of ordinary skill in the art in our Final Written Decision in IPR-970. We, therefore, adopt our finding in IPR-970 as to the level of ordinary skill in the art:

[A] person of ordinary skill in the art at the time of the '434 patent would have a Bachelor's degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.

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IPR-970 FWD, 10–11 (noting that our finding was “based on our review of the ’434 patent and the types of problems and solutions described in the ’434 patent and cited prior art”).

III. CLAIM CONSTRUCTION

In its Petition, Petitioner offers specific constructions for the claim terms “operatively coupled,” “ports,” “to generate”/“generating,” “data module,” “operable independently,” “cyclic data,” and “proximate.” Pet. 23–28. We determined that, for purposes of our Institution Decision, only the term “generate” required express construction. Dec. on Inst. 6. We interpreted “generate” as meaning “produce” or “cause.” *Id.* at 9.

In its Response, Patent Owner disagrees with our interpretation of “generate” as meaning “cause.” PO Resp. 18. Patent Owner contends our “Institution Decision also provides a construction for ‘configured to generate,’” and proposes its own construction for this phrase. *Id.* at 9 (citing Dec. on Inst. 8–9). In its Reply, Petitioner asserts that our interpretation of “generate” in the Institution Decision is correct (Reply 1), and offers a proposed construction for the term “configured” (*id.* at 9). Patent Owner’s and Petitioner’s proposed constructions for the disputed claim language are listed in the table below.

Claim Term/Phrase	Patent Owner’s Proposed Construction	Petitioner’s Proposed Construction
generate	produce (PO Resp. 18)	cause or initiate transmission (Pet. 24–25) produce or cause (Reply 1)
configured to generate	programmed to generate/produce (<i>see</i> PO Resp. 9)	arranged or prepared to produce or cause (Reply 9–10)

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In support of their proposed constructions of the disputed claim language, the parties rely on language in the '434 patent and extrinsic evidence. *See generally*, Pet. 24–25; PO Resp. 8–36; Reply 1–17. Neither party relies on prosecution history disclaimer in support of its proposed constructions.

In an *inter partes* review, the Board interprets claim terms in an unexpired patent according to the broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b); *see also In re Cuozzo Speed Techs., LLC*, 793 F.3d 1268, 1278–80 (Fed. Cir. 2015) (“Congress implicitly approved the broadest reasonable interpretation standard in enacting the AIA,” and “the standard was properly adopted by PTO regulation.”), *cert. granted, sub nom. Cuozzo Speed Techs. LLC v. Lee*, 136 S. Ct. 890 (mem.) (2016). Under that standard, and absent any special definitions, we give claim terms their ordinary and customary meaning, as would be understood by one of ordinary skill in the art at the time of the invention. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007); *see also Trivascular, Inc. v. Samuels*, No. 2015-1631, 2016 WL 463539, at *3 (Fed. Cir. Feb. 5, 2016) (“Under a broadest reasonable interpretation, words of the claim must be given their plain meaning, unless such meaning is inconsistent with the specification and prosecution history.”). Any special definitions for claim terms must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

“[T]he Board may rely on dictionaries ‘so long as the dictionary definition does not contradict any definition found in or ascertained by reading the patent document.’” *Belden Inc. v. Berk-Tek LLC*, 610 Fed.

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App’x 997, 1002 (Fed. Cir. 2015) (quoting *Phillips v. AWH Corp.*, 415 F.3d 1303, 1322–23 (Fed. Cir. 2005)). Expert testimony is useful to explain terms of art, and the state of the art at any given time, but cannot be used to prove “the proper or legal construction of any instrument of writing.” *Teva Pharm. USA, Inc. v. Sandoz, Inc.*, 135 S. Ct. 831, 841 (2015) (internal citations omitted); *see also* 37 C.F.R. § 42.65(a) (“Expert testimony that does not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.”).

Only terms which are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999). A court may revisit and alter its construction of claim terms as the record in a case develops. *See Pressure Prods. Med. Supplies, Inc. v. Greatbatch Ltd*, 599 F.3d 1308, 1316 (Fed. Cir. 2010).

A. Generate

Petitioner contends “[t]he Institution Decision correctly found that the intrinsic and extrinsic record supported construing ‘generate’ as ‘cause’ or ‘produce.’” Reply 1. Patent Owner contends “‘generate’ is properly construed to mean ‘produce,’” but “a construction of generate that includes ‘cause’ is unsupported by – and contradictory to – the intrinsic and extrinsic evidence, and fails to provide reasonable clarity.” PO Resp. 18. The same respective constructions of “generate” were advanced by Patent Owner and the petitioner, Sandisk Corporation, in IPR-970. *See* IPR-970 FWD, 25–27.

We provided an extensive analysis of the term “generate” in our Final Written Decision in IPR-970. *See id.* at 24–33. “[W]e agree[d] with Patent Owner that the broadest reasonable interpretation of the claim term

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‘generate’ is ‘produce’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’” *Id.* at 32.

We interpret[ed] the claim language “address and control signals *generated* by the control module and the data *generated* by the plurality of data handlers” (claim 1), and the corresponding limitations in claims 20 and 29, as encompassing signals and data that originated in these modules, including by transformation or modification of information and/or data received from another component.

Id. We also stated that “[w]e [did] not interpret this language as encompassing signals and data received by the data and control modules from another component, and merely provided, propagated, sent, or input to memory devices, without transformation or modification by the data and control modules.” *Id.*

In support of its contention that “generate” also should be interpreted as meaning “cause,” Petitioner relies on extrinsic evidence that was not before us in IPR-970, i.e., the Bagherzadeh Declaration (Ex. 1009) ¶ 28, ANSI/IEEE 100 - Standard Dictionary of Electrical and Electronic Terms, Fourth Edition, 1988 (Ex. 1016, “IEEE”), and *Webster’s II New College Dictionary*, 2001 (Ex. 1032, “Webster’s II”). See Pet. 24–25; Reply 1–9. We consider this evidence as it is relevant to whether the term “generate” has a particular meaning to one of ordinary skill in the art. See *Trivascular*, 2016 WL 463539, at *3 (“Construing individual words of a claim without considering the context in which those words appear is simply not ‘reasonable.’ Instead, it is the ‘use of the words in the context of the written description and customarily by those of skill in the relevant art that accurately reflects both the “ordinary” and “customary” meaning of the terms in the claims.” (quoting *Ferguson Beauregard/Logic Controls, Div.*

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of Dover Res., Inc. v. Mega Sys., LLC, 350 F.3d 1327, 1338 (Fed. Cir. 2003))).

The IEEE defines “generate (computing systems)” as: “[t]o produce a program by selection of subsets from a set of skeletal coding under the control of parameters.” Ex. 1016, 5. Dr. Bagherzadeh cites the IEEE definition in support of his testimony that “[t]he broadest reasonable interpretation of the claim [terms] ‘to generate’ and ‘generating’ in claims 1, 20, and 29, is ‘to cause or initiate transmission’ and ‘causing or initiating.’” Ex. 1009 ¶ 28 (also citing Ex. 1008, 6:3–5, 10:67–11:2, 6:14–17). Dr. Bagherzadeh testifies that “[t]his interpretation is consistent with the specification and extrinsic evidence because they describe generating as initiating or transmitting, which is also the broadest interpretation of ‘generate.’” *Id.*

We are not persuaded by Dr. Bagherzadeh’s testimony because it is devoid of specific facts and analysis as to what led him to conclude that a person of ordinary skill in the art would interpret the claim term “generate” in the manner Petitioner proposes. *See id.* Dr. Bagherzadeh does not explain how the IEEE definition supports an interpretation of generate as “cause or initiate transmission,” terms that do not appear in the IEEE definition. *See id.* Nor does Dr. Bagherzadeh identify other extrinsic evidence which he believes to be consistent with an interpretation of “generate” as meaning “cause or initiate transmission.” *See id.* Likewise, Dr. Bagherzadeh has not explained why he believes the ’434 patent “describe[s] generating as initiating or transmitting” (Ex. 1009 ¶ 28), as these terms do not appear in the cited portions of the ’434 patent. *See* Ex. 1008, 6:3–5, 10:67–11:2, 6:14–17. Accordingly, Dr. Bagherzadeh’s

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testimony does not persuade us to accept Petitioner's interpretation of "generate" as meaning "cause."

Petitioner contends "Dr. Bagherzadeh's use of the IEEE definition specifically illustrates that 'generate' includes 'selection . . . from a set,' which implies that 'generate' must not be limited to creating from previously undefined values, which may be a more traditional meaning, but holds little value in the intrinsic evidence in this proceeding." Reply 6. Petitioner has not identified corresponding testimony by Dr. Bagherzadeh or other evidentiary support for this statement and, therefore, this argument is likewise unpersuasive. *See id.*

In support of its construction of "generate" as meaning "cause," Petitioner also relies on a finding in our Institution Decision that such construction is supported by Merriam-Webster's Collegiate Dictionary, wherein one definition of "generate" is "to be the cause of" (Ex. 2009, 3). Reply 2 (citing Dec. on Inst. 9 [sic, 8]). This argument is not persuasive because the definition in Merriam-Webster's Collegiate Dictionary contradicts the definition of "generate" that we ascertained upon reading the '434 patent as discussed in our Final Written Decision in IPR-970. *See Belden*, 610 Fed. App'x at 1002.

In sum, we are not persuaded that Petitioner's extrinsic evidence supports a finding that a customary meaning of "generate" to one of ordinary skill in the art is "cause."¹ We, therefore, determine the broadest reasonable

¹ Petitioner relies on *Webster's II*, for a definition of "cause." *See* Reply 4, 6–8 (citing Ex. 1032, 3). Having determined the broadest reasonable interpretation of "generate" does not encompass "cause," we do not find this evidence relevant to our claim construction analysis.

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construction of “generate” is “produce” for the reasons stated above and in our Final Written Decision in IPR-970. We adopt and incorporate by reference our analysis and construction of the term “generate” in our Final Written Decision in IPR-970 (IPR-970 FWD, 24–33).

B. Configured to

In our Final Decision in IPR-970, we construed “configured to” as “designed to, adapted to, or arranged to.” *See* IPR-970 FWD, 20–24. Petitioner contends “configured” should be construed as “arranged” or “prepared.” Reply 9. Responsive to this Board panel’s questioning during the oral hearing, however, Petitioner agreed that “designed to, adapted to, or arranged to” was an acceptable definition for the claim term “configured to.” *See* Tr. 22:7–17.

Patent Owner contends the phrase “configured to generate” should be construed as “programmed to generate.” PO Resp. 9. Patent Owner does not propose a separate construction for the term “configured.” *See generally id.* at 9–18. In support of its proposed construction, Patent Owner relies on language in the ’434 patent and the testimony of its expert, Dr. Sechen. *See id.* Dr. Sechen testifies as to the meaning of “configured to generate,” but does not provide a separate discussion of the meaning of “configured to” outside the context of “configured to generate.” *See* Ex. 2019 ¶¶ 42–57.

As explained in our Final Decision in IPR-970, the term “configured to” is used in the ’434 patent in conjunction with numerous terms, such as “provide” and “selectively input,” and there is a presumption that the term “configured to” should carry the same meaning in each instance. *See* IPR-970 FWD, 20–21. The evidence relied on by Patent Owner in support of its proposed construction of the term “configured to” in the present *inter partes*

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review does not differ materially from the evidence relied upon by Patent Owner in IPR-970. *Compare* PO Resp. 9–18 with IPR-970, Paper 16, 12–17. Accordingly, we determine the broadest reasonable construction of “configured to” is “designed to, adapted to, or arranged to” for the reasons stated above and in our Final Written Decision in IPR-970. We adopt and incorporate by reference our analysis and construction of the term “configured to” in our Final Written Decision in IPR-970 (IPR-970 FWD, 20–24).

In summary, we construe the claim terms in controversy as follows:

Claim Term/Phrase	Interpretation
Generate	Produce
Configured to	Designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]
Configured to Generate	Designed, adapted, or arranged to produce

IV. ALLEGED ANTICIPATION OF CLAIMS 1–4, 14–20, 27, AND 29 BY AVERBUJ

A. Principles of Law

To prevail in its challenges to the patentability of the claims, a petitioner must establish facts supporting its challenges by a preponderance of the evidence. 35 U.S.C. § 316(e); 37 C.F.R. § 42.1(d).

To establish anticipation, each and every element in a claim, arranged as recited in the claim, must be found in a single prior art reference.

Net MoneyIN, Inc. v. VeriSign, Inc., 545 F.3d 1359, 1369 (Fed. Cir. 2008); *Karsten Mfg. Corp. v. Cleveland Golf Co.*, 242 F.3d 1376, 1383 (Fed. Cir. 2001). While the elements must be arranged or combined in the same way

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as in the claim, “the reference need not satisfy an *ipsissimis verbis* test.” *In re Gleave*, 560 F.3d 1331, 1334 (Fed. Cir. 2009).

A single prior art reference that discloses, either expressly or inherently, each limitation of a claim invalidates that claim by anticipation. Thus, a prior art reference without express reference to a claim limitation may nonetheless anticipate by inherency. “Under the principles of inherency, if the prior art necessarily functions in accordance with, or includes, the claims limitations, it anticipates.”

Perricone v. Medicis Pharm. Corp., 432 F.3d 1368, 1375–76 (Fed. Cir. 2005) (citations omitted). “In general, a limitation or the entire invention is inherent and in the public domain if it is the ‘natural result flowing from’ the explicit disclosure of the prior art.” *Schering Corp. v. Geneva Pharms., Inc.*, 339 F.3d 1373, 1379 (Fed. Cir. 2003).

“[T]he Board is not bound by any findings made in its Institution Decision. . . . The Board is free to change its view of the merits after further development of the record, and *should do so* if convinced its initial inclinations were wrong.” *Trivascular*, 2016 WL 463539, at *9.

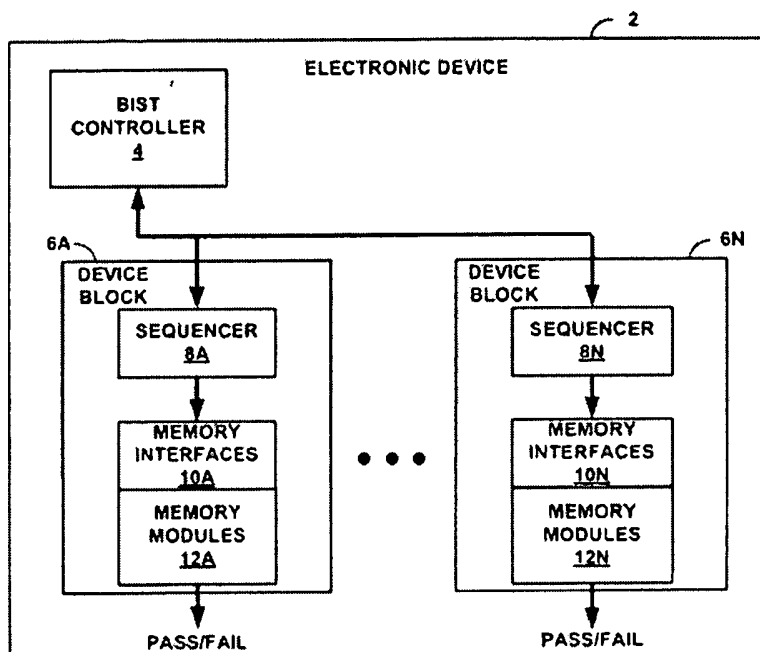
We analyze the instituted ground of unpatentability in accordance with the above-stated principles.

B. Averbuj (Ex. 1011)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1011, 1:64–65, 2:3–4. Figure 1 of Averbuj is reproduced below.

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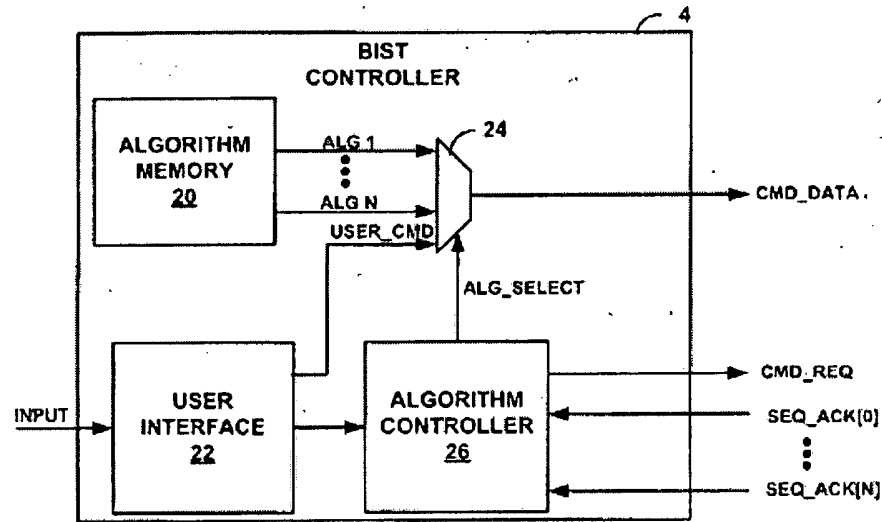


Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* at 3:62–64. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* at 3:64–67. “Each of device blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* at 3:67–4:3.

BIST controller 4 is illustrated in Figure 2, reproduced below.

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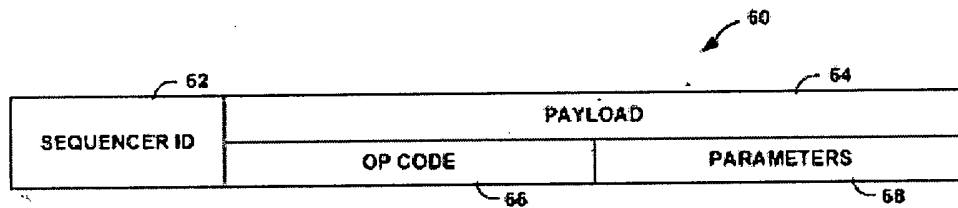
Averbuj Figure 2, above, “is a block diagram illustrating an example embodiment of a BIST controller.” *Id.* at 3:39–40. BIST controller 4 includes algorithm controller 26 that can be invoked either by a user through user interface 22 or automatically upon power-up of electronic device 2. *Id.* at 5:12–16. “Once invoked, algorithm controller 26 provides an algorithm select signal (ALG_SELECT) to multiplexer 24 to select one of the algorithms stored within algorithm memory 20.” *Id.* at 5:16–19. Alternatively, user interface 22 may programmably receive algorithms via external input, and deliver the received algorithms to multiplexer 24. *Id.* at 5:41–43. The algorithms, whether stored in algorithm memory 20 or received via external input, have a similar form, i.e., “a sequence of binary commands in which each command defines a test within the overall algorithm.” *Id.* at 5:43–47. BIST controller 4 provides and communicates the selected algorithm to sequencers 8 for application to device blocks 6 as a stream of binary commands (CMD_DATA), “each command specif[ying] an operational code [(OP CODE)] and a set of parameters that define one or

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more memory operations without regard to the physical characteristics or timing requirements of memory modules 12.” *Id.* at 4:4–12, 5:19–21.

Figure 8, reproduced below, “is a block diagram illustrating an example data structure of a command issued by BIST controller 4.” *Id.* at 9:15–16.

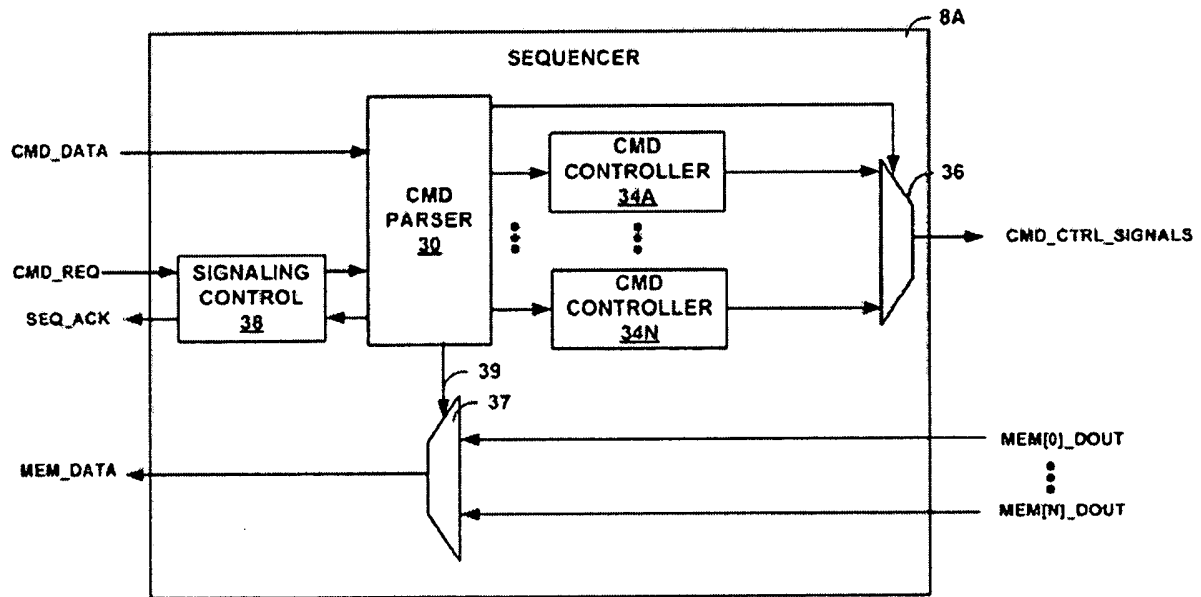


In the embodiment shown in Figure 8, above, “command 60 includes a sequencer identifier (ID) 62, and a payload 64. Sequencer ID 62 identifies a sequencer, e.g., sequencer 8A, to which command 60 is being issued.” *Id.* at 9:17–20. “Payload 64 of command 60 carries binary data that defines the command itself. In particular, payload 64 includes an operational code (OP CODE) 66 and a set of parameters 68.” *Id.* at 9:29–31. In one embodiment, for example, “OP CODE 66 and parameters 68 comprise three bits and twenty-nine bits, respectively, to form a 32-bit command.” *Id.* at 9:58–60. “OP CODE 66 specifies a particular function to be performed by the receiving sequencers 8.” *Id.* at 9:32–33. An exemplary OP CODE is SET ADDRESS (OP CODE 100), which “[s]ets a specific starting address as well as a maximum address limit for a test algorithm as applied to the memory modules.” *Id.* at Table 1.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* at 3:46–47.

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As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 that receives command data (CMD_DATA) from BIST controller 4. *Id.* at 6:24–26. Command parser 30 processes the commands from BIST controller 4 to identify a specified operation, e.g., by identifying an OP CODE specified by the command. *Id.* at 6:26–29. Based on the specified operation, command parser 30 may extract one or more parameters from the command, and pass the extracted parameters to a selected command controller (i.e., one of command controllers 34A–34N). *Id.* at 6:30–37. The invoked command controller, in turn, issues a sequence of one or more operations to each memory interface 10, sequentially driving the appropriate command control signals (CMD_CTRL_SIGNALS) to carry out each operation of the sequence. *Id.* at 6:41–43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with

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the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* at 4:45–49.

C. Arguments

Independent claims 1, 20, and 29 recite “a plurality of memory devices” and “a data module.” Petitioner contends the claimed “memory devices” read on Averbuj’s memory modules 12 and the claimed “data module” reads on the combination of sequencers 8 and memory interfaces 10. Pet. 28. Independent claims 1, 20, and 29 further recite “a control module configured to generate address and control signals for testing the memory devices.” *See also* claims 1 and 20 (reciting a circuit configured to test the memory devices using the address and control signals generated by the control module); claim 29 (reciting a step of “generating, by the control module, address and control signals for testing the memory devices”). With respect to this limitation, the Petition states:

the BIST controller 4 of Averbuj ‘442 represents circuitry of a control module that provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE* for testing the memory components 12A–N. Ex. 1011, Fig. 2 and Column 9: Table 1. The *SET ADDRESS* and *CMD_REQ* signals include address and control signals for the testing functions. Since the BIST controller 4 provides the *SET ADDRESS* and *CMD_REQ* signals as *OP CODE*, a person of ordinary skill in the art (“POSITA”) would understand that the BIST controller 4 is programmable to provide address and control signals as recited in part by the Challenged Claims. Ex. 1009, Bagherzadeh Decl., ¶ 35.

Pet. 29.

Patent Owner argues Averbuj’s BIST controller 4 does not “generate” the *OP CODE* communicated to sequencers 8, and Petitioner has not met its burden to show that Averbuj’s BIST controller 4 is “configured to generate

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address . . . signals for testing the memory devices” as recited in claims 1, 20, and 29 of the ’434 patent. PO Resp. 48–49. As discussed in greater detail below, we find this argument persuasive, and, for this reason, conclude Petitioner failed to meet its burden to prove unpatentability of claims 1–4, 14–20, 27, and 29 of the ’434 patent.²

D. Expert Testimony

In support of its argument that Averbuj’s BIST controller 4 is not configured to generate address signals as claimed in claims 1, 20, and 29, Patent Owner relies on paragraphs 133–141 of the Sechen Declaration. *Id.* at 48–51. Patent Owner also relies on Dr. Bagherzadeh’s cross-examination testimony (*id.* at 49–50 (quoting Ex. 2012, 55:6–56:9)) regarding Averbuj Figure 2 (*see* Ex. 2012, 53:4–5; *see generally id.* at 53:4–56:9).

With reference to Averbuj Figure 2 (*see* Section IV.B., above, wherein this figure is reproduced and described), Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 is generated by hand or by a tool (Ex. 2012, 55:20–56:9), and he identifies a compiler as a tool for generating such code (*id.* at 30:23–24; 41:1–4; *see id.* at 55:6–25).³ Dr. Bagherzadeh testifies that a compiler is not illustrated in

² Because we are persuaded by this argument that Petitioner has not met its burden to show unpatentability, we find it unnecessary to address the additional arguments advanced by Patent Owner in its Response.

³ Petitioner contends “[t]he discussion of a compiler, from PO’s deposition of [Dr. Bagherzadeh], was hypothetical and specific to an article coauthored by [Dr. Bagherzadeh].” Reply 22 (citing Ex. 2012, 15:2–19:3).³ We were unable to locate a discussion of a compiler or any reference to an article coauthored by Dr. Bagherzadeh in the cited portions of Dr. Bagherzadeh’s deposition transcript. Dr. Bagherzadeh’s first discussion of a compiler appears to be in connection with a use thereof in MorphoSys (*see* Ex. 2012,

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Averbuj Figure 2, and that he did not recall any mention of the word “compiler” in Averbuj. *See* Ex. 2012, 54:8–55:5. Dr. Bagherzadeh further testifies that “99 percent of the designs or chips do not have a compiler on-board.” *Id.* at 54:21–22. Dr. Bagherzadeh testifies that Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and “the output [] reflect[s] what’s in the input, based on the selection.” *Id.* at 54:8–13. Dr. Bagherzadeh testifies that assuming “no other hardware components between . . . [CMD_DATA] and the input,” CMD_DATA is stored in the algorithm memory. *Id.* at 54:15–19.

Dr. Sechen testifies that

the Petition fails to identify every element of claims 1, 20, and 29 for at least two reasons[:] (1) because the CMD_REQ and SET ADDRESS are not generated (“produced” or “caused”) by the purported control module (Averbuj’s BIST controller 4), and [(2)] because the OP CODE is not generated (“produced” or “caused”) by the alleged control module (Averbuj’s BIST controller 4).

30:12–20), a chip fabricated by Morpho Technologies (*id.* at 19:2–9), a company co-founded by Dr. Bagherzadeh (Ex. 1009 ¶ 7). An article relating to MorphoSys and co-authored by Dr. Bagherzadeh (*see* Ex. 2014, “MorphoSys: Case Study of A Reconfigurable Computing System Targeting Multimedia Applications”) is introduced on page 39 of the deposition transcript. *See* Ex. 2012, 39:19–22, cited in PO Resp. 27. In any event, Petitioner’s argument is not persuasive because we find the testimony on pages 54–56 of the Bagherzadeh transcript supports Patent Owner’s argument. *See* Section IV.F. *infra*.

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Ex. 2019 ¶ 136. Dr. Sechen testifies, more specifically, that the Petition fails to explain how Averbuj's BIST controller 4 "generate[s]/produce[s] OP CODE," "because the Petition does not state that Averbuj's BIST Controller includes a compiler, as described by Dr. Bagherzadeh." *Id.* ¶ 138. Dr. Sechen testifies that "[b]ecause SET ADDRESS (OP CODE 100) is a predefined 3-bit sequence, . . . it is not 'generated' by Averbuj's BIST Controller. Rather, it is stored in Algorithm Memory 20 of Fig. 2." *Id.* ¶ 140. Dr. Sechen further testifies that "command 60, which carries the OP CODE . . . (e.g. . . . the SET ADDRESS opcode), . . . is not 'generated' by Averbuj's BIST Controller," but "is merely stored in memory (Algorithm Memory 20 in Fig. 2) and is output as part of 'a stream of binary commands.' (Ex. 1011 at 5:1–21.)" Ex. 2019 ¶ 141.

E. Analysis

After considering the parties' arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses "a control module configured to generate address . . . signals for testing the memory devices" as recited in independent claims 1, 20, and 29.

As noted in Section III, above, in its Petition, Petitioner proposes construing the claim term "generate" as "cause or initiate transmission" (Pet. 24). The Petition cites the following disclosure in the '434 patent in support of this construction: "the data module 28 and/or the control module 22 are **configured to provide** memory signals (e.g., data, address and control signals)" Ex. 1008, '434 patent, 6:3-5 (emphasis added)." Pet. 24. The Petition states that Averbuj teaches "a control module configured to generate address and control signals for testing the memory devices,"

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because Averbuj’s “BIST controller 4 is programmable *to provide* address and control signals.” Pet. 29 (emphasis added); *see* PO Resp. 53.

For purposes of our Institution Decision, we interpreted “generate” as used in the ’434 patent claims as meaning “cause” or “produce.” *See* Dec. on Inst. 9. As indicated in Section III.A., above, subsequent to that decision and the oral hearing in the present *inter partes* review, we again considered the meaning of the claim term “generate” in our final decision in IPR-970, and determined that the broadest reasonable interpretation of the ’434 patent claim term “generate” is “produce,” and that “generate” does not mean “cause” or “cause to produce.” *See* IPR-970 FWD, 24–33. In Section III.A., above, we considered extrinsic evidence in support of Petitioner’s interpretation of “generate” as meaning “cause” that was not before us in IPR-970, but concluded the broadest reasonable interpretation of “generate” is “produce.” We, therefore, adopted and incorporated by reference in the present final decision our analysis and construction of the terms “generate” and “configured to generate” in our Final Written Decision in IPR-970. *See* Section III., above.

In construing the claim term “generate” in our final decision in IPR-970, we found that the terms “provide” and “generate” are used in the ’434 patent to describe different embodiments of the control module, and determined the claim term “generate” did not mean “provide.” *See* IPR-970 FWD, 28–30. We also determined that the phrases “address and control signals generated by the control module” (claims 1, 20) and “generating, by the control module, address and control signals” (claim 29) encompass signals that originated in the control module, including by transformation or modification of information received from another component. IPR-970

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FWD, 32. We further determined the broadest reasonable interpretation of this claim language does not encompass “signals received by the control module from another component, that are merely provided, propagated, sent, or input to the memory devices, without transformation or modification by the control module.” *Id.* We stated that “generate,” as used in the ’434 patent, does not encompass the selection function of a multiplexer. *Id.* at 29–30.

Petitioner’s contention that Averbuj teaches “a control module configured to generate address . . . signals for testing the memory devices,” as recited in independent claims 1, 20, and 29, is based on an interpretation of “generate” as meaning “cause.” Pet. 24, 29; *see also*, Reply 20–21 (asserting that Averbuj’s BIST controller 4 is “configured to generate address and control signals” as claimed in the ’434 patent because “the BIST controller of Averbuj *causes* the CMD_REQ and SET ADDRESS signals, representing the claimed ‘address and controls signals’” (emphasis added)) (citing Dec. on Inst. 15–16). Petitioner has not explained sufficiently, however, how Averbuj’s BIST controller 4 is designed, adapted, or arranged to produce address signals for testing the memory devices.⁴

⁴ During oral argument, Mr. Heafey, counsel for Petitioner, was asked the following question: “Assuming we adopt ‘to produce’ as the proper construction [of ‘to generate’], are [CMD]_REQ and SET ADDRESS produced by BIST controller 4?” Tr. 78:16–18. In response, Mr. Heafey stated: “[T]hose signals are definitely produced by BIST controller 4, yes, they are.” *Id.* at 78:19–20. Mr. Heafey did not elaborate, however, nor did he otherwise explain clearly during oral argument how Averbuj’s BIST controller 4 is designed, adapted, or arranged to produce address signals for testing the memory devices.

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Moreover, the evidence in this proceeding fails to support a finding that Averbuj's BIST controller 4 is configured to generate address signals for testing the memory devices in the manner claimed.

In Averbuj, the algorithms (i.e., CMD_DATA containing OP CODE such as SET ADDRESS (*see* Ex. 1011, 4:4–12, 5:19–21, Table 1)) communicated to sequencers 8 are either received by user interface 22 via external input (*id.* at 5:41–43) or generated by another component (i.e., a component that is not part of BIST controller 4 circuitry) and stored in algorithm memory 20 (*see* Ex. 2012, 30:23–24, 41:1–4, 54:21–22, 55:6–25, 55:20–56:9 (wherein Dr. Bagherzadeh testifies that OP CODE residing in the algorithm memory of BIST controller 4 are generated by hand or by a tool, e.g., a compiler, and that “99 percent of the designs or chips do not have a compiler on-board”)). Multiplexer 24 *selects* one of the algorithms (i.e., CMD_DATA) and BIST controller 4 *provides* to sequencers 8 the same CMD_DATA (e.g., address signals) without transformation or modification. *See* Ex. 1011, 4:4–12, 5:16–21; Ex. 2012, 54:8–13 (“Averbuj Figure 2, without consideration of any other disclosure in Averbuj, illustrates that a multiplexer selects an input from the algorithm memory and the output [] reflect[s] what’s in the input, based on the selection.”). Based on our interpretation of the terms “configured to” and “generate,” as discussed above, the claim recitation “a control module configured to generate address . . . signals for testing the memory devices” (claims 1, 20, and 29) does not read on Averbuj's BIST controller 4, because the SET ADDRESS signal output by BIST controller 4 is produced by a component external to BIST controller 4, and merely selected by multiplexer 24 and provided to sequencers 8 without transformation or modification by BIST controller 4.

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In sum, after considering the parties' arguments and evidence, we determine Petitioner has not shown, by a preponderance of the evidence, that Averbuj discloses "a control module configured to generate address . . . signals for testing the memory devices" as recited in independent claims 1, 20, and 29. Petitioner does not correct this deficiency in its challenges as to dependent claims 2–4, 14–19, and 27. *See* Pet. 32–39; 35 U.S.C. § 112, ¶ 4 ("A claim in dependent form shall be construed to incorporate by reference all the limitations of the claim to which it refers."). Because Petitioner has not identified in Averbuj a teaching of each and every limitation recited in claims 1–4, 14–20, 27, and 29, Petitioner has not met its burden to show these claims are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj.

V. MOTIONS TO EXCLUDE

Both Patent Owner and Petitioner filed Motions to Exclude.

A. Petitioner's Motion to Exclude

Petitioner moves to exclude Exhibit 2018 (the Deposition Transcript of Dr. Donald Alpert) and to strike Section IV(B)(5)(a) of Patent Owner's Response, wherein Patent Owner quotes Dr. Alpert's deposition testimony in support of its proposed construction of the claim term "generate" (*see* PO Resp. 34). Paper 26, 2. Petitioner argues Dr. Alpert is Sandisk Corporation's expert in IPR-970 and that Petitioner was not provided notice or otherwise invited to attend Dr. Alpert's deposition. *Id.* at 3. Petitioner contends Dr. Alpert's statements are not admissible as evidence in this proceeding "because they are hearsay and will unfairly prejudice Petitioner." *Id.* at 2.

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In construing the claim term “generate” in Section III.A., above, we adopted and incorporated by reference our analysis and construction of this term on pages 24–33 of our Final Written Decision in IPR-970. In our claim construction in IPR-970, we considered Dr. Alpert’s deposition and declaration testimony, but did not quote or cite to the testimony relied on by Patent Owner in Section IV(B)(5)(a) of its Patent Owner Response in the present proceeding. *Compare* IPR-970 FWD, 25–26 *with* PO Resp. 34. In IPR-970, we determined Dr. Alpert’s testimony that one of ordinary skill in the art would understand the claim term “generate” as meaning “cause” or “produce” was inconsistent with the ’434 patent’s implicit definition of this term as meaning only “produce.” *See* IPR-970 FWD, 30 n.4. We, therefore, did not rely on Dr. Alpert’s testimony in our construction of the claim term “generate” in IPR-970, or in the present case. *See id.*; *see also id.* at 32 (“Having considered the ordinary meaning of the term ‘generate’ in the context of both the claims and the ’434 patent as a whole, we agree with Patent Owner that the broadest reasonable interpretation of the claim term ‘generate’ is ‘produce,’ and that ‘generate’ does not mean ‘cause’ or ‘cause to produce.’”).

Because we did not rely on Exhibit 2018 or consider the argument in Section IV(B)(5)(a) of Patent Owner’s Response in rendering our decision in this proceeding, we dismiss Petitioner’s Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) as moot.

B. Patent Owner’s Motions to Exclude

Patent Owner filed a Motion to Exclude “Exhibits 1026, 1029, 1030, and 1032 (or portions thereof)” and requests that we strike the corresponding citations to these exhibits in Petitioner’s Reply. Paper 34, 2. We did not

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rely on these exhibits in rendering our decision. Accordingly, Patent Owner's Motion to Exclude these exhibits is dismissed as moot.

Patent Owner also filed a Motion to Exclude Portions of Petitioner's Reply, contending "Petitioner's Reply improperly presents new theories, never raised in its Petition or adopted by the Board's Institution Order, violating 37 CFR 42.22-23 and applicable law." Paper 35, 1. Patent Owner provides a table identifying the alleged new theories advanced by Petitioner in its Reply (*id.* at 2) and identifies by page and line number specific portions of sections II.A.3, II.B.1.a, II.B.5 & III.B.2 of the Reply that it contends should be excluded as advancing new theories (*see id.* at 2–8).

Our decision is not based on any of the alleged new theories identified by Patent Owner, and Patent Owner does not cite to any of the specific language in the Reply that Patent Owner contends should be excluded. Accordingly, Petitioner's Motion to Exclude portions of Petitioner's Reply is dismissed as moot.

VI. CONCLUSION

Petitioner has not shown by a preponderance of the evidence that claims 1–4, 14–20, 27, and 29 of the '434 patent are unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2).

VII. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–4, 14–20, 27, and 29 of U.S. Patent No. 8,001,434 B1 are not held unpatentable under 35 U.S.C. § 102(b) as anticipated by Averbuj (U.S. Patent No. 7,392,442 B2);

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FURTHER ORDERED that Petitioner's Motion to Exclude Exhibit 2018 and to strike Section IV(B)(5)(a) of Patent Owner's Response is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exhibits 1026, 1029, 1030, and 1032 to strike the corresponding citations to these exhibits in Petitioner's Reply is *dismissed*;

FURTHER ORDERED that Patent Owner's Motion to Exclude Portions of Petitioner's Reply is *dismissed*; and

FURTHER ORDERED that, because this is a final decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
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In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.):

DOCKET NO. 8:16-cv-01605	DATE FILED 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,756,364	June 17, 2014	Netlist, Inc.
2 8,516,185	August 20, 2013	Netlist, Inc.
3 8,001,434	August 16, 2011	Netlist, Inc.
4 8,359,501	January 22, 2013	Netlist, Inc.
5 8,689,064	April 1, 2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading		
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK	
1			
2			
3			
4			
5			

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT

CLERK	(BY) DEPUTY CLERK	DATE
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Copy 1—Upon initiation of action, mail this copy to Director Copy 3—Upon termination of action, mail this copy to Director
Copy 2—Upon filing document adding patent(s), mail this copy to Director Copy 4—Case file copy

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and
SK HYNIX MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00561
Patent 8,001,434 B1

Before BRYAN F. MOORE, MATHEW CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

DECISION
Instituting *Inter Partes* Review
37 C.F.R. § 42.108

I. INTRODUCTION

SK hynix Inc., SK hynix America Inc., and SK hynix memory solutions Inc., (collectively “Petitioner”) filed a Petition requesting an *inter partes* review of claims 1–7 of U.S. Patent No. 8,001,434 B1 (Ex. 1011, “the ‘434 Patent”). Paper 1 (“Pet.”). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response. Paper 6 (“Prelim. Resp.”).

We have jurisdiction under 35 U.S.C. § 314, which provides that an *inter partes* review may be authorized only if “the information presented in the petition . . . and any [preliminary] response . . . shows that there is a reasonable likelihood that the petitioner would prevail with respect to at least 1 of the claims challenged in the petition.” 35 U.S.C. § 314(a). Pursuant to 35 U.S.C. § 314, we determine that Petitioner has demonstrated a reasonable likelihood that it would prevail with respect to claims 1–7 of the ’434 Patent and, thus, institute an *inter partes* review as to the challenged claims.

A. Related Proceedings

Petitioner recites a list of District Court proceedings related to this *inter partes* review. Pet. 5. This *inter partes* review challenges the same patent at issue in the decision entered in IPR2014-00970 (the 970 IPR). *See Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB Dec. 16, 2014) (Paper 12). In the 970 IPR, as to the claims at issue here, claim 1 was held unpatentable as anticipated by Averbuj in a final written decision. *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB Dec. 14, 2015) (Paper 32). That decision is on appeal to Federal Circuit. In the 970 IPR, claims 2–7 were also challenged as obvious over Averbuj and Tsern, but that challenge was not instituted. *Sandisk*, IPR2014-00970 (Paper 12).

The ’434 Patent was also challenged in IPR2014-01372 (the 1372 IPR). In the 1372 IPR, as to the claims at issue here, claims 1–4 were held not unpatentable as anticipated by Averbuj in a final written decision. *Smart Modular Techs. Inc. v. Netlist, Inc.*, Case IPR2014-01372 (PTAB March 9, 2016) (Paper 45). The decision in the 1372 IPR was not appealed to Federal Circuit.

The '434 Patent was also challenged in IPR2014-01373, in which institution was denied as to all challenges on the merits. *Smart Modular Techs. Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB Mar. 13, 2015) (Paper 16). Neither Averbuj nor Tsern was asserted in the 1373 IPR.

We have discretion to institute, under 35 U.S.C. § 325(d), when considering “the same or substantially the same prior art or arguments previously . . . presented to the Office” and under 35 U.S.C. § 314(a) (authorizing institution of an inter partes review under particular circumstances, but not requiring institution under any circumstances). Considering the particular circumstances of this case, we address the merits of the Petition and do not exercise our discretion under 35 U.S.C. § 325(d) and/or 35 U.S.C. § 314(a). We consider the following factors relevant to exercising our discretion not to institute:

- (1) whether the same petitioner previously filed a petition directed to the same claims of the same patent;
- (2) whether the petitioner reasonably could have raised, in its earlier petition, the challenge now asserted;
- (3) the elapsed time between the petitioner’s current challenge and earlier proceedings, and the degree to which petitioner’s challenge, or the timing thereof, has been unfairly informed by information gleaned, by word or deed, from an earlier proceeding; and
- (4) whether petitioner identifies any mitigating factors that justify another challenge to a patent.¹

¹ These factors are a more generalized statement of factors analyzed by other panels of the Board. *See, e.g., NVIDIA Corp. v. Samsung Elec. Co.*, Case IPR2016-00134 (PTAB May 4, 2016) (Paper 9); *Xactware Sols., Inc. v.*

Applying these factors, and weighing the equities between Petitioner and Patent Owner, we do not exercise our discretion to not institute review for the following reasons. As noted above, Petitioner is not a party to any of the prior proceedings, and we have yet to review a challenge in a final written decision on the merits as to claims 5–7. The first and second factors therefore weigh against a discretionary denial. As to the third factor, we acknowledge that the third Petition was filed almost three years after the first petition. However, as a mitigating factor, Petitioner in this challenge was not sued until some significant amount of time after the petitioner in the 970 IPR was sued. Thus, the fourth factor weighs against a discretionary denial. On balance, we determine that the particular facts in this case weigh in favor of considering the Petition on the merits.

B. The '434 Patent (Ex. 1001)

The '434 Patent relates to self-testing electronic memory modules. Ex. 1001, 1:23–24. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '434 Patent, reproduced below.

Eagle View Techs., Inc., Case IPR2017-00027 (PTAB Apr. 19, 2017) (Paper 9).

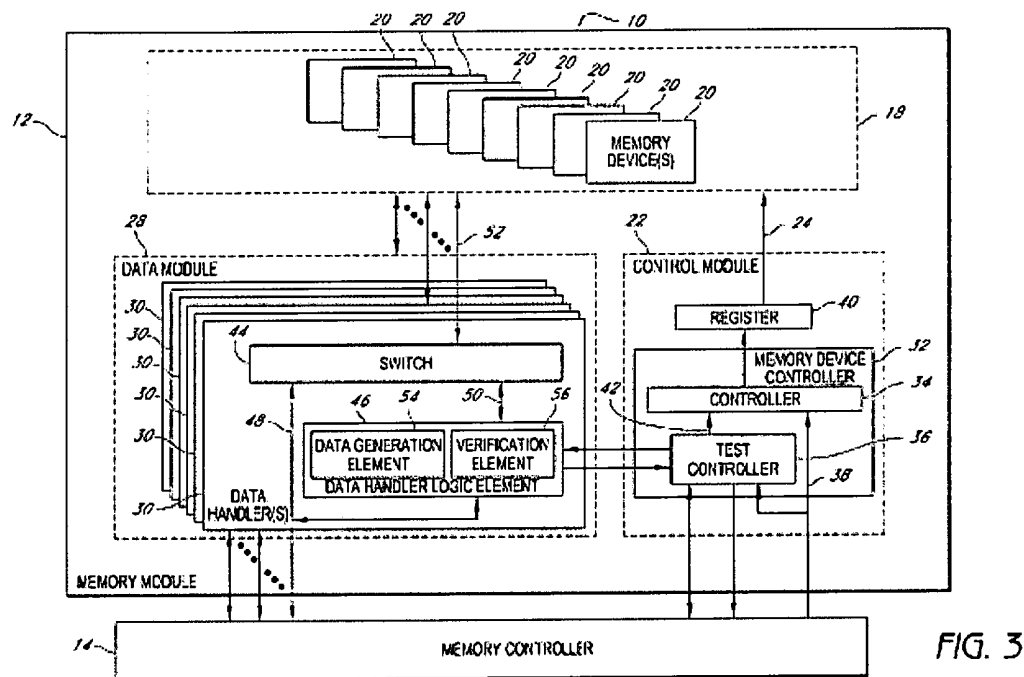


FIG. 3

As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 (PCB) configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:1–3. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:3–7. Memory module 10 further includes control module 22 and data module 28 comprising a plurality of independently operable data handlers 30. *Id.* at 5:7–11. Control module 22 includes memory device controller 34, which receives address and control signals 38 from system memory controller 14 and address and control signals 42 from test controller 36. *Id.* at 9:40–45. Similarly, data module 28 includes switch 44, which selectively inputs to memory devices 18 either data signals 48

from memory controller 14 or data signals 50 from data handler logic element 46. *Id.* at 10:7–11.

During test mode, memory device controller 34 sends address and control signals 38 from test controller 36 to register 40 for input to address and control ports of memory devices 18. *Id.* at 9:40–50. “[T]est controller 36 updates each of [] data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.)” *Id.* at 14:7–9. Data generation element 54 of data handler logic element 46 is configured to generate data signals and/or patterns of data signals based on the information received from test controller 36, for writing to memory devices 18. *Id.* at 10:28–37, 15:25–27.

C. Illustrative Claim

Of the challenged claims, claim 1 is independent. Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is configured to test the memory devices using the address

and control signals generated by the control module and the data generated by the plurality of data handlers.

D. Prior Art Relied Upon

Petitioner relies upon the following prior art references:

Reference	Patent/Printed Publication Number	Exhibit Number
Averbuj	U.S. Patent Application Publication No. 2005/0257109	Ex. 1005
Tsern	U.S. Patent Application Publication No. 2007/0070669	Ex. 1006

E. The Asserted Grounds

Petitioner asserts that the challenged claims are unpatentable based on the following grounds:

Reference[s]	Basis	Claims Challenged
Averbuj	§ 102	1–7
Averbuj	§ 103	1–7
Averbuj and Tsern	§ 103	1–7

II. ANALYSIS

A. Claim Construction

In an *inter partes* review, claim terms in an unexpired patent are given their broadest reasonable construction in light of the specification of the patent in which they appear. *See* 37 C.F.R. § 42.100(b). Under the broadest reasonable construction standard, claim terms are given their ordinary and customary meaning, as would be understood by one of ordinary skill in the art in the context of the entire disclosure. *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007). Any special definition for a claim term

must be set forth with reasonable clarity, deliberateness, and precision. *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994).

1. “*self-testing memory module*,” “*generate*,” “*operable independently*,” and “*configured to*”

Petitioner and Patent Owner request construction of several claim terms to be generally consistent with how those terms were previously construed in the 970 IPR. IPR2014-00970, Paper 12, 33; *see also id.* at 10–33 (analyzing those claims); Pet. 11–15; Prelim. Resp. 21–27. Patent Owner agrees with Petitioner’s proposed constructions, except for the term “self-testing memory module,” but also asserts that our analysis regarding the construction of the term “generate” should be considered to be a part of the construction. Prelim. Resp. 21–27. We have considered the claim term constructions and analysis, and discern no reason to alter the constructions used in the 970 IPR. The following chart indicates constructions we adopt for the purpose of this decision.

Claim Term	Construction
<i>Self-testing memory module</i>	A memory module that can be tested, at a minimum with only internal, or both internal and external test equipment
<i>Generate</i>	Produce
<i>Operable independently</i>	Operable without influence or control by another
<i>Configured to</i>	Designed to, adapted to, or arranged to [e.g., perform a function or be capable of performing a function]

2. “*proximate to*”

Petitioner proposes a construction for the term “proximate to.” Pet. 13–15. Patent Owner does not make an argument over the prior art based on that construction. On this record, we determine that the term does not require an explicit construction. *See Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)) (“[O]nly those terms need be construed that are in controversy, and only to the extent necessary to resolve the controversy.”).

3. “*memory controller*”

Patent Owner proposes a construction for the term “memory controller.” Prelim. Resp. 21, 26. However, this construction is not implicated in Patent Owner’s argument over the prior art. On this record, we determine that the term does not require an explicit construction.

4. “*memory module*”

Patent Owner argues that “memory module” should be construed as we construed the term in IPR2014-00882 (the 882 IPR), i.e., to mean “a carrier that contains one or more memory chips.” Prelim. Resp. 26; *see Diablo Tech. Inc. v. Netlist, Inc.*, Case IPR2014-00882 (PTAB Dec. 14, 2015) (Paper 33). The 882 IPR was filed by a different petitioner and involved a different patent (U.S. Patent No. 7,881,150, “the ’150 patent”) owned by Patent Owner but not related to the ’434 Patent.

Patent Owner argues that the Board’s construction from the 882 IPR should be read to mean “a removable circuit board, cartridge, or other carrier that contains one or more RAM memory chips.” Prelim Resp. 22, 26 (citing Microsoft Computer Dictionary, 5th ed. (Ex. 2004)). We disagree.

Our construction in the 882 IPR did not require that the carrier be “removable.” In the 882 IPR, Patent Owner argued the Board’s construction of “memory module” is inconsistent with the ’150 patent disclosure. 882 IPR, Paper 25, 11. Patent Owner asserted that claims 15, 22, and 31 recite “the memory module configured to be electrically coupled to a memory controller of a computer system so as to receive a set of input signals.” *Id.* According to Patent Owner, “due to the use of different terms in the ’150 patent, “memory module” would not be read as including the ’150 patent’s “memory controller” by a [person of ordinary skill in the art]” *Id.* Patent Owner also cited to its declarant and to the Specification of the ’150 patent. *See generally, id.* at 5–11. In this case, however, the ’434 Patent does not contain the claim language cited by Patent Owner in the 882 IPR and the Specification is different. Further, Patent Owner’s declarant in the ’882 IPR has not submitted a declaration in this case.

In the 882 IPR, Patent Owner argued that “memory module” should be construed to mean “a packaging arrangement of one or more memory device(s) for use in a computer socket.” *Id.* at 10. When rejecting Patent Owner’s construction, we stated “we decline to adopt Patent Owner’s claim construction as it would import limitations improperly from the specification into the claims and unnecessarily limit the scope of the claims.” 882 IPR, Paper 33, 10. Thus, we rejected Patent Owner’s previous attempt to limit “memory module” to a device for use in a computer socket, which implies incorporation of the concept of removability. In the present case, Patent Owner’s attempt to extrapolate from the term “carrier” a requirement that a “memory module” be “removable” is not persuasive for the same reasons.

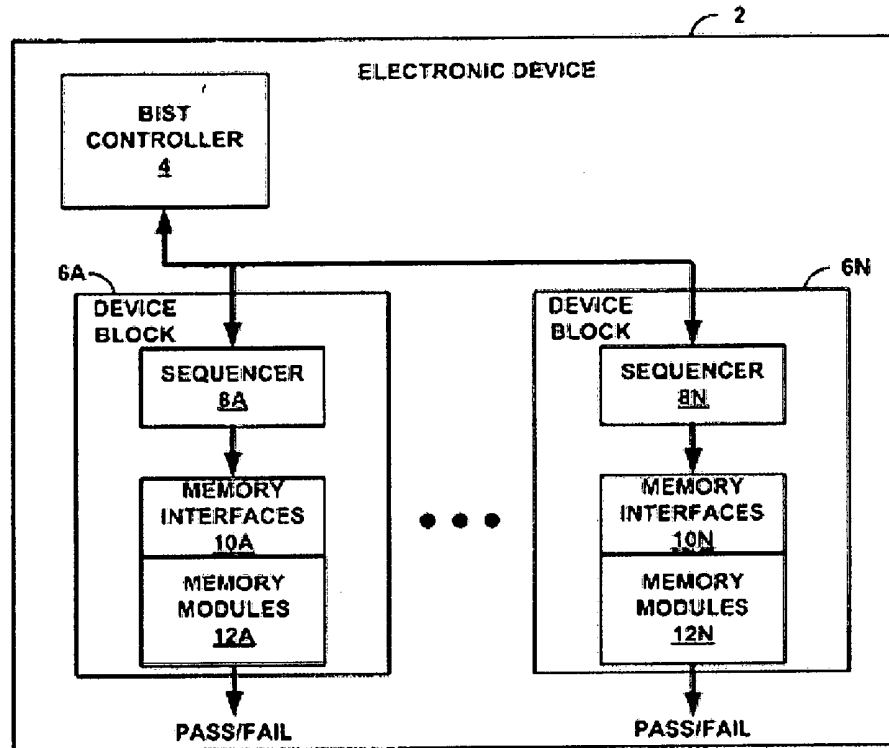
Finally, the issue of removability was not argued by the parties in the 882 IPR to distinguish a prior art reference from the “memory module” recited in the claims. *See generally* IPR 882, Paper 33. In the end, Patent Owner has not shown that the ’434 Patent limits “memory module” to a removable component as its analysis suggests. We construe “memory module” as “a carrier that contains one or more memory chips” but do not require that the carrier be removable.

B. Claims 1–7 — Anticipation by Averbuj (Ex. 1005)

Petitioner argues that claims 1–7 are unpatentable over Averbuj under 35 U.S.C. § 102. Pet. 24–48.

1. Averbuj (Ex. 1005)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1005 ¶¶ 7–8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* ¶ 28. “[E]lectronic device 2 may be any device that incorporates memory modules, such as an embedded computing system, a computer, server, personal digital assistant (PDA), mobile computing device, mobile communication device, digital recording device, network appliance, mobile positioning device, and the like.” *Id.* ¶ 32. Averbuj states that “electronic devices . . . are [normally] constructed from many integrated circuit chips and many supporting components mounted on a circuit board.” *Id.* ¶ 5. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks 6A through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device

blocks 6 includes a sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to sequencers 8 for application to device blocks 6. *Id.* ¶ 29.

Figure 5, below, illustrates an exemplary sequencer 8A. *Id.* ¶ 22

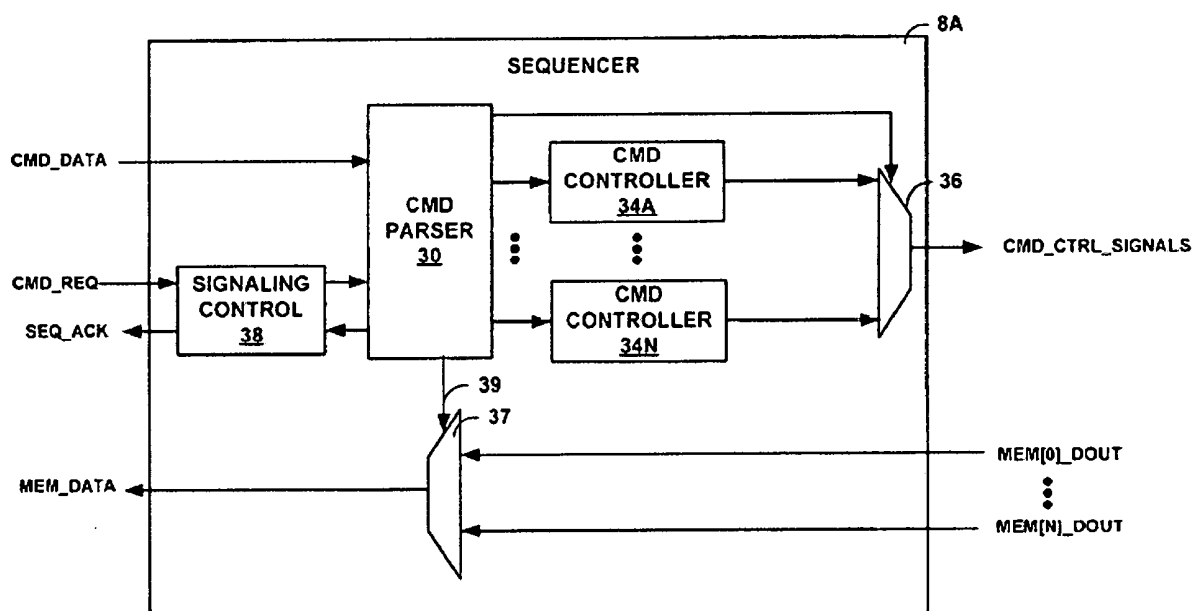


FIG. 5

As shown in Figure 5, above, exemplary sequencer 8A includes command parser 30 and command controllers 34. *Id.* ¶¶ 41–42. Command parser 30 processes the commands from BIST controller 4 and invokes one of command controllers 34, which, in turn, issues command control signals, such as signals to provide a memory address and data, to receiving memory interfaces 10; 41. *Id.* ¶¶ 43, 51. An exemplary memory interface is shown in Figure 6, below. *Id.* ¶ 23.

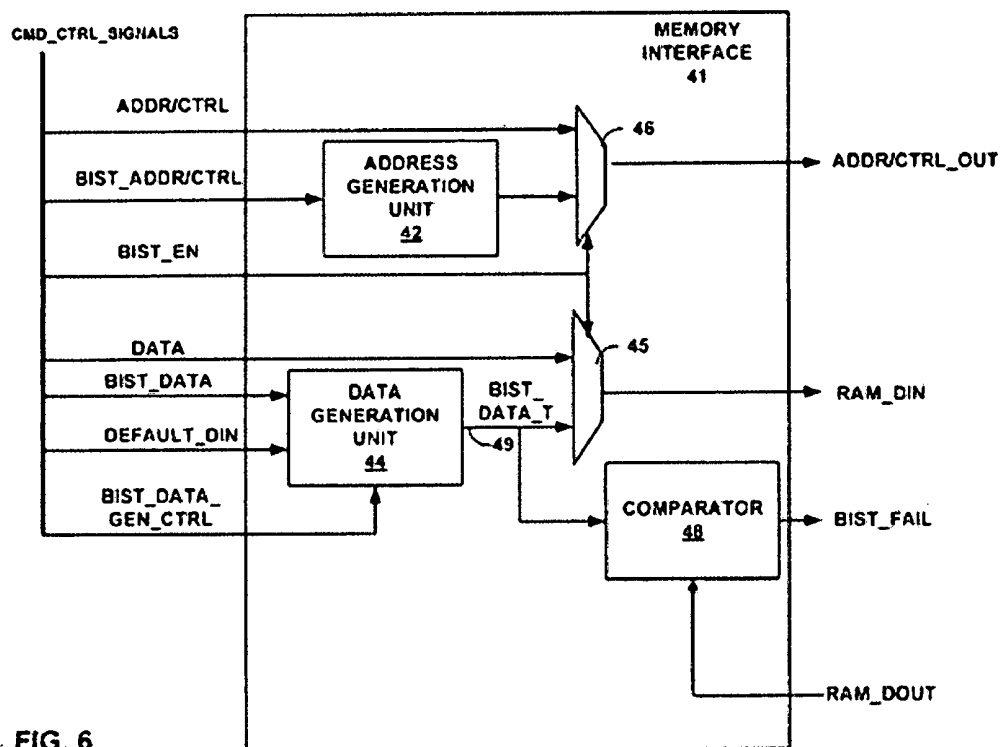


FIG. 6

As shown in the Figure 6 embodiment, above, memory interface 41 includes multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals (BIST_ADDR/CTRL) and the test data provided by a respective higher level sequencer[, e.g., sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level sequencer [8A], and transforms the data and

addresses provided by the sequencer as needed based on the particular physical characteristics of the memory module” (*id.* ¶ 54), using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “[A]ddress generation unit 42 generates the address applied to [] memory module 12 based on the addressing requirements specified by sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* ¶ 51. “[D]ata generation unit 44 [] receives BIST _DATA signals and default data (DEFAULT_DIN) as provided by [] sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the sequencer and the specific physical characteristics of the corresponding memory module.” *Id.* ¶ 49. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.*

2. Analysis

a. memory module

Petitioner contends Averbuj’s device blocks 6A–6N and electronic device 2 are “individually and collectively a ‘*self-testing memory module*,’” as recited in the preamble of claim 1. Pet. 24–25. Petitioner relies on the statement in Averbuj’s paragraph 5 that BIST units are normally constructed from integrated circuit chips mounted on a circuit board, to establish that device blocks 6 comprise a printed circuit board. *Id.* at 25–26. Petitioner further contends device blocks 6 are configured to be operatively coupled to a memory controller of a computer system, because Averbuj states that under normal operating conditions, address/control and data signals are provided by a programmable processor. *Id.* at 27 (citing Ex. 1005 ¶ 48). Petitioner contends Averbuj’s sequencer 8A, memory interfaces 10, 41, and

memory modules 12 correspond, respectively, to the claimed “control module,” “data handlers,” and “memory devices.” *See id.* at 28–32, 34–38.

As to this and all other limitations, Petitioner relies on the same aspects of Averbuj relied on in the petition and accepted by the Board in the final written decision of the IPR 970. IPR2014-970, Paper 1 (Petition), 7–18; Paper 32 (Final Written Decision), 35–44. In this decision, we have considered the fact that these aspects of Averbuj were determined to anticipate claim 1 in the previous final written decision by the Board. Petitioner also relies on several alternative bases of anticipation by Averbuj for various limitations for claim 1. Pet. 26–28, 32–34. At this juncture, we do not rely on those alternative bases in this Decision, but we do not exclude them from the trial.

Patent Owner contends Averbuj’s memory modules 12 are not the claimed “memory module” as recited in independent claim 1 because the memory modules 12 are “embedded” in an integrated circuit (“IC”). Prelim. Resp. 37. Patent Owner’s argument is not convincing in light of our interpretation (for purposes of this Decision) of “memory module” as “a carrier that contains one or more memory chips” but is not limited to a removable carrier as opposed to memory chips on a carrier that may include other elements. *See* Section II.A.4. above. Under this construction, Averbuj’s device blocks 6A–6N and electronic device 2, in which memory modules 12 are embedded, may be considered the “memory module” recited in claim 1.

b. printed circuit board

Petitioner asserts Averbuj discloses that electronic devices that are the subject of his invention may be constructed from many integrated circuit

chips and components mounted on a circuit board. Pet. 25–26 (citing Ex. 1005 ¶ 5; Ex. 1003 ¶ 94). Patent Owner argues Averbuj does not describe “a printed circuit board configured to be operatively coupled to a memory controller of a computer system.” Prelim. Resp. 38–41 (quoting Ex. 1001, claim 1). Patent Owner argues that because Averbuj’s memory module is embedded on an IC and is of a very small size, it is incompatible with being operatively coupled to a PCB and the BIST controller cannot be operably coupled to a PCB. Patent Owner’s argument is not convincing because Averbuj’s paragraph 32, relied on by Petitioner for disclosing this limitation (*see* Pet. 26), indicates electronic device 2 can be any of a number of conventional electronic devices. Additionally, Averbuj’s paragraph 5 describes electronic devices, in general, as constructed from integrated circuit chips and supporting components mounted on a circuit board. *See* Ex. 1003 ¶¶ 5, 32. Further, Averbuj’s paragraph 48 explicitly indicates multiplexers 45, 46 of memory interface 41, which are part of electronic device 2, are capable of receiving signals from a programmable processor. *See* Pet. 9; Ex. 1003 ¶ 48. Patent Owner’s “size” argument is based on its construction that requires that the memory module be on a removable PCB separate from other components, but we have rejected that construction for the reasons discussed above. Finally, claim 1’s use of the terms “comprising,” which is open ended, and “operably coupled,” which does not require a direct physical connection, belie Patent Owner’s suggested limitations on the claim.

c. plurality of memory devices

Petitioner asserts “Averbuj discloses that the electronic device (2) incorporates a number of memory modules (12), which may be any type of

memory, Ex. 1005 ¶ [0032], and are hence ‘a plurality of memory devices.’” Pet. 28. Patent Owner argues Averbuj does not describe “a plurality of memory devices on the printed circuit board,” as recited in claim 1. Prelim. Resp. 41–42 (quoting Ex. 1001, claim 1). Patent Owner again argues that due to the size of the memory modules of Averbuj, they cannot be “mounted” on a PCB. Claim 1 does not require that the memory devices be “mounted” on a PCB rather that they are simply “on” a PCB. On this record, we are persuaded that Averbuj’s memory modules 12 (i.e., the “plurality of memory devices”) are “on” the circuit board disclosed by Averbuj (i.e., “the printed circuit board”) based on the broadest reasonable construction of “on.”

d. control port

As to the claimed “control port,” Petitioner asserts “the memory interface exchanges address, data, and control signals with its corresponding memory module (12) through its ADDR/CTRL_OUT, RAM_DIN, and RAM_DOUT ports. Ex. 1005 at FIG. 6, ¶¶ [0049], [0053]; Ex. 1003 ¶¶ 112–113.” Pet. 29. Patent Owner argues Averbuj fails to describe a “control port” as recited in claim 1. *See* Prelim. Resp. 43–52. Patent Owner argues one of ordinary skill in the art would consider the ADDR/CTRL_OUT as a single signal. *Id.* at 45–46. As supporting evidence, Patent Owner relies on the Averbuj provisional application (Ex. 2001) as allegedly showing only one signal. *Id.* Patent Owner does not explain why the claimed address and control signal must be described as two signals beyond the name ADDR/CTRL_OUT. Contrary to Patent Owner’s argument, the ’434 Patent describes, similar to Averbuj, single signal line 24 or single line 38 respectively as “address and control signals” and does not provide disclosure

that separates that label in a meaningful way into two distinct “address” and “control” signals. Ex. 1001, 3:33–35, 4:41–44, 5:7–9, 37–42, 59–62, 6:3–6, 7:35–38, 9:25–56, 12:57–60, 13:17–20, 14:1–6, 15:23–25. Additionally, at this stage of the proceeding, we decline to impose a limited view of the disclosure of Averbuj based on an embodiment of the Averbuj provisional.

e. control module

Petitioner asserts “the sequencers (8) disclosed in Averbuj constitute ‘a control module’ as claimed” Pet. 30. Patent Owner asserts that Averbuj’s sequencer cannot be the claimed “control module configured to generate address and control signals for testing the memory devices” because it “does not input address and control signals to the alleged ‘memory devices.’” PO Resp. 47. Specifically, Patent Owner asserts that Petitioner incorrectly relies on a statement from the 970 IPR final written decision for the proposition that

in one configuration, the test address/control signals (BIST_ADDR/CTRL) are provided to the memory modules (12) “without modification or transformation, directly to a memory module, irrespective of its physical configuration.” IPR2014-00970, Paper 32 at 42 (citing Ex. 1005 at ¶ [0056]); Ex. 1003 ¶ 123.

Pet. 32. Patent Owner asserts the quoted statement is incorrect because “[p]aragraph 56 of Averbuj is only directed to the data generation—not address and control signals.” Prelim. Resp. 49. Therefore, according to Patent Owner, Petitioner has not shown that the alleged “control module” of Averbuj “generates” the address and control signals, as opposed to data signals, used to test the memory devices. *Id.* at 47–50.

To provide context, in the 970 IPR’s institution decision the Board construed the term “a control module configured to generate address and

control signals for testing the memory devices” as requiring the control module to “produce” or “cause [to be produced]” address and control signals. IPR2014-00970, Paper 12, 10. However, in the 970 IPR’s final written decision, the Board construed that term as limited to a control module that produces address and control signals. IPR2014-00970, Paper 32, 19–20, 32–33. Thus, the issue in the 970 IPR was whether Averbuj sent address and control signals that were produced by the sequencer and delivered unchanged to the memory devices rather than a signal that was transformed in some way after it leaves the sequencer but before it is used in testing the memory devices.

For further context, we describe in detail below Petitioner’s contention as to “control module.” Petitioner contends that Averbuj discloses a control module (sequencer 8) configured to generate address and control signals (CMD_CTRL_SIGNALS) used for testing memory devices (memory modules 12). Pet. 30–32. Averbuj discloses that “BIST controller 4 provides and communicates test algorithms to sequencers 8 for application to device blocks 6” (Ex. 1005 ¶ 29), and “[s]equencers 8 interpret” and “control the application of the test algorithms” to the memory modules (*id.* ¶ 30). *See also id.* ¶ 46 (“Sequencers 8 receive the generic BIST commands in accordance with the command protocol, and control the application of the commands by generating and issuing sequences of one or more memory operations for application to a set of respective memory modules 12.”).

For example, in the exemplary embodiment illustrated in Figure 5, command parser 30 of sequencer 8A processes a command received from BIST controller 4, and extracts one or more parameters from the command. *Id.* ¶¶ 41–42. Command parser 30 passes the extracted parameters to

command controller 34, which then issues a sequence of one or more operations to each memory interface 10. *Id.* ¶¶ 42–43. “Memory interfaces 10 handle specific interface requirements for each of memory modules 12. For example, each of memory interfaces 10 may be designed in accordance with the particular signal interface requirements and physical characteristics of the respective one of memory modules 12.” *Id.* ¶ 31.

In the 970 IPR, the Board relied on disclosure other than paragraph 56 to show the control module limitation was met. The 970 IPR’s final written decision states:

Averbuj further explains, in connection with the embodiment illustrated in Figure 6, that memory interface 41 “*transforms* the data and addresses provided by the sequencer *as needed* based on the particular physical characteristics of the memory module” (*id.* ¶ 54 (emphasis added)) using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51).

970 IPR Paper 32, 41. In other words, paragraph 54 of Averbuj suggests that data *and addresses* may be passed through the memory interface without being transformed or being transformed based on the necessity for transformation. The final written decision cited a specific example in which a data signal, BIST_DATA_T, is sent to the memory module explicitly “without modification.” *Id.* (citing IPR 970 Ex 1003 (Averbuj) ¶ 56). Even if Patent Owner is correct that the description in paragraph 56 of Averbuj is limited to data signals sent by the data generation unit, the statement in paragraph 54 of Averbuj sufficiently shows that address signals can also be sent to the memory module without being transformed by memory interface 41. Thus, we are not persuaded by Patent Owner’s argument on this issue.

Petitioner also presents alternative theories on how Averbuj meets the “control module” limitation of claims 1. Pet. 32–34. Patent Owner responds

to those theories. Prelim. Resp. 50–52. Although we do not limit the trial to Petitioner’s sequencer theory, we need not discuss Petitioner’s other theories for the purpose of institution because the sequencer theory is sufficient for institution.

For the reasons above, Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 is anticipated by Averbuj.

g. Claim 2

Claim 2 recites “[t]he self-testing memory module of claim 1, wherein the plurality of data handlers comprise at least two physically separate components mounted on the printed circuit board.” Petitioner asserts that

Averbuj further discloses that electronic devices, such as his electronic device (2), are constructed from many integrated circuit chips and other components mounted on a circuit board. Ex. 1005 at ¶ [0005] (“[The electronic devices] are constructed from many integrated circuit chips and many supporting components mounted on a circuit board.”); Ex. 1003 ¶ 154. Averbuj discloses that BIST units are commonly incorporated into chips or integrated circuits, which are synonyms. Ex. 1005 at ¶ [0003]; Ex. 1012 at 3, 5. Because the memory interface (12) is part of a BIST unit, it comprises a chip or an integrated circuit of an electronic device “mounted on a circuit board.” Ex. 1005 at ¶ [0005]; Ex. 1003 ¶ 154.

Pet. 41. In other words, Petitioner asserts that because a BIST unit can be incorporated into a chip mounted on a PCB and because memory module 12 is part of the BIST unit, memory unit 12 is mounted on a PCB. We disagree.

Unlike claim 1, which simply requires the memory unit is “on” the PCB, claim 2 requires that the memory module be a “physically separate component” and “mounted” on the PCB. Petitioner has not shown sufficient support for its broad reading that a module which is a circuit embedded in a

larger circuit included in a chip may be considered a physically separate component mounted on a PCB. Claims 3–7 depend from claim 2, and the Petition fails to provide additional prior art disclosures that cure the deficiency with respect to claim 2; that is, disclosure of separate components mounted on a printed circuit board in Averbuj. Pet. 41–48. Thus, Petitioner has failed to establish a reasonable likelihood of prevailing in its assertion that claims 2–7 are anticipated by Averbuj.

C. Claims 1–7 — Obviousness over Averbuj (Ex. 1005)

Petitioner argues that claims 1–7 are unpatentable over Averbuj under 35 U.S.C. § 103. Pet. 48–60. As to claim 1, Petitioner includes an obviousness ground to the extent we determine that Averbuj does not disclose the recited memory controller and “a plurality of memory devices on the printed circuit board.” *See id.* at 48, 50. We found above that both of these claim limitations were anticipated by Averbuj. Inasmuch as “anticipation is the epitome of obviousness” (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)), we determine that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj alone.²

As to claims 2–7, we are not persuaded that Petitioner has shown sufficiently that Averbuj anticipates those claims. Therefore, we discuss the obviousness challenges to dependent claim 2–7 below.

Petitioner’s argument regarding obviousness of claims 2–7 over Averbuj alone is based on Petitioner’s declarant’s testimony that it was

² There is no evidence of objective indicia of non-obviousness yet of record.

known to house data handlers in physically separate integrated circuit packages. Pet. 51–60 (citing Ex. 1003 ¶ 192–233)³. Petitioner argues that to do so would “have been the use of a known technique for its known purpose and would have achieved only predictable results, such as the protection of integrated circuits by the packaging.” Pet. 52. Petitioner asserts that memory modules at the time of the invention were composed of memory chips mounted on different portions of a memory board with BIST units in each memory chip. *Id.*

Given that memory chips mounted on different portions of a memory board with BIST units in each memory chip were known, according to Petitioner, Petitioner asserts that reducing redundancy is a motivation to modify Averbuj. Specifically, Petitioner asserts

[a] skilled artisan that seeks to reduce the redundant BIST circuitry of conventional memory chips and BIST units would have been motivated to modify the conventional design in such a way as to avoid unnecessary modifications to the conventional design. Ex. 1003 ¶ 200. Specifically, a skilled artisan would have understood that the BIST controller and sequencer portions of conventional BIST units would need to be carved out from each memory chip and combined to reduce redundant circuitry. Ex. 1003 ¶ 200. However, a skilled artisan would also have understood and recognized that the memory interface portions

³ Petitioner’s arguments regarding obviousness of claims 2–7 over Averbuj alone are based, in part, on disclosures in Tsern. Pet. 51–52 (citing Ex. 1006 ¶¶ 29, 52–53, Fig. 9A). Petitioner’s arguments based on the disclosure of Tsern rely on structures in Tsern to modify Averbuj rather than only Tsern’s general teachings to one of ordinary skill in the art, and, thus, those arguments are properly considered below in Petitioner’s ground based upon the combination of Averbuj and Tsern. As such we do not consider them here.

need not be carved out and/or combined since doing so would not reduce much, if any, circuitry. Ex. 1003 ¶ 200. Because a skilled artisan would have been motivated to make modifications in an efficient manner, he or she would have been motivated to keep the original design with respect to the memory interface portion. In other words, a skilled artisan would have been motivated to keep the memory interfaces (10, 41) (each of which includes a “*data handler*”) within each memory chip that was separately packaged and mounted on a printed circuit board of a memory module (“*physically separate integrated circuit packages*.”). Ex. 1003 ¶ 201.

Pet. 53–54. We disagree.

Patent Owner argues that “Averbuj, however, teaches away from the proposed combination because (as discussed above) Averbuj’s BIST unit and memory [modules] are in a single chip.” Prelim. Resp. 59. Patent Owner argues further that “modifying Averbuj in the proposed manner would render Averbuj inoperable for its intended purpose of verifying the functionality of internal circuitry of a chip while being included in the chip itself.” *Id.* at 59–60 (citing Ex. 1005 ¶ 5). Patent Owner’s arguments assume that Averbuj would not be modified based on the single chip structure of Averbuj. We do not rely on Patent Owner’s arguments. We focus, however, on the motivation given by Petitioner to modify Averbuj.

As to Petitioner’s arguments supported by its declarant, we find those to employ hindsight to piece together the device as claimed. The only evidence in the record that supports this argument, other than the long, complicated explanation of Petitioner’s declarant or the misplaced reliance on the structures of Tsern under this challenge, is in the ’434 Patent itself. “The inventor’s own path itself never leads to a conclusion of obviousness; that is hindsight. What matters is the path that the person of ordinary skill in

the art would have followed, as evidenced by the pertinent prior art.”

Otsuka Pharm. Co. v. Sandoz, Inc., 678 F.3d 1280, 1296 (Fed. Cir. 2012).

Thus, absent impermissible hindsight reconstruction, we are not persuaded that Petitioner has provided a sufficient rationale for modifying Averbuj as proposed in the Petition. Moreover, Petitioner’s declarant’s testimony lacks sufficient articulated reasoning with rational underpinning as to why an ordinary artisan, at the time of the invention, would have had a reason to modify the prior art in the manner required by the challenged claims. *See* Ex. 1003 ¶¶ 192–209.

Claims 3–7 depend from claim 2, and Petitioner’s arguments with respect to those claims fail to cure the deficiencies of the Petition with respect to claim 2; that is, Petitioner has not shown sufficiently that Averbuj teaches or suggests the physically separate components mounted on a printed circuit board limitation for the reasons discussed above. Thus, Petitioner has failed to establish a reasonable likelihood of prevailing in its assertion that claims 2–7 would have been obvious over Averbuj.

D. Claims 1–7 — Obviousness over Averbuj (Ex. 1005) and Tsern (Ex. 1006)

Petitioner argues that claims 1–7 are unpatentable over Averbuj and Tsern under 35 U.S.C. § 103. Pet. 60–75.

1. Tsern

Figure 1 of Tsern is reproduced below.

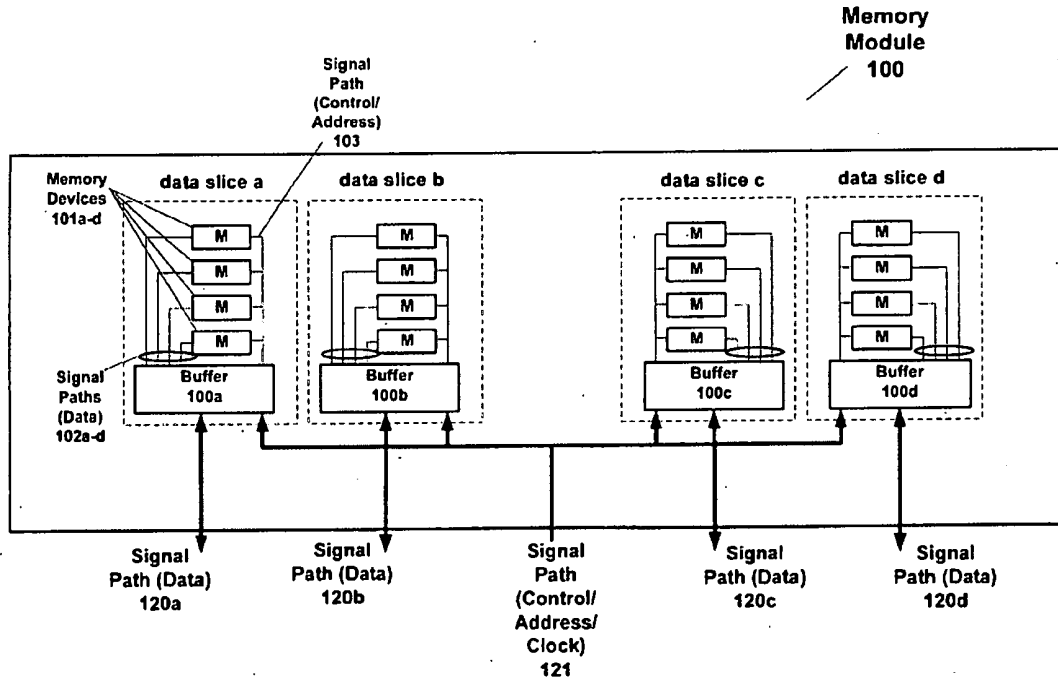


Fig. 1

Figure 1 of Tsern, above, illustrates an embodiment of a memory module with memory devices (101a-d) and corresponding buffer devices (100a-d). Figure 18 of Tsern is reproduced below.

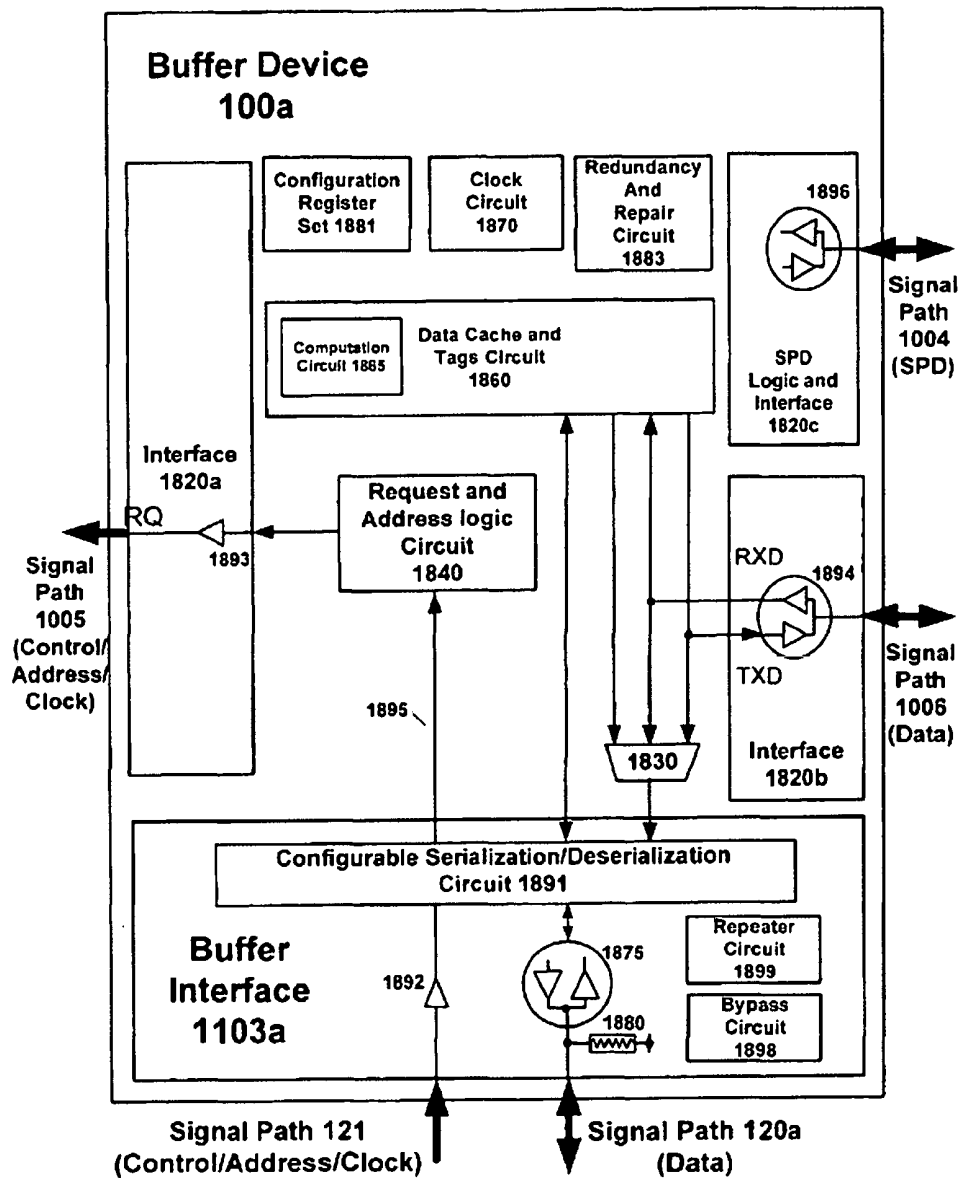


Fig. 18

As shown in Figure 18, above, Tsern discloses that, in one embodiment, each buffer device (100a) includes a redundancy and repair unit (1883), which tests and repairs the corresponding memory device(s). Ex. 1006, Fig. 18, ¶ 77. Each such buffer device (100a) includes data, address, and control interfaces (1820a and 1820b) that can be programmed

or configured to support memory modules with different number, size, width, and type of memory devices. *Id.* at Fig. 18, ¶ 99.

Figure 1 of Tsern shows that each buffer device (100a–d) communicates address, control, and data signals with its corresponding memory device(s) (101a–d) without being in communication with any of the other buffer devices (100a–d) or other data ports of the other memory devices. Ex. 1006, Fig. 1.

2. *Analysis*

As to claim 1, Petitioner includes an obviousness ground with Tsern to the extent we determine that Averbuj does not disclose, or teach or suggest, “a plurality of memory devices on the printed circuit board.” Pet. 61 (“To the extent one might argue that Averbuj does not disclose “*a printed circuit board*” as required by claim 1, it would have been obvious to include it in the system of Averbuj per the teachings of Tsern”). As discussed above, we determine that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj alone. We also determine that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj in combination with Tsern.

As to claims 2–7, we did not find that those claims are anticipated by, or would have been obvious over, Averbuj. Thus, below we discuss the combination of Averbuj and Tsern as it relates to claims 2–7.

Claim 2 recites the plurality of data handlers comprise at least two physically separate components mounted on the printed circuit board. Petitioner relies on Figure 9 of Tsern, reproduced below.

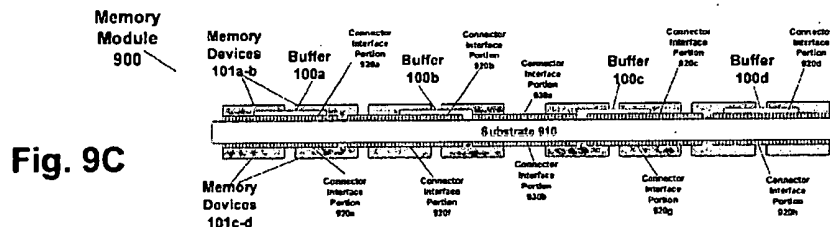
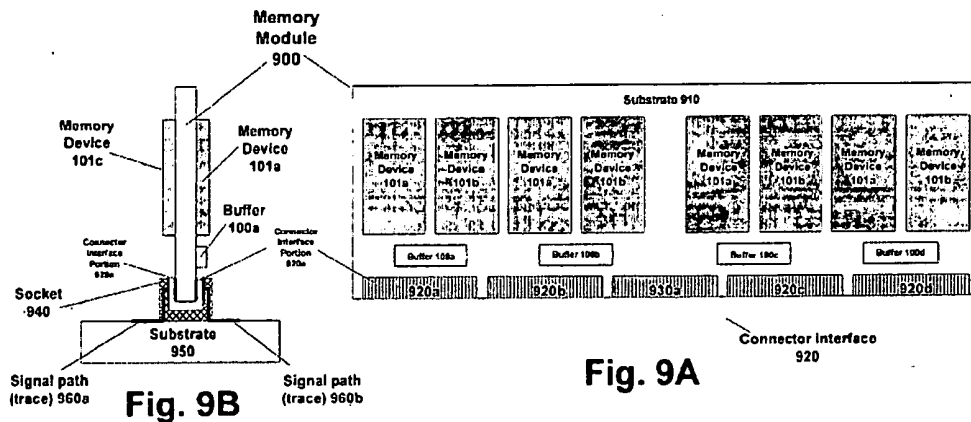


Figure 9 of Tsern, above, shows an implementation of a memory module in which each of the four buffer devices (100a–d) is implemented as a physically separate integrated circuit package and mounted on different portions of the printed circuit board that houses the memory module. Ex. 1006, Figs. 9A–C.

For purposes of this decision and based on Petitioner’s reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it also would have been obvious to modify the teachings of Averbuj with Tsern’s teachings, to include physically separate memory components mounted on a PCB. Pet. 60–62. For example, Petitioner asserts:

Tsern discloses a memory module including a number of memory devices such as DRAM or SRAM devices (Ex. 1006 at ¶ [0035]), and a number of buffer devices, in several embodiments. *See, e.g.*, Ex. 1006 at ¶¶ [0033-0042, 0052-0059]; FIGS. 1 & 9. Tsern further discloses that his memory module may consist of a number of buffer circuits and associated memory devices mounted on a printed circuit board (“PCB”). Ex. 1006 at ¶ [0052], FIG. 9A. Tsern also discloses that his buffer circuits implement the reading and writing of data from and to the memory devices, and also may include self-test functionality. Ex. 1006 at ¶ [0097]. The use of the printed board of Tsern in the system of Averbuj, in order to mount the memory devices and data generation circuitry for example, would therefore have been merely the use of a known structure for its known purpose to achieve the predictable result of mounting such electronic components in a sturdy and efficient substrate. Ex. 1003 ¶ 236.

Id.

Patent Owner asserts that Petitioner’s combination would be inoperative. Prelim Resp. 63–65. We find this argument is conclusory and relies on an expectation that Averbuj and Tsern must be bodily incorporated. Patent Owner also argues in conclusory fashion that the motivation to combine is “illogical and unsupported,” that there would be no benefit from the combination, that Averbuj and Tsern are from “different fields of art,” that Averbuj teaches away from the combinations, and that modifying Averbuj would change its principle of operation. Prelim. Resp. 65–67. We are not persuaded by those conclusory arguments.

Patent Owner asserts “[e]ven if these elements were capable of being physically combined into Tsern’s buffer, however, the combination would not have worked for its intended purpose. Averbuj’s sequencers require generalized command data, and the Petition fails to show how this could be compatible with the standards-based interface protocol in Tsern.” Prelim.

Resp. 68 (emphasis original). Patent Owner relies only on attorney argument and thus we are not persuaded.

We also are persuaded by Petitioner's contentions as to the limitations of claims 3–7. Patent Owner argues essentially that Petitioner has not shown that Tsern's buffer is a "data handler," as recited in claims 3–7, and that Petitioner has not shown that Tsern "generates" data, as required by the claims. However, Petitioner relies on the teachings of both Averbuj and Tsern in combination, so we are not persuaded by those arguments.

Accordingly, upon review of Petitioner's analysis and supporting evidence, we determine that Petitioner has demonstrated that there is a reasonable likelihood that it would prevail with respect to claims 1–7, on the ground that these claims are unpatentable over the combination of Averbuj and Tsern.

III. CONCLUSION

For the foregoing reasons, we determine that the information presented establishes a reasonable likelihood that Petitioner would prevail in showing that claims 1–7 of the '434 Patent are unpatentable. At this preliminary stage, we have not made a final determination with respect to the patentability of the challenged claims or any underlying factual and legal issues.

IV. ORDER

Accordingly, it is:

ORDERED that pursuant to 35 U.S.C. § 314(a), an *inter partes* review is hereby instituted on the following grounds:

1. claim 1 of the '434 Patent under 35 U.S.C. § 102 as anticipated by Averbuj;
2. claim 1 of the '434 Patent under 35 U.S.C. § 103(a) as obvious over Averbuj; and
3. claims 1–7 of the '434 Patent under 35 U.S.C. 103(a) as obvious over Averbuj and Tsern;

FURTHER ORDERED that pursuant to 35 U.S.C. § 314(c) and 37 C.F.R. § 42.4, notice is hereby given of the institution of a trial, which commences on the entry date of this decision; and

FURTHER ORDERED that the trial is limited to the grounds identified immediately above, and no other ground is authorized.

IPR2017-00561
Patent 8,001,434 B1

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UNITED STATES COURT OF APPEALS FOR THE FEDERAL CIRCUIT

**NOTICE OF ENTRY OF
JUDGMENT WITHOUT OPINION**

JUDGMENT ENTERED: 11/14/2017

The judgment of the court in your case was entered today pursuant to Rule 36. This Court affirmed the judgment or decision that was appealed. None of the relief sought in the appeal was granted. No opinion accompanied the judgment. The mandate will be issued in due course.

Information is also provided about petitions for rehearing and suggestions for rehearing en banc. The questions and answers are those frequently asked and answered by the Clerk's Office.

Costs are taxed against the appellant in favor of the appellee under Rule 39. The party entitled to costs is provided a bill of costs form and an instruction sheet with this notice.

The parties are encouraged to stipulate to the costs. A bill of costs will be presumed correct in the absence of a timely filed objection.

Costs are payable to the party awarded costs. If costs are awarded to the government, they should be paid to the Treasurer of the United States. Where costs are awarded against the government, payment should be made to the person(s) designated under the governing statutes, the court's orders, and the parties' written settlement agreements. In cases between private parties, payment should be made to counsel for the party awarded costs or, if the party is not represented by counsel, to the party pro se. Payment of costs should not be sent to the court. Costs should be paid promptly.

If the court also imposed monetary sanctions, they are payable to the opposing party unless the court's opinion provides otherwise. Sanctions should be paid in the same way as costs.

Regarding exhibits and visual aids: Your attention is directed to FRAP 34(g) which states that the clerk may destroy or dispose of the exhibits if counsel does not reclaim them within a reasonable time after the clerk gives notice to remove them. (The clerk deems a reasonable time to be 15 days from the date the final mandate is issued.)

FOR THE COURT

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court

cc: Wayne Bradley
Michael Francis Heafey
Scott P McBride
Gregory Schodde
Anup M. Shah
Ronald Spuhler
Thomas J. Wimbiscus

16-2666, -2667: Smart Modular Technologies v. Netlist, Inc.
United States Patent and Trademark Office, Case Nos. IPR2014-01372, IPR2014-01374

NOTE: This disposition is nonprecedential.

**United States Court of Appeals
for the Federal Circuit**

SMART MODULAR TECHNOLOGIES, INC.,
Appellant

v.

NETLIST, INC.,
Appellee

2016-2666, 2016-2667

Appeals from the United States Patent and Trade-
mark Office, Patent Trial and Appeal Board in Nos.
IPR2014-01372, IPR2014-01374.

JUDGMENT

MICHAEL FRANCIS HEAFEY, Troutman Sanders LLP,
San Francisco, CA, argued for appellant. Also represent-
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THOMAS J. WIMBISCUS, McAndrews, Held & Malloy,
Ltd., Chicago, IL, argued for appellee. Also represented
by WAYNE BRADLEY, SCOTT P. MCBRIDE, GREGORY
SCHODDE, RONALD SPUHLER.

THIS CAUSE having been heard and considered, it is

ORDERED and ADJUDGED:

PER CURIAM (PROST, *Chief Judge*, WALLACH and
HUGHES, *Circuit Judges*).

AFFIRMED. See Fed. Cir. R. 36.

ENTERED BY ORDER OF THE COURT

November 14, 2017

Date

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court

UNITED STATES COURT OF APPEALS
FOR THE FEDERAL CIRCUIT

Questions and Answers

Petitions for Rehearing (Fed. Cir. R. 40)
and
Petitions for Hearing or Rehearing En Banc (Fed. Cir. R. 35)

Q. When is a petition for rehearing appropriate?

A. Petitions for panel rehearing are rarely successful because they most often fail to articulate sufficient grounds upon which to grant them. For example, a petition for panel rehearing should not be used to reargue issues already briefed and orally argued; if a party failed to persuade the court on an issue in the first instance, a petition for panel rehearing should not be used as an attempt to get a second "bite at the apple." This is especially so when the court has entered a judgment of affirmance without opinion under Fed. Cir. R. 36. Such dispositions are entered if the court determines the judgment of the trial court is based on findings that are not clearly erroneous, the evidence supporting the jury verdict is sufficient, the record supports the trial court's ruling, the decision of the administrative agency warrants affirmance under the appropriate standard of review, or the judgment or decision is without an error of law.

Q. When is a petition for hearing or rehearing en banc appropriate?

A. En banc decisions are extraordinary occurrences. To properly answer the question, one must first understand the responsibility of a three-judge merits panel of the court. The panel is charged with deciding individual appeals according to the law of the circuit as established in the court's precedential opinions. While each merits panel is empowered to enter precedential opinions, the ultimate duty of the court en banc is to set forth the law of the Federal Circuit, which merit panels are obliged to follow.

Thus, as a usual prerequisite, a merits panel of the court must have entered a precedential opinion in support of its judgment for a suggestion for rehearing en banc to be appropriate. In addition, the party seeking rehearing en banc must show that either the merits panel has failed to follow identifiable decisions of the U.S. Supreme Court or

Federal Circuit precedential opinions or that the merits panel has followed circuit precedent, which the party seeks to have overruled by the court en banc.

Q. How frequently are petitions for rehearing granted by merits panels or petitions for rehearing en banc accepted by the court?

A. The data regarding petitions for rehearing since 1982 shows that merits panels granted some relief in only three percent of the more than 1900 petitions filed. The relief granted usually involved only minor corrections of factual misstatements, rarely resulting in a change of outcome in the decision.

En banc petitions were accepted less frequently, in only 16 of more than 1100 requests. Historically, the court itself initiated en banc review in more than half (21 of 37) of the very few appeals decided en banc since 1982. This sua sponte, en banc review is a by-product of the court's practice of circulating every precedential panel decision to all the judges of the Federal Circuit before it is published. No count is kept of sua sponte, en banc polls that fail to carry enough judges, but one of the reasons that virtually all of the more than 1100 petitions made by the parties since 1982 have been declined is that the court itself has already implicitly approved the precedential opinions before they are filed by the merits panel.

Q. Is it necessary to have filed either of these petitions before filing a petition for certiorari in the U.S. Supreme Court?

A. No. All that is needed is a final judgment of the Court of Appeals. As a matter of interest, very few petitions for certiorari from Federal Circuit decisions are granted. Since 1982, the U.S. Supreme Court has granted certiorari in only 31 appeals heard in the Federal Circuit. Almost 1000 petitions for certiorari have been filed in that period.

UNITED STATES COURT OF APPEALS FOR THE FEDERAL CIRCUIT

INFORMATION SHEET

FILING A PETITION FOR A WRIT OF CERTIORARI

There is no automatic right of appeal to the Supreme Court of the United States from judgments of the Federal Circuit. You must file a petition for a writ of certiorari which the Supreme Court will grant only when there are compelling reasons. (See Rule 10 of the Rules of the Supreme Court of the United States, hereinafter called Rules.)

Time. The petition must be filed in the Supreme Court of the United States within 90 days of the entry of judgment in this Court or within 90 days of the denial of a timely petition for rehearing. The judgment is entered on the day the Federal Circuit issues a final decision in your case. [The time does not run from the issuance of the mandate, which has no effect on the right to petition.] (See Rule 13 of the Rules.)

Fees. Either the \$300 docketing fee or a motion for leave to proceed in forma pauperis with an affidavit in support thereof must accompany the petition. (See Rules 38 and 39.)

Authorized Filer. The petition must be filed by a member of the bar of the Supreme Court of the United States or by the petitioner representing himself or herself.

Format of a Petition. The Rules are very specific about the order of the required information and should be consulted before you start drafting your petition. (See Rule 14.) Rules 33 and 34 should be consulted regarding type size and font, paper size, paper weight, margins, page limits, cover, etc.

Number of Copies. Forty copies of a petition must be filed unless the petitioner is proceeding in forma pauperis, in which case an original and ten copies of the petition for writ of certiorari and of the motion for leave to proceed in forma pauperis. (See Rule 12.)

Where to File. You must file your documents at the Supreme Court.

Clerk
Supreme Court of the United States
1 First Street, NE
Washington, DC 20543
(202) 479-3000

No documents are filed at the Federal Circuit and the Federal Circuit provides no information to the Supreme Court unless the Supreme Court asks for the information.

Access to the Rules. The current rules can be found in Title 28 of the United States Code Annotated and other legal publications available in many public libraries.

UNITED STATES COURT OF APPEALS FOR THE FEDERAL CIRCUIT

16-2666, 16-2667

SMART MODULAR TECHNOLOGIES, INC.,

Appellant

v.

NETLIST, INC.,

Appellee

Appeals from the United States Patent and Trademark Office in nos. IPR2014-01372, IPR2014-01374

Administrative Patent Judge Linda M. Gaudette, Administrative Patent

Judge Matthew R. Clements, Administrative Patent Judge Bryan F. Moore, Administrative Patent

Judge Peter P. Chen

MANDATE

In accordance with the judgment of this Court, entered November 14, 2017, and pursuant to Rule 41(a) of the Federal Rules of Appellate Procedure, the formal mandate is hereby issued.

FOR THE COURT

/s/ Peter R. Marksteiner

Peter R. Marksteiner
Clerk of Court



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY.DOCKET NO./TITLE	REQUEST ID
12/422,925	04/13/2009	Hyun Lee	129980-5050US	67137

Acknowledgement of Loss of Entitlement to Entity Status Discount

The entity status change request below filed through Private PAIR on 07/06/2018 has been accepted.

CERTIFICATIONS:

Change of Entity Status:

✕ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be notification of loss of entitlement to small or micro entity status, as applicable.

This portion must be completed by the signatory or signatories making the entity status change in accordance with 37 CFR 1.4(d)(4).

Signature:	/Jamie J. Zheng/
Name:	Jamie J. Zheng
Registration Number:	51167

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

SK HYNIX INC., SK HYNIX AMERICA INC., and
SK HYNIX MEMORY SOLUTIONS INC.,
Petitioner,

v.

NETLIST, INC.,
Patent Owner.

Case IPR2017-00561
Patent 8,001,434 B1

Before BRYAN F. MOORE, MATTHEW R. CLEMENTS, and
SHEILA F. McSHANE, *Administrative Patent Judges*.

MOORE, *Administrative Patent Judge*.

FINAL WRITTEN DECISION
35 U.S.C. 318(a)

I. INTRODUCTION

SK hynix Inc., SK hynix America Inc., and SK hynix memory solutions Inc., (collectively “Petitioner”) filed a Petition (Paper 1, “Pet.”) pursuant to 35 U.S.C. §§ 311–319 to institute an *inter partes* review of claims 1–7 (“the challenged claims”) of U.S. Patent No. 8,001,434 B1 (“the ’434 Patent,” Ex. 1001). The Petition is supported by the Declaration of Pinaki Mazumder, Ph.D. (“Mazumder Declaration,” “Mazumder Dec.,” Ex. 1003). Netlist, Inc. (“Patent Owner”) filed a Preliminary Response (“Prelim. Resp.,” Paper 6).

On July 7, 2017, we instituted an *inter partes* review of claims 1–7 of the ’434 Patent, but did not institute on all grounds. Paper 7, 7, 33 (“Inst. Dec.”). Patent Owner filed a Response. Paper 14 (“PO Resp.”). The Patent Owner Response is supported by the Declaration of R. Jacob Baker, Ph.D. (“Baker Declaration,” “Baker Dec.,” Ex. 2010). Petitioner filed a Reply. Paper 18 (“Reply”).

On February 28, 2018, Petitioner and Patent Owner filed motions to exclude. Paper 21 (“Pet. Mot. to Excl.”); Paper 23 (“PO Mot. to Excl.”). Petitioner and Patent Owner filed responses to their respective motions to exclude. Paper 25 (“Pet. Mot. to Excl. Resp.”); Paper 26 (“PO Mot. to Excl. Resp.”). Petitioner and Patent Owner filed replies to those respective responses. Paper 29 (“Pet. Mot. to Excl. Reply”); Paper 30 (“PO Mot. to Excl. Reply”).

On March 13, 2018, we entered a *sua sponte* Order to Show Cause why the *inter partes* review should not be terminated as to claim 1 because it has been finally adjudicated as unpatentable. Paper 24 (“Show Cause

Order”). Petitioner and Patent Owner responded to the order. Paper 27 (“Pet. Show Cause Resp.”); Paper 28 (“PO Show Cause Resp.”).

An oral hearing was held on April 6, 2018. Paper 34 (“Tr.”).

On April 24, 2018, the Supreme Court held that a decision on institution under 35 U.S.C. § 314 may not institute on less than all claims challenged in the petition. 138 S. Ct. 1348 (2018). We modified our Institution Decision to institute trial on all of the challenged claims and all of the grounds presented in the Petition. Paper 35 (“SAS Order”). We invited the parties to request briefing regarding the newly added grounds and no request was made. *Id.*

We have jurisdiction under 35 U.S.C. § 6. This Final Written Decision is issued pursuant to 35 U.S.C. § 318(a). For the reasons that follow, we determine that Petitioner has shown by a preponderance of the evidence that claims 2–7 are unpatentable. Based on collateral estoppel, we also terminate the trial pursuant to 37 C.F.R. § 42.72 as to claim 1.

A. Related Proceedings

Petitioner recites a list of District Court proceedings related to this *inter partes* review. Pet. 5. This *inter partes* review challenges the same patent at issue in the decision entered in IPR2014-00970 (the ’970 IPR). *See Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB Dec. 16, 2014) (Paper 12). In the ’970 IPR, as to the claims at issue here, claim 1 was held unpatentable as anticipated by Averbuj in a final written decision. *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB Dec. 14, 2015) (Paper 32). That decision was appealed to Federal Circuit, the decision was affirmed, and the mandate issued on December 21, 2017. *See Exs. 1029,*

1034. In the '970 IPR, claims 2–7 were also challenged as obvious over Averbuj and Tsern, but that challenge was not instituted. *Sandisk*, IPR2014-00970 (Paper 12).

The '434 Patent was also challenged in IPR2014-01372 (the '1372 IPR). In the 1372 IPR, as to the claims at issue here, claims 1–4 were held not unpatentable as anticipated by Averbuj (under a different theory of anticipation than the '970 IPR) in a final written decision. *Smart Modular Techs. Inc. v. Netlist, Inc.*, Case IPR2014-01372 (PTAB March 9, 2016) (Paper 45). The decision in the '1372 IPR was not appealed to Federal Circuit.

The '434 Patent was also challenged in IPR2014-01373 (the '1373 IPR), in which institution was denied as to all challenges on the merits. *Smart Modular Techs. Inc. v. Netlist, Inc.*, Case IPR2014-01373 (PTAB Mar. 13, 2015) (Paper 16). Neither Averbuj nor Tsern was asserted in the '1373 IPR.

B. The '434 Patent

The '434 Patent relates to self-testing electronic memory modules. Ex. 1001, 1:23–24. A block diagram of an exemplary self-testing memory module is shown in Figure 3 of the '434 Patent, reproduced below.

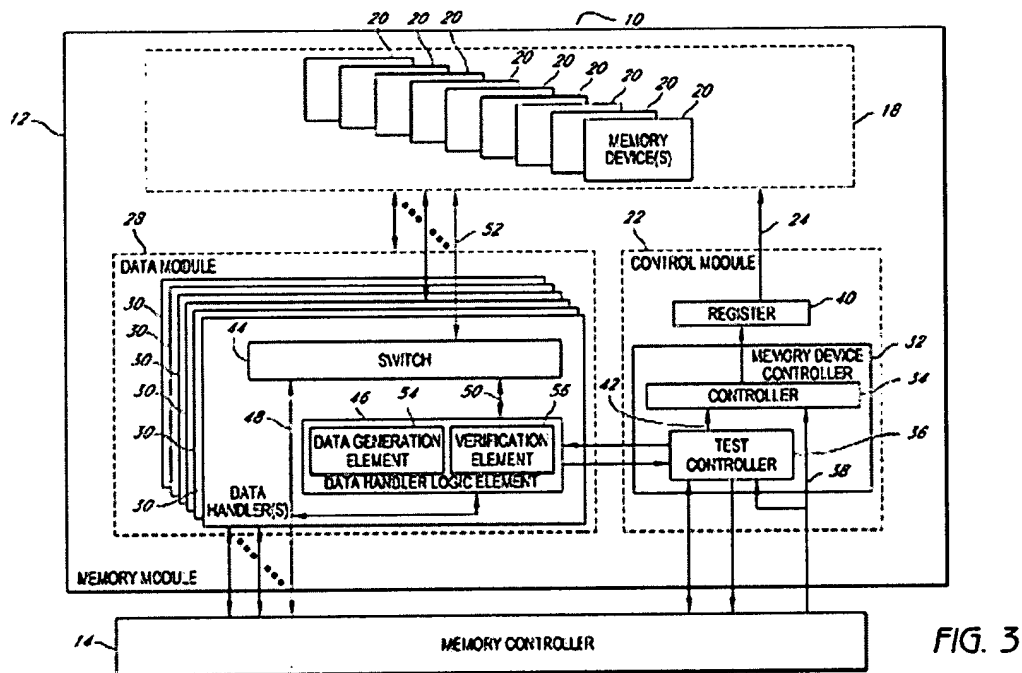


FIG. 3

As illustrated in Figure 3, above, “memory module 10 includes a printed circuit board 12 (PCB) configured to be operatively coupled to a memory controller 14 of a computer system 16.” *Id.* at 5:1–3. Memory module 10 includes a plurality of memory devices 18, each memory device 20 of the plurality of memory devices 18 comprising data, address, and control ports. *Id.* at 5:3–7. Memory module 10 further includes control module 22 and data module 28 comprising a plurality of independently operable data handlers 30. *Id.* at 5:7–11. Control module 22 includes memory device controller 34, which receives address and control signals 38 from system memory controller 14 and address and control signals 42 from test controller 36. *Id.* at 9:40–45. Similarly, data module 28 includes switch 44, which selectively inputs to memory devices 18 either data signals 48

from memory controller 14 or data signals 50 from data handler logic element 46. *Id.* at 10:7–11.

During test mode, memory device controller 34 sends address and control signals 38 from test controller 36 to register 40 for input to address and control ports of memory devices 18. *Id.* at 9:40–50. “[T]est controller 36 updates each of [] data handlers 30 (e.g., with new data patterns, write signal characteristics, etc.)” *Id.* at 14:7–9. Data generation element 54 of data handler logic element 46 is configured to generate data signals and/or patterns of data signals based on the information received from test controller 36, for writing to memory devices 18. *Id.* at 10:28–37, 15:25–27.

C. Illustrative Claim

Of the challenged claims, claim 1 is independent. Claim 1, reproduced below, is illustrative of the claimed subject matter:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports; and
 - a circuit comprising:
 - a control module configured to generate address and control signals for testing the memory devices; and
 - a data module comprising a plurality of data handlers, each data handler operable independently from each of the other data handlers of the plurality of data handlers and operatively coupled to a corresponding plurality of the data ports of one or more of the memory devices and configured to generate data for writing to the corresponding plurality of data ports, wherein the circuit is

configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers.

Ex. 1001, 16:32–53.

D. Instituted Grounds of Unpatentability

We instituted trial on the following ground (Inst. Dec. 34; SAS Order 2):

Reference[s]	Basis	Claims Challenged
Averbuj ¹	§ 102	1–7
Averbuj	§ 103	1–7
Averbuj and Tsern ²	§ 103	1–7

II. ANALYSIS

A. Collateral Estoppel

The parties raised an issue arising from challenges to the subject patent in related *inter partes* review proceedings brought by a different petitioner, namely the '970 IPR. In the '970 IPR, the Board held that claim 1 is unpatentable over art that is the same as the art considered in this proceeding as against a different petitioner (Sandisk) than this proceeding. *Sandisk Corp. v. Netlist, Inc.*, Case IPR2014-00970 (PTAB Dec. 14, 2015) (Paper 32). The Federal Circuit affirmed that decision. *Netlist, Inc. v.*

¹ U.S. Patent Application Publication No. 2005/0257109 (Ex. 1005, “Averbuj”)

² U.S. Patent Application Publication No. 2007/0070669 (Ex. 1006, “Tsern”)

SanDisk LLC, 702 F. App'x 986 (Fed. Cir. 2017). (per curiam).

Accordingly, claim 1, which is involved in this proceeding, has been held unpatentable. This raises the issue of whether we should terminate the proceeding as to claim 1 based on collateral estoppel.

“It is well established that collateral estoppel, also known as issue preclusion, applies in the administrative context.” *MaxLinear, Inc. v. CF CRESPE LLC*, 880 F.3d 1373, 1376 (Fed. Cir. 2018) (citations omitted); *Webpower, Inc.*, Case IPR2016-01239, slip. op. at 26–28 (PTAB Dec. 26, 2017) (Paper 21). A patentee is estopped from asserting the validity of a patent that has been declared invalid in a prior suit against a different defendant, unless patentee demonstrates that he did not have full and fair opportunity, procedurally, substantively, and evidentially, to litigate the validity of his patent in the prior suit. *MaxLinear, Inc.*, 880 F.3d at 1377 (citing *Blonder-Tongue Labs., Inc. v. Univ. of Illinois Found.*, 402 U.S. 313, 91 S. Ct. 1434, 28 L. Ed. 2d 788 (1971) (finding collateral estoppel applies to a patentee who had a full and fair opportunity to litigate the validity of a patent in a prior federal case.)) Thus, as to a different petitioner, when the prior decisions finding claims unpatentable have “subsequently been affirmed by [the Federal Circuit] those prior decisions, having been affirmed by [the Federal Circuit], are binding in this proceeding, as a matter of collateral estoppel.” *Id.* at 1376.

In response to our Order to Show Cause why the proceeding should not be terminated as to claim 1, Patent Owner argued that “[t]his IPR should be terminated as to claim 1” to avoid wasting the resources of the Board. PO Show Cause Resp. 1. Petitioner argued “[c]ollateral estoppel only binds

Patent Owner, not Petitioners, and Petitioners, in addition to advancing the same mapping of Averbuj to claim 1 as did SanDisk in IPR2014-0970, also present alternative mappings as well.” Pet. Show Cause Br. 1. Petitioner further argues “[g]iven that the Board may need to consider whether Petitioners’ alternative mappings read on the limitations of claim 1 when addressing the patentability of claims 2–7, there may be little efficiency gained by terminating this IPR as to claim 1.” *Id.* at 2. We recognize that Petitioner presents alternative mappings as to claim 1. Nevertheless, as explained below, we find that claims 2–7 are unpatentable under the mappings used in the ’970 IPR. Thus, given that we find that collateral estoppel applies, judicial efficiency is advanced by declining to reassess the patentability of claim 1 under alternative mappings.

Patent Owner asserts that this *inter partes* review relies on obviousness based on the combination of Averbuj and Tsern rather than the ground under which claim 1 was found unpatentable in the ’970 IPR, i.e. anticipation by Averbuj. Thus, according to Patent Owner, “Petitioners must also provide evidence of a reason why a POSITA would have modified/combined the limitations of claim 1 (as allegedly shown in Averbuj) with the limitations of dependent claims 2–7 (as allegedly shown in Tsern), to create the claimed inventions.” PO Show Cause Br. 1. We agree that Petitioner must show a motivation to combine Tsern with Averbuj, but this does not mean that Petitioner must prove again that Averbuj meets the limitations of claim 1; it means only that Petitioner need show a rationale for combining Tsern with those aspects of Averbuj that the Board has already determined meet the limitations of claim 1.

Patent Owner asserts that “collateral estoppel does not apply to the application of the claim 1 limitations in assessing dependent claims 2–7.” *Id.* at 2. Patent Owner further asserts “[a] party seeking to apply collateral estoppel based on a prior action must show that (1) the identical issue was actually litigated; (2) the issue was actually decided in a final decision on the merits; (3) the issue was necessary to the final decision; and (4) the party being estopped was adequately represented in the prior action [i.e. “the *United Access* test”].” *Id.* at 2 (citing *United Access Techs., LLC v. CenturyTel Broadband Services LLC*, 778 F.3d 1327, 1331 (Fed. Cir. 2015)). We agree that this is the proper test to use to assess the applicability of collateral estoppel. Nevertheless, we note *United Access* applies collateral estoppel in a different context and suggests that issue preclusion can be applied in an obviousness challenge, even if the final decision on the original issue was considered under an anticipation challenge. In particular, in *United Access* it was noted that “[i]ssue preclusion [traditionally called “collateral estoppel”³] applies to issues litigated in prior disputes . . . [and, thus, i]t thus makes no difference that the earlier no-damages-due-to-competition ruling arose in a case under the Truth in Negotiation Act and that this dispute arises under the False Claims Act and the common law.” *United Access*, 778 F.3d. at 726. The “issue” of whether Averbuj teaches the limitations of claim 1 is identical to the issue in and has been litigated in the ‘970 IPR which found that Averbuj anticipates the limitations of claim 1.

³ *Mother’s Rest., Inc. v. Mama’s Pizza, Inc.*, 723 F.2d 1566, 1569 (Fed. Cir. 1983)

The question becomes how does the fact that the limitations of claim 1 rely on an obviousness combination with Tsern effect the analysis. To that point, we recognize that “though anticipation is the epitome of obviousness, [they] are separate and distinct concepts.” *Jones v. Hardy*, 727 F.2d 1524, 1529 (Fed.Cir.1984). It is that difference that is important here.

The Federal Circuit has noted, however, that obviousness is different because: more than one reference can be used in obviousness; secondary considerations must be analyzed in obviousness; inherency in anticipation is different than obviousness (*Cohesive Techs., Inc. v. Waters Corp.*, 543 F.3d 1351, 1364 (Fed. Cir. 2008)); and single reference obviousness requires a showing of motivation to modify a reference when a party argues that such a modification must be made to meet the limitations of a claim (*Monsanto Tech. LLC v. E.I. DuPont de Nemours & Co.*, 878 F.3d 1336, 1346 (Fed. Cir. 2018)). Here, however, Petitioner relies on Averbuj alone, not on the combination of Averbuj and Tsern (or any modification of Avebuj), to meet any limitation of claim 1, Petitioner (here and in the '970 IPR) did not rely on inherency to show any limitations of claim 1, and Patent Owner has not argued secondary considerations. *See generally*, Pet.; PO Resp.; Ex. 1028 ('970 IPR FWD).

Thus, the relevant legal issue, in this essentially single reference obviousness situation as to claim 1, is the same as in anticipation, i.e. does the reference teach the limitations of the claim. In fact anticipation requires a higher level of teaching requiring that references teach limitations of a claim explicitly or inherently. Under obviousness, the reference need only render a limitation obvious to one of skill in the art at the time of the

invention. Thus, the adage “anticipation is the epitome of obviousness.” *Jones*, 727 F.2d at 1529. The Board is allowed to rely on findings regarding whether a reference anticipates a limitation in its obviousness analysis.⁴ See *Wasica Fin. GmbH v. Cont’l Auto. Sys., Inc.*, 853 F.3d 1272, 1278 (Fed. Cir. 2017) (stating approvingly in a footnote that the “Board noted that Oselin rendered [the challenged claims] obvious by virtue of its anticipation of them.”) (citing *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983)).

Also, as to the remaining prongs of the *United Access* test, the issue of whether Averbuj teaches the limitations of claim 1 was actually decided in a final decision on the merits, the issue was necessary to the final decision because it went to the ultimate issue of patentability, and there is no allegation that Patent Owner was not adequately represented in the prior action. Thus, for the reasons above, claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel and the issue of whether the limitations of claim 1 are met by Averbuj by obviousness over Averbuj and Tsern also is established under the doctrine of collateral estoppel. Thus, we terminate this proceeding as to claim 1.

⁴ In fact, the Board relied on this principle in the Decision to Institute. Dec. to Inst. 8 (“[W]e determine that claim 1 is anticipated by Averbuj. Inasmuch as ‘anticipation is the epitome of obviousness’ (*In re McDaniel*, 293 F.3d 1379, 1385 (Fed. Cir. 2002)), we determine that Petitioner has established a reasonable likelihood of prevailing in its assertion that claim 1 would have been obvious over Averbuj alone.”).

Our reviewing court has recently ordered the Board, on remand, to assess whether claims depending from a claim that has been finally adjudicated as unpatentable should not also be found unpatentable for the same reasons as the independent claim. *MaxLinear, Inc.*, 880 F.3d at 1377–1378. We apply this analysis to claims 2–7. Claims 2–7 have not been finally adjudicated as unpatentable in any previous case. Because the differences between the unadjudicated patent claims 2–7 and the adjudicated patent claim 1 materially alter the question of unpatentability, collateral estoppel does not apply. *Cf. In re Arunachalam*, 709 F. App’x 699, 702 (Fed. Cir. 2017) (unpublished) (finding collateral estoppel applies to an adjudicated claim that is not “materially different” than the adjudicated claims); *accord MaxLinear, Inc.*, 880 F.3d at 1377–1378. In this case, there are material differences between claims 2–7 and claim 1. For example, claim 2, from which claims 3–7 depend, recites a PCB and that the “data handlers” of claim 1 are separate components. It is still true, however, as Petitioner asserts, “Patent Owner is precluded from relitigating here whether Averbuj discloses all the limitations of claim 1, as arranged in the claim, including for purposes of determining the patentability of claims 2–7.” Pet. Show Cause Resp. 3. Thus, we do not terminate as to claims 2–7 under collateral estoppel.

B. Relevant Law

1. Obviousness Law

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject

matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and, (4) where in evidence, so-called secondary considerations, including commercial success, long-felt but unsolved needs, failure of others, and unexpected results.⁵ *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966) (“the *Graham* factors”).

2. *Level of Skill Law*

The level of ordinary skill in the art usually is evidenced by the references themselves. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978). For an obviousness analysis, prior art references must be “considered together with the knowledge of one of ordinary skill in the pertinent art.” *In re Paulsen*, 30 F.3d 1475, 1480 (Fed. Cir. 1994) (quoting *In re Samour*, 571 F.2d 559, 562 (CCPA 1978)). Moreover, “it is proper to take into account not only specific teachings of the reference but also the inferences which one skilled in the art would reasonably be expected to draw therefrom.” *In re Preda*, 401 F.2d 825, 826 (CCPA 1968). That is because an obviousness analysis “need not seek out

⁵ Patent Owner does not put forth evidence it alleges tend to show secondary considerations of non-obviousness in its Response.

precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ.” *KSR*, 550 U.S. at 418; *see also In re Translogic Tech., Inc.*, 504 F.3d at 1259.

3. *Claim Interpretation Law*

In an *inter partes* review, claim terms in an unexpired patent are given their broadest reasonable construction in light of the specification of the patent in which they appear. 37 C.F.R. § 42.100(b). Under this standard, we interpret claim terms using “the broadest reasonable meaning of the words in their ordinary usage as they would be understood by one of ordinary skill in the art, taking into account whatever enlightenment by way of definitions or otherwise that may be afforded by the written description contained in the applicant’s specification.” *In re Morris*, 127 F.3d 1048, 1054 (Fed. Cir. 1997).

We presume that claim terms have their ordinary and customary meaning. *See Trivascular, Inc. v. Samuels*, 812 F.3d 1056, 1062 (Fed. Cir. 2016) (“Under a broadest reasonable interpretation, words of the claim must be given their plain meaning, unless such meaning is inconsistent with the specification and prosecution history.”); *In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007) (“The ordinary and customary meaning is the meaning that the term would have to a person of ordinary skill in the art in question.” (internal citation and quotation marks omitted)). A patentee, however, may rebut this presumption by acting as his or her own lexicographer, providing a definition of the term in the specification with “reasonable clarity, deliberateness, and precision.” *In re Paulsen*, 30 F.3d

1475, 1480 (Fed. Cir. 1994). Only those terms that are in controversy need to be construed, and only to the extent necessary to resolve the controversy. *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999); *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017).

C. Level of Skill Analysis

Petitioner asserts “[a] a person of ordinary skill in the art at the time of the ‘434 Patent would have been someone with “a Bachelor’s degree in electrical engineering, computer engineering, or in a related field and at least one year of work experience relating to memory systems, and would be familiar with the design of memory devices, memory modules, and BIST.”

Pet. 9 (citing Ex. 1003 ¶ 51). Patent Owner asserts through its declarant:

[A Person of ordinary skill in the art at the time of the invention would have been] working on the design of memory devices, would have at least a Bachelor of Science degree in Electrical and/or Computer Engineering, and at least five years of industry experience designing memory devices and controllers. Alternatively, one of ordinary skill in the art would have a Master of Science degree in Electrical and/or Computer Engineering, and at least three years of industry experience designing memory devices and controllers. Also alternatively, one of ordinary skill in the art would have a Doctorate degree in Electrical and/or Computer Engineering, and have at least one year of industry experience designing memory devices and controllers.

I agree with Petitioner’s expert that the memory industry requires a POSITA with industry experience.

Ex. 2010 ¶¶ 32–33.

Neither party has argued that the difference between the parties' articulated level of skill should change the analysis as to patentability. Therefore, we adopt Petitioner's articulation of the level of skill, above, and acknowledge that, commensurate with that articulation of the level of skill, the level of ordinary skill in the art is also reflected by the prior art of record. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001); *In re GPAC Inc.*, 57 F.3d 1573, 1579 (Fed. Cir. 1995); *In re Oelrich*, 579 F.2d 86, 91 (CCPA 1978).

D. Claim Construction Analysis

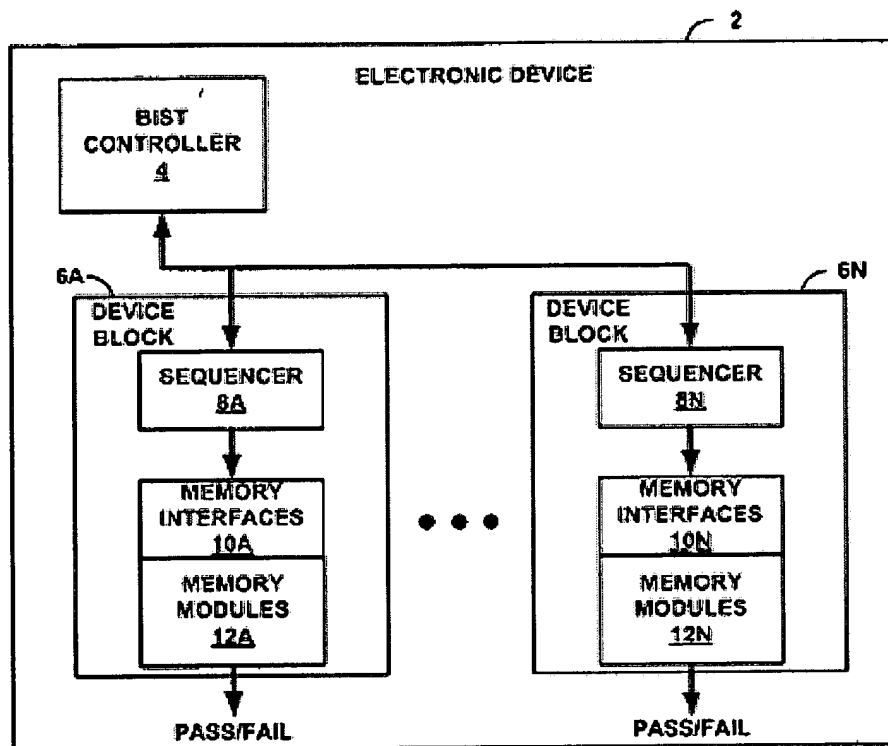
Claims 2–7 do not recite claim limitations whose constructions are in controversy. Patent Owner disputes only the construction for the term “memory module,” recited in claim 1. PO Resp. 17–25. The “memory module” is not argued as a reason that Averbuj or Tsern do not meet the limitations of claims 2–7. *Id.* at 60–71. Thus, we do not further construe any terms including memory module in this Final Written Decision.

E. Claims 1–7 — Anticipation by Averbuj (Ex. 1005) (Ground 1)

Petitioner argues that claims 1–7 are unpatentable over Averbuj under 35 U.S.C. § 102. Pet. 24–48. We refer to this as “Ground 1.”

1. Averbuj (Ex. 1005)

Averbuj describes a hierarchical built-in self-test (BIST) architecture wherein a BIST controller provides centralized, high level control of the testing of one or more memory modules. Ex. 1005 ¶¶ 7–8. Figure 1 of Averbuj is reproduced below.



Averbuj Figure 1, above, “is a block diagram illustrating an example electronic device 2 having a distributed, hierarchical built-in self-test (BIST) architecture.” *Id.* ¶ 28. “[E]lectronic device 2 may be any device that incorporates memory modules, such as an embedded computing system, a computer, server, personal digital assistant (PDA), mobile computing device, mobile communication device, digital recording device, network appliance, mobile positioning device, and the like.” *Id.* ¶ 32. Averbuj states that “electronic devices . . . are [normally] constructed from many integrated circuit chips and many supporting components mounted on a circuit board.” *Id.* ¶ 5. “[E]lectronic device 2 includes a built-in self-test (BIST) controller 4 that provides centralized, high-level control over testing of device blocks

6A through 6N (collectively ‘device blocks 6’).” *Id.* ¶ 28. “Each of device blocks 6 includes a Sequencer 8, and a set of one or more memory interfaces 10 and one or more respective memory modules 12.” *Id.* BIST controller 4 provides and communicates test algorithms as a set of commands to Sequencers 8 for application to device blocks 6. *Id.* ¶ 29.

Figure 5, below, illustrates an exemplary Sequencer 8A. *Id.* ¶ 22

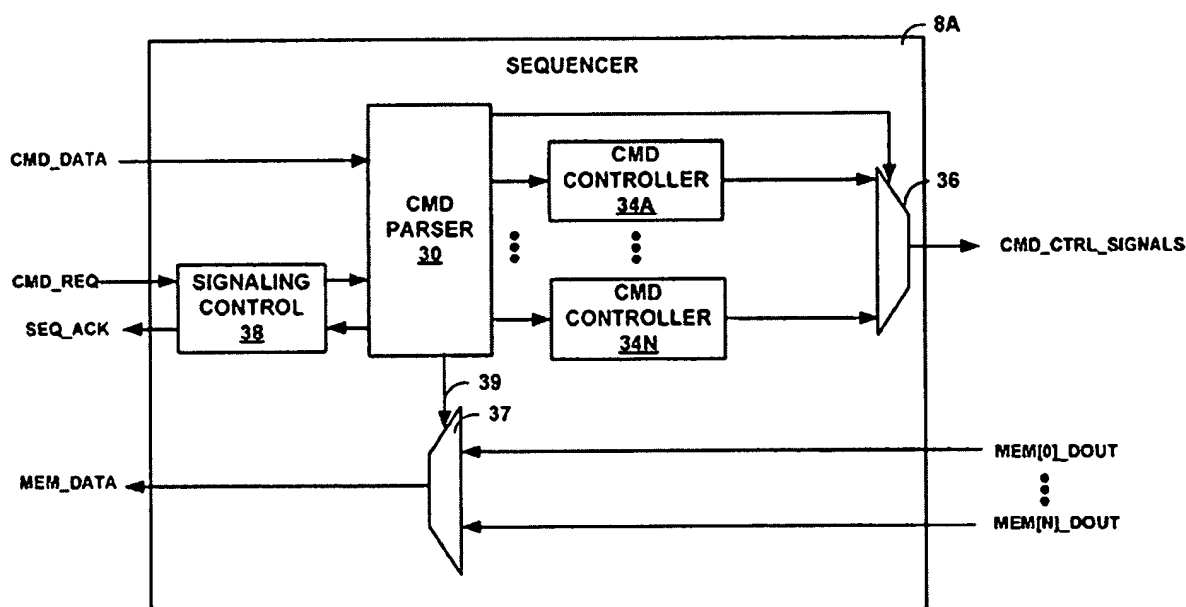


FIG. 5

As shown in Figure 5, above, exemplary Sequencer 8A includes command parser 30 and command controllers 34. *Id.* ¶¶ 41–42. Command parser 30 processes the commands from BIST controller 4 and invokes one of command controllers 34, which, in turn, issues command control signals, such as signals to provide a memory address and data, to receiving memory

interfaces 10, 41. *Id.* ¶¶ 43, 51. An exemplary memory interface is shown in Figure 6, below. *Id.* ¶ 23.

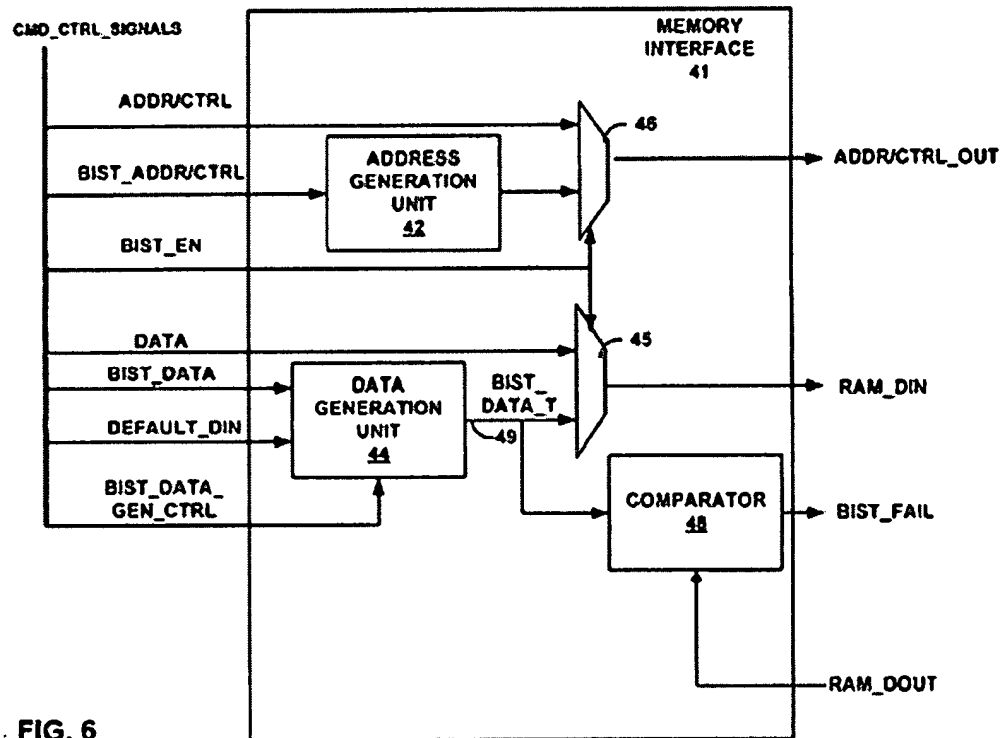


FIG. 6

As shown in the Figure 6 embodiment, above, memory interface 41 includes multiplexers 45, 46. *Id.* ¶ 48. “[U]nder normal operating conditions, BIST enable (BIST_EN) is de-asserted, causing multiplexers 45, 46 to select the address/control signals (ADDR/CTRL) and data signals (DATA), e.g., as provided by a programmable processor.” *Id.* “When electronic device 2 is operating in BIST mode, however, the BIST enable signal causes multiplexers 45, 46 to select the BIST address/control signals

(BIST_ADDR/CTRL) and the test data provided by a respective higher level Sequencer[, e.g., Sequencer 8A].” *Id.*

“[M]emory interface 41 processes the sequential memory operations issued by [] higher-level Sequencer [8A], and transforms the data and addresses provided by the Sequencer as needed based on the particular physical characteristics of the memory module” (*id.* ¶ 54), using address generation unit 42 and data generation unit 44 (*id.* ¶¶ 49, 51). “[A]ddress generation unit 42 generates the address applied to [] memory module 12 based on the addressing requirements specified by Sequencer 8A and the physical configuration of the rows and columns of the memory module.” *Id.* ¶ 51. “[D]ata generation unit 44 [] receives BIST_DATA signals and default data (DEFAULT_DIN) as provided by [] Sequencer 8[A], and generates transformed BIST data signals 49 (BIST_DATA_T) based on control signals (BIST_DATA_GEN_CTRL) provided by the Sequencer and the specific physical characteristics of the corresponding memory module.” *Id.* ¶ 49. “[D]ata generation unit 44 generates the exact data (RAM_DIN) applied to the memory inputs during each operation of the algorithm.” *Id.*

2. *Analysis*

a. *Claim 1*

Claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. *See supra* Section II. A.

b. *Claim 2*

In the Institution Decision, we found Petitioner has failed to establish a reasonable likelihood of prevailing in its assertion that claims 2–7 are anticipated by Averbuj. Inst. Dec. 22–23, 33. We now analyze this ground

based on the full record before us and under the preponderance of the evidence standard. Claim 2 recites “[t]he self-testing memory module of claim 1, wherein the plurality of data handlers comprise at least two physically separate components mounted on the printed circuit board.”

Petitioner asserts that

Averbuj further discloses that electronic devices, such as his electronic device (2), are constructed from many integrated circuit chips and other components mounted on a circuit board. Ex. 1005 at ¶ [0005] (“[The electronic devices] are constructed from many integrated circuit chips and many supporting components mounted on a circuit board.”); Ex. 1003 ¶ 154. Averbuj discloses that BIST units are commonly incorporated into chips or integrated circuits, which are synonyms. Ex. 1005 at ¶ [0003]; Ex. 1012 at 3, 5. Because the memory interface (12) is part of a BIST unit, it comprises a chip or an integrated circuit of an electronic device “mounted on a circuit board.” Ex. 1005 at ¶ [0005]; Ex. 1003 ¶ 154.

Pet. 41. In other words, Petitioner asserts that because a BIST unit can be incorporated into a chip mounted on a PCB and because memory module 12 is part of the BIST unit, memory unit 12 is mounted on a PCB. We disagree.

Unlike claim 1, which simply requires the memory unit is “on” the PCB, claim 2 requires that the memory module be a “physically separate component” and “mounted” on the PCB. Petitioner has not shown sufficient support for its broad reading that a module which is a circuit embedded in a larger circuit included in a chip may be considered a physically separate component mounted on a PCB. Claims 3–7 depend from claim 2, and the Petition fails to provide additional prior art disclosures that cure the deficiency with respect to claim 2; that is, disclosure of separate components

mounted on a printed circuit board in Averbuj. Pet. 41–48. Thus, Petitioner has failed to establish, by a preponderance of the evidence, its assertion that claims 2–7 are anticipated by Averbuj.

F. Claims 1–7 — Obviousness over Averbuj (Ex. 1005) (Ground 2)

Petitioner argues that claims 1–7 are unpatentable over Averbuj under 35 U.S.C. § 103. Pet. 48–60. We refer to this as “Ground 2.” As to claim 1, Petitioner includes an obviousness ground to the extent we determine that Averbuj does not disclose the recited memory controller and “a plurality of memory devices on the printed circuit board.” *See id.* at 48, 50. As explained above, claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. As also explained above, we determine that claim 1 can also be considered obvious over Averbuj.

As to claims 2–7, we explained above that we are not persuaded that Petitioner has shown sufficiently that Averbuj anticipates those claims. In the Institution Decision, we found Petitioner failed to establish a reasonable likelihood of prevailing in its assertion that claims 2–7 would have been obvious over Averbuj. Inst. Dec. 23–26. Therefore, we discuss the obviousness challenges to dependent claims 2–7 below under the preponderance of the evidence standard.

Petitioner’s argument regarding obviousness of claims 2–7 over Averbuj alone is based on Petitioner’s declarant’s testimony that it was known to house “data handlers” in physically separate integrated circuit

packages. Pet. 51–60 (citing Ex. 1003 ¶ 192–233)⁶. Petitioner argues that to do so would “have been the use of a known technique for its known purpose and would have achieved only predictable results, such as the protection of integrated circuits by the packaging.” Pet. 52. Petitioner asserts that memory modules at the time of the invention were composed of memory chips mounted on different portions of a memory board with BIST units in each memory chip. *Id.*

Given its assertion that memory chips mounted on different portions of a memory board with BIST units in each memory chip were known, Petitioner asserts that reducing redundancy is a motivation to modify Averbuj. Specifically, Petitioner asserts

[a] skilled artisan that seeks to reduce the redundant BIST circuitry of conventional memory chips and BIST units would have been motivated to modify the conventional design in such a way as to avoid unnecessary modifications to the conventional design. Ex. 1003 ¶ 200. Specifically, a skilled artisan would have understood that the BIST controller and Sequencer portions of conventional BIST units would need to be carved out from each memory chip and combined to reduce redundant circuitry. Ex. 1003 ¶ 200. However, a skilled artisan would also have

⁶ Petitioner’s arguments regarding obviousness of claims 2–7 over Averbuj alone are based, in part, on disclosures in Tsern. Pet. 51–52 (citing Ex. 1006 ¶¶ 29, 52–53, Fig. 9A). Petitioner’s arguments based on the disclosure of Tsern rely on structures in Tsern to modify Averbuj, rather than only Tsern’s general teachings to one of ordinary skill in the art, and, thus, those arguments are properly considered below in Petitioner’s ground based upon the combination of Averbuj and Tsern. As such we do not consider them here.

understood and recognized that the memory interface portions need not be carved out and/or combined since doing so would not reduce much, if any, circuitry. Ex. 1003 ¶ 200. Because a skilled artisan would have been motivated to make modifications in an efficient manner, he or she would have been motivated to keep the original design with respect to the memory interface portion. In other words, a skilled artisan would have been motivated to keep the memory interfaces (10, 41) (each of which includes a “*data handler*”) within each memory chip that was separately packaged and mounted on a printed circuit board of a memory module (“*physically separate integrated circuit packages*.”). Ex. 1003 ¶ 201.

Pet. 53–54. We disagree.

“There must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness.” *Innogenetics, N.V. v. Abbott Labs.*, 512 F.3d 1363, 1373 (Fed. Cir. 2008) (internal quotations omitted) (citing *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)). For example, in *ActiveVideo Networks, Inc. v. Verizon*, our reviewing court held that the following expert testimony on the motivation to combine was deficient as a matter of law:

[Y]ou wanted to build something better. You wanted a system that was more efficient, cheaper, or you wanted a system that had more features, makes it more attractive to your customers, because by combining these two things you could do something new that hadn’t been able to do before.

Active Video Networks, Inc. v. Verizon Commc’ns, Inc., 694 F.3d 1312, 1328 (Fed. Cir. 2012). The court noted that because the expert “failed to explain how specific references could be combined, which combination(s) of elements in specific references would yield a predictable result, or how any specific combination would operate or read on the asserted claims,” the

expert's testimony was conclusory and bore no relation to any specific combination of prior art elements. *Id.* at 1327. Although the declarant's testimony here goes further than in *Active Video*, it only fills in gaps about how the combination would be made with conclusory testimony that appears to be tainted with hindsight.

Patent Owner argued, in its Preliminary Response⁷, that “Averbuj, however, teaches away from the proposed combination because (as discussed above) Averbuj’s BIST unit and memory [modules] are in a single chip.” Prelim. Resp. 59. Patent Owner’s Preliminary Response argued further that “modifying Averbuj in the proposed manner would render Averbuj inoperable for its intended purpose of verifying the functionality of internal circuitry of a chip while being included in the chip itself.” *Id.* at 59–60 (citing Ex. 1005 ¶ 5). Patent Owner’s Preliminary Response arguments assume that Averbuj would not be modified based on the single chip structure of Averbuj. In this Final Written Decision, we do not need to, and do not rely on, Patent Owner’s arguments. Here, as in the preliminary Decision to Institute (Inst. Dec. 25) analysis, we focus on the motivation given by Petitioner to modify Averbuj.

As to Petitioner’s arguments supported by its declarant, we find those to employ hindsight to piece together the device as claimed. The expert

⁷ Patent Owner chose not to request additional briefing to address this ground and arguments in the Preliminary Response are generally waived (Paper 8, 2–3), however, we acknowledge these arguments due to the unusual circumstance created by the Supreme Court’s decision in *SAS* and to explain the background in the context of Patent Owners’ Motion to Strike.

provides a host of rationales for separating the components of Averbuj into separate ICs including: structures of Tsern (Ex. 1003 ¶ 203); reducing redundancy (Ex. 1003 ¶ 204); reducing busing area (Ex. 1003 ¶ 205 (citing Ex. 1017, 3:45–53)); reducing the load that would be imposed on an otherwise centralized memory interface chip (Ex. 1003 ¶ 206 (citing Ex. 1018, 1:5–23, Fig. 3; Ex. 1019, 1:32–2:3, Fig. 5)); easy replacement of BIST components (Ex. 1003 ¶ 207); the general best practices circuit design as understood by the art (Ex. 1003 ¶ 208 (citing Ex. 1021, 6; Ex. 1010, 207; Ex. 1021, 6; Ex. 1020, 2:27–3:6; Ex. 1010, 4–6, 56)). This litany of rationales indicates to us that this modification to Averbuj is a hindsight reconstruction that requires many pieces to come together in order to arrive at the claimed invention.

Nevertheless, we do acknowledge that Tsern contains structures that suggest the claimed configuration of the data handlers, but in this ground Petitioner does not assert Tsern as a reference, rather simply as evidence of the knowledge of one of skill in the art. Pet. 51–55. Relying on knowledge in the art does not obviate the need to prove a reference is prior art as of the critical date and show a motivation to combine existing components in the manner claimed free of hindsight motivation. This amounts to an argument that the individual elements of the claim were known and that combining them would have yielded the claimed invention, but Petitioner provides insufficient explanation why an ordinarily skilled artisan would have combined those elements in the claimed manner without the explicit reliance on the structures of Tsern. *See KSR*, 550 U.S. at 418 (“[A] patent composed of several elements is not proved obvious merely by demonstrating that each

of its elements was, independently, known in the prior art.); *Metalcraft of Mayville, Inc. v. The Toro Co.*, 848 F.3d 1358, 1366 (Fed. Cir. 2017) (“it is insufficient to simply conclude the combination would have been obvious without identifying any reason why a person of skill in the art would have made the combination”) (citation omitted).

For the above reasons, we find the evidence and argument presented on the ground based on Averbuj alone is insufficient to show by a preponderance of the evidence that claims 2–7 would have been obvious.

Claims 3–7 depend from claim 2, and Petitioner’s arguments with respect to those claims fail to cure the deficiencies of the Petition with respect to claim 2; that is, Petitioner has not shown sufficiently that Averbuj teaches or suggests the physically separate components mounted on a printed circuit board limitation for the reasons discussed above. Thus, Petitioner has failed to establish, by a preponderance of the evidence, its assertion that claims 2–7 would have been obvious over Averbuj.

G. Claims 1–7 — Obviousness over Averbuj (Ex. 1005) and Tsern (Ex. 1006) (Ground 3)

Petitioner argues that claims 1–7 are unpatentable over Averbuj and Tsern under 35 U.S.C. § 103. Pet. 60–75. We refer to this as “Ground 3.” Claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. Upon consideration the full record of evidence and arguments, we determine that Petitioner has shown by a preponderance of the evidence that claims 2–7 are unpatentable under 35 U.S.C. § 103(a) as obvious over Averbuj and Tsern.

1. Tsern

Figure 1 of Tsern is reproduced below.

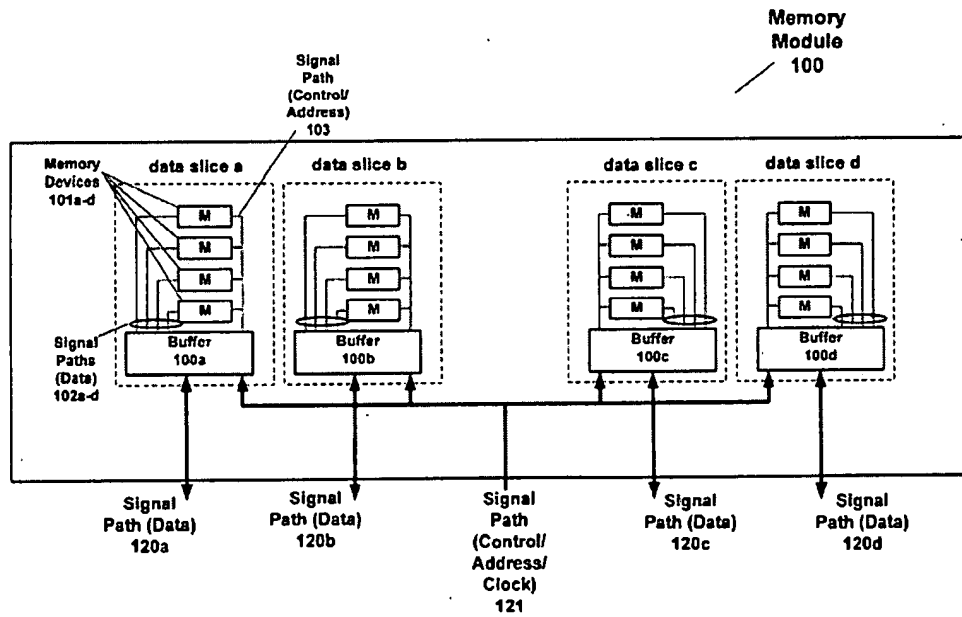


Fig. 1

Figure 1 of Tsern, above, illustrates an embodiment of a memory module with memory devices (101a-d) and corresponding buffer devices (100a-d). Figure 18 of Tsern is reproduced below.

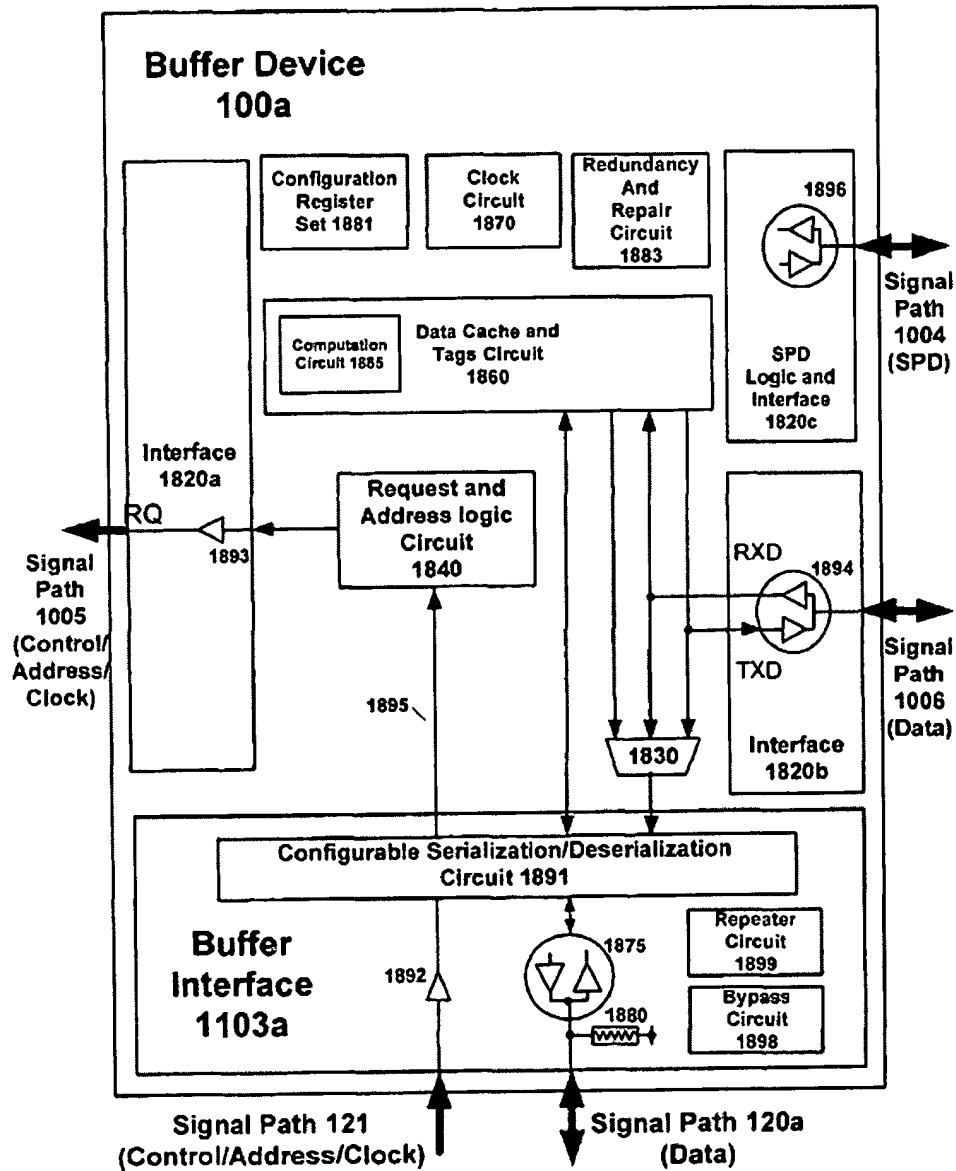


Fig. 18

As shown in Figure 18, above, Tsern discloses that, in one embodiment, each buffer device (100a) includes a redundancy and repair unit (1883), which tests and repairs the corresponding memory device(s). Ex. 1006, Fig. 18, ¶ 77. Each such buffer device (100a) includes data,

address, and control interfaces (1820a and 1820b) that can be programmed or configured to support memory modules with different number, size, width, and type of memory devices. *Id.* ¶ 99, Fig. 18.

Figure 1 of Tsern shows that each buffer device (100a–d) communicates address, control, and data signals with its corresponding memory device(s) (101a–d) without being in communication with any of the other buffer devices (100a–d) or other data ports of the other memory devices. Ex. 1006, Fig. 1.

2. Analysis – Claim 2

As to claim 1, Petitioner includes an obviousness ground with Tsern to the extent we determine that Averbuj does not disclose, or teach or suggest, “a plurality of memory devices on the printed circuit board.” Pet. 61 (“To the extent one might argue that Averbuj does not disclose “*a printed circuit board*” as required by claim 1, it would have been obvious to include it in the system of Averbuj per the teachings of Tsern”). As discussed above, claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel, and we determined that claim 1 is also rendered obvious by Averbuj.

As to claims 2–7, we did not find that those claims are anticipated by, or would have been obvious over, Averbuj. Thus, below we discuss the combination of Averbuj and Tsern as it relates to claims 2–7.

a. Claim 2 - PCB

Claim 2 recites “the plurality of data handlers comprise at least two physically separate components mounted on the printed circuit board.” Petitioner relies on Figure 9 of Tsern, reproduced below.

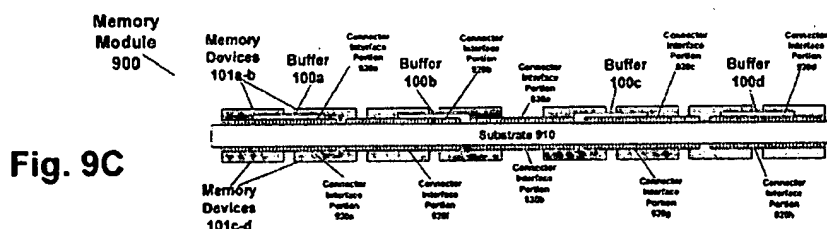
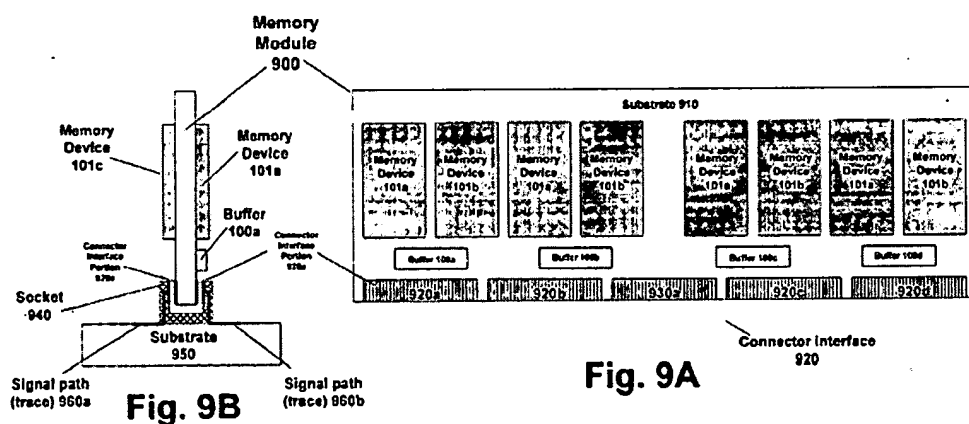


Figure 9 of Tsern, above, shows an implementation of a memory module in which each of the four buffer devices (100a–d) is implemented as a physically separate integrated circuit package and mounted on different portions of the printed circuit board that houses the memory module. Ex. 1006, Figs. 9A–C.

b. Claim 2 – “mounted on a printed circuit board”

As to claim 2’s limitation to data handlers mounted on the printed circuit board, Petitioner contends Tsern “discloses a memory module including a number of memory devices, such as DRAM or SRAM devices and a number of buffer devices, in several embodiments.” Pet. 61 (citing (Ex. 1006 ¶¶ 33, 35, 42, 52–59, Figs. 1, 9. Petitioner also assert “Tsern [] discloses that his memory module may consist of a number of buffer circuits

and associated memory devices mounted on a printed circuit board (“PCB”).” *Id.* (citing Ex. 1006 ¶ 52, Fig. 9A.) Petitioner also asserts “Tsern [] discloses that [] buffer circuits implement the reading and writing of data from and to the memory devices, and also may include self-test functionality.” *Id.* at 61–62 (citing Ex. 1006 ¶ 97). Petitioner asserts “[t]he use of the printed board of Tsern in the system of Averbuj, in order to mount the memory devices and data generation circuitry for example, would therefore have been merely the use of a known structure for its known purpose to achieve the predictable result of mounting such electronic components in a sturdy and efficient substrate.” Pet. 62 (citing Ex. 1003 ¶ 236). Petitioner cites other reasons for combining Tsern’s PCB with Averbuj. *Id.*

Based on Petitioner’s reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern’s teachings, to include memory components mounted on a PCB, as recited in claim 2. Pet. 60–62. For example, Petitioner asserts:

Tsern discloses a memory module including a number of memory devices such as DRAM or SRAM devices (Ex. 1006 at ¶ [0035]), and a number of buffer devices, in several embodiments. *See, e.g.*, Ex. 1006 at ¶¶ [0033-0042, 0052-0059]; FIGS. 1 & 9. Tsern further discloses that his memory module may consist of a number of buffer circuits and associated memory devices mounted on a printed circuit board (“PCB”). Ex. 1006 at ¶ [0052], FIG. 9A. Tsern also discloses that his buffer circuits implement the reading and writing of data from and to the memory devices, and also may include self-test functionality.

Ex. 1006 at ¶ [0097]. The use of the printed board of Tsern in the system of Averbuj, in order to mount the memory devices and data generation circuitry for example, would therefore have been merely the use of a known structure for its known purpose to achieve the predictable result of mounting such electronic components in a sturdy and efficient substrate. Ex. 1003 ¶ 236.

Id.

c. Claim 2 – “at least two physically separate components”

As to claim 2’s limitation to data handlers comprising at least two physically separate components, Petitioner asserts that “Tsern further discloses that [its] memory devices and buffer devices may be separate devices that can be housed in separate packages.” Pet. 64 (citing Ex. 1006 ¶¶ 31, 36, 53; Ex. 1003 ¶ 242). Thus, Petitioner contends that Averbuj’s data handler’s (memory devices 10A, 10B, and 10C) could be modified to reside in separate components as suggested by Tsern.

Petitioner further contends:

It would have been obvious to include the “separate components” configuration of Tsern, including the distribution of self-test circuitry, in the system of Averbuj for several reasons. First, to do so would have been merely the arrangement of old elements with each performing the same function it had been known to perform and yielding no more than one would expect from such an arrangement. Tsern demonstrates that, before the priority date of the ‘434 Patent, it was within the average skill of the art to include a plurality of data handler circuits in physically separate components, such as separate circuit packages, on a printed circuit board and that such circuits would operate as expected. Ex. 1003 ¶ 244.

A skilled artisan would have been further motivated to include the “separate components” configuration of Tsern in the system of Averbuj, *particularly for the Memory Interfaces of*

Averbuj, in order to reduce the busing area a centralized approach would require. *See, e.g.,* Ex. 1017 at 3:45-53 (“It has been proposed to reduce the busing area by using a separate pattern generator for each array to be tested and routing only a simple coded instruction from the controller to the pattern generator to instruct the pattern generator which of a set of canned tests stored in the pattern generator to execute. This approach saves on routing area at the expense of the area necessary to create individual pattern generators to test a plurality of memories.”) (emphasis added). Ex. 1003 ¶ 245.

Id. at 64–65. Thus, Petitioner asserts that the memory interfaces (the alleged data handlers) are specifically physically separate components. Petitioner also contends:

a skilled artisan, particularly a skilled artisan looking to add new BIST circuitry for a memory module according to the teachings of *Averbuj*, would have been motivated to incorporate each component of *Averbuj*’s BIST architecture—*i.e.*, BIST controller, Sequencers, and memory interfaces—into separate integrated circuit chips/packages. Ex. 1003 ¶ 202. This design approach had already been proposed and/or practiced by others (*see, e.g.,* Ex. 1006 at FIGS. 9A-C), and thus a skilled artisan would have known and have been motivated to try this approach. Ex. 1003 ¶ 203.

Pet. 53. Thus, Petitioner contends that *Averbuj*’s device could be modified such that the “data handlers” would be in separate ICs mounted on the PCB. Based on Petitioner’s reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it also would have been obvious to modify the teachings of *Averbuj* with Tsern’s teachings to include data handlers as separate components mounted on a PCB, as recited in claim 2. Pet. 62–64.

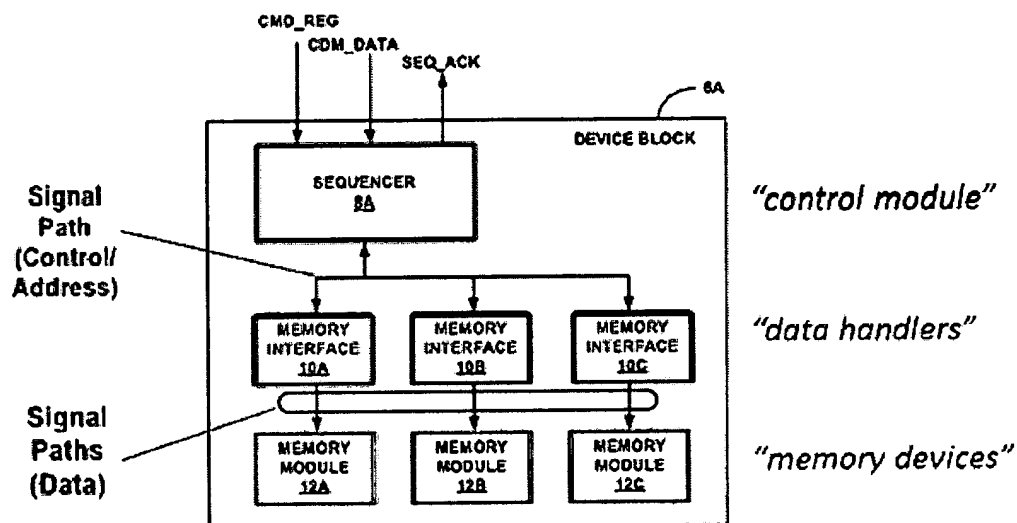
We also find that these findings regarding claim 2 are consistent with the findings of the '970 IPR that Averbuj meets the limitations of claim 1 such that those limitations, as they apply to claim 2, are met by collateral estoppel in this *inter partes* review.

d. Patent Owner Response –Mapping of claim 1

Patent Owner's Response to the Petition was directed to an alternative mapping in which the address generation unit is mapped to the "control module." See generally PO Resp. Thus, the arguments in the Patent Owner Response are not relevant to any issue we seek to resolve in this Final Written Decision. This is because Patent Owner contends that Petitioner did not assert in the Petition that the limitations of claims 2–7 were met under the same mapping of Averbuj to claim 1 as was used in the '970 IPR. Mot. To Strike, 2–5.

To summarize the argument, the Petition presents three mappings that are directed to which component of Averbuj is mapped to the "control module" recited in claim 1 "Mapping 1," which is the same mapping used in the '970 IPR, maps Averbuj's Sequencer to the "control module." Pet. 30–32. "Mapping 2," which was not at issue in the '970 IPR, maps the address generation unit to the "control module." Pet. 32–34. "Mapping 3," which was also not used in the '970 IPR, maps both the Sequencer and the address generation unit to the "control module." Pet. 34.

To illustrate Patent Owner's argument we include below several annotated figures provided by Patent Owner. Below is Patent Owner's annotated version of Figure 4 of Averbuj alleged to represent Mapping 1. PO Show Cause Resp. 3.



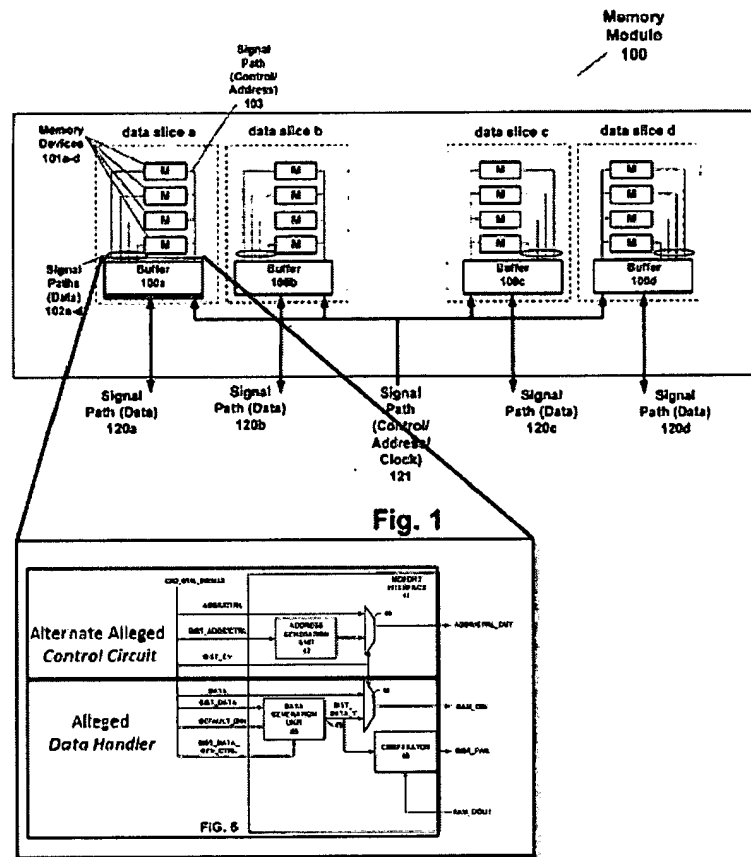
Averbuj, Fig. 4

MAPPING 1

The annotated Figure 4, above, shows Mapping 1 of claim 1 limitations to Averbuj, the mapping used in the '970 IPR, in which the Sequencer alone is the "control module."⁸ PO Show Cause Resp. 3; Pet. 30–32 (Petitioner's contention that the sequencer alone meet the "control module" limitation).

Below are Patent Owner's edited and annotated versions of Figures 1 and 6 of Averbuj. PO Resp. 62.

⁸ Unlike other Patent Owner annotated figures used in this Final Written Decision, this annotated figure does not show how each element would be placed on a separate IC as required by claim 3, however, in that case each element in the colored box would represent a different IC.



MAPPING 2

The annotated Figures 1 and 6, above, represents Mapping 2 in which the Address Generation Unit of Averbuj is mapped to the “control module.”

Patent Owner does not include an annotated figure for Mapping 3 but it would be similar to Mapping 2 but the Sequencer would be included in the red box containing the Address Generation Unit. *See* Pet. 66.

As noted above, in Ground 3 discussing the combination of Averbuj and Tsern, the Petition states: “A skilled artisan would have been further motivated to include the ‘separate components’ configuration of Tsern in the system of Averbuj, *particularly for the Memory Interfaces of Averbuj*, in

order to reduce the busing area a centralized approach would require.” Pet. 65 (emphasis added) (citations omitted). This statement does not mention at all where the “control module” is located or what component is mapped to the “control module.” Petitioner asserts that claim 2 is directed to separating the “data handlers” from each other and is, at best, agnostic as to whether the “control module” is in the same IC as the data handlers. Tr. 12:14–19. We find these facts are sufficient to find that Petitioner did not limit its contentions regarding the combination of Averbuj and Tsern to Mapping 2 or Mapping 3.

Nevertheless, Patent Owner presents a further argument that claim 2 requires a one-to-many relationship between the “control module” and the “data handlers” such that Petitioner’s theory under Ground 3 cannot meet the limitations of claim 1. We discuss that below.

e. Patent Owner Motion to Strike –One-to-Many

Patent Owner, in the Motion to Strike, asserts that because claim 1 recites that “the circuit is configured to test the memory devices using the address and control signals generated by the control module and the data generated by the plurality of data handlers” there must be a one to many relationship between the “control module” and the “data handlers.” Mot. to Strike 1–2. We disagree. The claim explicitly recites using signals from a “control module” and from a plurality of data handlers, but no explicit required connection between the plurality of “data handlers” and the “control module” is recited such that a one-to-many relationship is required. We also do not find any one-to-many relationship limitation was imported into the claims from the Specification by the Patentee. For that reason, as

explained above, Petitioner has set forth a sufficient contention that the combination of Avebuj and Tsern would render obvious putting the “data handlers” in physically separate components.

Nevertheless, we will analyze this issue assuming that a one-to-many relationship is required such that having one “control module” in a chip with only one “data handler” would not meet the limitations of claim 1, which are incorporated into claim 2. This exercise may be slightly difficult to follow because it appears that Petitioner did not anticipate this one-to-many argument and simply focused on showing that the “data handlers” each could be put in separate components regardless of the mapping of the “control module.” An additional difficulty is that this issue is separate and distinct from the mappings issue presented above.

i. Alleged Separate Component Theories

We begin by explaining Petitioner’s theories regarding where the “control module” and “data handlers” are mapped in the grounds presented in the Petition. In the Petition, as to anticipation by Avebuj, Petitioner argued that “a skilled artisan would understand that the various memory interfaces (10, 41) and their data generation circuitry (i.e., each data generation unit 44 and its associated multiplexor 45 and comparator 48) are ‘physically separate components.’” Pet. 41. Petitioner further argued that “the memory interfaces, each of which includes a data generation circuitry (“*data handler*”), are distributed among a plurality of memory chips mounted on a circuit board.” *Id.* at 42. Thus, Petitioner argued the memory interfaces 10A, 10B and 10C were physically separate. We disagreed that this contention, as presented in the Petition, met the reasonable likelihood

standard for institution, stating “claim 2 requires that the memory module be a ‘physically separate component’ and ‘mounted’ on the PCB. Petitioner has not shown sufficient support for its broad reading that a module which is a circuit embedded in a larger circuit included in a chip may be considered a physically separate component mounted on a PCB.” Inst. Dec. 22–23.

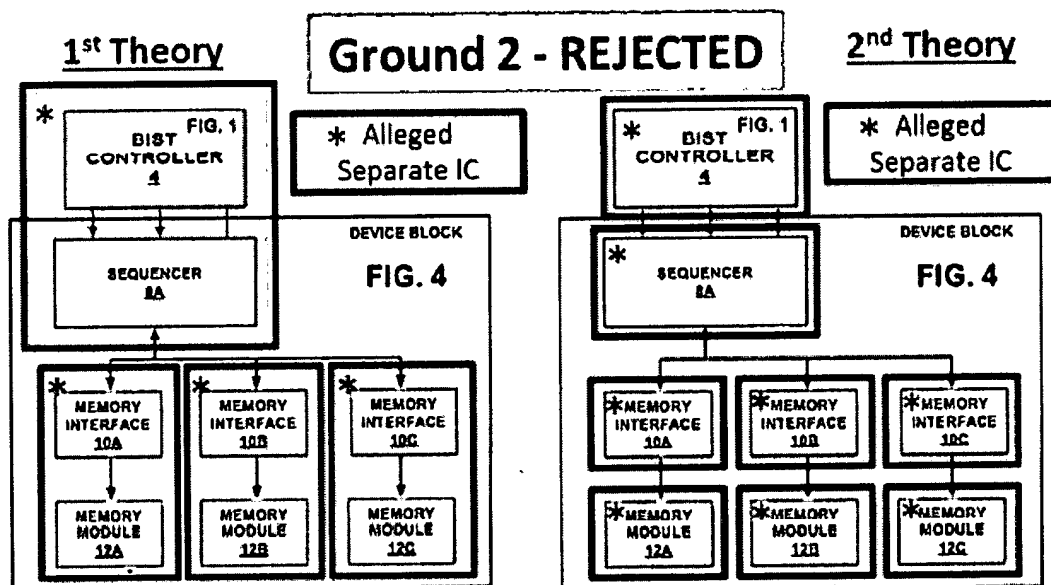
Thus, we found that Averbuj did not show sufficiently that memory interfaces 10A, 10B, and 10C were mounted on a PCB and physically separate under Ground 1 as to anticipation by Averbuj.

As to Ground 2, Petitioner argues that separating components of the BIST architecture into physically separate components would “have been the use of a known technique for its known purpose and would have achieved only predictable results, such as the protection of integrated circuits by the packaging.” Pet. 52. Petitioner asserts that memory modules at the time of the invention were composed of memory chips mounted on different portions of a memory board with BIST units in each memory chip. *Id.* Given its assertion that memory chips mounted on different portions of a memory board with BIST units in each memory chip were known, Petitioner asserts that reducing redundancy is a motivation to modify Averbuj.

Specifically, Petitioner presents a first alternative asserting “a skilled artisan would have understood that the BIST controller and Sequencer portions of conventional BIST units would need to be carved out from each memory chip [into its own chip] and combined to reduce redundant circuitry . . . [but] the memory interface portions need not be carved out and/or combined since doing so would not reduce much, if any, circuitry [thus remaining in a chip with the memory devices].” Pet. 53–54.

Later, also as to Ground 2, Petitioner presents a second alternative asserting “[a]lternatively, a skilled artisan, particularly a skilled artisan looking to add new BIST circuitry for a memory module according to the teachings of Averbuj, would have been motivated to incorporate each component of Averbuj’s BIST architecture—i.e., BIST controller, sequencers, and memory interfaces—into separate integrated circuit chips/packages.” Pet. 54.

Reproduced below are Patent Owner’s annotated versions of Figure 4 of Averbuj representing these two alternate theories as to Ground 2 which is obviousness based on Averbuj alone.



Patent Owner’s annotated versions of Figure 4, above, accurately shows two alternate theories which are associated with Ground 2 based on obviousness over Averbuj alone. These theories do not specify which

component is the “control module.” Pet. 53–54. In our Decision to Institute, we rejected these theories as relying on hindsight stating:

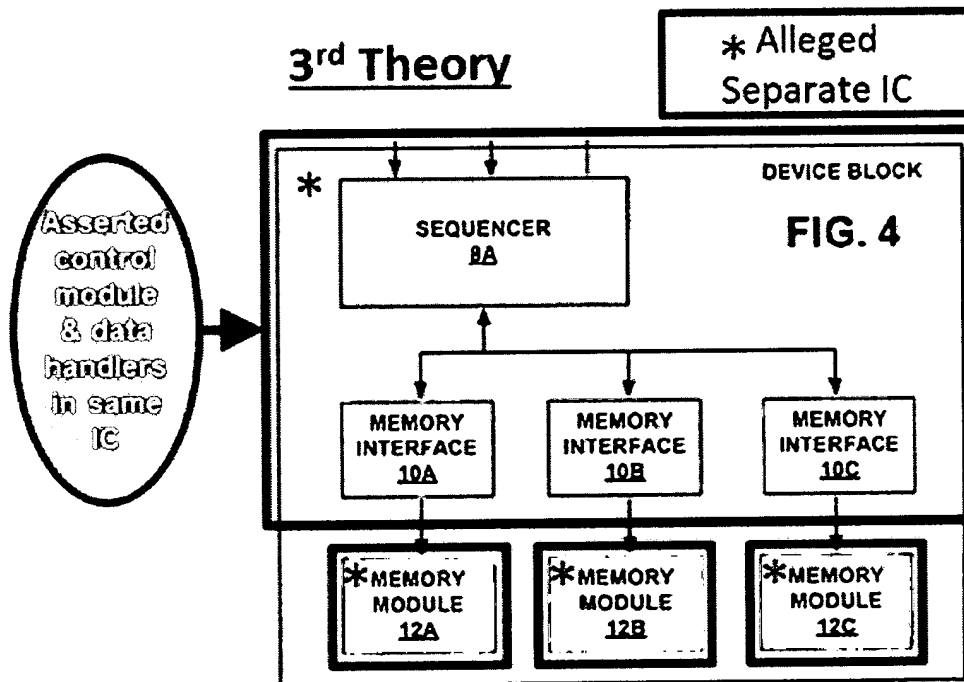
Patent Owner argues that “Averbuj, however, teaches away from the proposed combination because (as discussed above) Averbuj’s BIST unit and memory [modules] are in a single chip.” Prelim. Resp. 59. Patent Owner argues further that “modifying Averbuj in the proposed manner would render Averbuj inoperable for its intended purpose of verifying the functionality of internal circuitry of a chip while being included in the chip itself.” *Id.* at 59–60 (citing Ex. 1005 ¶ 5). Patent Owner’s arguments assume that Averbuj would not be modified based on the single chip structure of Averbuj. We do not rely on Patent Owner’s arguments. We focus, however, on the motivation given by Petitioner to modify Averbuj.

As to Petitioner’s arguments supported by its declarant, we find those to employ hindsight to piece together the device as claimed. The only evidence in the record that supports this argument, other than the long, complicated explanation of Petitioner’s declarant or the misplaced reliance on the structures of Tsern under this challenge, is in the ’434 Patent itself.

Inst. Dec. 25. Thus, we found only that there was insufficient motivation shown, absent a combination with Tsern, to break up Averbuj’s single chip architecture into an architecture in which the “data handlers” are physically separate.

Patent Owner represents what it considers Petitioner’s Ground 3 theory in the annotated Figure 4 of Averbuj below. Mot. To Strike, 3.

Ground 3 - INSTITUTED



As illustrated in the annotated Figure 4 of Averbuj, above, Patent Owner contends Petitioner is limited to a theory in which the Sequencer (the alleged “control module”) and the memory interfaces (the alleged data handlers) are in the same IC. Mot. To Strike, 3.

- ii. *Patent Owner Asserts Petitioner’s Ground 3 Theory does not include Sequencer Alone as the “Control Module”*

With that background, Patent Owner’s argument focuses on Ground 3 and the following statement in the Petition:

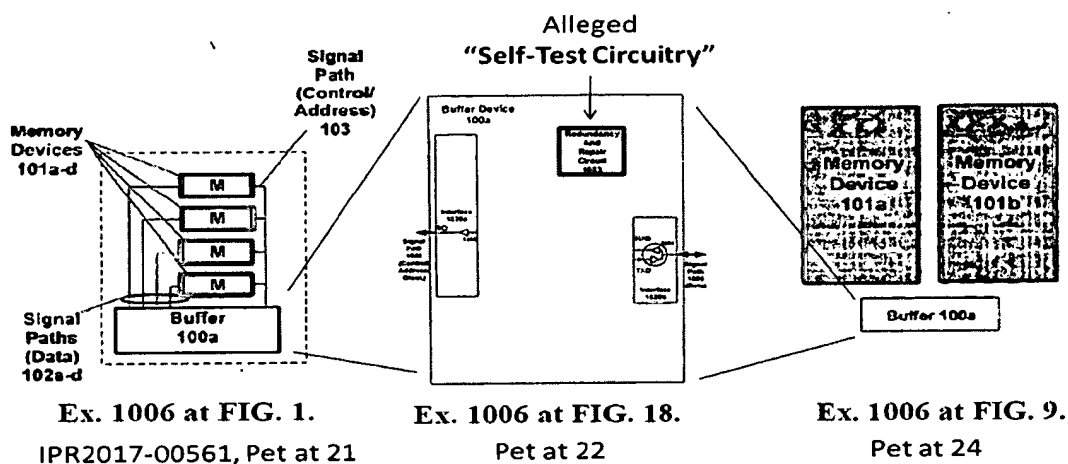
A skilled artisan also would also have been motivated to include the “separate components” configuration of Tsern in the system of Averbuj so that *the self-test circuitry could be conveniently placed in the same packaging as buffer circuitry* used to access

and isolate different portions of the memory array. By the priority date of the '434 Patent it was known that separate buffers, in separate packages, for different portions of the memory array could advantageously reduce the load experienced by the memory controllers and improve the memory timing. Ex. 1018 at 1:5-23, FIG. 3; Ex. 1019 at 1:32-2:3, FIG. 5. A skilled artisan would therefore have been motivated to *place the Sequencers and the Memory Interfaces of Averbuj within such buffer components* in order to reduce the load on the memory controller and also improve memory timing. Ex. 1003 ¶ 246.

Pet. 65–66 (emphasis added). Based on this statement, Patent Owner asserts Petitioner's theory under Ground 3 regarding the combination of Averbuj and Tsern is limited to a configuration in which the Sequencer and the memory interfaces are in the same chip such that the plurality of "data handlers" (memory interfaces) are not physically separate from the "control module." Mot. To Strike, 3–4. We note this paragraph does not mention the "control module" at all, but we will address that later. According to Patent Owner, this configuration would violate the so called "one-to-many" relationship required by claim 1. *Id.*

Petitioner responds that the statement that "the self-test circuitry could be conveniently placed in the same packaging as buffer circuitry" in the Petition was meant to express "that the components can be placed in the same type of packaging, not a single package." Mot. To Strike Resp. 5. We disagree. A reasonable reading of this statement is that a single buffer circuitry package would include "the self-test circuitry." Petitioner also responds that the statement "the Sequencers and the Memory Interfaces of Averbuj within such buffer components" in the Petition means "that each of the Sequencers (plural) and each of the Memory Interfaces can be placed in

a “buffer component” (i.e., “separate component”), not all in one.” *Id.* We disagree. A reasonable reading of this sentence is that the Sequencers and Memory Interfaces would both be in the buffer component of Tsern. This is supported by Patent Owner’s annotated figures, reproduced below, illustrating the Petition’s summary of Tsern.



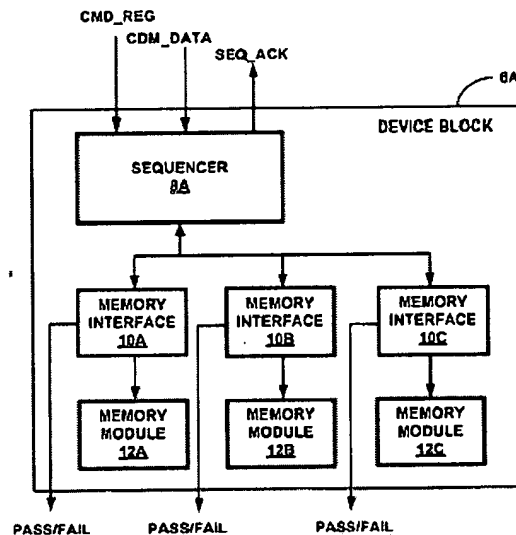
PO Show Cause Resp. 4. The annotated figures illustrate Patent Owner’s assertion that the buffer that Petitioner refers to as the “separate component” contains the “self-test circuitry” which contains the alleged “control module” (Sequencer) and the memory interface which is the alleged data handler. *Id.*

Nevertheless, Petitioner also responds that it had presented an argument that the “data handler” alone would be placed in separate chips. Resp. Mot. To Strike. 19–20. Petitioner ties the following statement from the Ground 2 Section of the Petition regarding obviousness over Averbuj alone to its argument regarding Averbuj in combination with Tsern.

It was known in the art to include circuit components of a memory system, *including those components responsible for handling the transfer of data to and from a memory device, in*

physically separate integrated circuit packages, and to do so would have been well within the average skill in the art. *See, e.g.*, Ex. 1006 at FIG. 9A; ¶¶ [0052-0053] (explaining that the depicted separate “buffer devices” are in one embodiment “housed in separate packages”); ¶ [0029] (noting that as used therein “an integrated circuit buffer device is also referred to as a buffer or buffer device.”); Ex. 1003 ¶ 192.

Pet. 53–54 (emphasis added). Petitioner’s declarant states in the cited portion of his testimony that “the idea to implement the components of a memory module, such as the components responsible for handling the transfer of data to and from a memory chip, in physically separate integrated circuit packages had already been proposed and adopted by the priority date of the ‘434 Patent.” Ex. 1003 ¶ 192 (citing Ex. 1006 at Fig. 9A, ¶¶ 52–53, 29). Reproduced below is an annotated version of Figure 4 of Averbuj from Petitioner’s declarant’s testimony. *Id.* ¶ 202.



Annotated Figure 4 above shows each memory interface in a separate IC. Ex. 1003 ¶ 202. This testimony is cited at page 54 of the Petition. Pet. 54. Thus, we find that Petitioner did present a contention that Averbuj, as

modified in light of the structures of Tsern, teaches a Sequencer in a separate component from its associated memory interfaces. We acknowledge that petitioners routinely rely on earlier anticipation or single reference obviousness grounds to support and build up later combination grounds. *See Silicon Labs, Inc. v. Cresta Tech. Corp.*, Case IPR2015-00626, slip op. at 38 (PTAB Aug. 11, 2016) (Paper 65) (“Patent Owner characterizes the ground of unpatentability as if Petitioner relies only on Grumman for a ‘signal processor’ as claimed. Such is not the case. The ground of unpatentability against claim 24 must be understood in the context that it builds upon the ground against claim 1.”).

Given that context, the Petition states later in the Ground 3 Section dealing with the combination of Averbuj and Tsern that

It was known in the art to include circuit components of a memory system, including those components responsible for handling the transfer of data to and from a memory device, in physically separate integrated circuit packages, and to do so would have been well within the average skill in the art. *See, e.g.*, Ex. 1006 at FIG. 9A; ¶¶ [0052-0053] (explaining that the depicted separate “buffer devices” are in one embodiment “housed in separate packages”); ¶ [0029] (noting that as used therein “an integrated circuit buffer device is also referred to as a buffer or buffer device.”); Ex. 1003 ¶ 192.

Pet. 64. In the quoted portion above, the Petition cites the same section of Tsern, i.e. Figure 9A and paragraphs 52 through 53, as cited in support of Petitioner’s contention that Averbuj alone rendered claim 2 obvious. The Petition also states “Tsern further discloses that his memory devices and buffer devices may be separate devices that can be housed in separate packages, *id.* at ¶¶ [0031, 0036, 0053]. Ex. 1003 ¶ 242.” *Id.* at 65. The

Petition also recites “[a] skilled artisan would have been further motivated to include the ‘separate components’ configuration of Tsern in the system of Averbuj, particularly for the Memory Interfaces of Averbuj, in order to reduce the busing area a centralized approach would require.” *Id.* (citing Ex. 1017 at 3:45–53.) Thus, as Petitioner points out in its response to the Motion to Strike, there are several statements in the Ground 3 portion of the Petition that can apply to Mapping 1, which was asserted in the ’970 IPR, and which places the Sequencer itself (as the alleged “control module”) in a separate IC from the “data handlers” thus satisfying the alleged one-to-many requirement of the claims. Mot. To Strike Resp., 2–5.

Patent Owner, in its Motion to Strike, argues that the contentions under this ground are either not sufficiently tied to Petitioner’s theory under Ground 3 or that they were specifically rejected as hindsight in the Institution Decision. Mot. to Strike. 2–5. We note Petitioner cannot expect the Board to read through the entire record, searching for relevant evidence that might support Petitioner’s arguments. *Cf., DeSilva v. DiLeonardi*, 181 F.3d 865, 866–67 (7th Cir 1999)(“[B]rief must make all arguments accessible to the judges, rather than ask them to play archaeologist with the record.”). Nevertheless, the reference to Tsern and its structures is plainly relevant to Ground 3 based on Averbuj and Tsern. We decline to limit Petitioner’s Ground 3 based on obviousness over Averbuj and Tsern to only the arguments made specifically in the Section of the Petition labeled Ground 3. We find that, based on all the parts of the Petition cited above, the Petition—without any *ex post facto* reconstruction by the Board—can be reasonably read to present two theories based on Averbuj and Tsern—one in

which each portion of the device is in a separate component (or IC) and possibly one in which the Sequencer and memory interfaces are together in the same component. *Cf. In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1380–81 (Fed. Cir. 2016) (rejecting an argument that the Board properly “ma[de] an obviousness argument on behalf of [petitioner]” that “could have been included in a properly drafted petition,” because “petitioner ... bears the burden of proof”). As noted above, we rely on collateral estoppel to show the limitations of claim 1. Thus, for the reasons above, Petitioner has shown sufficiently in the Petition its theory that each portion of the device, including the Sequencer and each memory interface, would be in separate components mounted on a printed circuit board, as recited by claim 2 even under Patent Owner’s “one-to-many” construction of claims 1 and 2. Finally, as explained above, Petitioner has shown sufficient motivation to combine Averbuj and Tsern.

3. Claim 3

Claim 3 recites “the plurality of data handlers comprise at least two physically separate integrated circuit packages.” Petitioner contends Tsern discloses that the buffer circuitry used to access his memory devices, including the circuitry for handling data, may comprise multiple “physically separate integrated circuit packages.” Pet. 66–67 (citing Ex. 1006 ¶¶ 31, 36, 53 (“In an embodiment, each memory device and buffer device are housed in separate packages.”)). We are persuaded by and adopt Petitioner’s contention that Tsern suggest placing components such as the “data handlers” in separate ICs in the context of BIST architecture. Based on Petitioner’s reliance upon specific prior art disclosures and supporting

evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern's teachings, to include "data handlers" in separate ICs, as recited in claim 3. Pet. 67–68. For example, Petitioner asserts:

It would have been obvious to include the "separate integrated circuit packages" configuration of Tsern in the system of Averbuj for reasons similar to those set forth above. For example, to do so would have been merely the arrangement of old elements with each performing the same function it had been known to perform and yielding no more than one would expect from such an arrangement for the same reasons demonstrated above, i.e., Tsern demonstrates that such a configuration was accomplished in the prior art without any unexpected results. Ex. 1003 ¶ 250.

A skilled artisan also would have been motivated to include the "separate packages" configuration of Tsern in the system of Averbuj so that BIST components could be easily replaced or upgraded by simply replacing a standard package on the printed circuit board, without the necessity to replace other BIST circuitry in other packages. Ex. 1003 ¶ 251.

Moreover, as demonstrated above, a skilled artisan would have been motivated to distribute the "data handlers" near the buffer circuitry used to access memory devices in order to reduce the routing area needed for bussing. Ex. 1017 at 3:45-53; Ex. 1003 ¶ 252. Physically distributing the "data handlers" in that manner would have further motivated, and likely required, a skilled artisan to place the "data handlers" in separate packages in order to protect them, as one package holding all data handlers distributed across the printed circuit board would have been infeasible. Ex. 1003 ¶ 252.

Also as demonstrate above, a skilled artisan would also have been motivated to use physically separate integrated circuit

packages for the data generation circuitry (44, 45, 48) (“data handlers”) of Averbuj in order to place each data handler at positions on the PCB close and perhaps closest to its associated memory devices, thereby simplifying the wiring plan on the circuit board (Ex. 1021 at 6; Ex. 1003 ¶ 224), reducing capacitance/propagation delay of the data lines (Ex. 1020 at 2:27-3:6; Ex. 1003 ¶ 225), and reducing the inductance/crosstalk of the data lines (Ex. 1010 at 23-25, 207; Ex. 1003 ¶ 226). Ex. 1003 ¶ 253.

Id. Thus, we agree that for ease of replacement and in order to place data generation circuitry near associated memory devices, *inter alia*, one of ordinary skill would be motivated to place Averbuj’s data generators in separate ICs as taught by Tsern.

4. Claim 4

Claim 4 recites “the physically separate integrated circuit packages are mounted on different portions of the printed circuit board.” Petitioner contends Tsern discloses this limitation and teaches configuring Averbuj to meet this limitation. Pet. 67 (citing Ex. 1006, Fig. 9A (Buffers 100a–d); ¶¶ 31, 36, 52, 53; Ex. 1003 ¶ 255). We are persuaded by and adopt Petitioner’s contention that Tsern suggests placing components of Averbuj such as the “data handlers” in separate ICs on different portions in the context of BIST architecture. Based on Petitioner’s reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern’s teachings, to include “data

handlers” in separate ICs, as recited in claim 3. Pet. 67–68. For example, Petitioner asserts:

It would have been obvious to adopt such a configuration for the “data handlers” of Averbuj because to do so would have been merely the arrangement of old elements with each performing the same function it had been known to perform and yielding no more than one would expect from such an arrangement, i.e. placing such circuitry closer to the memory devices with which it would interface. Ex. 1003 ¶ 257.

Moreover, as demonstrated above, it was known that distributing data handling circuitry in separate components or packaging placed closer to memory devices associated with that circuitry advantageously provided the ability to isolate the load associated with the memory device and control signal timing more effectively (Ex. 1018 at 1:5-23, FIG. 3; Ex. 1019 at 1:32-2:3, FIG. 5.), and reduced routing area for signal busses (Ex. 1017 at 3:45-53). Ex. 1003 ¶ 258. A skilled artisan would therefore have been motivated to place the data handlers of Averbuj in physically separate integrated circuit packages mounted on different portions of the printed circuit board for each of those reasons, particularly since Averbuj discloses the use of shared circuitry for applying BIST signals and normal operation memory access signals, as explained above. Ex. 1003 ¶ 258.

Further, as demonstrated above, there are only a limited number of places such components could be placed on a standard printed circuit board, so it would have been obvious to try to place each “data handler” at different portions of the board. And a skilled artisan would also have been motivated to place each “data handler” at different portions of the board in order to (i) simplify the wiring paths (so that all connections were not running to/from the same portion of the board) (*see, e.g.*, Ex. 1021 at 6); (ii) permit faster data transmission by reducing the capacitance of the signal lines (*see, e.g.*, Ex. 1020 at 2:27-3:6);

and (3) improve signal integrity by reducing impedance on signal lines (*see, e.g.*, Ex. 1010 at 4-6, 56)[]. Ex. 1003 ¶ 259.

Id. Thus, we agree that to simplify the wiring paths, provide faster data transmission by reducing the capacitance of the signal lines, and improve signal integrity by reducing impedance on signal lines, *inter alia*, one of ordinary skill would be motivated to place Averbuj's data generators in separate ICs as taught by Tsern.

5. *Claim 5*

Claim 5 recites "each of the plurality of data handlers is positioned on the printed circuit board proximate to the corresponding plurality of data ports." Petitioner contends Tsern discloses this limitation and teaches configuring Averbuj to meet this limitation. Pet. 67 (citing Ex. 1006, Figs. 1, 9A (Buffers 100a–d)). We are persuaded by and adopt Petitioner's contention that Tsern would have suggested Averbuj's data handlers in proximity to the corresponding plurality of data ports. Based on Petitioner's reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern's teachings, to include "data handlers" in separate ICs, as recited in claim 3. Pet. 67–68. For example, Petitioner asserts:

It would have been obvious to employ such placement with the data handlers of Averbuj because to do so would have been the arrangement of old elements (data handler placed near associated memory devices) with each performing the same function it had been known to perform (the expected ability to access the memory devices via the data handlers) and yielding no

more than one would expect from such an arrangement (efficient access to the memory device). Ex. 1003 ¶ 263.

Moreover, a skilled artisan would have been motivated to place the “*data handlers*” of Averbuj nearby and near the data ports of its associated “*memory device*” in order to (i) simplify the wiring (Ex. 1021 at 6; Ex. 1003 ¶ 224), reduce capacitance affecting signal transmission speed (Ex. 1020 at 2:27-3:6; Ex. 1003 ¶ 225), and reduce signal crosstalk affecting signal integrity (Ex. 1010 at 23-25, 207; Ex. 1003 ¶ 226). Ex. 1003 ¶ 264.

Id. Thus, we agree that to simplify the wiring paths, reduce capacitance affecting signal transmission speed, and reduce signal crosstalk affecting signal integrity, *inter alia*, one of ordinary skill would have been motivated to place Averbuj’s data generators in separate ICs as taught by Tsern.

6. *Claim 6*

Claim 6 recites “each of the plurality of data handlers is positioned closer to the corresponding plurality of data ports than to the other data ports of the plurality of memory devices.” Petitioner contends Tsern discloses this limitation and teaches configuring Averbuj to meet this limitation. Pet. 72 (Ex. 1006 at Figs. 1, 9A). Petitioner asserts, for example, buffer 100a of Fig. 1 of Tsern is closer to the ports of its associated memory devices (“data slice a”) than buffers 100b–d are to those data ports. *Id.* (citing Ex. 1003 ¶ 266). We are persuaded by and adopt Petitioner’s contention that the cited portion of Tsern suggests Averbuj’s data handlers being positioned closer to the corresponding plurality of data ports than to the other data ports of the plurality of memory devices. Based on Petitioner’s reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational

underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern's teachings, to include "data handlers" positioned closer to the corresponding plurality of data ports than to the other data ports of the plurality of memory devices, as recited in claim 6. Pet. 73. For example, Petitioner asserts:

It would have been obvious to employ such circuit placement with the data handlers of Averbuj because to do so would have been the arrangement of old elements (data handler placed closer to associated memory devices) with each performing the same function it had been known to perform (the expected ability to access the memory devices via the data handlers) and yielding no more than one would expect from such an arrangement (efficient access to the memory device). Ex. 1003 ¶ 269.

Moreover, a skilled artisan would have been motivated to place the "*data handlers*" of Averbuj nearby and closer to the data ports of its associated "*memory device*" in order to (i) simplify the wiring (Ex. 1021 at 6; Ex. 1003 ¶ 224), reduce capacitance affecting signal transmission speed (Ex. 1020 at 2:27-3:6; Ex. 1003 ¶ 225), and reduce signal crosstalk affecting signal integrity (Ex. 1010 at 23-25, 207; Ex. 1003 ¶ 226). Ex. 1003 ¶ 271.

Id. Thus, we agree that to simplify the wiring paths, reduce capacitance affecting signal transmission speed, and reduce signal crosstalk affecting signal integrity, *inter alia*, one of ordinary skill would have been motivated to place Averbuj's data generators closer to the corresponding plurality of data ports than to the other data ports of the plurality of memory devices as taught by Tsern.

7. *Claim 7*

Claim 7 recites “each of the data handlers is further configured to read from the corresponding plurality of data ports and further comprises a verification element for checking for failures in the operation of the memory devices by verifying that data read from the corresponding plurality of data ports corresponds to the data generated by the data handler for writing to the corresponding plurality of data ports.” Petitioner contends Tsern discloses this limitation and teaches configuring Averbuj to meet this limitation. Pet. 74 (citing Ex. 1006 at Figs. 1, 9A). Petitioner points to, for example, the inclusion of self-test circuitry in the Redundancy and Repair Circuit 1883 of Tsern’s buffer device. Pet. 74 (Ex. 1006 ¶ 97, Fig. 18.) Petitioner also asserts Tsern explains that such circuitry writes data into associated memory cells, reads that data out and then makes a comparison of the written to the read data in order to identify defective memory locations. *Id.* We are persuaded by and adopt Petitioner’s contention that cited portion of Tsern would suggest configuring Averbuj’s “data handlers” as recited in claim 7. Based on Petitioner’s reliance upon specific prior art disclosures and supporting evidence, we are persuaded that Petitioner sets forth sufficient articulated reasoning with rational underpinning to support the legal conclusion that it would have been obvious to modify the teachings of Averbuj with Tsern’s teachings, to configure “data handlers” as recited in claim 7. Pet. 75–76. For example, Petitioner asserts:

It would have been obvious to include this functionality in the system of Averbuj. Ex. 1003 ¶ 274.

First, to do would have been merely the use of a known technique for its known purpose to achieve a predictable result,

i.e., the identification of working and defective memory cells.
Ex. 1003 ¶ 275.

Moreover, a skilled artisan would have been motivated to include this functionality in Averbuj, since Averbuj is directed to the testing of memory cells and [] already discloses writing test data into those cells and the circuitry for comparing such data to data read from those cells. Thus, a skilled artisan would have been motivated to accomplish the goal of Averbuj – the testing of memory cells using Averbuj’s components. Ex. 1003 ¶ 275.

Id. Thus, we agree by using a known technique for its known purpose to achieve a predictable result, i.e., the identification of working and defective memory cells, *inter alia*, one of ordinary skill would have been motivated to configure Averbuj’s data generators as recited in claim 7.

8. Summary—Claims 2–7

For the reasons above, we are persuaded by Petitioner’s contentions as to the limitations of claims 2–7. Accordingly, upon review of the full record before us, we determine that Petitioner has demonstrated by a preponderance of the evidence that claims 2–7 are unpatentable over the combination of Averbuj and Tsern.

H. Patent Owner’s Motion to Strike

Patent Owner moves to strike pages 17–22 of Petitioners’ Reply and to exclude any argument based thereon, as violating Patent Owner’s due process and procedural rights. Mot. to Strike 1. Pages 17–22 of Petitioner’s Reply are directed to Ground 3 of the Petition which challenges claims 1–7 as rendered obvious by Averbuj and Tsern. In those pages, Petitioner argues that Patent Owner ignored Petitioner’s alternative theory that the Sequencer alone is the “control module.” Reply 17–22. As explained above (Sections

II. E. d.–e.), we do not find that Petitioner’s Reply introduces a new theory. *See* Mot. to Strike Resp. 1–5. Thus, Patent Owner’s Motion to Strike is denied.

I. Petitioner’s Motion to Exclude

Petitioner moves to exclude Patent Owners’s Exhibits 2005, 2006, 2012, 2013, 2014 and paragraphs 46–59 and 61–69 of Exhibit 2010 (Dr. Baker’s Declaration). Each of the exhibits and portions of exhibits that Petitioner moves to exclude relate to the construction of the term “memory module” and/or whether a PCB is included in the definition of memory module. This issue is moot because the limitations of claim 1 have been shown to be met by Averbuj by collateral estoppel and we explicitly find that Tsern teaches a PCB as recited in claims 2–7. Thus, it is unnecessary to construe the term “memory module” in this decision. Thus, we deny Petitioner’s Motion to Exclude.

J. Patent Owner’s Motion to Exclude

Patent Owner summarily moves to exclude paragraphs 1–50, 55–63, 71, 81, 88, 92, 93, 98, 105, 116, 117, 134, 137, 146, 147, 151, 152, 155–157, 161, 162, 164, 169, 170, 173, 178, 185, 191, 206, 211, 215, 220, 227–229, 234, 239, 243, 248, 249, 256, 267, 270 and 273 of Exhibit 1003, along with the entirety of Exhibits 1004 (Dr. Mazumder’s CV) and 1022 because those paragraphs/exhibits were submitted with, but were not cited in, the Petition. PO Mot. to Excl. 1.

Petitioner explains: why each of the paragraph/exhibits are referred to indirectly; why they are relevant to a foreseeable issue even though they are not explicitly cited; and why Petitioner’s reliance on these exhibits is not

unfairly prejudicial to Patent Owner. PO Mot. to Excl. Reply 1–3. Patent Owner does not respond to those specific arguments, except as to Exhibit 1022. The paragraphs cited in Exhibit 1003 are general background of the expert and how he conducted his analysis and summaries of analysis that is cited in the Petition. *Id.* Additionally, as explained above in relation to Petitioner’s Motion to Exclude, many of the relevant paragraphs are moot due to the collateral estoppel issue.

The Board acts as both the gatekeeper of evidence and as the weigher of evidence. Rather than excluding evidence that is allegedly confusing, misleading, untimely, and/or irrelevant, we will simply not rely on it or give it little weight, as appropriate, in our analysis. Similar to a district court in a bench trial, the Board, sitting as a non-jury tribunal with administrative expertise, is well positioned to determine and assign appropriate weight to evidence presented, including giving it no weight. *See, e.g., Donnelly Garment Co. v. NLRB*, 123 F.2d 215, 224 (8th Cir. 1941) (“One who is capable of ruling accurately upon the admissibility of evidence is equally capable of sifting it accurately after it has been received”). Thus, in this *inter partes* review, the better course is to have a complete record of the evidence to facilitate public access as well as appellate review. Thus, as to the relevant paragraphs of Exhibit 1003 and as to Exhibit 1004, Patent Owner’s Motion to Exclude is denied.

As to Exhibit 1022, this is a reference relied on by Dr. Mazumder to show the state of the art. PO Mot. to Excl. Reply 3. Patent Owner argues that this is an improper incorporation by reference. PO Mot. to Excl. Reply. 2–3 (citing 37 C.F.R. § 42.6(a)(3) (prohibiting incorporation by reference);

Oxford Nanopore Tech's Inc. v. University of Washington, Case IPR2014-00512, slip op. (PTAB Sept. 14, 2014) (Paper 12); *Biodelivery Sciences Intern., Inc. v. Monosol RX, LLC*, Case IPR2015-00167, slip op. (PTAB May 20, 2015) (Paper 6); *Conopco, Inc. v. Procter & Gamble Co.*, Case IPR2013-00510, slip op. at 8 (PTAB Feb. 12, 2014) (Paper 9) (“We decline to consider information presented in a supporting declaration, but not discussed in a petition, because, among other reasons, doing so would encourage the use of declarations to circumvent the page limits that apply to petitions.”). 37 C.F.R. § 42.6(a)(3) prohibits incorporating arguments into the Petition by reference to other documents, and 37 C.F.R. § 42.104(b)(5) gives us discretion to “exclude or give no weight to . . . evidence” where the Petition “fail[s] to state its relevance.” *See* 37 C.F.R. § 42.6(a)(3) (“Arguments must not be incorporated by reference from one document into another document.”); 37 C.F.R. § 42.104(b)(5) (“The Board may exclude or give no weight to the evidence where a party has failed to state its relevance or to identify specific portions of the evidence that support the challenge.”). Nevertheless, we are encouraged to consider all evidence of record that is probative of the background knowledge of a person of ordinary skill in the relevant technology. *Randall Mfg. v. Rea*, 733 F.3d 1355, 1362–63 (Fed. Cir. 2013). There is a balance between making sure the Petition is complete, as a stand-alone document, in stating everything necessary to meet the elements of establishing a preponderance of evidence of unpatentability, with the need for citation to, and reliance on, evidence to support each argument. Here, the balance tips in favor of Petitioner. The testimony regarding Exhibit 1022 is relevant to why Averbuj would be modified to the

configuration in claim 2 in which the components are on separate ICs. This argument is fully explored in the Petition despite the lack of specific citation to Exhibit 1022. *See* Pet. 51–55 (citing, *inter alia*, Ex. 1003 ¶¶ 197–198 (citing Exhibit 1022)). We do not find that Dr. Mazumder’s reference to this Exhibit introduces argument that was not contained in the Petition or introduces material that was either excessive or material that the relevance of which is not explained. Thus, we deny Patent Owner’s Motion to Exclude as to Exhibit 1022.

III. CONCLUSION

For the foregoing reasons, we determine claim 1 is anticipated by Averbuj under the doctrine of collateral estoppel. Thus, pursuant to 37 C.F.R. § 42.72, and the doctrine of collateral estoppel, this trial is terminated as to claim 1. Additionally, Petitioner has demonstrated, by a preponderance of the evidence, that claims 2–7 of the ’434 Patent are unpatentable as obvious in view of Averbuj and Tsern.

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 2–7 of the ’434 Patent have been shown to be unpatentable;

FURTHER ORDERED, pursuant to 37 C.F.R. § 42.72 and the doctrine of collateral estoppel, that trial is terminated as to claim 1; and

FURTHER ORDERED Patent Owner’s Motion to Strike is *denied*; Patent Owner’s Motion to Exclude is *denied*; and Petitioner’s Motion to Exclude is *denied*.

IPR2017-00561
Patent 8,001,434 B1

This is a final decision. Parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

PETITIONER:

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(12) **INTER PARTES REVIEW CERTIFICATE** (1077th)

United States Patent
Lee et al.

(10) **Number:** **US 8,001,434 K1**
(45) **Certificate Issued:** **Oct. 1, 2018**

(54) **MEMORY BOARD WITH SELF-TESTING
CAPABILITY**

(75) **Inventors:** **Hyun Lee; Jayesh R. Bhakta;
Soonju Choi**

(73) **Assignee:** **NETLIST, INC.**

Trial Numbers:

IPR2014-00970 filed Jun. 18, 2014
IPR2014-01372 filed Aug. 23, 2014

Inter Partes Review Certificate for:

Patent No.: 8,001,434
Issued: Aug. 16, 2011
Appl. No.: 12/422,925
Filed: Apr. 13, 2009

The results of IPR2014-00970 and IPR2014-01372 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,001,434 K1
Trial No. IPR2014-00970
Certificate Issued Oct. 1, 2018

1

2

AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claims **2-13, 16-18, 26** and **35** are found patentable.

5

Claims **1, 14, 15, 19-25** and **27-34** are cancelled.

* * * * *

NOTE: This disposition is nonprecedential.

**United States Court of Appeals
for the Federal Circuit**

NETLIST, INC.,
Appellant

v.

**SK HYNIX INC., SK HYNIX AMERICA INC., SK
HYNIX MEMORY SOLUTIONS INC.,**
Appellees

2018-2357, 2018-2358

Appeals from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in Nos. IPR2017-
00561, IPR2017-00562.

JUDGMENT

THOMAS J. WIMBISCUS, McAndrews, Held & Malloy,
Ltd., Chicago, IL, argued for appellant. Also represented
by WAYNE BRADLEY, RONALD SPUHLER.

JOSEPH A. MICALLEF, Sidley Austin LLP, Washington,
DC, argued for appellees. Also represented by CARTER
GLASGOW PHILLIPS, RYAN C. MORRIS, BRIAN R. NESTER.

THIS CAUSE having been heard and considered, it is

ORDERED and ADJUDGED:

PER CURIAM (PROST, *Chief Judge*, NEWMAN and
MOORE, *Circuit Judges*).

AFFIRMED. See Fed. Cir. R. 36.

ENTERED BY ORDER OF THE COURT

December 12, 2019
Date

/s/ Peter R. Marksteiner
Peter R. Marksteiner
Clerk of Court

**United States Court of Appeals
for the Federal Circuit**

NETLIST, INC.,
Appellant

v.

**SK HYNIX INC., SK HYNIX AMERICA INC., SK
HYNIX MEMORY SOLUTIONS INC.,**
Appellees

2018-2357, 2018-2358

Appeals from the United States Patent and Trademark
Office, Patent Trial and Appeal Board in Nos. IPR2017-
00561, IPR2017-00562.

MANDATE

In accordance with the judgment of this Court, entered
December 12, 2019, and pursuant to Rule 41 of the Federal
Rules of Appellate Procedure, the formal mandate is
hereby issued.

FOR THE COURT

January 21, 2020

/s/ Peter R. Marksteiner
Peter R. Marksteiner
Clerk of Court

(12) **INTER PARTES REVIEW CERTIFICATE** (1948th)

United States Patent
Lee et al.

(10) **Number:** **US 8,001,434 K2**
(45) **Certificate Issued:** **Feb. 26, 2021**

(54) **MEMORY BOARD WITH SELF-TESTING
CAPABILITY**

(75) **Inventors:** **Hyun Lee; Jayesh R. Bhakta;**
Soonju Choi

(73) **Assignee:** **NETLIST, INC.**

Trial Number:

IPR2017-00561 filed Jan. 5, 2017

Inter Partes Review Certificate for:

Patent No.: **8,001,434**
Issued: **Aug. 16, 2011**
Appl. No.: **12/422,925**
Filed: **Apr. 13, 2009**

The results of IPR2017-00561 are reflected in this inter partes review certificate under 35 U.S.C. 318(b).

INTER PARTES REVIEW CERTIFICATE
U.S. Patent 8,001,434 K2
Trial No. IPR2017-00561
Certificate Issued Feb. 26, 2021

1

2

AS A RESULT OF THE INTER PARTES
REVIEW PROCEEDING, IT HAS BEEN
DETERMINED THAT:

Claims 2-7 are cancelled.

5

* * * * *

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292.)

DOCKET NO. 8:16-cv-01605 JLS	DATE FILED 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,756,364	June 17, 2014	Netlist, Inc.
2 8,516,185	August 20, 2013	Netlist, Inc.
3 8,001,434	August 16, 2011	Netlist, Inc.
4 8,359,501	January 22, 2013	Netlist, Inc.
5 8,689,064	April 1, 2014	Netlist, Inc.

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

DATE INCLUDED	INCLUDED BY	☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
2		
3		
4		
5		

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT "SEE ATTACHED COPY OF ORDER DISMISSING CASE"

CLERK Kiry Gray	(BY) DEPUTY CLERK /s/ J. Lam	DATE 04/05/2021
--------------------	---------------------------------	--------------------

Copy 3—Upon termination of action,

AO 120 (Rev. 08/10)

TO: Mail Stop 8 Director of the U.S. Patent and Trademark Office P.O. Box 1450 Alexandria, VA 22313-1450	REPORT ON THE FILING OR DETERMINATION OF AN ACTION REGARDING A PATENT OR TRADEMARK
---	---

In Compliance with 35 U.S.C. § 290 and/or 15 U.S.C. § 1116 you are hereby advised that a court action has been filed in the U.S. District Court for the Central District of California, Santa Ana Division, on the following

☐ Trademarks or ☒ Patents. (☐ the patent action involves 35 U.S.C. § 292).

<u>DOCKET NO.</u> 8:16-cv-01605	<u>DATE FILED</u> 8/31/2016	U.S. DISTRICT COURT Central District of California, Santa Ana Division
PLAINTIFF NETLIST, INC.		DEFENDANT SK HYNIX INC., SK HYNIX AMERICA INC., and SK HYNIX MEMORY SOLUTIONS INC.
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1 8,489,837	July 16, 2013	Netlist, Inc.
2		
3		
4		
5		

In the above—entitled case, the following patent(s)/ trademark(s) have been included:

<u>DATE INCLUDED</u>	<u>INCLUDED BY</u> ☐ Amendment ☐ Answer ☐ Cross Bill ☐ Other Pleading	
PATENT OR TRADEMARK NO.	DATE OF PATENT OR TRADEMARK	HOLDER OF PATENT OR TRADEMARK
1		
2		
3		
4		
5		

In the above—entitled case, the following decision has been rendered or judgement issued:

DECISION/JUDGEMENT "SEE ATTACHED COPY OF ORDER DISMISSING CASE"

CLERK Kiry Gray	(BY) DEPUTY CLERK /s/ J. Lam	DATE 04/05/2021
--------------------	---------------------------------	--------------------

Copy 3—Upon termination of action,



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
12/422,925	04/13/2009	411127

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
12/422,925	04/13/2009	411127

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine , CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045PR
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
		☐ Party of Interest under 35 U.S.C. 118		
Prefix	Given Name	Middle Name	Family Name	Suffix
	To be provided later			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residence i	
Citizenship under 37 CFR 1.41(b) i				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Countryⁱ		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	20995
Email Address	efiling@kmob.com
	Add Email Remove Email

Application Information:

Title of the Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
Attorney Docket Number	NETL.045PR	Small Entity Status Claimed	☐
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045PR
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

Publication Information:
☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.
Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

			Remove
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

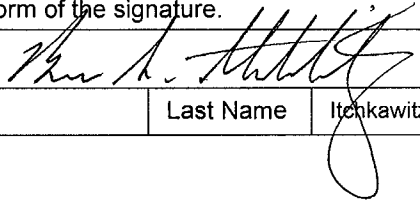
Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

Assignee 1

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045PR
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

If the Assignee is an Organization check here. ☒			
Organization Name	NETLIST, INC.		
Mailing Address Information:			
Address 1	51 Discovery, #150		
Address 2			
City	Irvine	State/Province	CA
Country	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	949-435-0031
Email Address	hlee@netlist.com		
Additional Assignee Data may be generated within this form by selecting the Add button.			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature			Date (YYYY-MM-DD)	2008-04-14	
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION**BACKGROUND**Field

[0001] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0002] As the demand on the performance of electronic systems increase, the systems require higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0003] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system which incorporates a multiplexer-demultiplexer to isolate unused subsystems. This arrangement can reduce the load on each driver when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver and the multiple destinations. However, the multiplexer and/or demultiplexer can present a transmission line discontinuity point in electronic systems with signal paths with electrical lengths longer than $\frac{1}{4}$ of the wavelength of the operational frequency or the signal transition rate. In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

[0004] In memory applications, some systems control “chip select” or device select signals, along with the memory address and control signals, based on the bank device select signals, in order to disable memory devices when they are not being accessed. Figure 2

schematically illustrates an example of such a system which isolates unused subsystems using device select signals. The devices are disabled by not activating (asserting) the address, control, and/or device select signals, thus logically isolating the memory devices which are not accessed. This arrangement generally requires a bank device select signal for each physical memory rank and the address and device select signals are generally controllable based on whether the corresponding memory device or group of memory devices is being accessed. Generating controllable address and device select signals can introduce timing delays on the bank device select, address, and control paths. In addition, this arrangement does not address the power dissipation and may degrade performance degradation by, for example, requiring that secondary control signals be generated for the memory devices.

SUMMARY

[0005] In certain embodiments, a memory module is provided which includes a printed circuit board having at least one connector configured to be electrically coupled to a memory controller of a computer system. The memory module also includes a plurality of memory devices on the printed circuit board and a circuit. The circuit includes a first set of ports having a plurality of bi-directional ports. Each port of the first set of ports is electrically coupled to at least one memory device of the plurality of memory devices. The circuit also includes a second set of ports having one or more bi-directional ports. Each port of the second set of ports is electrically coupled to the at least one connector. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0006] In certain embodiments, a method of using a memory module with a computer system is provided which involves providing a memory module including a printed circuit board having at least one connector configured to be electrically coupled to a memory controller of a computer system and to a plurality of memory devices. The memory module further includes a circuit having a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports electrically coupled to at least one memory device of

the plurality of memory devices. The circuit also includes a second set of ports comprising one or more bi-directional ports. Each port of the second set of ports electrically coupled to the at least one connector. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports. The method further includes activating the circuit to selectively electrically couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports. The method also involves activating the circuit to reduce noise in the one or more signals.

[0007] In certain embodiments, a circuit is provided which includes a first set of ports having a plurality of bi-directional ports. Each port of the first set of ports is configured to be electrically coupled to at least one memory device of a plurality of memory devices on a printed circuit board. The circuit also includes a second set of ports having one or more bi-directional ports. Each port of the second set of ports is configured to be electrically coupled to at least one connector on the printed circuit board. The at least one connector is configured to be electrically coupled to a memory controller of a computer system. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0008] In certain embodiments, a circuit is provided which includes a first set of ports having a plurality of bi-directional ports. Each port of the first set of ports is configured to be electrically coupled to at least one subsystem of a plurality of subsystems. The circuit also includes a second set of ports having one or more bi-directional ports. Each port of the second set of ports is configured to be electrically coupled to at least one subsystem of one or more subsystems. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports

of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Figure 1 schematically illustrates an example system which isolates unused subsystems and incorporates a multiplexer-demultiplexer.

[0010] Figure 2 schematically illustrates an example system which isolates unused subsystems using device select signals.

[0011] Figure 3 schematically illustrates an example system having an example circuit connecting two subsystems in accordance with certain embodiments described herein.

[0012] Figure 4 schematically illustrates an example circuit in accordance with certain embodiments described herein.

[0013] Figure 5 schematically illustrates an example programmable impedance matching circuit in accordance with certain embodiments described herein.

[0014] Figure 6 schematically illustrates an example filter/repeater/amplifier circuit in accordance with certain embodiments described herein.

BRIEF DESCRIPTION OF THE APPENDIX

[0015] Appendix: "Self-Adjusting Damper," incorporated in its entirety by reference herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0016] Certain embodiments described herein present a circuit for isolating inactive sub-systems and maintaining and improving signal integrity in a variety of systems. Certain embodiments described herein are implemented in a memory system of a computer system. Certain embodiments described herein are used to increase the performance of electronic systems without increasing the power dissipation or the memory access time.

[0017] Certain embodiments described herein describe a circuit for use in high performance electronic systems which may contain transmission lines and relatively heavy fan-outs. A circuit is described herein which, in certain embodiments, includes at least some of the following components: a signal router, a programmable coarse impedance matching circuit, a self-adjusting damper circuit, and an amplifier/repeater/filter.

[0018] In certain embodiments, the circuit includes multiple bi-directional ports. The signal router may comprise a multiplexer-demultiplexer. In certain embodiments, the signal router may be capable of interconnecting any number of the ports simultaneously. In certain embodiments, the circuit can pass signals (information) from one port to another port or from one port to multiple ports via multiplexer-demultiplexer. In certain embodiments, the multiplexer-demultiplexer isolates and provides a known static value to all un-driven ports, reducing any dynamic power consumption associated with floating or toggling electrical nodes. The multiplexer-demultiplexer can function as an isolation switch which isolates and inactivates individual subsystems not actively transferring information. In certain embodiments, the multiplexer-demultiplexer may function as a scalable isolation switch router which routes the information from one subsystem to another, establishing a one-to-one connection for transferring data from point-to-point and establishing a one-to-many connection for broadcasting data.

[0019] In certain embodiments, one or more of the ports in the circuit advantageously include a correction circuit for reducing noise. In certain embodiments, the correction circuit includes one or more elements to provide for coarse correction of signal noise due to impedance mismatches in the system. In some embodiments, the coarse correction elements may be a set of programmable coarse impedance matching circuits that are connected to the bi-directional ports. The impedance values of the impedance matching circuits may be dynamically controllable, configurable, and programmable, so as to match the characteristic impedances of the transmission lines in the system. The values of these impedances may differ when, for example, a particular signal path is connected or disconnected by the multiplexer-demultiplexer.

[0020] Advantageously, in certain embodiments, the correction circuit includes one or more fine correction elements which further reduce signal reflections due to impedance mismatches and/or reduce other noise. In certain embodiments the fine correction element is a damper circuit. For example, a self-adjusting damper circuit may be connected to each port of the multiplexer-demultiplexer. The damper circuit may be connected, for example, in parallel with the programmable coarse impedance matching circuit. In some embodiments, the damper circuit operates to reduce the impedance mismatches by introducing a current source

that damps to the slew rate of the input signal, reducing signal reflection at the interface of each port. In some embodiments, the current source is a negative feedback current source. In various embodiments, the self-adjusting damper includes an enable control input for enabling the self-adjusting damper circuit. Examples of the self-adjusting damper circuit in accordance with certain embodiments described herein are described in the Appendix which is incorporated in its entirety by reference herein.

[0021] In some embodiments, the circuit includes a filter/repeater/amplifier which provides wave reshaping functions and generally reduces signal noise. For example, the filter/repeater/amplifier may reduce signal ringing due to reflection at the multiplexer-demultiplexer. In certain embodiments, only the filter, only the repeater, only the amplifier, or any combination of the three elements may be included in the circuit. In certain embodiments, the filter/repeater/amplifier may also reduce noise from other sources such as, for example, signal coupling. In some embodiments, the filter/repeater/amplifier may include a band-pass filter and may reduce any low and/or high frequency noise components. The filter/repeater/amplifier may also, in certain embodiments enhance the signal slew rate to reduce the transient power dissipation at the target devices or components.

[0022] Figure 3 schematically illustrates an example system having an example circuit 200 connecting two subsystems in accordance with certain embodiments described herein. In some embodiments, the circuit 200 includes several independent, fully controllable functional blocks. For example, the circuit 200 may include a signal router, a programmable coarse impedance matching circuit, a self-adjusting damper circuit, and an amplifier/repeater/filter. In certain embodiments, any subset of one or more of these functional blocks may be implemented in order to enhance performance and reduce power dissipation. Figure 4 schematically illustrates a circuit 200 comprising a router 210 and which connects a subsystem 103 to another subsystem 104, or a subsystem 103 to multiple subsystems. The router 210 may include a router control input for controlling the connection paths of the router 210. The devices and methods described herein may be incorporated in any system (e.g. a board level system) or device that contains one or more connectors or wires. In certain embodiments the system has a signal delay which is more than $\frac{1}{2}$ the signal transition rate or the rise/fall time of the signal.

[0023] The system 100 of Figure 3 includes a driver 102, a receiver 101, and two subsystems 103, 104. For illustration purposes, the system 100 of Figure 3 is described as an open system that communicates with other systems via the driver 102 and the receiver 101. Furthermore, in some embodiments, a connector connects the system 100 with other systems by means of transmission lines. In some embodiments, a transmission line may be characterized as any wire or connector with an electrical length longer than the $\frac{1}{2}$ of the rise/fall time of the signal that is being transmitted. In various embodiments, the receiver 101 may or may not contain any active circuits. The receiver 101 may be a simple connector such as a socket or DIMM interface with a system board. The output signal from the receiver 101 may exhibit the characteristics of the signal on a transmission line such as, for example, wave dispersion, attenuation, and reflection. In certain embodiments, the output signal from the receiver goes through a number of refining stages in circuit 200 to improve the performance of the overall system. For example, the circuit 200 may reduce power consumption by selectively isolating the inactive (or un-accessed) subsystems and may enhance the access rate of a targeted subsystem.

[0024] Figure 4 schematically illustrates an example circuit 200 in accordance with certain embodiments described herein. In certain embodiments, the signal quality is enhanced by introducing the impedance matching circuit 230, the self-adjusting damper 240, and the filter/repeater/amplifier 250 on a signal path (e.g., from receiver 101 to subsystem 103). In certain embodiments, the performance is enhanced and power dissipation is reduced by removing the load associated with other subsystems (e.g., subsystem 104) and reshaping the signal waveform using the filter/repeater/amplifier 250. In some embodiments, there are two refining stages for reducing signal reflections due to impedance mismatches. In some embodiments, for example, the first refining stage is the impedance matching circuit 230 that matches the impedance of the transmission wire prior to the receiver 101 with the characteristic impedance of the wire in the system 100. The granularities of the configurable impedance values may be configured based on the application of the particular system. Advantageously, in some embodiments, the impedance matching circuit 230 is programmable. In some embodiments, the impedance matching circuit 230 may include a real time controllable impedance control input for pre-programming and/or dynamically controlling the

impedance matching circuit 230. In this manner, the impedance matching circuit 230 may, in certain embodiments, provide a dynamically controllable variable impedance which can be pre-set for matching the impedance of the subsystem that drives the isolation switch.

[0025] In the illustrated embodiment, the second refining stage comprises a self-adjusting damper circuit 240 which smoothes out the impedance-matched signal and removes the transmission line characteristics from the received signals. In other embodiments, the self-adjusting damper circuit 240 may be placed after the router 210 in order to achieve better results.

[0026] In certain embodiments, the second refining stage further comprises a filter/repeater/amplifier 250. As schematically shown in Figure 6, the filter/repeater/amplifier circuit 250 of certain embodiments includes a filter 253 which reduces high frequency noise that is induced by cross-coupling between wires or transmission lines. In certain embodiments, the filter 253 may reduce low frequency noise in addition to or instead of high frequency noise, and/or may reduce noise having a particular range of frequencies. In addition, in some embodiments, the filter/repeater/amplifier enable signal 251 puts the filter/repeater/amplifier 250 in a high impedance mode when the impedance matching circuit 230 is enabled or the self-adjusting damper 240 is active. In certain embodiments, for example, when the circuit 200 receives a signal (e.g., a signal traveling from receiver 101 to subsystem 103 in Figure 3, or from a first subsystem 220 to a second subsystem 420 in Figure 4), the impedance matching circuit 230, the damper circuit 240, and the filter/repeater/amplifier circuit 450, along with the dotted path in the router 210 are active and all other elements may be inactive, providing one-to-one connection between the first subsystem 220 and the second subsystem 420.

[0027] Figure 5 schematically illustrates an example programmable impedance matching circuit 230 in accordance with certain embodiments described herein. In certain embodiments, the impedance matching circuit 230 may be integrated with the router 210 and with the filter/repeater/amplifier 250. Alternatively, in other embodiments, the impedance matching circuit 230 may be implemented with discrete components.

[0028] Figure 6 schematically illustrates an example filter/repeater/amplifier circuit 250 in accordance with certain embodiments described herein. In certain embodiments, the

filter/repeater/amplifier 250 can be integrated with the router 210 and impedance matching circuit 230. In other embodiments, the filter/repeater/amplifier 250 may be implemented with discrete components. In certain embodiments, the filter/repeater/amplifier 250 comprises a filter 253 and a repeater/amplifier 252 with an output enable control input 251 for controlling the bi-directional signal path. In certain embodiments, the filter/repeater/amplifier 250 is programmable and dynamically controllable.

[0029] In some embodiments, the circuit 200 may be implemented in a memory subsystem of an electronic system (e.g., a personal computer, server computer, or data storage system). In general, such electronic systems include memory subsystems that contain multiple memory boards or modules that are driven by a single memory controller which synchronizes memory access and manages memory coherency. However, having a single memory controller may gradually degrade system performance and increase power consumption as the physical and the logical size of the memory subsystem increases. The increase in the memory speed may further impair the performance and increase the power consumption. The memory system may advantageously be used with a power supply which stabilizes the voltage level, and/or a thermo-mechanical component to moderate memory subsystem temperature, and/or software for throttling the processor to lower the memory access rate. Additionally, impedance mismatches between the system board, the memory boards, and the memory devices may also substantially degrade the performance and power efficiency of traditional memory systems.

[0030] In certain embodiments, the circuit 200 described herein addresses these speed and power issues in memory systems. For example, the circuit 200 of certain embodiments can comprise an isolation switch and can logically isolate inactive subsystems and reduce signal reflection and/or other noise at, for example, the interface between the system board and the memory board and at the interface between the memory board and the memory devices. In an embodiment in which the circuit 200 is incorporated in a memory system, the receiver 101 and the driver 102 of Figure 3 would represent data and address signals between the memory controller and the subsystems 103, 104 which may represent two sets of independently addressable memory devices. In certain embodiments, the circuit 200 may be disposed on the same memory board that is populated with memory devices 103, 104.

[0031] Referring to Figure 4, and with respect to an example memory application, the subsystem port connections 220, 320 represent a memory board interface to a system board while the subsystem port connections 420, 520 represent the isolation circuit 200 interface to the memory devices. In some embodiments, the impedance matching circuits 230, 330 and the self-adjusting dampers 240, 340 are employed to reduce the signal reflection due to the impedance mismatching between the system board and the memory board. In certain embodiments, the filter/repeater/amplifier 250 filters out, for example, cross-coupling between signals. For example, the filter 253 of the filter/repeater/amplifier 250 of Figure 6 may filter out cross-coupling between data, address, and/or control wires in a memory application. In certain embodiments, the repeater/amplifier 252 of the filter/repeater/amplifier of Figure 6 reshapes the signal which may be deteriorated due to transmission over lossy transmission lines having discontinuity points. In certain embodiments, the repeater/amplifier 252 may be configured to repeat the input signal, generating an output signal with earlier transition trigger points and faster slew rates in order to regain speed and to reduce the transient power.

[0032] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory module comprising:
 - a printed circuit board comprising at least one connector configured to be electrically coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board;
 - a circuit, comprising:
 - a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports electrically coupled to at least one memory device of the plurality of memory devices; and
 - a second set of ports comprising one or more bi-directional ports, each port of the second set of ports electrically coupled to the at least one connector;
 - wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.
2. The memory module of Claim 1, wherein the noise comprises signal reflections.
3. The memory module of Claim 1, wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.
4. The memory module of Claim 3, wherein the coarse correction element comprises a programmable impedance matching circuit.
5. The memory module of Claim 3, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.
6. The memory module of Claim 5, wherein the fine correction element comprises a damper circuit.
7. The memory module of Claim 6, wherein the damper circuit is a self-adjusting damper circuit.

8. The memory module of Claim 6, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

9. The memory module of Claim 8, wherein the filter circuit comprises a band-pass filter.

10. The memory module of Claim 8, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

11. The memory module of Claim 1, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

12. The memory module of Claim 1, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

13. A method of using a memory module with a computer system, the method comprising:

providing a memory module comprising a printed circuit board having at least one connector configured to be electrically coupled to a memory controller of a computer system and to a plurality of memory devices, the memory module further comprising a circuit, the circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports electrically coupled to at least one memory device of the plurality of memory devices; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports electrically coupled to the at least one connector; and

wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of

ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports; and

activating the circuit to selectively electrically couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports; and

activating the circuit to reduce noise in the one or more signals.

14. The method of Claim 13, wherein the noise comprises signal reflections.

15. The method of Claim 13, wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.

16. The method of Claim 15, wherein the coarse correction element comprises a programmable impedance matching circuit.

17. The method of Claim 15, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.

18. The method of Claim 17, wherein the fine correction element comprises a damper circuit.

19. The method of Claim 18, wherein the damper circuit is a self-adjusting damper circuit.

20. The method of Claim 18, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

21. The method of Claim 20, wherein the filter circuit comprises a band-pass filter.

22. The method of Claim 20, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

23. The method of Claim 13, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

24. A circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports configured to be electrically coupled to at least one memory device of a plurality of memory devices on a printed circuit board; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be electrically coupled to at least one connector on the printed circuit board, the at least one connector configured to be electrically coupled to a memory controller of a computer system;

wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

25. The circuit of Claim 24, wherein the noise comprises signal reflections.

26. The circuit of Claim 24, wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.

27. The circuit of Claim 26, wherein the coarse correction element comprises a programmable impedance matching circuit.

28. The circuit of Claim 26, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.

29. The circuit of Claim 28, wherein the fine correction element comprises a damper circuit.

30. The circuit of Claim 29, wherein the damper circuit is a self-adjusting damper circuit.

31. The circuit of Claim 29, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

32. The circuit of Claim 31, wherein the filter circuit comprises a band-pass filter.

33. The circuit of Claim 31, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

34. The circuit of Claim 24, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

35. The circuit of Claim 24, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

36. A circuit comprising:

- a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports configured to be electrically coupled to at least one subsystem of a plurality of subsystems; and

- a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be electrically coupled to at least one subsystem of one or more subsystems;

- wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

37. The circuit of Claim 36, wherein the noise comprises signal reflections.

38. The circuit of Claim 36 wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.

39. The circuit of Claim 38, wherein the coarse correction element comprises a programmable impedance matching circuit.

40. The circuit of Claim 38, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.

41. The circuit of Claim 40, wherein the fine correction element comprises a damper circuit.

42. The circuit of Claim 41, wherein the damper circuit is a self-adjusting damper circuit.

43. The circuit of Claim 41, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

44. The circuit of Claim 43, wherein the filter circuit comprises a band-pass filter.

45. The circuit of Claim 43, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

46. The circuit of Claim 36, wherein the correction circuits reduce impedance mismatches at nodes between the one or more subsystems and the plurality of subsystems.

47. The circuit of Claim 36, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

ABSTRACT

A memory module is provided which includes a printed circuit board having memory devices and a connector which can connect to a memory controller. The memory module also includes a circuit including a first set of ports comprising bi-directional ports. Each port of the first set of ports couples to at least one memory device. The circuit also includes a second set of ports comprising bi-directional ports. Each port of the second set of ports couples to the connector. The circuit is configured to selectively couple ports of the second set of ports to ports of the first set of ports. The ports of the first set of ports and the ports of the second set of ports include a correction circuit which reduces noise in signals transmitted between the ports of the first set of ports and the ports of the second set of ports.

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APPENDIX

SELF-ADJUSTING DAMPER**BACKGROUND**Field

[0001] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0002] Complex electronic systems often contain multiple sources of electrical noise. Such sources may include noise from signal cross coupling, transmission line reflections, ground bounce noise, noise due to an unstable power supply voltage, noise due to external EM waves, etc. In some cases these noise sources may be generally negligible depending on the noise tolerance of the system. However, many modern electronic systems incorporate relatively low supply voltages to accommodate state-of-art semiconductor processing technology. In such systems, the noise tolerance of the system is generally decreased. This phenomenon is clearly demonstrated in dynamic random-access memory (DRAM) systems where the operating voltage is sometimes substantially reduced in order to reduce the leakage current and optimize the performance.

[0003] The low power supply voltage can reduce the noise tolerance and can be compensated for, at least in part, by additional electronic components to ensure proper operation of the DRAM modules. For example, terminating resistors may be added in some cases (e.g., for DDR2 and DDR3 memory modules) at transmission line discontinuity points. Although terminating resistors may reduce the signal noise by lowering the amplitude of reflected waves, there may still remain substantial signal ringing since, in most cases, values of the terminating resistor on the memory module may not completely match the system board impedance. Furthermore, because each electrical component in a system exhibits its own unique electrical characteristics including the output driving capabilities, the signals generated from each electrical component produce their own unique reflected wave patterns, which cannot be completely matched with a set of impedance values.

[0004] Circuits often used to remove noises (e.g., to counteract ringing signals and/or reflected signals with small amplitudes) from the received signals include band-pass

filters, impedance matching circuits, echo cancellers, and error-correction circuits that reconstruct the transmitted signal. Each of these circuits exhibits specific characteristics and are used in well-defined application areas. In general, band-pass filters are analog circuits which generally pass frequencies within a certain range and reject frequencies outside that range. Impedance matching circuits are generally characterized by a fixed impedance value that exhibits impedance characteristics similar to those of the transmission line to which it is interfacing so as to reduce signal reflections. Echo cancellers and the error-correction circuits are typically digital circuits that interact with a micro-processor.

SUMMARY

[0005] In certain embodiments, a self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is, in certain embodiments, substantially representative of the signal component of the input signal.

[0006] In certain embodiments, a method of removing a noise component of an electrical signal is provided. The method involves receiving an input signal including a noise component and a signal component. The method further includes generating a first signal corresponding to the noise component of the input signal and generating a second signal corresponding to a delayed version of the input signal. The method also involves adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal which, in certain embodiments, is substantially representative of the signal component of the input signal.

[0007] In certain embodiments, a self-adjusting damper circuit is provided which includes a means for receiving an input signal comprising a noise component and a signal component, a first generation means for generating a first signal corresponding to the noise component of the input signal, and a second generation means for generating a second signal

corresponding to a delayed version of the input signal. The self-adjusting damper circuit also includes a means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal. In certain embodiments, the output signal is substantially representative of the signal component of the input signal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure 1 is a block diagram illustrating an example system incorporating self-adjusting damper circuits in accordance with certain embodiments described herein.

[0009] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0010] Figure 3 is a schematic diagram illustrating another example self-adjusting damper circuit in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0011] In many electronic systems, while multiple noise sources contribute to the distortion of the transmitted signal, for conventional systems there are usually one or two major noise sources while other noise sources make minor contributions. Thus, system designers have been able to implement one or two noise removing circuits in conventional electronic systems to remove the major noise sources and to ignore the minor noise sources. However, in some high performance systems such as memory subsystems, even relatively minor noises may contribute significant signal distortions due to, for example, the lower supply voltages. For example, a major noise contribution can be the reflected signals at the transmission line discontinuity points, and a minor noise contribution can be due to signal cross-coupling. Such systems may benefit from a noise filtering scheme that may include various types of noise removing circuits but may also potentially increase cost, complexity, and/or the time for system development.

[0012] Certain embodiments described herein present a self-adjusting electrical damper applicable to a wide frequency range. Certain embodiments described herein are implemented in an application-specific integrated circuit (ASIC) or in a System-on-Chip (SoC) device.

[0013] Certain embodiments described herein provide a self-adjusting damper configured to remove signal noise contributed by, for example signal cross-coupling and the

signal reflections. In certain embodiments, the self-adjusting damper is a component of a system comprising signal buses and transmission lines. In certain embodiments, the self-adjusting damper employs band-pass filtering and echo cancellation without any external control inputs.

[0014] In certain embodiments, the self-adjusting damper can be implemented either with discrete components or integrated into an ASIC device. Advantageously, the damper of certain embodiments is capable of reducing noise due to, for example, signal coupling between wires or signal reflection caused by mismatched terminations of transmission lines. In certain embodiments, the self-adjusting damper includes a delay component, a negative voltage controlled current source, and a positive voltage controlled current source. In certain embodiments, the signal is delayed through at least one of the current sources. In certain embodiments, the signal delay through the delay component is substantially similar to the signal delay through the current sources.

[0015] In certain embodiments, the voltage controlled current sources react to a change of the input voltage and have corresponding current source multiplication factors g_1 and g_2 . These multiplication factors are variables that are functions of the slew rate of the input signal such that a faster input transition produces higher transconductance on the current sources, smoothing out or reducing the signal ringing on the output. Advantageously, the damper circuit of certain embodiments actively counteracts signal noise, thereby allowing noise reduction over a broad range of noise frequencies rather than over, for example, a frequency window. Moreover, in certain embodiments, the damper circuit actively counteracts signal noise regardless of the periodicity of the input signal and/or of the noise component of the signal. In certain embodiments, the damper is implemented without analog circuit technology (e.g., within an ASIC without invoking analog circuit technology), thereby enabling easy design migration to various process technologies.

[0016] Figure 1 is a block diagram illustrating an example system incorporating example self-adjusting damper circuits 141, 142 in accordance with certain embodiments described herein. A plurality of drivers 101, 102 transmit signals via transmission lines or wires with a cross coupling impedance of Z_c 120. Z_c 120 includes cross-coupling capacitance and (mutual) inductance as well as any transconductance between the wires. In certain

embodiments, the impedance matching components 131, 132 may comprise a fixed termination resistor for the respective transmission line 111, 112. The self-adjusting damper circuits 141, 142 of certain embodiments advantageously reduce cross coupling noise due to Z_c and/or the remaining reflected wave after impedance matching is performed by the impedance matching components 131, 132.

[0017] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein. According to certain embodiments, the mathematical representation of the output 202 with load capacitance C_L can be expressed by the following equations:

$$V_o(t) = \int \left[\left(I(t) - \frac{g1\delta V_i(t - \Delta\tau)}{dt} - \frac{g2\delta V_i(t - \Delta\tau)}{dt} \right) C_L \right] dt$$

$$= \left[\int I(t) dt - (g1\delta V_i(t - \Delta\tau) + g2\delta V_i(t - \Delta\tau)) \right] C_L$$

Where,

$$I(t) = \frac{(V_o(t) - V_i(t))}{Z_d}$$

Thus,

$$Z_d * dV_o(t)/dt = V_i(t) - V_o(t) - (g1 + g2) * C_L * d(\delta V_i(t - \Delta\tau))/dt$$

And,

$$V_o(t) = V_i(t) - (g1 + g2) * C_L * d(\delta V_i(t - \Delta\tau))/dt - Z_d * dV_o(t)/dt$$

[0018] The above equations show that the output voltage of certain embodiments is a function of the change rate of the input signal transition rate and the change rate of the output itself. Because the change rate of the input transition rate and the output change rate have a negative effect on the output voltage, the damper of certain embodiments reduces noise having a slew rate that is faster than the slew rate of the non-noise component of the signal. Thus, certain embodiments provide a self-adjusting damper that effectively removes or reduces noises such as cross-coupling noises and reflected signals in complex electrical and electronic systems.

[0019] Figure 3 is a schematic diagram of another example of a self-adjusting damper circuit in accordance with certain embodiments described herein. The illustrated

embodiment comprises two voltage controlled current sources 340, 350 which are included in a first sub-circuit. The current source 340 of the illustrated embodiment is an AC-coupled, negative-gain pmos voltage-controlled current source for increasing the current flow from the power supply when the input sees a negative noise component, thus reducing the negative noise component on the output. The current source 340 includes two pmos transistors 341, 342. The pmos transistor 341 is a current-limiting transistor that ensures that the current supplied to the output has a relatively smaller magnitude than does the current to/from the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other pmos transistor 342 provides the negative-voltage-controlled current source function by AC coupling with the input signal V_i 301, on an input terminal, via a capacitor 311. Thus, the transconductance of this transistor 342 changes inversely proportionally to the slew rate of the input.

[0020] The current source 350 of the illustrated embodiment is an AC-coupled, positive-gain nmos current source for increasing the current flow to ground when the input sees a positive noise component, thus reducing the positive noise component on the output. The current source 350 includes two nmos transistors 351, 352. The nmos transistor 351 is a current-limiting transistor that ensures the current taken away from the output has a relatively smaller magnitude than does the current from/to the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other nmos transistor 352 provides the positive-voltage-controlled current source function by AC coupling with the input signal V_i 301 via a capacitor 311. Thus, the transconductance of this transistor 342 changes proportionally with the slew rate of the input.

[0021] In the illustrated embodiment, the resistor R_d 310 is included in a second sub-circuit. The resistor R_d 310 is a delay element which introduces a signal delay compatible with the current source delay from the first sub-circuit and also may reduce the reverse feedback of the output signal back to the input. Since the delay time from input V_i 301 on the input terminal to output V_o 302 on the output terminal is the function of the time constant, $R_d * C_L$, the resistance of the resistor R_d 310 can be selected based on the estimated value of C_L 360. Thus, in certain embodiments, the input signal can be delayed to match the reaction time of the voltage-controlled current sources that generate a negative feedback current to remove

signal noise. In certain embodiments the damper 141 can reduce cross coupling noise and reflections having various slew rate ranges.

[0022] In certain embodiments, the damper also includes a current-limiting, self-biasing op bias for biasing the reference voltage so that the operation point can be established in the middle of the linear range.

[0023] Advantageously, in certain embodiments, the damper is configured to be implemented in an application specific integrated circuit (“ASIC”) without requiring analog circuit technologies, enabling easy migration to other process technologies. For example, the damper is configured to be implemented in digital process technologies comprising a component resolution of approximately plus or minus 20 percent. Moreover, in certain embodiments, the damper may be implemented with no external control signals or memory elements.

[0024] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-adjusting damper circuit, comprising:
 - an input terminal adapted to receive an input signal comprising a noise component and a signal component;
 - an output terminal;
 - a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal; and
 - a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal, wherein the circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal, the output signal substantially representative of the signal component of the input signal.
2. The circuit of Claim 1, wherein the second sub-circuit comprises a resistor.
3. The circuit of Claim 1, wherein the first sub-circuit comprises a current source.
4. The circuit of Claim 3, wherein the current source comprises a voltage-controlled current source.
5. The circuit of Claim 4, wherein voltage-controlled current source comprises one or more transistors.
6. The circuit of Claim 1, wherein the first sub-circuit comprises a negative current source and a positive current source.
7. The circuit of Claim 6, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.
8. The circuit of Claim 7, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the trans-conductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.
9. The circuit of Claim 7, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

10. The circuit of Claim 9, wherein the first and second current limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

11. The circuit of Claim 10, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current limiting component, and the second current limiting component comprise one or more transistors.

12. The circuit of Claim 1, wherein the first sub-circuit is adapted to generate the first signal by generating a positive signal when the noise component of the input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

13. The circuit of Claim 12, wherein the first sub-circuit is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

14. The circuit of Claim 1, wherein the first signal comprises negative feedback current.

15. The circuit of Claim 1, wherein the circuit is configured to be operate on a digital semiconductor process technology.

16. The circuit of Claim 1, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution of less than plus or minus 20 percent.

17. The circuit of Claim 1, wherein the circuit is implemented on a memory module.

18. The circuit of Claim 1, wherein the noise-counteracting signal actively counteracts a noise component of the delayed version of the input signal.

19. A method of removing a noise component of an electrical signal, the method comprising:

receiving an input signal comprising a noise component and a signal component;

generating a first signal corresponding to the noise component of the input signal;

generating a second signal corresponding to a delayed version of the input signal; and

adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal substantially representative of the signal component of the input signal.

20. The method of Claim 19, wherein generating the first signal comprises generating a positive signal when the noise component of the input signal is negative and generating a negative signal when the noise component of the input signal is positive.

21. The method of Claim 20, wherein generating a positive signal comprises increasing the current flow from a power supply to the first signal and generating a negative signal comprises increasing the current flow from the first signal to ground.

22. The method of Claim 19, further comprising limiting the magnitude of the first signal in relation to the second signal.

23. The method of Claim 19, wherein adding the noise-counteracting signal to the second signal actively counteracts a noise component of the delayed version of the input signal.

24. A self-adjusting damper circuit, comprising:

means for receiving an input signal comprising a noise component and a signal component;

first generation means for generating a first signal corresponding to the noise component of the input signal; and

second generation means for generating a second signal corresponding to a delayed version of the input signal,

means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal, the output signal substantially representative of the signal component of the input signal.

25. The circuit of Claim 24, wherein the second generation means comprises a resistor.

26. The circuit of Claim 24, wherein the first generation means comprises a current source.

27. The circuit of Claim 26, wherein the current source comprises a voltage-controlled current source.

28. The circuit of Claim 27, wherein voltage-controlled current source comprises one or more transistors.

29. The circuit of Claim 24, wherein the first generation means comprises a negative current source and a positive current source.

30. The circuit of Claim 29, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.

31. The circuit of Claim 30, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the transconductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.

32. The circuit of Claim 31, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

33. The circuit of Claim 32, wherein the first and second current-limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

34. The circuit of Claim 33, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current-limiting component, and the second current-limiting component comprise one or more transistors.

35. The circuit of Claim 24, wherein the first generation means is adapted to generate the first signal by generating a positive signal when the noise component of the input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

36. The circuit of Claim 35, wherein the first generation means is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

37. The circuit of Claim 24, wherein the first signal comprises negative feedback current.

38. The circuit of Claim 24, wherein the circuit is configured to be operate on a digital semiconductor process technology.

39. The circuit of Claim 24, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution less than plus or minus 20 percent.

40. The circuit of Claim 24, wherein the circuit is implemented on a memory module.

41. The circuit of Claim 24, wherein the addition means actively counteracts a noise component of the delayed version of the input signal.

SELF-ADJUSTING DAMPER

ABSTRACT

A self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is substantially representative of the signal component of the input signal.

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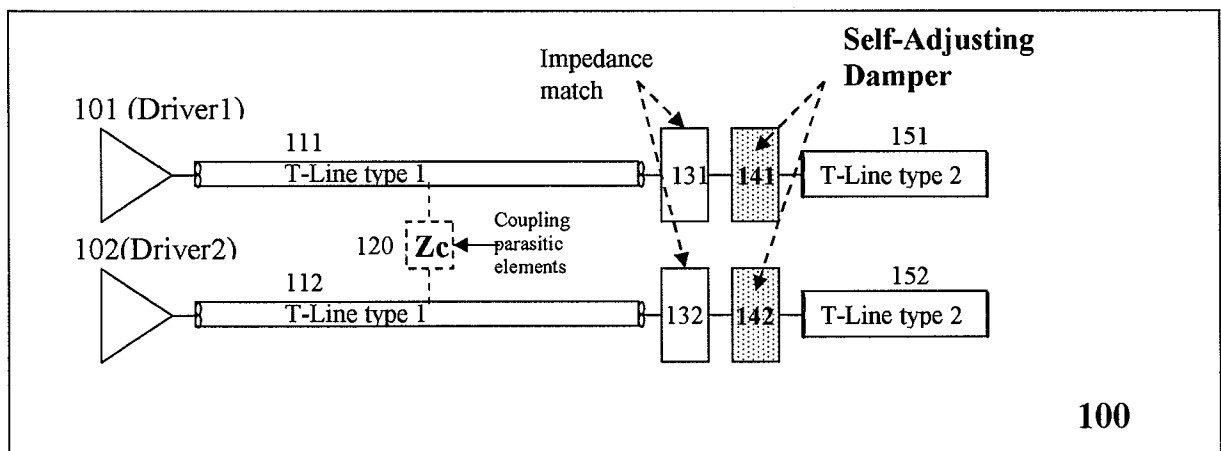


Figure 1

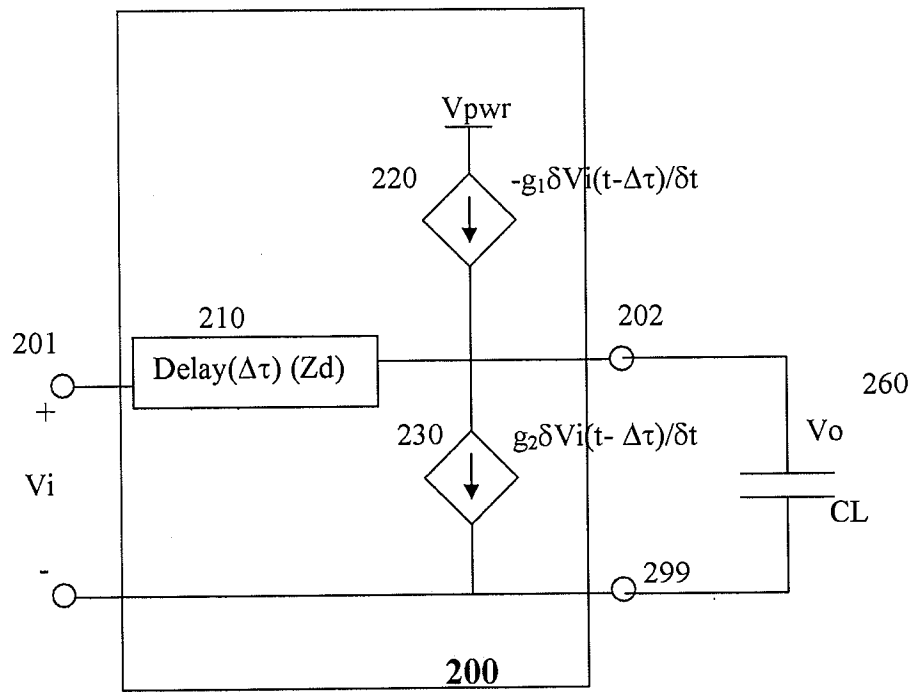


Figure2

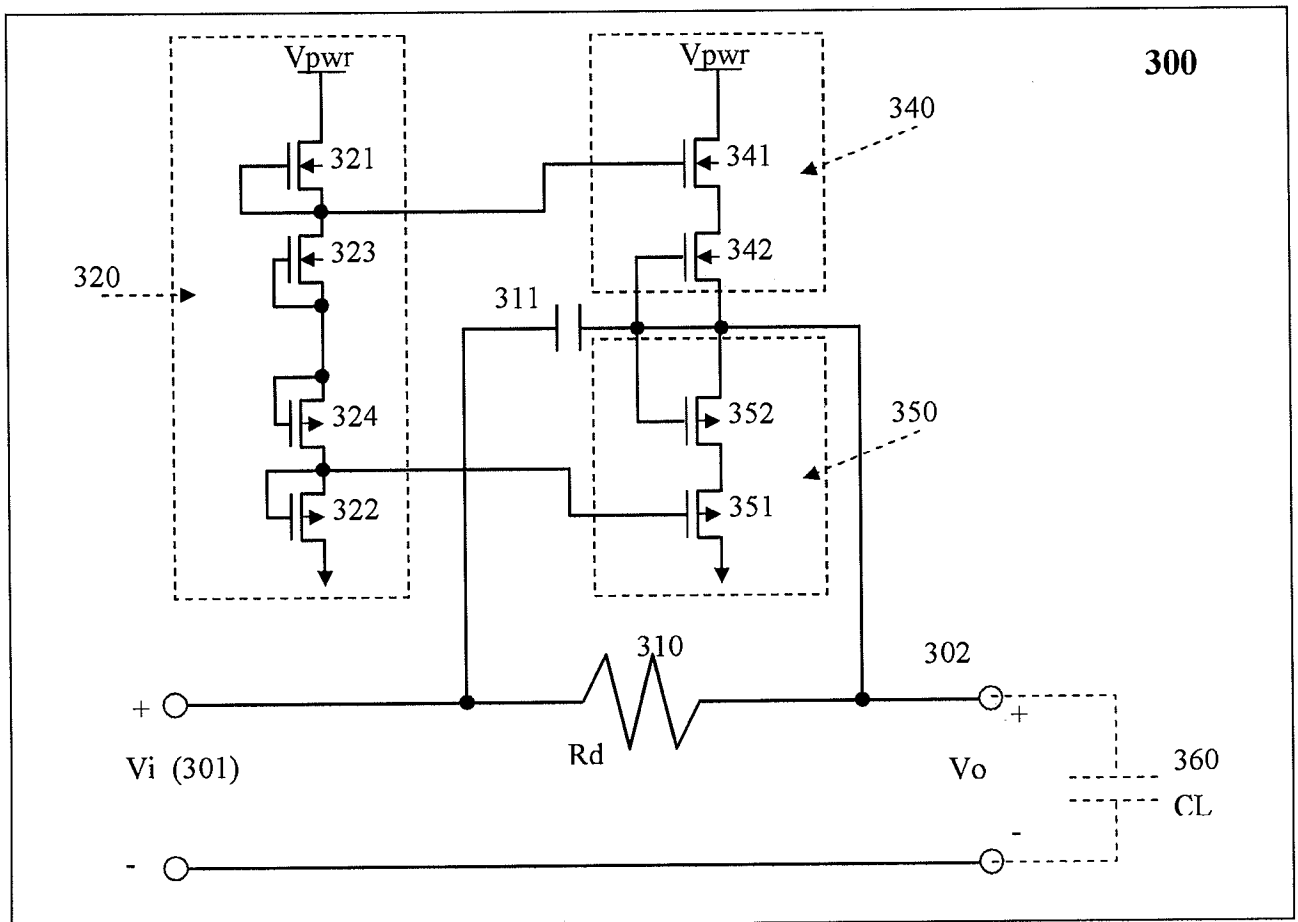


Figure 3

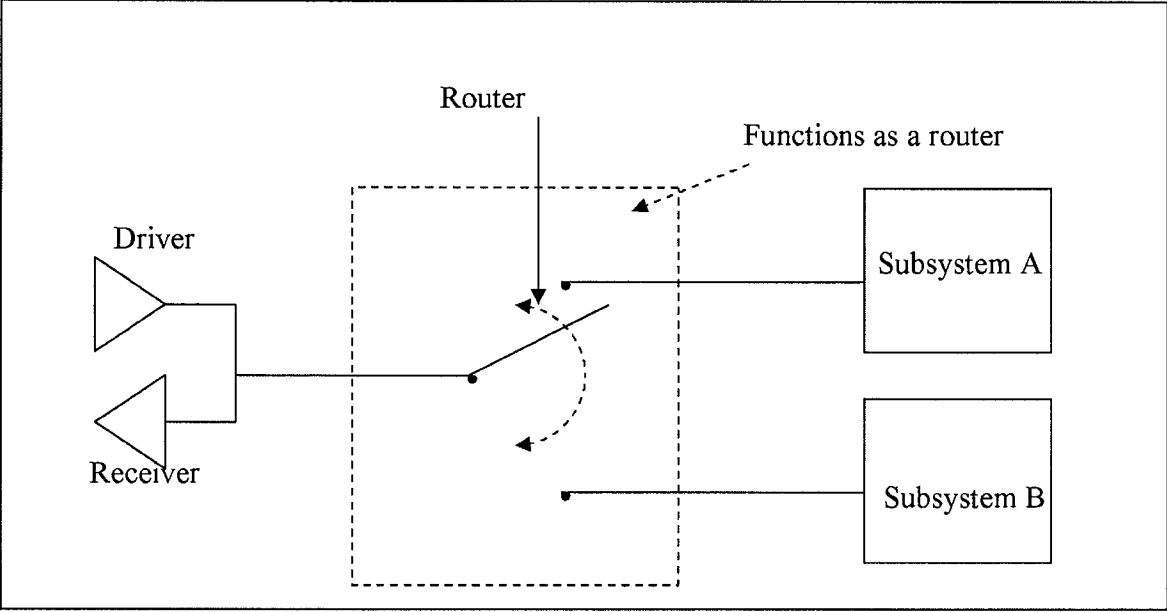


Figure 1

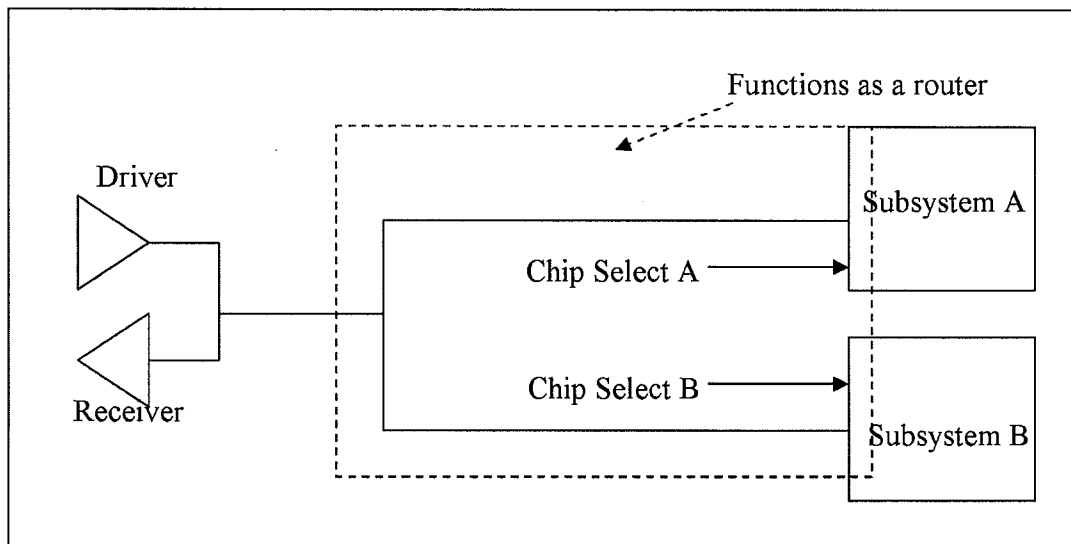


Figure 2

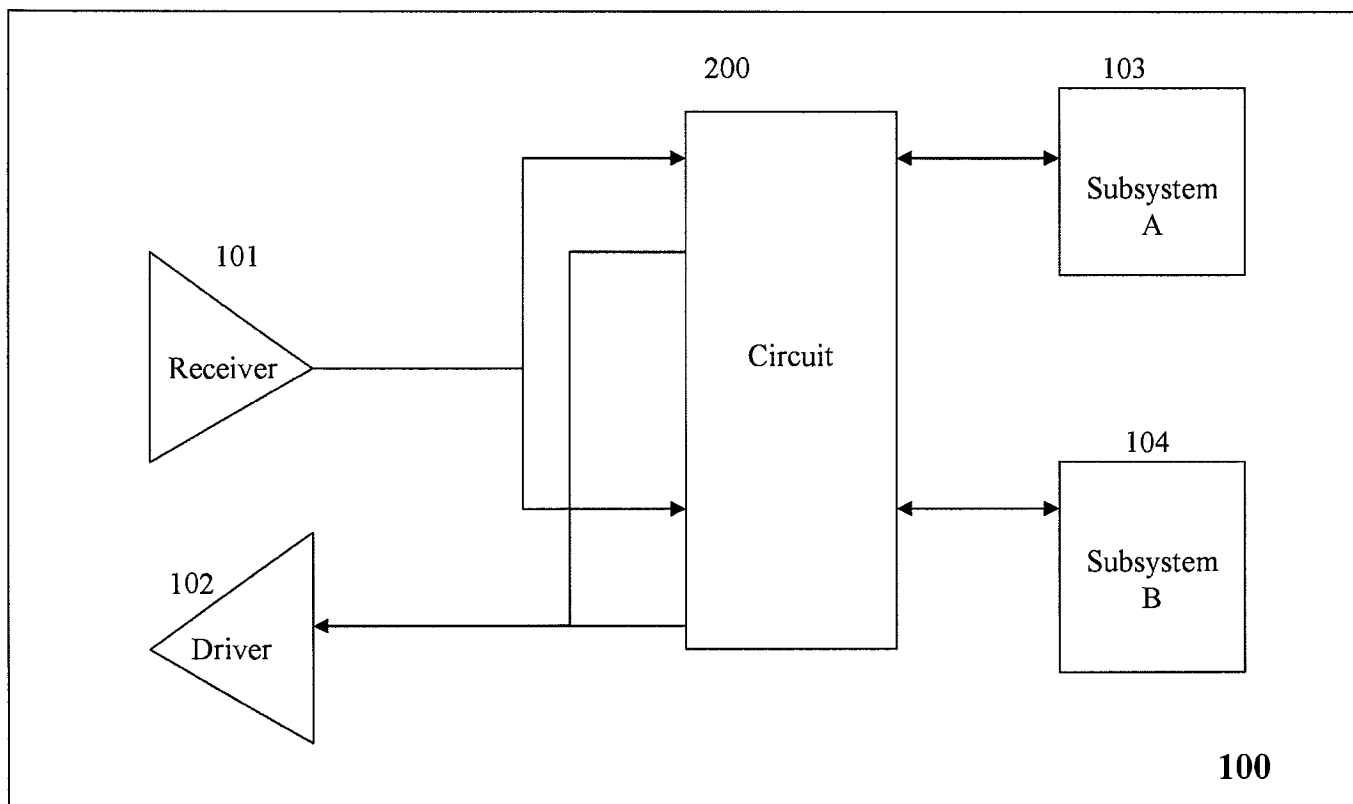


Figure 3

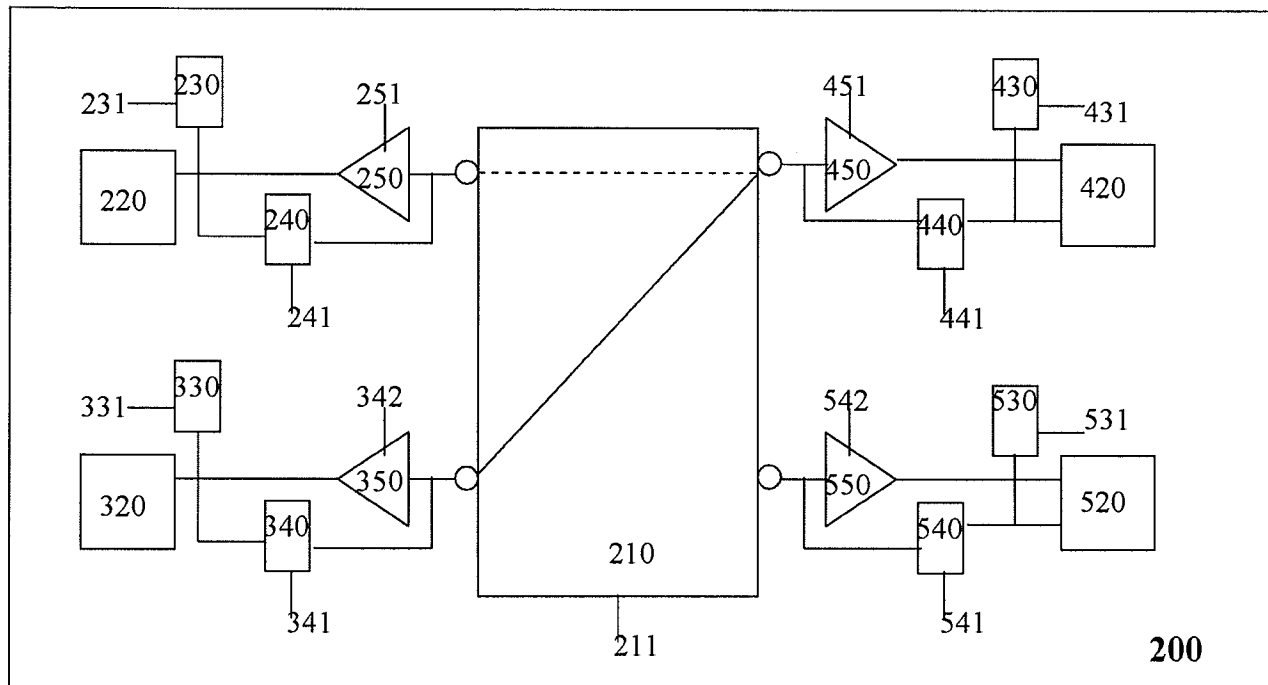


Figure 4

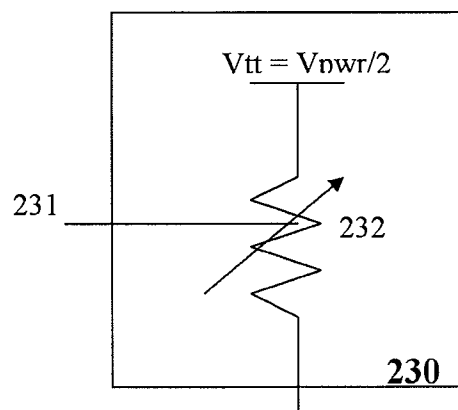
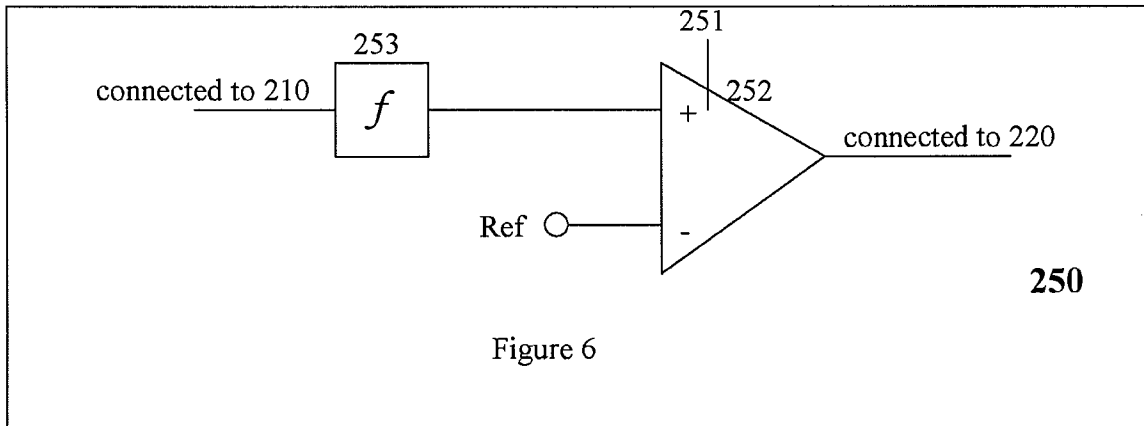


Figure 5



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Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

First Named Inventor/Applicant Name:

To be provided later .

Filer:

Bruce S. Itchkawitz/Dee Manena

Attorney Docket Number:

NETL.045PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
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Basic Filing:

Provisional application filing

1005

1

210

210

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				210

Electronic Acknowledgement Receipt

EFS ID:	3150693
Application Number:	61044839
International Application Number:	
Confirmation Number:	1764
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	To be provided later .
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Quyen Lieu
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045PR
Receipt Date:	14-APR-2008
Filing Date:	
Time Stamp:	19:31:38
Application Type:	Provisional

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NETL045PR_ADS.pdf	243412 6fe62a8c8c1875ea429c67ca1975c8e79e771125	no	4
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2		NETL045PR_APPLICATION.pdf	1484241 ae311f5ba5e9257dbc18cc928d5b60517f1bbeb36	yes	34
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	10	
	Claims		11	16	
	Abstract		17	17	
	Appendix to the Specification		18	34	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	NETL045PR_DRAWINGS.pdf	42457 70fcb3793d9dc2d11febe9ff14d311dcf1996cd9	no	6
Warnings:					
Information:					
4	Fee Worksheet (PTO-06)	fee-info.pdf	8142 af8709838e65915db8ed3d0e5379b7b16da146dc	no	2
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,839	04/14/2008		210	NETL.045PR		

CONFIRMATION NO. 1764

FILING RECEIPT

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IRVINE, CA 92614



Date Mailed: 04/22/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please write to the Office of Initial Patent Examination's Filing Receipt Corrections. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney: The patent practitioners associated with Customer Number 20995

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,839**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,839	04/14/2008		NETL.045PR

CONFIRMATION NO. 1764

FORMALITIES LETTER



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IRVINE, CA 92614

Date Mailed: 04/22/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of **\$50** for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$50** for a non-small entity

- **\$50** Surcharge.

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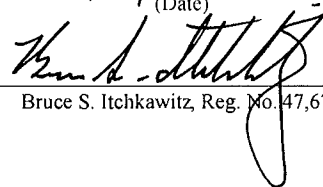
Applicant : Hyun Lee et al.
App. No : 61/044,839
Filed : April 14, 2008
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Art Unit : Unknown

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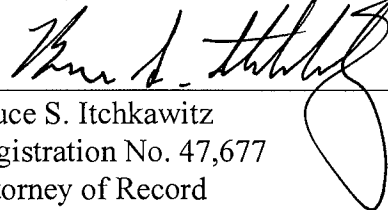
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Dear Sir:

The above-captioned application was filed without the inventors information and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following:

- (X) A Notice to File Missing Parts dated April 22, 2008.
- (X) Supplemental Application Data Sheet.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.



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Registration No. 47,677
Attorney of Record
Customer No. 20,995
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61/044,839	04/14/2008		NETL.045PR

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IRVINE, CA 92614

Date Mailed: 04/22/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of \$50 for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is \$50 for a non-small entity

- \$50 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

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<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

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/mduong/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/044,839
Filing Date:: April 14, 2008
Application Type:: Provisional
Subject Matter:: Utility
CD-ROM or CD-R?:: None
Title:: CIRCUIT PROVIDING LOAD ISOLATION AND
NOISE REDUCTION
Attorney Docket Number:: NETL.045PR
Request for Early Publication?:: No
Request for Non-Publication?:: No
Total Drawing Sheets:: 6
Small Entity?:: No
Petition Included?:: No
Secrecy Order in Parent Appl.?:: No

Inventor Information

Applicant Authority Type:: Inventor
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Representative Customer Number:: 20,995

Assignment Information

Assignee Name:: Netlist, Inc.
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State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92618

5519318
061208

Electronic Patent Application Fee Transmittal

Application Number:	61044839			
Filing Date:	14-Apr-2008			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/alexis swan			
Attorney Docket Number:	NETL.045PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	3464635
Application Number:	61044839
International Application Number:	
Confirmation Number:	1764
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Heide Young
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045PR
Receipt Date:	16-JUN-2008
Filing Date:	14-APR-2008
Time Stamp:	19:07:24
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	3759
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	NETL_045PR.pdf	201021	no	7
			35bb1b32a82863f1b229c06677111108a72ad1af		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	8170	no	2
			ce73b9b98f6cf828e46a4e6ef04d1e54abb57ba3		

Warnings:**Information:**

Total Files Size (in bytes):			209191
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,839	04/14/2008		260	NETL.045PR		

CONFIRMATION NO. 1764

UPDATED FILING RECEIPT

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614



OC000000030523670

Date Mailed: 06/23/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/16/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,839**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing

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Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.044PR
		Application Number	
Title of Invention	SELF-ADJUSTING DAMPER		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
		☐ Party of Interest under 35 U.S.C. 118		
Prefix	Given Name	Middle Name	Family Name	Suffix
	To be provided later			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residenceⁱ	
Citizenship under 37 CFR 1.41(b)ⁱ				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Countryⁱ		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				
				Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	SELF-ADJUSTING DAMPER		
Attorney Docket Number	NETL.044PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	3	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.044PR
		Application Number	
Title of Invention	SELF-ADJUSTING DAMPER		

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

Remove			
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

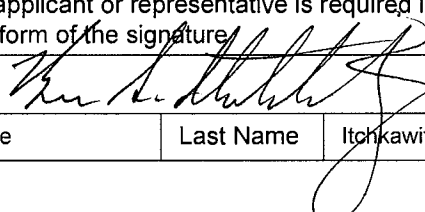
Assignee 1

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.044PR
		Application Number	
Title of Invention	SELF-ADJUSTING DAMPER		

If the Assignee is an Organization check here. ☒			
Organization Name	NETLIST, INC.		
Mailing Address Information:			
Address 1	51 Discovery, #150		
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City	Irvine	State/Province	CA
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Phone Number	949-435-0025	Fax Number	949-435-0031
Email Address	hlee@netlist.com		
Additional Assignee Data may be generated within this form by selecting the Add button.			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature			Date (YYYY-MM-DD)	2008-04-14	
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

SELF-ADJUSTING DAMPER**BACKGROUND**Field

[0001] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0002] Complex electronic systems often contain multiple sources of electrical noise. Such sources may include noise from signal cross coupling, transmission line reflections, ground bounce noise, noise due to an unstable power supply voltage, noise due to external EM waves, etc. In some cases these noise sources may be generally negligible depending on the noise tolerance of the system. However, many modern electronic systems incorporate relatively low supply voltages to accommodate state-of-art semiconductor processing technology. In such systems, the noise tolerance of the system is generally decreased. This phenomenon is clearly demonstrated in dynamic random-access memory (DRAM) systems where the operating voltage is sometimes substantially reduced in order to reduce the leakage current and optimize the performance.

[0003] The low power supply voltage can reduce the noise tolerance and can be compensated for, at least in part, by additional electronic components to ensure proper operation of the DRAM modules. For example, terminating resistors may be added in some cases (e.g., for DDR2 and DDR3 memory modules) at transmission line discontinuity points. Although terminating resistors may reduce the signal noise by lowering the amplitude of reflected waves, there may still remain substantial signal ringing since, in most cases, values of the terminating resistor on the memory module may not completely match the system board impedance. Furthermore, because each electrical component in a system exhibits its own unique electrical characteristics including the output driving capabilities, the signals generated from each electrical component produce their own unique reflected wave patterns, which cannot be completely matched with a set of impedance values.

[0004] Circuits often used to remove noises (e.g., to counteract ringing signals and/or reflected signals with small amplitudes) from the received signals include band-pass

filters, impedance matching circuits, echo cancellers, and error-correction circuits that reconstruct the transmitted signal. Each of these circuits exhibits specific characteristics and are used in well-defined application areas. In general, band-pass filters are analog circuits which generally pass frequencies within a certain range and reject frequencies outside that range. Impedance matching circuits are generally characterized by a fixed impedance value that exhibits impedance characteristics similar to those of the transmission line to which it is interfacing so as to reduce signal reflections. Echo cancellers and the error-correction circuits are typically digital circuits that interact with a micro-processor.

SUMMARY

[0005] In certain embodiments, a self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is, in certain embodiments, substantially representative of the signal component of the input signal.

[0006] In certain embodiments, a method of removing a noise component of an electrical signal is provided. The method involves receiving an input signal including a noise component and a signal component. The method further includes generating a first signal corresponding to the noise component of the input signal and generating a second signal corresponding to a delayed version of the input signal. The method also involves adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal which, in certain embodiments, is substantially representative of the signal component of the input signal.

[0007] In certain embodiments, a self-adjusting damper circuit is provided which includes a means for receiving an input signal comprising a noise component and a signal component, a first generation means for generating a first signal corresponding to the noise component of the input signal, and a second generation means for generating a second signal

corresponding to a delayed version of the input signal. The self-adjusting damper circuit also includes a means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal. In certain embodiments, the output signal is substantially representative of the signal component of the input signal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure 1 is a block diagram illustrating an example system incorporating self-adjusting damper circuits in accordance with certain embodiments described herein.

[0009] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0010] Figure 3 is a schematic diagram illustrating another example self-adjusting damper circuit in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0011] In many electronic systems, while multiple noise sources contribute to the distortion of the transmitted signal, for conventional systems there are usually one or two major noise sources while other noise sources make minor contributions. Thus, system designers have been able to implement one or two noise removing circuits in conventional electronic systems to remove the major noise sources and to ignore the minor noise sources. However, in some high performance systems such as memory subsystems, even relatively minor noises may contribute significant signal distortions due to, for example, the lower supply voltages. For example, a major noise contribution can be the reflected signals at the transmission line discontinuity points, and a minor noise contribution can be due to signal cross-coupling. Such systems may benefit from a noise filtering scheme that may include various types of noise removing circuits but may also potentially increase cost, complexity, and/or the time for system development.

[0012] Certain embodiments described herein present a self-adjusting electrical damper applicable to a wide frequency range. Certain embodiments described herein are implemented in an application-specific integrated circuit (ASIC) or in a System-on-Chip (SoC) device.

[0013] Certain embodiments described herein provide a self-adjusting damper configured to remove signal noise contributed by, for example signal cross-coupling and the

signal reflections. In certain embodiments, the self-adjusting damper is a component of a system comprising signal buses and transmission lines. In certain embodiments, the self-adjusting damper employs band-pass filtering and echo cancellation without any external control inputs.

[0014] In certain embodiments, the self-adjusting damper can be implemented either with discrete components or integrated into an ASIC device. Advantageously, the damper of certain embodiments is capable of reducing noise due to, for example, signal coupling between wires or signal reflection caused by mismatched terminations of transmission lines. In certain embodiments, the self-adjusting damper includes a delay component, a negative voltage controlled current source, and a positive voltage controlled current source. In certain embodiments, the signal is delayed through at least one of the current sources. In certain embodiments, the signal delay through the delay component is substantially similar to the signal delay through the current sources.

[0015] In certain embodiments, the voltage controlled current sources react to a change of the input voltage and have corresponding current source multiplication factors g_1 and g_2 . These multiplication factors are variables that are functions of the slew rate of the input signal such that a faster input transition produces higher transconductance on the current sources, smoothing out or reducing the signal ringing on the output. Advantageously, the damper circuit of certain embodiments actively counteracts signal noise, thereby allowing noise reduction over a broad range of noise frequencies rather than over, for example, a frequency window. Moreover, in certain embodiments, the damper circuit actively counteracts signal noise regardless of the periodicity of the input signal and/or of the noise component of the signal. In certain embodiments, the damper is implemented without analog circuit technology (e.g., within an ASIC without invoking analog circuit technology), thereby enabling easy design migration to various process technologies.

[0016] Figure 1 is a block diagram illustrating an example system incorporating example self-adjusting damper circuits 141, 142 in accordance with certain embodiments described herein. A plurality of drivers 101, 102 transmit signals via transmission lines or wires with a cross coupling impedance of Z_c 120. Z_c 120 includes cross-coupling capacitance and (mutual) inductance as well as any transconductance between the wires. In

certain embodiments, the impedance matching components 131, 132 may comprise a fixed termination resistor for the respective transmission line 111, 112. The self-adjusting damper circuits 141, 142 of certain embodiments advantageously reduce cross coupling noise due to Z_c and/or the remaining reflected wave after impedance matching is performed by the impedance matching components 131, 132.

[0017] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein. According to certain embodiments, the mathematical representation of the output 202 with load capacitance C_L can be expressed by the following equations:

$$V_o(t) = \int \left[\left(I(t) - \frac{g_1 \delta V_i(t - \Delta \tau)}{dt} - \frac{g_2 \delta V_i(t - \Delta \tau)}{dt} \right) C_L \right] dt$$

$$= \left[\int I(t) dt - (g_1 \delta V_i(t - \Delta \tau) + g_2 \delta V_i(t - \Delta \tau)) \right] C_L$$

Where,

$$I(t) = \frac{(V_o(t) - V_i(t))}{Z_d}$$

Thus,

$$Z_d * dV_o(t) / dt = V_i(t) - V_o(t) - (g_1 + g_2) * C_L * d(\delta V_i(t - \Delta \tau)) / dt$$

And,

$$V_o(t) = V_i(t) - (g_1 + g_2) * C_L * d(\delta V_i(t - \Delta \tau)) / dt - Z_d * dV_o(t) / dt$$

[0018] The above equations show that the output voltage of certain embodiments is a function of the change rate of the input signal transition rate and the change rate of the output itself. Because the change rate of the input transition rate and the output change rate have a negative effect on the output voltage, the damper of certain embodiments reduces noise having a slew rate that is faster than the slew rate of the non-noise component of the signal. Thus, certain embodiments provide a self-adjusting damper that effectively removes or reduces noises such as cross-coupling noises and reflected signals in complex electrical and electronic systems.

[0019] Figure 3 is a schematic diagram of another example of a self-adjusting damper circuit in accordance with certain embodiments described herein. The illustrated

embodiment comprises two voltage controlled current sources 340, 350 which are included in a first sub-circuit. The current source 340 of the illustrated embodiment is an AC-coupled, negative-gain pmos voltage-controlled current source for increasing the current flow from the power supply when the input sees a negative noise component, thus reducing the negative noise component on the output. The current source 340 includes two pmos transistors 341, 342. The pmos transistor 341 is a current-limiting transistor that ensures that the current supplied to the output has a relatively smaller magnitude than does the current to/from the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other pmos transistor 342 provides the negative-voltage-controlled current source function by AC coupling with the input signal V_i 301, on an input terminal, via a capacitor 311. Thus, the transconductance of this transistor 342 changes inversely proportionally to the slew rate of the input.

[0020] The current source 350 of the illustrated embodiment is an AC-coupled, positive-gain nmos current source for increasing the current flow to ground when the input sees a positive noise component, thus reducing the positive noise component on the output. The current source 350 includes two nmos transistors 351, 352. The nmos transistor 351 is a current-limiting transistor that ensures the current taken away from the output has a relatively smaller magnitude than does the current from/to the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other nmos transistor 352 provides the positive-voltage-controlled current source function by AC coupling with the input signal V_i 301 via a capacitor 311. Thus, the transconductance of this transistor 342 changes proportionally with the slew rate of the input.

[0021] In the illustrated embodiment, the resistor R_d 310 is included in a second sub-circuit. The resistor R_d 310 is a delay element which introduces a signal delay compatible with the current source delay from the first sub-circuit and also may reduce the reverse feedback of the output signal back to the input. Since the delay time from input V_i 301 on the input terminal to output V_o 302 on the output terminal is the function of the time constant, $R_d * C_L$, the resistance of the resistor R_d 310 can be selected based on the estimated value of C_L 360. Thus, in certain embodiments, the input signal can be delayed to match the reaction time of the voltage-controlled current sources that generate a negative feedback

current to remove signal noise. In certain embodiments the damper 141 can reduce cross coupling noise and reflections having various slew rate ranges.

[0022] In certain embodiments, the damper also includes a current-limiting, self-biasing op bias for biasing the reference voltage so that the operation point can be established in the middle of the linear range.

[0023] Advantageously, in certain embodiments, the damper is configured to be implemented in an application specific integrated circuit (“ASIC”) without requiring analog circuit technologies, enabling easy migration to other process technologies. For example, the damper is configured to be implemented in digital process technologies comprising a component resolution of approximately plus or minus 20 percent. Moreover, in certain embodiments, the damper may be implemented with no external control signals or memory elements.

[0024] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-adjusting damper circuit, comprising:
 - an input terminal adapted to receive an input signal comprising a noise component and a signal component;
 - an output terminal;
 - a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal; and
 - a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal, wherein the circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal, the output signal substantially representative of the signal component of the input signal.
2. The circuit of Claim 1, wherein the second sub-circuit comprises a resistor.
3. The circuit of Claim 1, wherein the first sub-circuit comprises a current source.
4. The circuit of Claim 3, wherein the current source comprises a voltage-controlled current source.
5. The circuit of Claim 4, wherein voltage-controlled current source comprises one or more transistors.
6. The circuit of Claim 1, wherein the first sub-circuit comprises a negative current source and a positive current source.
7. The circuit of Claim 6, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.
8. The circuit of Claim 7, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the trans-conductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.

9. The circuit of Claim 7, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

10. The circuit of Claim 9, wherein the first and second current limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

11. The circuit of Claim 10, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current limiting component, and the second current limiting component comprise one or more transistors.

12. The circuit of Claim 1, wherein the first sub-circuit is adapted to generate the first signal by generating a positive signal when the noise component of the input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

13. The circuit of Claim 12, wherein the first sub-circuit is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

14. The circuit of Claim 1, wherein the first signal comprises negative feedback current.

15. The circuit of Claim 1, wherein the circuit is configured to be operate on a digital semiconductor process technology.

16. The circuit of Claim 1, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution of less than plus or minus 20 percent.

17. The circuit of Claim 1, wherein the circuit is implemented on a memory module.

18. The circuit of Claim 1, wherein the noise-counteracting signal actively counteracts a noise component of the delayed version of the input signal.

19. A method of removing a noise component of an electrical signal, the method comprising:

receiving an input signal comprising a noise component and a signal component;

generating a first signal corresponding to the noise component of the input signal;

generating a second signal corresponding to a delayed version of the input signal; and

adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal substantially representative of the signal component of the input signal.

20. The method of Claim 19, wherein generating the first signal comprises generating a positive signal when the noise component of the input signal is negative and generating a negative signal when the noise component of the input signal is positive.

21. The method of Claim 20, wherein generating a positive signal comprises increasing the current flow from a power supply to the first signal and generating a negative signal comprises increasing the current flow from the first signal to ground.

22. The method of Claim 19, further comprising limiting the magnitude of the first signal in relation to the second signal.

23. The method of Claim 19, wherein adding the noise-counteracting signal to the second signal actively counteracts a noise component of the delayed version of the input signal.

24. A self-adjusting damper circuit, comprising:

means for receiving an input signal comprising a noise component and a signal component;

first generation means for generating a first signal corresponding to the noise component of the input signal; and

second generation means for generating a second signal corresponding to a delayed version of the input signal,

means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal, the output signal substantially representative of the signal component of the input signal.

25. The circuit of Claim 24, wherein the second generation means comprises a resistor.

26. The circuit of Claim 24, wherein the first generation means comprises a current source.

27. The circuit of Claim 26, wherein the current source comprises a voltage-controlled current source.

28. The circuit of Claim 27, wherein voltage-controlled current source comprises one or more transistors.

29. The circuit of Claim 24, wherein the first generation means comprises a negative current source and a positive current source.

30. The circuit of Claim 29, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.

31. The circuit of Claim 30, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the transconductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.

32. The circuit of Claim 31, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

33. The circuit of Claim 32, wherein the first and second current-limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

34. The circuit of Claim 33, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current-limiting component, and the second current-limiting component comprise one or more transistors.

35. The circuit of Claim 24, wherein the first generation means is adapted to generate the first signal by generating a positive signal when the noise component of the

input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

36. The circuit of Claim 35, wherein the first generation means is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

37. The circuit of Claim 24, wherein the first signal comprises negative feedback current.

38. The circuit of Claim 24, wherein the circuit is configured to be operate on a digital semiconductor process technology.

39. The circuit of Claim 24, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution less than plus or minus 20 percent.

40. The circuit of Claim 24, wherein the circuit is implemented on a memory module.

41. The circuit of Claim 24, wherein the addition means actively counteracts a noise component of the delayed version of the input signal.

SELF-ADJUSTING DAMPER

ABSTRACT

A self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is substantially representative of the signal component of the input signal.

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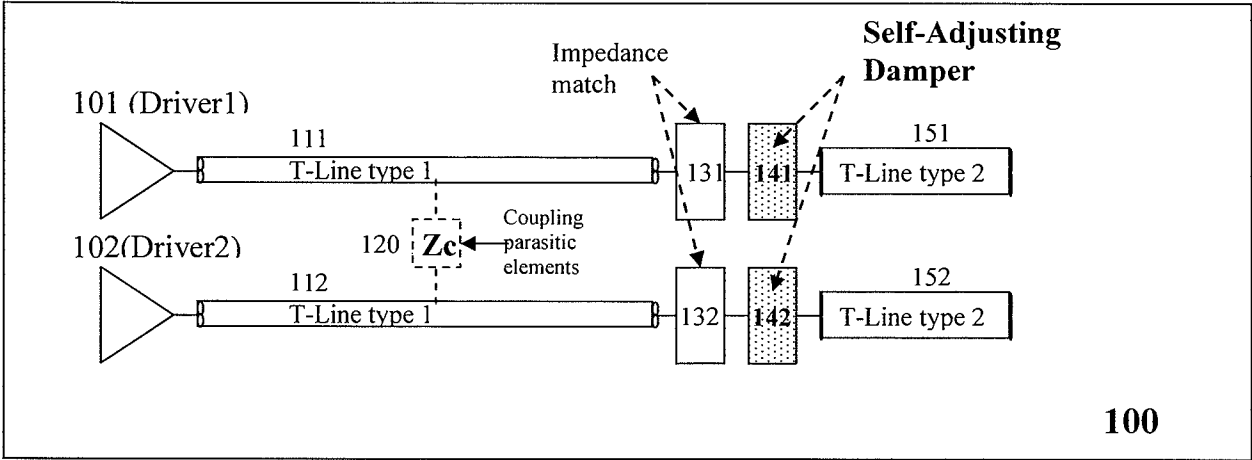


Figure 1

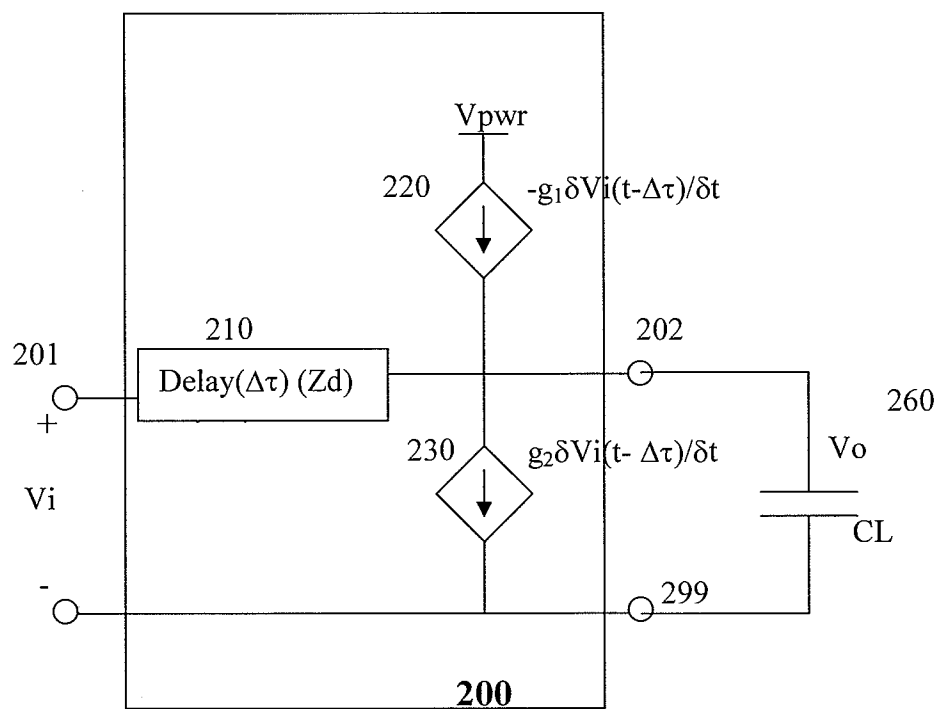


Figure2

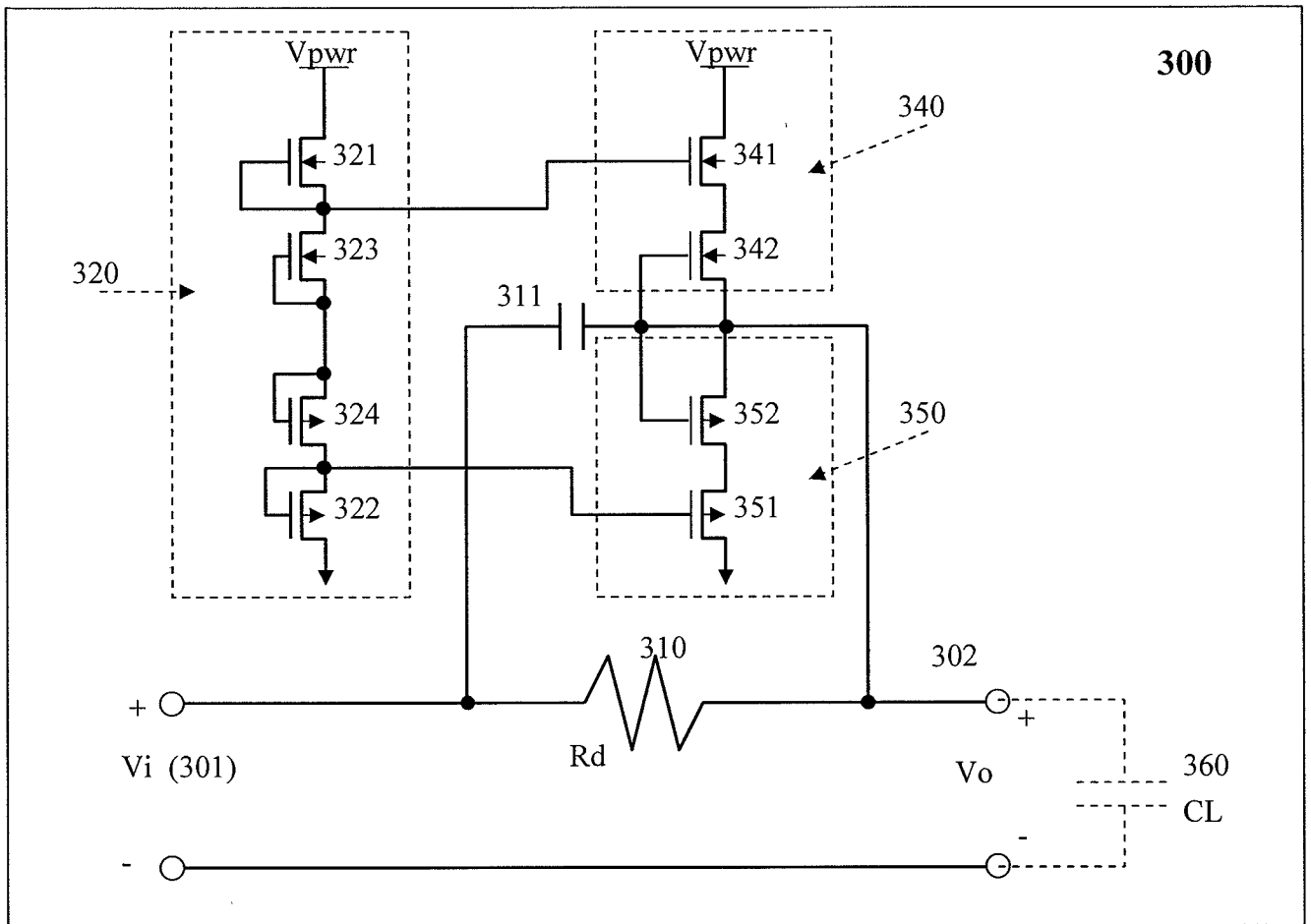


Figure 3

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

SELF-ADJUSTING DAMPER

First Named Inventor/Applicant Name:

To be provided later .

Filer:

Bruce S. Itchkawitz/Dee Manena

Attorney Docket Number:

NETL.044PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
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Basic Filing:

Provisional application filing

1005

1

210

210

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				210

Electronic Acknowledgement Receipt

EFS ID:	3150288
Application Number:	61044825
International Application Number:	
Confirmation Number:	1720
Title of Invention:	SELF-ADJUSTING DAMPER
First Named Inventor/Applicant Name:	To be provided later .
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Nicole Lauer
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.044PR
Receipt Date:	14-APR-2008
Filing Date:	
Time Stamp:	19:09:54
Application Type:	Provisional

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Payment Type	Credit Card
Payment was successfully received in RAM	\$210
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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NETL044PR_ADS.pdf	238840	no	4
			d8c233ab171200d8daae81b1a3fc40b7a5df15607		
Warnings:					
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2		NETL044PR_APPLICATION.pdf	612132	yes	13
			72bb8ada9928f27e6e6b1eb33298c875a35411226		
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	Document Description		Start	End	
	Specification		1	7	
	Claims		8	12	
	Abstract		13	13	
Warnings:					
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3	Drawings-only black and white line drawings	NETL044PR_DRAWINGS.pdf	26554	no	3
			1a63cf466dd06eb6b89db8757150145d5d600aa1		
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4	Fee Worksheet (PTO-06)	fee-info.pdf	8110	no	2
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,825	04/14/2008		210	NETL.044PR		

CONFIRMATION NO. 1720

FILING RECEIPT



OC000000029444807

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Date Mailed: 04/18/2008

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Applicant(s)

Power of Attorney: The patent practitioners associated with Customer Number 20995

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,825**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

SELF-ADJUSTING DAMPER

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,825	04/14/2008		NETL.044PR

CONFIRMATION NO. 1720

FORMALITIES LETTER



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FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of **\$50** for a non-small entity, must be submitted with the missing items identified in this notice.

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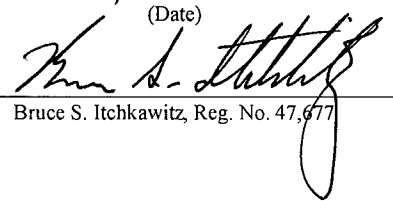
Applicant : Hyun Lee et al.
App. No : 61/044,825
Filed : April 14, 2008
For : SELF-ADJUSTING DAMPER
Art Unit : Unknown

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

6/16/08

(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

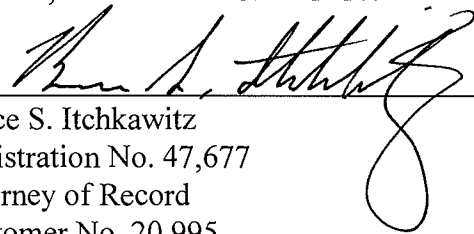
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without the inventors information and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following:

- (X) A Notice to File Missing Parts dated April 18, 2008.
- (X) Supplemental Application Data Sheet
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,825	04/14/2008		NETL.044PR

CONFIRMATION NO. 1720

FORMALITIES LETTER



0000000029444808

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/18/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of \$50 for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is \$50 for a non-small entity

- \$50 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

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<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

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/aabranys/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/044,825
Filing Date:: April 14, 2008
Application Type:: Provisional
Subject Matter:: Utility
CD-ROM or CD-R?:: None
Title:: SELF-ADJUSTING DAMPER
Attorney Docket Number:: NETL.044PR
Request for Early Publication?:: No
Request for Non-Publication?:: No
Total Drawing Sheets:: 3
Small Entity?:: No
Petition Included?:: No
Secrecy Order in Parent Appl.?:: No

Inventor Information

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Hyun
Family Name:: Lee
City of Residence:: Ladera Ranch
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 21 Thalia Street
City:: Ladera Ranch

State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92694

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Jayesh
Middle Name:: R.
Family Name:: Bhakta
City of Residence:: Cerritos
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 12220 Rose Street
City:: Cerritos
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 90703

Applicant Authority Type:: Inventor
Primary Citizenship Country:: Taiwan
Status:: Full Capacity
Given Name:: Chi-She
Family Name:: Chen
City of Residence:: Walnut
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 944 Crystal Water Lane

City:: Walnut
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 91789

Correspondence Information

Correspondence Customer Number:: 20,995
Phone Number:: (949) 760-0404
Fax Number:: (949) 760-9502
E-Mail Address:: efiling@kmob.com

Representative Information

Representative Customer Number:: 20,995

Assignment Information

Assignee Name:: Netlist, Inc.
Street:: 51 Discovery, Ste. #150
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92618

5519247
061208

Electronic Patent Application Fee Transmittal

Application Number:	61044825			
Filing Date:	14-Apr-2008			
Title of Invention:	SELF-ADJUSTING DAMPER			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/alexis swan			
Attorney Docket Number:	NETL.044PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	3464591
Application Number:	61044825
International Application Number:	
Confirmation Number:	1720
Title of Invention:	SELF-ADJUSTING DAMPER
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Lori Larson
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.044PR
Receipt Date:	16-JUN-2008
Filing Date:	14-APR-2008
Time Stamp:	19:54:53
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	4115
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	NETL_044PR.pdf	175330	no	6
			039d72533893d17831f3da11d1cf751b5fb01700		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	8138	no	2
			c96373900d3d8156a46644ab00c67617abbf1d56		

Warnings:**Information:**

Total Files Size (in bytes):			183468
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,825	04/14/2008		260	NETL.044PR		

CONFIRMATION NO. 1720

UPDATED FILING RECEIPT



OC000000030527948

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/19/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Chi-She Chen, Walnut, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/17/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,825**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

SELF-ADJUSTING DAMPER

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international

patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and

Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046PR
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name	Suffix
	To be provided later			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residence i	
Citizenship under 37 CFR 1.41(b) i				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Countryi		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
Attorney Docket Number	NETL.046PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	4	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046PR
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

			Remove
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

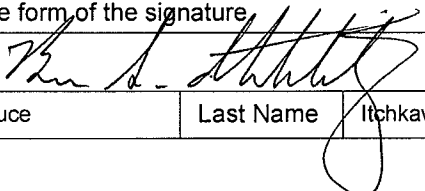
Assignee 1

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046PR
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

If the Assignee is an Organization check here. ☒			
Organization Name	NETLIST, INC.		
Mailing Address Information:			
Address 1	51 Discovery, #150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	949-435-0031
Email Address	hlee@netlist.com		
Additional Assignee Data may be generated within this form by selecting the Add button.			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature			Date (YYYY-MM-DD)	2008-04-14	
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

MEMORY BOARD WITH SELF-TESTING CAPABILITY**BACKGROUND**Field

[0001] The present invention relates to self-testing electronic modules and, more particularly, to self-testing electronic memory modules.

Description of the Related Art

[0002] The failure of memory components in an electronic system may result in the loss of valid data. Therefore, it is important to ensure proper memory operation in an electronic system. Memory integrated circuits ("memory chips") often go through a series of tests at various stages of system manufacture. Once memory chips are deployed in a system, they also generally go through a system level memory test each time the system is booted. In addition, memory chips may undergo a parity checking process during normal system operation.

[0003] There are typically at least three test phases which memories undergo during system manufacture. Each phase generally tests for memory defects and for the correct operation of the input/output interface. The first test phase is typically conducted by the memory chip manufacturer and generally involves checking for bit failures, correct memory access speed, etc. The second test phase is typically done by memory module manufacturers and generally involves testing the signal quality, the noise susceptibility, and the operational speed of the memory module as a single unit. The second test phase may also include checking for bit failures in individual memory chips. The third phase is usually carried out by the system manufacturer. During the third phase, the interaction of the memory subsystem with other components in the system is tested. During the third phase, the individual memory module operation is also tested again and the memory array is checked for defects. Because of the significant amount of testing that memories undergo during the manufacturing process, there is generally substantial test cost and test time associated with ensuring the proper memory operation. This test cost and test time translate into an increase in system cost and a decrease in system performance.

[0004] There are a number of memory test methodologies that employ either external test hardware, embedded self-test logic ("MBIST"), or both. However, the usefulness of these test methodologies is limited due to the high cost and other limitations associated with them. For instance, external test hardware such as automatic test equipment ("ATE") is very expensive. Moreover, the development time and cost associated with implementing MBIST is relatively high. These costs and limitations are especially significant when testing dynamic random access memory ("DRAM"). For example, technological developments, such as increases in DRAM speed, may require manufacturers to upgrade ATE machines relatively frequently. In addition, MBIST in DRAM chips generally cannot be fully utilized for system level testing of memory boards.

[0005] Because of the increasing cost, complexity, and time involved with fully testing DRAM chips, DRAM manufacturers often provide "effectively tested" ("ETT") DRAM chips to memory module manufacturers at a lower price rather than providing fully tested DRAM chips. Memory module manufacturers often prefer the ETT DRAM chips mainly due to their greater availability. Memory module manufacturers who receive ETT DRAM chips then have to assume a part of the responsibility of validating the DRAM chips, adding to the complexity of the memory module test process.

SUMMARY

[0006] In certain embodiments, a self-testing memory module is provided which includes a printed circuit board configured to be electrically coupled to a memory controller of a computer system. The memory module also includes a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices includes data, address, and control ports. The memory module also includes a test control module configured to generate address and control signals for testing the memory devices. The test control module is further configured to selectively input to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module. The memory module also includes a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices. The test data handler module is further configured to write data to the memory devices by selectively inputting to the data ports of the memory

devices either data signals from the system memory controller or data signals from the test data handler module.

[0007] In certain embodiments, a method of self-testing a memory module is provided which involves providing a memory module which includes a printed circuit board configured to be electrically coupled to a memory controller of a computer system. The memory module also includes a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices includes data, address, and control ports. The memory module further includes a test control module configured to generate address and control signals for testing the memory devices, and a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices. The method further includes selectively inputting to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module. The method also involves writing data to the memory devices by selectively inputting to the data ports of the memory devices either data signals from the system memory controller or data signals from the test data handler module.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein.

[0009] Figure 2 is a block diagram of example control circuitry of an example self-testing memory module in accordance with certain embodiments described herein.

[0010] Figure 3 is a flow diagram of an embodiment of a method for testing a self-testing memory module in accordance with certain embodiments described herein.

[0011] Figure 4 is a flow diagram of an embodiment of a method for executing a test mode of a self-testing memory module in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0012] Certain embodiments described herein present a self-testing electronic system, such as for example, a self-testing electronic memory module. Some embodiments described herein present a self-testing memory module that is populated with ETT DRAM

chips. Certain embodiments described herein present a self testing registered dual in-line memory module (“RDIMM”). Some embodiments described herein present a self-testing RDIMM that does not require any additional pins other than the pins on the standard JEDEC RDIMM connector. Certain embodiments described herein present a self-testing memory module that includes a control block and a data block which can generate memory addresses and data according to the JEDEC standard memory protocol.

[0013] Some embodiments described herein presents a self-testing memory module that can be configured through an I²C interface and that allows test results to be read out through the I²C interface. Certain embodiments described herein present a self-testing memory module that allows a test function to be configured, controlled, and executed without a system memory controller. Some embodiments present a self-testing memory module that can be tested without any external test equipment. For example, certain embodiments present a self-testing memory module that can be tested without any system driven test procedure. Various embodiments described herein present a self-testing memory module that can be self-tested at a target system speed. One embodiment described herein present a self-testing memory module that can generate data (DQ) and data-strobe (DQS) signals with wave characteristics that resemble the wave characteristics of DQ and DQS signals from a system memory controller.

[0014] MBIST is commonly used to test memories embedded in application specific integrated circuits (“ASICs”) or system on chip integrated circuits (“SoCs”) such as, for example, advanced memory buffers (“AMBs”). The MBIST implementation usually includes three distinct functional blocks: the address/control generator, the data generator/checker (sometimes referred to as a signature analyzer), and the test interface controller/register (“JTAG”). The test instructions and test patterns are generally loaded to the address/control generator and the data generator/checker through the JTAG interface. The test results are generally read out through the JTAG interface.

[0015] In memory module applications, there are obstacles associated with using MBIST. One of these obstacles is the large number of signals required to interface the three MBIST functional blocks. The inter-block timing requirements can present another obstacle. In addition, the ability to control the test is limited. For example, synchronizing the address/or

control signals with the data signal can be difficult. The relatively large amount of information needed to be gathered and stored while testing the memory presents yet another obstacle.

[0016] Unlike on an ASIC, in which the three MBIST functional blocks are in a single chip, the MBIST functional blocks on a memory module generally would be segregated into multiple chips on the memory module due to physical and electrical limitations and requirements. This makes implementing MBIST on a memory module difficult because, while there is virtually no limit on the number of available interface signals among the three MBIST functional blocks in an ASIC, the memory module can support only a limited amount of interface signals between memory chips. In addition, because the MBIST functional blocks are spread out to multiple chips on a memory module, the inter-block signal delay is generally much longer on a memory module than on an ASIC. It is also generally not feasible to implement cross-checking logic that operates the three MBIST functional blocks in lock-step. This is due to the limitation on the number of interface signals and to the relatively long inter-block signal delay time on a memory module versus on an ASIC. Finally, in most cases, since an ASIC cannot be repaired, the ASIC MBIST is generally only capable of detecting and reporting the pass/fail status of memory tests. Memory module test results, on the other hand, should generally include both addresses of the memory locations where failures occur and the data patterns that were read back from the failed memory locations. This type of reporting can help to facilitate the repair of the memory module by, for example, allowing for the identification and replacement of failed components.

[0017] In one embodiment, the self-testing memory module comprises logic including two blocks: a control block and a data block. The control block may also be referred to as, for example, a test control module. The data block may be referred to as, for example, a test data handler module. The control block may be implemented in the control register. The data block can be implemented as data path switches in the data path. In certain embodiments, the data block generates test data patterns to write to the memory module and checks the data patterns against corresponding data patterns that are expected to be read back from the memory locations. For example, in one embodiment, the data block generates random patterns to write to the memory module. In some embodiments, the data block also isolates the data path from the system board while the module is not accessed by the system.

For example, the data block may isolate the data path from the system board when the module is in a self-testing mode.

[0018] The control block may include, for example, a dual input register for registering address and control signals coming from either the self-testing logic or from the memory controller on the system board. In some embodiments, during testing, the control block generates address and control signals associated with memory locations to be tested and the data block generates corresponding test data patterns and provides them to the appropriate memory chips. For example, the data block may receive a read/write signal from the control block and provide data to be written to certain locations in the memory chips during a write operation. The data block may then read back the data from those locations and compare the read data with expected reference data. If there is a mismatch between the read data and the reference data, the data block may, for example, store the bit failure information and inform the control block about the failure. The control block may save the address of the memory location where the failure occurred.

[0019] In some embodiments, the control block and the data block produce memory addresses and data according to the JEDEC standard memory protocol. In some embodiments, the address sequences and the data patterns may be programmable either through the I²C interface or they may be defaulted to pre-defined values. In certain embodiments, the test speed, for example, may be defined by the speed of the clock. In some embodiments, for example, the control block and the data block generate the memory interface signals with proper edge relationships based on the JEDEC standard. In one embodiment, the data block is also able to generate data (DQ) and data-strobe (DQS) signals with programmable slew rates and programmable peak values so that the signals generally resemble the DQ and DQS signals generated by the system memory controller.

[0020] In some embodiments, the self-testing memory module does not require any additional DIMM connector pins. For example, the control block and the data block may utilize the address and control signals along with a test signal to enable and execute the self testing function.

[0021] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein. The self-testing memory module 100

may comprise, for example, a dual in-line memory module (“DIMM”). In one embodiment, the memory module 100 self-testing circuitry includes a control block 120 which includes a control mixer 122. In certain embodiments, the control mixer 122 generally controls the address and the control signals for the self-testing function. The data block 110, includes at least one data handler 111 which generally controls the data signals during self-testing operation. The data block 110 may include a plurality of data handlers 111. The data handlers 111 may operate handle different data segments. In some embodiments, for example, the data handler 111 may handle byte or nibble wise data segments. The control block 120 may further include a register 121 which, in some embodiments, may comprise a JEDEC standard component.

[0022] Figure 2 is a block diagram of example control circuitry 200 of an example self-testing memory module in accordance with certain embodiments described herein. The control circuitry 200 may include functional blocks and other features not shown in Figure 2. The data handler 111 may include a data multiplexer/demultiplexer (“data mux/demux”) 212 and a test data handler 213. The data mux/demux 212 may provide a bi-directional data multiplexer function. In some embodiments, the data mux/demux 212 selects the data to be written to the memories during memory write operations from either a system memory controller during normal operation, or from the test data handler 213 during self-test operation. The data mux/demux 212 may, in some embodiments, send the data read from the memories during memory read operations to both test data handler 213 and the system memory controller.

[0023] In some embodiments, the control block 220 includes a control mixer 122. The control mixer 122 may include a DRAM controller 223 and a test controller 224. The DRAM controller 223 generally pre-processes address and control information before it sends the information to the register 221. In one embodiment, the DRAM controller 223 receives address and control signals from both the system memory controller and the test controller 224. The DRAM controller 223 may send either the signals from the system memory controller or, alternatively, the signals from the test controller 224, to the register 221 depending on whether the memory module is in normal mode or test mode, respectively. In one embodiment, the DRAM controller 223 generates the address and control signals for

DRAM operations. The test controller 224 controls the generation of the address and control signal sequences to be used during self-testing operation and also communicates with the test data handler 213. The test data handler 213, based on control information it receives from the test controller 224, may generate data patterns to be written to the memory when the self-testing memory module performs a memory write test. In addition, the test data handler 213 may also check data patterns that are read back from the memories when the self-testing memory module performs a memory read test. The checking of the data patterns may also be done based on the control information the test data handler 213 receives from the test controller 224. In one embodiment, when memory failures occur, the test data handler 213 stores data patterns associated with failed memory locations and the test controller 224 stores the addresses of the failed memory locations. In certain embodiments, the test controller 224 may be updated through an I²C interface with new data patterns and/or with alternative memory access sequences to conduct AC tests.

[0024] Figure 3 is a flow diagram of an embodiment of a method 300 for testing a self-testing memory module in accordance with certain embodiments described herein. At step 310 the method 300 enters an idle state after power up or, in some embodiments, when the memory board is connected to a mother board or to a test board. In one embodiment, at step 310, the contents of the register (e.g., the register 121 of Figure 1) are undefined, and the controllers are in unknown state. After power up, the method 300 resets the board at step 311. At step 311, although the self-test control logic in the data handler and in the control mixer (e.g., the data handler 111 and the control mixer 122 of Figure 1) are set to a default value, the data mux/demux and the DRAM controller 223 (e.g., the data mux/demux 212 and the DRAM controller 223 of Figure 2) do not go into a self-test mode. The method 300 next configures the test mode at step 312 through, for example, an I²C interface.

[0025] The method 300 next initiates the test mode at step 313. At step 313, for example, the tester readies the data handler and the control mixer (e.g. data handler 111 and control mixer 122 of Figure 1) beginning the self-testing of the memory module. At step 313 the method 300 updates the test data handler (e.g., the test data handler 213 of Figure 2) using, for example, the test controller (e.g., the test controller 224 of Figure 2). The method 300 executes the test mode of the memory module at step 314. At step 314 the method, for

example, in one embodiment, asserts or toggles an input signal on a parity-in pin of the memory module. The method 300 continues executing the test mode at step 314 until the self-test is complete or until the method 300 detects a certain number of memory failures such that a failure count exceeds a preset number. If the failure count exceeds the preset number the method 300 enters step 315 and “errors out” of the test mode. If the self-test is completed at step 314 and the failure count does not exceed the preset number, the method also enters step 315. At step 315, the method 300 sends out a test completion indicator using the test controller. At step 316, the method 300 allows the fail status of the test to be read out through, for example, an I²C interface.

[0026] In some embodiments, the method 300 is generally interruptible. For example, the method 300 may abandon the self-test and return to step 310 if the signal going into the parity-in pin is, for example, de-asserted or removed. In other embodiments the interrupt signal or condition may be different and may not be a parity-in signal. For example, in one embodiment, the method 300 may abandon the self-test and return to step 310 if a timer having a pre-determined count expires. When the self-testing is interrupted, the method 300 will return to step 310 and the bit failure information in the control mixer and the data handler becomes invalid.

[0027] Figure 4 is a flow diagram of an embodiment of a method 400 for executing a test mode of a self-testing memory module in accordance with certain embodiments described herein. For example, in one embodiment, the test mode may be similar to step 314 of method 300. In certain embodiments, the method 400 executes various tasks which configure the self-test mode of the self-testing memory module using an I²C interface. For example, the method 400 may configure the test controller (e.g., test controller 224 of Figure 2) through the I²C interface at step 452 and activate the test controller through the I²C at step 453. Alternatively, the method 400 may activate the test controller through the I²C interface without first updating the test mode at step 451. The method 400 enters the self-test mode at step 454 by, for example, asserting or detecting the assertion of the parity-in signal on a parity-in pin.

[0028] At step 413, the method begins the execution of the self-test sequence by clearing, for example, a set of test status registers. Next, the method 400 performs a burst

write function at step 415. For example, in certain embodiments, the method 400 writes multiple memory locations in the memory module that are to be tested at step 415 in a relatively short amount of time (or in a burst). In certain embodiments, the method 400 can perform multiple write bursts at step 415. Next, at step 416, the method 400 performs a read & compare function. For example, in certain embodiments, the method 400 reads back certain memory locations at step 416 that have been written at step 415 and compares the values with certain expected reference data. In some embodiments, the method 400 reads from multiple memory locations in the memory module in a relatively short amount of time (or in a burst). In certain embodiments, the method 400 can execute multiple read bursts at step 416 as it compares the read data with expected reference data. In some embodiments, the method 400 stores any failed read data and the associated expected reference data at step 417 using, for example, the test data handler (e.g. test data handler 213 of Figure 2).

[0029] The method 400 stores the addresses of the memory locations associated with the failed read data using the test controller at step 417. In other embodiments the failed read data, expected reference data, and the addresses of the memory locations associated with the failed read data may be stored using the same functional block. For example, in one embodiment, the failed read data, expected reference data, and the addresses of the memory locations associated with the failed read data are all stored using the test controller at step 417. In some embodiments, the test controller is generally inaccessible through the I²C interface during self-test operation. For example, the test controller is inaccessible through the I2C interface until a test failure occurs at step 417 or until the test passes at step 419. When the test is complete, the test results can be read out through, for example, the I²C interface.

[0030] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to

any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be electrically coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;
 - a test control module configured to generate address and control signals for testing the memory devices, the test control module further configured to selectively input to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module; and
 - a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices, the test data handler module further configured to write data to the memory devices by selectively inputting to the data ports of the memory devices either data signals from the system memory controller or data signals from the test data handler module.
2. The memory module of Claim 1, wherein the test data handler module further comprises a plurality of switches which perform the selective electrical coupling of the data ports.
3. The memory module of Claim 3, wherein the switches comprise one or more data multiplexer-demultiplexers.
4. The memory module of Claim 1, wherein the test control module further comprises a DRAM controller which performs the selective electrical coupling of the address and control ports.
5. The memory module of Claim 1, wherein the memory module is configured to check for failures in the operation of the memory devices by comparing data read from the memory devices with data written to the memory devices.
6. The memory module of Claim 5, wherein the memory module is configured to report failures in the operation of the memory devices via an I²C interface.

7. The memory module of Claim 5, wherein the test data handler module is configured to check for failures in the operation of the memory devices.

8. The memory module of Claim 5, wherein data associated with failures in the operation of the memory devices is stored in the test data handler module and wherein memory addresses associated with failures in the operation of the memory devices are stored in the test control module.

9. The memory module of Claim 1, wherein the memory module can be controlled via an I²C interface.

10. The memory module of Claim 1, wherein the memory module is configured to be operated in a test mode in which the test control module selectively inputs to the address and control ports of the memory devices the address and control signals from the test control module and wherein the test data handler module writes data to the memory devices by selectively inputting to the data ports of the memory devices the data signals from the test data handler module.

11. The memory module of Claim 1, wherein the test data handler module is configured to generate data signals with programmable slew rates.

12. The memory module of Claim 1, wherein the test data handler module is configured to generate data signals having variable peak values.

13. A method of self-testing a memory module, comprising:

providing a memory module comprising:

a printed circuit board configured to be electrically coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;

a test control module configured to generate address and control signals for testing the memory devices;

a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices; and

selectively inputting to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module; and

writing data to the memory devices by selectively inputting to the data ports of the memory devices either data signals from the system memory controller or data signals from the test data handler module.

14. The method of Claim 13, wherein the test data handler module further comprises a plurality of switches which perform the selective electrical coupling of the data ports.

15. The method of Claim 14, wherein the switches comprise one or more data multiplexer-demultiplexers.

16. The method of Claim 13, wherein the test control module further comprises a DRAM controller which performs the selective electrical coupling of the address and control ports.

17. The method of Claim 13, further comprising checking for failures in the operation of the memory devices by comparing data read from the memory devices with data written to the memory devices.

18. The memory module of Claim 17, further comprising reporting failures in the operation of the memory devices via an I²C interface.

19. The method of Claim 17, wherein the test data handler module checks for errors in the memory devices.

20. The method of Claim 17, further comprising:

storing data associated with the failures in the operation of the memory devices in the data handler module; and

storing memory addresses associated with the failures in the operation of the memory devices in the test control module.

21. The method of Claim 13, further comprising configuring the memory module via an I²C interface.

22. The method of Claim 13, further comprising:

selectively inputting to the address and control ports of the memory devices the address and control signals from the test control module when the memory module is operating in a test mode; and

writing data to the memory devices by selectively inputting to the data ports of the memory devices the data signals from the test data handler module when the memory module is operating in the test mode.

23. The method of Claim 13, wherein the data handler module is configured to generate data signals with programmable slew rates.

24. The method of Claim 13, wherein the data handler module is configured to generate data signals having variable peak value.

MEMORY BOARD WITH SELF-TESTING CAPABILITY

ABSTRACT

A self-testing memory module is provided that includes a printed circuit board and is configured to couple to a memory controller. Memory devices are on the printed circuit board and include data, address, and control ports. The memory module includes a test control module which generates address and control signals to test the memories. The test control module is configured to selectively input to the address and control ports of the memories the address and control signals from the memory controller or the address and control signals from the test control module. The memory module also includes a test data handler module which generates and writes data to, and reads data from, the memories. The test data handler module writes data to the memories by selectively inputting to the data ports of the memories data signals from the system memory controller or data signals from the test data handler module.

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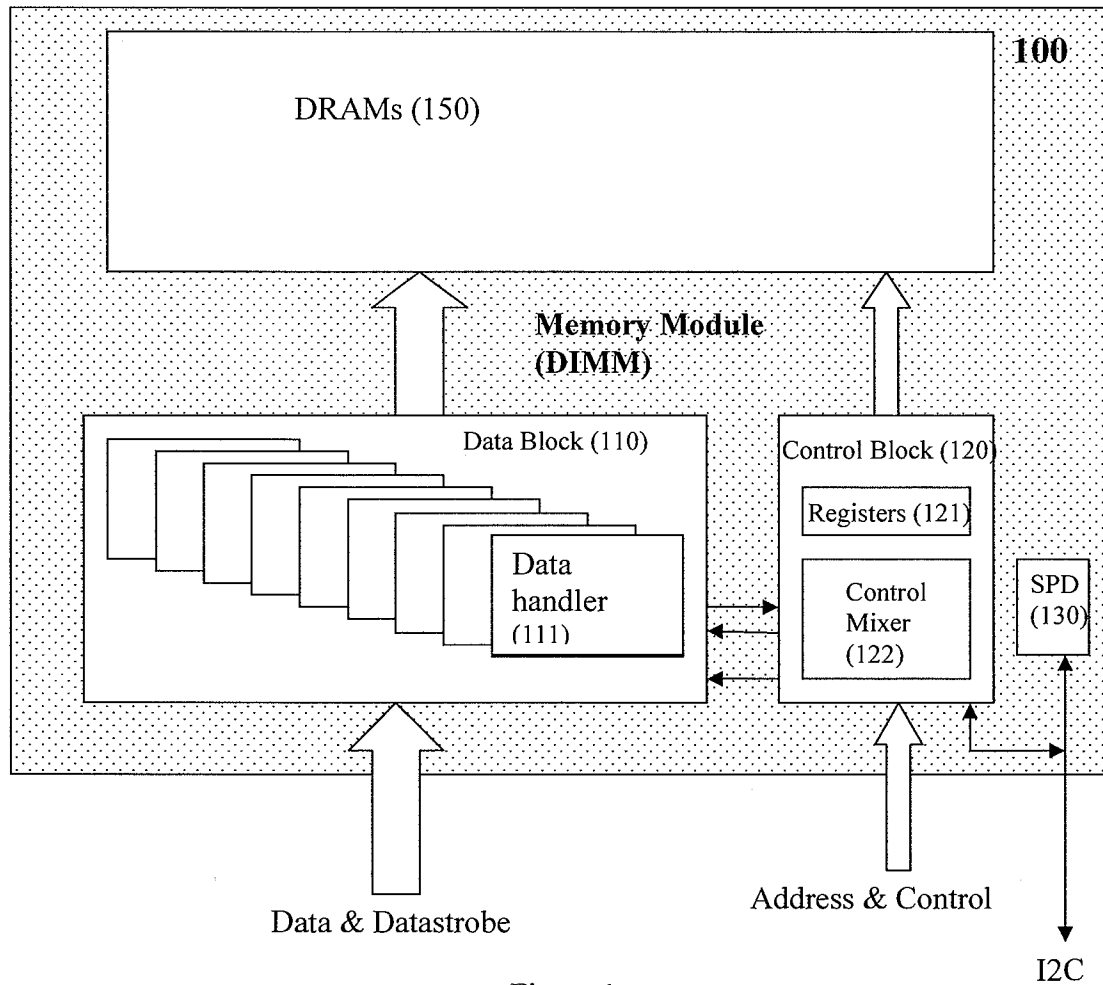


Figure 1

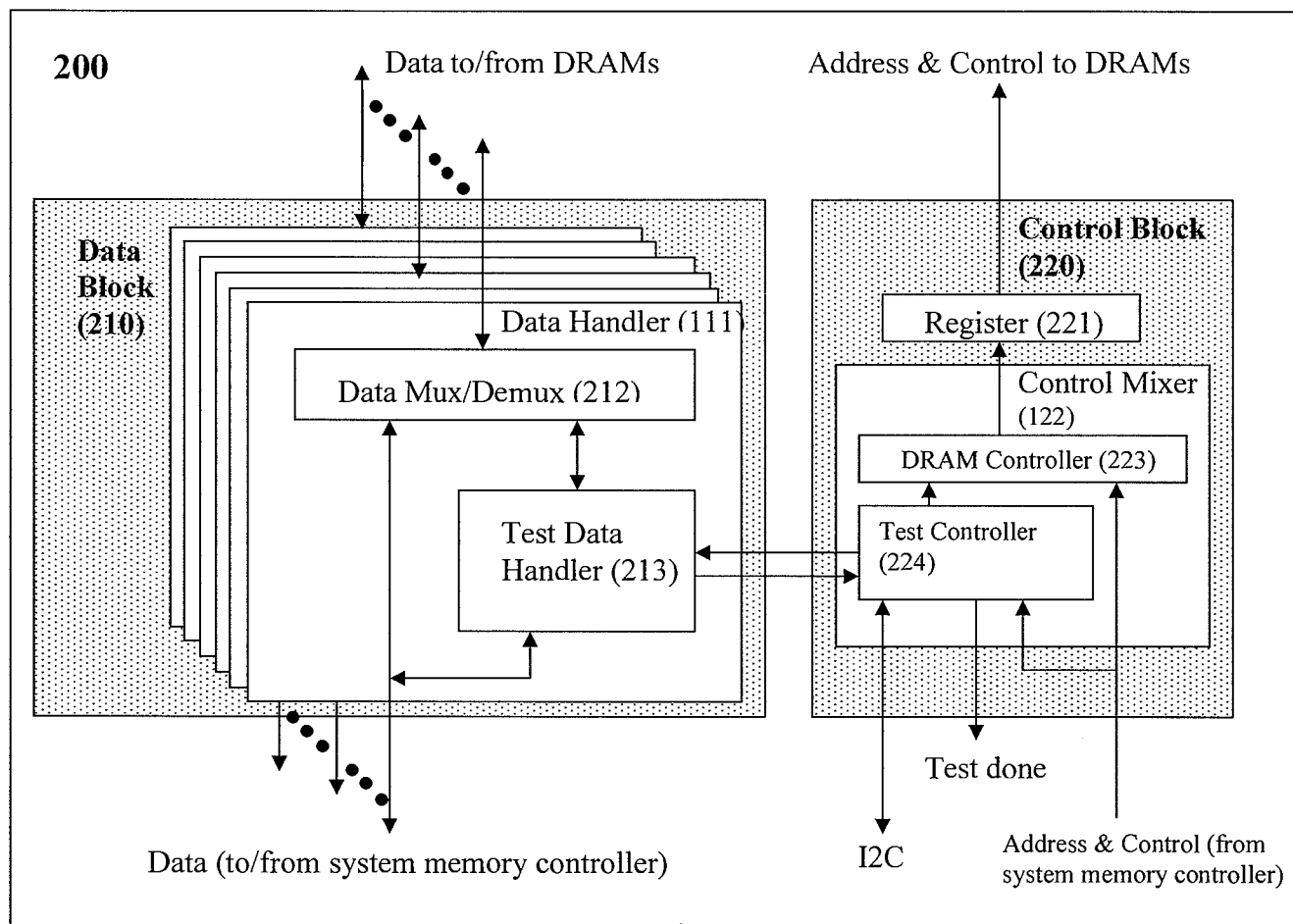


Figure 2

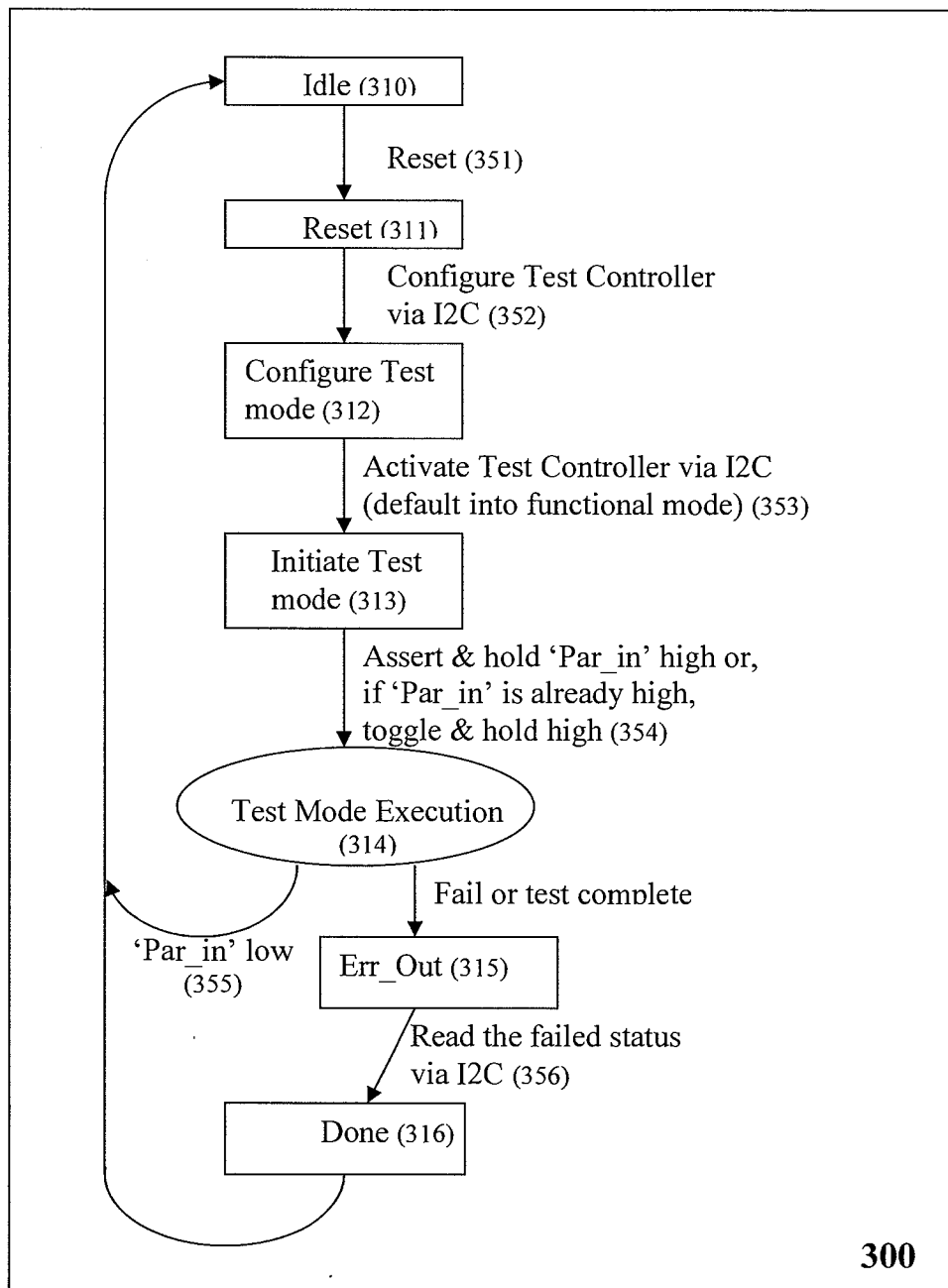


Figure 3: Example of the Test mode State Diagram

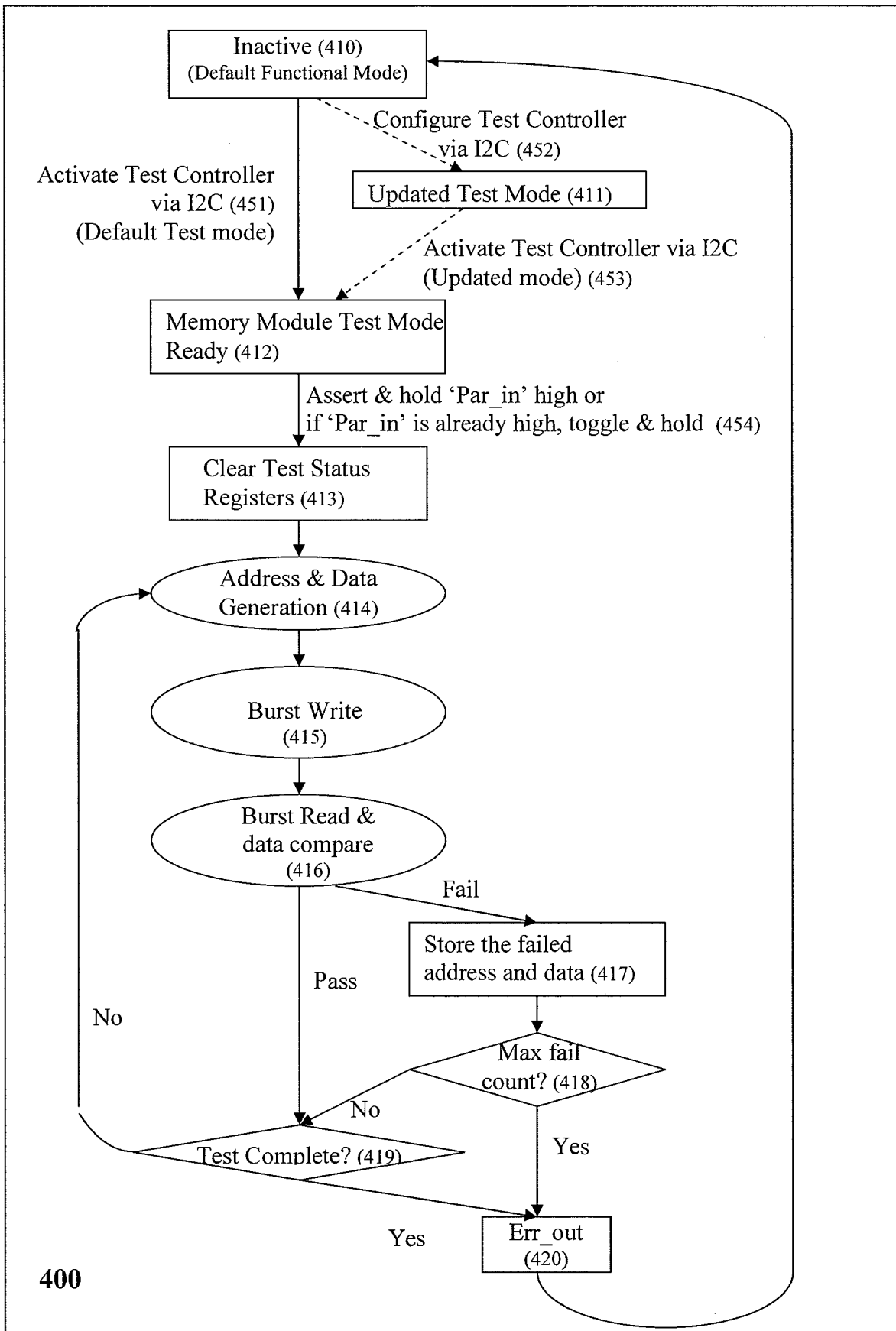


Figure 4

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

First Named Inventor/Applicant Name:

To be provided at a later date .

Filer:

Bruce S. Itchkawitz

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Provisional Filing Fees

Description

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Quantity

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Basic Filing:

Provisional application filing

1005

1

210

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Pages:

Claims:

Miscellaneous-Filing:

Petition:

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Post-Allowance-and-Post-Issuance:

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Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				210

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Application Number:	61044801
International Application Number:	
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First Named Inventor/Applicant Name:	To be provided at a later date .
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Filer:	Bruce S. Itchkawitz/Segovia Phan
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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NETL046PR_ADS.pdf	239940 b5b0f15d382b919d2c2f47e8c158dbdf9ce42f62	no	4
Warnings:					
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2		NETL046PR_APPLICATION.pdf	772497 e337f84847c62bf44922cc6aa248820682bf4bae	yes	16
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	Specification		1	11	
	Claims		12	15	
	Abstract		16	16	
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3	Drawings-only black and white line drawings	NETL046PR_DRAWINGS.pdf	99372 d70ca11eb8a6d6130f2ffa3a07c40c1b8ca69a6a	no	4
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4	Fee Worksheet (PTO-06)	fee-info.pdf	8132 c8d7f32cf135098bfe12c76dae9d3a9d9fbb477	no	2
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61/044,801	04/14/2008		NETL.046PR

CONFIRMATION NO. 1630

FORMALITIES LETTER



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Date Mailed: 04/23/2008

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FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of **\$50** for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$50** for a non-small entity

- **\$50** Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
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Alexandria VA 22313-1450

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/rerry/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,801	04/14/2008		210	NETL.046PR		

CONFIRMATION NO. 1630

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000029493456

Date Mailed: 04/23/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please write to the Office of Initial Patent Examination's Filing Receipt Corrections. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/21/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,801**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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Please Direct All Correspondence to Customer Number 20995

RESPONSE TO FORMALITIES NOTICE

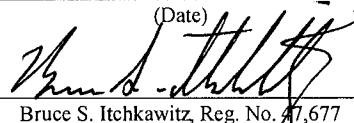
Applicant : Hyun Lee et al.
App. No : 61/044,801
Filed : April 14, 2008
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Art Unit : Unknown

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

6/16/08

(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

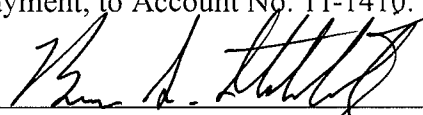
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without the inventors information and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following:

- (X) A Notice to File Missing Parts dated April 23, 2008.
- (X) Supplemental Application Data Sheet.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,801	04/14/2008		NETL.046PR

CONFIRMATION NO. 1630

FORMALITIES LETTER



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/23/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

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- \$50 Surcharge.

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/rerry/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/044,801
Filing Date:: April 14, 2008
Application Type:: Provisional
Subject Matter:: Utility
CD-ROM or CD-R?:: None
Title:: MEMORY BOARD WITH SELF-TESTING
CAPABILITY
Attorney Docket Number:: NETL.046PR
Request for Early Publication?:: No
Request for Non-Publication?:: No
Total Drawing Sheets:: 4
Small Entity?:: No
Petition Included?:: No
Secrecy Order in Parent Appl.?:: No

Inventor Information

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Hyun
Family Name:: Lee
City of Residence:: Ladera Ranch
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 21 Thalia Street

City:: Ladera Ranch
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92694

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Jayesh
Middle Name:: R.
Family Name:: Bhakta
City of Residence:: Cerritos
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 12220 Rose Street
City:: Cerritos
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 90703

Applicant Authority Type:: Inventor
Primary Citizenship Country:: Korea
Status:: Full Capacity
Given Name:: Soonju
Family Name:: Choi
City of Residence:: Irvine
State or Prov. of Residence:: CA
Country of Residence:: U.S.

Street:: 195 Woodbury
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92620

Correspondence Information

Correspondence Customer Number:: 20,995
Phone Number:: (949) 760-0404
Fax Number:: (949) 760-9502
E-Mail Address:: efiling@kmob.com

Representative Information

Representative Customer Number:: 20,995

Assignment Information

Assignee Name:: Netlist, Inc.
Street:: 51 Discovery, Ste. #150
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92618

5519888
061208

Electronic Patent Application Fee Transmittal

Application Number:	61044801			
Filing Date:	14-Apr-2008			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/alexis swan			
Attorney Docket Number:	NETL.046PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	3464654
Application Number:	61044801
International Application Number:	
Confirmation Number:	1630
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Wayne Mahoney
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046PR
Receipt Date:	16-JUN-2008
Filing Date:	14-APR-2008
Time Stamp:	19:41:33
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	4026
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	NETL_046PR.pdf	176994	no	6
			f336b1e8594d6ced85182e2b1b8b679b7a1db6b3		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	8164	no	2
			722b9f0ea5ec0de8cd1f6c163a9461ef9d3910a9		

Warnings:**Information:**

Total Files Size (in bytes):			185158
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,801	04/14/2008		260	NETL.046PR		

CONFIRMATION NO. 1630

UPDATED FILING RECEIPT



OC000000030525564

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/23/2008

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Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/21/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,801**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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